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# Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

Y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

The present document is part 4 of a multi-part Technical Specification (TS) covering the New Radio (NR) User Equipment (UE) conformance specification, which is divided in the following parts:

FFS.

# 1 Scope

The present document specifies the measurement procedures for the conformance test of the user equipment (UE) that contain performance requirements as part of 5G-NR.

The requirements are listed in different clauses only if the corresponding parameters deviate. More generally, tests are only applicable to those mobiles that are intended to support the appropriate functionality. To indicate the circumstances in which tests apply, this is noted in the "*definition and applicability*" part of the test.

For example only Release 15 and later UE declared to support 5G-NR shall be tested for this functionality. In the event that for some tests different conditions apply for different releases, this is indicated within the text of the test itself.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".

[3] 3GPP TS 38.101-2: "NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone"

[4] 3GPP TS 38.101-3: "NR; User Equipment (UE) radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios".

[5] 3GPP TS 38.101-4: "NR; User Equipment (UE) radio transmission and reception; Part 4: Performance requirements".

[6] 3GPP TS 38.508-1: "5GS; User Equipment (UE) conformance specification; Part 1: Common test environment"

[7] 3GPP TS 38.521-1: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Range 1 Standalone"

[8] 3GPP TS 38.521-2: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 2: Range 2 Standalone"

[9] 3GPP TS 38.211: "NR; Physical channels and modulation".

[10] 3GPP TS 38.212: "NR; Multiplexing and channel coding".

[11] 3GPP TS 38.213: "NR; Physical layer procedures for control".

[12] 3GPP TS 38.214: "NR; Physical layer procedures for data".

[13] 3GPP TS 37.340: "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity", Stage 2.

[14] 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities".

[15] 3GPP TR 38.901: "Study on channel model for frequencies from 0.5 to 100 GHz".

[16] 3GPP TS 36.521-1: "E-UTRA; User Equipment (UE) conformance specification; Radio transmission and reception; Part1: conformance testing"

[17] 3GPP TS 36.211: "Physical Channels and Modulation".

[18] Recommendation ITU-R M.1545: "Measurement uncertainty as it applies to test limits for the terrestrial component of International Mobile Telecommunications-2000".

[19] GPP TS 36.508: "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Packet Core (EPC); Common test environments for User Equipment (UE) conformance testing".

[20] 3GPP TS 38.331: “Radio Resource Control (RRC) protocol specification”.

[21] 3GPP TS 38.521-3: "NR; User Equipment (UE) conformance specification; Radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios"

# 3 Definition of terms, symbols and abbreviations

## 3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

**aggregated channel bandwidth:** The RF bandwidth in which a UE transmits and receives multiple contiguously aggregated carriers.

**carrier aggregation:** Aggregation of two or more component carriers in order to support wider transmission bandwidths.

**carrier aggregation band:** A set of one or more operating bands across which multiple carriers are aggregated with a specific set of technical requirements.

**carrier aggregation bandwidth class:** A class defined by the aggregated transmission bandwidth configuration and maximum number of component carriers supported by a UE.

**carrier aggregation configuration**: A combination of CA operating band(s) and CA bandwidth class(es) supported by a UE.

**DL BWP**: DL bandwidth part as defined in TS 38.213 [11].

**EN-DC**: E-UTRA-NR Dual Connectivity as defined in TS 37.340 [13, clause 4.1.2].

**FR1**: Frequency range 1 as defined in TS 38.101-3 [4] clause 5.1.

**FR2**: Frequency range 2 as defined in TS 38.101-3 [4] clause 5.1.

**PDSCH mapping type A or B:** A type of PDSCH allocation sent in the RRC message which defines the time domain allocation of PDSCH DMRS symbols. PDSCH mapping type A is slot based assignment with fixed starting OFDM symbol with variable length. PDSCH mapping type B is non-slot based assignment used for configuring min-slots.

**SSB:** SS/PBCH block as defined in TS 38.211 [9] clause 7.8.3.

## 3.2 Symbols

For the purposes of the present document, the following symbols apply:

$\mu$ Subcarrier spacing configuration as defined in TS 38.211 [9] clause 4.2]

$N_{oc}$ The power spectral density of a white noise source with average power per RE normalized to the subcarrier spacing as defined in Section 4.4.3 for conducted requirements and Section 4.5.3 for radiated requirements

## 3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

| | |
|---|---|
| CA | Carrier Aggregation |
| CC | Component Carrier |
| CCE | Control Channel Element |
| CORESET | Control Resource Set |
| CP | Cyclic Prefix |
| CSI | Channel-State Information |
| CSI-IM | CSI Interference Measurement |
| CSI-RS | CSI Reference Signal |
| CW | Codeword |
| CQI | Channel Quality Indicator |
| CRC | Cyclic Redundancy Check |
| CRI | CSI-RS Resource Indicator |
| DC | Dual Connectivity |
| DCI | Downlink Control Information |
| DL | Downlink |
| DMRS | Demodulation Reference Signal |
| DPS | Dynamic Point Selection |
| EPRE | Energy Per Resource Element |
| EN-DC | E-UTRA-NR Dual Connectivity |
| FR | Frequency Range |
| FRC | Fixed Reference Channel |
| HARQ | Hybrid Automatic Repeat Request |
| HST | High Speed Train |
| HST-SFN | High Speed Train Single Frequency Network |
| LI | Layer Indicator |
| MAC | Medium Access Control |
| MCS | Modulation and Coding Scheme |
| MIB | Master Information Block |
| NR | New Radio |
| NE-DC | NR-E-UTRA Dual Connectivity |
| NR/5GC | NR connected to 5GC NSA Non-Standalone Operation Mode |
| OCNG | OFDMA Channel Noise Generator |
| OFDM | Orthogonal Frequency Division Multiplexing |
| OFDMA | Orthogonal Frequency Division Multiple Access |
| PBCH | Physical Broadcast Channel |
| Pcell | Primary Cell |
| PDCCH | Physical Downlink Control Channel |
| PDSCH | Physical Downlink Shared Channel |
| PMI | Precoding Matrix Indicator |
| PRB | Physical Resource Block |
| PRG | Physical resource block group |
| PSS | Primary Synchronization Signal |
| PTRS | Phase Tracking Reference Signal |
| PUCCH | Physical Uplink Control Channel |
| PUSCH | Physical Uplink Shared Channel |
| QCL | Quasi Co-location |
| RB | Resource Block |
| RBG | Resource Block Group |
| RE | Resource Element |
| REG | Resource Element Group |
| RI | Rank Indicator |
| RRC | Radio Resource Control |
| SA | Standalone operation mode |
| SCS | Subcarrier Spacing |
| SINR | Signal-to-Interference-and-Noise Ratio |

| | |
|---|---|
| SNR | Signal-to-Noise Ratio |
| SS | Synchronization Signal |
| SSB | Synchronization Signal Block |
| SSS | Secondary Synchronization Signal |
| TCI | Transmission Configuration Indicator |
| TDM | Time division multiplexing |
| TTI | Transmission Time Interval |
| UL | Uplink |
| VRB | Virtual Resource Block |

# 4 General

## 4.1 Relationship between minimum requirements and test requirements

TS 38.101-4 [5] is a Single-RAT and interwork specification for NR UE, covering minimum performance requirements of both conducted and radiated requirements. Conformance to the TS 38.101-4 [5] is demonstrated by fulfilling the test requirements specified in the present document.

The Minimum Requirements given in TS 38.101-4 [5]makes no allowance for measurement uncertainty (MU). The present document defines test tolerances (TT). These test tolerances are individually calculated for each test. The test tolerances are used to relax the minimum requirements in TS 38.101-4 [5] to create test requirements. For some requirements, including regulatory requirements, the test tolerance is set to zero.

The measurement results returned by the test system are compared - without any modification - against the test requirements as defined by various levels of "Shared Risk" principle as described below

a) Core specification value is not relaxed by any relaxation value (TT=0). For each single measurement, the probability of a borderline good UE being judged as FAIL equals the probability of a borderline bad UE being judged as PASS.

   - Test tolerances equal to 0 (TT=0) are considered in this specification.

b) Core specification value is relaxed by a relaxation value (TT>0). For each single measurement, the probability of a borderline bad UE being judged as PASS is greater than the probability of a borderline good UE being judged as FAIL.

   - Test tolerances lower than measurement uncertainty and greater than 0 (0 < TT < MU) are considered in this specification.

   - Test tolerances high up to measurement uncertainty (TT = MU) are considered in this specification which is also known as “Never fail a good DUT” principle.

c) Core specification value is tightened by a stringent value (TT<0). For each single measurement, the probability of a borderline good UE being judged as FAIL is greater than the probability of a borderline bad UE being judged as PASS.

   Test tolerances lower than 0 (TT<0) are not considered in this specification..

The “Never fail a good DUT” and the “Shared Risk” principles are defined in Recommendation ITU-R M.1545 [18].

## 4.2 Applicability of minimum requirements

The applicability of each requirement is described under each clause in 5.1, 6.1, 7.1, 8.1, 9.1 and 10.1 of TS 38.101-4.

The conducted minimum requirements specified in the present document shall be met in all applicable scenarios for FR1. The radiated minimum requirements specified in the present document shall be met in all applicable scenarios for FR2. The interwork minimum requirement specified in the present document shall be met in all applicable scenarios for NR interworking operation.

All minimum performance requirements defined in Sections 5-8 are applicable to NR/5GC, EN-DC and NE-DC unless otherwise explicitly stated in Section 9 and 10.

All minimum performance requirements defined in Sections 5-10 are applicable to all UE power classes unless otherwise stated.

For radiated minimum requirements specified in the specification, if maximum achievable SNR in the TE chamber for certain test conditions is less than the defined SNR requirement for those tests, those tests will not be tested.

# 4.3 Specification suffix information

Unless stated otherwise the following suffixes are used for indicating at 2nd level clause, shown in table 4.3-1.

**Table 4.3-1: Definition of suffixes**

| Clause suffix | Variant |
|---|---|
| None | Single Carrier |
| A | Carrier Aggregation (CA) |
| B | Dual-Connectivity (DC) |
| C | Supplement Uplink (SUL) |

A terminal which supports the above features needs to meet the requirement defined in the additional clause (suffix A, B, C) in clauses 5, 6, 7, 8, 9, 10.

# 4.4 Conducted requirements

4.4.0 Introduction

The requirements are defined for the following modes:

- Mode 1: Conditions with external noise source
  - Wanted signal with power level Es is transmitted.
  - External white noise source with power spectral density Noc is used.
  - *Es* and *Noc* levels are selected to achieve target SNR as described in Clause 4.4.2.
- Mode 2: Noise free conditions
  - Wanted signal with power level Es is transmitted.
  - No external noise transmitted.

## 4.4.1 Reference point

The reference point for SNR, *Es* and Noc of DL signal is the UE antenna connector or connectors.

## 4.4.2 SNR definition

For Mode 1 conditions conducted UE demodulation and CSI requirements, the SNR is defined as:

$$SNR = \frac{\sum_{j=1}^{N_{RX}} E_s^{(j)}}{\sum_{j=1}^{N_{RX}} N_{oc}^{(j)}}$$

Where:

- $N_{RX}$ denotes the number of receiver antenna connectors and the superscript receiver antenna connector *j*.

- The above SNR definition assumes that the REs are not precoded, and does not account for any gain which can be associated to the precoding operation.
- Unless otherwise stated, the SNR refers to the SSS wanted signal.
- The downlink SSS transmit power is defined as the linear average over the power contributions in [W] of all resource elements that carry the SSS within the operating system bandwidth.
- The power ratio of other wanted signals to the SSS is defined in clause C.3.1..

## 4.4.3 Noc

### 4.4.3.1 Introduction

This clause describes the Noc power level for Mode 1 conditions conducted testing of demodulation and CSI requirements.

Unless otherwise stated for CA and EN-DC testing, the same Noc level shall be provided on different component carriers.

### 4.4.3.2 Noc for NR operating bands in FR1

The Noc power spectrum density shall be larger or equal to the minimum Noc power level for each operating band supported by the UE as defined in clause 4.4.3.2.1.

Unless otherwise stated, a fixed Noc power level of -134 dBm/Hz shall be used for all operating bands.

#### 4.4.3.2.1 Derivation of Noc values for NR operating bands in FR1

The minimum Noc power level for an operating band, subcarrier spacing and channel bandwidth is derived based on the following equation:

$$Noc_{Band_X,\ SCS_Y,\ CBW_Z} = REFSENS_{Band_X,\ SCS_Y,\ CBW_Z} - 10*\log10(12*SCS_Y*nPRB) + D - SNR_{REFSENS} + \Delta_{thermal}$$

where

- $REFSENS_{Band_X,\ SCS_Y,\ CBW_Z}$ is the REFSENS value in dBm for Band X, SCS Y and CBW Z specified in Table 7.3.2-1 of TS 38.101-1 [2]
- 12 is the number of subcarriers in a PRB
- SCS Y is the subcarrier spacing associated with the REFSENS value
- nPRB is the maximum number of PRB for SCS Y and CBW Z associated with the REFSENS value, and is specified in Table 5.3.2-1 of TS 38.101-1 [2]
- D is diversity gain equal to 3 dB
- $SNR_{REFSENS}$ = -1 dB is the SNR used for simulation of REFSENS
- $\Delta_{thermal}$ is the amount of dB that the wanted noise is set above UE thermal noise, giving a defined rise in total noise. $\Delta_{thermal}$ = 16dB, giving a rise in total noise of 0.1dB, regarded as insignificant.

The calculated Noc value for the baseline of Band n12, 15 kHz SCS, 15 MHz CBW is -135.5 dBm/Hz.

An allowance of 1.5dB is made for CA and for future bands, giving an Noc power level of -134 dBm/Hz.

## 4.4.4 Es

### 4.4.4.1 Introduction

This clause describes the Es power level for Mode 2 conditions conducted testing of demodulation and CSI requirements.

Unless otherwise stated for CA and EN-DC testing, the same Es level shall be provided on different component carriers.

#### 4.4.4.2 Es for NR operating bands in FR1

The Es power spectrum density shall be larger or equal to the minimum Es power level for each operating band supported by the UE as defined in Clause 4.4.4.2.1.

Unless otherwise stated, a fixed Es power level of -112 dBm/Hz shall be used for all operating bands.

##### 4.4.4.2.1 Derivation of Es values for NR operating bands in FR1

The minimum Es power level for an operating band, subcarrier spacing and channel bandwidth is derived based on the following equation:

$Es_{Band_X,\ SCS_Y,\ CBW_Z} = REFSENS_{Band_X,\ SCS_Y,\ CBW_Z} - 10*\log10(12*SCS_Y*nPRB) + D - SNR_{REFSENS} + dB_{EVM} + \Delta_{thermal}$

where:

- $REFSENS_{Band_X,\ SCS_Y,\ CBW_Z}$ is the REFSENS value in dBm for Band X, SCS Y and CBW Z specified in Table 7.3.2-1 of TS 38.101-1 [2]
- 12 is the number of subcarriers in a PRB
- SCS Y is the subcarrier spacing associated with the REFSENS value
- nPRB is the maximum number of PRB for SCS Y and CBW Z associated with the REFSENS value, and is specified in Table 5.3.2-1 of TS 38.101-1 [2]
- D is diversity gain equal to 3 dB
- $SNR_{REFSENS}$ = -1 dB is the SNR used for simulation of REFSENS
- $dB_{EVM}$ is the SNR of the applied signal due to EVM impairment on the wanted Es. An allowed EVM of 3% gives a $dB_{EVM}$ of 30.5dB, derived as 20*log10(1/0.03)**.**
- $\Delta_{thermal}$ is the amount of dB that the impairment due to EVM on the wanted Es is set above UE thermal noise, giving a defined rise in total impairment. $\Delta_{thermal}$ = 7.6dB, giving a rise in total impairment of 0.7dB, regarded as acceptable**.**

The calculated Es value for the baseline of Band n12, 15kHz SCS, 15MHz CBW is -113.5 dBm/Hz.

An allowance of 1.5dB is made for CA and for future bands, giving an Es power level of -112 dBm/Hz.

## 4.5 Radiated requirements

4.5.0 Introduction

The requirements are defined for the following modes:

- Mode 1: conditions with external noise source
  - Wanted signal with power level Es is transmitted.
- External white noise source with power spectral density Noc is used.
- *Es* and *Noc* levels are selected to achieve target SNR as described in Clause 4.5.2.
- Mode 2: Noise free conditions
  - Wanted signal with power level Es is transmitted.
  - No external noise transmitted.

### 4.5.1 Reference point

The reference point for SNR, Es and Noc of DL signal from the UE perspective is the input of UE antenna array.

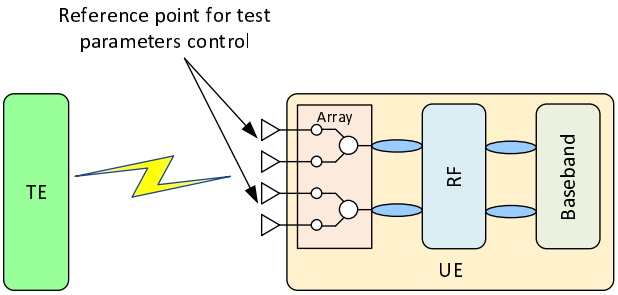



**Figure 4.5.1-1: Reference point for radiated Demodulation and CSI requirements**

### 4.5.2 SNR definition

For Mode 1 conditions UE demodulation and CSI requirements, the Minimum performance requirement in clause 7, 8, 9 and 10 are defined relative to the baseband SNR level $SNR_{BB}$. The SNR at the reference point is defined as

$SNR = SNR_{BB} + \Delta_{BB}$

where $\Delta_{BB}$ is specified in clause 4.5.3.

The reference point SNR is defined as:

$$SNR = \frac{\sum_{j=1}^{N_{RX}} E_s^{(j)}}{\sum_{j=1}^{N_{RX}} N_{oc}^{(j)}}$$

- $N_{RX}$ denotes the number of receiver reference points, and the super script receiver reference point $j$.
- The above SNR definition assumes that the REs are not precoded, and does not account for any gain which can be associated to the precoding operation.
- Unless otherwise stated, the SNR refers to the SSS wanted signal.
- The downlink SSS transmit power is defined as the linear average over the power contributions in [W] of all resource elements that carry the SSS within the operating system bandwidth.
- The power ratio of other wanted signals to the SSS is defined in clause C.3.1.

### 4.5.3 Noc

#### 4.5.3.1 Introduction

For Mode 1 conditions radiated testing of demodulation and CSI requirements it is not feasible in practice to use signal levels high enough to make the noise contribution of the UE negligible. Demodulation requirements are therefore specified with the applied noise higher than the UE peak EIS level in TS 38.101-2 [3] by a defined amount, so that the impact of UE noise floor is limited to no greater than a value $\Delta_{BB}$ at the specified Noc level. As UEs have EIS levels that are dependent on operating band and power class, Noc level is dependent on operating band and power class.

The Noc power level for test case execution shall be further increased by 5.19dB for UE power class 3 on top of the Noc power level defined in 4.5.3.2.

#### 4.5.3.2 Noc for NR operating bands in FR2

Values for Noc according to operating band and power class for single carrier requirements are specified in Table 4.5.3.2-1 for $\Delta_{BB}$ =1dB.

**Table 4.5.3.2-1: Noc power level for different UE power classes and frequency bands**

| Operating band | UE Power class | | | |
|---|---|---|---|---|
| | 1 | 2 | 3 | 4 |
| n257 | -166.8 | -163.8 | -157.6 | -166.3 |
| n258 | -166.8 | -163.8 | -157.6 | -166.3 |
| n260 | -163.8 | | -155.0 | -164.3 |
| n261 | -166.8 | -163.8 | -157.6 | -166.3 |
| Note 1: Noc levels are specified in dBm/Hz | | | | |

For PC3 multi-band devices, the Noc power level ($Noc_{MB}$) shall increase by multi-band relaxation defined in TS 38.101-2 [3] Table 6.2.1.3-4.

$Noc_{MB} = Noc_{SB} + \Sigma MB_P$

- $Noc_{SB}$ is the Noc defined in Table 4.5.3.2-1
- $\Sigma MB_P$ values are specified in TS 38.101-2 [3].

For CA case, the Noc power level ($Noc_{CA}$) shall increase by a relaxation factor defined in TS 38.101-2 [3] Table 7.3A.2.1-1:

$Noc_{CA} = Noc_{SC} + \Delta R_{IB}$

- $Noc_{SC}$ is derived by assuming UE supports single carrier.
- $\Delta R_{IB}$ values are specified in TS 38.101-2 [3].

### 4.5.3.3 Derivation of Noc values for NR operating bands in FR2

The Noc values in Table 4.5.3.2-1 are based on REFSENS for the operating band and on the UE Power class, and taking a baseline of UE Power class 3 in Band n260.

$$Noc = REFSENS_{PC3,\ n260,\ 50MHz} - 10Log_{10}(SCS_{REFSENS} \times PRB_{REFSENS} \times 12) - SNR_{REFSENS} + \Delta_{thermal}$$

where:

- $REFSENS_{PC3,\ n260,\ 50MHz}$ is the REFSENS value in dBm specified for Power Class 3 UE in Band n260 for 50MHz Channel bandwidth in TS 38.101-2 [3] Table 7.3.2.3-1.
- $SCS_{REFSENS}$ is a subcarrier spacing associated with $N_{RB}$ for 50MHz in TS 38.101-2 [3] Table 5.3.2-1, chosen as 120 kHz.
- $PRB_{REFSENS}$ is $N_{RB}$ associated with subcarrier spacing 120 kHz for 50MHz in TS 38.101-2 [3] Table 5.3.2-1 and is 32.
- 12 is the number of subcarriers in a PRB
- $SNR_{REFSSENS}$ = -1 dB is the SNR used for simulation of R EFSENS.
- $\Delta_{thermal}$ is the amount of dB that the wanted noise is set above UE thermal noise, giving a rise in total noise of $\Delta_{BB}$. $\Delta_{thermal}$ = 6 dB, giving a rise in total noise of 1 dB.

The calculated Noc value for the baseline of UE Power class 3 in Band n260 is rounded to -155 dBm/Hz.

The following methodology to define the Noc level for UE power class X (PC_X) and operating band Y (Band_Y) is used for the single carrier case and single band devices:

$$Noc(PC_X, Band_Y) = -155\ dBm/Hz + REFSENS_{PC_X,\ Band_Y,\ 50MHz} - REFSENS_{PC3,\ n260,\ 50MHz}$$

where REFSENS values are specified in TS 38.101-2 [3].

### 4.5.4 Angle of arrival

Unless otherwise stated, the downlink signal and noise are aligned to arrive in the UE Rx beam peak direction as defined in TS 38.101-2 [3].

### 4.5.5 Es

For Mode 2 the test system shall transmit the wanted signal with power level Es which is the best achievable power level by the test system.

The test system shall be able to determine achievable Es level and the maximum achievable SNR level

## 4.6 Test coverage across 5G NR connectivity options

The test cases in the present document cover both NR/5GC (including FR1+FR2 CA or FR1+FR2 NR-DC) as well as EN-DC, NE-DC and NGEN-DC testing. Below shall be the understanding with respect to coverage across 5G NR connectivity options:

1) Unless otherwise stated within the test case, it shall be understood that test requirements are agnostic of the NR/5GC, EN-DC, NE-DC and NGEN-DC connectivity options configured within the test. The test coverage across the NR/5GC, EN-DC, NE-DC and NGEN-DC connectivity options shall be considered fulfilled by executing the test case in one of these connectivity options.

2) Except for sustained data rate test cases, NR/5GC,EN-DC, NE-DC and NGEN-DC connectivity option types are utilized in the definition of each test case within this test specification. NR/5GC is the default connectivity option if supported.

Editor's Note: Generic procedure parameter to be used in Initial Conditions for NE-DC and NGEN-DC is FFS

3) If UE supports NR/5GC in addition to other connectivity options, it suffices to test the requirements using NR/5GC connectivity option for all test cases. Additionally for sustained data rate test case, if UE supports EN-DC and NE-DC, test coverage is fulfilled by testing the UE using EN-DC connectivity option.

**Table 4.6-1: Void**

**Table 4.6-2: Void**

# 5 Demodulation performance requirements (Conducted requirements)

## 5.1 General

### 5.1.1 Applicability of requirements

#### 5.1.1.1 General

The minimum performance requirements are applicable to all FR1 operating bands defined in TS 38.101-1 [2].

The minimum performance requirements in Clause 5 are mandatory for UE supporting NR operation, except test cases listed in Clauses 5.1.1.3, 5.1.1.4.

### 5.1.1.2 Applicability of requirements for different number of RX antenna ports

The number of RX antenna ports for different RF operating bands is up to UE declaration.

The UE shall support 2 or 4 RX antenna ports for different RF operating bands. The operating bands, where 4 RX antenna ports shall be the baseline, are defined in Clause 7.2 of TS 38.101-1 [3]. The UE requirements applicability for UEs with different number of RX antenna ports is defined in Table 5.1.1.2-1.

**Table 5.1.1.2-1: Requirements applicability**

| Supported RX antenna ports | Test type | Test list |
|---|---|---|
| UE supports only 2RX | PDSCH | All tests in Clause 5.2.2 |
| | PDCCH | All tests in Clause 5.3.2 |
| | PBCH | All tests in Clause 5.4.2 |
| UE supports only 4RX or both 2RX and 4RX | PDSCH | All tests in Clause 5.2.3 |
| | PDCCH | All tests in Clause 5.3.3 |
| | PBCH | All tests in Clause 5.4.2 or 5.4.3 Note 1 |
| Note 1: Requirements for PBCH with 4Rx is up to UE declaration | | |

### 5.1.1.3 Applicability of requirements for optional UE features

The performance requirements in Table 5.1.1.3-1 shall apply for UEs which support optional UE features only.

**Table 5.1.1.3-1: Requirements applicability for optional UE features**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| SU-MIMO Interference Mitigation advanced receiver | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 3-1)<br><br>Clause 5.2.3.1.1 (Test 5-1) | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Test 3-1)<br><br>Clause 5.2.3.2.1 (Test 5-1) | |
| Alternative additional DMRS position for co-existence with LTE CRS (*additionalDMRS-DL-Alt)* | FR1 FDD | PDSCH | Clause 5.2.2.1.4 (Test 1-2)<br>Clause 5.2.3.1.4 (Test 1-2) | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.4 (Test 1-2)<br><br>Clause 5.2.3.2.4 (Test 1-2) | |
| Basic DL NR-NR CA operation (*supportedBandCombinationList*) | NR CA | SDR | Clause 5.5A.1 | 1) Up to 16 DL carriers<br>2) Same numerology across carrier for data/control channel at a given time |
| Enhanced demodulation processing for HST-SFN joint transmission scheme with velocity up to 500km/h | FR1 FDD | PDSCH | Clause 5.2.2.1.9 (Test 1-1)<br><br>Clause 5.2.3.1.9 (Test 1-1) | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.9 (Test 1-1)<br><br>Clause 5.2.3.2.9 (Test 1-1) | |

| | | | | |
|---|---|---|---|---|
| Single DCI based SDM transmission for multi-TRxP (singleDCI-SDM-scheme-r16) | FR1 FDD | PDSCH | Clause 5.2.2.1.11<br>Clause 5.2.3.1.11 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.11<br>Clause 5.2.3.2.11 | |
| Multi DCI based multi-TRxP support (multiDCI-MultiTRP-r16) | FR1 FDD | PDSCH | Clause 5.2.2.1.12<br>Clause 5.2.3.1.12 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.12<br>Clause 5.2.3.2.12 | |
| Single DCI based FDM Scheme-A for multi-TRxP(supportFDM-SchemeA-r16) | FR1 FDD | PDSCH | Clause 5.2.2.1.13<br>Clause 5.2.3.1.13 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.13<br>Clause 5.2.3.2.13 | |
| Single DCI based inter-slot TDM for multi-TRxP (supportInter-slotTDM-r16) | FR1 FDD | PDSCH | Clause 5.2.2.1.14<br>Clause 5.2.3.1.14 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.14<br>Clause 5.2.3.2.14 | |
| Maximum number of TCI states in Single-DCI based inter-slot TDM (maxNumberTCI-states-r16) | FR1 FDD | PDSCH | Clause 5.2.2.1.14<br>Clause 5.2.3.1.14 | The requirements apply only when maxNumberTCI-states-r16 = 2. |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.14<br>Clause 5.2.3.2.14 | |
| Alternative 64QAM MCS table for PDSCHNew 64QAM MCS table for PDSCH (*dl-64QAM-MCS-TableAlt*) | FR1 FDD | PDSCH | Clause 5.2.2.1.5<br>Clause 5.2.3.1.5<br>Clause 5.2.2.1.6<br>Clause 5.2.3.1.6 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.5<br>Clause 5.2.3.2.5<br>Clause 5.2.2.2.6<br>Clause 5.2.3.2.6 | |
| CQI table with target BLER of 10^-5New CQI table (cqi-TableAlt) | FR1 FDD | PDSCH | Clause 5.2.2.1.5<br>Clause 5.2.3.1.5 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.5<br>Clause 5.2.3.2.5 | |
| PDSCH repetitions over multiple slots *(pdsch-RepetitionMultiSlots)* | FR1 FDD | PDSCH | Clause 5.2.2.1.6<br>Clause 5.2.3.1.6 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.6<br>Clause 5.2.3.2.6 | |
| UE PDSCH processing capability #2 *(pdsch-ProcessingType2)* | FR1 FDD | PDSCH | Clause 5.2.2.1.7<br>Clause 5.2.3.1.7 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.7<br>Clause 5.2.3.2.7 | |
| Pre-emption indication for DL *(pre-EmptIndication-DL)* | FR1 FDD | PDSCH | Clause 5.2.2.1.8<br>Clause 5.2.3.1.8 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.8<br>Clause 5.2.3.2.8 | |

### 5.1.1.4 Applicability of requirements for mandatory UE features with capability signalling

The performance requirements in Table 5.1.1.4-1 shall apply for UEs which support mandatory UE features with capability signalling only.

**Table 5.1.1.4-1: Requirements applicability for mandatory features with UE capability signalling**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| 256QAM modulation scheme for PDSCH for FR1 (*pdsch-256QAM-FR1*) | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-3) Clause 5.2.3.1.1 (Test 1-3) | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Test 1-3) Clause 5.2.3.2.1 (Test 1-3) | |
| PDSCH mapping type B (*pdsch-MappingTypeB*) | FR1 FDD | PDSCH | Clause 5.2.2.1.3 Clause 5.2.3.1.3 Clause 5.2.2.1.7 Clause 5.2.3.1.7 | |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.3 Clause 5.2.3.2.3 Clause 5.2.2.2.7 Clause 5.2.3.2.7 | |
| Rate-matching around LTE CRS (rateMatchingLTE-CRS) | FR1 FDD | PDSCH | Clause 5.2.2.1.4 Clause 5.2.3.1.4 | For UEs supporting “Alternative additional DMRS position for co-existence with LTE CRS”, if Test 1-2 is tested, the test coverage can be considered fulfilled without executing Test 1-1. Otherwise, only Test 1-1 is tested. |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.4 Clause 5.2.3.2.4 | |
| Supported maximum number of ports across all configured NZP-CSI-RS resources per CC (maxConfigNumberPortsAcrossNZP-CSI-RS-PerCC) | FR1 FDD | PDSCH | Clause 5.2.2.1.4 (Tests 1-1, 1-2) Clause 5.2.3.1.1 (Tests 3-1, 4-1, 5-1) Clause 5.2.3.1.4 (Tests 1-1, 1-2) | The requirements apply only in case the number of NZP-CSI-RS ports in the test case satisfies UE capability on maximum number of NZP-CSI-RS ports |
| | FR1 TDD | PDSCH | Clause 5.2.3.2.1 (Tests 3-1, 4-1, 5-1) | |

| | | | | |
|---|---|---|---|---|
| Supported maximum number of PDSCH MIMO layers (*maxNumberMIMO-LayersPDSCH*) | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Tests 2-1, 2-2, 3-1) Clause 5.2.2.1.2 Clause 5.2.3.1.1 (Tests 2-1, 2-2, 3-1, 4-1, 5-1) Clause 5.2.3.1.2 | The requirements apply only in case the PDSCH MIMO rank in the test case does not exceed UE PDSCH MIMO layers capability |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Tests 2-1, 2-2, 3-1) Clause 5.2.2.2.2 Clause 5.2.3.2.1 (Tests 2-1, 2-2, 3-1, 4-1, 5-1) Clause 5.2.3.2.2 | |
| Support number of active TCI states per BWP per CC, including control and data *(maxNumberActiveTCI-PerBWP)* | FR1 FDD | PDSCH | Clause 5.2.2.1.10 (Test 1-2) Clause 5.2.3.1.10 (Test 1-2) | The requirements apply only when *maxNumberActiveTCI-PerBWP* is other than n1. |
| | FR1 TDD | PDSCH | Clause 5.2.2.2.10 (Test 1-2) Clause 5.2.3.2.10 (Test 1-2) | |
| Support for maximum number of TRS resource sets per CC which the UE can track simultaneously (*maxSimultaneousResourceSetsPerCC*) | FR1 FDD | PDSCH | Clause 5.2.2.1.10 (Test 1-2) Clause 5.2.3.1.10 (Test 1-2) Clause 5.2.2.1.11 Clause 5.2.2.1.12 Clause 5.2.2.1.13 Clause 5.2.2.1.14 Clause 5.2.3.1.11 Clause 5.2.3.1.12 Clause 5.2.3.1.13 Clause 5.2.3.1.14 | The requirements apply only when *maxSimultaneousResourceSetsPerCC* ≥ 2 |

| | | | | |
|---|---|---|---|---|
| | FR1 TDD | PDSCH | Clause 5.2.2.2.10 (Test 1-2)<br>Clause 5.2.3.2.10 (Test 1-2)<br>Clause 5.2.2.2.11<br>Clause 5.2.2.2.12<br>Clause 5.2.2.2.13<br>Clause 5.2.2.2.14<br>Clause 5.2.3.2.11<br>Clause 5.2.3.2.12<br>Clause 5.2.3.2.13<br>Clause 5.2.3.2.14 | |

## 5.1.1.5 Applicability of CA requirements

### 5.1.1.5.1 Definition of CA capability

The definition with respect to CA capabilities is given as in Table 5.1.1.5.1-1.

**Table 5.1.1.5.1-1: Definition of CA capability**

| CA Capability | CA Capability Description |
|---|---|
| CA_C | Intra-band contiguous CA |
| CA_N | Intra-band non-contiguous CA |
| CA_AX | Inter-band CA (X bands) |
| NOTE 1: CA_C corresponds to NR CA configurations and bandwidth combination sets defined in Section 5.5A.1 of TS 38.101-1[2].<br>CA_N corresponds to NR CA configurations and bandwidth combination sets defined in Section 5.5A.2 of TS 38.101-1[2].<br>CA_AX corresponds to NR CA configurations and bandwidth combination sets defined in Section 5.5A.3 of TS 38.101-1[2]. | |

### 5.1.1.5.2 Applicability and test rules for different CA configurations and bandwidth combination sets

The performance requirement for CA UE demodulation tests in Section 5.2A are defined independent of CA configurations and bandwidth combination sets specified in Section 5.5A of TS 38.101-1[2]. For UEs supporting different CA configurations and bandwidth combination sets, the applicability and test rules are defined in Table 5.1.1.5.2-1 and Table 5.1.1.5.2-2. For simplicity, CA configuration below refers to combination of CA configuration and bandwidth combination set.

**Table 5.1.1.5.2-1: Applicability and test rules for CA UE demodulation tests**

| Tests | CA capability where the tests apply | CA configuration from the selected CA capability where the tests apply | CA Bandwidth combination to be tested in priority order | PCell CC configuration |
|---|---|---|---|---|
| Test 1 in Section 5.2A.2.1 and 5.2A.3.1 | CA_C, CA_N, CA_AX | Table 5.1.1.5.2-2 | Largest aggregated CA bandwidth combination | Any of CCs |
| Test 2 in Section 5.2A.2.1 and 5.2A.3.1 | CA_C, CA_N, CA_AX | Table 5.1.1.5.2-2 | Largest aggregated CA bandwidth combination | Any of CCs |
| Test 3 in Section 5.2A.2.1 and 5.2A.3.1 | CA_AX | Table 5.1.1.5.2-2 | Largest aggregated CA bandwidth combination | TDD CC if supported, otherwise FDD CC |
| Test 4 in Clause 5.2A.2.1 and 5.2A.3.1 (NOTE 2) | CA_AX | Table 5.1.1.5.2-2 | Largest aggregated CA bandwidth combination | Any of CCs |
| Test 5 in Section 5.2A.2.1 and 5.2A.3.1 (NOTE 3) | CA_AX | Table 5.1.1.5.2-2 | Largest aggregated CA bandwidth combination | 15 kHz CC is supported, otherwise 30 kHz CC |

NOTE 1: In case CA_AX with different number of X is supported, [scenarios with maximum number of X and with the largest aggregated channel bandwidth are tested].
NOTE 2: These scenarios are only tested for UEs which are not verified with Test 3 in Section 5.2A.2.1 and 5.2A.3.1
NOTE 3: These scenarios are only tested for UEs which are not verified with Test 4 in Section 5.2A.2.1 and 5.2A.3.1

**Table 5.1.1.5.2-2: Selection of CA configurations**

| CA capability | Step 1 | Step 2 | Step 3 | Step 4 |
|---|---|---|---|---|
| CA_C or CA_N | Select the CA configurations with the maximum number of CCs, for which the supported maximum number of MIMO layers is not lower than 2. | Select any one of CA configurations, which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate is not lower than the tested date rate, among all the selected CA configurations from Step 1. | N/A | N/A |
| CA_AX | Select the CA configurations with the maximum number of CCs, for which the supported maximum number of MIMO layers is not lower than 2. | Select any one of CA configurations, which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate is not lower than the tested date rate, among all the selected CA configurations from Step 1. | Select the CA configurations with the largest number of bands and with the maximum number of CCs, for which the supported maximum number of MIMO layers is not lower than 2. | Select any one of CA configurations, which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate is not lower than the tested date rate, among all the selected CA configurations from Step 3. |

NOTE 1: For CA_AX capability, if CA configuration from step 2 is CA configuration with the largest number of bands then Step 3 and Step 4 are skipped. Otherwise, the two CA configurations selected from Step 2 and Step 4 are used for testing.
NOTE 2: Maximum supported data rate for Step 2 and Step 4 is calculated based clause 4.1.2 of TS 38.306 [14].
NOTE 3: Tested data rate for Step 2 and Step 4 is calculated based on the equation $DataRate = 10^{-3}\sum_{j=1}^{J} TBS_j 2^{\mu_j}$ and FRCs used in the test.

#### 5.1.1.5.3 Antenna connection for CA tests with 4 RX

FFS

### 5.1.1.6 Applicability and test rules for PDSCH performance requirements with power imbalance for intra-band contiguous CA

For UE passing the FDD and TDD CA power imbalance performance requirements with 2 DL CCs as defined in sections 5.2A.2.2 and 5.2A.3.2, the test coverage can be considered fulfilled with FDD or TDD intra-band contiguous CA with 3 or more DL CCs supported by the UE. During the test, UE is required to test the supported intra-band contiguous CA configurations with 2 DL CCs covering the lowest and highest operating bands.

The channel bandwidth combination for testing is determined by following procedure:

- First select the bandwidth combinations with the same bandwidth in each carrier.
  - If there is no such bandwidth combination, select the bandwidth combinations with smallest bandwidth difference between the two carriers, and the carrier with smaller bandwidth will be used for test.
- Among the bandwidth combinations selected, select the CA combination with largest aggregated bandwidth combination.

### 5.1.1.7 Applicability of different requirements for HST

The applicability rules for different HST requirements in section 5 are specified in Table 5.1.1.7-1.

**Table 5.1.1.7-1: Applicability of requirements for HST**

| If UE has passed | | | UE can skip | | | Applicability notes |
|---|---|---|---|---|---|---|
| Test type | | Test list | Test type | | Test list | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-6) | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-5) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Test 1-11) | FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Test 1-7) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.1 (Test 1-6) | FR1 FDD | PDSCH | Clause 5.2.3.1.1 (Test 1-5) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.1 (Test 1-11) | FR1 TDD | PDSCH | Clause 5.2.3.2.1 (Test 1-7) | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.9 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-5 ) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.9 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Test 1-7 and 1-11) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.9 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.3.1.1 (Test 1-5) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.9 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.3.2.1 (Test 1-7 and 1-11) | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-7) | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.1 (Test 1-7) | FR1 FDD | PDSCH | Clause 5.2.3.1.1 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.10 (Test 1-1 or 1-2) | FR1 FDD | PDSCH | Clause 5.2.2.1.1 (Test 1-5 ) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.10 (Test 1-1 or 1-2) | FR1 TDD | PDSCH | Clause 5.2.2.2.1 (Test 1-7 and 1-11) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.10 (Test 1-1 or 1-2) | FR1 FDD | PDSCH | Clause 5.2.3.1.1 (Test 1-5) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.10 (Test 1-1 or 1-2) | FR1 TDD | PDSCH | Clause 5.2.3.2.1 (Test 1-7 and 1-11) | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.10 (Test 1-2) | FR1 FDD | PDSCH | Clause 5.2.2.1.10 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.10 (Test 1-2) | FR1 TDD | PDSCH | Clause 5.2.2.2.10 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.10 (Test 1-2) | FR1 FDD | PDSCH | Clause 5.2.3.1.10 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.10 (Test 1-2) | FR1 TDD | PDSCH | Clause 5.2.3.2.10 (Test 1-1) | |

### 5.1.1.8 Applicability of different requirements with Multi-TRxP

The applicability rules for requirements with multi-TRxP transmission schemes in section 5 are specified in Table 5.1.1.8-1.

**Table 5.1.1.8-1: Applicability of requirements with Multi-TRxP Transmission**

| If UE has passed | | | UE can skip | | | Applicability notes |
|---|---|---|---|---|---|---|
| Test type | | Test list | Test type | | Test list | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.12 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.2.1.11 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.12 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.2.1.13 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.2.1.6 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.2.1.14 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.12 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.2.2.11 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.12 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.2.2.13 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.2.2.6 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.2.2.14 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.12 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.3.1.11 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.12 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.3.1.13 (Test 1-1) | |
| FR1 FDD | PDSCH | Clause 5.2.3.1.6 (Test 1-1) | FR1 FDD | PDSCH | Clause 5.2.3.1.14 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.12 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.3.2.11 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.12 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.3.2.13 (Test 1-1) | |
| FR1 TDD | PDSCH | Clause 5.2.3.2.6 (Test 1-1) | FR1 TDD | PDSCH | Clause 5.2.3.2.14 (Test 1-1) | |

# 5.2 PDSCH demodulation requirements

The parameters specified in Table 5.2-1 are valid for all PDSCH tests unless otherwise stated.

**Table 5.2-1: Common test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| Carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 2) | RBs | 0 |
| | Subcarrier spacing | kHz | 15 or 30 |
| DL BWP configuration #1 | Cyclic prefix | | Normal |
| | RB offset | RBs | 0 |
| | Number of contiguous PRB | PRBs | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | Symbols | 0, 1 |
| | Number of PRBs in CORESET | | Table 5.2-2 for tested channel bandwidth and subcarrier spacing |
| | Number of PDCCH candidates and aggregation levels | | 1/AL8 |
| | CCE-to-REG mapping type | | Non-interleaved |
| | DCI format | | 1_1 |
| | TCI state | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | Single Panel Type I, Random per slot with equal probability of each applicable $i_1$, $i_2$ combination, and with REG bundling granularity for number of Tx larger than 1 |
| Cross carrier scheduling | | | Not configured |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | $k_0$=0 for CSI-RS resource 1,2,3,4 |
| | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 1 and 3<br>$l_0$ = 10 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4 |
| | Density (ρ) | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | Slots | 15 kHz SCS: 20 for CSI-RS resource 1,2,3,4<br>30 kHz SCS: 40 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | Slots | 15 kHz SCS:<br>10 for CSI-RS resource 1 and 2<br>11 for CSI-RS resource 3 and 4<br><br>30 kHz SCS:<br>20 for CSI-RS resource 1 and 2<br>21 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | First subcarrier index in the PRB used for CSI-RS | | $k_0$ = 0 |
| | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 12 |
| | Number of CSI-RS ports (X) | | Same as number of transmit antenna |
| | CDM Type | | 'No CDM' for 1 transmit antenna<br>'FD-CDM2' for 2 and 4 transmit antenna |
| | Density (ρ) | | 1 |
| | CSI-RS periodicity | Slots | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | Slots | 0 |
| | Frequency Occupation | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | TCI state #1 |

| | | | | |
|---|---|---|---|---|
| ZP CSI-RS for CSI acquisition | First subcarrier index in the PRB used for CSI-RS | | | $k_0 = 4$ |
| | First OFDM symbol in the PRB used for CSI-RS | | | $l_0 = 12$ |
| | Number of CSI-RS ports (X) | | | 4 |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | Slots | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | Slots | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| PDSCH DMRS configuration | Antenna ports indexes | | | {1000} for Rank 1 tests<br>{1000, 1001} for Rank 2 tests<br>{1000-1002} for Rank 3 tests<br>{1000-1003} for Rank 4 tests |
| | Position of the first DMRS for PDSCH mapping type A | | | 2 |
| | Number of PDSCH DMRS CDM group(s) without data | | | 1 for Rank 1 and Rank 2 tests<br>2 for Rank 3 and Rank 4 tests |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| PT-RS configuration | | | | PT-RS is not configured |
| Maximum number of code block groups for ACK/NACK feedback | | | | 1 |
| Maximum number of HARQ transmission | | | | 4 |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| PDSCH & PDSCH DMRS Precoding configuration | | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with PRB bundling granularity |
| Symbols for all unused REs | | | | OCNG Annex A.5 |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |
| Note 1: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission.<br>Note 2: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing. | | | | |

**Table 5.2-2: Number of PRBs in CORESET**

| SCS (kHz) | 5 MHz | 10 MHz | 15 MHz | 20 MHz | 25 MHz | 30 MHz | 40 MHz | 50 MHz | 60 MHz | 80 MHz | 100 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 24 | 48 | 78 | 102 | 132 | 156 | 216 | 270 | N/A | N/A | N/A |
| 30 | 6 | 24 | 36 | 48 | 60 | 78 | 102 | 132 | 162 | 216 | 270 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2

## 5.2.1 1RX requirements (Void)

## 5.2.2 2RX requirements

### 5.2.2.1 FDD

#### 5.2.2.1.1 2Rx FDD FR1 PDSCH mapping Type A performance

##### 5.2.2.1.1.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.1.0-3 and Table 5.2.2.1.1.0-4, with the test parameters defined in table 5.2.2.1.1.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.1.1.0-1.

**Table 5.2.2.1.1.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers | 1-1, 1-2, 1-3, 1-5, 1-6, 1-7, 2-1, 2-2 |
| Verify the PDSCH mapping Type A HARQ soft combining performance under 2 receive antenna conditions. | 1-4 |
| Verify the PDSCH mapping Type A performance requirements for Enhanced Receiver Type 1 under 2 receive antenna conditions. | 3-1 |

**Table 5.2.2.1.1.0-2: Test Parameters for Testing**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 4 for Test 1-1<br>2 for other tests |
| | Resource allocation type | | Test 1-2: Type 1 with start RB = 23, $L_{RBs}$ = 6<br>Other tests: Type 0 |
| | RBG size | | Test 1-2: N/A<br>Other tests: Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 for Tests 1-1, 1-5, 1-6, 1-7<br>1 for other tests |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | CSI-RS periodicity | Slots | Test 1-5, 1-6, 1-7:<br>10 for CSI-RS resource 1,2,3,4.<br><br>Other tests: Table 5.2-1. |
| | CSI-RS offset | Slots | Test 1-5, 1-6, 1-7:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4.<br><br>Other tests: Table 5.2-1. |

| | | |
|---|---|---|
| Number of HARQ Processes | | 8 for Test 1-4<br>4 for other tests |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | 2 |

**Table 5.2.2.1.1.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 FDD | 10 / 15 | QPSK, 0.30 | TDLB100-400 | 2x2, ULA Low | 70 | -0.8 |
| 1-2 | R.PDSCH.1-1.2 FDD | 10 / 15 | QPSK, 0.30 | TDLC300-100 | 2x2, ULA Low | 70 | 0.2 |
| 1-3 | R.PDSCH.1-4.1 FDD | 10 / 15 | 256QAM, 0.82 | TDLA30-10 | 2x2, ULA Low | 70 | 24.6 |
| 1-4 | R.PDSCH.1-2.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 2x2, ULA Low | 30 | 1.1 |
| 1-5 | R.PDSCH.1-8.1 FDD | 10 / 15 | 16QAM, 0.48 | HST-750 | 1x2 | 70 | 6.2 |
| 1-6 | R.PDSCH.1-8.2 FDD | 10 / 15 | 64QAM, 0.43 | HST-972 | 1x2 | 70 | [9.9] |
| 1-7 | R.PDSCH.1-8.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-600 | 2x2 | 70 | [8.6] |

**Table 5.2.2.1.1.0-4: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 2-1 | R.PDSCH.1-3.1 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 19.4 |
| 2-2 | R.PDSCH.2-1.1 FDD | 20 / 30 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 19.7 |

**Table 5.2.2.1.1.0-5: Minimum performance for Rank 2 and Enhanced Receiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.1-2.2 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Medium | 70 | 17.6 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.1.

5.2.2.1.1_1 2Rx FDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's Note: This test cases is incomplete in following aspects:

- SNR in test requirements table is within square brackets for test point 1-6, 1-7.

#### 5.2.2.1.1_1.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 1 and Rank 2 scenarios.

#### 5.2.2.1.1_1.2 Test applicability

This test applies to all types of UE release 15 and forward supporting NR/5GC.

This test also applies to all types of UE release 15 and forward supporting EN-DC.

#### 5.2.2.1.1_1.3 Test description

##### 5.2.2.1.1_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.1.1.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for NR/5GC with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for EN-DC according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.1_1.3.3.

##### 5.2.2.1.1_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.2.1.1_1.4-1 and 5.2.2.1.1_1.4-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.1.1_1.4-1 and 5.2.2.1.1_1.4-2 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Table s 5.2.2.1.1_1.4-1 and 5.2.2.1.1_1.4-2 as appropriate.

##### 5.2.2.1.1_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

5.2.2.1.1_1.3.3_1 Message exceptions for NR/5GC

**Table 5.2.2.1.1_1.3.3_1-1: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | n4, n2<br>Not present | n4 for test 1-1<br>n2 will be used by default | test 1-1<br>test point other than test 1-1 |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.1_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | For test 1-1, 1-5, 1-6 and 1-7 | |
| | pos1 | For other tests | |
| } | | | |

**Table 5.2.2.1.1_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n8, n4 | n8 for Test 1-4<br>n4 for other tests | |
| } | | | |

**Table 5.2.2.1.1_1.3.3_1-4: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots10 | 1 (for CSI-RS resources 1 and 2)<br>2 (for CSI-RS resources 3 and 4) | For test 1-5, 1-6, 1-7:<br>CSI-RS offset:<br>1 for CSI-RS resources 1 and 2<br>2 for CSI-RS resources 3 and 4<br>CSI-RS periodicity: 10 slots | |
| slots40 | 20 (for CSI-RS resources 1 and 2)<br>21 (for CSI-RS resources 3 and 4) | For test 2-2:<br>CSI-RS offset:<br>20 for CSI-RS resources 1 and 2<br>21 for CSI-RS resources 3 and 4<br>CSI-RS periodicity: 40 slots | |
| slots20 | 10 (for CSI-RS resources 1 and 2)<br>11 (for CSI-RS resources 3 and 4) | For other tests:<br>CSI-RS offset:<br>10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4<br>CSI-RS periodicity: 20 slots | |
| } | | | |

### 5.2.2.1.1_1.3.3_2 Message exceptions for EN-DC

Same as 5.2.2.1.1_1.3.3_1

### 5.2.2.1.1_1.4 Test requirement

Tables 5.2.2.1.1_1.4-1 and 5.2.2.1.1_1.4-2 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.1_1.4-1 and Table 5.2.2.1.1_1.4-2 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.1_1.4-1: Test Requirements for Rank 1**

| Test num. | Reference channel | Modulation format | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 FDD | QPSK, 0.30 | TDLB100-400 | 2x2, ULA Low | 70 | 0.1 |
| 1-2 | R.PDSCH.1-1.2 FDD | QPSK, 0.30 | TDLC300-100 | 2x2, ULA Low | 70 | 1.1 |
| 1-3 | R.PDSCH.1-4.1 FDD | 256AM, 0.82 | TDLA30-10 | 2x2, ULA Low | 70 | 25.6 |
| 1-4 | R.PDSCH.1-2.1 FDD | 16QAM, 0.48 | TDLC300-100 | 2x2, ULA Low | 30 | 2 |
| 1-5 | R.PDSCH.1-8.1 FDD | 16QAM, 0.48 | HST-750 | 1x2 | 70 | 7.1 |
| 1-6 | R.PDSCH.1-8.2 FDD | 64QAM, 0.43 | HST-972 | 1x2 | 70 | [10.5] |

**Table 5.2.2.1.1_1.4-2: Test Requirements for Rank 2**

| Test num. | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 2-1 | R.PDSCH.1-3.1 FDD | 64QAM, 0.51 | TDLA30-10 | 2x2, ULA Low | 70 | 20.4 |
| 2-2 | R.PDSCH.2-1.1 FDD | 64QAM, 0.51 | TDLA30-10 | 2x2, ULA Low | 70 | 20.7 |

### 5.2.2.1.1_2 2Rx FDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with enhanced receiver type 1 for both SA and NSA

#### 5.2.2.1.1_2.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with enhanced receiver type 1 configuration, for Rank 2 scenarios.

#### 5.2.2.1.1_2.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting NR enhanced receiver type 1.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and NR enhanced receiver type 1.

#### 5.2.2.1.1_2.3 Test description

Same test description as in clause 5.2.2.1.1_1.3.

##### 5.2.2.1.1_2.3.1 Initial conditions

Same initial conditions as in clause 5.2.2.1.1_1.3.1.

##### 5.2.2.1.1_2.3.2 Test procedure

Same test procedure as in clause 5.2.2.1.1_1.3.2.

##### 5.2.2.1.1_2.3.3 Message contents

Same message contents as in clause 5.2.2.1.1_1.3.3.

###### 5.2.2.1.1_2.3.3_1 Message exceptions for SA

Same message exceptions for SA as in clause 5.2.2.1.1_1.3.3_1.

###### 5.2.2.1.1_2.3.3_2 Message exceptions for NSA

Same message exceptions for NSA as in clause 5.2.2.1.1_1.3.3_2.

##### 5.2.2.1.1_2.3.4 Test requirement

Same test requirement as in clause 5.2.2.1.1_1.3.4.

**Table 5.2.2.1.1_2.3.4-1: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3-1 | R.PDSCH.1-2.2 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Medium | 70 | 18.6 |

### 5.2.2.1.2 2Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance

#### 5.2.2.1.2.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.2.0-3, with the addition of test parameters in table 5.2.2.1.2.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.1.2.0-1.

**Table 5.2.2.1.2.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| [Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and CSI-RS overlapped with PDSCH] | 1-1 |

**Table 5.2.2.1.2.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | PRB size | | Config2 |
| | Resource allocation type | | Type 0 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Length | | 1 |
| NZP CSI-RS for CSI acquisition | OFDM symbols in the PRB used for CSI-RS | | $l_0 = 13$ |
| | CSI-RS periodicity | Slots | 5 |
| ZP CSI-RS for CSI acquisition | Subcarrier index in the PRB used for CSI-RS | | $(k_0, k_1, k_2, k_3)=(2, 4, 6, 8)$ |
| | Number of CSI-RS ports (X) | | 8 |
| | CSI-RS periodicity | Slots | 5 |
| Number of HARQ Processes | | | 4 |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | 2 |

**Table 5.2.2.1.2.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-5.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 2x2, ULA Low | 70 | 14.8 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.2.1.2.

### 5.2.2.1.2_1 2Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.1.2_1.1 Test purpose

Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and CSI-RS overlapped with PDSCH

#### 5.2.2.1.2_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 5.2.2.1.2_1.3 Test description

##### 5.2.2.1.2_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.6.2 for TE diagram and section A.3.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.1.1.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.2_1.3.3.

##### 5.2.2.1.2_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.2_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.2_1.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Annex G.1.4.

##### 5.2.2.1.2_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

5.2.2.1.2_1.3.3_1 Message exceptions for SA

Same as for test number 1-2 in 5.2.2.1.1_1.3.3_1 with following exceptions:

**Table 5.2.2.1.2_1.3.3_1-1: NZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 13 | $l_0 = 13$ | |
| } | | | |

**Table 5.2.2.1.2_1.3.3_1-2: CSI-ResourcePeriodicityAndOffset for ZP and NZP CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots5 | 0 | CSI-RS offset: 0 CSI-RS periodicity: 5 slots | |
| } | | | |

**Table 5.2.2.1.2_1.3.3_1-3: ZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], clause5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | (k0, k1, k2, k3)=(2, 4, 6, 8) | |
| } | | | |
| nrofPorts | P8 | Number of CSI-RS ports (X) = 8 | |
| firstOFDMSymbolInTimeDomain | 12 | $l_0 = 12$ | |
| cdm-Type | fd-CDM2 | | |
| density CHOICE { | | | |
| one | NULL | Density (ρ) = 1 | |
| } | | | |
| freqBand | CSI-FrequencyOccupation | Frequency Occupation: Start PRB 0 (see Table 4.6.3-33 in TS 38.508-1) Number of PRB = 52 (see Table 5.4.2.0-23:in TS 38.508-1 [6]. | |
| } | | | |

**Table 5.2.2.1.2_1.3.3_1-4: Void**

5.2.2.1.2_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.2_1.3.3_1

#### 5.2.2.1.2_1.4 Test requirement

Table 5.2.2.1.2.0-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.2_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.2_1.4-1: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-5.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 2x2, ULA Low | 70 | 15.7 |

### 5.2.2.1.3 2Rx FDD FR1 PDSCH mapping Type B performance

#### 5.2.2.1.3.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.3.0-3, with the addition of test parameters in Table 5.2.2.1.3.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.3.0-1.

**Table 5.2.2.1.3.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.1.3.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 5 |
| | Length (L) | | 7 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.2.1.3.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.3 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x2, ULA Low | 70 | -0.9 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.3.

### 5.2.2.1.3_1 2Rx FDD FR1 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.1.3_1.1 Test purpose

To verify PDSCH mapping Type B performance under 2 receive antenna conditions.

#### 5.2.2.1.3_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting PDSCH mapping type B.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and PDSCH mapping type B.

#### 5.2.2.1.3_1.3 Test description

##### 5.2.2.1.3_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.3.4 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.1.3.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.2.1.3_1.3.3.

##### 5.2.2.1.3_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.3_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.3_1.4-1 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.2.1.3_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 5.2.2.1.3_1.3.3_1 Message exceptions for SA

**Table 5.2.2.1.3_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | | |
| } | | | |

**Table 5.2.2.1.3_1.3.3_1-2: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeB | | |
| startSymbolAndLength | 89 | Start symbol(S)=5, Length(L)=7 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.3_1.3.3_1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| dmrs-DownlinkForPDSCH-MappingTypeB CHOICE { | | | |
| setup | DMRS-DownlinkConfig | | |
| } | | | |
| } | | | |

##### 5.2.2.1.3_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.3_1.3.3_1

#### 5.2.2.1.3_1.4 Test requirement

Table 5.2.2.1.3_1.4-1 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.3_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.3_1.4-1: Test Requirements for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.3 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x2, ULA Low | 70 | 0.1 |

### 5.2.2.1.4 2Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance

#### 5.2.2.1.4.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.4.0-3, with the addition of test parameters in Table 5.2.2.1.4.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.1.4.0-1.

**Table 5.2.2.1.4.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions with CRS rate matching configured | 1-1, 1-2 |

**Table 5.2.2.1.4.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| NR UL transmission with a 7.5 kHz shift to the LTE raster | | | true |
| PDCCH configuration | Symbols with PDCCH | | Symbol# 2 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 3 |
| | Length (L) | | 9 for Test 1-1<br>11 for Test 1-2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Position of the first DM-RS for downlink | | 3 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CRS for rate matching (Note 1) | LTE carrier centre subcarrier location | | Same as NR carrier centre subcarrier location |
| | LTE carrier BW | MHz | 10 |
| | Number of antenna ports | | 4 |
| | v-shift | | 0 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |
| Note 1: No MBSFN is configured on LTE carrier | | | |

**Table 5.2.2.1.4.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-7.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x2, ULA Low | 70 | -1.0 |
| 1-2 | R.PDSCH.1-7.2 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x2, ULA Low | 70 | -1.0 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.4.

5.2.2.1.4_1 2Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA

5.2.2.1.4_1.1 Test purpose

To verify the Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions with CRS rate matching configured.

#### 5.2.2.1.4_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 15 and forward supporting capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test1-1 also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test 1-2 applies to all types of NR UE release 15 and forward supporting capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

Test 1-2 also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

#### 5.2.2.1.4_1.3 Test description

##### 5.2.2.1.4_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 5.2.2.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.6 for TE diagram and section A.3.2.3 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.4.0-2 and Table 5.2.2.1.4.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.4_1.3.3.

##### 5.2.2.1.4_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.4.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.4_1.3.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

5.2.2.1.4_1.3.3 Message contents

5.2.2.1.4_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.4_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 94 | Start symbol(S)=3, Length(L)=9 for Test 1-1 | |
| | 66 | Start symbol(S)=3, Length(L)=11 for Test 1-2 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 66 | Start symbol(S)=3, Length(L)=11 for Test 1-2 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.4_1.3.3_1-2: *SearchSpace***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 and5.4.2.0-7 using condition USS, FR1_10MHz, Long_DCI | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| monitoringSymbolsWithinSlot | 00100000000000 | | |
| } | | | |

**Table 5.2.2.1.4_1.3.3_1-3: ServingCellConfigCommon**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ServingCellConfigCommon ::= SEQUENCE { | | | |
| dmrs-TypeA-Position | pos3 | | |
| lte-CRS-ToMatchAround | RateMatchPatternLTE-CRS | | |
| } | | | |

**Table 5.2.2.1.4_1.3.3_1-4: RateMatchPatternLTE-CRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-20 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RateMatchPatternLTE-CRS ::= SEQUENCE { | | | |
| carrierFreqDL | Same as NR carrier centre subcarrier location | | |
| carrierBandwidthDL | n50 | 10MHz | |
| mbsfn-SubframeConfigList | Not present | | |
| nrofCRS-Ports | n4 | | |
| v-Shift | n0 | | |
| } | | | |

**Table 5.2.2.1.4_1.3.3_1-5: *Void***

**Table 5.2.2.1.4_1.3.3_1-6: *FrequencyInfoUL-SIB***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-62 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| FrequencyInfoUL-SIB SEQUENCE { | | | |
| frequencyShift7p5khz | true | | |
| } | | | |

**Table 5.2.2.1.4_1.3.3_1-7: PDCCH-ControlResourceSet**

| Derivation Path: Table TS 38.508-1 [6], 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| duration | 1 | SearchSpace duration of 1 symbol from third symbol | |
| } | | | |

**Table 5.2.2.1.4_1.3.3_1-8: *Void***

**Table 5.2.2.1.4_1.3.3_1-9: *SearchSpace for CSS***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 and 5.4.2-4 using condition CSS, FR1_10MHz, Long_DCI | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| searchSpaceId | SearchSpaceId with condition CSS | | CSS |
| controlResourceSetId | 1 | | |
| monitoringSlotPeriodicityAndOffset CHOICE { | | | |
| sl1 | NULL | | |
| } | | | |
| duration | Not present | 1 slot per default | |
| monitoringSymbolsWithinSlot | 00100000000000 | | |
| } | | | |

5.2.2.1.4_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.4_1.3.3_1 with the following exceptions:

Table 5.2.2.1.4_1.3.3_2-1: *SearchSpace*

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 and 5.4.2.0-7 using condition USS, FR1_10MHz, Long_DCI | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| monitoringSymbolsWithinSlot | 00100000000000 | | |
| } | | | |

Table 5.2.2.1.4_1.3.3_2-2: PDCCH-ControlResourceSet

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| duration | 1 | SearchSpace duration of 1 symbol from third symbol | |
| } | | | |

#### 5.2.2.1.4_1.3.4 Test requirement

Table 5.2.2.1.4.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.4_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.4_1.3.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-7.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x2, ULA Low | 70 | 0.0 |
| 1-2 | R.PDSCH.1-7.2 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x2, ULA Low | 70 | 0.0 |

### 5.2.2.1.5 2Rx FDD FR1 PDSCH 0.001% BLER performance

#### 5.2.2.1.5.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.5.0-3, with the addition of test parameters in Table 5.2.2.1.5.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.5.0-1.

**Table 5.2.2.1.5.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH 0.001% BLER performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.1.5.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.2.1.5.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-1.4 FDD | 10 / 15 | QPSK, 0.59 | AWGN | 1x2, ULA Low | 0.001% | 3.2 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.5.

### 5.2.2.1.5_1 2Rx FDD FR1 PDSCH 0.001% BLER performance - 1x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.1.5_1.1 Test purpose

To verify the PDSCH 0.001% BLER performance under 2 receive antenna conditions.

#### 5.2.2.1.5_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *dl-64QAM-MCS-TableAlt* and capability IE *cqi-TableAlt*.

#### 5.2.2.1.5_1.3 Test description

##### 5.2.2.1.5_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.5.0-2 and Table 5.2.2.1.5.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On*, *Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.5_1.3.3.

### 5.2.2.1.5_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.5.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.5_1.3.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.4.3-1 in Annex G.

### 5.2.2.1.5_1.3.3 Message contents

#### 5.2.2.1.5_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.5_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| mcs-Table | qam64LowSE | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

#### 5.2.2.1.5_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.5_1.3.3_1.

#### 5.2.2.1.5_1.3.4 Test requirement

Table 5.2.2.1.5.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.5_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.5_1.3.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-1.4 FDD | 10 / 15 | QPSK, 0.59 | AWGN | 1x2, ULA Low | 0.001% | 3.8 |

### 5.2.2.1.6 2Rx FDD FR1 PDSCH repetitions over multiple slots performance

#### 5.2.2.1.6.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.6.0-3, with the addition of test parameters in Table 5.2.2.1.6-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.6.0-1.

**Table 5.2.2.1.6.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.1.6.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.2.1.6.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-11.1 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.6 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.6.

### 5.2.2.1.6_1 2Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- MU factor due to finite test time needs further study

#### 5.2.2.1.6_1.1 Test purpose

To Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions.

#### 5.2.2.1.6_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

#### 5.2.2.1.6_1.3 Test description

##### 5.2.2.1.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.6.0-2 and Table 5.2.2.1.6.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On*, *Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.6_1.3.3.

#### 5.2.2.1.6_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.5.0-3. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.5_1.3.4-1.

3. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport bloks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to clause G.1.5 and Table G.1.5-1a in Annex G clause G.1.5.

#### 5.2.2.1.6_1.3.3 Message contents

##### 5.2.2.1.6_1.3.3_1 Message exceptions for SA & NSA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.6_1.3.3_1-1: PDSCH-Config**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| pdsch-AggregationFactor | 2 | | |
| } | | | |

#### 5.2.2.1.6_1.3.4 Test requirement

Table 5.2.2.1.6.0-3 defines the primary level settings.

The target BLER percentage for the downlink reference measurement channels specified in Annex 3.2.1 for each BLER test shall meet or exceed the specified value in Table 5.2.2.1.6_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.6_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-11.1 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.6 + TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

### 5.2.2.1.7 2Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance

#### 5.2.2.1.7.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.7.0-3, with the addition of test parameters in Table 5.2.2.1.7.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.7.0-1.

**Table 5.2.2.1.7.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance and UE processing capability 2 under two receive antenna conditions | 1-1 |

**Table 5.2.2.1.7.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 0 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 2 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 0 |

**Table 5.2.2.1.7.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-12.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x2, ULA Low | 70 | [0.8] |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.7.

### 5.2.2.1.7_1 2Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Minimum requirements are within brackets.

#### 5.2.2.1.7_1.1 Test purpose

To verify PDSCH mapping Type B performance and UE processing capability 2 under two receive antenna conditions.

#### 5.2.2.1.7_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-ProcessingType2*.

#### 5.2.2.1.7_1.3 Test description

##### 5.2.2.1.7_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.7.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.7_1.3.3.

##### 5.2.2.1.7_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.7.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.7_1.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

##### 5.2.2.1.7_1.3.3 Message contents

###### 5.2.2.1.7_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.7_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entries | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeB | | |
| startSymbolAndLength | 16 | Start symbol(S)=2, Length(L)=2 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.7_1.3.3_1-2: *PUCCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-112 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PUCCH-Config ::= SEQUENCE { | | | FR1 |
| dl-DataToUL-ACK SEQUENCE (SIZE (1)) OF INTEGER { | 1 entry | | |
| INTEGER[1] | 0 | entry 1 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.7_1.3.3_1-3: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PDSCH-to-HARQ_feedback timing indicator | K1=0 as per dl-DataToUL-ACK in Table 5.2.2.1.7_1.3.3_1-3 | "000" | |

### 5.2.2.1.7_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.7_1.3.3_1.

### 5.2.2.1.7_1.4 Test requirement

Table 5.2.2.1.7.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.7_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.7_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-12.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x2, ULA Low | 70 | [1.8] |

### 5.2.2.1.8 2Rx FDD FR1 PDSCH pre-emption performance

#### 5.2.2.1.8.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.8.0-3, with the addition of test parameters in Table 5.2.2.1.8.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.8.0-1.

**Table 5.2.2.1.8.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH pre-emption performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.1.8.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDCCH configuration (Note 4) | Symbols with PDCCH | | 0, 1 |
| | DCI format | | 2_1 |
| | timeFrequencySet | | 14x1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Pre-emption configuration (Note 2) | Starting symbol (S) | | 3 |
| | Length (L) | | 2 |
| | Pre-emption periodicity and offset (Note 3) | Slots | 10/1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |
| Note 1: Void<br>Note 2: Interference modelled as random data on pre-empted REs.<br>Note 3: Pre-emption is scheduled with a fixed scheduling with 10% probability within 10ms periodicity.<br>Note 4: In addition to PDCCH configuration in Table 5.2-1. | | | |

**Table 5.2.2.1.8.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH. 1-2.5 FDD | 10 / 15 | 16QAM 0.64 | TDLA30-10 | 2x2, ULA Low | 70 | 10.5 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.8.

### 5.2.2.1.8_1 2Rx FDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.1.8_1.1 Test purpose

To Verify the PDSCH pre-emption performance under 2 receive antenna conditions.

#### 5.2.2.1.8_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pre-EmptIndication-DL-r16*.

#### 5.2.2.1.6_1.3 Test description

##### 5.2.2.1.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.8.0-2 and Table 5.2.2.1.8.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.8_1.3.3.

##### 5.2.2.1.8_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.8.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. SS transmits PDCCH DCI format 2_1 for int_RNTI with 10% probability to transmit the DL Preemption indication according to Table 5.2.2.1.8.0-2. In the time and frequency set indicated by PDCCH DCI format 2_1, SS stops transmission of PDSCH.
3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.8_1.3.4-1.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

5.2.2.1.8_1.3.3 Message contents

5.2.2.1.8_1.3.3_1 Message exceptions for SA

**Table 5.2.2.1.8_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.8_1.3.3_1-2: PDCCH-Config**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config ::= SEQUENCE { | | | |
| DownlinkPreemption ::= SEQUENCE { | SS arbitrarily selects a value between '0001'H and 'FFEF'H different from the MCG (and SCG) RNTI-Value. | | |
| int-RNTI | | | |
| timeFrequencySet | set0 | | |
| dci-PayloadSize | 14 | | |
| Int-ConfigurationPerServingCell SEQUENCE (SIZE (1..maxNrofServingCells)) OF SEQUENCE { | | | |
| servingCellId | ServCellIndex | | |
| positionInDCI | 0 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.8_1.3.3_1-3: Physical layer parameters for DCI format 2_1**

| Parameter | Value | Value in binary | Condition |
|---|---|---|---|
| Pre-emption indication 1 | Indicating symbols 3 and 4 | 01100000000000 | |

5.2.2.1.8_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.8_1.3.3_1.

5.2.2.1.8_1.3.4 Test requirement

Table 5.2.2.1.8.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.8_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.8_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 1-2.5 FDD | 10 / 15 | 16QAM 0.64 | TDLA30-10 | 2x2, ULA Low | 70 | 11.5 |

## 5.2.2.1.9 2Rx FDD FR1 HST-SFN performance

### 5.2.2.1.9.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.9.0-3, with the test parameters defined in Table 5.2.2.1.9.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.1.9.0-1.

**Table 5.2.2.1.9.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 2 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when highSpeedDemodFlag-r16 IE [20] is configured | 1-1 |

**Table 5.2.2.1.9.0-2: Test Parameters for Testing**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | CSI-RS periodicity | Slots | 10 for CSI-RS resource 1,2,3,4. |
| | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4. |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.2.1.9.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.3 FDD | 10 / 15 | 16QAM, 0.48 | HST-SFN | 2x2 | 70 | 13.0 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.9.

### 5.2.2.1.9_1 2Rx FDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: The minimum test time value is in []

#### 5.2.2.1.9_1.1 Test purpose

To verify the PDSCH performance under 2 receive antennas conditions in the HST-SFN scenario defined in B.3.2 when *highSpeedDemodFlag-r16* IE [20] is configured and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.2.1.9_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC that supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

#### 5.2.2.1.9_1.3 Test description

##### 5.2.2.1.9_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.1.9.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On*, *Test Mode* On or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release*

*On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.9_1.3.3.

### 5.2.2.1.9_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.2.1.9_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.1.9_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.2.1.9_1.4-1 as appropriate.

### 5.2.2.1.9_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

### 5.2.2.1.9_1.3.3_1 Message exceptions for SA

**Table 5.2.2.1.9_1.3.3_1-1: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 for test 1-1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.9_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1 | |
| } | | | |

**Table 5.2.2.1.9_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | for test 1-1 | |
| } | | | |

**Table 5.2.2.1.9_1.3.3_1-4: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots10 | 1 for CSI-RS resource #1 and #2<br><br>2 for CSI-RS resource #3 and #4 | For test 1-1:<br>offset = 1 for CSI-RS resource 1 and 2<br>offset =2 for CSI-RS resource 3 and 4. | |
| } | | | |

### 5.2.2.1.9_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.9_1.3.3_1

### 5.2.2.1.9_1.4 Test requirement

Tables 5.2.2.1.9_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.9_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.9_1.4-1: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.3 FDD | 10 / 15 | 16QAM, 0.48 | HST-SFN | 2x2 | 70 | 13.6 |

## 5.2.2.1.10 2Rx FDD FR1 HST DPS performance

### 5.2.2.1.10.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.10.0-3, with the test parameters defined in Table 5.2.2.1.10.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.1.10.0-1.

**Table 5.2.2.1.10.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify UE performance in the HST-DPS scenario defined in B.3.3 | 1-1, 1-2 |

**Table 5.2.2.1.10.0-2: Test Parameters for Testing**

| Parameter | | | Unit | Value |
|---|---|---|---|---|
| Duplex mode | | | | FDD |
| Active DL BWP index | | | | 1 |
| PDCCH configuration | TCI state | | | Note 1 |
| PDSCH configuration | Mapping type | | | Type A |
| | k0 | | | 0 |
| | Starting symbol (S) | | | 2 |
| | Length (L) | | | 12 |
| | PDSCH aggregation factor | | | 1 |
| | PRB bundling type | | | Static |
| | PRB bundling size | | | 2 |
| | Resource allocation type | | | Type 0 |
| | RBG size | | | Config2 |
| | VRB-to-PRB mapping type | | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A |
| | TCI state | | | Note 1 |
| PDSCH DMRS configuration | DMRS Type | | | Type 1 |
| | Number of additional DMRS | | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 |
| CSI-RS for tracking | Resource set #1 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 5 for CSI-RS resource 1 and 3<br>$l_0$ = 9 for CSI-RS resource 2 and 4 |
| | | CSI-RS periodicity | Slots | 10 for CSI-RS resource 1,2,3,4. |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4 |
| | | QCL info | | TCI state #2 |
| | Resource set #2 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 5 and 6<br>$l_0$ = 10 for CSI-RS resource 7 and 8 |
| | | CSI-RS periodicity | Slots | 10 for CSI-RS resource 5,6,7,8. |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 5 and 6<br>2 for CSI-RS resource 7 and 8 |
| | | QCL info | | TCI state #3 |
| NZP CSI-RS for CSI acquisition | Resource set #3 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 12 |
| | | CSI-RS periodicity | Slots | 20 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #0 |
| | Resource set #4 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 13 |
| | | CSI-RS periodicity | Slots | 20 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #1 |
| TCI state #0 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking Resource set #1' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 5 from 'CSI-RS for tracking Resource set #2' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #2 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |

| | | | | |
|---|---|---|---|---|
| | | QCL Type | | N/A |
| TCI state #3 | Type 1 QCL information | SSB index | | SSB #1 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| Number of HARQ Processes | | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 |

Note 1: SSB # (k mod 2) , CSI-RS (for tracking) resource set # ((k mod 2) + 1) and CSI-RS (for CSI acquisition) resource set # ((k mod 2) + 3) are transmitted by k[th] RRH.

For Test 1-1, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by k[th] RRH from slot#

$$\max[(2k-1)n + 1 + T_{HARQ} + T_{MAC\,proc} + T_{firstTRS} + T_{TRS\,proc}, 0]$$

to slot#

$(2k+1)n + T_{HARQ} + T_{MAC\,proc}$,

PDCCH and PDSCH are DTXed in other slots in which throughput statistics are not considered.

For Test 1-2, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by k[th] RRH from slot#

$$\max[(2k-1)n + 1 + T_{HARQ} + T_{MAC\,proc}, 0]$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\,proc}$$

Where k=0, 1, 2… is the RRH number, n = 2520 is half of the number of slots between two RRH, $T_{HARQ}$ = 2 is the number of slots between PDSCH and corresponding HARQ-ACK information, $T_{MAC\,proc}$ = 3 is the number of slots for MAC CE processing, $T_{firstTRS}$ = 6 is the number of slots to first TRS transmission occasion after MAC CE command is decoded by the UE, $T_{TRS\,proc}$ = 2 is the number of slots for TRS processing.

**Table 5.2.2.1.10.0-3: Minimum performance for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 1 | 2x2 | 70 | 13.4 |
| 1-2 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 2 | 2x2 | 70 | 13.4 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.10.

### 5.2.2.1.10_1 2Rx FDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.1.10_1.1 Test purpose

To verify UE performance in the HST-DPS scenario defined in B.3.3 and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.2.1.10_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 5.2.2.1.10_1.3 Test description

#### 5.2.2.1.10_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.1.10.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.10_1.3.3.

#### 5.2.2.1.10_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.2.1.10_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.1.10_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.2.1.10_1.4-1 as appropriate.

#### 5.2.2.1.10_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.2.2.1.10_1.3.3_1 Message exceptions for SA

**Table 5.2.2.1.10_1.3.3_1-1: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 is used | test 1-1, 1-2 |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-4: NZP-CSI-RS-Resource for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | i-1 for CSI-RS resource #i, i=1,2,3,4,5,6,7,8 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 2 for CSI-RS resource #1, #2, #3, #4<br>3 for CSI-RS resource #5, #6, #7, #8 | for test 1-1, 1-2:<br>TCI-StateId for TCI-State #2 for CSI-RS resource #1, #2, #3, #4<br>TCI-StateId for TCI-State #3 for CSI-RS resource #5, #6, #7, #8 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-5: CSI-RS-ResourceMapping for TRS (Table 5.2.2.1.10_1.3.3_1-4)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 with condition TRS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 5 for CSI-RS resource #1 and #3<br>9 for CSI-RS resource #2 and #4<br>6 for CSI-RS resource #5 and #6<br>10 for CSI-RS resource #7 and #8 | for test 1-1, 1-2:<br><br>$l_0$ = 5 for CSI-RS resource 1 and 3<br>$l_0$ = 9 for CSI-RS resource 2 and 4<br>$l_0$ = 6 for CSI-RS resource 5 and 6<br>$l_0$ = 10 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-5: CSI-ResourcePeriodicityAndOffset for CSI Tracking (Table 5.2.2.1.10_1.3.3_1-4)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots10 | 1 for CSI-RS resource #1, #2, #5, #6<br><br>2 for CSI-RS resource #3 #4, #7, #8 | For test 1-1, 1-2:<br>periodicity:<br>10 slots.<br>offset:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4<br>1 for CSI-RS resource 5 and 6<br>2 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-6: NZP-CSI-RS-ResourceSet for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 0 for Resource set #1<br>1 for Resource set #2 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #1 |
| NZP-CSI-RS-ResourceId[1] | 0 | entry 1<br>CSI-RS resource #1 | |
| NZP-CSI-RS-ResourceId[2] | 1 | entry 2<br>CSI-RS resource #2 | |
| NZP-CSI-RS-ResourceId[3] | 2 | entry 3<br>CSI-RS resource #3 | |
| NZP-CSI-RS-ResourceId[4] | 3 | entry 4<br>CSI-RS resource #4 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #2 |
| NZP-CSI-RS-ResourceId[1] | 4 | entry 1<br>CSI-RS resource #5 | |
| NZP-CSI-RS-ResourceId[2] | 5 | entry 2<br>CSI-RS resource #6 | |
| NZP-CSI-RS-ResourceId[3] | 6 | entry 3<br>CSI-RS resource #7 | |
| NZP-CSI-RS-ResourceId[4] | 7 | entry 4<br>CSI-RS resource #8 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-7: NZP-CSI-RS-Resource for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | 8 for CSI-RS resource #9<br>9 for CSI-RS resource #10 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 0 for CSI-RS resource #9<br>1 for CSI-RS resource #10 | for test 1-1, 1-2:<br>TCI-State #0 for CSI-RS resource #9<br><br>TCI-State #1 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-8: CSI-RS-ResourceMapping for CSI Acquisition (Table 5.2.2.1.10_1.3.3_1-7)**

| **Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15** | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 12 for CSI-RS resource #9<br>13 for CSI-RS resource #10 | for test 1-1, 1-2<br><br>$l_0$=12 for CSI-RS resource #9<br><br>$l_0$=13 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-9: CSI-ResourcePeriodicityAndOffset for CSI Acquisition (Table 5.2.2.1.10_1.3.3_1-7)**

| **Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16** | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 0 | For test 1-1, 1-2:<br>periodicity = 20 slots.<br>offset = 0 slots | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-10: NZP-CSI-RS-ResourceSet for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-18 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 2 for Resource set #3<br>3 for Resource set #4 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #3 |
| NZP-CSI-RS-ResourceId[1] | 8 | entry 1<br>CSI-RS resource #9 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #4 |
| NZP-CSI-RS-ResourceId[1] | 9 | entry 1<br>CSI-RS resource #10 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.10_1.3.3_1-10: *TCI-State***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 for TCI state #0<br>1 for TCI state #1<br>2 for TCI state #2<br>3 for TCI state #3 | For test 1-1, 1-2 | |
| qcl-Type1 SEQUENCE { | | | |
| bwp-Id | BWP-Id of active BWP | | TCI state #0, TCI state #1 |
| | Not present | | TCI state #2, TCI state #3 |
| referenceSignal CHOICE { | | | |
| csi-rs | 0 | CSI-RS resource #1 | TCI state #0 |
| | 4 | CSI-RS resource #5 | TCI state #1 |
| ssb | 0 | SSB #0 | TCI state #2 |
| | 1 | SSB #1 | TCI state #3 |
| } | | | |
| qcl-Type | typeA | | TCI state #0, TCI state #1 |
| | typeC | | TCI state #2, TCI state #3 |
| } | | | |
| } | | | |

### 5.2.2.1.10_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.10_1.3.3_1

### 5.2.2.1.10_1.4 Test requirement

Tables 5.2.2.1.10_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.10_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.10_1.4-1: Test Requirements for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 1 | 2x2 | 70 | 14.0 |
| 1-2 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 2 | 2x2 | 70 | 14.0 |

## 5.2.2.1.11 2Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance

### 5.2.2.1.11.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.11.0-3, with the addition of test parameters in Table 5.2.2.1.11.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.11.0-1.

**Table 5.2.2.1.11.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance with Single-DCI based SDM scheme under 2 receive antenna conditions | 1-1,1-2 |

**Table 5.2.2.1.11.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | 0 | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 20 | |
| | | CSI-RS offset | Slots | 10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6<br>11 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |

| | | | | | |
|---|---|---|---|---|---|
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000 | 1002 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Full-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.5 for test 1-1<br>2 for test 1-2 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 for test 1-1<br>0 for test 1-2 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs (PDSCH Layer 0 is transmitted from TRxP #1 and PDSCH layer 1 is transmitted from TRxP #2) | | | | | |

**Table 5.2.2.1.11.0-3: Minimum performance**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 20.7 |
| 1-2 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 20.1 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.11.

### 5.2.2.1.11_1 2Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.1.11_1.1 Test purpose

To verify the PDSCH performance with Single-DCI based SDM scheme under 2 receive antenna conditions.

#### 5.2.2.1.11_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *singleDCI-SDM-scheme-r16*.

#### 5.2.2.1.11_1.3 Test description

##### 5.2.2.1.11_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.11.0-2 and Table 5.2.2.1.11.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.11_1.3.3.

##### 5.2.2.1.11_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.11_1.3.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.11_1.3.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Table 5.2.2.1.11_1.3.4-1 as appropriate.

##### 5.2.2.1.11_1.3.3 Message contents

###### 5.2.2.1.11_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.11_1.3.3_1-1: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PDSCH-to-HARQ_feedback timing indicator | K1 = 2 | “010” | |
| Antenna port(s) | DMRS port 0 and 2 | “1011” | |
| Transmission configuration indication | TCI state 1 and 2 | “000” | |

**Table 5.2.2.1.11_1.3.3_1-2: *CellGroupConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| simultaneousTCI-UpdateList1-r16 SEQUENCE { | | | |
| ServCellIndex [1] | ServCellIndex | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.11_1.3.3_1-3: *ControlResourceSet***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-28 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.2.1.11_1.3.3_1-4: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] | *TCI-State* with condition TCI-state-0 | | |
| TCI-State[2] | *TCI-State* with condition TCI-state-1 | | |
| TCI-State[3] | *TCI-State* with condition TCI-state-2 | | |
| } | | | |
| rbg-Size | config2 | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.11_1.3.3_1-5: *TCI-State***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 | | TCI-state-0 |
| | 1 | | TCI-state-1 |
| | 2 | | TCI-state-2 |
| qcl-Type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | TCI-state-0 |
| csi-rs | 1 | | TCI-state-1 |
| | 5 | | TCI-state-2 |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| qcl-Type2 | Not present | | |
| } | | | |

**Table 5.2.2.1.11_1.3.3_1-6: *NZP-CSI-RS-Resource***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-85 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| resourceMapping SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| cdm-Type | noCDM | | |
| density CHOICE { | | | |
| three | NULL | | |
| } | | | |
| } | | | |
| periodicityAndOffset CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |
| qcl-InfoPeriodicCSI-RS | 0 | | |
| } | | | |

5.2.2.1.11_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.11_1.3.3_1.

#### 5.2.2.1.11_1.3.4 Test requirement

Table 5.2.2.1.11.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.11_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.11_1.3.4-1: Test requirement**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 21.7 |
| 1-2 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 21.1 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | |

### 5.2.2.1.12 2Rx FDD FR1 PDSCH Multi-DCI based transmission scheme performance

#### 5.2.2.1.12.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.12.0-3, with the addition of test parameters in Table 5.2.2.1.12.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.12.0-1.

**Table 5.2.2.1.12.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs | 1-1 |

**Table 5.2.2.1.12.0-2: Test parameters**

| Parameter | | Unit | Value | |
|---|---|---|---|---|
| | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | TRxP #1 | |
| PDCCH configuration | TCI state | | TCI State #1 | TCI State #2 |
| | CORESETPoolIndex | | 0,1 | |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | 3 | |
| | CSI-RS periodicity | Slots | 20 | |

| | | | | | |
|---|---|---|---|---|---|
| | CSI-RS offset | | Slots | 10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6<br>11 for CSI-RS resources 7 and 8 |
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | {1000,1001} | {1002,1003} |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Non-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.5 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs. Transmission from TRxP #1 uses CORESETPoolIndex 0 and transmission from TRxP #2 uses CORESETPoolIndex 1 | | | | | |

**Table 5.2.2.1.12.0-3: Minimum performance**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | TRxP #1 | TRxP #2 | | | | | | |
| 1-1 | R.PDSCH. 1-3.3 FDD | R.PDSCH. 1-3.4 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 20.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.12.

### 5.2.2.1.12_1 2Rx FDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.1.12_1.1 Test purpose

To verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs.

#### 5.2.2.1.12_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *multiDCI-MultiTRP-r16*.

#### 5.2.2.1.12_1.3 Test description

##### 5.2.2.1.12_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.9 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.12.0-2 and Table 5.2.2.1.12.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.12_1.3.3.

##### 5.2.2.1.12_1.3.2 Test procedure

1. SS transmits PDSCH in TRxP#1 via PDCCH DCI format 1_1 for C_RNTI in ControlResourceSetid1 (Table 5.2.2.1.12_1.3.3_1-2), and transmits PDSCH in TRxP#2 via PDCCH DCI format 1_1 for C_RNTI in ControlResourceSetid2 (Table 5.2.2.1.12_1.3.3_1-3), to transmit the DL RMC according to Table 5.2.2.1.12_1.3.4-1. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR for TRxP#1 and TRxP#2 according to Table 5.2.2.1.12_1.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

### 5.2.2.1.12_1.3.3 Message contents

#### 5.2.2.1.12_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.12_1.3.3_1-1: *PDCCH-Config* (Preamble)**

| Derivation Path: TS 38.508-1 [4],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet { | 2 entries | | |
| ControlResourceSet[1] | ControlResourceSetid1 | entry 1 | |
| ControlResourceSet[2] | ControlResourceSetid2 | entry 2 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.12_1.3.3_1-2: *ControlResourceSetId1* (Table 5.2.2.1.12_1.3.3_1-1: *PDCCH-Config*)**

| Derivation Path: TS 38.508-1 [4], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11110000 00000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 24 RBs of the BWP | |
| tci-StatesPDCCH-ToAddList SEQUENCE(SIZE (1..maxNrofTCI-StatesPDCCH)) OF TCI-StateId { | | | |
| TCI-StateId[1] | 1 | | |
| } | | | |
| tci-PresentInDCI | enabled | | |
| coresetPoolIndex-r16 | 0 | | |
| } | | | |

**Table 5.2.2.1.12_1.3.3_1-3: *ControlResourceSetId2* (Table 5.2.2.1.12_1.3.3_1-1: *PDCCH-Config*)**

| Derivation Path: TS 38.508-1 [4], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 00001111 00000000 00000000 00000000 00000000 00000 | CORESET to use the RBs 24~47 of the BWP | |
| tci-StatesPDCCH-ToAddList SEQUENCE(SIZE (1..maxNrofTCI-StatesPDCCH)) OF TCI-StateId { | | | |
| TCI-StateId[1] | 2 | | |
| } | | | |
| tci-PresentInDCI | enabled | | |
| coresetPoolIndex-r16 | 1 | | |
| } | | | |

**Table 5.2.2.1.12_1.3.3_1-4: Physical layer parameters for DCI format 1_1 in ControlResourceSetid1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 and 1 | “0111” | |
| Transmission configuration indication | TCI State #1 | “000” | |

**Table 5.2.2.1.12_1.3.3_1-5: Physical layer parameters for DCI format 1_1 in ControlResourceSetid2**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 2 and 3 | “1000” | |
| Transmission configuration indication | TCI State #2 | “001” | |

**Table 5.2.2.1.12_1.3.3_1-6: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] SEQUENCE { | | TCI-state-0 | |
| tci-StateId | 0 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | |
| } | | | |
| qcl-Type | typeC | | |
| } | | | |
| } | | | |
| TCI-State[2] | | TCI-state-1 | |
| tci-StateId | 1 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 1 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| TCI-State[3] | | TCI-state-2 | |
| tci-StateId | 2 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 5 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.12_1.3.3_1-7: *CSI-RS-ResourceMapping* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| } | | | |

**Table 5.2.2.1.12_1.3.3_1-8: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.2.1.12_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.12_1.3.3_1.

### 5.2.2.1.12_1.4 Test requirement

Table 5.2.2.1.12.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.12_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.12_1.4-1: Test requirement**

| Test num. | Reference channel | | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| | TRxP #1 | TRxP #2 | | | | | | |
| 1-1 | R.PDSCH.1-3.3 FDD | R.PDSCH.1-3.4 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x2, ULA Low | 70 | 21.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

### 5.2.2.1.13 2Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance

#### 5.2.2.1.13.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.13.0-3, with the addition of test parameters in Table 5.2.2.1.13.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.13.0-1.

**Table 5.2.2.1.13.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 2 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16” defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.2.1.13.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | Not configured | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | ‘No CDM’ for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 20 | |
| | | CSI-RS offset | Slots | 10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6<br>11 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | Wideband | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000,1001 | 1000,1001 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |

| | | | | | |
|---|---|---|---|---|---|
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.5 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs | | | | | |

**Table 5.2.2.1.13.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.1-2.5 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 70 | 17.3 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.13.

### 5.2.2.1.13_1 2Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.1.13_1.1 Test purpose

To verify the PDSCH performance under 2 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16”.

#### 5.2.2.1.13_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportFDM-SchemeA-r16*.

#### 5.2.2.1.13_1.3 Test description

##### 5.2.2.1.13_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.9 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.13.0-2 and Table 5.2.2.1.13.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On*, *Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.13_1.3.3.

### 5.2.2.1.13_1.3.2 Test procedure

1. SS transmits PDSCH in TRxP#1 and TRxP#2 via PDCCH DCI format 1_1 for C_RNTI (Table 5.2.2.1.13_1.3.3_1-2), to transmit the DL RMC according to Table 5.2.2.1.13_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR for TRxP#1 and TRxP#2 according to Table 5.2.2.1.13_1.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

### 5.2.2.1.13_1.3.3 Message contents

### 5.2.2.1.13_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.13_1.3.3_1-1: *PDCCH-ControlResourceSet* (Preamble)**

| Derivation Path: TS 38.508-1 [4],Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.2.1.13_1.3.3_1-2: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 and 1 | “0111” | |
| Transmission configuration indication | TCI codepoint 0, corresponding to TCI State #1 and #2 | “000” | |

**Table 5.2.2.1.13_1.3.3_1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] SEQUENCE { | | TCI-state-0 | |
| tci-StateId | 0 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | |
| } | | | |
| qcl-Type | typeC | | |
| } | | | |
| } | | | |
| TCI-State[2] | | TCI-state-1 | |
| tci-StateId | 1 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 1 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| TCI-State[3] | | TCI-state-2 | |
| tci-StateId | 2 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 5 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| } | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | wideband | | |
| } | | | |
| } | | | |
| repetitionSchemeConfig-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| fdm-TDM-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| repetitionScheme-r16 | fdmSchemeA | | |
| startingSymbolOffsetK-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.13_1.3.3_1-4: *CSI-RS-ResourceMapping* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| } | | | |

**Table 5.2.2.1.13_1.3.3_1-5: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.2.1.13_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.13_1.3.3_1.

### 5.2.2.1.13_1.4 Test requirement

Table 5.2.2.1.13.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.13_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.13_1.4-1: Test requirement for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.1-2.5 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 70 | 18.3 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

### 5.2.2.1.14 2Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance

#### 5.2.2.1.14.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.14.0-3, with the addition of test parameters in Table 5.2.2.1.14.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.14.0-1.

**Table 5.2.2.1.14.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 2 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.2.1.14.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | Not configured | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 20 | |
| | | CSI-RS offset | Slots | 10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6<br>11 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | | Mapping type | | Type A | |
| | | k0 | | 0 | |
| | | Starting symbol (S) | | 2 | |
| | | Length (L) | | 12 | |
| | | Repetition number | | 2 | |
| | | PRB bundling type | | Static | |
| | | PRB bundling size | | 2 | |
| | | Resource allocation type | | Type 0 | |
| | | RBG size | | Config2 | |
| | | VRB-to-PRB mapping type | | Non-interleaved | |
| | | VRB-to-PRB mapping interleaver bundle size | | N/A | |
| PDSCH DMRS configuration | | Antenna port indexes | | 1000 | 1000 |
| | | TCI state | | TCI State #1 | TCI State #2 |
| | | DMRS Type | | Type 1 | |
| | | Number of additional DMRS | | 1 | |
| | | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 | |

| | | | | | |
|---|---|---|---|---|---|
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | 2 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs | | | | | |

**Table 5.2.2.1.14.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition (Note 1) | Correlation matrix and antenna configuration Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR (dB) (Note 4) |
| 1-1 | R.PDSCH.1-11.2 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 1 (Note 3) | 2.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.<br>Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.14.

### 5.2.2.1.14_1 2Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.1.14_1.1 Test purpose

To verify the PDSCH performance under 2 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states.

#### 5.2.2.1.14_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportTDM-SchemeA-r16*.

#### 5.2.2.1.14_1.3 Test description

##### 5.2.2.1.14_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.9 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.14.0-2 and Table 5.2.2.1.14.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.14_1.3.3.

### 5.2.2.1.14_1.3.2 Test procedure

1. SS transmits PDSCH in TRxP#1 and TRxP#2 via PDCCH DCI format 1_1 for C_RNTI (Table 5.2.2.1.14_1.3.3_1-2), to transmit the DL RMC according to Table 5.2.2.1.14_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR for TRxP#1 and TRxP#2 according to Table 5.2.2.1.14_1.4-1.
3. Measure the residual BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport blocks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to clause G.1.4 and Table G.1.5-1a in Annex G clause G.1.5.

### 5.2.2.1.14_1.3.3 Message contents

### 5.2.2.1.14_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.14_1.3.3_1-1: *PDCCH-ControlResourceSet* (Preamble)**

| Derivation Path: TS 38.508-1 [4],Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.2.1.14_1.3.3_1-2: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 | “0000” | |
| Transmission configuration indication | TCI codepoint 0, corresponding to TCI State #1 and #2 | “000” | |

**Table 5.2.2.1.14_1.3.3_1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] SEQUENCE { | | TCI-state-0 | |
| tci-StateId | 0 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | |
| } | | | |
| qcl-Type | typeC | | |
| } | | | |
| } | | | |
| TCI-State[2] | | TCI-state-1 | |
| tci-StateId | 1 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 1 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| TCI-State[3] | | TCI-state-2 | |
| tci-StateId | 2 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 5 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| } | | | |
| pdsch-TimeDomainAllocationList | Not present | | |
| pdsch-TimeDomainAllocationList-r16 CHOICE { | | | |
| setup SEQUENCE (SIZE(1..maxNrofDL-Allocations)) OF PDSCH-TimeDomainResourceAllocation-r16 { | | | |
| PDSCH-TimeDomainResourceAllocation-r16[1] SEQUENCE { | | | |
| k0-r16 | Not present | | |
| mappingType-r16 | typeA | | |
| startSymbolAndLength-r16 | 44 | Start symbol(S) =2, Length(L)= 4 | For Slot i, if mod(i, 10) = 7 for i from {0,…,39} |
| repetitionNumber-r16 | 2 | | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation-r16[2] SEQUENCE { | | | |
| k0-r16 | Not present | | |
| mappingType-r16 | typeA | | |
| startSymbolAndLength-r16 | 53 | Start symbol(S) =2, Length(L)= 12 | For Slot i, if mod(i, 10) = {0,1,2,3,4,5,}) for i from {1,…,39} |
| repetitionNumber-r16 | 2 | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.1.14_1.3.3_1-4: *CSI-RS-ResourceMapping* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| } | | | |

**Table 5.2.2.1.14_1.3.3_1-5: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.2.1.14_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.14_1.3.3_1.

### 5.2.2.1.14_1.4 Test requirement

Table 5.2.2.1.14.0-3 defines the primary level settings.

The residual BLER specified in Note 3 of Table 5.2.2.1.14_1.4-1 test shall meet or be lower than the specified value in Table 5.2.2.1.14_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.14_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition (Note 1) | Correlation matrix and antenna configuration Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR (dB) (Note 4) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.1-11.2 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 1 (Note 3) | 3.9 |

Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.
Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.
Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.
Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2

## 5.2.2.2 TDD

### 5.2.2.2.1 2Rx TDD FR1 PDSCH mapping Type A performance

#### 5.2.2.2.1.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.1.0-3 and Table 5.2.2.2.1.0-4, with the addition of test parameters in Table 5.2.2.2.1.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.2.1.0-1.

**Table 5.2.2.2.1.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers | 1-1, 1-2, 1-3, 1-5, 1-6, 1-7, 1-8, 1-9, 1-10, 1-11, 2-1, 2-2 |
| Verify the PDSCH mapping Type A HARQ soft combining performance under 2 receive antenna conditions. | 1-4 |
| Verify the PDSCH mapping Type A performance requirements for Enhanced Receiver Type 1 under 2 receive antenna conditions. | 3-1 |

**Table 5.2.2.2.1.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | Specific to each Reference channel |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 4 for Tests 1-1, 1-8, 1-9<br>2 for other tests |
| | Resource allocation type | | Test 1-2: Type 1 with start RB = 50, $L_{RBs}$ = 6<br><br>Other tests: Type 0 |
| | RBG size | | Test 1-2: N/A<br><br>Other tests: Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 for Tests 1-1 , 1-7, 1-8, 1-9, 1-10, 1-11<br>1 for other tests |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |

| | | | |
|---|---|---|---|
| CSI-RS for tracking | First OFDM symbol in the PRB used for CSI-RS | | Tests 1-8, 1-9:<br>l0 = 4 for CSI-RS resource 1 and 3<br>l0 = 8 for CSI-RS resource 2 and 4<br><br>Other tests; Table 5.2-1. |
| | CSI-RS periodicity | Slots | Test 1-7, 1-10, 1-11:<br>20 for CSI-RS resource 1,2,3,4.<br><br>Other tests: Table 5.2-1. |
| | CSI-RS offset | Slots | Test 1-7:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4.<br><br>Other tests: Table 5.2-1. |
| | Frequency Occupation | | Test 1-7, 1-10, 1-11:<br>Start PRB 0<br>Number of PRB = 52<br><br>Other tests: Table 5.2-1. |
| Number of HARQ Processes | | | 16 for Test 1-4<br>10 for Test 1-9<br>8 for other tests |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.2.2.1.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-1.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1A | TDLB100-400 | 2x2, ULA Low | 70 | -1.1 |
| 1-2 | R.PDSCH.2-1.2 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLC300-100 | 2x2, ULA Low | 70 | 0.2 |
| 1-3 | R.PDSCH.2-4.1 TDD | 40 / 30 | 256QAM, 0.82 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 25.3 |
| 1-4 | R.PDSCH.2-2.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x2, ULA Low | 30 | 1.6 |
| 1-5 | R.PDSCH.2-5.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x2, ULA Low | 70 | -0.9 |
| 1-6 | R.PDSCH.2-6.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-3 | TDLA30-10 | 2x2, ULA Low | 70 | -0.8 |
| 1-7 | R.PDSCH.2-10.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-1000 | 1x2 | 70 | 6.4 |
| 1-8 | R.PDSCH.2-11.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-5 | TDLB100-400 | 2x2, ULA Low | 70 | -1.0 |
| 1-9 | R.PDSCH.2-12.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-6 | TDLB100-400 | 2x2, ULA Low | 70 | -1.1 |
| 1-10 | R.PDSCH.2-10.2 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-1200 | 2x2 | 70 | 9.5 |
| 1-11 | R.PDSCH.2-10.3 TDD | 40 / 30 | 64QAM, 0.43 | FR1.30-1 | HST-1667 | 1x2 | 70 | 9.6 |

**Table 5.2.2.2.1.0-4: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2-1 | R.PDSCH.2-3.1 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 19.8 |
| 2-2 | R.PDSCH.2-9.1 TDD | 20 / 30 | 64QAM, 0.50 | FR1.30-4 | TDLA30-10 | 2x2, ULA Low | 70 | 19.8 |

**Table 5.2.2.2.1.0-5: Minimum performance for Rank 2 and EnhancedReceiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.2-2.2 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 2x2, ULA Medium | 70 | 18.0 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.2.2.1.0.

### 5.2.2.2.1_1 2Rx TDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.2.1_1.1 Test Purpose

Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers

#### 5.2.2.2.1_1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 5.2.2.2.1_1.3 Test Description

##### 5.2.2.2.1_1.3.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of TS 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 5.2-1 and Table 5.2.2.2.1.0-2 and as appropriate.
3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions for the NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release*

*On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.1_1.4.3.

### 5.2.2.2.1_1.3.2 Test Procedure

1. SS transmits PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.1_1.4-1 and Table 5.2.2.2.1_1.4-2. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.2.1_1.4-1 and 5.2.2.2.1_1.4-2 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.1.5-2 in Annex G clause G.1.5.

4. Repeat steps from 1 to 3 for each subtest in Table 5.2.2.2.1_1.4-1 and Table 5.2.2.2.1_1.4-2 as appropriate.

### 5.2.2.2.1_1.3.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclauses 4.6.1 and 5.4.2.

### 5.2.2.2.1_1.3.3_1 Message exceptions for SA

**Table 5.2.2.2.1_1.3.3_1-1: Void**

**Table 5.2.2.2.1_1.3.3_1-2: Void**

**Table 5.2.2.2.1_1.3.3_1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| dmrs-DownlinkForPDSCH-MappingTypeA CHOICE { | | | |
| setup | DMRS-DownlinkConfig | | |
| } | | | |
| mcs-Table | qam256 | 256qam table for test 1-3 | |
| | Not present | 64qam table for all tests except test 1-3 | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | n4<br><br>Not present | n4 for test 1-1<br>n2 for other tests<br>n2 is used by default | test 1-1<br><br>all test points except test 1-1 |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-4: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | For tests 1-1, 1-7, 1-8, 1-9, 1-10, and 1-11 | |
| | pos1 | For other tests | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-5: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n16, n10, n8 | n16 for Test 1-4, n10 for Test 1-9 n8 for other tests | |
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-6: RACH-ConfigGeneric**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-130 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigGeneric ::= SEQUENCE { | | | |
| prach-ConfigurationIndex | 163 | Only for test 2-2 | |
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-7: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots20 | 1 (for CSI-RS resources 1 and 2)<br>2 (for CSI-RS resources 3 and 4) | For test 1-7, 1-10, 1-11:<br>CSI-RS offset:<br>1 for CSI-RS resources 1 and 2<br>2 for CSI-RS resources 3 and 4<br>CSI-RS periodicity: 20 slots | |
| Slots40 | 20 (for CSI-RS resources 1 and 2)<br>21 (for CSI-RS resources 3 and 4) | For other tests:<br>CSI-RS offset:<br>20 for CSI-RS resources 1 and 2<br>21 for CSI-RS resources 3 and 4<br>CSI-RS periodicity: 40 slots | |
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-8: CSI-FrequencyOccupation for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-11 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-FrequencyOccupation ::= SEQUENCE { | | | |
| nrofRBs | 52 | 52 for tests 1-7, 1-10, 1-11, 2-2 | |
| | 108 | 108 for other tests | |
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-9: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl20 | 7 | For test 1-9 | |
| sl20 | 5 | For test 2-2 | |
| } | | | |
| } | | | |

**Table 5.2.2.2.1_1.3.3_1-10: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PUCCH resource indicator | *PUCCH-ResourceId[1]* = 6 in pucch-ResourceSetID[1] or *PUCCH-ResourceId[1]* = 14 in pucch-ResourceSetID[2] as defined in Table 4.6.3-112 (Mapping as per Table 9.2.3-2 in TS 38.213) | '110'B | Slot S1 for test 1-9 |

**Table 5.2.2.2.1_1.3.3_1-11: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-27 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 3 entry | | Test 1-5, Test 1-6 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 44 | Start symbol(S)=2, Length(L)=4 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[3] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 5 entry | | Test 1-9 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 100 | Start symbol(S)=2, Length(L)=8 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[3] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 81 | Start symbol(S)=2, Length(L)=10 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[4] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[5] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |

| | | | |
|---|---|---|---|
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

5.2.2.2.1_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.1_1.3.3_1.

5.2.2.2.1_1.4 Test Requirements

Table 5.2.2.2.1_1.3-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.1_1.4-1 and 1 and Table 5.2.2.2.1_1.4-2 for the specified SNR including test tolerances for all throughput tests

**Table 5.2.2.2.1_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-1.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1A | TDLB100-400 | 2x2, ULA Low | 70 | -0.2 |
| 1-2 | R.PDSCH.2-1.2 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLC300-100 | 2x2, ULA Low | 70 | 1.1 |
| 1-3 | R.PDSCH.2-4.1 TDD | 40 / 30 | 256QAM, 0.82 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 26.3 |
| 1-4 | R.PDSCH.2-2.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x2, ULA Low | 30 | 2.5 |
| 1-5 | R.PDSCH.2-5.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x2, ULA Low | 70 | 0.1 |
| 1-6 | R.PDSCH.2-6.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-3 | TDLA30-10 | 2x2, ULA Low | 70 | 0.2 |
| 1-7 | R.PDSCH.2-10.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-1000 | 1x2 | 70 | 7.3 |
| 1-8 | R.PDSCH.2-11.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-5 | TDLB100-400 | 2x2, ULA Low | 70 | -0.1 |
| 1-9 | R.PDSCH.2-12.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-6 | TDLB100-400 | 2x2, ULA Low | 70 | -0.2 |
| 1-10 | R.PDSCH.2-10.2 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-1200 | 2x2 | 70 | 10.4 |
| 1-11 | R.PDSCH.2-10.3 TDD | 40 / 30 | 64QAM, 0.43 | FR1.30-1 | HST-1667 | 1x2 | 70 | 10.2 |

**Table 5.2.2.2.1_1.4-2: Test requirement for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 2-1 | R.PDSCH.2-3.1 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 20.8 |
| 2-2 | R.PDSCH.2-9.1 TDD | 20 / 30 | 64QAM, 0.50 | FR1.30-4 | TDLA30-10 | 2x2, ULA Low | 70 | 20.8 |

#### 5.2.2.2.1_2 2Rx TDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with enhanced receiver type 1 for both SA and NSA

##### 5.2.2.2.1_2.1 Test Purpose

Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers

##### 5.2.2.2.1_2.2 Test Applicability

This test applies to all types of NR UE release 15 and forward supporting NR enhanced receiver type 1.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and NR enhanced receiver type 1.

##### 5.2.2.2.1_2.3 Test Description

Same test description as in clause 5.2.2.2.1_1.4 with the following exception:

- Table 5.2.2.2.1_2.4-1 instead of 5.2.2.2.1_1.4-1

##### 5.2.2.2.1_2.4 Test Requirements

Table 5.2.2.2.1_1.3-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.1.4.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.1_2.4-1: Test requirement for Rank 2 and EnhancedReceiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.2-2.2 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 2x2, ULA Medium | 70 | 19.0 |

### 5.2.2.2.2 2Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance

#### 5.2.2.2.2_1 2Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA

##### 5.2.2.2.2_1.1 Test Purpose

Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and CSI-RS overlapped with PDSCH

##### 5.2.2.2.2_1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 5.2.2.2.2_1.3 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.2_1.3-3, with the addition of test parameters in table 5.2.2.2.2_1.3-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.2.2_1.3-1.

**Table 5.2.2.2.2_1.3-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and CSI-RS overlapped with PDSCH | 1-1 |

**Table 5.2.2.2.2_1.3-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | Specific to each Reference channel |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Length | | 1 |
| NZP CSI-RS for CSI acquisition | OFDM symbols in the PRB used for CSI-RS | | $l_0 = 13$ |
| | CSI-RS periodicity | Slots | 5 |
| ZP CSI-RS for CSI acquisition | Subcarrier index in the PRB used for CSI-RS | | $(k_0, k_1, k_2, k_3)=(2, 4, 6, 8)$ |
| | Number of CSI-RS ports (X) | | 8 |
| | CSI-RS periodicity | Slots | 5 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.2.2.2_1.3-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-7.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x2, ULA Low | 70 | 14.8 |

The normative reference for this requirement is TS 38.101-4 [2] clause 5.2.2.1.2

### 5.2.2.2.2_1.4 Test Description

#### 5.2.2.2.2_1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 5.1.2.1 and 5.1.2.2.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 5.2-1 and Table 5.2.2.2.2_1.3-2 and as appropriate.
3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions for the NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.2_1.4.3.

#### 5.2.2.2.2_1.4.2 Test Procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.2_1.1-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.2_1.5-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.1.5-2 in Annex G clause G.1.5.

#### 5.2.2.2.2_1.4.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclauses 4.6.1 and 5.4.2.

#### 5.2.2.2.2_1.4.3_1 Message exceptions for SA

**Table 5.2.2.2.2_1.4.3_1-1: Void**

**Table 5.2.2.2.2_1.4.3_1-2: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| dataScramblingIdentityPDSCH | 0 | | |
| dmrs-DownlinkForPDSCH-MappingTypeA CHOICE { | | | |
| Setup | DMRS-DownlinkConfig | | |
| } | | | |
| resourceAllocation | resourceAllocationType0 | | Used_for_Type0 |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | | If a bundleSize(Set) value is absent, the UE applies the value n2. | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.2_1.4.3_1-3: Void**

**Table 5.2.2.2.2_1.4.3_1-4: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| tci-StatesPDCCH-ToAddList { | | | |
| | 0 | TCI State #0 | |
| | 1 | TCI State #1 | |
| } | | | |
| } | | | |

**Table 5.2.2.2.2_1.4.3_1-5: Void**

**Table 5.2.2.2.2_1.4.3_1-6: NZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 13 | $l_0 = 13$ | |
| } | | | |

**Table 5.2.2.2.2_1.4.3_1-7: CSI-ResourcePeriodicityAndOffset for CSI Acquisition for NZP CSI-RS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots5 | 0 | Periodicity 5 slots and offset 0 | |
| } | | | |

**Table 5.2.2.2.2_1.4.3_1-8: ZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| nrofPorts | P8 | Eight Ports | |
| } | | | |

**Table 5.2.2.2.2_1.4.3_1-9: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | Not present | pos2<br>If the field is absent, the UE applies the value pos2 | FR1_TDD, |
| } | | | |

**Table 5.2.2.2.2_1.4.3_1-10: CSI-ResourcePeriodicityAndOffset for CSI Acquisition for ZP CSI-RS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-22 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots5 | 0 | Periodicity 5 slots and offset 0 | |
| } | | | |

### 5.2.2.2.2_1.4.3_2 Message exceptions for NSA

Same as 5.2.2.2.2_1.4.3_2

### 5.2.2.2.2_1.5 Test Requirements

Table 5.2.2.2.2_1.3-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.2_1.3-3 for the specified SNR including test tolerances for all throughput tests

**Table 5.2.2.2.2_1.5-1: Test requirement for Rank 2**

| Test num. | Reference channel | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.2-7.1 TDD | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x2, ULA Low | 70 | 15.7 |

### 5.2.2.2.3 2Rx TDD FR1 PDSCH mapping Type B performance

#### 5.2.2.2.3.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.3.0-3, with the addition of test parameters in Table 5.2.2.2.3.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.3.0-1.

**Table 5.2.2.2.3.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.2.3.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 5 |
| | Length (L) | | 7 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.2.2.3.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-1.3 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | -0.9 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.2.2.3.

### 5.2.2.2.3_1 2Rx TDD FR1 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.2.3_1.1 Test purpose

To verify the PDSCH mapping Type B normal performance under 2 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput with baseline receiver configuration.

#### 5.2.2.2.3_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting PDSCH mapping type B.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and PDSCH mapping type B.

#### 5.2.2.2.3_1.3 Test description

##### 5.2.2.2.3_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.2.2.3.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.3.2.3_1.3.3.

##### 5.2.2.2.3_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.3.0-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.2.3_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

##### 5.2.2.2.3_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

5.2.2.2.3_1.3.3_1 Message exceptions for SA

**Table 5.2.2.2.3_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-102 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | Not present | | |
| } | | | |

**Table 5.2.2.2.3_1.3.3_1-2: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-103 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeB | | |
| startSymbolAndLength | 89 | Start symbol(S)=5, Length(L)=7 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

5.2.2.2.3_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.3_1.3.3_1

5.2.2.2.3_1.4 Test requirement

Table 5.2.2.2.3.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.3_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.3_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-1.3 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 0.1 |

### 5.2.2.2.4 2Rx TDD FR1 PDSCH mapping Type A and LTE-NR coexistence performance

#### 5.2.2.2.4.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.4.0-3, with the addition of test parameters in Table 5.2.2.2.4.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.2.2.4.0-1.

**Table 5.2.2.2.4.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions with CRS rate matching configured | 1-1, 1-2 |

**Table 5.2.2.2.4.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| **NR UL transmission with a 7.5 kHz shift to the LTE raster** | | | true |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 3 |
| | Length (L) | | 9 for Test 1-1<br>11 for Test 1-2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Position of the first DM-RS for downlink | | 3 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CRS for rate matching (Note 1) | LTE carrier centre subcarrier location | | Same as NR carrier centre subcarrier location |
| | LTE carrier BW | MHz | 10 |
| | Number of antenna ports | | 4 |
| | v-shift | | 0 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |
| Note 1: No MBSFN is configured on LTE carrier | | | |

**Table 5.2.2.2.4.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.1-1.1 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x2, ULA Low | 70 | -0.8 |
| 1-2 | R.PDSCH.1-1.2 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x2, ULA Low | 70 | -0.8 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.2.2.4.

### 5.2.2.2.4_1 2Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.2.4_1.1 Test purpose

To verify the PDSCH mapping Type A coexistence performance under 2 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput with baseline receiver configuration.

#### 5.2.2.2.4_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 15 and forward supporting capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test 1-1 also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC and capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test 1-2 applies to all types of NR UE release 15 and forward supporting capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

Test 1-2 also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC and capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

#### 5.2.2.2.4_1.3 Test description

##### 5.2.2.2.4_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.6 for TE diagram and clause A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.2.2.4.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.2.2.4_1.3.3.

#### 5.2.2.2.4_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.4.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.2.4_1.4-1 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.2.2.4_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

#### 5.2.2.2.4_1.3.3_1 Message exceptions for SA

**Table 5.2.2.2.4_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-102 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n8 | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-2: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-103 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 94 | Start symbol(S)=3, Length(L)=9 | Test 1-1 |
| | 66 | Start symbol(S)=3, Length(L)=11 | Test 1-2 |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 66 | Start symbol(S)=3, Length(L)=11 | Test 1-2 |
| } | | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-3: *SearchSpace***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 and5.4.2.0-7 using condition USS, FR1_10MHz, Long_DCI | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-4: ServingCellConfigCommon**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ServingCellConfigCommon ::= SEQUENCE { | | | |
| dmrs-TypeA-Position | pos3 | | |
| lte-CRS-ToMatchAround | RateMatchPatternLTE-CRS | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-5: RateMatchPatternLTE-CRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-28 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RateMatchPatternLTE-CRS ::= SEQUENCE { | | | |
| carrierFreqDL | Same as NR carrier centre subcarrier location | | |
| carrierBandwidthDL | n50 | 10MHz | |
| mbsfn-SubframeConfigList | Not present | | |
| nrofCRS-Ports | n4 | | |
| v-Shift | n0 | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-6: FrequencyInfoUL-SIB**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-62 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| FrequencyInfoUL-SIB SEQUENCE { | | | |
| frequencyShift7p5khz | true | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-7: PDCCH-ControlResourceSet**

| Derivation Path: Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| duration | 1 | SearchSpace duration of 1 symbol from third symbol | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-8: *PDCCH-ConfigCommon***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-96 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-ConfigCommon ::= SEQUENCE { | | | |
| commonControlResourceSet ::= SEQUENCE { | | | SA |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11110000 00000000 00000000 00000000 00000000 00000 | | |
| Duration | 1 | | |
| cce-REG-MappingType CHOICE { | | | |
| nonInterleaved | Null | | |
| } | | | |
| precoderGranularity | sameAsREG-bundle | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_1-9: *SearchSpace for CSS***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 and 5.4.2.0-7 using condition CSS, FR1_10MHz, Long_DCI | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| searchSpaceId | SearchSpaceId with condition CSS | | CSS |
| controlResourceSetId | 1 | | |
| monitoringSlotPeriodicityAndOffset CHOICE { | | | |
| sl1 | NULL | | |
| } | | | |
| duration | Not present | 1 slot per default | |
| } | | | |

5.2.2.2.4_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.4_1.3.3_1 with the following exceptions:

**Table 5.2.2.2.4_1.3.3_2-1: *SearchSpace***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 and 5.4.2.0-7 using condition USS, FR1_10MHz, Long_DCI | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| } | | | |

**Table 5.2.2.2.4_1.3.3_2-2: PDCCH-ControlResourceSet**

| Derivation Path: Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| duration | 1 | SearchSpace duration of 1 symbol from third symbol | |
| } | | | |

#### 5.2.2.2.4_1.4 Test requirement

Table 5.2.2.2.4.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.4_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.4_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x2, ULA Low | 70 | 0.1 |
| 1-2 | R.PDSCH.1-1.2 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x2, ULA Low | 70 | 0.1 |

### 5.2.2.2.5 2Rx TDD FR1 PDSCH 0.001% BLER performance

#### 5.2.2.2.5.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.5.0-3, with the addition of test parameters in Table 5.2.2.2.5.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.5.0-1.

**Table 5.2.2.2.5.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH 0.001% BLER performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.2.5.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Defined in Annex A.1.2 for TDD pattern FR1.30-1 |

**Table 5.2.2.2.5.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.2-1.4 TDD | 40 / 30 | QPSK, 0.59 | FR1.30-1 | AWGN | 1x2, ULA Low | 0.001% | 3.3 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.5.

### 5.2.2.2.5_1 2Rx TDD FR1 PDSCH 0.001% BLER performance - 1x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.2.5_1.1 Test purpose

To verify the PDSCH 0.001% BLER performance under 2 receive antenna conditions.

#### 5.2.2.2.5_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *dl-64QAM-MCS-TableAlt* and capability IE *cqi-TableAlt*.

#### 5.2.2.2.5_1.3 Test description

##### 5.2.2.2.5_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.2.5.0-2 and Table 5.2.2.2.5.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.5_1.3.3.

##### 5.2.2.2.5_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.5.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.5_1.3.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.4.3-1 in Annex G.

#### 5.2.2.2.5_1.3.3 Message contents

##### 5.2.2.2.5_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.2.5_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| mcs-Table | qam64LowSE | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

##### 5.2.2.2.5_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.5_1.3.3_1.

#### 5.2.2.2.5_1.3.4 Test requirement

Table 5.2.2.2.5.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.5_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.5_1.3.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.2-1.4 TDD | 40 / 30 | QPSK, 0.59 | FR1.30-1 | AWGN | 1x2, ULA Low | 0.001% | 3.9 |

### 5.2.2.2.6 2Rx TDD FR1 PDSCH repetitions over multiple slots performance

#### 5.2.2.2.6.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.6.0-3, with the addition of test parameters in Table 5.2.2.2.6.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.6.0-1.

**Table 5.2.2.2.6.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.2.6.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 (Note 1) |
| Note 1: ACK/NACK feedback is generated for PDSCH on slot i, where mod(i,10) = {2, 4, 6}. | | | |

**Table 5.2.2.1.6.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-16.1 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.4 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.6.

### 5.2.2.2.6_1 2Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- MU factor due to finite test time needs further study

#### 5.2.2.2.6_1.1 Test purpose

To Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions.

#### 5.2.2.2.6_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

#### 5.2.2.2.6_1.3 Test description

##### 5.2.2.2.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.6.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On*, *Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.6_1.3.3.

##### 5.2.2.2.6_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.6.0-3. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.6_1.3.4-1.
3. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport bloks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to clause G.1.5 and Table G.1.5-1a in Annex G clause G.1.5.

##### 5.2.2.2.6_1.3.3 Message contents

###### 5.2.2.2.6_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

###### 5.2.2.2.6_1.3.3_2 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

##### 5.2.2.2.6_1.3.4 Test requirement

Table 5.2.2.2.6.0-3 defines the primary level settings.

The target BLER percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each BLER test shall meet or exceed the specified value in Table 5.2.2.2.6_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.6_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.2-16.1 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.4 + TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

## 5.2.2.2.7 2Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance

### 5.2.2.2.7.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.7.0-3, with the addition of test parameters in Table 5.2.2.2.7.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.7.0-1.

**Table 5.2.2.2.7.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance and UE processing capability 2 under two receive antenna conditions | 1-1 |

**Table 5.2.2.2.7.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 0 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 2 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 0 |

**Table 5.2.2.2.7.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-17.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x2, ULA Low | 70 | 0.6 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.7.

### 5.2.2.2.7_1 2Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.2.7_1.1 Test purpose

To verify PDSCH mapping Type B performance and UE processing capability 2 under two receive antenna conditions.

#### 5.2.2.2.7_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-ProcessingType2*.

#### 5.2.2.2.7_1.3 Test description

##### 5.2.2.2.7_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.2.7.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.7_1.3.3.

##### 5.2.2.2.7_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.7.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.7_1.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

### 5.2.2.2.7_1.3.3 Message contents

#### 5.2.2.2.7_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.2.7_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entries | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeB | | |
| startSymbolAndLength | 16 | Start symbol(S)=2, Length(L)=2 | |
| } | | | |
| } | | | |

**Table 5.2.2.2.7_1.3.3_1-2: *PUCCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-112 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PUCCH-Config ::= SEQUENCE { | | | FR1 |
| dl-DataToUL-ACK SEQUENCE (SIZE (1)) OF INTEGER { | 1 entry | | |
| INTEGER[1] | 0 | entry 1 | |
| } | | | |
| } | | | |

**Table 5.2.2.2.7_1.3.3_1-3: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PDSCH-to-HARQ_feedback timing indicator | K1=0 as per dl-DataToUL-ACK in Table 5.2.2.2.7_1.3.3_1-3 | “000” | |

#### 5.2.2.2.7_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.7_1.3.3_1.

### 5.2.2.2.7_1.4 Test requirement

Table 5.2.2.2.7.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.7_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.7_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-17.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x2, ULA Low | 70 | 1.6 |

## 5.2.2.2.8 2Rx TDD FR1 PDSCH pre-emption performance

### 5.2.2.2.8.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.8.0-3, with the addition of test parameters in Table 5.2.2.2.8.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.8.0-1.

**Table 5.2.2.2.8.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH pre-emption performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.2.8.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDCCH configuration (Note 4) | Symbols with PDCCH | | 0, 1 |
| | DCI format | | 2_1 |
| | timeFrequencySet | | 14x1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Pre-emption configuration (Note 2) | Starting symbol (S) | | 3 |
| | Length (L) | | 2 |
| | Pre-emption periodicity and offset | Slots | 40/(1,12,23,34) (Note 3) |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | FR1.30-1 |

Note 1: Void
Note 2: Interference modelled as random data on pre-empted REs.
Note 3: Pre-emption is scheduled with 10% probability within 20ms periodicity.
Note 4: In addition to PDCCH configuration in Table 5.2-1.

**Table 5.2.2.2.8.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 2-2.5 TDD | 40 / 30 | 16QAM 0.48 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 12.5 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.8.

### 5.2.2.2.8_1 2Rx TDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA

#### 5.2.2.2.8_1.1 Test purpose

To Verify the PDSCH pre-emption performance under 2 receive antenna conditions.

#### 5.2.2.2.8_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pre-EmptIndication-DL-r16*.

#### 5.2.2.2.6_1.3 Test description

##### 5.2.2.2.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.2.8.0-2 and Table 5.2.2.2.8.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.8_1.3.3.

##### 5.2.2.2.8_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.8.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. SS transmits PDCCH DCI format 2_1 for int_RNTI with 10% probability to transmit the DL Preemption indication according to Table 5.2.2.2.8.0-2. In the time and frequency set indicated by PDCCH DCI format 2_1, SS stops transmission of PDSCH.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.8_1.3.4-1.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.2.2.8_1.3.3 Message contents

##### 5.2.2.2.8_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.8_1.3.3_1

##### 5.2.2.1.8_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.8_1.3.3_1

#### 5.2.2.2.8_1.3.4 Test requirement

Table 5.2.2.2.8.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.8_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.8_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 2-2.5 TDD | 40 / 30 | 16QAM 0.48 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 13.5 |

### 5.2.2.2.9 2Rx TDD FR1 HST-SFN performance

#### 5.2.2.2.9.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.9.0-3, with the test parameters defined in Table 5.2.2.2.9.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.9.0-1.

**Table 5.2.2.2.9.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 2 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when *highSpeedDemodFlag-r16* [17] is configured | 1-1 |

**Table 5.2.2.2.9.0-2: Test Parameters for Testing**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | CSI-RS periodicity | Slots | 20 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | Start PRB 0<br>Number of PRB = 52 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.2.2.9.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.4 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-SFN | 2x2 | 70 | 14.2 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.9.

### 5.2.2.2.9_1 2Rx TDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: The minimum test time value is in []

#### 5.2.2.2.9_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when *highSpeedDemodFlag-r16* IE [20] is configured and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.2.2.9_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC that supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

### 5.2.2.2.9_1.3 Test description

#### 5.2.2.2.9_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.2.9.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.9_1.3.3.

#### 5.2.2.2.9_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.2.2.9_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.2.9_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.2.2.9_1.4-1 as appropriate.

#### 5.2.2.2.9_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

#### 5.2.2.2.9_1.3.3_1 Message exceptions for SA

**Table 5.2.2.2.9_1.3.3_1-1: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 for test 1-1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.9_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1 | |
| } | | | |

**Table 5.2.2.2.9_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n8 | for test 1-1 | |
| } | | | |

**Table 5.2.2.2.9_1.3.3_1-4: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots20 | 1 for CSI-RS resource #1 and #2<br><br>2 for CSI-RS resource #3 and #4 | For test 1-1:<br>offset = 1 for CSI-RS resource 1 and 2<br>offset =2 for CSI-RS resource 3 and 4. | |
| } | | | |

### 5.2.2.2.9_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.9_1.3.3_1

### 5.2.2.2.9_1.4 Test requirement

Tables 5.2.2.2.9_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.9_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.9_1.4-1: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.4 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-SFN | 2x2 | 70 | 14.8 |

## 5.2.2.2.10 2Rx TDD FR1 HST DPS performance

### 5.2.2.2.10.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.10.0-3, with the test parameters defined in Table 5.2.2.2.10.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.10.0-1.

**Table 5.2.2.2.10.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify UE performance in the HST-DPS scenario defined in B.3.3 | 1-1, 1-2 |

**Table 5.2.2.2.10.0-2: Test Parameters for Testing**

| Parameter | | | Unit | Value |
|---|---|---|---|---|
| Duplex mode | | | | TDD |
| Active DL BWP index | | | | 1 |
| PDCCH configuration | TCI state | | | Note 1 |
| PDSCH configuration | Mapping type | | | Type A |
| | k0 | | | 0 |
| | Starting symbol (S) | | | 2 |
| | Length (L) | | | Specific to each Reference channel |
| | PDSCH aggregation factor | | | 1 |
| | PRB bundling type | | | Static |
| | PRB bundling size | | | 2 |
| | Resource allocation type | | | Type 0 |
| | RBG size | | | Config2 |
| | VRB-to-PRB mapping type | | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A |
| | TCI state | | | Note 1 |
| PDSCH DMRS configuration | DMRS Type | | | Type 1 |
| | Number of additional DMRS | | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 |
| CSI-RS for tracking | Resource set #1 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 5 for CSI-RS resource 1 and 3 |
| | | | | $l_0$ = 9 for CSI-RS resource 2 and 4 |
| | | CSI-RS periodicity | Slots | 20 for CSI-RS resource 1,2,3,4 |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2 |
| | | | | 2 for CSI-RS resource 3 and 4 |
| | | QCL info | | TCI state #2 |
| | | Frequency Occupation | | Start PRB 0 |
| | | | | Number of PRB = 52 |
| | Resource set #2 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 5 and 6 |
| | | | | $l_0$ = 10 for CSI-RS resource 7 and 8 |
| | | CSI-RS periodicity | Slots | 20 for CSI-RS resource 5,6,7,8. |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 5 and 6 |
| | | | | 2 for CSI-RS resource 7 and 8 |
| | | QCL info | | TCI state #3 |
| | | Frequency Occupation | | Start PRB 0 |
| | | | | Number of PRB = 52 |
| NZP CSI-RS for CSI acquisition | Resource set #3 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 12 |
| | | CSI-RS periodicity | Slots | 40 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #0 |
| | Resource set #4 | First OFDM symbol in the PRB used for CSI-RS | | l0 = 13 |
| | | CSI-RS periodicity | Slots | 40 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #1 |
| TCI state #0 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking Resource set #1' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 5 from 'CSI-RS for tracking Resource set #2' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #2 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #3 | Type 1 QCL information | SSB index | | SSB #1 |
| | | QCL Type | | Type C |
| | | SSB index | | N/A |

| | | | | |
|---|---|---|---|---|
| | Type 2 QCL information | QCL Type | | N/A |
| Number of HARQ Processes | | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

Note 1: SSB # (k mod 2) , CSI-RS (for tracking) resource set # ((k mod 2) + 1) and CSI-RS (for CSI acquisition) resource set # ((k mod 2) + 3) are transmitted by $k^{th}$ RRH.
For Test 1-1, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by $k^{th}$ RRH from slot#

$$\max[(2k-1)n + 1 + T_{HARQ} + T_{MAC\ proc} + T_{firstTRS} + T_{TRS\ proc}, 0]$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\ proc},$$

PDCCH and PDSCH are DTXed in other slots in which throughput statistics are not considered.
For Test 1-2, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by $k^{th}$ RRH from slot#

$$\max[(2k-1)n + 1 + T_{HARQ} + T_{MAC\ proc}, 0]$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\ proc}$$

Where k=0, 1, 2… is the RRH number, n = 5040 is half of the number of slots between two RRH, $T_{HARQ}$ = 8 is the number of slots between PDSCH and corresponding HARQ-ACK information, $T_{MAC\ proc}$ = 6 is the number of slots for MAC CE processing, $T_{firstTRS}$ = 7 is the number of slots to first TRS transmission occasion after MAC CE command is decoded by the UE, $T_{TRS\ proc}$ = 4 is the number of slots for TRS processing.

**Table 5.2.2.2.10.0-3: Minimum performance for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 1 | 2x2 | 70 | 13.0 |
| 1-2 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 2 | 2x2 | 70 | 13.0 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.10.

### 5.2.2.2.10_1 2Rx TDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: The minimum test time value is in []

#### 5.2.2.2.10_1.1 Test purpose

To verify UE performance in the HST-DPS scenario defined in B.3.3 and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.2.2.10_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 5.2.2.2.10_1.3 Test description

##### 5.2.2.2.10_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.2.2.10.0-2 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.10_1.3.3.

### 5.2.2.2.10_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.2.2.10_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.2.2.10_1.4-1 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.2.2.10_1.4-1 as appropriate.

### 5.2.2.2.10_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

#### 5.2.2.2.10_1.3.3_1 Message exceptions for SA

**Table 5.2.2.2.10_1.3.3_1-1: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 is used | test 1-1, 1-2 |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n8 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-4: NZP-CSI-RS-Resource for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | i-1 for CSI-RS resource #i, i=1,2,3,4,5,6,7,8 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 2 for CSI-RS resource #1, #2, #3, #4<br>3 for CSI-RS resource #5, #6, #7, #8 | for test 1-1, 1-2:<br>TCI-StateId for TCI-State #2 for CSI-RS resource #1, #2, #3, #4<br>TCI-StateId for TCI-State #3 for CSI-RS resource #5, #6, #7, #8 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-5: CSI-RS-ResourceMapping for TRS (Table 5.2.2.2.10_1.3.3_1-4)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 with condition TRS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 5 for CSI-RS resource #1 and #3<br>9 for CSI-RS resource #2 and #4<br>6 for CSI-RS resource #5 and #6<br>10 for CSI-RS resource #7 and #8 | for test 1-1, 1-2:<br><br>$l_0$ = 5 for CSI-RS resource 1 and 3<br>$l_0$ = 9 for CSI-RS resource 2 and 4<br>$l_0$ = 6 for CSI-RS resource 5 and 6<br>$l_0$ = 10 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-5: CSI-ResourcePeriodicityAndOffset for CSI Tracking (Table 5.2.2.2.10_1.3.3_1-4)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots20 | 1 for CSI-RS resource #1, #2, #5, #6<br><br>2 for CSI-RS resource #3 #4, #7, #8 | For test 1-1, 1-2:<br>periodicity:<br>20 slots.<br>offset:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4<br>1 for CSI-RS resource 5 and 6<br>2 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-6: NZP-CSI-RS-ResourceSet for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 0 for Resource set #1<br>1 for Resource set #2 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #1 |
| NZP-CSI-RS-ResourceId[1] | 0 | entry 1<br>CSI-RS resource #1 | |
| NZP-CSI-RS-ResourceId[2] | 1 | entry 2<br>CSI-RS resource #2 | |
| NZP-CSI-RS-ResourceId[3] | 2 | entry 3<br>CSI-RS resource #3 | |
| NZP-CSI-RS-ResourceId[4] | 3 | entry 4<br>CSI-RS resource #4 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #2 |
| NZP-CSI-RS-ResourceId[1] | 4 | entry 1<br>CSI-RS resource #5 | |
| NZP-CSI-RS-ResourceId[2] | 5 | entry 2<br>CSI-RS resource #6 | |
| NZP-CSI-RS-ResourceId[3] | 6 | entry 3<br>CSI-RS resource #7 | |
| NZP-CSI-RS-ResourceId[4] | 7 | entry 4<br>CSI-RS resource #8 | |
| } | | | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-7: NZP-CSI-RS-Resource for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | 8 for CSI-RS resource #9<br>9 for CSI-RS resource #10 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 0 for CSI-RS resource #9<br>1 for CSI-RS resource #10 | for test 1-1, 1-2:<br>TCI-State #0 for CSI-RS resource #9<br><br>TCI-State #1 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-8: CSI-RS-ResourceMapping for CSI Acquisition (Table 5.2.2.2.10_1.3.3_1-7)**

| **Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15** | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 12 for CSI-RS resource #9<br>13 for CSI-RS resource #10 | for test 1-1, 1-2<br><br>$l_0$=12 for CSI-RS resource #9<br><br>$l_0$=13 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-9: CSI-ResourcePeriodicityAndOffset for CSI Acquisition (Table 5.2.2.2.10_1.3.3_1-7)**

| **Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16** | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 0 | For test 1-1, 1-2:<br>periodicity = 40 slots.<br>offset = 0 slots | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-10: NZP-CSI-RS-ResourceSet for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-18 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 2 for Resource set #3<br>3 for Resource set #4 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #3 |
| NZP-CSI-RS-ResourceId[1] | 8 | entry 1<br>CSI-RS resource #9 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #4 |
| NZP-CSI-RS-ResourceId[1] | 9 | entry 1<br>CSI-RS resource #10 | |
| } | | | |
| } | | | |

**Table 5.2.2.2.10_1.3.3_1-11: *TCI-State***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 for TCI state #0<br>1 for TCI state #1<br>2 for TCI state #2<br>3 for TCI state #3 | For test 1-1, 1-2 | |
| qcl-Type1 SEQUENCE { | | | |
| bwp-Id | BWP-Id of active BWP | | TCI state #0, TCI state #1 |
| | Not present | | TCI state #2, TCI state #3 |
| referenceSignal CHOICE { | | | |
| csi-rs | 0 | CSI-RS resource #1 | TCI state #0 |
| | 4 | CSI-RS resource #5 | TCI state #1 |
| ssb | 0 | SSB #0 | TCI state #2 |
| | 1 | SSB #1 | TCI state #3 |
| } | | | |
| qcl-Type | typeA | | TCI state #0, TCI state #1 |
| | typeC | | TCI state #2, TCI state #3 |
| } | | | |
| } | | | |

### 5.2.2.2.10_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.10_1.3.3_1

### 5.2.2.2.10_1.4 Test requirement

Tables 5.2.2.2.10_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.10_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.10_1.4-1: Test Requirements for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 1 | 2x2 | 70 | 13.6 |
| 1-2 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 2 | 2x2 | 70 | 13.6 |

## 5.2.2.2.11 2Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance

### 5.2.2.2.11.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.11.0-3, with the addition of test parameters in Table 5.2.2.2.11.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.11.0-1.

**Table 5.2.2.2.11.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance with Single-DCI based SDM scheme under 2 receive antenna conditions | 1-1,1-2 |

**Table 5.2.2.2.11.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | 0 | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 40 | |
| | | CSI-RS offset | Slots | 20 for CSI-RS resources 1 and 2<br>21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6<br>21 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |

| | | | | | |
|---|---|---|---|---|---|
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000 | 1002 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Full-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.25 for test 1-1<br>1 for test 1-2 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 for test 1-1<br>0 for test 1-2 | |
| Number of HARQ Processes | | | | 8 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs (PDSCH Layer 0 is transmitted from TRxP #1 and PDSCH layer 1 is transmitted from TRxP #2) | | | | | |

**Table 5.2.2.2.11.0-3: Minimum performance**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 20.2 |
| 1-2 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 20.0 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.11.

### 5.2.2.2.11_1 2Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.2.11_1.1 Test purpose

To verify the PDSCH performance with Single-DCI based SDM scheme under 2 receive antenna conditions.

#### 5.2.2.2.11_1.2 Test applicability

Test applies to all types of NR UE release 16 and forward supporting capability IE *singleDCI-SDM-scheme-r16*.

#### 5.2.2.2.11_1.3 Test description

##### 5.2.2.2.11_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.2.11.0-2 and Table 5.2.2.2.11.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.11_1.3.3.

##### 5.2.2.2.11_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.11_1.3.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.11_1.3.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Table 5.2.2.2.11_1.3.4-1 as appropriate.

##### 5.2.2.2.11_1.3.3 Message contents

###### 5.2.2.2.11_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.2.11_1.3.3_1-1: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 and 2 | “1011” | |
| Transmission configuration indication | TCI state 1 and 2 | “000” | |

**Table 5.2.2.2.11_1.3.3_1-2: *CellGroupConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| simultaneousTCI-UpdateList1-r16 SEQUENCE { | | | |
| ServCellIndex [1] | ServCellIndex | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.11_1.3.3_1-3: *ControlResourceSet***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-28 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.2.2.11_1.3.3_1-4: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] | *TCI-State* with condition TCI-state-0 | | |
| TCI-State[2] | *TCI-State* with condition TCI-state-1 | | |
| TCI-State[3] | *TCI-State* with condition TCI-state-2 | | |
| } | | | |
| rbg-Size | config2 | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.2.2.11_1.3.3_1-5: *TCI-State***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 | | TCI-state-0 |
| | 1 | | TCI-state-1 |
| | 2 | | TCI-state-2 |
| qcl-Type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | TCI-state-0 |
| csi-rs | 1 | | TCI-state-1 |
| | 5 | | TCI-state-2 |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| qcl-Type2 | Not present | | |
| } | | | |

**Table 5.2.2.2.11_1.3.3_1-6: *NZP-CSI-RS-Resource***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-85 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| resourceMapping SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| cdm-Type | noCDM | | |
| density CHOICE { | | | |
| three | NULL | | |
| } | | | |
| } | | | |
| periodicityAndOffset CHOICE { | | | |
| slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |
| qcl-InfoPeriodicCSI-RS | 0 | | |
| } | | | |

5.2.2.2.11_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.2.11_1.3.3_1.

#### 5.2.2.2.11_1.3.4 Test requirement

Table 5.2.2.2.11.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.11_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.11_1.3.4-1: Test requirement**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 21.2 |
| 1-2 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 21.0 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | | |

### 5.2.2.2.12 2Rx TDD FR1 PDSCH Multi-DCI based transmission scheme performance

#### 5.2.2.2.12.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.12.0-3, with the addition of test parameters in Table 5.2.2.2.12.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.12.0-1.

**Table 5.2.2.2.12.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs | 1-1 |

**Table 5.2.2.2.12.0-2: Test parameters**

| Parameter | | Unit | Value | |
|---|---|---|---|---|
| | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | TRxP #1 | |
| PDCCH configuration | TCI state | | TCI State #1 | TCI State #2 |
| | CORESETPoolIndex | | 0,1 | |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | 3 | |
| | CSI-RS periodicity | Slots | 40 | |

| | | | | | |
|---|---|---|---|---|---|
| | CSI-RS offset | | Slots | 20 for CSI-RS resources 1 and 2<br>21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6<br>21 for CSI-RS resources 7 and 8 |
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | {1000,1001} | {1002,1003} |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Non-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.25 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 | |
| Number of HARQ Processes | | | | 8 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs. Transmission from TRxP #1 uses CORESETPoolIndex 0 and transmission from TRxP #2 uses CORESETPoolIndex 1 | | | | | |

**Table 5.2.2.2.12.0-3: Minimum performance**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition( Note 1) | Correlation matrix and antenna configuration( Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | TRxP #1 | TRxP #2 | | | | | | | |
| 1-1 | R.PDSCH.2-3.3 TDD | R.PDSCH.2-3.4 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 20.4 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.12.

### 5.2.2.2.12_1 2Rx TDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.2.12_1.1 Test purpose

To verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs.

#### 5.2.2.2.12_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *multiDCI-MultiTRP-r16*.

#### 5.2.2.2.12_1.3 Test description

Same test description as in clause 5.2.2.1.12_1.3 with the following exception:

- Table 5.2.2.2.12_1.4-1 instead of 5.2.2.1.12_1.4-1
- Table 5.2.2.2.12_1.3-1 instead of Table 5.2.2.1.12_1.3.3_1-8

**Table 5.2.2.2.12_1.3-1: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| Slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

#### 5.2.2.2.12_1.4 Test requirement

Table 5.2.2.2.12.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.12_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.12_1.4-1: Test requirement**

| **est um.** | **Reference channel** | **Bandwidth (MHz) / Subcarrier spacing (kHz)** | **Modulation format and code rate** | **TDD UL-DL pattern** | **Propagation condition(Note 1)** | **Correlation matrix and antenna configuration(Note 2)** | **Reference value** | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | **Fraction of maximum throughput (%)** | **SNR (dB)(Not 3)** |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | TRxP #1 | TRxP #2 | | | | | | | |
| l-1 | R.PDSCH.2-3.3 TDD | R.PDSCH.2-3.4 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 21.4 |

ote 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent
ote 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2
ote 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2

### 5.2.2.2.13 2Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance

#### 5.2.2.2.13.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.13.0-3, with the addition of test parameters in Table 5.2.2.2.13.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.13.0-1.

**Table 5.2.2.2.13.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 2 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16” defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.2.2.13.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | Not configured | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3 l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7 l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | ‘No CDM’ for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 40 | |
| | | CSI-RS offset | Slots | 20 for CSI-RS resources 1 and 2 21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6 21 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | Wideband | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |

| | | | | | |
|---|---|---|---|---|---|
| PDSCH DMRS configuration | Antenna port indexes | | | 1000,1001 | 1000,1001 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.25 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 | |
| Number of HARQ Processes | | | | 8 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs | | | | | |

**Table 5.2.2.2.13.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.2-2.5 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 17.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.13.

### 5.2.2.2.13_1 2Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.2.13_1.1 Test purpose

To verify the PDSCH performance under 2 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16”.

#### 5.2.2.2.13_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportFDM-SchemeA-r16*.

#### 5.2.2.2.13_1.3 Test description

Same test description as in clause 5.2.2.1.13_1.3 with the following exception:

- Table 5.2.2.2.13_1.4-1 instead of 5.2.2.1.13_1.4-1

- Table 5.2.2.2.13_1.3-1 instead of Table 5.2.2.1.13_1.3.3_1-5

**Table 5.2.2.2.13_1.3-1: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| Slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.2.2.13_1.4 Test requirement

Table 5.2.2.2.13.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.13_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.13_1.4-1: Test requirement**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.2-2.5 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 18.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

## 5.2.2.2.14 2Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance

### 5.2.2.2.14.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.14.0-3, with the addition of test parameters in Table 5.2.2.2.14.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.14.0-1.

**Table 5.2.2.2.14.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 2 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.2.2.14.0-2: Test parameters**

| Parameter | | Unit | Value | |
|---|---|---|---|---|
| | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | TRxP #1 | |
| PDCCH configuration | TCI state | | TCI State #1 | |
| | CORESETPoolIndex | | Not configured | |

| | | | | | |
|---|---|---|---|---|---|
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | | l0 = 6 for CSI-RS resources 1 and 3 l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7 l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | | 3 | |
| | CSI-RS periodicity | | Slots | 40 | |
| | CSI-RS offset | | Slots | 20 for CSI-RS resources 1 and 2 21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6 21 for CSI-RS resources 7 and 8 |
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | Repetition number | | | 2 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000 | 1000 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | 1 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 (Note 2) | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs<br>Note 2: ACK/NACK feedback is generated for PDSCH on slot i, where mod(i,10) = {2, 4, 6}. | | | | | |

**Table 5.2.2.2.14.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | BLER (%) | SNR (dB) (Note 4) |
| 1-1 | R.PDSCH.2-16.2 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 1 (Note 3) | 2.8 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.<br>Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.14.

### 5.2.2.2.14_1 2Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.2.2.14_1.1 Test purpose

To verify the PDSCH performance under 2 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states.

#### 5.2.2.2.14_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportTDM-SchemeA-r16*.

#### 5.2.2.2.14_1.3 Test description

Same test description as in clause 5.2.2.1.14_1.3 with the following exception:

- Table 5.2.2.2.14_1.4-1 instead of 5.2.2.1.14_1.4-1
- Table 5.2.2.2.14_1.3-1 instead of Table 5.2.2.1.14_1.3.3_1-5

**Table 5.2.2.2.14_1.3-1: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| Slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

#### 5.2.2.2.14_1.4 Test requirement

Table 5.2.2.2.14.0-3 defines the primary level settings.

The residual BLER specified in Note 3 of Table 5.2.2.2.14_1.4-1 test shall meet or be lower than the specified value in Table 5.2.2.2.14_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.14_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | BLER (%) | SNR (dB) (Note 4) |
| 1-1 | R.PDSCH.2-16.2 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 1 (Note 3) | 3.8 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.<br>Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

## 5.2.3 4RX requirements

### 5.2.3.1 FDD

#### 5.2.3.1.1 4Rx FDD FR1 PDSCH mapping Type A performance

##### 5.2.3.1.1.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.1.0-3, Table 5.2.3.1.1.0-4, Table 5.2.3.1.1.0-5, Table 5.2.3.1.1.0-6 and Table 5.2.3.1.1.0-7, with the addition of test parameters in Table 5.2.3.1.1.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.3.1.1.0-1.

**Table 5.2.3.1.1.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and with different channel models, MCSs and number of MIMO layers | 1-1, 1-2, 1-3, 1-5, 1-6, 1-7, 2-1, 2-2, 3-1, 4-1 |
| Verify the PDSCH mapping Type A HARQ soft combining performance under 4 receive antenna conditions. | 1-4 |
| Verify the PDSCH mapping Type A performance requirements for Enhanced Receiver Type 1 under 4 receive antenna conditions. | 5-1 |

**Table 5.2.3.1.1.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 4 for Test 1-1<br>WB for Test 3-1<br>2 for other tests |
| | Resource allocation type | | Test 1-2: Type 1 with start RB = 23, $L_{RBs}$ = 6<br><br>Other test: Type 0 |
| | RBG size | | Test 1-2: N/A<br><br>Other tests: Config2 |

| Parameter | | Unit | Value |
|---|---|---|---|
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 for Test 1-1, 1-5, 1-6, 1-7<br>1 for other tests |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | CSI-RS periodicity | Slots | Test 1-5, 1-6, 1-7:<br>10 for CSI-RS resource 1,2,3,4.<br><br>Other tests: Table 5.2-1. |
| | CSI-RS offset | Slots | Test 1-5, 1-6, 1-7:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4.<br><br>Other tests: Table 5.2-1. |
| Number of HARQ Processes | | | 8 for Test 1-4, 2-1<br>4 for other tests |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.3.1.1.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 FDD | 10 / 15 | QPSK, 0.30 | TDLB100-400 | 2x4, ULA Low | 70 | -3.5 |
| 1-2 | R.PDSCH.1-1.2 FDD | 10 / 15 | QPSK, 0.30 | TDLC300-100 | 2x4, ULA Low | 70 | -2.9 |
| 1-3 | R.PDSCH.1-4.1 FDD | 10 / 15 | 256QAM, 0.82 | TDLA30-10 | 2x4, ULA Low | 70 | 21.0 |
| 1-4 | R.PDSCH.1-2.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 2x4, ULA Low | 30 | -1.5 |
| 1-5 | R.PDSCH.1-8.1 FDD | 10 / 15 | 16QAM, 0.48 | HST-750 | 1x4 | 70 | 3.3 |
| 1-6 | R.PDSCH.1-8.2 FDD | 10 / 15 | 64QAM, 0.43 | HST-972 | 1x4 | 70 | [6.8] |
| 1-7 | R.PDSCH.1-8.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-600 | 2x4 | 70 | [5.8] |

**Table 5.2.3.1.1.0-4: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2-1 | R.PDSCH.1-3.1 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 13.5 |
| 2-2 | R.PDSCH.2-1.1 FDD | 20 / 30 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 13.7 |

**Table 5.2.3.1.1.0-5: Minimum performance for Rank 3**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.1-2.3 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 4x4, ULA Low | 70 | 11.0 |

**Table 5.2.3.1.1.0-6: Minimum performance for Rank 4**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 4-1 | R.PDSCH.1-2.4 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 4x4, ULA Low | 70 | 15.6 |

**Table 5.2.3.1.1.0-7: Minimum performance for Rank 3 and Enhanced Receiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5-1 | R.PDSCH.1-2.3 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 4x4, ULA Medium A | 70 | 22.3 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.1.1.

### 5.2.3.1.1_1 4Rx FDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with baseline receiver for both SA and NSA

Editor's note: Minimum test time is FFS for test point 1-5, 1-6, 1-7. SNR in test requirements table is within square brackets for test point 1-6, 1-7.

#### 5.2.3.1.1_1.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 1 and Rank 2 scenarios.

#### 5.2.3.1.1_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

#### 5.2.3.1.1_1.3 Test description

##### 5.2.3.1.1_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.3.1.1.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.1_1.3.3.

##### 5.2.3.1.1_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.1.0-3 and Table 5.2.3.1.1.0-4. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.1.1_1.4-1 and 5.2.3.1.1_1.4-2 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 4 for each subtest in Tables 5.2.3.1.1_1.4-1 and 5.2.3.1.1_1.4-2 as appropriate.

##### 5.2.3.1.1_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.2.3.1.1_1.3.3_1 Message exceptions for SA

**Table 5.2.3.1.1_1.3.3_1-1: *BWP***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| BWP ::= SEQUENCE { | | | |
| locationAndBandwidth | 13750 | For Test 2-2 (20MHz BW, SCS 30kHz) | |
| | 14025 | For other tests (10MHz BW, SCS 15kHz) | |
| } | | | |

**Table 5.2.3.1.1_1.3.3_1-2: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType0 | resourceAllocation Type0 for all tests except test 1-2 | |
| | resourceAllocationType1 | resourceAllocation Type1 for test 1-2 | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | n4 | n4 for test 1-1 | |
| | wideband | wideband for test 3-1 | |
| | Not present | n2 for other tests | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.1_1.3.3_1-3: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos1 | pos1 for all tests except test 1-1, 1-5, 1-6, 1-7 | |
| | Not present | pos2 for test 1-1, 1-5, 1-6, 1-7 | |
| } | | | |

**Table 5.2.3.1.1_1.3.3_1-4: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | Not present | n8 for test 1-4, 2.1 | |
| | n4 | n4 for other tests | |
| } | | | |

**Table 5.2.3.1.1_1.3.3_1-5: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots10 | 1 (for CSI-RS resources 1 and 2)<br>2 (for CSI-RS resources 3 and 4) | Periodicity 10 slots and offset 1/2 for test 1-5, 1-6, 1-7 | |
| } | | | |

### 5.2.3.1.1_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.1_1.3.3_1

### 5.2.3.1.1_1.4 Test requirement

Table 5.2.3.1.1.0-3 and Table 5.2.3.1.1.0-4 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.1_1.4-1 and Table 5.2.3.1.1_1.4-2 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.1_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 FDD | 10 / 15 | QPSK, 0.30 | TDLB100-400 | 2x4, ULA Low | 70 | -2.6 |
| 1-2 | R.PDSCH.1-1.2 FDD | 10 / 15 | QPSK, 0.30 | TDLC300-100 | 2x4, ULA Low | 70 | -2.0 |
| 1-3 | R.PDSCH.1-4.1 FDD | 10 / 15 | 256QAM, 0.82 | TDLA30-10 | 2x4, ULA Low | 70 | 22.0 |
| 1-4 | R.PDSCH.1-2.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 2x4, ULA Low | 30 | -0.6 |
| 1-5 | R.PDSCH.1-8.1 FDD | 10 / 15 | 16QAM, 0.48 | HST-750 | 1x4 | 70 | 4.2 |
| 1-6 | R.PDSCH.1-8.2 FDD | 10 / 15 | 64QAM, 0.43 | HST-972 | 1x4 | 70 | [7.7] |
| 1-7 | R.PDSCH.1-8.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-600 | 2x4 | 70 | [6.7] |

**Table 5.2.3.1.1_1.4-2: Test Requirement for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 2-1 | R.PDSCH.1-3.1 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 14.5 |
| 2-2 | R.PDSCH.2-1.1 FDD | 20 / 30 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 14.7 |

### 5.2.3.1.1_2 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.1.1_2.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 3 and Rank 4 scenarios.

#### 5.2.3.1.1_2.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

#### 5.2.3.1.1_2.3 Test description

Same test description as in clause 5.2.3.1.1_1.3 with the following exception:

- Step 1 of test procedure to call for Tables 5.2.3.1.1.0-5 and 5.2.3.1.1.0-6 instead of Tables 5.2.3.1.1.0-3 and 5.2.3.1.1.0-4

Table 5.2.3.1.1_2.4-1 instead of 5.2.3.1.1_1.4-1

- Table 5.2.3.1.1_2.4-2 instead of 5.2.3.1.1_1.4-2
- Figure A.3.1.7.5 instead of A.3.1.7.4

### 5.2.3.1.1_2.4 Test requirement

Table 5.2.3.1.1.0-5 and Table 5.2.3.1.1.0-6 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.1_2.4-1 and Table 5.2.3.1.1_2.4-2 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.1_2.4-1: Test Requirement for Rank 3**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.1-2.3 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 4x4, ULA Low | 70 | 12.0 |

**Table 5.2.3.1.1_2.4-2: Test Requirement for Rank 4**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 4-1 | R.PDSCH.1-2.4 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 4x4, ULA Low | 70 | 16.6 |

### 5.2.3.1.1_3 FFS

### 5.2.3.1.1_4 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with enhanced receiver type 1 for both SA and NSA

### 5.2.3.1.1_4.1 Test purpose

To verify the PDSCH mapping Type A enhanced performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default enhanced receiver type 1 configuration, for Rank 3 scenario.

### 5.2.3.1.1_4.2 Test applicability

This test applies to all types of NR UE Rel-15 and forward supporting 4 Rx antenna ports and NR enhanced receiver type 1.

This test also applies to all types of EUTRA UE Rel-15 and forward supporting EN-DC, 4 Rx antenna ports and NR enhanced receiver type 1.

### 5.2.3.1.1_4.3 Test description

Same test description as in clause 5.2.3.1.1_1.3 with the following exception:

- Figure A.3.1.7.5 instead of A.3.1.7.4

Step 1 and 2 of Test procedure as in clause 5.2.3.1.1_1.3.2 are replaced by:

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.1.0-7. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.1.1_4.4-1 as appropriate.

#### 5.2.3.1.1_4.4 Test requirement

Table 5.2.3.1.1.0-7 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.1_4.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.1_4.4-1: Test Requirement for Rank 3 and Enhanced Receiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5-1 | R.PDSCH.1-2.3 FDD | 10 / 15 | 16QAM, 0.48 | TDLA30-10 | 4x4, ULA Medium A | 70 | 23.3 |

### 5.2.3.1.2 4Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance

#### 5.2.3.1.2.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.2.0-3, with the addition of test parameters in Table 5.2.3.1.2.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.2.0-1.

**Table 5.2.3.1.2.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and CSI-RS overlapped with PDSCH | 1-1 |

**Table 5.2.3.1.2.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |

| | | | |
|---|---|---|---|
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| NZP CSI-RS for CSI acquisition | OFDM symbols in the PRB used for CSI-RS | | $l_0$ = 13 |
| | CSI-RS periodicity | Slots | 5 |
| ZP CSI-RS for CSI acquisition | Subcarrier index in the PRB used for CSI-RS | | ($k_0$, $k_1$, $k_2$, $k_3$)=(2, 4, 6, 8) |
| | Number of CSI-RS ports (X) | | 8 |
| | CSI-RS periodicity | Slots | 5 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.3.1.2.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-5.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 4x4, ULA Low | 70 | 9.1 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.1.2.

### 5.2.3.1.2_1 4Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 4x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.1.2_1.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration for CSI-RS overlapped with PDSCH scenario.

#### 5.2.3.1.2_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

#### 5.2.3.1.2_1.3 Test description

##### 5.2.3.1.2_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.5 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.3.1.2.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.3.1.2_1.3.3.

### 5.2.3.1.2_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.2.0-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.1.2_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-12 in Annex G clause G.1.5.

### 5.2.3.1.2_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

#### 5.2.3.1.2_1.3.3_1 Message exceptions for SA

**Table 5.2.3.1.2_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | | |
| } | | | |

**Table 5.2.3.1.2_1.3.3_1-2: NZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row4 | 001 | $k_0$=0 | |
| } | | | |
| firstOFDMSymbolInTimeDomain | 13 | $l_0$ = 13 | |
| } | | | |

**Table 5.2.3.1.2_1.3.3_1-3: CSI-ResourcePeriodicityAndOffset for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots5 | 0 | Periodicity 5 slots and offset 0 | |
| } | | | |

**Table 5.2.3.1.2_1.3.3_1-4: ZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], clause5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | (k0, k1, k2, k3)=(2, 4, 6, 8) | |
| } | | | |
| nrofPorts | P8 | Eight Ports | |
| firstOFDMSymbolInTimeDomain | 12 | $l_0 = 12$ | |
| cdm-Type | fd-CDM2 | | |
| density CHOICE { | | | |
| one | NULL | | |
| } | | | |
| freqBand | CSI-FrequencyOccupation | | |
| } | | | |

### 5.2.3.1.2_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.2_1.3.3_1

### 5.2.3.1.2_1.4 Test requirement

Table 5.2.3.1.2.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.2_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.2_1.4-1: Test Requirement for Rank 2**

| **Test num.** | **Reference channel** | **Bandwidth (MHz) / Subcarrier spacing (kHz)** | **Modulation format and code rate** | **Propagation condition** | **Correlation matrix and antenna configuration** | **Reference value** | |
|---|---|---|---|---|---|---|---|
| | | | | | | **Fraction of maximum throughput (%)** | **SNR (dB)** |
| 1-1 | R.PDSCH.1-5.1 FDD | 10 / 15 | 16QAM, 0.48 | TDLC300-100 | 4x4, ULA Low | 70 | 10 |

### 5.2.3.1.3 4Rx FDD FR1 PDSCH mapping Type B performance

#### 5.2.3.1.3.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.3.0-3, with the addition of test parameters in Table 5.2.3.1.3.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.3.0-1.

**Table 5.2.3.1.3.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| PDSCH mapping Type B performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.1.3.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 5 |
| | Length (L) | | 7 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.3.1.3.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.3 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x4, ULA Low | 70 | -3.8 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.1.3.

### 5.2.3.1.3_1 4Rx FDD FR1 PDSCH mapping Type B performance - 2x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.1.3_1.1 Test purpose

To verify the PDSCH mapping Type B normal performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput with baseline receiver configuration.

#### 5.2.3.1.3_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and PDSCH mapping type B.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports and PDSCH mapping type B.

#### 5.2.3.1.3_1.3 Test description

##### 5.2.3.1.3_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.3.1.3.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.3.1.3_1.3.3.

##### 5.2.3.1.3_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.3.0-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.1.3_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

##### 5.2.3.1.3_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

5.2.3.1.3_1.3.3_1 Message exceptions for SA

**Table 5.2.3.1.3_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | | |
| } | | | |

**Table 5.2.3.1.3_1.3.3_1-2: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeB | | |
| startSymbolAndLength | 89 | Start symbol(S)=5, Length(L)=7 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | TypeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

5.2.3.1.3_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.3_1.3.3_1

5.2.3.1.3_1.4 Test requirement

Table 5.2.3.1.3.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.3_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.3_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.3 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x4, ULA Low | 70 | -2.8 |

### 5.2.3.1.4 4Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance

#### 5.2.3.1.4.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.4.0-3, with the addition of test parameters in Table 5.2.3.1.4.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.3.1.4.0-1.

**Table 5.2.3.1.4.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions with CRS rate matching configured | 1-1, 1-2 |

**Table 5.2.3.1.4.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| **NR UL transmission with a 7.5 kHz shift to the LTE raster** | | | **true** |
| PDCCH configuration | Symbols with PDCCH | | Symbol# 2 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 3 |
| | Length (L) | | 9 for Test 1-1<br>11 for Test 1-2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Position of the first DM-RS for downlink | | 3 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CRS for rate matching (Note 1) | LTE carrier centre subcarrier location | | Same as NR carrier centre subcarrier location |
| | LTE carrier BW | MHz | 10 |
| | Number of antenna ports | | 4 |
| | v-shift | | 0 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |
| Note 1: No MBSFN is configured on LTE carrier | | | |

**Table 5.2.3.1.4.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.1-7.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x4, ULA Low | 70 | -4.0 |
| 1-2 | R.PDSCH.1-7.2 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x4, ULA Low | 70 | -4.0 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.4.

### 5.2.3.1.4_1 4Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.1.4_1.1 Test purpose

Same as 5.2.2.1.4_1.1.

#### 5.2.3.1.4_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test1-1 also applies to all types of EUTRA UE release 15 and forward supporting EN-DC supporting 4 Rx antenna ports and capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test 1-2 applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

Test 1-2 also applies to all types of EUTRA UE release 15 and forward supporting EN-DC supporting 4 Rx antenna ports and capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

#### 5.2.3.1.4_1.3 Test description

##### 5.2.3.1.4_1.3.1 Initial conditions

Same as 5.2.2.1.4_1.3.1 with the following exceptions:

- Use Figure A.3.1.7.4 for TE diagram
- Use Figure A.3.2.5 for UE diagram
- Instead of 5.2.2.1.4.x → refer 5.2.2.3.4.x

##### 5.2.3.1.4_1.3.2 Test procedure

Same as 5.2.2.3.4_1.3.2 with the following exceptions:

- Instead of 5.2.2.1.4.x → refer 5.2.2.3.4.x

##### 5.2.3.1.4_1.3.3 Message contents

Same as 5.2.2.1.4_1.3.3.

##### 5.2.3.1.4_1.3.4 Test requirement

Table 5.2.3.1.4.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.4_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.4_1.3.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-7.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x4, ULA Low | 70 | -3.0 |
| 1-2 | R.PDSCH.1-7.2 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 4x4, ULA Low | 70 | -3.0 |

## 5.2.3.1.5 4Rx FDD FR1 PDSCH 0.001% BLER performance

### 5.2.3.1.5.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.5.0-3, with the addition of test parameters in Table 5.2.3.1.5.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.5.0-1.

**Table 5.2.3.1.5.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH 0.001% BLER performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.1.5.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.3.1.5.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.1-1.4 FDD | 10 / 15 | QPSK, 0.59 | AWGN | 1x4, ULA Low | 0.001% | 0.7 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.5.

### 5.2.3.1.5_1 4Rx FDD FR1 PDSCH 0.001% BLER performance - 1x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.1.5_1.1 Test purpose

To verify the PDSCH 0.001% BLER performance under 4 receive antenna conditions.

#### 5.2.3.1.5_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *dl-64QAM-MCS-TableAlt* and capability IE *cqi-TableAlt*.

#### 5.2.3.1.5_1.3 Test description

##### 5.2.3.1.5_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.5.0-2 and Table 5.2.3.1.5.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.5_1.3.3.

##### 5.2.3.1.5_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.5.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.1.5_1.3.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.4.3-1 in Annex G.

#### 5.2.3.1.5_1.3.3 Message contents

##### 5.2.3.1.5_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.1.5_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| mcs-Table | qam64LowSE | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

##### 5.2.3.1.5_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.5_1.3.3_1.

#### 5.2.3.1.5_1.3.4 Test requirement

Table 5.2.3.1.5.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.5_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.5_1.3.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-1.4 FDD | 10 / 15 | QPSK, 0.59 | AWGN | 1x4, ULA Low | 0.001% | 1.3 |

### 5.2.3.1.6 4Rx FDD FR1 PDSCH repetitions over multiple slots performance

#### 5.2.3.1.6.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.6.0-3, with the addition of test parameters in Table 5.2.3.1.6.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.6.0-1.

**Table 5.2.3.1.6.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.1.6.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.3.1.6.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-11.1 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x4, ULA Low | 1% (Note 1) | -2.3 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.6.

### 5.2.3.1.6_1 4Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x4 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- MU factor due to finite test time needs further study

#### 5.2.3.1.6_1.1 Test purpose

To Verify the PDSCH repetitions over multiple slots performance under 4 receive antenna conditions.

#### 5.2.3.1.6_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

#### 5.2.3.1.6_1.3 Test description

##### 5.2.3.1.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.6.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.6_1.3.3.

### 5.2.3.1.6_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.6.0-3. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.1.6_1.3.4-1.
3. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport bloks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to clause G.1.5 and Table G.1.5-1a in Annex G clause G.1.5.

### 5.2.3.1.6_1.3.3 Message contents

#### 5.2.3.1.6_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

#### 5.2.3.1.6_1.3.3_2 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

### 5.2.3.1s.6_1.3.4 Test requirement

Table 5.2.3.1.6.0-3 defines the primary level settings.

The target BLER for the downlink reference measurement channels specified in Annex A.3.2.1 for each BLER test shall meet or exceed the specified value in Table 5.2.3.1.6_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.6_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-11.1 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x4, ULA Low | 1% (Note 1) | -2.3 + TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

## 5.2.3.1.7 4Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance

### 5.2.3.1.7.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.7.0-3, with the addition of test parameters in Table 5.2.3.1.7.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.7.0-1.

**Table 5.2.3.1.7.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance and UE processing capability 2 under four receive antenna conditions | 1-1 |

**Table 5.2.3.1.7.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 0 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 2 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 0 |

**Table 5.2.3.1.7.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-12.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x4, ULA Low | 70 | [-2.3] |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.7.

### 5.2.3.1.7_1 4Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x4 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Minimum requirements are within brackets.

#### 5.2.3.1.7_1.1 Test purpose

To verify PDSCH mapping Type B performance and UE processing capability 2 under four receive antenna conditions.

#### 5.2.3.1.7_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-ProcessingType2*.

#### 5.2.3.1.7_1.3 Test description

##### 5.2.3.1.7_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.7.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On*, *Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.7_1.3.3.

5.2.3.1.7_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.7.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.1.7_1.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

5.2.3.1.7_1.3.3 Message contents

5.2.3.1.7_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.1.7_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entries | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeB | | |
| startSymbolAndLength | 16 | Start symbol(S)=2, Length(L)=2 | |
| } | | | |
| } | | | |

**Table 5.2.3.1.7_1.3.3_1-2: *PUCCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-112 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PUCCH-Config ::= SEQUENCE { | | | FR1 |
| dl-DataToUL-ACK SEQUENCE (SIZE (1)) OF INTEGER { | 1 entry | | |
| INTEGER[1] | 0 | entry 1 | |
| } | | | |
| } | | | |

**Table 5.2.3.1.7_1.3.3_1-3: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PDSCH-to-HARQ_feedback timing indicator | K1=0 as per dl-DataToUL-ACK in Table 5.2.3.1.7_1.3.3_1-3 | “000” | |

5.2.3.1.7_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.7_1.3.3_1.

5.2.3.1.7_1.4 Test requirement

Table 5.2.3.1.7.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.7_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.7_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-12.1 FDD | 10 / 15 | QPSK, 0.30 | TDLA30-10 | 2x4, ULA Low | 70 | [-1.4] |

## 5.2.3.1.8 4Rx FDD FR1 PDSCH pre-emption performance

### 5.2.3.1.8.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.8.0-3, with the addition of test parameters in Table 5.2.3.1.8.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.8.0-1.

**Table 5.2.3.1.8.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH pre-emption performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.1.8.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDCCH configuration (Note 4) | Symbols with PDCCH | | 0, 1 |
| | DCI format | | 2_1 |
| | timeFrequencySet | | 14x1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Pre-emption configuration (Note 2) | Starting symbol (S) | | 3 |
| | Length (L) | | 2 |
| | Pre-emption periodicity and offset (Note 3) | Slots | 10/1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |
| Note 1: Void<br>Note 2: Interference modelled as random data on pre-empted REs.<br>Note 3: Pre-emption is scheduled with a fixed scheduling with 10% probability within 10ms periodicity.<br>Note 4: In addition to PDCCH configuration in Table 5.2-1. | | | |

**Table 5.2.3.1.8.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 1-2.6 FDD | 10 / 15 | 16QAM 0.64 | TDLA30-10 | 2x4, ULA Low | 70 | 6.6 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.8.

5.2.3.1.8_1 4Rx FDD FR1 PDSCH pre-emption performance - 2x4 MIMO with baseline receiver for both SA and NSA

5.2.3.1.8_1.1 Test purpose

To Verify the PDSCH pre-emption performance under 4 receive antenna conditions.

#### 5.2.3.1.8_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pre-EmptIndication-DL-r16*.

#### 5.2.3.1.8_1.3 Test description

##### 5.2.3.1.8_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.8.0-2 and Table 5.2.3.1.8.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.8_1.3.3.

##### 5.2.3.1.8_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.8.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. SS transmits PDCCH DCI format 2_1 for int_RNTI with 10% probability to transmit the DL Preemption indication according to Table 5.2.3.1.8.0-2. In the time and frequency set indicated by PDCCH DCI format 2_1, SS stops transmission of PDSCH.
3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.1.8_1.3.4-1.
4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

##### 5.2.3.1.8_1.3.3 Message contents

###### 5.2.3.1.8_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.8_1.3.3_1.

###### 5.2.3.1.8_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.8_1.3.3_1.

#### 5.2.3.1.8_1.3.4 Test requirement

Table 5.2.3.1.8.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.8_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.8_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 1-2.6 FDD | 10 / 15 | 16QAM 0.64 | TDLA30-10 | 2x4, ULA Low | 70 | 7.6 |

### 5.2.3.1.9 4Rx FDD FR1 HST-SFN performance

#### 5.2.3.1.9.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.9.0-3, with the test parameters defined in Table 5.2.3.1.9.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.3.1.9.0-1.

**Table 5.2.3.1.9.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 4 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when highSpeedDemodFlag-r16 IE [20] is configured | 1-1 |

**Table 5.2.3.1.9.0-2: Test Parameters for Testing**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | CSI-RS periodicity | Slots | 10 for CSI-RS resource 1,2,3,4. |
| | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4. |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.3.1.9.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.3 FDD | 10 / 15 | 16QAM, 0.48 | HST-SFN | 2x4 | 70 | 10.4 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.9.

### 5.2.3.1.9_1 4Rx FDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA

The minimum test time value is in []5.2.3.1.9_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when *highSpeedDemodFlag-r16* IE [20] is configured and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.3.1.9_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

#### 5.2.3.1.9_1.3 Test description

##### 5.2.3.1.9_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.3.1.9.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.9_1.3.3.

#### 5.2.3.1.9_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.3.1.9_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.1.9_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.3.1.9_1.4-1 as appropriate.

#### 5.2.3.1.9_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

#### 5.2.3.1.9_1.3.3_1 Message exceptions for SA

**Table 5.2.3.1.9_1.3.3_1-1: PDSCH-Config**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 for test 1-1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.9_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1 | |
| } | | | |

**Table 5.2.3.1.9_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | for test 1-1 | |
| } | | | |

**Table 5.2.3.1.9_1.3.3_1-4: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots10 | 1 for CSI-RS resource #1 and #2<br><br>2 for CSI-RS resource #3 and #4 | For test 1-1:<br>offset = 1 for CSI-RS resource 1 and 2<br>offset =2 for CSI-RS resource 3 and 4. | |
| } | | | |

#### 5.2.3.1.9_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.9_1.3.3_1

### 5.2.3.1.9_1.4 Test requirement

Tables 5.2.3.1.9_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.9_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.9_1.4-1: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.3 FDD | 10 / 15 | 16QAM, 0.48 | HST-SFN | 2x4 | 70 | 11 |

## 5.2.3.1.10 4Rx FDD FR1 HST DPS performance

### 5.2.3.1.10.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.10.0-3, with the test parameters defined in Table 5.2.3.1.10.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.3.1.10.0-1.

**Table 5.2.3.1.10.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify UE performance in the HST-DPS scenario defined in B.3.3 | 1-1, 1-2 |

**Table 5.2.3.1.10.0-2: Test Parameters for Testing**

| Parameter | | | Unit | Value |
|---|---|---|---|---|
| Duplex mode | | | | FDD |
| Active DL BWP index | | | | 1 |
| PDCCH configuration | TCI state | | | Note 1 |
| PDSCH configuration | Mapping type | | | Type A |
| | k0 | | | 0 |
| | Starting symbol (S) | | | 2 |
| | Length (L) | | | 12 |
| | PDSCH aggregation factor | | | 1 |
| | PRB bundling type | | | Static |
| | PRB bundling size | | | 2 |
| | Resource allocation type | | | Type 0 |
| | RBG size | | | Config2 |
| | VRB-to-PRB mapping type | | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A |
| | TCI state | | | Note 1 |
| PDSCH DMRS configuration | DMRS Type | | | Type 1 |
| | Number of additional DMRS | | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 |
| CSI-RS for tracking | Resource set #1 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 5 for CSI-RS resource 1 and 3<br>$l_0$ = 9 for CSI-RS resource 2 and 4 |
| | | CSI-RS periodicity | Slots | 10 for CSI-RS resource 1,2,3,4. |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4 |
| | | QCL info | | TCI state #2 |
| | Resource set #2 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 5 and 6<br>$l_0$ = 10 for CSI-RS resource 7 and 8 |
| | | CSI-RS periodicity | Slots | 10 for CSI-RS resource 5,6,7,8. |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 5 and 6<br>2 for CSI-RS resource 7 and 8 |
| | | QCL info | | TCI state #3 |
| NZP CSI-RS for CSI acquisition | Resource set #3 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 12 |
| | | CSI-RS periodicity | Slots | 20 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #0 |
| | Resource set #4 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 13 |
| | | CSI-RS periodicity | Slots | 20 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #1 |
| TCI state #0 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking Resource set #1' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 5 from 'CSI-RS for tracking Resource set #2' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #2 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |

| | | | | |
|---|---|---|---|---|
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #3 | Type 1 QCL information | SSB index | | SSB #1 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| Number of HARQ Processes | | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 |

Note 1: SSB # (k mod 2) , CSI-RS (for tracking) resource set # ((k mod 2) + 1) and CSI-RS (for CSI acquisition) resource set # ((k mod 2) + 3) are transmitted by k[th] RRH.

For Test 1-1, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by k[th] RRH from slot#

$$\max\hat{a}_i\,[(2k-1)n + 1 + T_{HARQ} + T_{MAC\ proc} + T_{firstTRS} + T_{TRS\ proc}, 0$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\ proc},$$

PDCCH and PDSCH are DTXed in other slots in which throughput statistics are not considered.

For Test 1-2, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by k[th] RRH from slot#

$$\max\hat{a}_i\,[(2k-1)n + 1 + T_{HARQ} + T_{MAC\ proc}, 0]$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\ proc}$$

Where k=0, 1, 2… is the RRH number, n = 2520 is half of the number of slots between two RRH, $T_{HARQ}$= 2 is the number of slots between PDSCH and corresponding HARQ-ACK information, $T_{MAC\ proc}$ = 3 is the number of slots for MAC CE processing, $T_{firstTRS}$= 6 is the number of slots to first TRS transmission occasion after MAC CE command is decoded by the UE, $T_{TRS\ proc}$ = 2 is the number of slots for TRS processing.

**Table 5.2.3.1.10.0-3: Minimum performance for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 1 | 2x4 | 70 | 10.6 |
| 1-2 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 2 | 2x4 | 70 | 10.6 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.10.

### 5.2.3.1.10_1 4Rx FDD FR1 HST-DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA

The minimum test time value is in []

#### 5.2.3.1.10_1.1 Test purpose

To verify UE performance in the HST-DPS scenario defined in B.3.3 and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.3.1.10_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 5.2.3.1.10_1.3 Test description

#### 5.2.3.1.10_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.3.1.10.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.10_1.3.3.

#### 5.2.3.1.10_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.3.1.10_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.1.10_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.3.1.10_1.4-1 as appropriate.

#### 5.2.3.1.10_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.2.3.1.10_1.3.3_1 Message exceptions for SA

**Table 5.2.3.1.10_1.3.3_1-1: PDSCH-Config**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 is used | test 1-1, 1-2 |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n4 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-4: NZP-CSI-RS-Resource for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | i-1 for CSI-RS resource #i, i=1,2,3,4,5,6,7,8 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 2 for CSI-RS resource #1, #2, #3, #4<br>3 for CSI-RS resource #5, #6, #7, #8 | for test 1-1, 1-2:<br>TCI-StateId for TCI-State #2 for CSI-RS resource #1, #2, #3, #4<br>TCI-StateId for TCI-State #3 for CSI-RS resource #5, #6, #7, #8 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-5: CSI-RS-ResourceMapping for TRS (Table 5.2.3.1.10_1.3.3_1-4)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 with condition TRS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 5 for CSI-RS resource #1 and #3<br>9 for CSI-RS resource #2 and #4<br>6 for CSI-RS resource #5 and #6<br>10 for CSI-RS resource #7 and #8 | for test 1-1, 1-2:<br>$l_0$ = 5 for CSI-RS resource 1 and 3<br>$l_0$ = 9 for CSI-RS resource 2 and 4<br>$l_0$ = 6 for CSI-RS resource 5 and 6<br>$l_0$ = 10 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-6: CSI-ResourcePeriodicityAndOffset for TRS (Table 5.2.3.1.10_1.3.3_1-4)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots10 | 1 for CSI-RS resource #1, #2, #5, #6<br><br>2 for CSI-RS resource #3 #4, #7, #8 | For test 1-1, 1-2:<br>periodicity:<br>10 slots.<br>offset:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4<br>1 for CSI-RS resource 5 and 6<br>2 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-7: NZP-CSI-RS-ResourceSet for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 0 for Resource set #1<br>1 for Resource set #2 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #1 |
| NZP-CSI-RS-ResourceId[1] | 0 | entry 1<br>CSI-RS resource #1 | |
| NZP-CSI-RS-ResourceId[2] | 1 | entry 2<br>CSI-RS resource #2 | |
| NZP-CSI-RS-ResourceId[3] | 2 | entry 3<br>CSI-RS resource #3 | |
| NZP-CSI-RS-ResourceId[4] | 3 | entry 4<br>CSI-RS resource #4 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #2 |
| NZP-CSI-RS-ResourceId[1] | 4 | entry 1<br>CSI-RS resource #5 | |
| NZP-CSI-RS-ResourceId[2] | 5 | entry 2<br>CSI-RS resource #6 | |
| NZP-CSI-RS-ResourceId[3] | 6 | entry 3<br>CSI-RS resource #7 | |
| NZP-CSI-RS-ResourceId[4] | 7 | entry 4<br>CSI-RS resource #8 | |
| } | | | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-8: NZP-CSI-RS-Resource for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | 8 for CSI-RS resource #9<br>9 for CSI-RS resource #10 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 0 for CSI-RS resource #9<br>1 for CSI-RS resource #10 | for test 1-1, 1-2:<br>TCI-State #0 for CSI-RS resource #9<br><br>TCI-State #1 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-9: CSI-RS-ResourceMapping for CSI Acquisition (Table 5.2.3.1.10_1.3.3_1-8)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 12 for CSI-RS resource #9<br>13 for CSI-RS resource #10 | for test 1-1, 1-2<br>$l_0$=12 for CSI-RS resource #9<br>$l_0$=13 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-10: CSI-ResourcePeriodicityAndOffset for CSI Acquisition (Table 5.2.3.1.10_1.3.3_1-8)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 0 | For test 1-1, 1-2:<br>periodicity = 20 slots.<br>offset = 0 slots | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-11: NZP-CSI-RS-ResourceSet for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-18 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 2 for Resource set #3<br>3 for Resource set #4 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #3 |
| NZP-CSI-RS-ResourceId[1] | 8 | entry 1<br>CSI-RS resource #9 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #4 |
| NZP-CSI-RS-ResourceId[1] | 9 | entry 1<br>CSI-RS resource #10 | |
| } | | | |
| } | | | |

**Table 5.2.3.1.10_1.3.3_1-12: TCI-State**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 for TCI state #0<br>1 for TCI state #1<br>2 for TCI state #2<br>3 for TCI state #3 | For test 1-1, 1-2 | |
| qcl-Type1 SEQUENCE { | | | |
| bwp-Id | BWP-Id of active BWP | | TCI state #0, TCI state #1 |
| | Not present | | TCI state #2, TCI state #3 |
| referenceSignal CHOICE { | | | |
| csi-rs | 0 | CSI-RS resource #1 | TCI state #0 |
| | 4 | CSI-RS resource #5 | TCI state #1 |
| ssb | 0 | SSB #0 | TCI state #2 |
| | 1 | SSB #1 | TCI state #3 |
| } | | | |
| qcl-Type | typeA | | TCI state #0, TCI state #1 |
| | typeC | | TCI state #2, TCI state #3 |
| } | | | |
| } | | | |

### 5.2.3.1.10_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.10_1.3.3_1

### 5.2.3.1.10_1.4 Test requirement

Tables 5.2.3.1.10_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.10_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.10_1.4-1: Test Requirements for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 1 | 2x4 | 70 | 11.2 |
| 1-2 | R.PDSCH.1-8.4 FDD | 10 / 15 | 64QAM, 0.43 | HST-DPS | 2 | 2x4 | 70 | 11.2 |

## 5.2.3.1.11 4Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance

### 5.2.3.1.11.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.11.0-3, with the addition of test parameters in Table 5.2.3.1.11.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.11.0-1.

**Table 5.2.3.1.11.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance with Single-DCI based SDM scheme under 4 receive antenna conditions | 1-1,1-2 |

**Table 5.2.3.1.11.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | 0 | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3 l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7 l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 20 | |
| | | CSI-RS offset | Slots | 10 for CSI-RS resources 1 and 2 11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6 11 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |

| | | | | | |
|---|---|---|---|---|---|
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000 | 1002 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Full-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.5 for test 1-1<br>2 for test 1-2 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 for test 1-1<br>0 for test 1-2 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs (PDSCH Layer 0 is transmitted from TRxP #1 and PDSCH layer 1 is transmitted from TRxP #2) | | | | | |

**Table 5.2.3.1.11.0-3: Minimum performance**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 14.6 |
| 1-2 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 13.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.11.

### 5.2.3.1.11_1 4Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.1.11_1.1 Test purpose

To verify the PDSCH performance with Single-DCI based SDM scheme under 4 receive antenna conditions.

#### 5.2.3.1.11_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *singleDCI-SDM-scheme-r16*.

#### 5.2.3.1.11_1.3 Test description

##### 5.2.3.1.11_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.11.0-2 and Table 5.2.3.1.11.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.11_1.3.3.

##### 5.2.3.1.11_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.1.11_1.3.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.1.11_1.3.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Table 5.2.3.1.11_1.3.4-1 as appropriate.

##### 5.2.3.1.11_1.3.3 Message contents

###### 5.2.3.1.11_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.1.11_1.3.3_1-1: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PDSCH-to-HARQ_feedback timing indicator | K1 = 2 | “010” | |
| Antenna port(s) | DMRS port 0 and 2 | “1011” | |
| Transmission configuration indication | TCI state 1 and 2 | “000” | |

**Table 5.2.3.1.11_1.3.3_1-2: *CellGroupConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| simultaneousTCI-UpdateList1-r16 SEQUENCE { | | | |
| ServCellIndex [1] | ServCellIndex | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.11_1.3.3_1-3: *ControlResourceSet***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-28 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.3.1.11_1.3.3_1-4: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] | *TCI-State* with condition TCI-state-0 | | |
| TCI-State[2] | *TCI-State* with condition TCI-state-1 | | |
| TCI-State[3] | *TCI-State* with condition TCI-state-2 | | |
| } | | | |
| rbg-Size | config2 | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.11_1.3.3_1-5: *TCI-State***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 | | TCI-state-0 |
| | 1 | | TCI-state-1 |
| | 2 | | TCI-state-2 |
| qcl-Type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | TCI-state-0 |
| csi-rs | 1 | | TCI-state-1 |
| | 5 | | TCI-state-2 |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| qcl-Type2 | Not present | | |
| } | | | |

**Table 5.2.3.1.11_1.3.3_1-6: *NZP-CSI-RS-Resource***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-85 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| resourceMapping SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| cdm-Type | noCDM | | |
| density CHOICE { | | | |
| three | NULL | | |
| } | | | |
| } | | | |
| periodicityAndOffset CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |
| qcl-InfoPeriodicCSI-RS | 0 | | |
| } | | | |

5.2.3.1.11_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.11_1.3.3_1.

#### 5.2.3.1.11_1.3.4 Test requirement

Table 5.2.3.1.11.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.11_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.11_1.3.4-1: Test requirement**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 15.6 |
| 1-2 | R.PDSCH.1-3.2 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 14.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | |

### 5.2.3.1.12 4Rx FDD FR1 PDSCH Multi-DCI based transmission scheme performance

#### 5.2.3.1.12.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.12.0-3, with the addition of test parameters in Table 5.2.3.1.12.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.12.0-1.

**Table 5.2.3.1.12.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs | 1-1 |

**Table 5.2.3.1.12.0-2: Test parameters**

| Parameter | | Unit | Value | |
|---|---|---|---|---|
| | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | TRxP #1 | |
| PDCCH configuration | TCI state | | TCI State #1 | TCI State #2 |
| | CORESETPoolIndex | | 0,1 | |
| | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |

| | | | | | |
|---|---|---|---|---|---|
| CSI-RS for tracking | CDM Type | | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | | 3 | |
| | CSI-RS periodicity | | Slots | 20 | |
| | CSI-RS offset | | Slots | 10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6<br>11 for CSI-RS resources 7 and 8 |
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | {1000,1001} | {1002,1003} |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Non-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.5 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs. Transmission from TRxP #1 uses CORESETPoolIndex 0 and transmission from TRxP #2 uses CORESETPoolIndex 1 | | | | | |

**Table 5.2.3.1.12.0-3: Minimum performance**

| Test num. | Reference channel | | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| | TRxP #1 | TRxP #2 | | | | | | |
| 1-1 | R.PDSCH.1-3.3 FDD | R.PDSCH.1-3.4 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 14.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.12.

### 5.2.3.1.12_1 4Rx FDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.1.12_1.1 Test purpose

To verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs.

#### 5.2.3.1.12_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *multiDCI-MultiTRP-r16*.

#### 5.2.3.1.12_1.3 Test description

##### 5.2.3.1.12_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.9 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.12.0-2 and Table 5.2.3.1.12.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.12_1.3.3.

#### 5.2.3.1.12_1.3.2 Test procedure

1. SS transmits PDSCH in TRxP#1 via PDCCH DCI format 1_1 for C_RNTI in ControlResourceSetid1 (Table 5.2.3.1.12_1.3.3_1-2), and transmits PDSCH in TRxP#2 via PDCCH DCI format 1_1 for C_RNTI in ControlResourceSetid2 (Table 5.2.3.1.12_1.3.3_1-3), to transmit the DL RMC according to Table 5.2.3.1.12_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR for TRxP#1 and TRxP#2 according to Table 5.2.3.1.12_1.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.3.1.12_1.3.3 Message contents

##### 5.2.3.1.12_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.1.12_1.3.3_1-1: *PDCCH-Config* (Preamble)**

| Derivation Path: TS 38.508-1 [4],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet { | 2 entries | | |
| ControlResourceSet[1] | ControlResourceSetid1 | entry 1 | |
| ControlResourceSet[2] | ControlResourceSetid2 | entry 2 | |
| } | | | |
| } | | | |

**Table 5.2.3.1.12_1.3.3_1-2: *ControlResourceSetId1* (Table 5.2.3.1.12_1.3.3_1-1: *PDCCH-Config*)**

| Derivation Path: TS 38.508-1 [4], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11110000 00000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 24 RBs of the BWP | |
| tci-StatesPDCCH-ToAddList SEQUENCE(SIZE (1..maxNrofTCI-StatesPDCCH)) OF TCI-StateId { | | | |
| TCI-StateId[1] | 1 | | |
| } | | | |
| tci-PresentInDCI | enabled | | |
| coresetPoolIndex-r16 | 0 | | |
| } | | | |

**Table 5.2.3.1.12_1.3.3_1-3: *ControlResourceSetId2* (Table 5.2.3.1.12_1.3.3_1-1: *PDCCH-Config*)**

| Derivation Path: TS 38.508-1 [4], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 00001111 00000000 00000000 00000000 00000000 00000 | CORESET to use the RBs 24~47 of the BWP | |
| tci-StatesPDCCH-ToAddList SEQUENCE(SIZE (1..maxNrofTCI-StatesPDCCH)) OF TCI-StateId { | | | |
| TCI-StateId[1] | 2 | | |
| } | | | |
| tci-PresentInDCI | enabled | | |
| coresetPoolIndex-r16 | 1 | | |
| } | | | |

**Table 5.2.3.1.12_1.3.3_1-4: Physical layer parameters for DCI format 1_1 in ControlResourceSetid1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 and 1 | “0111” | |
| Transmission configuration indication | TCI State #1 | “000” | |

**Table 5.2.3.1.12_1.3.3_1-5: Physical layer parameters for DCI format 1_1 in ControlResourceSetid2**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 2 and 3 | “1000” | |
| Transmission configuration indication | TCI State #2 | “001” | |

**Table 5.2.3.1.12_1.3.3_1-6: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] SEQUENCE { | | TCI-state-0 | |
| tci-StateId | 0 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | |
| } | | | |
| qcl-Type | typeC | | |
| } | | | |
| } | | | |
| TCI-State[2] | | TCI-state-1 | |
| tci-StateId | 1 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 1 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| TCI-State[3] | | TCI-state-2 | |
| tci-StateId | 2 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 5 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.12_1.3.3_1-7: *CSI-RS-ResourceMapping* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| } | | | |

**Table 5.2.3.1.12_1.3.3_1-8: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.3.1.12_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.12_1.3.3_1.

### 5.2.3.1.12_1.4 Test requirement

Table 5.2.3.1.12.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.12_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.12_1.4-1: Test requirement**

| Test num. | Reference channel | | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| | TRxP #1 | TRxP #2 | | | | | | |
| 1-1 | R.PDSCH.1-3.3 FDD | R.PDSCH.1-3.4 FDD | 10 / 15 | 64QAM, 0.50 | TDLA30-10 | 2x4, ULA Low | 70 | 15.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

## 5.2.3.1.13 4Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance

### 5.2.3.1.13.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.13.0-3, with the addition of test parameters in Table 5.2.3.1.13.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.13.0-1.

**Table 5.2.3.1.13.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 4 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16” defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.3.1.13.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | TCI state | | | TCI State #1 | |
| | CORESETPoolIndex | | | Not configured | |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | | l0 = 6 for CSI-RS resources 1 and 3 l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7 l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | | 3 | |
| | CSI-RS periodicity | | Slots | 20 | |
| | CSI-RS offset | | Slots | 10 for CSI-RS resources 1 and 2 11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6 11 for CSI-RS resources 7 and 8 |
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | Wideband | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000,1001 | 1000,1001 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |

| | | |
|---|---|---|
| Timing offset of the second TRxP from the first TRxP | us | -0.5 |
| Frequency offset of the second TRxP from the first TRxP | Hz | 200 |
| Number of HARQ Processes | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | 2 |
| Precoding configuration | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity |
| Note 1: PDSCH transmission is done from both TRxPs | | |

**Table 5.2.3.1.13.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.1-2.5 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x4, ULA Low | 70 | 10.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.13.

### 5.2.3.1.13_1 4Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.1.13_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16”.

#### 5.2.3.1.13_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportFDM-SchemeA-r16*.

#### 5.2.3.1.13_1.3 Test description

##### 5.2.3.1.13_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.9 for TE diagram and section A.3.2.5 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.13.0-2 and Table 5.2.3.1.13.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.13_1.3.3.

#### 5.2.3.1.13_1.3.2 Test procedure

1. SS transmits PDSCH in TRxP#1 and TRxP#2 via PDCCH DCI format 1_1 for C_RNTI (Table 5.2.3.1.13_1.3.3_1-2), to transmit the DL RMC according to Table 5.2.3.1.13_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR for TRxP#1 and TRxP#2 according to Table 5.2.3.1.13_1.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.3.1.13_1.3.3 Message contents

#### 5.2.3.1.13_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.1.13_1.3.3_1-1: *PDCCH-ControlResourceSet* (Preamble)**

| Derivation Path: TS 38.508-1 [4],Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.3.1.13_1.3.3_1-2: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 and 1 | “0111” | |
| Transmission configuration indication | TCI codepoint 0, corresponding to TCI State #1 and #2 | “000” | |

**Table 5.2.3.1.13_1.3.3_1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] SEQUENCE { | | TCI-state-0 | |
| tci-StateId | 0 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | |
| } | | | |
| qcl-Type | typeC | | |
| } | | | |
| } | | | |
| TCI-State[2] | | TCI-state-1 | |
| tci-StateId | 1 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 1 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| TCI-State[3] | | TCI-state-2 | |
| tci-StateId | 2 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 5 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| } | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | wideband | | |
| } | | | |
| } | | | |
| repetitionSchemeConfig-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| fdm-TDM-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| repetitionScheme-r16 | fdmSchemeA | | |
| startingSymbolOffsetK-r16 | Not present | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.13_1.3.3_1-4: *CSI-RS-ResourceMapping* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| } | | | |

**Table 5.2.3.1.13_1.3.3_1-5: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.3.1.13_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.13_1.3.3_1.

### 5.2.3.1.13_1.4 Test requirement

Table 5.2.3.1.13.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.1.13_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.13_1.4-1: Test requirement for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition(Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.1-2.5 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 70 | 11.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

### 5.2.3.1.14 4Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance

#### 5.2.3.1.14.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.1.14.0-3, with the addition of test parameters in Table 5.2.3.1.14.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.1.14.0-1.

**Table 5.2.3.1.14.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 4 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.3.1.14.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | |
| | | CORESETPoolIndex | | Not configured | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 20 | |
| | | CSI-RS offset | Slots | 10 for CSI-RS resources 1 and 2<br>11 for CSI-RS resources 3 and 4 | 10 for CSI-RS resources 5 and 6<br>11 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | FDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | Repetition number | | | 2 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000 | 1000 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |

| | | | | | |
|---|---|---|---|---|---|
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | 2 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 200 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | 2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs | | | | | |

**Table 5.2.3.1.14.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition (Note 1) | Correlation matrix and antenna configuration Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR (dB) (Note 4) |
| 1-1 | R.PDSCH.1-11.2 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x4, ULA Low | 1 (Note 3) | -0.4 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.<br>Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.1.14.

### 5.2.3.1.14_1 4Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.1.14_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states.

#### 5.2.3.1.14_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportTDM-SchemeA-r16*.

#### 5.2.3.1.14_1.3 Test description

##### 5.2.3.1.14_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.9 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.1.14.0-2 and Table 5.2.3.1.14.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.1.14_1.3.3.

### 5.2.3.1.14_1.3.2 Test procedure

1. SS transmits PDSCH in TRxP#1 and TRxP#2 via PDCCH DCI format 1_1 for C_RNTI (Table 5.2.3.1.14_1.3.3_1-2), to transmit the DL RMC according to Table 5.2.3.1.14_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR for TRxP#1 and TRxP#2 according to Table 5.2.3.1.14_1.4-1.
3. Measure the residual BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport blocks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to clause G.1.4 and Table G.1.5-1a in Annex G clause G.1.5.

### 5.2.3.1.14_1.3.3 Message contents

### 5.2.3.1.14_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.1.14_1.3.3_1-1: *PDCCH-ControlResourceSet* (Preamble)**

| Derivation Path: TS 38.508-1 [4],Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.3.1.14_1.3.3_1-2: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 | “0000” | |
| Transmission configuration indication | TCI codepoint 0, corresponding to TCI State #1 and #2 | “000” | |

**Table 5.2.3.1.14_1.3.3_1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1..maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] SEQUENCE { | | TCI-state-0 | |
| tci-StateId | 0 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | |
| } | | | |
| qcl-Type | typeC | | |
| } | | | |
| } | | | |
| TCI-State[2] | | TCI-state-1 | |
| tci-StateId | 1 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 1 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| TCI-State[3] | | TCI-state-2 | |
| tci-StateId | 2 | | |
| qcl-type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| csi-rs | 5 | | |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| } | | | |
| } | | | |
| pdsch-TimeDomainAllocationList | Not present | | |
| pdsch-TimeDomainAllocationList-r16 CHOICE { | | | |
| setup SEQUENCE (SIZE(1..maxNrofDL-Allocations)) OF PDSCH-TimeDomainResourceAllocation-r16 { | | | |
| PDSCH-TimeDomainResourceAllocation-r16[1] SEQUENCE { | | | |
| k0-r16 | Not present | | |
| mappingType-r16 | typeA | | |
| startSymbolAndLength-r16 | 44 | Start symbol(S) =2, Length(L)= 4 | For Slot i, if mod(i, 10) = 7 for i from {0,…,39} |
| repetitionNumber-r16 | 2 | | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation-r16[2] SEQUENCE { | | | |
| k0-r16 | Not present | | |
| mappingType-r16 | typeA | | |
| startSymbolAndLength-r16 | 53 | Start symbol(S) =2, Length(L)= 12 | For Slot i, if mod(i, 10) = {0,1,2,3,4,5,}) for i from {1,…,39} |
| repetitionNumber-r16 | 2 | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.1.14_1.3.3_1-4: *CSI-RS-ResourceMapping* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| } | | | |

**Table 5.2.3.1.14_1.3.3_1-5: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 10 | For CSI-RS resources 1,2,5,6 | |
| slots20 | 11 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

### 5.2.3.1.14_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.1.14_1.3.3_1.

### 5.2.3.1.14_1.4 Test requirement

Table 5.2.3.1.14.0-3 defines the primary level settings.

The residual BLER specified in Note 3 of Table 5.2.3.1.14_1.4-1 test shall meet or be lower than the specified value in Table 5.2.3.1.14_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.1.14_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | BLER (%) | SNR (dB) (Note 4) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.1-11.2 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x4, ULA Low | 1 (Note 3) | 0.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.<br>Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | |

## 5.2.3.2 TDD

### 5.2.3.2.1 4Rx TDD FR1 PDSCH mapping Type A performance

#### 5.2.3.2.1.0 Minimum conformance requirements for PDSCH Mapping Type A

The performance requirements are specified in Table 5.2.3.2.1.0-3, Table 5.2.3.2.1.0-4, Table 5.2.3.2.1.0-5 and Table 5.2.3.2.1.0-6, with the test parameters defined in Table 5.2.3.2.1.0-2 and the downlink physical channel setup according to Annex C.2.1.

The test purposes are specified in Table 5.2.3.2.1.0-1.

**Table 5.2.3.2.1.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under4 receive antenna conditions and with different channel models, MCSs and number of MIMO layers | 1-1, 1-2, 1-3, 1-5, 1-6, 1-7, 1-8, 1-9, 1-10, 1-11, 2-1, 2-2, 3-1, 4-1 |
| Verify the PDSCH mapping Type A HARQ soft combining performance under 4 receive antenna conditions. | 1-4 |
| Verify the PDSCH mapping Type A performance requirements for Enhanced Receiver Type 1 under 4 receive antenna conditions. | 5-1 |

**Table 5.2.3.2.1.0-2: Test Parameters for Testing**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **TDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | Specific to each Reference channel |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 4 for Tests 1-1, 1-8, 1-9<br>WB for Test 3-1<br>2 for other tests |
| | Resource allocation type | | Test 1-2: Type 1 with start RB = 50, $L_{RBs}$ = 6<br>Other tests: Type 0 |
| | RBG size | | Test 1-2: N/A<br>Other tests: Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |

| | | | |
|---|---|---|---|
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 for Tests 1-1, 1-7, 1-8, 1-9, 1-10, 1-11<br>1 for other tests |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | First OFDM symbol in the PRB used for CSI-RS | | Tests 1-8, 1-9:<br>$l_0$ = 4 for CSI-RS resource 1 and 3<br>$l_0$ = 8 for CSI-RS resource 2 and 4<br><br>Other tests; Table 5.2-1. |
| | CSI-RS periodicity | Slots | Test 1-7, 1-10, 1-11:<br>20 for CSI-RS resource 1,2,3,4.<br><br>Other tests: Table 5.2-1. |
| | CSI-RS offset | Slots | Test 1-7, 1-10, 1-11:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4.<br><br>Other tests: Table 5.2-1. |
| | Frequency Occupation | | Test 1-7, 1-10, 1-11:<br>Start PRB 0<br>Number of PRB = 52<br><br>Other tests: Table 5.2-1. |
| Number of HARQ Processes | | | 16 for Test 1-4<br>10 for Test 1-9<br><br>8 for other tests |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.3.2.1.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.2-1.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1A | TDLB100-400 | 2x4, ULA Low | 70 | -4.1 |
| 1-2 | R.PDSCH.2-1.2 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLC300-100 | 2x4, ULA Low | 70 | -2.7 |
| 1-3 | R.PDSCH.2-4.1 TDD | 40 / 30 | 256QAM, 0.82 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 21.6 |
| 1-4 | R.PDSCH.2-2.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x4, ULA Low | 30 | -1.2 |
| 1-5 | R.PDSCH.2-5.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x4, ULA Low | 70 | -3.8 |
| 1-6 | R.PDSCH.2-6.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-3 | TDLA30-10 | 2x4, ULA Low | 70 | -3.6 |
| 1-7 | R.PDSCH.2-10.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-1000 | 1x4 | 70 | 3.4 |
| 1-8 | R.PDSCH.2-11.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-5 | TDLB100-400 | 2x4, ULA Low | 70 | -4.0 |
| 1-9 | R.PDSCH.2-12.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-6 | TDLB100-400 | 2x4, ULA Low | 70 | -4.0 |
| 1-10 | R.PDSCH.2-10.2 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-1200 | 2x4 | 70 | 5.8 |
| 1-11 | R.PDSCH.2-10.3 TDD | 40 / 30 | 64QAM, 0.43 | FR1.30-1 | HST-1667 | 1x4 | 70 | 6.8 |

**Table 5.2.3.2.1.0-4: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 2-1 | R.PDSCH.2-3.1 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 13.6 |
| 2-2 | R.PDSCH.2-9.1 TDD | 20 / 30 | 64QAM, 0.50 | FR1.30-4 | TDLA30-10 | 2x4, ULA Low | 70 | 13.7 |

**Table 5.2.3.2.1.0-5: Minimum performance for Rank 3**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.2-2.3 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 4x4, ULA Low | 70 | 11.1 |

**Table 5.2.3.2.1.0-6: Minimum performance for Rank 4**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 4-1 | R.PDSCH.2-2.4 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 4x4, ULA Low | 70 | 15.4 |

**Table 5.2.3.2.1.0-7: Minimum performance for Rank 3 and EnhancedReceiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5-1 | R.PDSCH.2-2.3 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 4x4, ULA Medium A | 70 | 22.9 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.2.1.

### 5.2.3.2.1_1 4Rx TDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.1_1.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 1 and Rank 2 scenarios.

#### 5.2.3.2.1_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward, supporting 4Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4Rx antenna ports.

#### 5.2.3.2.1_1.3 Test description

##### 5.2.3.2.1_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.3.2.1.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [] clause 4.5. Message contents are defined in clause 5.2.3.2.1_1.4.3.

5.2.3.2.1_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.1.0-3 and Table 5.2.3.2.1.0-4. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.2.1_1.3.4-1 and 5.2.3.2.1_1.3.4-2 as appropriate.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.3.2.1_1.3.4-1 and 5.2.3.2.1_1.3.4-2 as appropriate.

5.2.3.2.1_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

5.2.3.2.1_1.3.3_1 Message exceptions for SA

**Table 5.2.3.2.1_1.3.3_1-1: *BWP***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| BWP ::= SEQUENCE { | | | |
| locationAndBandwidth | 13750 | For Test 2-2 (20MHz BW, SCS 30kHz) | |
| | 28875 | For other tests (40MHz BW, SCS 30kHz) | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-2: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | n4 | n4 for tests 1-1, 1-8, 1-9 | |
| | wideband | wideband for test 3-1 | |
| | Not present | n2 for other tests | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-3: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos1 | pos1 for all tests except tests 1-1, 1-7, 1-8, 1-9 | |
| | Not present | pos2 for tests 1-1, 1-7, 1-8, 1-9, 1-10, 1-11 | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-4: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | Not present | n8 for other tests | |
| | n16 | n16 for test 1-4 | |
| | n10 | n10 for test 1-9 | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-5: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots20 | 1 (for CSI-RS resources 1 and 2)<br>2 (for CSI-RS resources 3 and 4) | Periodicity 20 slots and offset 1/2 for test 1-7, 1-10, 1-11 | |
| | | | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-5A: CSI-RS-ResourceMapping for TRS**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 4 | For Tests 1-8, 1-9:<br>$l_0$ = 4 for CSI-RS resource 1 and 3 | TRS |
| | 8 | For Tests 1-8, 1-9: $l_0$ = 8 for CSI-RS resource 2 and 4 | TRS |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-6: CSI-FrequencyOccupation for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-11 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-FrequencyOccupation ::= SEQUENCE { | | | |
| nrofRBs | 52 | 52 for tests 1-7, 1-10, 1-11, 2-2 | TRS |
| | 108 | 108 for other tests | TRS |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-7: RACH-ConfigGeneric**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-130 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RACH-ConfigGeneric ::= SEQUENCE { | | | |
| prach-ConfigurationIndex | 163 | Only for test 2-2 | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-8: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl20 | 7 | For test 1-9 | |
| sl20 | 5 | For test 2-2 | |
| } | | | |
| } | | | |

**Table 5.2.3.2.1_1.3.3_1-9: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PUCCH resource indicator | *PUCCH-ResourceId[1]* = 6 in pucch-ResourceSetID[1] or *PUCCH-ResourceId[1]* = 14 in pucch-ResourceSetID[2] as defined in Table 4.6.3-112 (Mapping as per Table 9.2.3-2 in TS 38.213) | '110'B | Slot S1 for test 1-9 |

**Table 5.2.3.2.1_1.3.3_1-10: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-27 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 3 entry | | Test 1-5, Test 1-6 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 44 | Start symbol(S)=2, Length(L)=4 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[3] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | Test 1-8 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 81 | Start symbol(S)=2, Length(L)=10 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 5 entry | | Test 1-9 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 100 | Start symbol(S)=2, Length(L)=8 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[3] SEQUENCE { | | | |

| | | | |
|---|---|---|---|
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 81 | Start symbol(S)=2, Length(L)=10 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[4] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[5] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

#### 5.2.3.2.1_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.1_1.3.3_1

### 5.2.3.2.1_1.3.4 Test requirement

Table 5.2.3.2.1.0-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A clause A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.1_1.3.4-1 and Table 5.2.3.2.1_1.3.4-2 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.1_1.3.4-1: Test Requirements for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.2-1.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1A | TDLB100-400 | 2x4, ULA Low | 70 | -3.1 |
| 1-2 | R.PDSCH.2-1.2 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLC300-100 | 2x4, ULA Low | 70 | -1.7 |
| 1-3 | R.PDSCH.2-4.1 TDD | 40 / 30 | 256QAM, 0.82 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 22.5 |
| 1-4 | R.PDSCH.2-2.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x4, ULA Low | 30 | -0.3 |
| 1-5 | R.PDSCH.2-5.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x4, ULA Low | 70 | -2.8 |
| 1-6 | R.PDSCH.2-6.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-3 | TDLA30-10 | 2x4, ULA Low | 70 | -2.6 |
| 1-7 | R.PDSCH.2-10.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-1000 | 1x4 | 70 | 4.3 |
| 1-8 | R.PDSCH.2-11.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-5 | TDLB100-400 | 2x4, ULA Low | 70 | -3.1 |
| 1-9 | R.PDSCH.2-12.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-6 | TDLB100-400 | 2x4, ULA Low | 70 | -3.1 |
| 1-10 | R.PDSCH.2-10.2 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-1200 | 2x4 | 70 | 6.7 |
| 1-11 | R.PDSCH.2-10.3 TDD | 40 / 30 | 64QAM, 0.43 | FR1.30-1 | HST-1667 | 1x4 | 70 | 7.7 |

**Table 5.2.3.2.1_1.3.4-2: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 2-1 | R.PDSCH.2-3.1 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 14.6 |
| 2-2 | R.PDSCH.2-9.1 TDD | 20 / 30 | 64QAM, 0.50 | FR1.30-4 | TDLA30-10 | 2x4, ULA Low | 70 | 14.7 |

### 5.2.3.2.1_2 4Rx TDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.1_2.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 3 and Rank 4 scenarios.

#### 5.2.3.2.1_2.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

#### 5.2.3.2.1_2.3 Test description

Same test description as in clause 5.2.3.2.1_1.3 with the following exception:

- Figure A.3.1.7.5 instead of A.3.1.7.4
- Step 1 of Test procedure as in clause 5.2.3.2.1_1.3.2 to call for Tables 5.2.3.2.1.0-5 and 5.2.3.2.1.0-6 instead of Table 5.2.3.2.1.0-3 and 5.2.3.2.1.0-4.
- Step 2 and 4 of Test procedure as in clause 5.2.3.2.1_1.3.2 to call for Tables 5.2.3.2.1_2.3.4-1 and 5.2.3.2.1_2.3.4-2 instead of Tables 5.2.3.2.1_1.3.4-1 and 5.2.3.2.1_1.3.4-2.

5.2.3.2.1_2.3.1 Void

5.2.3.2.1_2.3.2 Void

5.2.3.2.1_2.3.3 Void

5.2.3.2.1_2.3.4 Test requirement

Table 5.2.3.2.1.0-5 and Table 5.2.3.2.1.0-6 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.1_2.3.4-1 andTable 5.2.3.2.1_2.3.4-2 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.1_2.3.4-1: Test Requirements for Rank 3**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 3-1 | R.PDSCH.2-2.3 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 4x4, ULA Low | 70 | 12.1 |

**Table 5.2.3.2.1_2.3.4-2: Test Requirements for Rank 4**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 4-1 | R.PDSCH.2-2.4 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 4x4, ULA Low | 70 | 16.4 |

5.2.3.2.1_3 4Rx TDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with enhanced receiver type 1 for both SA and NSA

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5.2.3.2.1_4 4Rx TDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with enhanced receiver type 1 for both SA and NSA

5.2.3.2.1_4.1 Test purpose

To verify the PDSCH mapping Type A enhanced performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default enhanced receiver type 1 configuration, for Rank 3 scenario.

5.2.3.2.1_4.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and NR enhanced receiver type 1.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC, 4 Rx antenna ports and NR enhanced receiver type1.

5.2.3.2.1_4.3 Test description

Same test description as in clause 5.2.3.2.1_2.3 with the following exception:

- Step 1 of Test procedure as in clause 5.2.3.2.1_1.3.2 to call for Table 5.2.3.2.1.0-7 instead of Table 5.2.3.2.1.0-3 and 5.2.3.2.1.0-4.

- Step 2 and 4 of Test procedure as in clause 5.2.3.2.1_1.3.2 to call for Table 5.2.3.2.1_4.3.4-1 instead of Tables 5.2.3.2.1_1.3.4-1 and 5.2.3.2.1_1.3.4-2.

5.2.3.2.1_4.3.1 Void

5.2.3.2.1_4.3.2 Void

5.2.3.2.1_4.3.3 Void

5.2.3.2.1_4.3.4 Test requirement

Table 5.2.3.2.1.0-7 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A A.3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.1_4.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.1_4.3.4-1: Test Requirements for Rank 3 and Enhanced Receiver Type 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5-1 | R.PDSCH.2-2.3 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLA30-10 | 4x4, ULA Medium A | 70 | 23.9 |

## 5.2.3.2.2 4Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance

### 5.2.3.2.2.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.2.0-3, with the addition of test parameters in Table 5.2.3.2.2.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.2.0-1.

**Table 5.2.3.2.2.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions and CSI-RS overlapped with PDSCH | 1-1 |

**Table 5.2.3.2.2.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **TDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |

| | | | |
|---|---|---|---|
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| NZP CSI-RS for CSI acquisition | OFDM symbols in the PRB used for CSI-RS | | $l_0 = 13$ |
| | CSI-RS periodicity | Slots | 5 |
| ZP CSI-RS for CSI acquisition | Subcarrier index in the PRB used for CSI-RS | | $(k_0, k_1, k_2, k_3)=(2, 4, 6, 8)$ |
| | Number of CSI-RS ports (X) | | 8 |
| | CSI-RS periodicity | Slots | 5 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.3.2.2.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-7.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x4, ULA Low | 70 | 9.0 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.2.2.

### 5.2.3.2.2_1 4Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.2_1.1 Test purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration for CSI-RS overlapped with PDSCH scenario.

#### 5.2.3.2.2_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

#### 5.2.3.2.2_1.3 Test description

##### 5.2.3.2.2_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.3.2.2.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.3.2.2_1.3.3.

### 5.2.3.2.2_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.2.0-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.2.2_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-12 in Annex G clause G.1.5.

### 5.2.3.2.2_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

#### 5.2.3.2.2_1.3.3_1 Message exceptions for SA

**Table 5.2.3.2.2_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | Not present | | |
| } | | | |

**Table 5.2.3.2.2_1.3.3_1-2: NZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 13 | $l_0 = 13$ | |
| } | | | |

**Table 5.2.3.2.2_1.3.3_1-3: CSI-ResourcePeriodicityAndOffset for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots5 | 0 | Periodicity 5 slots and offset 0 | |
| } | | | |

**Table 5.2.3.2.2_1.3.3_1-4: ZP CSI-RS-ResourceMapping for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], clause5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | (k0, k1, k2, k3)=(2, 4, 6, 8) | |
| } | | | |
| nrofPorts | P8 | Eight Ports | |
| freqBand | CSI-FrequencyOccupation | | |
| } | | | |

**Table 5.2.3.2.2_1.3.3_1-4A: ZP CSI-ResourcePeriodicityAndOffset for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots5 | 0 | Periodicity 5 slots and offset 0 | |
| } | | | |

#### 5.2.3.2.2_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.2_1.3.3_1

#### 5.2.3.2.2_1.4 Test requirement

Table 5.2.3.2.2.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.2_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.2_1.4-1: Test Requirement for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-7.1 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | TDLC300-100 | 2x4, ULA Low | 70 | 9.9 |

### 5.2.3.2.3 4Rx TDD FR1 PDSCH mapping Type B performance

#### 5.2.3.2.3.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.3.0-3, with the addition of test parameters in Table 5.2.3.2.3.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.3.0-1.

**Table 5.2.3.2.3.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| PDSCH mapping Type B performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.2.3.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **TDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 5 |
| | Length (L) | | 7 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.3.2.3.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH,2-1.3 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | -3.9 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.2.3.

### 5.2.3.2.3_1 4Rx TDD FR1 PDSCH mapping Type B performance - 2x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.3_1.1 Test purpose

To verify the PDSCH mapping Type B normal performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput with baseline receiver configuration.

#### 5.2.3.2.3_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and PDSCH mapping type B.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports and PDSCH mapping type B.

### 5.2.3.2.3_1.3 Test description

#### 5.2.3.2.3_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Tables 5.2-1 and 5.2.3.2.3.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 5.2.3.2.3_1.3.3.

#### 5.2.3.2.3_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.3.0-2. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.2.3_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.3.2.3_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

##### 5.2.3.2.3_1.3.3_1 Message exceptions for SA

**Table 5.2.3.2.3_1.3.3_1-1: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | Not present | | |
| } | | | |

**Table 5.2.3.2.3_1.3.3_1-2: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeB | | |
| startSymbolAndLength | 89 | Start symbol(S)=5, Length(L)=7 | |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

### 5.2.3.2.3_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.3_1.3.3_1

### 5.2.3.2.3_1.4 Test requirement

Table 5.2.3.2.3.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.3_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.3_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH,2-1.3 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | -2.9 |

## 5.2.3.2.4 4Rx TDD FR1 PDSCH mapping Type A performance

### 5.2.3.2.4.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.4.0-3, with the addition of test parameters in Table 5.2.3.2.4.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.4.0-1.

**Table 5.2.3.2.4.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions with CRS rate matching configured | 1-1, 1-2 |

**Table 5.2.3.2.4.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| NR UL transmission with a 7.5 kHz shift to the LTE raster | | | true |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 3 |
| | Length (L) | | 9 for Test 1-1<br>11 for Test 1-2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Position of the first DM-RS for downlink | | 3 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CRS for rate matching (Note 1) | LTE carrier centre subcarrier location | | Same as NR carrier centre subcarrier location |
| | LTE carrier BW | MHz | 10 |
| | Number of antenna ports | | 4 |
| | v-shift | | 0 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |
| Note 1: No MBSFN is configured on LTE carrier | | | |

**Table 5.2.3.2.4.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x4, ULA Low | 70 | -3.6 |
| 1-2 | R.PDSCH.1-1.2 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x4, ULA Low | 70 | -3.5 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.2.4.

### 5.2.3.2.4_1 4Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.4_1.1 Test purpose

To verify the PDSCH mapping Type A coexistence performance under 4 receive antenna conditions for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput with baseline receiver configuration.

#### 5.2.3.2.4_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test 1-1 also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports and capability IE *rateMatchingLTE-CRS* but not supporting capability IE *additionalDMRS-DL-Alt*.

Test 1-2 applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports and capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

Test 1-2 also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports and and capability IE *additionalDMRS-DL-Alt* and *rateMatchingLTE-CRS*.

#### 5.2.3.2.4_1.3 Test description

##### 5.2.3.2.4_1.3.1 Initial conditions

Same as 5.2.2.2.4_1.3.1 with the following exceptions:

- Use Figure A.3.1.7.5 for TE diagram
- Use Figure A.3.2.5 for UE diagram
- Instead of 5.2.2.2.4.x → refer 5.2.3.2.4.x

##### 5.2.3.2.4_1.3.2 Test procedure

Same as 5.2.2.2.4_1.3.2 with the following exceptions:

- Instead of 5.2.2.2.4.x → refer 5.2.3.2.4.x

##### 5.2.3.2.4_1.3.3 Message contents

Same as 5.2.2.2.4_1.3.3

#### 5.2.3.2.4_1.4 Test requirement

Table 5.2.3.2.4.0-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.4_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.4_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-1.1 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x4, ULA Low | 70 | -2.6 |
| 1-2 | R.PDSCH.1-1.2 TDD | 10 / 15 | QPSK, 0.30 | FR1.15-1 | TDLA30-10 | 4x4, ULA Low | 70 | -2.5 |

### 5.2.3.2.5 4Rx TDD FR1 PDSCH 0.001% BLER performance

#### 5.2.3.2.5.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.5.0-3, with the addition of test parameters in Table 5.2.3.2.5.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.5.0-1.

**Table 5.2.3.2.5.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH 0.001% BLER performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.2.5.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **TDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Defined in Annex A.1.2 for TDD pattern FR1.30-1 |

**Table 5.2.3.2.5.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.2-1.4 TDD | 40 / 30 | QPSK, 0.59 | FR1.30-1 | AWGN | 1x4, ULA Low | 0.001% | 0.7 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.5.

### 5.2.3.2.5_1 4Rx TDD FR1 PDSCH 0.001% BLER performance - 1x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.5_1.1 Test purpose

To verify the PDSCH 0.001% BLER performance under 4 receive antenna conditions.

#### 5.2.3.2.5_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *dl-64QAM-MCS-TableAlt* and capability IE *cqi-TableAlt*.

#### 5.2.3.2.5_1.3 Test description

##### 5.2.3.2.5_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.2.5.0-2 and Table 5.2.3.2.5.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.5_1.3.3.

##### 5.2.3.2.5_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.5.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.2.5_1.3.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.4. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.4.3-1 in Annex G.

##### 5.2.3.2.5_1.3.3 Message contents

###### 5.2.3.2.5_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.2.5_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| mcs-Table | qam64LowSE | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

5.2.3.2.5_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.5_1.3.3_1.

5.2.3.2.5_1.3.4 Test requirement

Table 5.2.3.2.5.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.5_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.5_1.3.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.2-1.4 TDD | 40 / 30 | QPSK, 0.59 | FR1.30-1 | AWGN | 1x4, ULA Low | 0.001% | 1.3 |

## 5.2.3.2.6 4Rx TDD FR1 PDSCH repetitions over multiple slots performance

### 5.2.3.2.6.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.6.0-3, with the addition of test parameters in Table 5.2.3.2.6.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.6.0-1.

**Table 5.2.3.2.6.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.2.6.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |

| | | | |
|---|---|---|---|
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 (Note 1) |
| Note 1: ACK/NACK feedback is generated for PDSCH on slot i, where mod(i,10) = {2, 4, 6}. | | | |

**Table 5.2.3.2.6.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-16.1 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 1% (Note 1) | -2.6 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.6.

### 5.2.3.2.6_1 4Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x4 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- MU factor due to finite test time needs further study

#### 5.2.3.2.6_1.1 Test purpose

To Verify the PDSCH repetitions over multiple slots performance under 4 receive antenna conditions.

#### 5.2.3.2.6_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

#### 5.2.3.2.6_1.3 Test description

##### 5.2.3.2.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.2.3.2.6.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.6_1.3.3.

### 5.2.3.2.6_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.6.0-3. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.2.6_1.3.4-1.
3. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport bloks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to clause G.1.5 and Table G.1.5-1a in Annex G clause G.1.5.

### 5.2.3.2.6_1.3.3 Message contents

#### 5.2.3.2.6_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

#### 5.2.2.2.6_1.3.3_2 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

### 5.2.3.2.6_1.3.4 Test requirement

Table 5.2.3.2.6.0-3 defines the primary level settings.

The target BLER percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each BLER test shall meet or exceed the specified value in Table 5.2.2.2.6_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.6_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.2-16.1 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 1% (Note 1) | -2.6 +TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

### 5.2.3.2.7 4Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance

#### 5.2.3.2.7.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.7.0-3, with the addition of test parameters in Table 5.2.3.2.7.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.7.0-1.

**Table 5.2.3.2.7.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance and UE processing capability 2 under four receive antenna conditions | 1-1 |

**Table 5.2.3.2.7.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **TDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 0 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Maximum number of HARQ transmission | | | 1 |
| Number of HARQ Processes | | | 2 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 0 |

**Table 5.2.3.2.7.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-17.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x4, ULA Low | 70 | -2.5 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.7.

### 5.2.3.2.7_1 4Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.7_1.1 Test purpose

To verify PDSCH mapping Type B performance and UE processing capability 2 under four receive antenna conditions.

#### 5.2.3.2.7_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-ProcessingType2*.

#### 5.2.3.2.7_1.3 Test description

##### 5.2.3.2.7_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.2.7.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.7_1.3.3.

##### 5.2.3.2.7_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.7.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.2.7_1.4-1.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

##### 5.2.3.2.7_1.3.3 Message contents

###### 5.2.3.2.7_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.2.7_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeB | | |
| startSymbolAndLength | 16 | Start symbol(S)=2, Length(L)=2 | |
| } | | | |
| } | | | |

**Table 5.2.3.2.7_1.3.3_1-2: *PUCCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-112 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PUCCH-Config ::= SEQUENCE { | | | FR1 |
| dl-DataToUL-ACK SEQUENCE (SIZE (1)) OF INTEGER { | 1 entry | | |
| INTEGER[1] | 0 | entry 1 | |
| } | | | |
| } | | | |

**Table 5.2.3.2.7_1.3.3_1-3: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| PDSCH-to-HARQ_feedback timing indicator | K1=0 as per dl-DataToUL-ACK in Table 5.2.3.2.7_1.3.3_1-3 | "000" | |

### 5.2.3.2.7_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.7_1.3.3_1.

### 5.2.3.2.7_1.4 Test requirement

Table 5.2.3.2.7.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.7_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.7_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-17.1 TDD | 40 / 30 | QPSK, 0.30 | FR1.30-2 | TDLA30-10 | 2x4, ULA Low | 70 | -1.5 |

### 5.2.3.2.8 4Rx TDD FR1 PDSCH pre-emption performance

#### 5.2.3.2.8.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.8.0-3, with the addition of test parameters in Table 5.2.3.2.8.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.8.0-1.

**Table 5.2.3.2.8.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH pre-emption performance under 4 receive antenna conditions | 1-1 |

**Table 5.2.3.2.8.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDCCH configuration (Note 4) | Symbols with PDCCH | | 0, 1 |
| | DCI format | | 2_1 |
| | timeFrequencySet | | 14x1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Pre-emption configuration (Note 2) | Starting symbol (S) | | 3 |
| | Length (L) | | 2 |
| | Pre-emption periodicity and offset | Slots | 40/(1,12,23,34) (Note 3) |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | FR1.30-1 |
| Note 1: Void<br>Note 2: Interference modelled as random data on pre-empted REs.<br>Note 3: Pre-emption is scheduled with 10% probability within 20ms periodicity.<br>Note 4: In addition to PDCCH configuration in Table 5.2-1. | | | |

**Table 5.2.3.2.8.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 2-2.6 TDD | 40 / 30 | 16QAM 0.64 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 8.7 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.8.

### 5.2.3.2.8_1 4Rx TDD FR1 PDSCH pre-emption performance - 2x4 MIMO with baseline receiver for both SA and NSA

#### 5.2.3.2.8_1.1 Test purpose

To Verify the PDSCH pre-emption performance under 4 receive antenna conditions.

#### 5.2.3.2.8_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pre-EmptIndication-DL-r16*.

#### 5.2.3.2.8_1.3 Test description

##### 5.2.3.2.8_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.2.8.0-2 and Table 5.2.3.2.8.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.8_1.3.3.

##### 5.2.3.2.8_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.8.0-3. The SS sends downlink MAC padding bits on the DL RMC.
2. SS transmits PDCCH DCI format 2_1 for int_RNTI with 10% probability to transmit the DL Preemption indication according to Table 5.2.3.2.8.0-2. In the time and frequency set indicated by PDCCH DCI format 2_1, SS stops transmission of PDSCH.
3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.2.8_1.3.4-1.
4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

#### 5.2.3.2.8_1.3.3 Message contents

##### 5.2.3.2.8_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.8_1.3.3_1

##### 5.2.3.2.8_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.8_1.3.3_1

#### 5.2.3.2.8_1.3.4 Test requirement

Table 5.2.3.2.8.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.8_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.8_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH. 2-2.6 TDD | 40 / 30 | 16QAM 0.64 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 9.7 |

### 5.2.3.2.9 4Rx TDD FR1 HST-SFN performance

#### 5.2.3.2.9.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.9.0-3, with the addition of test parameters in Table 5.2.3.2.9.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.9.0-1.

**Table 5.2.3.2.9.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 4 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when *highSpeedDemodFlag-r16* [17] is configured. | 1-1 |

**Table 5.2.3.2.9.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **TDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |

| | | | |
|---|---|---|---|
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| CSI-RS for tracking | CSI-RS periodicity | Slots | 20 for CSI-RS resource 1,2,3,4. |
| | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 22 for CSI-RS resource 3 and 4. |
| | Frequency Occupation | | Start PRB 0<br>Number of PRB = 52 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

**Table 5.2.3.2.9.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.4 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-SFN | 2x4 | 70 | 11.7 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.2.3.2.9.

### 5.2.3.2.9_1 4Rx TDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA

Editor's note: The minimum test time value is in [].

#### 5.2.3.2.9_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions in the HST-SFN scenario defined in B.3.2 when *highSpeedDemodFlag-r16* IE [20] is configured and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.3.2.9_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC supporting enhanced demodulation processing for HST-SFN joint transmission scheme.

#### 5.2.3.2.9_1.3 Test description

##### 5.2.3.2.9_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.3.2.9.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.9_1.3.3.

### 5.2.3.2.9_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.3.2.9_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.2.9_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.3.2.9_1.4-1 as appropriate.

### 5.2.3.2.9_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

#### 5.2.3.2.9_1.3.3_1 Message exceptions for SA

**Table 5.2.3.2.9_1.3.3_1-1: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | n2 for test 1-1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.2.9_1.3.3_1-2: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1 | |
| } | | | |

**Table 5.2.3.2.9_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n8 | for test 1-1 | |
| } | | | |

**Table 5.2.3.2.9_1.3.3_1-4: CSI-ResourcePeriodicityAndOffset for CSI Tracking**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots20 | 1 for CSI-RS resource #1 and #2<br><br>2 for CSI-RS resource #3 and #4 | For test 1-1:<br>offset = 1 for CSI-RS resource 1 and 2<br>offset =2 for CSI-RS resource 3 and 4. | |
| } | | | |

### 5.2.3.2.9_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.9_1.3.3_1

### 5.2.3.2.9_1.4 Test requirement

Tables 5.2.3.2.9_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.9_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.9_1.4-1: Test Requirements for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.4 TDD | 40 / 30 | 16QAM, 0.48 | FR1.30-1 | HST-SFN | 2x4 | 70 | 11.7 + TT |

## 5.2.3.2.10 4Rx TDD FR1 HST DPS performance

### 5.2.3.2.10.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.10.0-3, with the addition of test parameters in Table 5.2.3.2.10.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.10.0-1.

**Table 5.2.3.2.10.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify UE performance in the HST-DPS scenario defined in B.3.3 | 1-1, 1-2 |

**Table 5.2.3.2.10.0-2: Test parameters**

| Parameter | | | Unit | Value |
|---|---|---|---|---|
| Duplex mode | | | | TDD |
| Active DL BWP index | | | | 1 |
| PDCCH configuration | TCI state | | | Note 1 |
| PDSCH configuration | Mapping type | | | Type A |
| | k0 | | | 0 |
| | Starting symbol (S) | | | 2 |
| | Length (L) | | | Specific to each Reference channel |
| | PDSCH aggregation factor | | | 1 |
| | PRB bundling type | | | Static |
| | PRB bundling size | | | 2 |
| | Resource allocation type | | | Type 0 |
| | RBG size | | | Config2 |
| | VRB-to-PRB mapping type | | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A |
| | TCI state | | | Note 1 |
| PDSCH DMRS configuration | DMRS Type | | | Type 1 |
| | Number of additional DMRS | | | 2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 |
| CSI-RS for tracking | Resource set #1 | First OFDM symbol in the PRB used for CSI-RS | | l0 = 5 for CSI-RS resource 1 and 3 |
| | | | | l0 = 9 for CSI-RS resource 2 and 4 |
| | | CSI-RS periodicity | Slots | 20 for CSI-RS resource 1,2,3,4 |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 1 and 2 |
| | | | | 2 for CSI-RS resource 3 and 4 |
| | | QCL info | | TCI state #2 |
| | | Frequency Occupation | | Start PRB 0 |
| | | | | Number of PRB = 52 |
| | Resource set #2 | First OFDM symbol in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 5 and 6 |
| | | | | $l_0$ = 10 for CSI-RS resource 7 and 8 |
| | | CSI-RS periodicity | Slots | 20 for CSI-RS resource 5,6,7,8. |
| | | CSI-RS offset | Slots | 1 for CSI-RS resource 5 and 6 |
| | | | | 2 for CSI-RS resource 7 and 8 |
| | | QCL info | | TCI state #3 |
| | | Frequency Occupation | | Start PRB 0 |
| | | | | Number of PRB = 52 |
| NZP CSI-RS for CSI acquisition | Resource set #3 | First OFDM symbol in the PRB used for CSI-RS | | l0 = 12 |
| | | CSI-RS periodicity | Slots | 40 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #0 |
| | Resource set #4 | First OFDM symbol in the PRB used for CSI-RS | | l0 = 13 |
| | | CSI-RS periodicity | Slots | 40 |
| | | CSI-RS offset | Slots | 0 |
| | | QCL info | | TCI state #1 |
| TCI state #0 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking Resource set #1' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 5 from 'CSI-RS for tracking Resource set #2' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| TCI state #2 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #3 | Type 1 QCL information | SSB index | | SSB #1 |
| | | QCL Type | | Type C |

| | | | | |
|---|---|---|---|---|
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| Number of HARQ Processes | | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |

Note 1: SSB # (k mod 2) , CSI-RS (for tracking) resource set # ((k mod 2) + 1) and CSI-RS (for CSI acquisition) resource set # ((k mod 2) + 3) are transmitted by k[th] RRH.
For Test 1-1, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by k[th] RRH from slot#

$$\max[(2k-1)n + 1 + T_{HARQ} + T_{MAC\ proc} + T_{firstTRS} + T_{TRS\ proc}, 0]$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\ proc},$$

PDCCH and PDSCH are DTXed in other slots in which throughput statistics are not considered.
For Test 1-2, TCI state switching command scheduled by MAC CE with MCS 4 is transmitted in slot #i that satisfy $\mathrm{mod}(i, 2n) = n$. PDCCH and PDSCH associated with TCI # (k mod 2) is transmitted by k[th] RRH from slot#

$$\max[(2k-1)n + 1 + T_{HARQ} + T_{MAC\ proc}, 0]$$

to slot#

$$(2k+1)n + T_{HARQ} + T_{MAC\ proc}$$

Where k=0, 1, 2… is the RRH number, n = 5040 is half of the number of slots between two RRH, $T_{HARQ}$ = 8 is the number of slots between PDSCH and corresponding HARQ-ACK information, $T_{MAC\ proc}$ = 6 is the number of slots for MAC CE processing, $T_{firstTRS}$ = 7 is the number of slots to first TRS transmission occasion after MAC CE command is decoded by the UE, $T_{TRS\ proc}$ = 4 is the number of slots for TRS processing.

**Table 5.2.3.2.10.0-3: Minimum performance for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 1 | 2x4 | 70 | 10.2 |
| 1-2 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 2 | 2x4 | 70 | 10.2 |

### 5.2.3.2.10_1 4Rx TDD FR1 HST DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- MU and TT are FFS

#### 5.2.3.2.10_1.1 Test purpose

To verify UE performance in the HST-DPS scenario defined in B.3.3 and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput and as well verify the HARQ soft combining with default baseline receiver configuration, for Rank 2 scenarios.

#### 5.2.3.2.10_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 5.2.3.2.10_1.3 Test description

#### 5.2.3.2.10_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and clause A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1 and Table 5.2.3.1.10.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.10_1.3.3.

#### 5.2.3.2.10_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2.3.2.10_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 5.2.3.2.10_1.4-1 as appropriate.
3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.
4. Repeat steps from 1 to 3 for each subtest in Tables 5.2.3.2.10_1.4-1 as appropriate.

#### 5.2.3.2.10_1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.2.3.2.10_1.3.3_1 Message exceptions for SA

**Table 5.2.3.2.10_1.3.3_1-1: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-24 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-AdditionalPosition | pos2 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-2: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | n8 | for test 1-1, 1-2 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-3: NZP-CSI-RS-Resource for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-8 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | i-1 for CSI-RS resource #i, i=1,2,3,4,5,6,7,8 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 2 for CSI-RS resource #1, #2, #3, #4<br>3 for CSI-RS resource #5, #6, #7, #8 | for test 1-1, 1-2:<br>TCI-StateId for TCI-State #2 for CSI-RS resource #1, #2, #3, #4<br>TCI-StateId for TCI-State #3 for CSI-RS resource #5, #6, #7, #8 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-4: CSI-RS-ResourceMapping for TRS (Table 5.2.3.2.10_1.3.3_1-3)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-9 with condition TRS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 5 for CSI-RS resource #1 and #3<br>9 for CSI-RS resource #2 and #4<br>6 for CSI-RS resource #5 and #6<br>10 for CSI-RS resource #7 and #8 | for test 1-1, 1-2:<br>$l_0$ = 5 for CSI-RS resource 1 and 3<br>$l_0$ = 9 for CSI-RS resource 2 and 4<br>$l_0$ = 6 for CSI-RS resource 5 and 6<br>$l_0$ = 10 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-5: CSI-ResourcePeriodicityAndOffset for TRS (Table 5.2.3.2.10_1.3.3_1-3)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 1 for CSI-RS resource #1, #2, #5, #6<br><br>2 for CSI-RS resource #3 #4, #7, #8 | For test 1-1, 1-2:<br>periodicity:<br>10 slots.<br>offset:<br>1 for CSI-RS resource 1 and 2<br>2 for CSI-RS resource 3 and 4<br>1 for CSI-RS resource 5 and 6<br>2 for CSI-RS resource 7 and 8 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-6: NZP-CSI-RS-ResourceSet for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 0 for Resource set #1<br>1 for Resource set #2 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #1 |
| NZP-CSI-RS-ResourceId[1] | 0 | entry 1<br>CSI-RS resource #1 | |
| NZP-CSI-RS-ResourceId[2] | 1 | entry 2<br>CSI-RS resource #2 | |
| NZP-CSI-RS-ResourceId[3] | 2 | entry 3<br>CSI-RS resource #3 | |
| NZP-CSI-RS-ResourceId[4] | 3 | entry 4<br>CSI-RS resource #4 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 4 entries | For test 1-1, 1-2 | Resource set #2 |
| NZP-CSI-RS-ResourceId[1] | 4 | entry 1<br>CSI-RS resource #5 | |
| NZP-CSI-RS-ResourceId[2] | 5 | entry 2<br>CSI-RS resource #6 | |
| NZP-CSI-RS-ResourceId[3] | 6 | entry 3<br>CSI-RS resource #7 | |
| NZP-CSI-RS-ResourceId[4] | 7 | entry 4<br>CSI-RS resource #8 | |
| } | | | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-7: NZP-CSI-RS-Resource for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| nzp-CSI-RS-ResourceId | 8 for CSI-RS resource #9<br>9 for CSI-RS resource #10 | for test 1-1, 1-2 | |
| qcl-InfoPeriodicCSI-RS | 0 for CSI-RS resource #9<br>1 for CSI-RS resource #10 | for test 1-1, 1-2:<br>TCI-State #0 for CSI-RS resource #9<br><br>TCI-State #1 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-8: CSI-RS-ResourceMapping for CSI Acquisition (Table 5.2.3.2.10_1.3.3_1-7)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 12 for CSI-RS resource #9<br>13 for CSI-RS resource #10 | for test 1-1, 1-2<br>$l_0$=12 for CSI-RS resource #9<br>$l_0$=13 for CSI-RS resource #10 | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-9: CSI-ResourcePeriodicityAndOffset for CSI Acquisition (Table 5.2.3.2.10_1.3.3_1-7)**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-16 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots20 | 0 | For test 1-1, 1-2:<br>periodicity = 20 slots.<br>offset = 0 slots | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-10: NZP-CSI-RS-ResourceSet for CSI Acquisition**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-18 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-ResourceSet ::= SEQUENCE { | | | |
| nzp_CSI_ResourceSetId | 2 for Resource set #3<br>3 for Resource set #4 | For test 1-1, 1-2 | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #3 |
| NZP-CSI-RS-ResourceId[1] | 8 | entry 1<br>CSI-RS resource #9 | |
| } | | | |
| nzp-CSI-RS-Resources SEQUENCE (SIZE (1..maxNrofNZP-CSI-RS-ResourcesPerSet)) OF NZP-CSI-RS-ResourceId { | 1 entry | For test 1-1, 1-2 | Resource set #4 |
| NZP-CSI-RS-ResourceId[1] | 9 | entry 1<br>CSI-RS resource #10 | |
| } | | | |
| } | | | |

**Table 5.2.3.2.10_1.3.3_1-11: TCI-State**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 for TCI state #0<br>1 for TCI state #1<br>2 for TCI state #2<br>3 for TCI state #3 | For test 1-1, 1-2 | |
| qcl-Type1 SEQUENCE { | | | |
| bwp-Id | BWP-Id of active BWP | | TCI state #0, TCI state #1 |
| | Not present | | TCI state #2, TCI state #3 |
| referenceSignal CHOICE { | | | |
| csi-rs | 0 | CSI-RS resource #1 | TCI state #0 |
| | 4 | CSI-RS resource #5 | TCI state #1 |
| ssb | 0 | SSB #0 | TCI state #2 |
| | 1 | SSB #1 | TCI state #3 |
| } | | | |
| qcl-Type | typeA | | TCI state #0, TCI state #1 |
| | typeC | | TCI state #2, TCI state #3 |
| } | | | |
| } | | | |

#### 5.2.3.2.10_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.10_1.3.3_1

### 5.2.3.2.10_1.4 Test requirement

Tables 5.2.3.2.10_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.10_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.10_1.4-1: Test Requirements for HST-DPS**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Number of active PDSCH TCI states | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 1 | 2x4 | 70 | 10.2 + TT |
| 1-2 | R.PDSCH.2-10.5 TDD | 40 / 30 | 64QAM, 0.43 | HST-DPS | 2 | 2x4 | 70 | 10.2 + TT |

## 5.2.3.2.11 4Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance

### 5.2.3.2.11.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.11.0-3, with the addition of test parameters in Table 5.2.3.2.11.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.11.0-1.

**Table 5.2.3.2.11.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance with Single-DCI based SDM scheme under 4 receive antenna conditions | 1-1,1-2 |

**Table 5.2.3.2.11.0-2: Test parameters**

| Parameter | | | | Unit | Value | |
|---|---|---|---|---|---|---|
| | | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | | TRxP #1 | |
| PDCCH configuration | | | TCI state | | TCI State #1 | |
| | | | CORESETPoolIndex | | 0 | |
| CSI-RS for tracking | | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3 l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7 l0 = 10 for CSI-RS resources 6 and 8 |
| | | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | | Density | | 3 | |
| | | | CSI-RS periodicity | Slots | 40 | |
| | | | CSI-RS offset | Slots | 20 for CSI-RS resources 1 and 2 21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6 21 for CSI-RS resources 7 and 8 |
| | | | QCL info | | TCI state #0 | |
| Duplex mode | | | | | TDD | |
| Active DL BWP index | | | | | 1 | |
| PDSCH configuration | Mapping type | | | | Type A | |
| | k0 | | | | 0 | |
| | Starting symbol (S) | | | | 2 | |
| | Length (L) | | | | 12 | |
| | PRB bundling type | | | | Static | |
| | PRB bundling size | | | | 2 | |
| | Resource allocation type | | | | Type 1 | |
| | RBG size | | | | Config2 | |
| | VRB-to-PRB mapping type | | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | | 1000 | 1002 |
| | TCI state | | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | | Type 1 | |
| | Number of additional DMRS | | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | | 1 | |
| TCI State #1 | | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | | QCL Type | | Type A | N/A |
| | | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | | QCL Type | | N/A | N/A |
| TCI State #2 | | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | | QCL Type | | N/A | Type A |
| | | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | | QCL Type | | N/A | N/A |

| | | |
|---|---|---|
| Resource allocation | | Full-overlapping |
| Timing offset of the second TRxP from the first TRxP | us | -0.25 for test 1-1<br>1 for test 1-2 |
| Frequency offset of the second TRxP from the first TRxP | Hz | 300 for test 1-1<br>0 for test 1-2 |
| Number of HARQ Processes | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 |
| Precoding configuration | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity |
| Note 1: PDSCH transmission is done from both TRxPs (PDSCH Layer 0 is transmitted from TRxP #1 and PDSCH layer 1 is transmitted from TRxP #2) | | |

**Table 5.2.3.2.11.0-3: Minimum performance**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| 1-1 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 14.5 |
| 1-2 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 13.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.11.

### 5.2.3.2.11_1 2Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA

#### 5.2.3.2.11_1.1 Test purpose

To verify the PDSCH performance with Single-DCI based SDM scheme under 4 receive antenna conditions.

#### 5.2.3.2.11_1.2 Test applicability

Test applies to all types of NR UE release 16 and forward supporting capability IE *singleDCI-SDM-scheme-r16*.

#### 5.2.3.2.11_1.3 Test description

##### 5.2.3.2.11_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.3 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.3.2.11.0-2 and Table 5.2.3.2.11.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.3.2.11_1.3.3.

### 5.2.3.2.11_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.3.2.11_1.3.4-1. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.3.2.11_1.3.4-1.

3. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

4. Repeat steps from 1 to 3 for each subtest in Table 5.2.3.2.11_1.3.4-1 as appropriate.

### 5.2.3.2.11_1.3.3 Message contents

#### 5.2.3.2.11_1.3.3_1 Message exceptions for SA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.3.2.11_1.3.3_1-1: Physical layer parameters for DCI format 1_1**

| Derivation Path: TS 38.508-1 [6], Table 4.3.6.1.2.2-1 | | | |
|---|---|---|---|
| **Parameter** | **Value** | **Value in binary** | **Condition** |
| Antenna port(s) | DMRS port 0 and 2 | “1011” | |
| Transmission configuration indication | TCI state 1 and 2 | “000” | |

**Table 5.2.3.2.11_1.3.3_1-2: *CellGroupConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CellGroupConfig ::= SEQUENCE { | | | |
| simultaneousTCI-UpdateList1-r16 SEQUENCE { | | | |
| ServCellIndex [1] | ServCellIndex | | |
| } | | | |
| } | | | |

**Table 5.2.3.2.11_1.3.3_1-3: *ControlResourceSet***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-28 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| tci-PresentInDCI | enabled | | |
| } | | | |

**Table 5.2.3.2.11_1.3.3_1-4: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| tci-StatesToAddModList SEQUENCE(SIZE (1.. maxNrofTCI-States)) OF TCI-State { | 2 entries | | |
| TCI-State[1] | *TCI-State* with condition TCI-state-0 | | |
| TCI-State[2] | *TCI-State* with condition TCI-state-1 | | |
| TCI-State[3] | *TCI-State* with condition TCI-state-2 | | |
| } | | | |
| rbg-Size | config2 | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.2.3.2.11_1.3.3_1-5: *TCI-State***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-190 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| TCI-State ::= SEQUENCE { | | | |
| tci-StateId | 0 | | TCI-state-0 |
| | 1 | | TCI-state-1 |
| | 2 | | TCI-state-2 |
| qcl-Type1 SEQUENCE { | | | |
| cell | Not present | | |
| bwp-Id | Not present | | |
| referenceSignal CHOICE { | | | |
| ssb | SSB-Index | | TCI-state-0 |
| csi-rs | 1 | | TCI-state-1 |
| | 5 | | TCI-state-2 |
| } | | | |
| qcl-Type | typeA | | |
| } | | | |
| qcl-Type2 | Not present | | |
| } | | | |

**Table 5.2.3.2.11_1.3.3_1-6: *NZP-CSI-RS-Resource***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-85 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| resourceMapping SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | For CSI-RS resources 1, 2, 3, 4 | |
| | 0001 | For CSI-RS resources 5,6,7,8 | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 6 | For CSI-RS resources 1,3,5,7 | |
| | 10 | For CSI-RS resources 2,4,6,8 | |
| cdm-Type | noCDM | | |
| density CHOICE { | | | |
| three | NULL | | |
| } | | | |
| } | | | |
| periodicityAndOffset CHOICE { | | | |
| slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |
| qcl-InfoPeriodicCSI-RS | 0 | | |
| } | | | |

### 5.2.3.2.11_1.3.3_2 Message exceptions for NSA

Same as 5.2.3.2.11_1.3.3_1.

### 5.2.3.2.11_1.3.4 Test requirement

Table 5.2.3.2.11.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.11_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.11_1.3.4-1: Test requirement**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition(Note 1) | Correlation matrix and antenna configuration(Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1-1 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.3 0-1 | TDLA30-10 | 2x4, ULA Low | 70 | 15.5 |
| 1-27 | R.PDSCH.2-3.2 TDD | 40 / 30 | 64QAM, 0.50 | FR1.3 0-1 | TDLA30-10 | 2x4, ULA Low | 70 | 14.9 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 with scaling factor as 1/sqrt(2) for transmitted signal from each TRxP | | | | | | | | |

### 5.2.3.2.12 4Rx TDD FR1 PDSCH Multi-DCI based transmission scheme performance

#### 5.2.3.2.12.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.12.0-3, with the addition of test parameters in Table 5.2.3.2.12.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.12.0-1.

**Table 5.2.3.2.12.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs | 1-1 |

**Table 5.2.3.2.12.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | | TCI state | | TCI State #1 | TCI State #2 |
| | | CORESETPoolIndex | | 0,1 | |
| CSI-RS for tracking | | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | | Density | | 3 | |
| | | CSI-RS periodicity | Slots | 40 | |
| | | CSI-RS offset | Slots | 20 for CSI-RS resources 1 and 2<br>21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6<br>21 for CSI-RS resources 7 and 8 |
| | | QCL info | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 1 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |

| | | | | | |
|---|---|---|---|---|---|
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | {1000,1001} | {1002,1003} |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Resource allocation | | | | Non-overlapping | |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.25 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 | |
| Number of HARQ Processes | | | | 8 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs. Transmission from TRxP #1 uses CORESETPoolIndex 0 and transmission from TRxP #2 uses CORESETPoolIndex 1 | | | | | |

**Table 5.2.3.2.12.0-3: Minimum performance**

| Test num. | Reference channel | | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition( Note 1) | Correlation matrix and antenna configuration( Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| | TRxP #1 | TRxP #2 | | | | | | | |
| 1-1 | R.PDSCH.2-3.3 TDD | R.PDSCH.2-3.4 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 14.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.12.

### 5.2.3.2.12_1 4Rx TDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.2.12_1.1 Test purpose

To verify the PDSCH performance when UE is configured two different values of CORESETPoolIndex in ControlResourceSet and when UE receives multiple PDCCHs scheduling PDSCHs.

#### 5.2.3.2.12_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *multiDCI-MultiTRP-r16*.

#### 5.2.3.2.12_1.3 Test description

Same test description as in clause 5.2.3.1.12_1.3 with the following exception:

- Table 5.2.3.2.12_1.4-1 instead of 5.2.3.1.12_1.4-1
- Table 5.2.3.2.12_1.3-1 instead of Table 5.2.3.1.12_1.3.3_1-8

**Table 5.2.3.2.12_1.3-1: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| Slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

#### 5.2.3.2.12_1.4 Test requirement

Table 5.2.3.2.12.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.12_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.12_1.4-1: Test requirement**

| Test num. | Reference channel | | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition( Note 1) | Correlation matrix and antenna configuration( Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Fraction of maximum throughput (%) | SNR (dB)(Note 3) |
| | TRxP #1 | TRxP #2 | | | | | | | |
| 1-1 | R.PDSCH.2-3.3 TDD | R.PDSCH.2-3.4 TDD | 40 / 30 | 64QAM, 0.50 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 15.6 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | | |

### 5.2.3.2.13 4Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance

#### 5.2.3.2.13.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.13.0-3, with the addition of test parameters in Table 5.2.3.2.13.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.13.0-1.

**Table 5.2.3.2.13.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 4 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16” defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.3.2.13.0-2: Test parameters**

| Parameter | | | Unit | Value | |
|---|---|---|---|---|---|
| | | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | | TRxP #1 | |
| PDCCH configuration | TCI state | | | TCI State #1 | |
| | CORESETPoolIndex | | | Not configured | |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | | l0 = 6 for CSI-RS resources 1 and 3 l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7 l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | CDM Type | | | ‘No CDM’ for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | | 3 | |
| | CSI-RS periodicity | | Slots | 40 | |
| | CSI-RS offset | | Slots | 20 for CSI-RS resources 1 and 2 21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6 21 for CSI-RS resources 7 and 8 |
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | Wideband | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000,1001 | 1000,1001 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking’ configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking’ configuration |
| | | QCL Type | | N/A | Type A |
| | | CSI-RS resource | | N/A | N/A |

| | | | | | |
|---|---|---|---|---|---|
| | Type 2 QCL information | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | -0.25 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 | |
| Number of HARQ Processes | | | | 8 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs | | | | | |

**Table 5.2.3.2.13.0-3: Minimum performance for Rank 2**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.2-2.5 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 10.5 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.13.

### 5.2.3.2.13_1 4Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.2.13_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions when UE is configured with “FDMSchemeA” in “RepetitionScheme-r16”.

#### 5.2.3.2.13_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportFDM-SchemeA-r16*.

#### 5.2.3.2.13_1.3 Test description

Same test description as in clause 5.2.3.1.13_1.3 with the following exception:

- Table 5.2.3.2.13_1.4-1 instead of 5.2.3.1.13_1.4-1
- Table 5.2.3.2.13_1.3-1 instead of Table 5.2.3.1.13_1.3.3_1-5

**Table 5.2.3.2.13_1.3-1: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| Slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

#### 5.2.3.2.13_1.4 Test requirement

Table 5.2.3.2.13.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.3.2.13_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.13_1.4-1: Test requirement**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) (Note 3) |
| 1-1 | R.PDSCH.2-2.5 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 70 | 11.5 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

### 5.2.3.2.14 4Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance

#### 5.2.3.2.14.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.3.2.14.0-3, with the addition of test parameters in Table 5.2.3.2.14.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.3.2.14.0-1.

**Table 5.2.3.2.14.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH performance under 4 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states defined in clause 5.1 of TS 38.214 [12] | 1-1 |

**Table 5.2.3.2.14.0-2: Test parameters**

| Parameter | | Unit | Value | |
|---|---|---|---|---|
| | | | TRxP #1(Note 1) | TRxP #2(Note 1) |
| Transmit TRxP of SSB | | | TRxP #1 | |
| PDCCH configuration | TCI state | | TCI State #1 | |
| | CORESETPoolIndex | | Not configured | |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS | | k0=0 for CSI-RS resources 1,2,3,4 | k0=1 for CSI-RS resources 5,6,7,8 |
| | First OFDM symbol in the PRB used for CSI-RS | | l0 = 6 for CSI-RS resources 1 and 3<br>l0 = 10 for CSI-RS resources 2 and 4 | l0 = 6 for CSI-RS resources 5 and 7<br>l0 = 10 for CSI-RS resources 6 and 8 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 | 1 for CSI-RS resource 5,6,7,8 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4,5,6,7,8 | |
| | Density | | 3 | |
| | CSI-RS periodicity | Slots | 40 | |
| | CSI-RS offset | Slots | 20 for CSI-RS resources 1 and 2<br>21 for CSI-RS resources 3 and 4 | 20 for CSI-RS resources 5 and 6<br>21 for CSI-RS resources 7 and 8 |

| | | | | | |
|---|---|---|---|---|---|
| | QCL info | | | TCI state #0 | |
| Duplex mode | | | | TDD | |
| Active DL BWP index | | | | 1 | |
| PDSCH configuration | Mapping type | | | Type A | |
| | k0 | | | 0 | |
| | Starting symbol (S) | | | 2 | |
| | Length (L) | | | 12 | |
| | Repetition number | | | 2 | |
| | PRB bundling type | | | Static | |
| | PRB bundling size | | | 2 | |
| | Resource allocation type | | | Type 0 | |
| | RBG size | | | Config2 | |
| | VRB-to-PRB mapping type | | | Non-interleaved | |
| | VRB-to-PRB mapping interleaver bundle size | | | N/A | |
| PDSCH DMRS configuration | Antenna port indexes | | | 1000 | 1000 |
| | TCI state | | | TCI State #1 | TCI State #2 |
| | DMRS Type | | | Type 1 | |
| | Number of additional DMRS | | | 1 | |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | | 1 | |
| TCI State #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration | N/A |
| | | QCL Type | | Type A | N/A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| TCI State #2 | Type 1 QCL information | CSI-RS resource | | N/A | CSI-RS resource 5 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | N/A | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A | N/A |
| | | QCL Type | | N/A | N/A |
| Timing offset of the second TRxP from the first TRxP | | | us | 1 | |
| Frequency offset of the second TRxP from the first TRxP | | | Hz | 300 | |
| Number of HARQ Processes | | | | 4 | |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 (Note 2) | |
| Precoding configuration | | | | SP Type I, independent precoding generation is applied for both TRxPs, random per slot with PRB bundling granularity | |
| Note 1: PDSCH transmission is done from both TRxPs<br>Note 2: ACK/NACK feedback is generated for PDSCH on slot i, where mod(i,10) = {2, 4, 6}. | | | | | |

**Table 5.2.3.2.14.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | BLER (%) | SNR (dB) (Note 4) |
| 1-1 | R.PDSCH.2-16.2 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 1 (Note 3) | -0.5 |
| Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.<br>Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.<br>Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.<br>Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2 | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.3.2.14.

### 5.2.3.2.14_1 4Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x4 MIMO for both SA and NSA

#### 5.2.3.2.14_1.1 Test purpose

To verify the PDSCH performance under 4 receive antenna conditions when UE is configured with repetitionNumber-r16 with multiple slot level PDSCH transmission occasions of the same TB with two TCI states.

#### 5.2.3.2.14_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *supportTDM-SchemeA-r16*.

#### 5.2.3.2.14_1.3 Test description

Same test description as in clause 5.2.3.1.14_1.3 with the following exception:

- Table 5.2.3.2.14_1.4-1 instead of 5.2.3.1.14_1.4-1
- Table 5.2.3.2.14_1.3-1 instead of Table 5.2.3.1.14_1.3.3_1-5

**Table 5.2.3.2.14_1.3-1: *CSI-ResourcePeriodicityAndOffset* for TRS**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-10 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots40 | 20 | For CSI-RS resources 1,2,5,6 | |
| Slots40 | 21 | For CSI-RS resources 3,4,7,8 | |
| } | | | |

#### 5.2.3.2.14_1.4 Test requirement

Table 5.2.3.2.14.0-3 defines the primary level settings.

The residual BLER specified in Note 3 of Table 5.2.3.2.14_1.4-1 test shall meet or be lower than the specified value in Table 5.2.3.2.14_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.3.2.14_1.4-1: Test requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition (Note 1) | Correlation matrix and antenna configuration (Note 2) | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | BLER (%) | SNR (dB) (Note 4) |
| 1-1 | R.PDSCH.2-16.2 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x4, ULA Low | 1 (Note 3) | 0.5 |

Note 1: The propagation conditions apply to each of TRxP #1 and TRxP #2 and are statistically independent.
Note 2: Correlation matrix and antenna configuration parameters apply to each of TRxP #1 and TRxP #2.
Note 3: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block.
Note 4: SNR corresponds to SNR of TRxP #1 and TRxP #2 as defined in 4.4.2

## 5.2A PDSCH demodulation requirements for CA

The parameters specified in Table 5.2-1 for PDSCH single carrier tests are reused for PDSCH CA tests unless otherwise stated.

**Table 5.2A-1: Common test parameters for CA**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD and TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | FDD: 12<br>TDD: Specific to each Reference channel |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | As defined in Table 5.2A-2 |
| TDD UL-DL pattern | | | 15kHz SCS: FR1.15-1<br>30kHz SCS: FR1.30-1 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | As defined in Table 5.2A-3 |
| Number of PUCCH ResourceGroups | | | 1 |
| PUCCH format for HARQ-ACK feedback | | | PUCCH format 1 for cases with no more chan 2 DL CCs<br>PUCCH format 3 for cases with more than 2 DL CCs |

**Table 5.2A-2: Test parameters for number of HARQ processes**

| HARQ process number | | CCs with the same duplex mode & SCS with Pcell | CCs with different duplex mode / SCS with Pcell |
|---|---|---|---|
| FDD 15 kHz + TDD 30 kHz CA | FDD PCell | 4 | 8 |
| | TDD PCell | 8 | 8 |
| FDD 15 kHz + TDD 15 kHz CA | FDD PCell | 4 | 4 |
| | TDD PCell | 8 | 8 |
| TDD 15 kHz + TDD 30 kHz CA | 15kHz PCell | 8 | 12 |
| | 30kHz PCell | 8 | 8 |
| FDD 15 kHz + FDD 15 kHz CA | FDD PCell | 4 | N/A |
| TDD 30 kHz + TDD 30 kHz CA | TDD PCell | 8 | N/A |

**Table 5.2A-3: Test parameters for K1 values**

| The number of slots between PDSCH and corresponding HARQ-ACK information | | CCs with the same duplex mode and SCS with Pcell | CCs with different duplex mode and/or SCS with Pcell |
|---|---|---|---|
| FDD 15 kHz + TDD 30 kHz CA | FDD PCell | {2} | {2} |
| | TDD PCell | {8,7,6,5,5,4,3,2} | {7,5,4,11,9} |
| FDD 15 kHz + TDD 15 kHz CA | FDD PCell | {2} | {2} |
| | TDD PCell | {4,3,2,6} | {4,3,2,6,5} |
| TDD 15 kHz + TDD 30 kHz CA | 15kHz PCell | {4,3,2,6} | {4,4,3,3,2,2,6,6} |
| | 30kHz PCell | {8,7,6,5,5,4,3,2} | {7,5,4,11} |

| | | | |
|---|---|---|---|
| FDD 15 kHz + FDD 15 kHz CA | FDD PCell | {2} | N/A |
| TDD 30 kHz + TDD 30 kHz CA | TDD PCell | {8,7,6,5,5,4,3,2} | N/A |

## 5.2A.1 1RX requirements (Void)

## 5.2A.2 2RX requirements

### 5.2A.2.1 Requirements for 2RX normal PDSCH

#### 5.2A.2.1.0 Minimum conformance requirements for 2RX normal PDSCH

For CA with different numbers of DL component carriers, the requirements are defined in Table 5.2A.2.1.0-4 based on the single carrier requirements for different SCSs and different bandwidth specified in Table 5.2A.2.1.0-1 to Table 5.2A.2.1.0-3, with the parameters in Table 5.2A-1 to Table 5.2A-3 and the downlink physical channel setup according to Annex C.3.1. The performance requirements specified in this sub-clause do not apply for UE single carrier test.

**Table 5.2A.2.1.0-1: Single carrier performance for FDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-9.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 10 | R.PDSCH.1-2.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 15 | R.PDSCH.1-9.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 20 | R.PDSCH.1-9.3 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.8 |
| 25 | R.PDSCH.1-9.4 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.0 |
| 30 | R.PDSCH.1-9.5 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.8 |
| 40 | R.PDSCH.1-10.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.0 |
| 50 | R.PDSCH.1-10.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.4 |

**Table 5.2A.2.1.0-2: Single carrier performance for TDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.1-2.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 10 | R.PDSCH.1-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.8 |
| 15 | R.PDSCH.1-2.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.8 |
| 20 | R.PDSCH.1-2.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.9 |
| 25 | R.PDSCH.1-2.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.0 |
| 30 | R.PDSCH.1-3.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.9 |
| 40 | R.PDSCH.1-3.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.2 |
| 50 | R.PDSCH.1-3.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.5 |

**Table 5.2A.2.1.0-3: Single carrier performance for TDD 30 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.2-13.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 10 | R.PDSCH.2-13.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 15 | R.PDSCH.2-13.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 20 | R.PDSCH.2-13.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.7 |
| 25 | R.PDSCH.2-13.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.7 |
| 30 | R.PDSCH.2-14.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.7 |
| 40 | R.PDSCH.2-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.9 |
| 50 | R.PDSCH.2-14.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.1 |
| 60 | R.PDSCH.2-14.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.0 |
| 80 | R.PDSCH.2-14.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.5 |
| 90 | R.PDSCH.2-14.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.3 |
| 100 | R.PDSCH.2-15.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |

**Table 5.2A.2.1.0-4: Minimum performance for multiple CA configurations**

| Test number | CA duplex mode | Minimum performance requirements |
|---|---|---|
| 1 | FDD 15 kHz + FDD 15 kHz | As defined in Table 5.2A.2.1.0-1 |
| 2 | TDD 30 kHz + TDD 30 kHz | As defined in Table 5.2A.2.1.0-3 |
| 3 | FDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.2.1.0-1 and Table 5.2A.2.1.0-3 per CC |
| 4 | FDD 15 kHz + TDD 15 kHz | As defined in Table 5.2A.2.1.0-1 and Table 5.2A.2.1.0-2 per CC |
| 5 | TDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.2.1.0-2 and Table 5.2A.2.1.0-3 per CC |
| Note 1: The applicability of requirements for different CA duplex modes, SCSs, CA configurations and bandwidth combination sets is defined in 5.1.1.5. | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2A.2.1.

### 5.2A.2.1.1 2Rx Normal PDSCH Demodulation Performance for CA (2DL CA)

#### 5.2A.2.1.1.1 Test Purpose

To verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions for multiple CA configurations and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput per CC.

#### 5.2A.2.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 2DL CA.

#### 5.2A.2.1.1.3 Test description

##### 5.2A.2.1.1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

Channel BW to be tested: largest aggregated bandwidth combination as per Table 5.1.1.5.2-2.

CA capability to be tested: As per table 5.1.1.5.2-1

**Table 5.2A.2.1.1.3.1.1-1: Test point selection table**

| Test number | CA duplex mode | Configuration |
|---|---|---|
| 1 | FDD 15 kHz + FDD 15 kHz | As defined in Table 5.2A.2.1.0-1 |
| 2 | TDD 30 kHz + TDD 30 kHz | As defined in Table 5.2A.2.1.0-3 |
| 3 | FDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.2.1.0-1 and Table 5.2A.2.1.0-3 per CC |
| 4 (note 2) | FDD 15 kHz + TDD 15 kHz | As defined in Table 5.2A.2.1.0-1 and Table 5.2A.2.1.0-2 per CC |
| 5 (note 3) | TDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.2.1.0-2 and Table 5.2A.2.1.0-3 per CC |
| Note 1: For each test point, select any one of the CA configurations which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate based on the equation $DataRate = 10^{-3}\sum_{j=1}^{J} TBS_j 2^{\mu_j}$<br>Note 2: Test point 4 can be skipped if test point 3 is verified.<br>Note 3: Test point 5 can be skipped if test point 3 or 4 is verified. | | |

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.x for TE diagram and clause A.3.2.6 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2A-1 to Table 5.2A-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2A.2.1.1.3.3.

#### 5.2A.2.1.1.3.2 Test procedure

1. Configure SCC according to Annex C.0, C.1 and C.2 for all downlink physical channels.
2. The SS shall configure SCC as per TS 38.508-1 [6] clause 5.5.1. Message contents are defined in clause 5.2.2.1.1_1.3.3.
3. SS activates SCC by sending the activation MAC-CE (Refer TS 38.321 [18], clauses 5.9, 6.1.3.10). Wait for at least 1 second (Refer TS 38.133[19], clause9.3).
4. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 5.2A.2.1.0-1 to 5.2A.2.1.0-4 as appropriate on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.
5. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to 5.2A.2.1.0-1 to 5.2A.2.1.0-4 as appropriate on both PCC and SCC.
6. Measure the average throughput per each component carrier for a duration sufficient to achieve statistical significance according to Annex G clause TBD. Count the number of NACKs, ACKs and statDTXs on the UL and decide pass or fail according to Table TBD in Annex G clause TBD.
7. Repeat steps from 1 to 6 for each test points in Table 5.2A.2.1.0-4 as appropriate.

#### 5.2A.2.1.1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

#### 5.2A.2.1.1.3.4 Test Requirement

Tables 5.2A.2.1.1.3.4-1, 5.2A.2.1.1.3.4-2 and 5.2A.2.1.1.3.4-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 and A.3.2.2 for each component carrier for throughput test point combination shall meet or exceed the specified value in Table 5.2A.2.1.1.3.4-1, 5.2A.2.1.1.3.4-2 and 5.2A.2.1.1.3.4-3 for the specified SNR including test tolerances for the test points listed in Table 5.2A.2.1.1.3.1.1-1.

**Table 5.2A.2.1.1.3.4-1: Test requirements for FDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.1-9.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 10 | R.PDSCH.1-2.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 15 | R.PDSCH.1-9.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 20 | R.PDSCH.1-9.3 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.8 |
| 25 | R.PDSCH.1-9.4 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.0 |
| 30 | R.PDSCH.1-9.5 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.8 |
| 40 | R.PDSCH.1-10.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.0 |
| 50 | R.PDSCH.1-10.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.4 |

**Table 5.2A.2.1.1.3.4-2: Test requirements for TDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-2.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 10 | R.PDSCH.1-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.8 |
| 15 | R.PDSCH.1-2.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.8 |
| 20 | R.PDSCH.1-2.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.9 |
| 25 | R.PDSCH.1-2.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.0 |
| 30 | R.PDSCH.1-3.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.9 |
| 40 | R.PDSCH.1-3.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.2 |
| 50 | R.PDSCH.1-3.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.5 |

**Table 5.2A.2.1.1.3.4-3: Test requirements for TDD 30 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.2-13.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 10 | R.PDSCH.2-13.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 15 | R.PDSCH.2-13.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 20 | R.PDSCH.2-13.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |
| 25 | R.PDSCH.2-13.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |
| 30 | R.PDSCH.2-14.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |
| 40 | R.PDSCH.2-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.9 |
| 50 | R.PDSCH.2-14.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.1 |
| 60 | R.PDSCH.2-14.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.0 |
| 80 | R.PDSCH.2-14.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.5 |
| 90 | R.PDSCH.2-14.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.3 |
| 100 | R.PDSCH.2-15.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.7 |

#### 5.2A.2.1.2 2Rx Normal PDSCH Demodulation Performance for CA (3DL CA)

FFS

#### 5.2A.2.1.3 2Rx Normal PDSCH Demodulation Performance for CA (4DL CA)

FFS

### 5.2A.2.2 Requirements for 2RX PDSCH carrier aggregation with power imbalance

#### 5.2A.2.2.0 Minimum conformance requirements for 2RX PDSCH CA with power imbalance

The performance requirements are specified in Table 5.2A.2.2.0-3 and Table 5.2A.2.2.0-4, with the addition of test parameters in Table 5.2A.2.2.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2A.2.2.0-1.

**Table 5.2A.2.2.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the ability of an intra-band adjacent carrier aggregation UE to demodulate the signal transmitted by the PCell or SCell in the presence of a stronger SCell or PCell signal on an adjacent frequency. Throughput is measured on the PCell or SCell only | |

**Table 5.2A.2.2.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD and TDD |
| Active DL BWP index | | | 1 |
| Propagation condition | | | Static propagation condition<br>No external noise sources are applied |
| Antenna configuration | | | 1x2 |
| PDSCH configuration | Length (L) | | FDD: 12TDD: 12 for DL slot, 4 for special slot |
| | PRB bundling size | | WB |

| | | |
|---|---|---|
| Modulation and code rate | | 64QAM, MCS 26 |
| Number of HARQ Processes | | FDD: 4<br>TDD: 8 |
| Maximum number of HARQ transmission | | 1 |
| Redundancy version coding sequence | | {0} |
| TDD UL-DL pattern | | 30kHz SCS: FR1.30-1 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | As defined in Table A.1.2-2 for FR1.30-1 |
| PUCCH format for HARQ-ACK feedback | | PUCCH format 1 |
| Overhead for TBS determination | | 0 |
| SSB transmission | | Slot#0 with periodicity 20ms |
| RB assignment | | Full applicable test bandwidth as defined in Table 5.3.5-1 of TS 38.101-1 [6] |

**Table 5.2A.2.2.0-3: Minimum performance for FDD CA with 15 kHz SCS**

| Test Number | Bandwidth (MHz) | | Reference channel | | Power at antenna port (dBm/Hz) | | Reference value Fraction of Maximum Throughput (%) | |
|---|---|---|---|---|---|---|---|---|
| | PCell | SCell | PCell | SCell | $\hat{E}_{s_PCell}$ for PCell | $\hat{E}_{s_SCell}$ for Scell | PCell | SCell |
| 1 | Selected Channel bandwidth as per section 5.1.1.6 | | Derived as per section 5.1.3.2 of TS 38.214 [12] | NA | -112 | -106 | 85 | NA |

**Table 5.2A.2.2.0-4: Minimum performance for TDD CA with 30 kHz SCS**

| Test Number | Bandwidth (MHz) | | Reference channel | | Power at antenna port (dBm/Hz) | | Reference value Fraction of Maximum Throughput (%) | |
|---|---|---|---|---|---|---|---|---|
| | PCell | SCell | PCell | SCell | $\hat{E}_{s_PCell}$ for PCell | $\hat{E}_{s_SCell}$ for Scell | PCell | SCell |
| 1 | Selected Channel bandwidth as per section 5.1.1.6 | | Derived as per section 5.1.3.2 of TS 38.214 [12] | NA | -112 | -106 | 85 | NA |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2A.2.2.

### 5.2A.2.2.1 2Rx PDSCH Demodulation Performance for CA with power imbalance (2DL CA)

Editor's note: Following aspects in this clause are incomplete

- Connection Diagram is in []
- Message contents is TBD
- Annex F update is TBD

#### 5.2A.2.2.1.1 Test Purpose

To verify the ability of an intra-band adjacent carrier aggregation UE to demodulate the signal transmitted by the PCell or SCell in the presence of a stronger SCell or PCell signal on an adjacent frequency. Throughput is measured on the PCell or SCell only.

#### 5.2A.2.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 2DL CA.

#### 5.2A.2.2.1.3 Test description

##### 5.2A.2.2.1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

Band selection: Supported intra-band contiguous CA configurations covering the lowest and highest operating bands,

Channel BW combination to be tested: Select bandwidth combination with same bandwidth in each carrier. If not supported, select bandwidth combination with smallest bandwidth difference between the two carriers and the carrier with smaller bandwidth will be used for the test.

CA capability to be tested: Either FDD or TDD intra-band contiguous CA

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure [A.3.1.7.1] for TE diagram and clause [A.3.2.2] for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2A-1 to Table 5.2A-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.1.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode* On according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2A.2.1.1.3.3.

##### 5.2A.2.2.1.3.2 Test procedure

1. Configure SCC according to Annex C.0, C.1 and C.2 for all downlink physical channels.
2. The SS shall configure SCC as per TS 38.508-1 [6] clause 5.5.1. Message contents are defined in clause 5.2A.2.2.1.3.3.
3. SS activates SCC by sending the activation MAC-CE (Refer TS 38.321 [18], clauses 5.9, 6.1.3.10). Wait for at least 1 second (Refer TS 38.133[19], clause9.3).
4. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the MCS according to Table 5.2A.2.2.0-2 on PCC. The SS sends downlink MAC padding bits on the DL RMC.
5. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to 5.2A.2.2.0-2 to 5.2A.2.2.0-4 as appropriate on both PCC and SCC.
6. Measure the average throughput on PCC for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL and decide pass or fail according to Table G.1.5-1 in Annex G clause G.1.5.

##### 5.2A.2.2.1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

FFS

#### 5.2A.2.2.1.4 Test Requirement

Table 5.2A.2.2.0-2, Table 5.2A.2.2.1.4-1 and Table 5.2A.2.2.1.4-2 define the primary level settings.

The fraction of maximum throughput percentage for the MCS scheduled as specified in Table 5.2A.2.2.0-2 for the throughput test shall meet or exceed the specified value in Table 5.2A.2.2.1.4-1 for the specified cell power levels.

**Table 5.2A.2.2.1.4-1: Minimum performance for FDD CA with 15 kHz SCS**

| Test Number | Bandwidth (MHz) | | Reference channel | | Power at antenna port (dBm/Hz) | | Reference value Fraction of Maximum Throughput (%) | |
|---|---|---|---|---|---|---|---|---|
| | PCell | SCell | PCell | SCell | $\hat{E}_{s_PCell}$ for PCell | $\hat{E}_{s_SCell}$ for Scell | PCell | SCell |
| 1 | Selected Channel bandwidth as per section 5.1.1.6 | | Derived as per section 5.1.3.2 of TS 38.214 [12] | NA | -112 | -106 | 85 | NA |

**Table 5.2A.2.2.1.4-2: Minimum performance for TDD CA with 30 kHz SCS**

| Test Number | Bandwidth (MHz) | | Reference channel | | Power at antenna port (dBm/Hz) | | Reference value Fraction of Maximum Throughput (%) | |
|---|---|---|---|---|---|---|---|---|
| | PCell | SCell | PCell | SCell | $\hat{E}_{s_PCell}$ for PCell | $\hat{E}_{s_SCell}$ for Scell | PCell | SCell |
| 1 | Selected Channel bandwidth as per section 5.1.1.6 | | Derived as per section 5.1.3.2 of TS 38.214 [12] | NA | -112 | -106 | 85 | NA |

### 5.2A.2.2.2 2Rx PDSCH Demodulation Performance for CA with power imbalance (3DL CA)

FFS

### 5.2A.2.2.3 2Rx PDSCH Demodulation Performance for CA with power imbalance (4DL CA)

FFS

## 5.2A.3 4RX requirements

### 5.2A.3.1 Requirements for 4RX normal PDSCH

#### 5.2A.3.1.0 Minimum conformance requirements for 4RX normal PDSCH

For CA with different numbers of DL component carriers, the requirements are defined in Table 5.2A.3.1.0-4 based on the single carrier requirements for different SCSs and different bandwidth specified in Table 5.2A.3.1.0-1 ~ Table 5.2A.3.1.0-3, with the parameters in Table 5.2A-1 ~ Table 5.2A-3 and the downlink physical channel setup according to Annex C.3.1. The performance requirements specified in this sub-clause do not apply for UE single carrier test.

**Table 5.2A.3.1.0-1: Single carrier performance for FDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.1-9.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] |
| 10 | R.PDSCH.1-2.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] |
| 15 | R.PDSCH.1-9.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 20 | R.PDSCH.1-9.3 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 25 | R.PDSCH.1-9.4 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] |
| 30 | R.PDSCH.1-9.5 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 40 | R.PDSCH.1-10.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] |
| 50 | R.PDSCH.1-10.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.9] |

**Table 5.2A.3.1.0-2: Single carrier performance for TDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-2.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] |
| 10 | R.PDSCH.1-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 15 | R.PDSCH.1-2.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] |
| 20 | R.PDSCH.1-2.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 25 | R.PDSCH.1-2.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] |
| 30 | R.PDSCH.1-3.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 40 | R.PDSCH.1-3.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] |
| 50 | R.PDSCH.1-3.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.0] |

**Table 5.2A.3.1.0-3: Single carrier performance for TDD 30 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.2-13.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] |
| 10 | R.PDSCH.2-13.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] |
| 15 | R.PDSCH.2-13.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] |
| 20 | R.PDSCH.2-13.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 25 | R.PDSCH.2-13.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 30 | R.PDSCH.2-14.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] |
| 40 | R.PDSCH.2-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] |
| 50 | R.PDSCH.2-14.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.9] |
| 60 | R.PDSCH.2-14.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] |
| 80 | R.PDSCH.2-14.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.1] |
| 90 | R.PDSCH.2-14.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.0] |
| 100 | R.PDSCH.2-15.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.3] |

**Table 5.2A.3.1.0-4: Minimum performance for multiple CA configurations**

| Test number | CA duplex mode | Minimum performance requirements |
|---|---|---|
| 1 | FDD 15 kHz + FDD 15 kHz | As defined in Table 5.2A.3.1.0-1 |
| 2 | TDD 30 kHz + TDD 30 kHz | As defined in Table 5.2A.3.1.0-3 |
| 3 | FDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.3.1.0-1 and Table 5.2A.3.1.0-3 per CC |
| 4 | FDD 15 kHz + TDD 15 kHz | As defined in Table 5.2A.3.1.0-1 and Table 5.2A.3.1.0-2 per CC |
| 5 | TDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.3.1.0-2 and Table 5.2A.3.1.0-3 per CC |
| Note 1: The applicability of requirements for different CA duplex modes, SCSs, CA configurations and bandwidth combination sets is defined in 5.1.1.5. | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2A.3.1.

### 5.2A.3.1.1 4Rx Normal PDSCH Demodulation Performance for CA (2DL CA)

Editor's note: Following aspects in this clause are incomplete

- Test requirement SNR is in square bracket

#### 5.2A.3.1.1.1 Test Purpose

To verify the PDSCH mapping Type A normal performance under 4 receive antenna conditions for multiple CA configurations and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput per CC.

#### 5.2A.3.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 2DL CA and 4Rx antenna ports on each CC.

#### 5.2A.3.1.1.3 Test description

##### 5.2A.3.1.1.3.1 Initial conditions

Same initial conditions as specified in clause 5.2A.2.1.1.3.1 with the following exception

Channel BW to be tested: largest aggregated bandwidth combination as per Table 5.1.1.5.2-2.

CA capability to be tested: As per table 5.1.1.5.2-1

**Table 5.2A.3.1.1.3.1-1: Test point selection table**

| Test number | CA duplex mode | Minimum performance requirements |
|---|---|---|
| 1 | FDD 15 kHz + FDD 15 kHz | As defined in Table 5.2A.3.1.0-1 |
| 2 | TDD 30 kHz + TDD 30 kHz | As defined in Table 5.2A.3.1.0-3 |
| 3 | FDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.3.1.0-1 and Table 5.2A.3.1.0-3 per CC |
| 4 (note 2) | FDD 15 kHz + TDD 15 kHz | As defined in Table 5.2A.3.1.0-1 and Table 5.2A.3.1.0-2 per CC |
| 5 (note 3) | TDD 15 kHz + TDD 30 kHz | As defined in Table 5.2A.3.1.0-2 and Table 5.2A.3.1.0-3 per CC |
| Note 1: For each test point, select any one of the CA configurations which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate based on the equation $DataRate = 10^{-3}\sum_{j=1}^{J} TBS_j 2^{\mu_j}$<br>Note 2: Test point 4 can be skipped if test point 3 is verified.<br>Note 3: Test point 5 can be skipped if test point 3 or 4 is verified. | | |

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.y for TE diagram and clause A.3.2.6 for UE diagram.

### 5.2A.3.1.1.3.2 Test procedure

Same test procedure as specified in clause 5.2A.2.1.1.3.2 with the following exception:

Instead of Table 5.2A.2.1.0-4 → 5.2A.3.1.0-4

### 5.2A.3.1.1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

### 5.2A.3.1.1.3.4 Test Requirement

Tables 5.2A.3.1.1.3.4-1, 5.2A.3.1.1.3.4-2 and 5.2A.3.1.1.3.4-3 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 and A.3.2.2 for each component carrier for throughput test point combination shall meet or exceed the specified value in Table 5.2A.3.1.1.3.4-1, 5.2A.3.1.1.3.4-2 and 5.2A.3.1.1.3.4-3 for the specified SNR including test tolerances for the combination selected following the test rules outlined in 5.1.1.5.2-2.

**Table 5.2A.3.1.1.3.4-1: Test requirements for FDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.1-9.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5]+TT |
| 10 | R.PDSCH.1-2.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 15 | R.PDSCH.1-9.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 20 | R.PDSCH.1-9.3 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 25 | R.PDSCH.1-9.4 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 30 | R.PDSCH.1-9.5 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 40 | R.PDSCH.1-10.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 50 | R.PDSCH.1-10.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.9] +TT |

**Table 5.2A.3.1.1.3.4-2: Test requirements for TDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-2.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 10 | R.PDSCH.1-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 15 | R.PDSCH.1-2.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 20 | R.PDSCH.1-2.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 25 | R.PDSCH.1-2.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] +TT |
| 30 | R.PDSCH.1-3.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 40 | R.PDSCH.1-3.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] +TT |
| 50 | R.PDSCH.1-3.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.0] +TT |

**Table 5.2A.3.1.1.3.4-3: Test requirements for TDD 30 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.2-13.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 10 | R.PDSCH.2-13.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 15 | R.PDSCH.2-13.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 20 | R.PDSCH.2-13.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 25 | R.PDSCH.2-13.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 30 | R.PDSCH.2-14.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 40 | R.PDSCH.2-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 50 | R.PDSCH.2-14.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.9] +TT |
| 60 | R.PDSCH.2-14.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] +TT |
| 80 | R.PDSCH.2-14.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.1] +TT |
| 90 | R.PDSCH.2-14.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.0] +TT |
| 100 | R.PDSCH.2-15.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.3] +TT |

#### 5.2A.3.1.2 2Rx Normal PDSCH Demodulation Performance for CA (3DL CA)

FFS

#### 5.2A.3.1.3 2Rx Normal PDSCH Demodulation Performance for CA (4DL CA)

FFS

## 5.2A.3A 2Rx-4RX requirements

### 5.2A.3A.1 Requirements for 2Rx-4RX normal PDSCH

#### 5.2A.3A.1.0 Minimum conformance requirements for 2Rx-4RX normal PDSCH

For UE supporting a combination of 2Rx and 4Rx antenna port per component carrier,

The 2Rx requirements are defined in Table 5.2A.2.1.0-4 based on the single carrier requirements for different SCSs and different bandwidth specified in Table 5.2A.2.1.0-1 to Table 5.2A.2.1.0-1.

The 4Rx requirements are defined in Table 5.2A.3.1.0-4 based on the single carrier requirements for different SCSs and different bandwidth specified in Table 5.2A.3.1.0-1 to Table 5.2A.3.1.0-1.

Parameters are specified in Table 5.2A-1 ~ Table 5.2A-3 and the downlink physical channel setup according to Annex C.3.1.

#### 5.2A.3A.1.1 2Rx-4Rx Normal PDSCH Demodulation Performance for CA (2DL CA)

Editor's note: Following aspects in this clause are incomplete

- Test requirement SNR is in square bracket

##### 5.2A.3A.1.1.1 Test Purpose

To verify the PDSCH mapping Type A normal performance under combination of 2 and 4 receive antenna conditions for multiple CA configurations and with different channel models, MCSs and number of MIMO layers for a specified downlink Reference Measurement Channel (RMC) to achieve a certain throughput per CC.

#### 5.2A.3A.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 2DL CA and 4Rx antenna ports on some of the CC.

#### 5.2A.3A.1.1.3 Test description

##### 5.2A.3A.1.1.3.1 Initial conditions

Same initial conditions as specified in clause 5.2A.2.1.1.3.1 with the following exception

Channel BW to be tested: largest aggregated bandwidth combination as per Table 5.1.1.5.2-2.

CA capability to be tested: As per table 5.1.1.5.2-1

**Table 5.2A.3.1.1.3.1-1: Test point selection table**

| Test number | CA duplex mode | Configuration |
|---|---|---|
| 1 | FDD 15 kHz + FDD 15 kHz | For 2Rx CC, As defined in Table 5.2A.2.1.0-1<br>For 4Rx CC, as defined in Table 5.2A.3.1.0-1 |
| 2 | TDD 30 kHz + TDD 30 kHz | For 2Rx CC, As defined in Table 5.2A.2.1.0-3<br>For 4Rx CC, as defined in Table 5.2A.3.1.0-3 |
| 3 | FDD 15 kHz + TDD 30 kHz | For 2Rx CC, As defined in Table 5.2A.2.1.0-1 and Table 5.2A.2.1.0-3 per CC<br>For 4Rx CC, as defined in Table 5.2A.3.1.0-1 and Table 5.2A.3.1.0-3 per CC |
| 4 (note 2) | FDD 15 kHz + TDD 15 kHz | For 2Rx CC, As defined in Table 5.2A.2.1.0-1 and Table 5.2A.2.1.0-2 per CC<br>For 4Rx CC, as defined in Table 5.2A.3.1.0-1 and Table 5.2A.3.1.0-2 per CC |
| 5 (note 3) | TDD 15 kHz + TDD 30 kHz | For 2Rx CC, As defined in Table 5.2A.2.1.0-2 and Table 5.2A.2.1.0-3 per CC<br>For 4Rx CC, as defined in Table 5.2A.3.1.0-2 and Table 5.2A.3.1.0-3 per CC |
| Note 1: For each test point, select any one of the CA configurations which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate based on the equation $DataRate = 10^{-3}\sum_{j=1}^{J} TBS_j 2^{\mu_j}$<br>Note 2: Test point 4 can be skipped if test point 3 is verified.<br>Note 3: Test point 5 can be skipped if test point 3 or 4 is verified. | | |

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.x and A.3.1.7.y for TE diagram for 2Rx and 4Rx CC respectively and clause A.3.2.6 for UE diagram

##### 5.2A.3A.1.1.3.2 Test procedure

Same test procedure as specified in clause 5.2A.2.1.1.3.2 for 2Rx CC and with the following exception for 4Rx CC

Instead of Table 5.2A.2.1.0-4 → 5.2A.3.1.0-4 for 4Rx CC

##### 5.2A.3.1.1.3.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.2A.3.1.1.3.4 Test Requirement

Tables 5.2A.3A.1.1.3.4-1, 5.2A.3A.1.1.3.4-2, 5.2A.3A.1.1.3.4-3, 5.2A.3A.1.1.3.4-4, 5.2A.3A.1.1.3.4-5, 5.2A.3A.1.1.3.4-6 define the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A 3.2.1 and A.3.2.2 for each component carrier for throughput test point combination shall meet or exceed the specified value in Table 5.2A.3A.1.1.3.4-1, 5.2A.3A.1.1.3.4-2, 5.2A.3A.1.1.3.4-3, 5.2A.3A.1.1.3.4-4, 5.2A.3A.1.1.3.4-5,

5.2A.3A.1.1.3.4-6 for the specified SNR including test tolerances for the combination selected following the test rules outlined in 5.1.1.5.2-2.

**Table 5.2A.3A.1.1.3.4-1: Test requirements for FDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-9.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 10 | R.PDSCH.1-2.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 15 | R.PDSCH.1-9.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.6 |
| 20 | R.PDSCH.1-9.3 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.8 |
| 25 | R.PDSCH.1-9.4 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.0 |
| 30 | R.PDSCH.1-9.5 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 13.8 |
| 40 | R.PDSCH.1-10.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.0 |
| 50 | R.PDSCH.1-10.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.4 |

**Table 5.2A.3A.1.1.3.4-2: Test requirements for TDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-2.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 10 | R.PDSCH.1-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.8 |
| 15 | R.PDSCH.1-2.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.8 |
| 20 | R.PDSCH.1-2.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.9 |
| 25 | R.PDSCH.1-2.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.0 |
| 30 | R.PDSCH.1-3.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.9 |
| 40 | R.PDSCH.1-3.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.2 |
| 50 | R.PDSCH.1-3.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.5 |

**Table 5.2A.3A.1.1.3.4-3: Test requirements for TDD 30 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.2-13.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 10 | R.PDSCH.2-13.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 15 | R.PDSCH.2-13.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.6 |
| 20 | R.PDSCH.2-13.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |
| 25 | R.PDSCH.2-13.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |
| 30 | R.PDSCH.2-14.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.7 |
| 40 | R.PDSCH.2-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 14.9 |
| 50 | R.PDSCH.2-14.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.1 |
| 60 | R.PDSCH.2-14.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.0 |
| 80 | R.PDSCH.2-14.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.5 |
| 90 | R.PDSCH.2-14.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.3 |
| 100 | R.PDSCH.2-15.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x2, ULA Low | 70 | 15.7 |

**Table 5.2A.3A.1.1.3.4-4: Test requirements for FDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.1-9.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 10 | R.PDSCH.1-2.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 15 | R.PDSCH.1-9.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 20 | R.PDSCH.1-9.3 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 25 | R.PDSCH.1-9.4 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 30 | R.PDSCH.1-9.5 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 40 | R.PDSCH.1-10.1 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 50 | R.PDSCH.1-10.2 FDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.9] +TT |

**Table 5.2A.3A.1.1.3.4-5: Test requirements for TDD 15 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |

| | | | | | | |
|---|---|---|---|---|---|---|
| 5 | R.PDSCH.1-2.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 10 | R.PDSCH.1-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 15 | R.PDSCH.1-2.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 20 | R.PDSCH.1-2.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 25 | R.PDSCH.1-2.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] +TT |
| 30 | R.PDSCH.1-3.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 40 | R.PDSCH.1-3.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] +TT |
| 50 | R.PDSCH.1-3.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.0] +TT |

**Table 5.2A.3A.1.1.3.4-6: Test requirements for TDD 30 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 5 | R.PDSCH.2-13.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 10 | R.PDSCH.2-13.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 15 | R.PDSCH.2-13.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.5] +TT |
| 20 | R.PDSCH.2-13.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 25 | R.PDSCH.2-13.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 30 | R.PDSCH.2-14.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.6] +TT |
| 40 | R.PDSCH.2-2.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.7] +TT |
| 50 | R.PDSCH.2-14.2 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.9] +TT |
| 60 | R.PDSCH.2-14.3 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [8.8] +TT |
| 80 | R.PDSCH.2-14.4 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.1] +TT |
| 90 | R.PDSCH.2-14.5 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.0] +TT |
| 100 | R.PDSCH.2-15.1 TDD | 16QAM, 0.48 | TDLA30-10 | 2x4, ULA Low | 70 | [9.3] +TT |

### 5.2A.3A.1.2 2Rx-4Rx Normal PDSCH Demodulation Performance for CA (3DL CA)

FFS

### 5.2A.3A.1.3 2Rx-4Rx Normal PDSCH Demodulation Performance for CA (4DL CA)

FFS

## 5.3 PDCCH demodulation requirements

The receiver characteristics of the PDCCH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg).

The parameters specified in Table 5.3-1 are valid for all PDCCH tests unless otherwise stated.

**Table 5.3-1: Common test Parameters**

| Parameter | | | Unit | Value |
|---|---|---|---|---|
| Carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 1) | | | 0 |
| DL BWP configuration #1 | Cyclic prefix | | | Normal |
| | RB offset | | RBs | 0 |
| Common serving cell parameters | Physical Cell ID | | | 0 |
| | SSB position in burst | | | 1 |
| | SSB periodicity | | ms | 20 |
| PDCCH configuration | Slots for PDCCH monitoring | | | Each slot |
| | Number of PDCCH candidates | | | 1 |
| | Frequency domain resource allocation for CORESET | | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | | TCI state #1 |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | | 0 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | | CSI-RS resource 1: 4<br>CSI-RS resource 2: 8<br>CSI-RS resource 3: 4<br>CSI-RS resource 4: 8 |
| | Number of CSI-RS ports ($X$) | | | 1 |
| | CDM Type | | | No CDM |
| | Density ($\rho$) | | | 3 |
| | CSI-RS periodicity | | Slots | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | Slots | 15 kHz SCS:<br>10 for CSI-RS resource 1 and 2<br>11 for CSI-RS resource 3 and 4<br><br>30 kHz SCS:<br>20 for CSI-RS resource 1 and 2<br>21 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type D |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type D |

| | | |
|---|---|---|
| PDCCH & PDCCH DMRS Precoding configuration | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination with REG bundling granularity for number of Tx larger than 1 |
| Physical signals, channels mapping and precoding | | As specified in Annex B.4.1 |
| Symbols for all unused REs | | OP.1 FDD as defined in Annex A.5.1.1<br>OP.1 TDD as defined in Annex A.5.2.1 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | 2 for FDD.<br>For TDD, specific to each TDD UL-DL pattern and as defined in Annex A.1.2. |
| Note 1: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing. | | |

The normative reference for this requirement is TS 38.101-4 [2] clause 5.3.

## 5.3.1 1RX requirements

(Void)

## 5.3.2 2RX requirements

### 5.3.2.1 FDD

The parameters specified in Table 5.3.2.1-1 are valid for all FDD tests unless otherwise stated.

**Table 5.3.2.1-1: Test Parameters**

| Parameter | Unit | 1 Tx Antenna | 2 Tx Antenna |
|---|---|---|---|
| CCE to REG mapping type | | nonInterleaved | |
| REG bundle size | | 6 | |
| Shift index | | 0 | |

#### 5.3.2.1.1 2Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA

##### 5.3.2.1.1.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.2.1.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

##### 5.3.2.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 5.3.2.1.1.3 Minimum conformance requirements

For the parameters specified in Table 5.3.2.1-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.1.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.2.1.1.3-1: Minimum performance for 1 Tx PDCCH with 15 kHz SCS**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 24 | 2 | 2 | R.PDCCH.1-2.1 FDD | TDLA30-10 | 1x2 Low | 1 | 8.1 |
| 2 | 10 MHz | 24 | 2 | 2 | R.PDCCH.1-2.3 FDD | TDLC300-100 | 1x2 Low | 1 | 8.2 |
| 3 | 10 MHz | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x2 Low | 1 | 5.5 |
| 4 | 10 MHz | 48 | 1 | 4 | R.PDCCH.1-1.1 FDD | TDLA30-10 | 1x2 Low | 1 | 4.4 |
| 5 | 10 MHz | 48 | 2 | 16 | R.PDCCH.1-2.6 FDD | TDLA30-10 | 1x2 Low | 1 | -2.1 |

The normative reference for this requirement is TS 38.101-4 [2] clause 5.3.

#### 5.3.2.1.1.4 Test description

##### 5.3.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A in Figure A.3.1.7.2 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1 and Table 5.3.2.1.1-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.1.4.3.

##### 5.3.2.1.1.4.2 Test procedure

1. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.2.1.1.3-1. The details of PDCCH are specified in Table 5.3.1, Table 5.3.2.1-1, Table 5.3.2.1.1.3-1 respectively. The details of PDSCH are specified in Table A.3.3.1.1-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.2.1.1.3-1 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.2.1.1.4.4-1, pass the UE. Otherwise fail the UE.

4. Repeat steps from 1 to 3 for each subtest in Table 5.3.2.1.1-1 as appropriate.

#### 5.3.2.1.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.3.2.1.1.4.3.1 Message exceptions for SA

**Table 5.3.2.1.1.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 00000000 00000000 00000000 00000 | CORESET to use the least significant 48 RBs of the BWP Test 3, 4, 5 | |
| | 11110000 00000000 00000000 00000000 00000 | CORESET to use the least significant 24 RBs of the BWP Test 1, 2 | |
| Duration | 2 | SearchSpace duration of 2 symbols Test 1, 2, 3, 5 | |
| | 1 | SearchSpace duration of 1 symbol Test 4 | |
| } | | | |

**Table 5.3.2.1.1.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel1 | n0 | | |
| aggregationLevel2 | n1 | AL2 | Test 1, Test 2 |
| aggregationLevel4 | n1 | AL4 | Test 3, Test 4 |
| aggregationLevel8 | n0 | | |
| aggregationLevel16 | n1 | AL16 | Test 5 |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 for tests 2 and 3 | Long_DCI |
| | formats0-0-And-1-0 | DCI Format 1_0 for tests 1, 4, 5 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.1.4.3.1-3: Void**

**Table 5.3.2.1.1.4.3.1-3A: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.2-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType1 | | Test 1,<br>Test 4,<br>Test 5 |
| } | | | |

#### 5.3.2.1.1.4.3.2 Message exceptions for NSA

Same as 5.3.2.1.1.4.3.1

#### 5.3.2.1.1.4.4 Test requirement

Table 5.3.2.1.1.4.4-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.1.1.4.4-1.

**Table 5.3.2.1.1.4.4-1: Test Requirement for 1Tx PDCCH with 15 kHz SCS**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 24 | 2 | 2 | R.PDCCH.1-2.1 FDD | TDLA30-10 | 1x2 Low | 1 | 9.0 |
| 2 | 10 MHz | 24 | 2 | 2 | R.PDCCH.1-2.3 FDD | TDLC300-100 | 1x2 Low | 1 | 9.1 |
| 3 | 10 MHz | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x2 Low | 1 | 6.4 |
| 4 | 10 MHz | 48 | 1 | 4 | R.PDCCH.1-1.1 FDD | TDLA30-10 | 1x2 Low | 1 | 5.3 |
| 5 | 10MHz | 48 | 2 | 16 | R.PDCCH.1-2.6 FDD | TDLA30-10 | 1x2 Low | 1 | -1.2 |

### 5.3.2.1.2 2Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA

#### 5.3.2.1.2.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.2.1.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

#### 5.3.2.1.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 5.3.2.1.2.3 Minimum conformance requirements

For the parameters specified in Table 5.3.2.1-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.1.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.2.1.2.3-1: Minimum performance for 2 Tx PDCCH with 15 kHz SCS**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 24 | 2 | 4 | R.PDCCH.1-2.2 FDD | TDLC300-100 | 2x2 Low | 1 | 2.0 |
| 2 | 10 MHz | 48 | 2 | 8 | R.PDCCH.1-2.5 FDD | TDLC300-100 | 2x2 Low | 1 | -1.3 |
| 3 | 10 MHz | 48 | 1 | 8 | R.PDCCH.1-1.3 FDD | TDLA30-10 | 2x2 Low | 1 | -0.2 |

The normative reference for this requirement is TS 38.101-4 [2] clause 5.3.

### 5.3.2.1.2.4 Test description

#### 5.3.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.2.1-1 and Table 5.3.2.1.2.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.2.1.2.4.3.

#### 5.3.2.1.2.4.2 Test procedure

1. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channelfor C_RNTI to transmit the DL RMC according to Table 5.3.2.1.2.3-1. The details of PDCCH are specified in Table 5.3.1, Table 5.3.2.1-1, Table 5.3.2.1.2.3-1 respectively. The details of PDSCH are specified in Table A.3.3.1.1-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.2.1.2.3-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.2.1.2.4.4-1, pass the UE. Otherwise fail the UE.

4. Repeat steps from 1 to 3 for each subtest in Table 5.3.2.1.2.3-1 as appropriate.

#### 5.3.2.1.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.3.2.1.2.4.3.1 Message exceptions for SA

**Table 5.3.2.1.2.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 00000000 00000000 00000000 00000 | CORESET to use the least significant 48 RBs of the BWP Test 2, 3 | |
| | 11110000 00000000 00000000 00000000 00000 | CORESET to use the least significant 24 RBs of the BWP Test 1 | |
| Duration | 2 | SearchSpace duration of 2 symbols Test 1, 2 | |
| | 1 | SearchSpace duration of 1 symbol Test 3 | |
| } | | | |

**Table 5.3.2.1.2.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel1 | n0 | | |
| aggregationLevel2 | n0 | | |
| aggregationLevel4 | n1 | AL4 | Test 1 |
| aggregationLevel8 | n1 | AL8 | Test 2, 3 |
| aggregationLevel16 | n0 | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 for tests 2 and 3 | Long_DCI |
| | formats0-0-And-1-0 | DCI Format 1_0 for test 1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.2.4.3.1-3: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType1 | | Test 1 |
| } | | | |

#### 5.3.2.1.2.4.3.2 Message exceptions for NSA

Same as 5.3.2.1.2.4.3.1

### 5.3.2.1.2.4.4 Test requirement

Table 5.3.2.1.2.4.4-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.1.2.4.4-1.

**Table 5.3.2.1.2.4.4-1: Test Requirements for 2 Tx PDCCH with 15 kHz SCS**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 MHz | 24 | 2 | 4 | R.PDCCH.1-2.2 FDD | TDLC300-100 | 2x2 Low | 1 | 3.0 |
| 2 | 10 MHz | 48 | 2 | 8 | R.PDCCH.1-2.5 FDD | TDLC300-100 | 2x2 Low | 1 | -0.3 |
| 3 | 10 MHz | 48 | 1 | 8 | R.PDCCH.1-1.3 FDD | TDLA30-10 | 2x2 Low | 1 | 0.8 |

## 5.3.2.1.3 2Rx FDD FR1 PDCCH 1 Tx antenna performance for power saving

### 5.3.2.1.3.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.2.1.3.3-2. The downlink physical setup is in accordance with Annex C.2.1.

### 5.3.2.1.3.2 Test applicability

This test applies to all types of NR UE release 16 and forward that supporting DRX adaptation

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC that supporting DRX adaptation.

### 5.3.2.1.3.3 Minimum conformance requirements

The parameters specified in Table 5.3.2.1.3.3-1 are valid for FDD test unless otherwise stated.

**Table 5.3.2.1.3.3-1: Test Parameters**

| Parameter | | Unit | 1 Tx Antenna |
|---|---|---|---|
| CCE to REG mapping type | | | nonInterleaved |
| REG bundle size | | | 6 |
| Shift Index | | | 0 |
| DRX cycle | | ms | 10 |
| ps-WakeUp-r16 | | | absent |
| Wake-up indication bit in DCI format 2_6 | | | 1 |
| PDCCH DCI format 2_6 configuration | PS-offset | | $(T_{minimumTimeGap} + 1)/2^{\mu}/0.125$ |
| | Number of PDCCH candidates | | 1 |
| | Frequency domain resource allocation for CORESET | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | TCI state #1 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot during DRX-on period |
| | | | |
| Note: $T_{minimumTimeGap}$ is signalled as a part of *drx-Adaptation-r16* UE capability. | | | |

For the parameters specified in Table 5.3.2.1.3.3-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.1.3.3-2. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.2.1.3.3-2: Minimum performance for PDCCH with 15 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x2 Low | 1 | 5.5 |
| | | | 2 | 8 | R.PDCCH.1-2.7 FDD | | | | |

The normative reference for this requirement is TS 38.101-4 [2] clause 5.3.2.1.3.

### 5.3.2.1.3.4 Test description

#### 5.3.2.1.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.2.1-1, Table 5.3.2.1.3.3-1 and Table 5.3.2.1.3.3-2 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.2.1.3.4.3.

#### 5.3.2.1.3.4.2 Test procedure

1. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.2.1.3.3-1. The details of PDCCH are specified in Table 5.3.1, Table 5.3.2.1-1, Table 5.3.2.1.3.3-1 and Table 5.3.2.1.3.3-2 respectively. The details of PDSCH are specified in Table A.3.3.1.1-3. The SS sends downlink MAC padding bits on the DL RMC. During the test the UE shall monitor the DCI format 2_6 PDCCH in DRX off state and decide whether to receive the following PDCCH in DRX on period.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.2.1.3.3-2 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.2.1.3.4.4-1, pass the UE. Otherwise fail the UE.

4. Repeat steps from 1 to 3 for each subtest in Table 5.3.2.1.3.3-2 as appropriate.

#### 5.3.2.1.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.3.2.1.3.4.3.1 Message exceptions for SA

**Table 5.3.2.1.3.4.3.1-1: DRX-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-56 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DRX-Config ::= SEQUENCE { | | | |
| drx-onDurationTimer CHOICE { | | | |
| milliSeconds | ms1 | | |
| } | | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | 0 | | |
| drx-HARQ-RTT-TimerUL | 0 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms10 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-2: DCP-Config**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-106 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| dcp-Config-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| ps-Offset-r16 | 40 | | |
| sizeDCI-2-6-r16 | 2 | | |
| ps-PositionDCI-2-6-r16 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-3: PDCCH-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet ::= SEQUENCE { | 2 entry | | |
| ControlResourceSet[1] | ControlResourceSet1 | | |
| ControlResourceSet[2] | ControlResourceSet2 | | |
| } | | | |
| searchSpacesToAddModList SEQUENCE(SIZE (1..10)) OF SearchSpace ::= SEQUENCE { | 1 entry | | |
| SearchSpace[1] | SearchSpace | | |
| } | | | |
| searchSpacesToAddModListExt-r16 SEQUENCE(SIZE (1..10)) OF SearchSpace { | 1 entry | | |
| searchSpaceExt-r16[1] | SearchSpaceExt | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-4: PDCCH-ControlResourceSet1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n2 | | |
| interleaverSize | n3 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-5: PDCCH-ControlResourceSet2**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n2 | | |
| interleaverSize | n3 | | |
| } | | | |
| } | | | |
| tci-StatesPDCCH-ToAddList { | 1 | TCI State #1 | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-6: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| searchSpaceId | 2 | SearchSpaceId with condition USS | |
| controlResourceSetId | 1 | ControlResourceSetId | |
| monitoringSlotPeriodicityAndOffset CHOICE { | | | |
| sl1 | NULL | | |
| } | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel4 | n1 | AL4 | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-7: PDCCH Search Space Ext**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpaceExt-r16 ::= SEQUENCE { | | | |
| controlResourceSetId-r16 | 2 | | |
| searchSpaceType-r16 SEQUENCE { | | | |
| common SEQUENCE { | | | |
| dci-Format2-6-r16 SEQUENCE { | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.1.3.4.3.1-8: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| TCI-State[1] | TCI-StateId 0 | | |
| qcl-type1 { | | Type 1 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type C | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| TCI-State[2] | TCI-StateId 1 | Type 1 QCL information | |
| qcl-type1 { | | | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | CSI-RS # 0 | |
| Qcl-Type | Type A | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| } | | | |
| } | | | |

### 5.3.2.1.3.4.3.2 Message exceptions for NSA

Same as 5.3.2.1.3.4.3.1

### 5.3.2.1.3.4.4 Test requirement

Table 5.3.2.1.3.4.4-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.1.3.4.4-1.

**Table 5.3.2.1.3.4.4-1: Test Requirements for PDCCH with 15 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x2 Low | 1 | 6.4 |
| | | | 2 | 8 | R.PDCCH.1-2.7 FDD | | | | |

### 5.3.2.2 TDD

The parameters specified in Table 5.3.2.2-1 are valid for all TDD tests unless otherwise stated.

**Table 5.3.2.2-1: Test Parameters**

| Parameter | Unit | 1 Tx Antenna | 2 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | FR1.30-1 | |
| CCE to REG mapping type | | Test 3: non-interleaved<br>Other tests: interleaved | interleaved |
| Interleaver size | | 3 | |
| REG bundle size | | Test 3: 6<br>Other tests: 2 | 6 |
| Shift Index | | 0 | |

#### 5.3.2.2.1 2Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA

##### 5.3.2.2.1.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.2.2.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

##### 5.3.2.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 5.3.2.2.1.3 Minimum conformance requirements

For the parameters specified in Table 5.3.2.2-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.2.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.2.2.1.3-1: Minimum performance for PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 2 | R.PDCCH.2-1.1 TDD | TDLA30-10 | 1x2 Low | 1 | 7.0 |
| 2 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x2 Low | 1 | 3.0 |
| 3 | 40 | 48 | 2 | 16 | R.PDCCH.2-2.1 TDD | TDLC300-100 | 1x2 Low | 1 | -3.8 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.2.2.

##### 5.3.2.2.1.4 Test description

###### 5.3.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.2.2-1 and Table 5.3.2.2.1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.2.2.1.4.3.

#### 5.3.2.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.2.2.1.3-1. The details of PDCCH are specified in Table 5.3-1, Table 5.3.2.2-1 and Table 5.3.2.2.1.3-1 respectively. The details of PDSCH are specified in Table A.3.3.2.2-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.2.2.1.3-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.2.2.1.5-1, pass the UE. Otherwise fail the UE.
4. Repeat steps from 1 to 3 for each subtest in Table 5.3.2.2.1.3-1 as appropriate.

#### 5.3.2.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

##### 5.3.2.2.1.4.3.1 Message exceptions for SA

**Table 5.3.2.2.1.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 00000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 48 RBs of the BWP Test 3 | |
| | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP Test 1, 2 | |

| | | | |
|---|---|---|---|
| Duration | 2 | SearchSpace duration of 2 symbols Test 3 | |
| | 1 | SearchSpace duration of 1 symbol Test 1, 2 | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | Test 1, Test 2 |
| reg-BundleSize | n2 | | 1 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| nonInterleaved | null | | Test 3 |
| } | | | |
| } | | | |

**Table 5.3.2.2.1.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel2 | n1 | AL2 | Test 1 |
| aggregationLevel4 | n1 | AL4 | Test 2 |
| aggregationLevel16 | n1 | AL16 | Test 3 |
| } | | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 for test 2 | Long_DCI |
| | formats0-0-And-1-0 | DCI Format 1_0 for tests 1 and 3 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.2.1.4.3.1-3: Void**

**Table 5.3.2.2.1.4.3.1-3A: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.2-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType1 | | Test 1, Test 3 |
| } | | | |

### 5.3.2.2.1.4.3.2 Message exceptions for NSA

Same as 5.3.2.2.1.4.3.1.

### 5.3.2.2.1.5 Test requirement

Table 5.3.2.2.1.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.2.1.5-1.

**Table 5.3.2.2.1.5-1: Test Requirement for 1Tx PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 2 | R.PDCCH.2-1.1 TDD | TDLA30-10 | 1x2 Low | 1 | 7.9 |
| 2 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x2 Low | 1 | 3.9 |
| 3 | 40 | 48 | 2 | 16 | R.PDCCH.2-2.1 TDD | TDLC300-100 | 1x2 Low | 1 | -2.9 |

## 5.3.2.2.2 2Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA

### 5.3.2.2.2.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.2.2.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

### 5.3.2.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 5.3.2.2.2.3 Minimum conformance requirements

For the parameters specified in Table 5.3.2.2-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.2.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.2.2.2.3-1: Minimum performance for PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 90 | 1 | 8 | R.PDCCH.2-1.3 TDD | TDLC300-100 | 2x2 Low | 1 | -1.2 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.2.2.

### 5.3.2.2.2.4 Test description

#### 5.3.2.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.2.2-1 and Table 5.3.2.2.2.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.2.2.2.4.3.

#### 5.3.2.2.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH with DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.3.2.2.2.3-1. The details of PDCCH are specified in Table 5.3-1, Table 5.3.2.2-1 and Table 5.3.2.2.2.3-1 respectively. The details of PDSCH are specified in Table A.3.3.2.2-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.2.2.2.3-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.2.2.2.5-1, pass the UE. Otherwise fail the UE.

#### 5.3.2.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.3.2.2.2.4.3.1 Message exceptions for SA

**Table 5.3.2.2.2.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 11111110 00000000 00000000 00000000 00000 | CORESET to use the least significant 90 RBs of the BWP Test 1 | |
| Duration | 1 | SearchSpace duration of 1 symbols Test 1 | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n6 | | 2 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| } | | | |

| | | | |
|---|---|---|---|
| } | | | |

**Table 5.3.2.2.2.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel8 | n1 | AL8 | Test 1 |
| } | | | |
| } | | | |

**Table 5.3.2.2.2.4.3.1-3: Void**

#### 5.3.2.2.2.4.3.2 Message exceptions for NSA

Same as 5.3.2.2.2.4.3.1.

#### 5.3.2.2.2.5 Test requirement

Table 5.3.2.2.2.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.2.2.5-1.

**Table 5.3.2.2.2.5-1: Test Requirement for 2Tx PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 90 | 1 | 8 | R.PDCCH.2-1.3 TDD | TDLC300-100 | 2x2 Low | 1 | -0.2 |

### 5.3.2.2.3 2Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving

#### 5.3.2.2.3.1 Test Purpose

This test verifies the demodulation performance of *DCI format 2_6* PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.2.2.3.3-2 after receipt wake-up indication in the *DCI format 2_6* PDCCH in DRX off state. The downlink physical setup is in accordance with Annex C.2.1.

#### 5.3.2.2.3.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting Long DRX Cycle and DRX adaptation.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and Long DRX Cycle and DRX adaptation.

#### 5.3.2.2.3.3 Minimum conformance requirements

During the test the UE shall monitor the *DCI format 2_6* PDCCH in DRX off state and decide whether to receive the following PDCCH in DRX on period.

The parameters specified in Table 5.3.2.2.3.3-1 are valid for all TDD tests for power saving unless otherwise stated.

**Table 5.3.2.2.3.3-1: Test Parameters**

| Parameter | | Unit | 1 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | | FR1.30-1 |
| CCE to REG mapping type | | | interleaved |
| Interleaver size | | | 3 |
| REG bundle size | | | 2 |
| Shift Index | | | 0 |
| DRX cycle | | ms | 10 |
| ps-WakeUp-r16 | | | absent |
| Wake-up indication bit in DCI format 2_6 | | | 1 |
| PDCCH DCI format 2_6 configuration | PS-offset | | $(T_{minimumTimeGap}+1)/2^{\mu}/0.125$ |
| | Number of PDCCH candidates | | 1 |
| | Frequency domain resource allocation for CORESET | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | TCI state #1 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot during DRX-on period |
| | | | |
| Note: $T_{minimumTimeGap}$ is signalled as a part of *drx-Adaptation-r16* UE capability. | | | |

For the parameters specified in Table 5.3.2.2.3.3-1, the average probability of a missed downlink scheduling grant (Pm-dsg) observed on PDCCH during DRX on shall be below the specified value in Table 5.3.2.2.3.3-2. The downlink physical setup is in accordance with Annex C.3.1.

**Table 5.3.2.2.3.3-2: Minimum performance with 30 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x2 Low | 1 | 3.0 |
| | | | | 8 | R.PDCCH.2-1.4 TDD | | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.2.2.

### 5.3.2.2.3.4 Test description

#### 5.3.2.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of DRX, DCP, PDSCH and PDCCH before measurement are specified in 5.3.2.2.3.4.3.1and Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.2.2-1 and Table 5.3.2.2.3.3-2 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.2.2.3.4.3.

#### 5.3.2.2.3.4.2 Test procedure

1. SS transmits PDCCH with DCI format2_6 as specified in PDCCH Reference Channel for C_RNTI within DRX off state. The Wake-up indication bit in PDCCH is set to 1.

2. SS transmits PDSCH via PDCCH with DCI format1_1 as specified in PDCCH Reference Channel R.PDCCH.5-1.2 TDD for C_RNTI to transmit the DL RMC according to Table 5.3.2.2.3.3-2 in DRX on period. The details of PDCCH are specified in Table 5.3-1, Table 5.3.2.2-1 and Table 5.3.2.2.3.3-2 respectively. The details of PDSCH are specified in Table A.3.3.2.2-3. The SS sends downlink MAC padding bits on the DL RMC.

3. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.2.2.3.3-2 as appropriate.

4. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.2.2.3.5-1, pass the UE. Otherwise fail the UE.

5. Repeat steps from 2 to 4 for each subtest in Table 5.3.2.2.3.3-1 as appropriate.

#### 5.3.2.2.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

##### 5.3.2.2.3.4.3.1 Message exceptions for SA

**Table 5.3.2.2.3.4.3.1-1: DRX-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-56 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DRX-Config ::= SEQUENCE { | | | |
| drx-onDurationTimer CHOICE { | | | |
| milliSeconds | ms1 | | |
| } | | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | 0 | | |
| drx-HARQ-RTT-TimerUL | 0 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms10 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-2: DCP-Config**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-106 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| dcp-Config-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| ps-Offset-r16 | 40 | | |
| sizeDCI-2-6-r16 | 2 | | |
| ps-PositionDCI-2-6-r16 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-3: PDCCH-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet ::= SEQUENCE { | 2 entry | | |
| ControlResourceSet[1] | ControlResourceSet1 | | |
| ControlResourceSet[2] | ControlResourceSet2 | | |
| } | | | |
| searchSpacesToAddModList SEQUENCE (SIZE (1..10)) OF SearchSpace ::= SEQUENCE { | 1 entry | | |
| SearchSpace[1] | SearchSpace | | |
| } | | | |
| searchSpacesToAddModListExt-r16 SEQUENCE(SIZE (1..10)) OF SearchSpace { | 1 entry | | |
| searchSpaceExt-r16[1] | SearchSpaceExt | | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-4: PDCCH-ControlResourceSet1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n2 | | 1 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-5: PDCCH-ControlResourceSet2**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n2 | | 1 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| } | | | |
| tci-StatesPDCCH-ToAddList { | 1 | TCI State #1 | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-6: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| searchSpaceId | 2 | SearchSpaceId with condition USS | USS |
| controlResourceSetId | 1 | ControlResourceSetId | |
| monitoringSlotPeriodicityAndOffset CHOICE { | | | |
| sl1 | NULL | | |
| } | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel4 | n1 | AL4 | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-7: PDCCH *Search Space Ext***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpaceExt-r16 ::= SEQUENCE { | | | |
| controlResourceSetId-r16 | 2 | | |
| searchSpaceType-r16 SEQUENCE { | | | |
| common SEQUENCE { | | | |
| dci-Format2-6-r16 SEQUENCE { | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.2.2.3.4.3.1-8: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| TCI-State[1] | TCI-StateId 0 | | |
| qcl-type1 { | | Type 1 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type C | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| TCI-State[2] | TCI-StateId 1 | Type 1 QCL information | |
| qcl-type1 { | | | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | CSI-RS # 0 | |
| Qcl-Type | Type A | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| } | | | |
| } | | | |

5.3.2.2.3.4.3.2 Message exceptions for NSA

Same as 5.3.2.2.3.4.3.1.

5.3.2.2.3.5 Test requirement

Table 5.3.2.2.3.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.2.2.3.5-1.

**Table 5.3.2.2.3.5-1: Minimum performance with 30 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x2 Low | 1 | 3.9 |
| | | | | 8 | R.PDCCH.2-1.4 TDD | | | | |

## 5.3.3 4RX requirements

### 5.3.3.1 FDD

The parameters specified in Table 5.3.3.1-1 are valid for all FDD tests unless otherwise stated.

**Table 5.3.3.1-1: Test Parameters**

| Parameter | Unit | 1 Tx Antenna | 2 Tx Antenna |
|---|---|---|---|
| CCE to REG mapping type | | nonInterleaved | |
| REG bundle size | | 6 | |
| Shift index | | 0 | |

#### 5.3.3.1.1 4Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA

##### 5.3.3.1.1.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 4 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.3.1.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

##### 5.3.3.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

##### 5.3.3.1.1.3 Minimum conformance requirements

For the parameters specified in Table 5.3.3.1-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.1.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.3.1.1.3-1: Minimum performance for PDCCH with 15 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 24 | 2 | 2 | R.PDCCH.1-2.1 FDD | TDLA30-10 | 1x4 Low | 1 | 2.2 |
| 2 | 10 | 24 | 2 | 2 | R.PDCCH.1-2.3 FDD | TDLC300-100 | 1x4 Low | 1 | 2.7 |
| 3 | 10 | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x4 Low | 1 | 0.2 |
| 4 | 10 | 48 | 1 | 4 | R.PDCCH.1-1.1 FDD | TDLA30-10 | 1x4 Low | 1 | -0.4 |
| 5 | 10 | 48 | 2 | 16 | R.PDCCH.1-2.6 FDD | TDLA30-10 | 1x4 Medium A | 1 | -3.2 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.3.1.

##### 5.3.3.1.1.4 Test description

###### 5.3.3.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.3.1-1 and Table 5.3.3.1.1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.3.1.1.4.3.

#### 5.3.3.1.1.4.2 Test procedure

1. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.3.1.1.3-1. The details of PDCCH are specified in Table 5.3-1, Table 5.3.3.1-1 and Table 5.3.3.1.1.3-1 respectively. The details of PDSCH are specified in Table A.3.3.1.1-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.3.1.1.3-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.3.1.1.5-1, pass the UE. Otherwise fail the UE.
4. Repeat steps from 1 to 3 for each subtest in Table 5.3.3.1.1.3-1 as appropriate.

#### 5.3.3.1.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.3.3.1.1.4.3.1 Message exceptions for SA

**Table 5.3.3.1.1.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 00000000 00000000 00000000 00000 | CORESET to use the least significant 48 RBs of the BWP Test 3, 4, 5 | |
| | 11110000 00000000 00000000 00000000 00000 | CORESET to use the least significant 24 RBs of the BWP Test 1, 2 | |

| | | | |
|---|---|---|---|
| Duration | 2 | SearchSpace duration of 2 symbols Test 1, 2, 3, 5 | |
| | 1 | SearchSpace duration of 1 symbol Test 4 | |
| } | | | |

**Table 5.3.3.1.1.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel2 | n1 | AL2 | Test 1, Test 2 |
| aggregationLevel4 | n1 | AL4 | Test 3, Test 4 |
| aggregationLevel16 | n1 | AL16 | Test 5 |
| } | | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 for tests 2 and 3 | Long_DCI |
| | formats0-0-And-1-0 | DCI Format 1_0 for tests 1, 4, 5 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.1.4.3.1-3: Void**

**Table 5.3.3.1.1.4.3.1-3A: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.2-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType1 | | Test 1, Test 4, Test 5 |
| } | | | |

### 5.3.3.1.1.4.3.2 Message exceptions for NSA

Same as 5.3.3.1.1.4.3.1.

### 5.3.3.1.1.5 Test requirement

Table 5.3.3.1.1.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.1.1.5-1.

**Table 5.3.3.1.1.5-1: Test Requirement for 1Tx PDCCH with 15 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 24 | 2 | 2 | R.PDCCH.1-2.1 FDD | TDLA30-10 | 1x4 Low | 1 | 3.1 |
| 2 | 10 | 24 | 2 | 2 | R.PDCCH.1-2.3 FDD | TDLC300-100 | 1x4 Low | 1 | 3.6 |
| 3 | 10 | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x4 Low | 1 | 1.1 |
| 4 | 10 | 48 | 1 | 4 | R.PDCCH.1-1.1 FDD | TDLA30-10 | 1x4 Low | 1 | 0.5 |
| 5 | 10 | 48 | 2 | 16 | R.PDCCH.1-2.6 FDD | TDLA30-10 | 1x4 Medium A | 1 | -2.3 |

## 5.3.3.1.2 4Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA

### 5.3.3.1.2.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 4 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.3.1.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

### 5.3.3.1.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

### 5.3.3.1.2.3 Minimum conformance requirements

For the parameters specified in Table 5.3.3.1-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.1.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.3.1.2.3-1: Minimum performance for PDCCH with 15 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 24 | 2 | 4 | R.PDCCH.1-2.2 FDD | TDLC300-100 | 2x4 Low | 1 | -1.9 |
| 2 | 10 | 48 | 2 | 8 | R.PDCCH.1-2.5 FDD | TDLC300-100 | 2x4 Low | 1 | -4.5 |
| 3 | 10 | 48 | 1 | 4 | R.PDCCH.1-1.2 FDD | TDLA30-10 | 2x4 Low | 1 | -1.0 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.3.1.

### 5.3.3.1.2.4 Test description

#### 5.3.3.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.3.1-1 and Table 5.3.3.1.2.3-1 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On*)* for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.3.1.2.4.3.

#### 5.3.3.1.2.4.2 Test procedure

1. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.3.1.2.3-1. The details of PDCCH are specified in Table 5.3-1, Table 5.3.3.1-1 and Table 5.3.3.1.2.3-1 respectively. The details of PDSCH are specified in Table A.3.3.1.1-3. The SS sends downlink MAC padding bits on the DL RMC.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.3.1.2.3-1 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.3.1.2.5-1, pass the UE. Otherwise fail the UE.

4. Repeat steps from 1 to 3 for each subtest in Table 5.3.3.1.2.3-1 as appropriate.

#### 5.3.3.1.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

##### 5.3.3.1.2.4.3.1 Message exceptions for SA

**Table 5.3.3.1.2.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 00000000 00000000 00000000 00000 | CORESET to use the least significant 48 RBs of the BWP Test 2, 3 | |
| | 11110000 00000000 00000000 00000000 00000 | CORESET to use the least significant 24 RBs of the BWP Test 1 | |

| | | | |
|---|---|---|---|
| Duration | 2 | SearchSpace duration of 2 symbols Test 1, 2 | |
| | 1 | SearchSpace duration of 1 symbol Test3 | |
| } | | | |

**Table 5.3.3.1.2.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel4 | n1 | AL4 | Test 1, Test 3 |
| aggregationLevel8 | n1 | AL8 | Test 2 |
| } | | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 for tests 2 and 3 | Long_DCI |
| | formats0-0-And-1-0 | DCI Format 1_0 for test 1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.2.4.3.1-3: Void**

**Table 5.3.3.1.2.4.3.1-3A: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.2-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType1 | | Test 1 |
| } | | | |

### 5.3.3.1.2.4.3.2 Message exceptions for NSA

Same as 5.3.3.1.2.4.3.1.

### 5.3.3.1.2.5 Test requirement

Table 5.3.3.1.2.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.1.2.5-1.

**Table 5.3.3.1.2.5-1: Test Requirement for 2Tx PDCCH with 15 kHz SCS**

| Test number | Band width (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 10 | 24 | 2 | 4 | R.PDCCH.1-2.2 FDD | TDLC300-100 | 2x4 Low | 1 | -0.9 |
| 2 | 10 | 48 | 2 | 8 | R.PDCCH.1-2.5 FDD | TDLC300-100 | 2x4 Low | 1 | -3.5 |
| 3 | 10 | 48 | 1 | 4 | R.PDCCH.1-1.2 FDD | TDLA30-10 | 2x4 Low | 1 | 0 |

### 5.3.3.1.3 4Rx FDD FR1 PDCCH 1 Tx antenna performance for power saving

#### 5.3.3.1.3.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 4 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.3.1.3.3-2. The downlink physical setup is in accordance with Annex C.2.1.

#### 5.3.3.1.3.2 Test applicability

This test applies to all types of NR UE release 16 and forward that supporting 4 Rx antenna ports and DRX adaptation.

This test also applies to all types of EUTRA UE release 16 and forward that supporting EN-DC, 4 Rx antenna ports and DRX adaptation.

#### 5.3.3.1.3.3 Minimum conformance requirements

The parameters specified in Table 5.3.3.1.3.3-1 are valid for FDD test unless otherwise stated.

**Table 5.3.3.1.3.3-1: Test Parameters**

| Parameter | | Unit | 1 Tx Antenna |
|---|---|---|---|
| CCE to REG mapping type | | | nonInterleaved |
| REG bundle size | | | 6 |
| Shift Index | | | 0 |
| DRX cycle | | ms | 10 |
| ps-WakeUp-r16 | | | absent |
| Wake-up indication bit in DCI format 2_6 | | | 1 |
| PDCCH DCI format 2_6 configuration | PS-offset | | $(T_{minimumTimeGap} + 1)/2^{\mu}/0.125$ |
| | Number of PDCCH candidates | | 1 |
| | Frequency domain resource allocation for CORESET | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | TCI state #1 |
| Slots for PDCCH monitoring | | | Each slot during DRX-on period |
| Note: $T_{minimumTimeGap}$ is signalled as a part of *drx-Adaptation-r16* UE capability. | | | |

For the parameters specified in Table 5.3.3.1.3.3-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.1.3.3-2. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.3.1.3.3-2: Minimum performance for PDCCH with 15 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x4 Low | 1 | 0.2 |
| | | | 2 | 8 | R.PDCCH.1-2.7 FDD | | | | |

The normative reference for this requirement is TS 38.101-4 [2] clause 5.3.3.1.3.

#### 5.3.3.1.3.4 Test description

##### 5.3.3.1.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and clause A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.3.1-1, Table 5.3.3.1.3.3-1 and Table 5.3.3.1.3.3-2 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.3.1.3.4.3.

##### 5.3.3.1.3.4.2 Test procedure

1. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.3.1.3.3-1. The details of PDCCH are specified in Table 5.3.1, Table 5.3.3.1-1, Table 5.3.3.1.3.3-1 and Table 5.3.3.1.3.3-2 respectively. The details of PDSCH are specified in Table A.3.3.1.1-3. The SS sends downlink MAC padding bits on the DL RMC. During the test the UE shall monitor the DCI format 2_6 PDCCH in DRX off state and decide whether to receive the following PDCCH in DRX on period.

2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.3.1.3.3-2 as appropriate.

3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.3.1.3.4.4-1, pass the UE. Otherwise fail the UE.

4. Repeat steps from 1 to 3 for each subtest in Table 5.3.3.1.3.3-2 as appropriate.

##### 5.3.3.1.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

5.3.3.1.3.4.3.1 Message exceptions for SA

**Table 5.3.3.1.3.4.3.1-1: DRX-Config**

| Derivation Path: TS 38.508-1 [4],Table 4.6.3-56 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DRX-Config ::= SEQUENCE { | | | |
| drx-onDurationTimer CHOICE { | | | |
| milliSeconds | ms1 | | |
| } | | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | 0 | | |
| drx-HARQ-RTT-TimerUL | 0 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms10 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-2: DCP-Config**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-106 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| dcp-Config-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| ps-Offset-r16 | 40 | | |
| sizeDCI-2-6-r16 | 2 | | |
| ps-PositionDCI-2-6-r16 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-3: PDCCH-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet ::= SEQUENCE { | 2 entry | | |
| ControlResourceSet[1] | ControlResourceSet1 | | |
| ControlResourceSet[2] | ControlResourceSet2 | | |
| } | | | |
| searchSpacesToAddModList SEQUENCE(SIZE (1..10)) OF SearchSpace ::= SEQUENCE { | 1 entry | | |
| SearchSpace[1] | SearchSpace | | |
| } | | | |
| searchSpacesToAddModListExt-r16 SEQUENCE(SIZE (1..10)) OF SearchSpace { | 1 entry | | |
| searchSpaceExt-r16[1] | SearchSpaceExt | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-4: PDCCH-ControlResourceSet1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n2 | | |
| interleaverSize | n3 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-5: PDCCH-ControlResourceSet2**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n2 | | |
| interleaverSize | n3 | | |
| } | | | |
| } | | | |
| tci-StatesPDCCH-ToAddList { | 1 | TCI State #1 | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-6: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| searchSpaceId | 2 | SearchSpaceId with condition USS | |
| controlResourceSetId | 1 | ControlResourceSetId | |
| monitoringSlotPperiodicityAndOffset CHOICE { | | | |
| sl1 | NULL | | |
| } | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel4 | n1 | AL4 | |
| } | | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | |
| ue-Specific SEQUENCE { | | | |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-7: PDCCH *Search Space Ext***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpaceExt-r16 ::= SEQUENCE { | | | |
| controlResourceSetId-r16 | 2 | | |
| searchSpaceType-r16 SEQUENCE { | | | |
| common SEQUENCE { | | | |
| dci-Format2-6-r16 SEQUENCE { | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.1.3.4.3.1-8: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| TCI-State[1] | TCI-StateId 0 | | |
| qcl-type1 { | | Type 1 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type C | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| TCI-State[2] | TCI-StateId 1 | Type 1 QCL information | |
| qcl-type1 { | | | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | CSI-RS # 0 | |
| Qcl-Type | Type A | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| } | | | |
| } | | | |

5.3.3.1.3.4.3.2 Message exceptions for NSA

Same as 5.3.3.1.3.4.3.1

5.3.3.1.3.4.4 Test requirement

Table 5.3.3.1.3.4.4-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.1.3.4.4-1.

**Table 5.3.3.1.3.4.4-1: Test Requirements for PDCCH with 15 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 10 | 48 | 2 | 4 | R.PDCCH.1-2.4 FDD | TDLA30-10 | 1x4 Low | 1 | 1.1 |
| | | | 2 | 8 | R.PDCCH.1-2.7 FDD | | | | |

### 5.3.3.2 TDD

The parameters specified in Table 5.3.3.2-1 are valid for all TDD tests unless otherwise stated.

**Table 5.3.3.2-1: Common Test Parameters**

| Parameter | Unit | 1 Tx Antenna | 2 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | FR1.30-1 | |
| CCE to REG mapping type | | Test 3: Non-interleaved<br>Other tests: interleaved | interleaved |
| | | | |
| Interleaver size | | 3 | |
| REG bundle size | | Test 3: 6<br>Other tests: 2 | 6 |
| Shift Index | | 0 | |

#### 5.3.3.2.1 4Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA

##### 5.3.3.2.1.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 4 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.3.2.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

##### 5.3.3.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

##### 5.3.3.2.1.3 Minimum conformance requirements

For the parameters specified in Table 5.3.3.2-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.2.1.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.3.2.1.3-1: Minimum performance for PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 2 | R.PDCCH.2-1.1 TDD | TDLA30-10 | 1x4 Low | 1 | 2.1 |
| 2 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x4 Low | 1 | -0.9 |
| 3 | 40 | 48 | 2 | 16 | R.PDCCH.2-2.1 TDD | TDLA30-10 | 1x4 Medium A | 1 | -3.6 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.3.2.

#### 5.3.3.2.1.4 Test description

##### 5.3.3.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.3.2-1 and Table 5.3.3.2.1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On*)* for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.3.2.1.4.3.

##### 5.3.3.2.1.4.2 Test procedure

1. SS transmits PDSCH via PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.3.2.1.3-1. The details of PDCCH are specified in Table 5.3-1, Table 5.3.3.2-1 and Table 5.3.3.2.1.3-1 respectively. The details of PDSCH are specified in Table A.3.3.2.2-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.3.2.1.3-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.3.2.1.5-1, pass the UE. Otherwise fail the UE.
4. Repeat steps from 1 to 3 for each subtest in Table 5.3.3.2.1.3-1 as appropriate.

##### 5.3.3.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2.

5.3.3.2.1.4.3.1 Message exceptions for SA

**Table 5.3.3.2.1.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 00000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 48 RBs of the BWP Test 3 | |
| | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP Test 1, 2 | |
| Duration | 2 | SearchSpace duration of 2 symbols Test 3 | |
| | 1 | SearchSpace duration of 1 symbol Test 1, 2 | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | Test 1, Test 2 |
| reg-BundleSize | n2 | | 1 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| nonInterleaved | null | | Test 3 |
| } | | | |
| } | | | |

**Table 5.3.3.2.1.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel2 | n1 | AL2 | Test 1 |
| aggregationLevel4 | n1 | AL4 | Test 2 |
| aggregationLevel16 | n1 | AL16 | Test 3 |
| } | | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 for test 2 | Long_DCI |
| | formats0-0-And-1-0 | DCI Format 1_0 for test 1 and 3 | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.1.4.3.1-3: Void**

**Table 5.3.3.2.1.4.3.1-3A: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.2-3 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| resourceAllocation | resourceAllocationType1 | | Test 1, Test 3 |
| } | | | |

#### 5.3.3.2.1.4.3.2 Message exceptions for NSA

Same as 5.3.3.2.1.4.3.1.

### 5.3.3.2.1.5 Test requirement

Table 5.3.3.2.1.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.2.1.5-1.

**Table 5.3.3.2.1.5-1: Test Requirement for 1Tx PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 2 | R.PDCCH.2-1.1 TDD | TDLA30-10 | 1x4 Low | 1 | 3 |
| 2 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x4 Low | 1 | 0 |
| 3 | 40 | 48 | 2 | 16 | R.PDCCH.2-2.1 TDD | TDLA30-10 | 1x4 Medium A | 1 | -2.7 |

## 5.3.3.2.2 4Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA

### 5.3.3.2.2.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 4 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.3.2.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

### 5.3.3.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

### 5.3.3.2.2.3 Minimum conformance requirements

For the parameters specified in Table 5.3.3.2-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.2.2.3-1. The downlink physical setup is in accordance with Annex C.2.1.

**Table 5.3.3.2.2.3-1: Minimum performance for PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 90 | 1 | 8 | R.PDCCH.2-1.3 | TDLC300-100 | 2x4 Low | 1 | -4.3 |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.3.2.

### 5.3.3.2.2.4 Test description

#### 5.3.3.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.3.2-1 and Table 5.3.3.2.2.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.3.2.2.4.3.

#### 5.3.3.2.2.4.2 Test procedure

1. SS transmits PDSCH via PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 5.3.3.2.2.3-1. The details of PDCCH are specified in Table 5.3-1, Table 5.3.3.2-1 and Table 5.3.3.2.2.3-1. The details of PDSCH are specified in Table A.3.3.2.2-3. The SS sends downlink MAC padding bits on the DL RMC.
2. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.3.2.2.3-1 as appropriate.
3. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.3.2.2.5-1, pass the UE. Otherwise fail the UE.
4. Repeat steps from 1 to 3 for each subtest in Table 5.3.3.2.2.3-1 as appropriate.

5.3.3.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

5.3.3.2.2.4.3.1 Message exceptions for SA

**Table 5.3.3.2.2.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 11111110 00000000 00000000 00000000 00000 | CORESET to use the least significant 90 RBs of the BWP Test 1 | |
| Duration | 1 | SearchSpace duration of 1 symbols Test 1 | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n6 | | 2 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.2.4.3.1-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 with condition USS | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel8 | n1 | AL8 | Test 1 |
| } | | | |
| } | | | |

**Table 5.3.3.2.2.4.3.1-3: Void**

5.3.3.2.2.4.3.2 Message exceptions for NSA

FFS

5.3.3.2.2.5 Test requirement

Table 5.3.3.2.2.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.2.2.5-1.

**Table 5.3.3.2.2.5-1: Test Requirement for 2Tx PDCCH with 30 kHz SCS**

| Test number | Band width (MHz) | CORE SET RB | CORES ET duration | Aggregati on level | Reference Channel | Propagation Condition | Antenna configurati on and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 40 | 90 | 1 | 8 | R.PDCCH.2-1.3 | TDLC300-100 | 2x4 Low | 1 | -3.3 |

### 5.3.3.2.3 4Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving

#### 5.3.3.2.3.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 4 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 5.3.3.2.3.3-2 after receipt wake-up indication in the *DCI format 2_6* PDCCH in DRX off state. The downlink physical setup is in accordance with Annex C.2.1.

#### 5.3.3.2.3.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting 4 Rx antenna ports and Long DRX Cycle and DRX adaptation.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and 4 Rx antenna ports and Long DRX Cycle and DRX adaptation.

#### 5.3.3.2.3.3 Minimum conformance requirements

During the test the UE shall monitor the *DCI format 2_6* PDCCH in DRX off state and decide whether to receive the following PDCCH in DRX on period.

For the parameters specified in Table 5.3.3.2.3.3-1, the average probability of a missed downlink scheduling grant (Pm-dsg) observed on PDCCH during DRX on shall be below the specified value in Table 5.3.3.2.3.3-2. The downlink physical setup is in accordance with Annex C.3.1.

**Table 5.3.3.2.3.3-1: Test Parameters**

| Parameter | | Unit | 1 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | | FR1.30-1 |
| CCE to REG mapping type | | | interleaved |
| Interleaver size | | | 3 |
| REG bundle size | | | 2 |
| Shift Index | | | 0 |
| DRX cycle | | ms | 10 |
| ps-WakeUp-r16 | | | absent |
| Wake-up indication bit in DCI format 2_6 | | | 1 |
| PDCCH DCI format 2_6 configuration | PS-offset | | $(T_{minimumTimeGap}+1)/2^{\mu}/0.125$ |
| | Number of PDCCH candidates | | 1 |
| | Frequency domain resource allocation for CORESET | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | TCI state #1 |
| Slots for PDCCH monitoring | | | Each slot during DRX-on period |
| Note: $T_{minimumTimeGap}$ is signalled as a part of *drx-Adaptation-r16* UE capability. | | | |

**Table 5.3.3.2.3.3-2: Minimum performance with 30 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x4 Low | 1 | -0.9 |
| | | | | 8 | R.PDCCH.2-1.4 TDD | | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 5.3.3.2.

#### 5.3.3.2.3.4 Test description

##### 5.3.3.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-2 of 38.521-1 [7].

Configurations of DRX, DCP, PDSCH and PDCCH before measurement are specified in 5.3.3.2.3.4.3.1 and Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise sources to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2.5 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.3-1, Table 5.3.3.2-1 and Table 5.3.3.2.1.3-2 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without Release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode* On*)* for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.3.3.2.1.4.3.

##### 5.3.3.2.3.4.2 Test procedure

1. SS transmits PDCCH with DCI format2_6 as specified in PDCCH Reference Channel for C_RNTI within DRX off state. The Wake-up indication bit in PDCCH is set to 1.

2. SS transmits PDSCH via PDCCH with DCI format1_1 as specified in PDCCH Reference Channel for C_RNTI R.PDCCH.5-1.2 TDD to transmit the DL RMC according to Table 5.3.3.2.3.3-2 in DRX on period. The details of PDCCH are specified in Table 5.3-1, Table 5.3.3.2-1 and Table 5.3.3.2.3.3-2 respectively. The details of PDSCH are specified in Table A.3.3.2.2-3. The SS sends downlink MAC padding bits on the DL RMC.

3. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 5.3.3.2.3.3-2 as appropriate.

4. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 5.3.3.2.3.5-1, pass the UE. Otherwise fail the UE.

5. Repeat steps from 2 to 4 for each subtest in Table 5.3.3.2.3.3-2 as appropriate.

##### 5.3.3.2.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1 and 5.4.2.

5.3.3.2.3.4.3.1 Message exceptions for SA

**Table 5.3.3.2.3.4.3.1-1: DRX-Config**

| Derivation Path: TS 38.508-1 [4],Table 4.6.3-56 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DRX-Config ::= SEQUENCE { | | | |
| drx-onDurationTimer CHOICE { | | | |
| milliSeconds | ms1 | | |
| } | | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | 0 | | |
| drx-HARQ-RTT-TimerUL | 0 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms10 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-2: DCP-Config**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-106 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| dcp-Config-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| ps-Offset-r16 | 40 | | |
| sizeDCI-2-6-r16 | 2 | | |
| ps-PositionDCI-2-6-r16 | 0 | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-3: PDCCH-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet ::= SEQUENCE { | 2 entry | | |
| ControlResourceSet[1] | ControlResourceSet1 | | |
| ControlResourceSet[2] | ControlResourceSet2 | | |
| } | | | |
| searchSpacesToAddModList SEQUENCE(SIZE (1..10)) OF SearchSpace ::= SEQUENCE { | 1 entry | | |
| SearchSpace[1] | SearchSpace | | |
| } | | | |
| searchSpacesToAddModListExt-r16 SEQUENCE(SIZE (1..10)) OF SearchSpace { | 1 entry | | |
| searchSpaceExt-r16[1] | SearchSpaceExt | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-4: PDCCH-ControlResourceSet1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 1 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n2 | | 1 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-5: PDCCH-ControlResourceSet2**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 11111111 11111111 10000000 00000000 00000000 00000 | CORESET to use the least significant 102 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n2 | | 1 Tx |
| interleaverSize | n3 | | TDD |
| } | | | |
| } | | | |
| tci-StatesPDCCH-ToAddList { | 1 | TCI State #1 | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-6: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| searchSpaceId | 2 | SearchSpaceId with condition USS | USS |
| controlResourceSetId | 1 | ControlResourceSetId | |
| monitoringSlotPperiodicityAndOffset CHOICE { | | | |
| sl1 | NULL | | |
| } | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel4 | n1 | AL4 | |
| } | | | |
| } | | | |
| searchSpaceType CHOICE { | | | |
| common SEQUENCE { | | | CSS, SISS |
| ue-Specific SEQUENCE { | | | USS |
| dci-Formats | formats0-1-And-1-1 | DCI Format 1_1 | Long_DCI |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-7: PDCCH *Search Space Ext***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpaceExt-r16 ::= SEQUENCE { | | | |
| controlResourceSetId-r16 | 2 | | |
| searchSpaceType-r16 SEQUENCE { | | | |
| common SEQUENCE { | | | |
| dci-Format2-6-r16 SEQUENCE { | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 5.3.3.2.3.4.3.1-8: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| TCI-State[1] | TCI-StateId 0 | | |
| qcl-type1 { | | Type 1 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type C | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | Ssb : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| TCI-State[2] | TCI-StateId 1 | Type 1 QCL information | |
| qcl-type1 { | | | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | CSI-RS # 0 | |
| Qcl-Type | Type A | | |
| } | | | |
| qcl-type2 { | | Type 2 QCL information | |
| Cell | ServCellIndex | | |
| Bwp-id | 1 | BWP ID | |
| referenceSignal | csi-rs : 0 | SSB # 0 | |
| Qcl-Type | Type D | | |
| } | | | |
| } | | | |
| } | | | |

5.3.3.2.3.4.3.2 Message exceptions for NSA

Same as 5.3.3.2.3.4.3.1.

5.3.3.2.3.5 Test requirement

Table 5.3.3.2.3.5-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 5.3.3.2.3.5-1.

**Table 5.3.3.2.3.5-1: Minimum performance with 30 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | SNR (dB) |
| 1 | 40 | 102 | 1 | 4 | R.PDCCH.2-1.2 TDD | TDLC300-100 | 1x4 Low | 1 | 0 |
| | | | | 8 | R.PDCCH.2-1.4 TDD | | | | |

# 5.4 PBCH demodulation requirements

TS 38.101-4 shall specify the PBCH performance requirements and has recommended that these requirements do not need to be tested.

# 5.5 Sustained downlink data rate provided by lower layers

## 5.5.1 FR1 Sustained downlink data rate performance for single carrier

### 5.5.1.1 Test Purpose

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum data rate indicated by UE capabilities. The sustained downlink data rate shall be verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the RF conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement

### 5.5.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

### 5.5.1.3 Minimum conformance requirements

The requirements in this clause are applicable to the FR1 single carrier case.

The TB success rate shall be higher than 85% when PDSCH is scheduled with MCS defined for the channel bandwidth with the downlink physical channel setup according to Annex C.3.1.

The TB success rate is defined as $100\%*N_{DL_correct_rx} / (N_{DL_newtx} + N_{DL_retx})$, where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

The common test parameters are specified in Table 5.5.1.3-1. The parameters specified in Table 5.5.1.3-2 are applicable for tests on FDD bands and parameters specified in Table 5.5.1.3-3 are applicable for tests on TDD bands.

Unless otherwise stated, no user data is scheduled on slot #0, 10 and 11 within 20 ms for SCS 15 kHz.

Unless otherwise stated, no user data is scheduled on slot #0, 20 and 21 within 20 ms for SCS 30 kHz.

**Table 5.5.1.3-1: Common test parameters for FDD and TDD bands**

| Parameter | | Unit | Value |
|---|---|---|---|
| **PDSCH transmission scheme** | | | **Transmission scheme 1** |
| **EPRE ratio of PTRS to PDSCH** | | **dB** | **N/A** |
| Channel bandwidth | | MHz | Channel bandwidth from selected CA bandwidth combination |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| | First DMRS position for Type A PDSCH mapping | | 2 |
| Cross carrier scheduling | | | Not configured |
| Active DL BWP index | | | 1 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 2) | RBs | 0 |
| | Subcarrier spacing | kHz | 15 or 30 |
| DL BWP configuration #1 | RB offset | RBs | 0 |
| | Number of contiguous PRB | | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing |
| | Subcarrier spacing | kHz | 15 or 30 |

| | | | |
|---|---|---|---|
| | Cyclic prefix | | Normal |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | Symbols #0 |
| | Number of PRBs in CORESET | | Table 5.5.1.3-4 |
| | Number of PDCCH candidates and aggregation levels | | 2/AL2 for 15 kHz / 5 MHz and 30 kHz / 15 MHz<br>2/AL4 for 15 kHz / 10 MHz, 30 kHz / 10 MHz and 30 kHz / 20 MHz<br>2/AL8 for other greater combinations |
| | CCE-to-REG mapping type | | Non-interleaved |
| | DCI format | | 1_1 |
| | TCI State | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | For 2Tx:<br>Single Panel Type I, Random precoder chosen from precoder index 0 and 2, selection updated per slot<br><br>For 4Tx:<br>Single Panel Type I, Random precoder chosen from precoders with i_1,1 in {1,2,3,5,6,7} and i_2 in {0,2}, selection updated per slot |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | WB |
| | Resource allocation type | | Type 0 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Length | | 1 |
| | Antenna ports indexes | | {1000} for 1 Layer CCs<br>{1000, 1001} for 2 Layers CCs<br>{1000 – 1003} for 4 Layers CCs |
| | Number of PDSCH DMRS CDM group(s) without data | | 1 for 1 layer and 2 layers CCs<br>2 for 4 Layers CCs |
| PTRS configuration | | | PTRS is not configured |
| CSI-RS for tracking | Subcarrier indexes in the PRB used for CSI-RS | | $k_0 = 3$ for CSI-RS resource 1,2,3,4 |
| | OFDM symbols in the PRB used for CSI-RS | | $l_0 = 6$ for CSI-RS resource 1 and 3<br>$l_0 = 10$ for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4 |
| | Density (ρ) | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | Slots | 15 kHz SCS: 20 for CSI-RS resource 1,2,3,4<br>30 kHz SCS: 40 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | Slots | 15 kHz SCS:<br>10 for CSI-RS resource 1 and 2<br>11 for CSI-RS resource 3 and 4<br><br>30 kHz SCS:<br>20 for CSI-RS resource 1 and 2<br>21 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | $k_0 = 4$ |
| | OFDM symbols in the PRB used for CSI-RS | | $l_0 = 12$ |
| | Number of CSI-RS ports (X) | | Same as number of transmit antenna |
| | CDM Type | | 'FD-CDM2' |

| | | | | |
|---|---|---|---|---|
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0 = 0$ |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0 = 12$ |
| | Number of CSI-RS ports (X) | | | 4 |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| Maximum number of code block groups for ACK/NACK feedback | | | | 1 |
| Maximum number of HARQ transmission | | | | 4 |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| PDSCH & PDSCH DMRS Precoding configuration | | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination with PRB bundling granularity |
| Symbols for all unused REs | | | | OCNG Annex A.5 |
| Propagation condition | | | | Static propagation condition<br>No external noise sources are applied |
| Antenna configuration | 1 layer CCs | | | 1x2 or 1x4 |
| | 2 layers CCs | | | 2x2 or 2x4 |
| | 4 layers CCs | | | 4x4 |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |
| Note 1: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission<br>Note 2: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing | | | | |

**Table 5.5.1.3-2: Additional test parameters for FDD band**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| PDSCH configuration | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| Number of HARQ Processes | | | 4 |
| K1 value | | | 2 |

**Table 5.5.1.3-3: Additional test parameters for TDD band**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| PDSCH configuration | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |

| | | |
|---|---|---|
| Number of HARQ Processes | | 8 |
| K1 value | | Specific to each UL-DL pattern |
| TDD UL-DL pattern | | 15 kHz SCS: FR1.15-1<br>30 kHz SCS: FR1.30-1 |
| Note 1: PDSCH is scheduled only on full DL slots | | |

**Table 5.5.1.3-4: Number of PRBs in CORESET**

| SCS (kHz) | 5MHz | 10MHz | 15MHz | 20 MHz | 25 MHz | 30 MHz | 40 MHz | 50MHz | 60 MHz | 80 MHz | 100 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 24 | 48 | 78 | 102 | 132 | 156 | 216 | 270 | N/A | N/A | N/A |
| 30 | 6 | 24 | 36 | 48 | 60 | 78 | 102 | 132 | 162 | 216 | 270 |

**Table 5.5.1.3-5: MCS indexes for indicated UE capabilities**

| Maximum number of PDSCH MIMO layers | Maximum modulation format | Scaling factor | MCS |
|---|---|---|---|
| 1 | 8 | 1 | 26 |
| 1 | 8 | 0.8 | 21 |
| 1 | 8 | 0.75 | 20 |
| 1 | 8 | 0.4 | 11 |
| 1 | 6 | 1 | 27 |
| 1 | 6 | 0.8 | 23 |
| 1 | 6 | 0.75 | 22 |
| 1 | 6 | 0.4 | 14 |
| 1 | 4 | 1 | 16 |
| 1 | 4 | 0.8 | 16 |
| 1 | 4 | 0.75 | 16 |
| 1 | 4 | 0.4 | 10 |
| 1 | 2 | 1 | 9 |
| 1 | 2 | 0.8 | 9 |
| 1 | 2 | 0.75 | 9 |
| 1 | 2 | 0.4 | 4 |
| 2 | 8 | 1 | 26 |
| 2 | 8 | 0.8 | 21 |
| 2 | 8 | 0.75 | 20 |
| 2 | 8 | 0.4 | 11 |
| 2 | 6 | 1 | 27 |
| 2 | 6 | 0.8 | 23 |
| 2 | 6 | 0.75 | 22 |
| 2 | 6 | 0.4 | 14 |
| 2 | 4 | 1 | 16 |
| 2 | 4 | 0.8 | 16 |
| 2 | 4 | 0.75 | 16 |
| 2 | 4 | 0.4 | 10 |
| 2 | 2 | 1 | 9 |
| 2 | 2 | 0.8 | 9 |
| 2 | 2 | 0.75 | 9 |
| 2 | 2 | 0.4 | 4 |
| 4 | 8 | 1 | 26 |
| 4 | 8 | 0.8 | 23 |
| 4 | 8 | 0.75 | 22 |
| 4 | 8 | 0.4 | 12 |
| 4 | 6 | 1 | 27 |
| 4 | 6 | 0.8 | 24 |
| 4 | 6 | 0.75 | 23 |
| 4 | 6 | 0.4 | 14 |
| 4 | 4 | 1 | 16 |
| 4 | 4 | 0.8 | 16 |
| 4 | 4 | 0.75 | 16 |
| 4 | 4 | 0.4 | 11 |
| 4 | 2 | 1 | 9 |
| 4 | 2 | 0.8 | 9 |
| 4 | 2 | 0.75 | 9 |
| 4 | 2 | 0.4 | 5 |
| Note 1: MCS Index for maximum modulation format 2,4 and 6 is based on MCS index table 1 defined in clause 5.1.3.1 of TS 38.214 [12]<br>Note 2: MCS Index for maximum modulation format 8 is based on MCS index table 2 defined in clause 5.1.3.1 of TS 38.214 [12] | | | |

### 5.5.1.3.1 Procedure for test parameter selection

Below test parameter selection procedure is from 38.101-4 [5] by replacing CA configuration with operating band, and bandwidth instead of bandwidth combination.

The test parameters are determined by the following procedure:

- Select one operating band among all supported operating bands and set of per band UE capabilities among all supported UE capabilities that provides the largest data rate [TS 38.306 [14, Section 4.1.2]].
- Set of per band UE capabilities includes channel bandwidth, subcarrier spacing, number of PDSCH MIMO layers, modulation format and scaling factor [TS 38.306 [14, Section 4.1.2]].
- When there are multiple sets of bandwidths and UE capabilities (channel bandwidth, subcarrier spacing, number of MIMO layer, modulation format, scaling factor) with same largest data rate, select one among sets with the smallest channel bandwidth.
- For each operating band, use Table 5.5.1.3-5 to determine MCS based on test parameters and indicated UE capabilities

Pasting relevant portion of max data rate equation from TS 38.306 [14] section 4.1

For NR, the approximate data rate for a given number of aggregated carriers in a band or band combination is computed as follows.

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left( v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

wherein

J is the number of aggregated component carriers in a band or band combination

$R_{max}$ = 948/1024

For the j-th CC,

$v_{Layers}^{(j)}$ is the maximum number of supported layers given by higher layer parameter *maxNumberMIMO-LayersPDSCH* for downlink and maximum of higher layer parameters *maxNumberMIMO-LayersCB-PUSCH* and *maxNumberMIMO-LayersNonCB-PUSCH* for uplink.

$Q_m^{(j)}$ is the maximum supported modulation order given by higher layer parameter *supportedModulationOrderDL* for downlink and higher layer parameter *supportedModulationOrderUL* for uplink.

$f^{(j)}$ is the scaling factor given by higher layer parameter *scalingFactor* and can take the values 1, 0.8, 0.75, and 0.4.

$\mu$ is the numerology (as defined in TS 38.211 [6])

$T_s^{\mu}$ is the average OFDM symbol duration in a subframe for numerology $\mu$, i.e. $T_s^{\mu} = \frac{10^{-3}}{14 \cdot 2^{\mu}}$. Note that normal cyclic prefix is assumed.

$N_{PRB}^{BW(j),\mu}$ is the maximum RB allocation in bandwidth $BW^{(j)}$ with numerology $\mu$, as defined in 5.3 TS 38.101-1 [2] and 5.3 TS 38.101-2 [3], where $BW^{(j)}$ is the UE supported maximum bandwidth in the given band or band combination.

$OH^{(j)}$ is the overhead and takes the following values

0.14, for frequency range FR1 for DL

0.18, for frequency range FR2 for DL

0.08, for frequency range FR1 for UL

0.10, for frequency range FR2 for UL

NOTE: Only one of the UL or SUL carriers (the one with the higher data rate) is counted for a cell operating SUL.

The approximate maximum data rate can be computed as the maximum of the approximate data rates computed using the above formula for each of the supported band or band combinations.

The normative reference for this requirement is TS 38.101-4 [5], clause 5.5.1.

### 5.5.1.4 Test description

#### 5.5.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of TS 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

1. Connect the SS to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram (without fader and AWGN) and clause A.3.2.2 for UE diagram.
2. The parameter settings for the NR cell are initially set up according to Table 5.5.1.3-1 as appropriate.
3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions for the NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR with *Connected without release On, Test Loop Function On with UE Test Loop Mode A with UL_PDCP_SDU_SIZE = 0* according to TS 38.508-1 [6] clause 4.5.4. Message content are defined in clause 5.5.1.4.3.
6. SS shall transmit UECapabilityEnquiry message.
7. The UE shall transmit UECapabilityInformation message.
8. Using the UE capabilities advertised in the *UE-CapabilityRAT-Container* of the type *UE-NR-Capability*, and the procedure outlined in 5.5.1.3.1 determine one set of parameters that would provide the largest data rate.
9. Setup up the NR cell using these parameters for the test.
10. Configure the TBsize, DL RMC, UL RMC, PDCP size from Annex A.3.2_1 and Annex A.2.2 for UL as appropriate.

#### 5.5.1.4.2 Test procedure

1. SS configures T-reordering timer to be infinity.
2. SS sends a PDCP reestablishment via RRC Reconfiguration message requesting for PDCP Status Report.
3. SS sets the counters $N_{DL_newtx}$ $N_{DL_retx}$ to 0.
4. For each new DL HARQ transmission the SS generates sufficient PDCP SDUs (max PDCP SDU size and minimum number of consecutive PDCP SDUs) to fill up the TB in accordance with Annex A.3.2_1. The SS ciphers the PDCP SDUs, concatenates the resultant PDCP PDUs to form an RLC PDU and then a MAC PDU. The SS transmits the MAC PDU. The SS increments then $N_{DL_newtx}$ by one
5. If PHY requests a DL HARQ retransmission, the SS performs a HARQ retransmission and increments $N_{DL_retx}$ by one.

6. Steps 5 to 6 are repeated at every TTI for at least 300 frames and the SS waits for 300ms to let any HARQ retransmissions and RLC retransmissions to finish.

7. SS sends a PDCP reestablishment via RRC Reconfiguration message requesting for PDCP Status Report.

8. The SS calculates the TB success rate as A = 100%* $N_{DL_correct_rx}$ / ($N_{DL_newtx}$ + $N_{DL_retx}$).

10. SS computes the PDCP SDU loss by looking into the FMC and Bitmap field in the PDCP Status Report. PDCP SDU loss B = COUNT reported in the Bitmap field of PDCP Stata Report.

11. The UE passes the test if A $\geq$ 85% TB success rates and B = 0.

Note 1: In case of RLC PDU retransmission, the number of new required PDCP SDUs is as many as to fill the rest of TB.

#### 5.5.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions

**Table 5.5.1.4.3-0: CLOSE UE TEST LOOP (in the preamble)**

| Derivation Path: 38.509 clause 6.3.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 1 1 1 1 | | |
| Skip indicator | 0 0 0 0 | | |
| Message type | 1 0 0 0 0 0 0 0 | | |
| UE test loop mode | 0 0 0 0 0 0 0 0 | UE test loop mode A | |
| UE test loop mode A LB setup | | | |
| Length of UE test loop mode A LB setup list in bytes | 0 0 0 0 0 0 1 1 | Length of one LB setup DRB (3 bytes) | |
| LB setup DRB | 0 0 0 0 0 0 0 0,<br>0 0 0 0 0 0 0 0,<br>0 0 Q5 Q4 Q3 Q2 Q1 Q0 | UL PDCP SDU size = 0<br>Q5 = 1 (for NR Data Radio Bearers)<br>Q4..Q0 = Data Radio Bearer identity number -1 for the radio bearer. See 38.509 clause 6.3.1 | |
| UE test loop mode B LB setup | Not present | | |

**Table 5.5.1.4.3-1 to -6: Void**

**Table 5.5.1.4.3-7: RadioBearerConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3-132 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioBearerConfig ::= SEQUENCE { | | | |
| drb-ToAddModList SEQUENCE (SIZE (1..maxDRB)) OF SEQUENCE { | 1 entry | | DRB1 |
| cnAssociation CHOICE { | | | |
| sdap-Config | SDAP-Config | | |
| } | | | |
| drb-Identity | DRB-Identity using condition DRB1 | | |
| reestablishPDCP | true | | DRB1 AND Re-establish_PDCP |
| pdcp-Config | PDCP-Config | | |
| } | | | |

**Table 5.5.1.4.3-8: *PDCP-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-99 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCP-Config ::= SEQUENCE { | | | |
| drb SEQUENCE { | | | |
| discardTimer | infinity | | |
| pdcp-SN-Size-UL | len18bits | | |
| pdcp-SN-Size-DL | len18bits | | |
| headerCompression CHOICE { | | | |
| notUsed | Null | | |
| } | | | |
| integrityProtection | Not present | | |
| statusReportRequired | true | | |
| outOfOrderDelivery | Not present | | |
| } | | | |
| t-Reordering | Not present | | |
| } | | | |

#### 5.5.1.5 Test requirement

The PDCP SDU success rate of greater than 85% shall be sustained during at least 300 frames.

## 5.5A.1 FR1 Sustained downlink data rate performance for carrier aggregation

Editor's note: Following aspects in this clause are incomplete

- Connection Diagram is in []
- Procedure for parameter selection is TBD
- Message contents is TBD
- Test procedure is TBD

### 5.5A.1.1 FR1 SDR performance for CA (2DLCA)

#### 5.5A.1.1.1 Test Purpose

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum data rate indicated by UE capabilities. The sustained downlink data rate shall be

verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the RF conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement.

#### 5.5A.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 2DL CA

#### 5.5A.1.1.3 Minimum conformance requirements

The Sustained Data Rate (SDR) requirements in this clause are applicable to the FR1 CA.

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum data rate indicated by UE capabilities. The sustained downlink data rate shall be verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the RF conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement.

The test parameters are determined by the following procedure:

- Select one CA bandwidth combination among all supported CA configurations and set of per component carrier (CC) UE capabilities among all supported UE capabilities that provides the largest data rate in accordance with clause 4.1.2 of TS 38.306 [14].
  - Set of per CC UE capabilities includes channel bandwidth, subcarrier spacing, number of PDSCH MIMO layers, modulation format and scaling factor in accordance with clause 4.1.2 of TS 38.306 [14].
  - When there are multiple sets of CA bandwidth combinations and UE capabilities (channel bandwidth, subcarrier spacing, number of MIMO layer, modulation format, scaling factor) with same largest data rate, select one among sets with the smallest aggregated channel bandwidth.
- For each CC in CA bandwidth combination, use Table 5.5A-5 to determine MCS based on test parameters and indicated UE capabilities.

The TB success rate shall be higher than 85% when PDSCH is scheduled with MCS defined for the selected CA bandwidth combination and with the downlink physical channel setup according to Annex C.3.1.

The TB success rate is defined as $100\%*N_{DL_correct_rx} / (N_{DL_newtx} + N_{DL_retx})$, where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

The common test parameters are specified in Table 5.5A-1. The parameters specified in Table 5.5A-2 are applicable for tests on FDD CCs and parameters specified in Table 5.5A-3 are applicable for tests on TDD CCs.

Unless otherwise stated, no user data is scheduled on slot #0, 10 and 11 within 20 ms for SCS 15 kHz.

Unless otherwise stated, no user data is scheduled on slot #0, 20 and 21 within 20 ms for SCS 30 kHz.

**Table 5.5A-1: Common test parameters for FDD and TDD component carriers**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| EPRE ratio of PTRS to PDSCH | | dB | N/A |
| Channel bandwidth | | MHz | Channel bandwidth from selected CA bandwidth combination |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| | First DMRS position for Type A PDSCH mapping | | 2 |
| Cross carrier scheduling | | | Not configured |
| Active DL BWP index | | | 1 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 2) | RBs | 0 |
| | Subcarrier spacing | kHz | 15 or 30 |

| | | | |
|---|---|---|---|
| DL BWP configuration #1 | RB offset | RBs | 0 |
| | Number of contiguous PRB | | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-1 [6] for tested channel bandwidth and subcarrier spacing |
| | Subcarrier spacing | kHz | 15 or 30 |
| | Cyclic prefix | | Normal |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | Symbols #0 |
| | Number of PRBs in CORESET | | Table 5.5A-4 |
| | Number of PDCCH candidates and aggregation levels | | 1/AL 1 for 30 kHz / 5 MHz<br>1/AL4 for 15 kHz / 5 MHz, 30 kHz / 10 MHz and 30 kHz / 15 MHz<br>1/AL 8 for other combinations |
| | CCE-to-REG mapping type | | Non-interleaved |
| | DCI format | | 1_1 |
| | TCI State | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | For 2Tx:<br>Single Panel Type I, Random precoder chosen from precoder index 0 and 2, selection updated per slot<br><br>For 4Tx:<br>Single Panel Type I, Random precoder chosen from precoders with i_1,1 in {1,2,3,5,6,7} and i_2 in {0,2}, selection updated per slot |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | wideband |
| | Resource allocation type | | Type 0 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Length | | 1 |
| | Antenna ports indexes | | {1000} for 1 Layer CCs<br>{1000, 1001} for 2 Layers CCs<br>{1000 – 1003} for 4 Layers CCs |
| | Number of PDSCH DMRS CDM group(s) without data | | 1 for 1 layer and 2 layers CCs<br>2 for 4 Layers CCs |
| PTRS configuration | | | PTRS is not configured |
| CSI-RS for tracking | Subcarrier indexes in the PRB used for CSI-RS | | $k_0$ = 3 for CSI-RS resource 1,2,3,4 |
| | OFDM symbols in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 1 and 3<br>$l_0$ = 10 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4 |
| | Density (ρ) | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | Slots | 15 kHz SCS: 20 for CSI-RS resource 1,2,3,4<br>30 kHz SCS: 40 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | Slots | 15 kHz SCS:<br>10 for CSI-RS resource 1 and 2<br>11 for CSI-RS resource 3 and 4<br><br>30 kHz SCS:<br>20 for CSI-RS resource 1 and 2<br>21 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | Start PRB 0<br>Number of PRB = ceil(BWP size /4)*4 |
| | QCL info | | TCI state #0 |

| | | | | |
|---|---|---|---|---|
| NZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0 = 4$ |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0 = 12$ |
| | Number of CSI-RS ports (X) | | | Same as number of transmit antenna |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = ceil(BWP size /4)*4 |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0 = 0$ |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0 = 12$ |
| | Number of CSI-RS ports (X) | | | 4 |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = ceil(BWP size/4)*4 |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| Maximum number of code block groups for ACK/NACK feedback | | | | 1 |
| Maximum number of HARQ transmission | | | | 4 |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| PDSCH & PDSCH DMRS Precoding configuration | | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination with PRB bundling granularity |
| Symbols for all unused REs | | | | OP.1 FDD as defined in Annex A.5.1.1<br>OP.1 TDD as defined in Annex A.5.2.1 |
| Propagation condition | | | | Static propagation condition<br>No external noise sources are applied |
| Antenna configuration | 1 layer CCs | | | 1x2 or 1x4 |
| | 2 layers CCs | | | 2x2 or 2x4 |
| | 4 layers CCs | | | 4x4 |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |
| Note 1: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission<br>Note 2: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-1 [6] for tested channel bandwidth and subcarrier spacing | | | | |

**Table 5.5A-2: Additional test parameters for FDD CC**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| PDSCH configuration | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| Number of HARQ Processes | | | 4 |
| K1 value | | | 2 |

**Table 5.5A-3: Additional test parameters for TDD CC**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| PDSCH configuration | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| Number of HARQ Processes | | | 8 |
| K1 value | | | Specific to each UL-DL pattern |
| TDD UL-DL pattern | | | 15 kHz SCS: FR1.15-1<br>30 kHz SCS: FR1.30-1 |
| Note 1: PDSCH is scheduled only on full DL slots | | | |

**Table 5.5A-4: Number of PRBs in CORESET**

| SCS (kHz) | 5 MHz | 10 MHz | 15 MHz | 20 MHz | 25 MHz | 30 MHz | 35 MHz | 40 MHz | 45 MHz | 50 MHz | 60 MHz | 80 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 24 | 48 | 78 | 102 | 132 | 156 | 186 | 216 | 240 | 270 | N/A | N/A |
| 30 | 6 | 24 | 36 | 48 | 60 | 78 | 90 | 102 | 114 | 132 | 162 | 216 |

**Table 5.5A-5: MCS indexes for indicated UE capabilities**

| Maximum number of PDSCH MIMO layers | Maximum modulation format | Scaling factor | MCS |
|---|---|---|---|
| 1 | 8 | 1 | 26 |
| 1 | 8 | 0.8 | 21 |
| 1 | 8 | 0.75 | 20 |
| 1 | 8 | 0.4 | 11 |
| 1 | 6 | 1 | 27 |
| 1 | 6 | 0.8 | 23 |
| 1 | 6 | 0.75 | 22 |
| 1 | 6 | 0.4 | 14 |
| 1 | 4 | 1 | 16 |
| 1 | 4 | 0.8 | 16 |
| 1 | 4 | 0.75 | 16 |
| 1 | 4 | 0.4 | 10 |
| 1 | 2 | 1 | 9 |
| 1 | 2 | 0.8 | 9 |
| 1 | 2 | 0.75 | 9 |
| 1 | 2 | 0.4 | 4 |
| 2 | 8 | 1 | 26 |
| 2 | 8 | 0.8 | 21 |
| 2 | 8 | 0.75 | 20 |
| 2 | 8 | 0.4 | 11 |
| 2 | 6 | 1 | 27 |
| 2 | 6 | 0.8 | 23 |
| 2 | 6 | 0.75 | 22 |
| 2 | 6 | 0.4 | 14 |
| 2 | 4 | 1 | 16 |
| 2 | 4 | 0.8 | 16 |
| 2 | 4 | 0.75 | 16 |
| 2 | 4 | 0.4 | 10 |
| 2 | 2 | 1 | 9 |
| 2 | 2 | 0.8 | 9 |
| 2 | 2 | 0.75 | 9 |
| 2 | 2 | 0.4 | 4 |
| 4 | 8 | 1 | 26 |
| 4 | 8 | 0.8 | 23 |
| 4 | 8 | 0.75 | 22 |
| 4 | 8 | 0.4 | 12 |
| 4 | 6 | 1 | 27 |
| 4 | 6 | 0.8 | 24 |
| 4 | 6 | 0.75 | 23 |
| 4 | 6 | 0.4 | 14 |
| 4 | 4 | 1 | 16 |
| 4 | 4 | 0.8 | 16 |
| 4 | 4 | 0.75 | 16 |
| 4 | 4 | 0.4 | 11 |
| 4 | 2 | 1 | 9 |
| 4 | 2 | 0.8 | 9 |
| 4 | 2 | 0.75 | 9 |
| 4 | 2 | 0.4 | 5 |
| Note 1: MCS Index for maximum modulation format 2,4 and 6 is based on MCS index Table 1 defined in clause 5.1.3.1 of TS 38.214 [12]<br>Note 2: MCS Index for maximum modulation format 8 is based on MCS index Table 2 defined in clause 5.1.3.1 of TS 38.214 [12] | | | |

5.5A.1.1.3.1 Procedure for test parameter selection

FFS

5.5A.1.1.4 Test description

5.5A.1.1.4.1 Initial conditions

FFS

5.5.1.4.2 Test procedure

FFS

5.5A.1.4.3 Message contents

FFS

5.5.1.5 Test requirement

The PDCP SDU success rate of greater than 85% shall be sustained during at least 300 frames on each CC.

## 5.5.2

## 5.5.2.1

### 5.2.2.1.1 to 5.2.2.1.5

### 5.2.2.1.6 2Rx FDD FR1 PDSCH repetitions over multiple slots performance

#### 5.2.2.1.6.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.6.0-3, with the addition of test parameters in Table 5.2.2.1.6-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.6.0-1.

**Table 5.2.2.1.6.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.1.6.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | 2 |

**Table 5.2.2.1.6.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-11.1 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.6 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.6.

## 5.2.2.1.6_1 2Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Statistical significance in Annex G is FFS
- MU factor due to finite test time needs further study
- Test Procedure is incomplete

### 5.2.2.1.6_1.1 Test purpose

To Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions.

### 5.2.2.1.6_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

### 5.2.2.1.6_1.3 Test description

#### 5.2.2.1.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.6.0-2 and Table 5.2.2.1.6.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.6_1.3.3.

#### 5.2.2.1.6_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.5.0-3. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.5_1.3.4-1.
3. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause FFS. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table FFS in Annex G clause FFS.

#### 5.2.2.1.6_1.3.3 Message contents

##### 5.2.2.1.6_1.3.3_1 Message exceptions for SA & NSA

As defined in clause 5.4.2 of TS 38.508-1 [6] with the following exceptions:

**Table 5.2.2.1.6_1.3.3_1-1: PDSCH-Config**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-26 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| pdsch-AggregationFactor | 2 | | |
| } | | | |

#### 5.2.2.1.6_1.3.4 Test requirement

Table 5.2.2.1.6.0-3 defines the primary level settings.

The target BLER percentage for the downlink reference measurement channels specified in Annex FFS for each BLER test shall meet or exceed the specified value in Table 5.2.2.1.6_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.6_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-11.1 FDD | 10 / 15 | 16QAM, 0.54 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.6 + TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | |

### 5.2.2.1.7

### 5.2.2.1.8 2Rx FDD FR1 PDSCH pre-emption performance

#### 5.2.2.1.8.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.1.8.0-3, with the addition of test parameters in Table 5.2.2.1.8.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.1.8.0-1.

**Table 5.2.2.1.8.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH pre-emption performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.1.8.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| **Duplex mode** | | | **FDD** |
| **Active DL BWP index** | | | **1** |
| PDCCH configuration (Note 4) | Symbols with PDCCH | | 0, 1 |
| | DCI format | | 2_1 |
| | timeFrequencySet | | 14x1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |

| | | | |
|---|---|---|---|
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Pre-emption configuration (Note 2) | Starting symbol (S) | | 3 |
| | Length (L) | | 2 |
| | Pre-emption periodicity and offset (Note 3) | Slots | 10/1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | 2 |
| Note 1: Void<br>Note 2: Interference modelled as random data on pre-empted REs.<br>Note 3: Pre-emption is scheduled with a fixed scheduling with 10% probability within 10ms periodicity.<br>Note 4: In addition to PDCCH configuration in Table 5.2-1. | | | |

**Table 5.2.2.1.8.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-2.5 FDD | 10 / 15 | 16QAM 0.64 | TDLA30-10 | 2x2, ULA Low | 70 | 10.5 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.1.8.

### 5.2.2.1.8_1 2Rx FDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Statistical significance in Annex G is FFS
- MU factor due to finite test time needs further study
- Test Procedure is incomplete

#### 5.2.2.1.8_1.1 Test purpose

To Verify the PDSCH pre-emption performance under 2 receive antenna conditions.

#### 5.2.2.1.8_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pre-EmptIndication-DL-r16*.

#### 5.2.2.1.6_1.3 Test description

#### 5.2.2.1.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.8.0-2 and Table 5.2.2.1.8.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.1.8_1.3.3.

### 5.2.2.1.8_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.1.8.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. SS transmits PDCCH DCI format 2_1 for int_RNTI with 10% probability to transmit the DL Pre-emption indication according to Table 5.2.2.1.8.0-2. In the time and frequency set indicated by PDCCH DCI format 2_1, SS stops transmission of PDSCH.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.1.8_1.3.4-1.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause FFS. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table FFS in Annex G clause FFS.

### 5.2.2.1.8_1.3.3 Message contents

### 5.2.2.1.8_1.3.3_1 Message exceptions for SA

**Table 5.2.2.1.8_1.3.3_1-1: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | FR1 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| k0 | 0 | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | |
| } | | | |
| } | | | |

**Table 5.2.2.1.8_1.3.3_1-2: DownlinkPreemption**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-54 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DownlinkPreemption ::= SEQUENCE { | | | |
| FFS | | | |
| } | | | |

**Table 5.2.2.1.8_1.3.3_1-3: Physical layer parameters for DCI format 2_1**

FFS

### 5.2.2.1.8_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.8_1.3.3_1.

### 5.2.2.1.8_1.3.4 Test requirement

Table 5.2.2.1.8.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.1.8_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.1.8_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|
| | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.1-2.5 FDD | 10 / 15 | 16QAM 0.64 | TDLA30-10 | 2x2, ULA Low | 70 | 10.5 + TT |

## 5.2.2.2

### 5.2.2.2.1 to 5.2.2.2.5

### 5.2.2.2.6 2Rx TDD FR1 PDSCH repetitions over multiple slots performance

#### 5.2.2.2.6.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.6.0-3, with the addition of test parameters in Table 5.2.2.2.6.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.6.0-1.

**Table 5.2.2.2.6.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.2.6.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 4 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.2 (Note 1) |
| Note 1: ACK/NACK feedback is generated for PDSCH on slot i, where mod(i,10) = {2, 4, 6}. | | | |

**Table 5.2.2.1.6.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-16.1 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.4 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.6.

### 5.2.2.2.6_1 2Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Statistical significance in Annex G is FFS
- MU factor due to finite test time needs further study
- Test Procedure is incomplete

#### 5.2.2.2.6_1.1 Test purpose

To Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions.

#### 5.2.2.2.6_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

### 5.2.2.2.6_1.3 Test description

#### 5.2.2.2.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.6.0-2 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On,* Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.6_1.3.3.

#### 5.2.2.2.6_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.6.0-3. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.
2. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.6_1.3.4-1.
3. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause FFS. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table FFS in Annex G clause FFS.

#### 5.2.2.2.6_1.3.3 Message contents

##### 5.2.2.2.6_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

##### 5.2.2.2.6_1.3.3_2 Message exceptions for SA

Same as 5.2.2.1.6_1.3.3_1.

#### 5.2.2.2.6_1.3.4 Test requirement

Table 5.2.2.2.6.0-3 defines the primary level settings.

The target BLER percentage for the downlink reference measurement channels specified in Annex FFS for each BLER test shall meet or exceed the specified value in Table 5.2.2.2.6_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.6_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.1-16.1 TDD | 40 / 30 | 16QAM, 0.54 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 1% (Note 1) | 1.4 + TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

5.2.2.2.7

## 5.2.2.2.8 2Rx TDD FR1 PDSCH pre-emption performance

### 5.2.2.2.8.0 Minimum conformance requirements

The performance requirements are specified in Table 5.2.2.2.8.0-3, with the addition of test parameters in Table 5.2.2.2.8.0-2 and the downlink physical channel setup according to Annex C.3.1.

The test purposes are specified in Table 5.2.2.2.8.0-1.

**Table 5.2.2.2.8.0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH pre-emption performance under 2 receive antenna conditions | 1-1 |

**Table 5.2.2.2.8.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDCCH configuration (Note 4) | Symbols with PDCCH | | 0, 1 |
| | DCI format | | 2_1 |
| | timeFrequencySet | | 14x1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Pre-emption configuration (Note 2) | Starting symbol (S) | | 3 |
| | Length (L) | | 2 |
| | Pre-emption periodicity and offset | Slots | 40/(1,12,23,34) (Note 3) |

| | | |
|---|---|---|
| Number of HARQ Processes | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | FR1.30-1 |
| Note 1: Void<br>Note 2: Interference modelled as random data on pre-empted REs.<br>Note 3: Pre-emption is scheduled with 10% probability within 20ms periodicity.<br>Note 4: In addition to PDCCH configuration in Table 5.2-1. | | |

**Table 5.2.2.2.8.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-2.5 TDD | 40 / 30 | 16QAM 0.48 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 12.5 |

The normative reference for this requirement is TS 38.101-4 [5], clause 5.2.2.2.8.

### 5.2.2.2.8_1 2Rx TDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Statistical significance in Annex G is FFS
- MU factor due to finite test time needs further study
- Test Procedure is incomplete

#### 5.2.2.2.8_1.1 Test purpose

To Verify the PDSCH pre-emption performance under 2 receive antenna conditions.

#### 5.2.2.2.8_1.2 Test applicability

Test 1-1 applies to all types of NR UE release 16 and forward supporting capability IE *pre-EmptIndication-DL-r16*.

#### 5.2.2.2.6_1.3 Test description

##### 5.2.2.2.6_1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.2.8.0-2 and Table 5.2.2.2.8.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On*, Test Mode *On* or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On, Test Mode* On, for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 5.2.2.2.8_1.3.3.

#### 5.2.2.2.8_1.3.2 Test procedure

1. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 5.2.2.2.8.0-3. The SS sends downlink MAC padding bits on the DL RMC.

2. SS transmits PDCCH DCI format 2_1 for int_RNTI with 10% probability to transmit the DL Pre-emption indication according to Table 5.2.2.2.8.0-2. In the time and frequency set indicated by PDCCH DCI format 2_1, SS stops transmission of PDSCH.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Table 5.2.2.2.8_1.3.4-1.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G clause FFS. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Table FFS in Annex G clause FFS.

#### 5.2.2.2.8_1.3.3 Message contents

##### 5.2.2.2.8_1.3.3_1 Message exceptions for SA

Same as 5.2.2.1.8_1.3.3_1

##### 5.2.2.1.8_1.3.3_2 Message exceptions for NSA

Same as 5.2.2.1.8_1.3.3_1

#### 5.2.2.2.8_1.3.4 Test requirement

Table 5.2.2.2.8.0-3 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A for each throughput test shall meet or exceed the specified value in Table 5.2.2.2.8_1.3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 5.2.2.2.8_1.3.4-1: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.2-2.5 TDD | 40 / 30 | 16QAM 0.48 | FR1.30-1 | TDLA30-10 | 2x2, ULA Low | 70 | 12.5 + TT |

# 6 CSI reporting requirements (Conducted requirements)

## 6.1 General

This Clause includes conducted requirements for the reporting of channel state information (CSI).

### 6.1.1 Applicability of requirements

#### 6.1.1.1 General

The minimum performance requirements are applicable to all FR1 operating bands defined in TS 38.101-1 [2].

The minimum performance requirements in Clause 6 are mandatory for UE supporting NR operation, except test cases listed in Clause 6.1.1.3, 6.1.1.4.

#### 6.1.1.2 Applicability of requirements for different number of RX antenna ports

The number of RX antenna ports for different RF operating bands is up to UE declaration.

The UE shall support 2 or 4 RX antenna ports for different RF operating bands. The operating bands, where 4 RX antenna ports shall be the baseline, are defined in Clause 7.2 of TS 38.101-1 [2]. The UE requirements applicability for UEs with different number of RX antenna ports is defined in Table 6.1.1.2-1.

**Table 6.1.1.2-1: Requirements applicability**

| Supported RX antenna ports | Test type | Test list |
|---|---|---|
| UE supports only 2RX | CQI | All tests in Clause 6.2.2 |
| | PMI | All tests in Clause 6.3.2 |
| | RI | All tests in Clause 6.4.2 |
| UE supports only 4RX or both 2RX and 4RX | CQI | All tests in Clause 6.2.3 |
| | PMI | All tests in Clause 6.3.3 |
| | RI | All tests in Clause 6.4.3 |

#### 6.1.1.3 Applicability of requirements for optional UE features

The performance requirements in Table 6.1.1.3-1 shall apply for UEs which support optional UE features with capability signalling only.

**Table 6.1.1.3-1: Requirements applicability for optional features with UE capability signalling**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| CQI table with target BLER of 10^-5New CQI table (cqi-TableAlt) | FR1 FDD | CQI | Clause 6.2.2.1.1.2<br>Clause 6.2.3.1.1.2 | |
| | FR1 TDD | CQI | Clause 6.2.2.2.1.2<br>Clause 6.2.3.2.1.2 | |
| Alternative 64QAM MCS table for PDSCH New 64QAM MCS table for PDSCH (*dl-64QAM-MCS-TableAlt*) | FR1 FDD | CQI | Clause 6.2.2.1.1.2<br>Clause 6.2.3.1.1.2 | |
| | FR1 TDD | CQI | Clause 6.2.2.2.1.2<br>Clause 6.2.3.2.1.2 | |

The performance requirements in Table 6.1.1.3-2 shall apply for UEs which support optional UE features only.

**Table 6.1.1.3-2: Requirements applicability for optional UE features**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| Support of Type II codebook (*CodebookParameters* contains *type2, supportedCSI-RS-ResourceList, parameterLx, amplitudeScalingType, amplitudeSubsetRestriction*) | FR1 FDD | PMI | Clause 6.3.2.1.5<br>Clause 6.3.3.1.5 | |
| | FR1 TDD | PMI | Clause 6.3.2.2.5<br>Clause 6.3.3.2.5 | |
| Support of Enhanced Type II codebook with at least 16 ports per CSI-RS resource(*codebookParametersAddition-r16* contains *etype2R1-r16, supportedCSI-RS-ResourceListAdd-r16, maxNumberTxPortsPerResource*) | FR1 FDD | PMI | Clause 6.3.2.1.6<br>Clause 6.3.3.1.6 | |
| | FR1 TDD | PMI | Clause 6.3.2.2.6<br>Clause 6.3.3.2.6 | |

## 6.1.1.4 Applicability of requirements for mandatory UE features with capability signalling

The performance requirements in Table 6.1.1.4-1 shall apply for UEs which support mandatory UE features with capability signalling only.

**Table 6.1.1.4-1: Requirements applicability for mandatory features with UE capability signalling**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| PDSCH MIMO layers (*maxNumberMIMO-LayersPDSCH*) | FR1 FDD | CQI | Clause 6.2.3.1.1.1 | The requirements apply only in case the PDSCH MIMO rank in the test case does not exceed UE PDSCH MIMO layers capability |
| | | PMI | Clause 6.3.3.1.2 | |
| | | RI | Clause 6.4.2.1<br>Clause 6.4.3.1 | |
| | FR1 TDD | CQI | Clause 6.2.3.2.1.1 | |
| | | PMI | Clause 6.3.3.2.2 | |
| | | RI | Clause 6.4.2.2<br>Clause 6.4.3.2 | |
| Supported maximum number of ports across all configured NZP-CSI-RS resources per CC (*maxConfigNumberPortsAcross NZP-CSI-RS-PerCC*) | FR1 FDD | PMI | Clause 6.3.2.1.1<br>Clause 6.3.2.1.2<br>Clause 6.3.3.1.1<br>Clause 6.3.3.1.2 | The requirements apply only in case the number of NZP-CSI-RS ports in the test case satisfies UE capability on maximum number of NZP-CSI-RS ports |
| | | RI | Clause 6.4.3.1 (Test 4) | |
| | FR1 TDD | PMI | Clause 6.3.2.2.1<br>Clause 6.3.2.2.2<br>Clause 6.3.3.2.1<br>Clause 6.3.3.2.2 | |
| | | RI | Clause 6.4.3.2 (Test 4) | |

## 6.1.1.5 Applicability of Channel Quality Indicator (CQI) reporting requirements for CA

### 6.1.1.5.1 Applicability and test rules for different duplex modes and SCS combinations

The applicability and test rules for different duplex modes and SCS combinations are defined in Table 6.1.1.5.1-1.

**Table 6.1.1.5.1-1: Applicability for different duplex modes and SCS combinations**

| Tests | PCell CC configuration |
|---|---|
| Test 1 in Clause 6.2A.3.1.1 | TDD CC if supported, otherwise FDD CC |
| Test 2 in Clause 6.2A.3.1.1 (NOTE 2) | Any of CCs |
| Test 3 in Clause 6.2A.3.1.1 | Any of CCs |
| NOTE 1: The test coverage can be considered fulfilled if UE passes one of the CC as PCell in Test 1.<br>NOTE 2: These scenarios are only tested for UEs which are not verified with Test 1 in Clause 6.2A.3.1.1. | |

### 6.1.1.5.2 Applicability and test rules for different CA configurations and bandwidth combination sets

The performance requirement for CA CQI tests in clause 6.2A are defined independent of CA configurations and bandwidth combination sets specified in clause 5.5A in TS 38.101-1 [6].

For UEs supporting multiple CA capabilities, test any one of the supported CA capabilities with largest aggregated CA bandwidth combination. The categorization of CA capability is specified in clause 5.1.1.7.1.

For UEs supporting multiple CA configurations from the selected CA capability, test any one of the supported CA configurations with largest aggregated CA bandwidth combination. For simplicity, the CA configuration refers to combination of CA configuration and bandwidth combination set.

A single uplink CC is configured for all tests.

### 6.1.1.5.3 Test coverage for different number of component carriers

For CA CQI tests specified in clause 6.2A, among all supported CA capabilities, if corresponding CA tests with the largest number of CCs supported by the UE are tested, the test coverage can be considered fulfilled without executing the CA tests with less than the largest number of CCs supported by the UE.

### 6.1.1.5.4 Applicability rule and antenna connection for CA tests with 4 RX

All the requirements specified in clause 6.2A for CA with 2 RX are applied for 4 RX capable UEs by connecting all 4 RX with data source from system simulator and reducing the signal power density by 3 dB compared to the signal power density for 2 RX in the test configurations.

## 6.1.2 Common test parameters

Parameters specified in Table 6.1.2-1 are applied for all test cases in this section unless otherwise stated.

**Table 6.1.2-1: Test parameters for CSI test cases**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 3) | RBs | 0 |
| | Subcarrier spacing | kHz | 15 or 30 |
| DL BWP configuration #1 | Cyclic prefix | | Normal |
| | RB offset | RBs | 0 |
| | Number of contiguous PRB | PRBs | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing |
| Active DL BWP index | | | 1 |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | 0,1 |
| | Number of PDCCH candidates and aggregation levels | | 1/AL8 |
| | DCI format | | 1_1 |
| | TCI state | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | Multi-path fading propagation conditions:<br>Single Panel Type I, Random per slot with equal probability of each applicable i1, i2 combination, and with REG bundling granularity for number of Tx larger than 1<br><br>Static propagation conditions:<br>Single Panel Type I, Random precoder chosen from precoder index 0 an 2, selection updated per slot |
| Cross carrier scheduling | | | Not configured |
| PDSCH configuration | Mapping type | | Type A |
| | *k0* | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | type 0 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| | DMRS ports indexes | | {1000} for Rank1<br>{1000,1001} for Rank2<br>{1000,1001,1002} for Rank3<br>{1000,1001,1002,1003} for Rank4 |
| | Number of PDSCH DMRS CDM group(s) without data | | 2 |
| PTRS configuration | Frequency density ($K_{PT-RS}$) | | N/A |
| | Time density ($L_{PT-RS}$) | | N/A |

| | | | | |
|---|---|---|---|---|
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | | 0 for CSI-RS resource 1,2,3,4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | | 4 for CSI-RS resource 1 and 3<br>8 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports ($X$) | | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2,3,4 |
| | Density ($\rho$) | | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | | slot | 15 kHz SCS: 20 for CSI-RS resource 1,2,3,4<br>30 kHz SCS: 40 for CSI-RS resource |
| | CSI-RS offset | | slot | 15 kHz SCS:<br>10 for CSI-RS resource 1 and 2<br>11 for CSI-RS resource 3 and 4<br><br>30 kHz SCS:<br>20 for CSI-RS resource 1 and 2<br>21 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| Number of HARQ Processes | | | | 4 For FDD<br>8 for TDD |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | | 2 for FDD<br>For FR1.30-1:<br>8 if mod(i,10) = 0<br>6 if mod(i,10) = 2<br>5 if mod(i,10) = 3<br>5 if mod(i,10) = 4<br>4 if mod(i,10) = 5<br>3 if mod(i,10) = 6<br>Where i is slot index per radio frame with 0~19 |
| Symbols for unused REs | | | | OCNG as specified in A.5 |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |
| Note 1: PDSCH is not scheduled on slots containing CSI-RS or slots which are not full DL.<br>Note 2: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission.<br>Note 3: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing. | | | | |

# 6.2 Reporting of Channel Quality Indicator (CQI)

## 6.2.1 1RX requirements (Void)

## 6.2.2 2RX requirements

### 6.2.2.1 FDD

#### 6.2.2.1.1 CQI reporting definition under AWGN conditions

The reporting accuracy of the channel quality indicator (CQI) under frequency non-selective conditions is determined by the reporting variance and the BLER performance using the transport format indicated by the reported CQI median. The purpose is to verify that the reported CQI values are in accordance with the CQI definition given in TS 38.214 [12]. To account for sensitivity of the input SNR the reporting definition is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB

##### 6.2.2.1.1.1 2Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA

###### 6.2.2.1.1.1.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

###### 6.2.2.1.1.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

###### 6.2.2.1.1.1.3 Minimum requirement for periodic CQI reporting

For the parameters specified in Table 6.2.2.1.1.1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to 0.1.

**Table 6.2.2.1.1.1.3-1: CQI reporting definition test**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Duplex Mode | | | FDD | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| SNR | | dB | 8 | 9 | 14 | 15 |
| Propagation channel | | | AWGN | | | |
| Antenna configuration | | | 2×2 with static channel specified in Annex B.1 | | | |
| Beamforming Model | | | As specified in Section Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| CSI-reportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 5/0 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 010000 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 8 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-2 | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.1.1.1.

#### 6.2.2.1.1.1.4 Test Description

##### 6.2.2.1.1.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7 for TE diagram and section A.3.2.3 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.2.1.1.1.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.2.1.1.1.4.3.

##### 6.2.2.1.1.1.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.2.1.1.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and [1800] or more of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

   For the filtered ACK and NACK responses if the ratio (NACK / ACK + NACK) $\leq$ 0.1 then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

If the ratio (NACK /ACK + NACK) > 0.1

then pass the UE for this test and go to step 9, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) ≤ 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. Repeat step 1 to 8 for Test2.

#### 6.2.2.1.1.1.4.3 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

#### 6.2.2.1.1.1.4.3_1 Message exceptions for SA

**Table 6.2.2.1.1.1.4.4_1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | Periodic | |
| periodic SEQUENCE { | | | |
| reportSlotConfig | CSI-ReportPeriodicityAndOffset | 5/1 | |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE{ | | | |
| Subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.2.2.1.1.1.4.4_1-2: *CodebookConfig***

| Derivation Path: TS38.508-1 [6], Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 010000 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

#### 6.2.2.1.1.1.4.3_2 Message exceptions for NSA

Same as specified in 6.2.2.1.1.1.4.4_1.

### 6.2.2.1.1.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.2.1.1.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.2.1.1.2 2Rx FDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Whether to adopt minimum sample for pass/fail verdict needs further study.

### 6.2.2.1.1.2.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of $10^{-5}$ falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

### 6.2.2.1.1.2.2 Test Applicability

This test applies to all types of NR UE release 16 supporting *cqi-TableAlt*.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and *cqi-TableAlt*.

### 6.2.2.1.1.2.3 Minimum requirement for periodic CQI reporting with Table 3

For the parameters specified in Table 6.2.2.1.1.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to $10^{-5}$, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than $10^{-5}$. If the PDSCH BLER using the transport format indicated by the median CQI is greater than $10^{-5}$, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to $10^{-5}$.

c) The reported CQI value according to the reference channel shall be $\geq 1$.

**Table 6.2.2.1.1.2.3-1: CQI reporting test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| Bandwidth | | MHz | 10 | |
| Duplex Mode | | | FDD | |
| Subcarrier spacing | | kHz | 15 | |
| SNR | | dB | 1 | 2 |
| Propagation channel | | | AWGN | |
| Antenna configuration | | | 1×2 with static channel specified in Annex B.1 | |
| Beamforming Model | | | As specified in Annex B.4.1 | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | |
| | Number of CSI-RS ports (*X*) | | 4 | |
| | CDM Type | | FD-CDM2 | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | |

| | | | |
|---|---|---|---|
| | CSI-RS periodicity and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 1 |
| | CDM Type | | No CDM |
| | Density (ρ) | | 3 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 1,(0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 3 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| Csi-ReportingBand | | | 1111111 |
| CSI-Report periodicity and offset | | slot | 5/0 |
| aperiodicTriggeringOffset | | | Not configured |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUCCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-4, TBS.4-1 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.1.1.2.

#### 6.2.2.1.1.2.4 Test Description

##### 6.2.2.1.1.2.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.2.1.1.2.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.2.1.1.2.4.3.

#### 6.2.2.1.1.2.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.1.1.2.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 5000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and 4500 or more of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   For the filtered ACK and NACK responses if the ratio (NACK / (ACK + NACK)) $\leq 10^{-5}$ then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   If the ratio (NACK /(ACK + NACK)) > $10^{-5}$

   then pass the UE for this test, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   If the ratio (NACK /ACK + NACK) $\leq 10^{-5}$

   then pass the UE for this test, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

#### 6.2.2.1.1.2.4.3 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

6.2.2.1.1.2.4.3_1 Message exceptions for SA

**Table 6.2.2.1.1.2.4.3_1-1: NZP CSI-RS-ResourceMapping**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | K0=0, row 1, | 1Tx test cases |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| CDM Type | noCDM | | |
| density CHOICE { | | | |
| three | NULL | | |
| } | | | |
| } | | | |

**Table 6.2.2.1.1.2.4.4_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-IM-Resource ::= SEQUENCE { | | | |
| periodicityAndOffset SEQUENCE { | | | |
| slot5 | 0 | | |
| } | | | |
| } | | | |

**Table 6.2.2.1.1.2.4.3_1-2: *NZP-CSI-RS-Resource***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| slot5 | 1 | | |
| } | | | |
| } | | | |

**Table 6.2.2.1.1.2.4.4_1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| cqi-Table | table3 | | |
| } | | | |

**Table 6.2.2.1.1.2.4.4_1-4: *CodebookConfig***

| Derivation Path: TS38.508-1 [6], Table 5.4.2.4-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 000001 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

6.2.2.1.1.2.4.3_2 Message exceptions for NSA

Same as specified in 6.2.2.1.1.2.4.3_1.

6.2.2.1.1.2.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.2.1.1.2.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.2.1.2 CQI reporting under fading conditions

The reporting accuracy of CQI under frequency non-selective fading conditions is determined by the reporting variance, the relative increase of the throughput obtained when the transport format is indicated by the reported CQI compared to the throughput obtained when a fixed transport format is configured according to the reported median CQI, and a minimum BLER using the transport formats indicated by the reported CQI. To account for sensitivity of the input SNR the sub-band CQI reporting under frequency selective fading conditions is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of [1] dB.

### 6.2.2.1.2.1 2Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA

6.2.2.1.2.1.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

6.2.2.1.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

6.2.2.1.2.1.3 Minimum conformance requirements

For the parameters specified in Table 6.2.2.1.2.1.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A CQI index not in the set {median CQI -1, median CQI, median CQI +1} shall be reported at least $\alpha$% of the time where $\alpha$% is specified in Table 6.2.2.1.2.1-2;

b) The ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.2.1.2.1.3-2;

c) When transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater than or equal to 0.02.

**Table 6.2.2.1.2.1.3-1: Wideband CQI reporting test under frequency non-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| Duplex Mode | | | FDD | | | |
| SNR | | dB | 6 | 7 | 12 | 13 |
| Propagation channel | | | TDLA30-5 | | | |
| Antenna configuration | | | 2×2 | | | |
| Correlation configuration | | | ULA high | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| Csi-ReportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 5/0 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 8 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-1 | | | |

**Table 6.2.2.1.2.1.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.1.2.1.

### 6.2.2.1.2.1.4 Test description

#### 6.2.2.1.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and Figure A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.2.2.1.2.1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with Connected without release On, Test Mode On or EN-DC, DC bearer MCG and SCG, Connected without release On, Test Mode On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.2.1.2.1.4.3.

#### 6.2.2.1.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.1.2.1.5-1.
2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 6000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.
3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as Median CQI value.
4. If Median CQI value is not equal to 1 or 15 and 1200 ($\alpha$%)or more of the wideband CQI values are outside the range (Median CQI - 1) ≤ Median CQI ≤ (Median CQI + 1) then continue with step 5, otherwise go to step 7.
5. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the Median CQI value from step 3 and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. Measure the average throughput according to Annex G.3.3 and G.3.4.

   Declare the throughput as $t_{mediar}$

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the UE's reported wideband CQI value. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data, record the BLER (NACK / ACK + NACK) and measure the average throughput according to Annex G.3.3 and G.3.4. Declare the throughput as t.

   If the recorded BLER $\geq$ 0.02 and t / $t_{media} \geq \gamma$ then pass the UE for this test and go to step 8.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. Repeat step 1 to 7, with test conditions according to the table 6.2.2.1.2.1.5 -1, for Test2 as appropriate.

#### 6.2.2.1.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

#### 6.2.2.1.2.1.4.3_1 Message exceptions for SA

**Table 6.2.2.1.2.1.4.3_1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | | |
| } | | | |
| nrofPorts | p2 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.2.2.1.2.1.4.3_1-2: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.2.2.1.2.1.4.3_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.2.2.1.2.1.4.3_1-4: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots5 | 1 | | |
| } | | | |

**Table 6.2.2.1.2.1.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-CodebookSubsetRestriction | 000001 | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | | |

**Table 6.2.2.1.2.1.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| reportSlotConfig | CSI-ReportPeriodicityAndOffset | | |
| pucch-CSI-ResourceList | PUCCH-CSI-Resource | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| subbandSize | value2 | | |
| } | | | |

### 6.2.2.1.2.1.4.3_2 Message exceptions for NSA

Same as in 6.2.2.1.2.1.4.3_1.

### 6.2.2.1.2.1.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.2.1.2.1.4.2.

**Table 6.2.2.1.2.1.5-1: Wideband CQI reporting test under frequency non-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| Duplex Mode | | | FDD | | | |
| SNR | | dB | 6 | 7 | 12 | 13 |
| Propagation channel | | | TDLA30-5 | | | |
| Antenna configuration | | | 2×2 | | | |
| Correlation configuration | | | ULA high | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| Csi-ReportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 5/5 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 8 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-1 | | | |

**Table 6.2.2.1.2.1.5-2: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.05 -TT | 1.05 -TT |
| Note1 : TT = 0.01 | | |

### 6.2.2.1.2.2 2Rx FDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA

#### 6.2.2.1.2.2.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

#### 6.2.2.1.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

#### 6.2.2.1.2.2.3 Minimum conformance requirements

The accuracy of sub-band channel CQI reporting under the frequency-selective fading conditions is determined by a double-sided percentile of the reported differential CQI offset level 0 per sub-band, and the relative increase of the throughput obtained when transmitting the transport format indicated by the corresponding reported sub-band CQI on a randomly selected sub-band among the sub-bands with the highest reported differential CQI offset level compared to the throughput when transmitting a fixed transport format according to the wideband CQI median on a randomly selected sub-band among all the sub-bands.

For the parameters specified in Table 6.2.2.1.2.2.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A sub-band differential CQI offset level of 0 shall be reported at least $\alpha$% of the time but less than $\beta$% of the time for each sub-band, where $\alpha$ and $\beta$ are specified in Table 6.2.2.1.2.2.3-2.

b) The ratio of the throughput obtained when transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level and that obtained when transmitting the transport format indicated by the reported wideband CQI median on a randomly selected sub-band among all the sub-bands shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.2.1.2.2.3-2.

c) When transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level, the average BLER for the indicated transport format shall be greater than or equal to 0.02.

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each TTI for FDD.

**Table 6.2.2.1.2.2.3-1: Sub-band CQI reporting test under frequency-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| Duplex Mode | | | FDD | | | |
| SNR | | dB | 8 | 9 | 14 | 15 |
| Propagation channel | | | Two tap model specified in Annex B.2.4 with $a$=1, $f_D$ = 5Hz, and $\tau_d$=0.45μs | | | |
| Antenna configuration | | | 2×2 | | | |
| Correlation configuration | | | As per Annex B.1 | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Aperiodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Subband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report interval and offset | | slot | Not configured | | | |
| Aperiodic Report Slot Offset | | | 5 | | | |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 | | | |
| reportTriggerSize | | | 1 | | | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUSCH | | | |

| | | |
|---|---|---|
| CQI/RI/PMI delay | ms | 8 |
| Maximum number of HARQ transmission | | 1 |
| Measurement channel | | As specified in Table A.4-2, TBS.2-5 |

**Table 6.2.2.1.2.2.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.1.2.2.

### 6.2.2.1.2.2.4 Test description

#### 6.2.2.1.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.6.1 for TE diagram and Figure A.3.2.3.1 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.2.2.1.2.2.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with Connected without release On, Test Mode On or EN-DC, DC bearer MCG and SCG, Connected without release On, Test Mode On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.2.1.2.2.4.3.

#### 6.2.2.1.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.1.2.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband and subband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband CQI reports and full-size subband CQI reports for each full-size subband have been gathered. In this process the SS collects sub-band CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as subband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. For each subband, if subband differential CQI offset level of 0 is reported, at least $\alpha$% but less than $\beta$% of 6000 full-size subband CQI report, then continue to step 5, otherwise, go to step 7.

5. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC in an each available downlink transmission instance randomly selected full-size subband using the transport format according to the wideband median CQI value regardless of UE wideband or subband CQI report. Note that each full-size subband shall be selected in the equal probability. The SS sends downlink MAC padding bits on the DL RMC. Measure the average throughput according to Annex G.3.3 and G.3.4. Declare the throughput as $t_{median}$.

6. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the highest UE reported full-size subband CQI value in one full-size subband selected among the sub-bands in which UE report the highest full-size subband CQI. Subband differential CQI offset level is selected from {0, 1, 2, -1}. Note that the SS shall send PDSCH in the same full-size subband until next UE report is available. In case when same full-size subbands are reported subsequently as subbands with highest full-size subband CQI, the SS shall select for transmission a different subband with respect to the last selection. The SS sends downlink MAC padding bits on the DL RMC Measure the average throughput and (NACK /(ACK + NACK)) according to Annex G.3.3 and G.3.4. Declare the throughput as $t_{subband}$. If the ratio ($t_{subband}$ / $t_{median}$) $\geq \gamma$ and (NACK /(ACK + NACK)) $\geq$ 0.02, pass the UE and go to step 8. Otherwise, go to step 7.

7. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 6) for the other SNR point as appropriate. Otherwise fail the UE.

8. If both tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the table 6.2.2.1.2.2.3-1 for the other test as appropriate.

#### 6.2.2.1.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

#### 6.2.2.1.2.2.4.3_1 Message exceptions for SA

**Table 6.2.2.1.2.2.4.3_1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | | |
| } | | | |
| nrofPorts | p2 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.2.2.1.2.2.4.3_1-2: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.2.2.1.2.2.4.3_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.2.2.1.2.2.4.3_1-4: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots5 | 1 | | |
| } | | | |

**Table 6.2.2.1.2.2.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-CodebookSubsetRestriction | 000001 | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | | |

**Table 6.2.2.1.2.2.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportFreqConfiguration SEQUENCE { | | | |
| cqi-FormatIndicator | subbandCQI | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |

6.2.2.1.2.2.4.3_2 Message exceptions for NSA

Same as in 6.2.2.1.2.2.4.3_1.

6.2.2.1.2.2.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.2.1.2.2.4.2.

**Table: 6.2.2.1.2.2.5-1: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.04 | 1.04 |

## 6.2.2.2 TDD

### 6.2.2.2.1 CQI Reporting definition under AWGN conditions

#### 6.2.2.2.1.1 2Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA

##### 6.2.2.2.1.1.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

##### 6.2.2.2.1.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 6.2.2.2.1.1.3 Minimum requirement for periodic CQI reporting

The purpose of the requirements is to verify that the reported CQI values are in accordance with the CQI definition given in TS 38.214 [12]. The reporting accuracy of CQI under AWGN condition is determined by the reporting variance and BLER performance using the transport format indicated by the reported CQI median.

For the parameters specified in Table 6.2.2.2.1.1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to 0.1.

**Table 6.2.2.2.1.1.3-1: CQI reporting definition test**

| Parameter | | Unit | Test 1 | Test 2 | | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | | | |
| Subcarrier spacing | | kHz | 30 | | | |
| Duplex Mode | | | TDD | | | |
| TDD UL-DL pattern | | | FR1.30-1 | | | |
| SNR | | dB | 8 | 9 | 14 | 15 |
| Propagation channel | | | AWGN | | | |
| Antenna configuration | | | 2×2 with static channel specified in Annex B.1 | | | |
| Beamforming Model | | | As specified in Section Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 10/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 16 | | | |
| CSI-reportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 10/9 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 010000 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 9.5 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-4 | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.2.1.1.

#### 6.2.2.2.1.1.4 Test Description

##### 6.2.2.2.1.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7 for TE diagram and section A.3.2.3 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.2.2.1.1.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.2.2.1.1.4.3.

##### 6.2.2.2.1.1.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.2.1.1.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and [1800] or more of the wideband CQI values are in the range (Median CQI - 1) ≤ Median CQI ≤ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

   For the filtered ACK and NACK responses if the ratio (NACK / ACK + NACK) ≤ 0.1 then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends

downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

If the ratio (NACK /ACK + NACK) > 0.1

then pass the UE for this test and go to step 9, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) ≤ 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. Repeat step 1 to 8 for Test2.

#### 6.2.2.2.1.1.4.4 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

#### 6.2.2.2.1.1.4.4_1 Message exceptions for SA

**Table 6.2.2.2.1.1.4.4_1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | Periodic | |
| periodic SEQUENCE { | | | |
| reportSlotConfig | CSI-ReportPeriodicityAndOffset | 10/1 | |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| | widebandCQI | | |
| | widebandPMI | | |
| csi-ReportingBand CHOICE{ | | | |
| Subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.2.2.2.1.1.4.4_1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 010000 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

### 6.2.2.2.1.1.4.4_2 Message exceptions for NSA

Same as specified in 6.2.2.2.1.1.4.4_1.

### 6.2.2.2.1.1.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.2.2.1.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.2.2.1.2 2Rx TDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Whether to adopt minimum sample for pass/fail verdict needs further study.

### 6.2.2.2.1.2.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of $10^{-5}$ falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

### 6.2.2.2.1.2.2 Test Applicability

This test applies to all types of NR UE release 16 supporting *cqi-TableAlt*.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and *cqi-TableAlt*.

### 6.2.2.2.1.2.3 Minimum requirement for periodic CQI reporting with Table 3

For the parameters specified in Table 6.2.2.2.1.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to $10^{-5}$, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than $10^{-5}$. If the PDSCH BLER using the transport format indicated by the median CQI is greater than $10^{-5}$, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to $10^{-5}$.

c) The reported CQI value according to the reference channel shall be ≥ 1.

**Table 6.2.2.2.1.2.3-1: CQI reporting test parameters**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD UL-DL pattern | | | FR1.30-1 |
| SNR | | dB | 1 \| 2 |
| Propagation channel | | | AWGN |
| Antenna configuration | | | 1×2 with static channel specified in Annex B.1 |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 |
| | CSI-RS periodicity and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 1 |
| | CDM Type | | No CDM |
| | Density (ρ) | | 3 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 1,(0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 1 |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 3 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| Csi-ReportingBand | | | 1111111 |
| CSI-Report periodicity and offset | | slot | 10/9 |
| aperiodicTriggeringOffset | | | Not configured |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUCCH |
| CQI/RI/PMI delay | | ms | 9.5 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-4, TBS.4-2 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.2.1.2.

#### 6.2.2.2.1.2.4 Test Description

##### 6.2.2.2.1.2.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.2 for TE diagram and section A.3.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.2.2.1.2.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.2.2.1.2.4.3.

##### 6.2.2.2.1.2.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.2.1.2.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 5000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 10 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and 4500 or more of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   For the filtered ACK and NACK responses if the ratio (NACK / (ACK + NACK)) $\leq 10^{-5}$ then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the

ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

If the ratio (NACK / (ACK + NACK)) > $10^{-5}$

then pass the UE for this test, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   If the ratio (NACK / ACK + NACK) $\leq 10^{-5}$

   then pass the UE for this test, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

#### 6.2.2.2.1.2.4.3 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

#### 6.2.2.2.1.2.4.3_1 Message exceptions for SA

**Table 6.2.2.2.1.2.4.3_1-1: NZP CSI-RS-ResourceMapping**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0000 | K0=0, row 1, | 1Tx test cases |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 1 | | |
| CDM Type | noCDM | | |
| density CHOICE { | | | |
| three | NULL | | |
| } | | | |
| } | | | |

**Table 6.2.2.2.1.2.4.3_1-2: *NZP-CSI-RS-Resource***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| NZP-CSI-RS-Resource ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| slot10 | 1 | | |
| } | | | |
| } | | | |

**Table 6.2.2.2.1.2.4.4_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-IM-Resource ::= SEQUENCE { | | | |
| periodicityAndOffset SEQUENCE { | | | |
| slot10 | 1 | | |
| } | | | |
| } | | | |

**Table 6.2.2.2.1.2.4.4_1-4: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-12 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| cqi-Table | table3 | | |
| } | | | |

**Table 6.2.2.2.1.2.4.4_1-5: *CodebookConfig***

| Derivation Path: TS38.508-1 [6], Table 5.4.2.4-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 000001 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

6.2.2.2.1.2.4.3_2 Message exceptions for NSA

Same as specified in 6.2.2.2.1.2.4.3_1.

6.2.2.2.1.2.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.2.2.1.2.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.2.2.2 CQI reporting under fading conditions

### 6.2.2.2.2.1 2Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA

#### 6.2.2.2.2.1.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

#### 6.2.2.2.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

#### 6.2.2.2.2.1.3 Minimum conformance requirements

For the parameters specified in Table 6.2.2.2.2.1.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A CQI index not in the set {median CQI -1, median CQI, median CQI +1} shall be reported at least $\alpha$% of the time where $\alpha$% is specified in Table 6.2.2.2.2.1.3-2;

b) The ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.2.2.2.1.3-2;

c) When transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater than or equal to 0.02.

**Table 6.2.2.2.2.1.3-1: Wideband CQI reporting test under frequency non-selective fading conditions**

| Parameter | | Unit | Test 1 | Test 2 | | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | | | |
| Subcarrier spacing | | kHz | 30 | | | |
| Duplex Mode | | | TDD | | | |
| TDD UL-DL pattern | | | FR1.30-1 | | | |
| SNR | | dB | 6 | 7 | 12 | 13 |
| Propagation channel | | | TDLA30-5 | | | |
| Antenna configuration | | | 2×2 | | | |
| Correlation configuration | | | ULA high | | | |
| Beamforming Model | | | As specified in AnnexB.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 10/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | | | |
| CSI-IM configuration | CSI-RS resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 16 | | | |
| Csi-ReportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 10/9 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 9.5 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-1, TBS.2-3 | | | |

**Table 6.2.2.2.2.1.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.2.2.1.

### 6.2.2.2.2.1.4 Test description

#### 6.2.2.2.2.1.4.1 Initial conditions

Same initial conditions as specified in clause 6.2.2.1.2.1.4.1 with the following exceptions:

Instead of Table 6.2.2.1.2.1.3-1 → use Table 6.2.2.2.2.1.3-1.

#### 6.2.2.2.2.1.4.2 Test procedure

Same test procedure as specified in clause 6.2.2.1.2.1.4.2 with the following exceptions:

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 6000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 10 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

Instead of Table 6.2.2.1.2.1.5-1 → use Table 6.2.2.2.2.1.3-1.

#### 6.2.2.2.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

#### 6.2.2.2.2.1.4.3_1 Message exceptions for SA

Same as 6.2.2.1.2.1.4.3_1 with following exceptions:

**Table 6.2.2.2.2.1.4.3_1-1: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots10 | 1 | | |
| } | | | |

#### 6.2.2.2.2.1.4.3_2 Message exceptions for NSA

Same as 6.2.2.2.2.1.4.3_1.

### 6.2.2.2.2.1.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.2.2.2.1.4.2.

**Table 6.2.2.2.2.1.5-1: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 20 | 20 |
| $\gamma$ | 1.04 | 1.04 |
| | | |

#### 6.2.2.2.2.2 2Rx TDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA

##### 6.2.2.2.2.2.1 Test purpose

To verify the variance of the subband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2 % for the indicated transport format.

##### 6.2.2.2.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

##### 6.2.2.2.2.2.3 Minimum conformance requirements

The purpose of the requirements is to verify that the preferred sub-bands can be used for frequency-selective scheduling under the frequency-selective fading conditions.

The accuracy of sub-band channel CQI reporting under the frequency-selective fading conditions is determined by a double-sided percentile of the reported differential CQI offset level 0 per sub-band, and the relative increase of the throughput obtained when transmitting the transport format indicated by the corresponding reported sub-band CQI on a randomly selected sub-band among the sub-bands with the highest reported differential CQI offset level compared to the throughput when transmitting a fixed transport format according to the wideband CQI median on a randomly selected sub-band among all the sub-bands.

For the parameters specified in Table 6.2.2.2.2.2.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A sub-band differential CQI offset level of 0 shall be reported at least $\alpha$% of the time but less than $\beta$% of the time for each sub-band, where $\alpha$ and $\beta$ are specified in Table 6.2.2.2.2.2.3-2;

b) The ratio of the throughput obtained when transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level and that obtained when transmitting the transport format indicated by the reported wideband CQI median on a randomly selected sub-band among all the sub-bands shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.2.2.2.2.3-2;

c) When transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level, the average BLER for the indicated transport format shall be greater than or equal to 0.02.

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each available downlink transmission instance for TDD.

**Table 6.2.2.2.2.2.3-1: Sub-band CQI reporting test under frequency-selective fading conditions**

| Parameter | | Unit | Test 1 | Test 2 | | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | | | |
| Subcarrier spacing | | kHz | 30 | | | |
| Duplex Mode | | | TDD | | | |
| TDD UL-DL pattern | | | FR1.30-1 | | | |
| SNR | | dB | 8 | 9 | 14 | 15 |
| Propagation channel | | | Two tap model specified in Annex B.2.4 with $a$=1, $f_D$ = 5Hz, and $\tau_d$=0.1125μs | | | |
| Antenna configuration | | | 2×2 | | | |
| Correlation configuration | | | As per Annex B.1 | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 10/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 | | | |
| ReportConfigType | | | Aperiodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Subband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 16 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report interval and offset | | slot | Not configured | | | |
| Aperiodic Report Slot Offset | | | 8 | | | |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 | | | |
| reportTriggerSize | | | 1 | | | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |

| | | |
|---|---|---|
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 9.5 |
| Maximum number of HARQ transmission | | 1 |
| Measurement channel | | As specified in Table A.4-2, TBS.2-6 |

**Table 6.2.2.2.2.2.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.2.2.2.2.

### 6.2.2.2.2.2.4 Test description

#### 6.2.2.2.2.2.4.1 Initial conditions

Same initial conditions as specified in clause 6.2.2.1.2.2.4.1 with the following exceptions:

Instead of Table 6.2.2.1.2.2.3-1 → use Table 6.2.2.2.2.2.3 -1.

Instead of clause 6.2.2.1.2.2.4.3 → use clause 6.2.2.2.2.2.4.3.

#### 6.2.2.2.2.2.4.2 Test procedure

Same test procedure as specified in clause 6.2.2.1.2.2.4.2 with the following exceptions:

Instead of Table 6.2.2.1.2.2.3-1 → use Table 6.2.2.2.2.2.3-1.

#### 6.2.2.2.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

##### 6.2.2.2.2.2.4.3_1 Message exceptions for SA

**Table 6.2.2.2.2.2.4.3_1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | | |
| } | | | |
| nrofPorts | p2 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.2.2.2.2.2.4.3_1-2: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.2.2.2.2.2.4.3_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.2.2.2.2.2.4.3_1-4: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots10 | 1 | | |
| } | | | |

**Table 6.2.2.2.2.2.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-CodebookSubsetRestriction | 000001 | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | | |

**Table 6.2.2.2.2.2.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportFreqConfiguration SEQUENCE { | | | |
| cqi-FormatIndicator | subbandCQI | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |

### 6.2.2.2.2.2.4.3_2 Message exceptions for NSA

Same as in 6.2.2.2.2.2.4.3_1.

#### 6.2.2.2.2.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.2.2.2.4.2.

**Table 6.2.2.2.2.5-1: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.04 | 1.04 |

## 6.2.3 4RX requirements

### 6.2.3.1 FDD

#### 6.2.3.1.1 CQI reporting definition under AWGN conditions

The reporting accuracy of the channel quality indicator (CQI) under frequency non-selective conditions is determined by the reporting variance and the BLER performance using the transport format indicated by the reported CQI median. The purpose is to verify that the reported CQI values are in accordance with the CQI definition given in TS 38.214 [12]. To account for sensitivity of the input SNR the reporting definition is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB

##### 6.2.3.1.1.1 4Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA

###### 6.2.3.1.1.1.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

###### 6.2.3.1.1.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

###### 6.2.3.1.1.1.3 Minimum requirement for periodic CQI reporting

For the parameters specified in Table 6.2.3.1.1.1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to 0.1.

**Table 6.2.3.1.1.1.3-1: CQI reporting definition test**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| Duplex Mode | | | FDD | | | |
| SNR | | dB | 5 | 6 | 11 | 12 |
| Propagation channel | | | AWGN | | | |
| Antenna configuration | | | 2×4 with static channel specified in Annex B.1 | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 5/0 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 010000 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 8 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-2 | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.1.1.1.

#### 6.2.3.1.1.1.4 Test Description

##### 6.2.3.1.1.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.5 for TE diagram and section A.3.2.5 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.3.1.1.1.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.3.1.1.1.4.3.

##### 6.2.3.1.1.1.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.2.1.1.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and [1800] or more of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

   For the filtered ACK and NACK responses if the ratio (NACK / ACK + NACK) $\leq$ 0.1 then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

If the ratio (NACK /ACK + NACK) > 0.1

then pass the UE for this test and go to step 9, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) ≤ 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. Repeat step 1 to 8 for Test2.

#### 6.2.3.1.1.1.4.4 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

#### 6.2.3.1.1.1.4.4_1 Message exceptions for SA

Same as specified in clause 6.2.2.1.1.1.4.4_1

#### 6.2.3.1.1.1.4.4_2 Message exceptions for NSA

Same as specified in clause 6.2.3.1.1.1.4.4_1.

### 6.2.3.1.1.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.3.1.1.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.3.1.2 4Rx FDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Whether to adopt minimum sample for pass/fail verdict needs further study.

### 6.2.3.1.2.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of $10^{-5}$ falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

### 6.2.3.1.2.2 Test Applicability

This test applies to all types of NR UE release 16 supporting *cqi-TableAlt*.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and *cqi-TableAlt*.

### 6.2.3.1.2.3 Minimum requirement for periodic CQI reporting with Table 3

For the parameters specified in Table 6.2.3.1.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to $10^{-5}$, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than $10^{-5}$. If the PDSCH BLER using the transport format indicated by the median CQI is greater than $10^{-5}$, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to $10^{-5}$.

c) The reported CQI value according to the reference channel shall be $\geq 1$.

**Table 6.2.3.1.1.2.3-1: CQI reporting test parameters**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| SNR | | dB | -2 \| -1 |
| Propagation channel | | | AWGN |
| Antenna configuration | | | 1×4 with static channel specified in Annex B.1 |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 |
| | CSI-RS periodicity and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 1 |
| | CDM Type | | No CDM |
| | Density (ρ) | | 3 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 1,(0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 1 |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 3 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| Csi-ReportingBand | | | 1111111 |
| CSI-Report periodicity and offset | | slot | 5/0 |
| aperiodicTriggeringOffset | | | Not configured |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUCCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-4, TBS.4-1 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.1.1.2.

#### 6.2.3.1.1.2.4 Test Description

##### 6.2.3.1.1.2.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.3.1.1.2.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.3.1.1.2.4.3.

##### 6.2.3.1.1.2.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.3.1.1.2.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 5000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and 4500 or more of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   For the filtered ACK and NACK responses if the ratio (NACK / (ACK + NACK)) $\leq 10^{-5}$ then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the

ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

If the ratio (NACK /(ACK + NACK)) > $10^{-5}$

then pass the UE for this test, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   If the ratio (NACK /ACK + NACK) $\leq 10^{-5}$

   then pass the UE for this test, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

#### 6.2.3.1.1.2.4.3 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

#### 6.2.3.1.1.2.4.3_1 Message exceptions for SA

Same as specified in clause 6.2.2.1.1.2.4.3_1.

#### 6.2.3.1.1.2.4.3_2 Message exceptions for NSA

Same as specified in 6.2.3.1.1.2.4.3_1.

#### 6.2.3.1.1.2.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.3.1.1.2.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

### 6.2.3.1.2 CQI reporting definition under fading conditions

#### 6.2.3.1.2.1 4Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA

#### 6.2.3.1.2.1.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

#### 6.2.3.1.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

#### 6.2.3.1.2.1.3 Minimum conformance requirements

The purpose of the requirements is to verify that the UE is tracking the channel variations and selecting the largest transport format possible according to the prevailing channel state for the frequency non-selective scheduling.

The reporting accuracy of CQI under frequency non-selective fading conditions is determined by the reporting variance, the relative increase of the throughput obtained when the transport format is indicated by the reported CQI compared to

the throughput obtained when a fixed transport format is configured according to the reported median CQI, and a minimum BLER using the transport formats indicated by the reported CQI.

For the parameters specified in Table 6.2.3.1.2.1.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A CQI index not in the set {median CQI -1, median CQI, median CQI +1} shall be reported at least $\alpha$% of the time where $\alpha$% is specified in Table 6.2.3.1.2.1.3-2;

b) The ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.3.1.2.1.3-2;

c) When transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater than or equal to 0.02.

**Table 6.2.3.1.2.1.3-1: Wideband CQI reporting test under frequency non-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| Duplex Mode | | | FDD | | | |
| SNR | | dB | 3 | 4 | 9 | 10 |
| Propagation channel | | | TDLA30-5 | | | |
| Antenna configuration | | | 2×4 | | | |
| Correlation configuration | | | XP High | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 5/0 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 8 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-1 | | | |

**Table 6.2.3.1.2.1.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 5 | 5 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.1.2.1.

### 6.2.3.1.2.1.4 Test description

#### 6.2.3.1.2.1.4.1 Initial conditions

Same initial conditions as specified in clause 6.2.2.1.2.1.4.1 with the following exceptions:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.

Instead of Table 6.2.2.1.2.1.3-1→ use Table 6.2.3.1.2.1.3-1.

#### 6.2.3.1.2.1.4.2 Test procedure

Same test procedure as specified in clause 6.2.2.1.2.1.4.2 with the following exceptions:

4. If Median CQI value is not equal to 1 or 15 and 300 ($\alpha$%)or more of the wideband CQI values are outside the range (Median CQI - 1) ≤ Median CQI ≤ (Median CQI + 1) then continue with step 5, otherwise go to step 7.

Instead of Table 6.2.2.1.2.1.5-1 → use Table 6.2.3.1.2.1.3-1.

#### 6.2.3.1.2.1.4.3 Message contents

Same message contests as specified in clause 6.2.2.1.2.1.4.3 with the following exceptions:

##### 6.2.3.1.2.1.4.3_1 Message exceptions for SA

**Table 6.2.3.1.2.1.4.3_1-1: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots5 | 1 | | |
| } | | | |

##### 6.2.3.1.2.1.4.3_2 Message exceptions for NSA

Same as in 6.2.3.1.2.1.4.3_1.

### 6.2.3.1.2.1.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.3.1.2.1.4.2.

**Table 6.2.3.1.2.1.3-1: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 5 | 5 |
| $\gamma$ | 1.04 | 1.04 |

#### 6.2.3.1.2.2 4Rx FDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA

##### 6.2.3.1.2.2.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

##### 6.2.3.1.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

##### 6.2.3.1.2.2.3 Minimum conformance requirements

The purpose of the requirements is to verify that the preferred sub-bands can be used for frequency-selective scheduling under the frequency-selective fading conditions.

The accuracy of sub-band channel CQI reporting under the frequency-selective fading conditions is determined by a double-sided percentile of the reported differential CQI offset level 0 per sub-band, and the relative increase of the throughput obtained when transmitting the transport format indicated by the corresponding reported sub-band CQI on a randomly selected sub-band among the sub-bands with the highest reported differential CQI offset level compared to the throughput when transmitting a fixed transport format according to the wideband CQI median on a randomly selected sub-band among all the sub-bands.

For the parameters specified in Table 6.2.3.1.2.2.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A sub-band differential CQI offset level of 0 shall be reported at least $\alpha$% of the time but less than $\beta$% of the time for each sub-band, where $\alpha$ and $\beta$ are specified in Table 6.2.3.1.2.2.3-2;

b) The ratio of the throughput obtained when transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level and that obtained when transmitting the transport format indicated by the reported wideband CQI median on a randomly selected sub-band among all the sub-bands shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.3.1.2.2.3-2;

c) When transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level, the average BLER for the indicated transport format shall be greater than or equal to 0.02.

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each TTI for FDD.

**Table 6.2.3.1.2.2.3-1: Sub-band CQI reporting test under frequency-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | | | |
| Subcarrier spacing | | kHz | 15 | | | |
| Duplex Mode | | | FDD | | | |
| SNR | | dB | 5 | 6 | 11 | 12 |
| Propagation channel | | | Two tap model specified in Annex B.2.4 with $a$=1, $f_D$ = 5Hz, and $\tau_d$=0.45μs | | | |
| Antenna configuration | | | 2×4 | | | |
| Correlation configuration | | | As per Annex B.1 | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 5/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | | | |
| ReportConfigType | | | Aperiodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Subband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 8 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report interval and offset | | slot | Not configured | | | |
| Aperiodic Report Slot Offset | | | 5 | | | |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 | | | |
| reportTriggerSize | | | 1 | | | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUSCH | | | |

| | | |
|---|---|---|
| CQI/RI/PMI delay | ms | 8 |
| Maximum number of HARQ transmission | | 1 |
| Measurement channel | | As specified in Table A.4-2, TBS.2-5 |

**Table 6.2.3.1.2.2.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.1.2.2.

### 6.2.3.1.2.2.4 Test description

#### 6.2.3.1.2.2.4.1 Initial conditions

Same initial conditions as specified in clause 6.2.2.1.2.2.4.1 with the following exceptions:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.

Instead of Table 6.2.2.1.2.2.3-1→ use Table 6.2.3.1.2.2.3-1.

#### 6.2.3.1.2.2.4.2 Test procedure

Same test procedure as specified in clause 6.2.2.1.2.2.4.2 with the following exceptions:

Instead of Table 6.2.2.1.2.2.5-1 → use Table 6.2.3.1.2.2.3-1.

#### 6.2.3.1.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

##### 6.2.3.1.2.2.4.3_1 Message exceptions for SA

Same message exceptions as in 6.2.2.1.2.2.4.3_1.

##### 6.2.3.1.2.2.4.3_2 Message exceptions for NSA

Same as in 6.2.3.1.2.2.4.3_1.

### 6.2.3.1.2.2.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.3.1.2.2.4.2.

**Table 6.2.3.1.2.2.5-1: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.04 | 1.04 |

## 6.2.3.2 TDD

### 6.2.3.2.1 CQI reporting definition under AWGN conditions

The reporting accuracy of the channel quality indicator (CQI) under frequency non-selective conditions is determined by the reporting variance and the BLER performance using the transport format indicated by the reported CQI median. The purpose is to verify that the reported CQI values are in accordance with the CQI definition given in TS 38.214 [12]. To account for sensitivity of the input SNR the reporting definition is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB

#### 6.2.3.2.1.1 4Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA

##### 6.2.3.2.1.1.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

##### 6.2.3.2.1.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 6.2.3.2.1.1.3 Minimum requirement for periodic CQI reporting

For the parameters specified in Table 6.2.3.2.1.1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to 0.1.

**Table 6.2.3.2.1.1.3-1: CQI reporting definition test**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | | | |
| Subcarrier spacing | | kHz | 30 | | | |
| Duplex Mode | | | TDD | | | |
| TDD UL-DL pattern | | | FR1.30-1 | | | |
| SNR | | dB | 5 | 6 | 11 | 12 |
| Propagation channel | | | AWGN | | | |
| Antenna configuration | | | 2×4 with static channel specified in Annex B.1 | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 10/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |

| | | | |
|---|---|---|---|
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 2 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report periodicity and offset | | slot | 10/9 |
| aperiodicTriggeringOffset | | | Not configured |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured |
| | CodebookSubsetRestriction | | 010000 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUCCH |
| CQI/RI/PMI delay | | ms | 9.5 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-4 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.2.1.1.

#### 6.2.3.2.1.1.4 Test Description

##### 6.2.3.2.1.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.5 for TE diagram and section A.3.2.5 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.3.2.1.1.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.3.2.1.1.4.3.

#### 6.2.3.2.1.1.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.2.2.1.1.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 5 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and [1800] or more of the wideband CQI values are in the range (Median CQI - 1) ≤ Median CQI ≤ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

   For the filtered ACK and NACK responses if the ratio (NACK / ACK + NACK) ≤ 0.1 then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) > 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) ≤ 0.1

   then pass the UE for this test and go to step 9, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. Repeat step 1 to 8 for Test2.

#### 6.2.3.2.1.1.4.4 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

#### 6.2.3.2.1.1.4.4_1 Message exceptions for SA

Same as specified in 6.2.2.2.1.1.4.4_1.

#### 6.2.3.2.1.1.4.4_2 Message exceptions for NSA

Same as specified in 6.2.3.2.1.1.4.4_1.

### 6.2.3.2.1.1.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.3.2.1.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.3.2.1.2 4Rx TDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Whether to adopt minimum sample for pass/fail verdict needs further study.

### 6.2.3.2.1.2.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of $10^{-5}$ falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

### 6.2.3.2.1.2.2 Test Applicability

This test applies to all types of NR UE release 16 supporting *cqi-TableAlt*.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and *cqi-TableAlt*.

### 6.2.3.2.1.2.3 Minimum requirement for periodic CQI reporting with Table 3

For the parameters specified in Table 6.2.3.2.1.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) The reported CQI value according to the reference channel shall be in the range of ±1 of the reported median more than 90% of the time.

b) If the PDSCH BLER using the transport format indicated by median CQI is less than or equal to $10^{-5}$, then the BLER using the transport format indicated by the (median CQI+1) shall be greater than $10^{-5}$. If the PDSCH BLER using the transport format indicated by the median CQI is greater than $10^{-5}$, then the BLER using transport format indicated by (median CQI-1) shall be less than or equal to $10^{-5}$.

c) The reported CQI value according to the reference channel shall be $\geq 1$.

**Table 6.2.3.2.1.2.3-1: CQI reporting test parameters**

| Parameter | | Unit | Test 1 | |
|---|---|---|---|---|
| Bandwidth | | MHz | 40 | |
| Subcarrier spacing | | kHz | 30 | |
| Duplex Mode | | | TDD | |
| TDD UL-DL pattern | | | FR1.30-1 | |
| SNR | | dB | -2 | -1 |
| Propagation channel | | | AWGN | |
| Antenna configuration | | | 1×4 with static channel specified in Annex B.1 | |
| Beamforming Model | | | As specified in Annex B.4.1 | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | |
| | Number of CSI-RS ports (*X*) | | 4 | |
| | CDM Type | | FD-CDM2 | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | |
| | First OFDM symbol in the PRB used | | 9 | |

| | | | |
|---|---|---|---|
| | for CSI-RS ($l_0$) | | |
| | CSI-RS periodicity and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 1 |
| | CDM Type | | No CDM |
| | Density (ρ) | | 3 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 1,(0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 1 |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 3 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| Csi-ReportingBand | | | 1111111 |
| CSI-Report periodicity and offset | | slot | 10/9 |
| aperiodicTriggeringOffset | | | Not configured |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUCCH |
| CQI/RI/PMI delay | | ms | 9.5 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-4, TBS.4-2 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.2.1.2.

#### 6.2.3.2.1.2.4 Test Description

##### 6.2.3.2.1.2.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D.

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.3 for TE diagram and section A.3.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 6.1.2-1 and 6.2.3.2.1.2.3-1 as appropriate.

3. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 , and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode* On for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2.3.2.1.2.4.3.

#### 6.2.3.2.1.2.4.2 Test Procedure

1. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 6.2.3.2.1.2.3-1.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 5000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 10 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

3. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

4. If Median CQI is not equal to 1 or 15 and 4500 or more of the wideband CQI values are in the range (Median CQI - 1) ≤ Median CQI ≤ ( Median CQI + 1) then continue with step 5, otherwise go to step 8.

5. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   For the filtered ACK and NACK responses if the ratio (NACK / (ACK + NACK)) $\leq 10^{-5}$ then go to step 6, otherwise go to step 7.

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   If the ratio (NACK / (ACK + NACK)) > $10^{-5}$

   then pass the UE for this test, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 5, and measure the BLER for a duration sufficient to achieve statistical significance according to Annex G.4.

   If the ratio (NACK / ACK + NACK) $\leq 10^{-5}$

   then pass the UE for this test, otherwise go to step 8.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

#### 6.2.3.2.1.2.4.3 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

#### 6.2.3.2.1.2.4.3_1 Message exceptions for SA

Same as specified in clause 6.2.2.2.1.2.4.3_1.

#### 6.2.3.2.1.2.4.3_2 Message exceptions for NSA

Same as specified in 6.2.3.2.1.2.4.3_1.

### 6.2.3.2.1.2.5 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2.3.2.1.2.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

## 6.2.3.2.2 CQI reporting under fading conditions

### 6.2.3.2.2.1 4Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA

#### 6.2.3.2.2.1.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

#### 6.2.3.2.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

#### 6.2.3.2.2.1.3 Minimum conformance requirements

The purpose of the requirements is to verify that the UE is tracking the channel variations and selecting the largest transport format possible according to the prevailing channel state for the frequency non-selective scheduling.

The reporting accuracy of CQI under frequency non-selective fading conditions is determined by the reporting variance, the relative increase of the throughput obtained when the transport format is indicated by the reported CQI compared to the throughput obtained when a fixed transport format is configured according to the reported median CQI, and a minimum BLER using the transport formats indicated by the reported CQI.

For the parameters specified in Table 6.2.3.2.2.1.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A CQI index not in the set {median CQI -1, median CQI, median CQI +1} shall be reported at least $\alpha$% of the time where $\alpha$% is specified in Table 6.2.3.2.2.1.3-2;

b) The ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.3.2.2.1.3-2;

c) When transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater than or equal to 0.02.

**Table 6.2.3.2.2.1.3-1: Wideband CQI reporting test under frequency non-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | | | |
| Subcarrier spacing | | kHz | 30 | | | |
| Duplex Mode | | | TDD | | | |
| TDD UL-DL pattern | | | FR1.30-1 | | | |
| SNR | | dB | 3 | 4 | 9 | 10 |
| Propagation channel | | | TDLA30-5 | | | |
| Antenna configuration | | | 2×4 | | | |
| Correlation configuration | | | XP High | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 10/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 | | | |
| ReportConfigType | | | Periodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Wideband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 16 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report periodicity and offset | | slot | 10/9 | | | |
| aperiodicTriggeringOffset | | | Not configured | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |
| Physical channel for CSI report | | | PUCCH | | | |
| CQI/RI/PMI delay | | ms | 9.5 | | | |
| Maximum number of HARQ transmission | | | 1 | | | |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-3 | | | |

**Table 6.2.3.2.2.1.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 5 | 5 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.2.2.1.

### 6.2.3.2.2.1.4 Test description

#### 6.2.3.2.2.1.4.1 Initial conditions

Same initial conditions as specified in clause 6.2.2.1.2.1.4.1 with the following exceptions:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.

Instead of Table 6.2.2.1.2.1.3-1 → use Table 6.2.3.2.2.1.3-1.

#### 6.2.3.2.2.1.4.2 Test procedure

Same test procedure as specified in clause 6.2.2.1.2.1.4.2 with the following exceptions:

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 6000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 10 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

4. If Median CQI value is not equal to 1 or 15 and 300 ($\alpha$%)or more of the wideband CQI values are outside the range (Median CQI - 1) ≤ Median CQI ≤ (Median CQI + 1) then continue with step 5, otherwise go to step 7.

Instead of Table 6.2.2.1.2.1.5-1 → use Table 6.2.3.2.2.1.3-1.

#### 6.2.3.2.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

##### 6.2.3.2.2.1.4.3_1 Message exceptions for SA

Same as 6.2.2.1.2.1.4.3_1 with following exceptions:

**Table 6.2.3.2.2.1.4.3_1-1: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots10 | 1 | | |
| } | | | |

##### 6.2.3.2.2.1.4.3_2 Message exceptions for NSA

Same as 6.2.3.2.2.1.4.3_1.

### 6.2.3.2.2.1.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.3.2.2.1.4.2.

**Table 6.2.3.2.2.1.5-1: Test requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 5 | 5 |
| $\gamma$ | 1.04 | 1.04 |

### 6.2.3.2.2.2 4Rx TDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA

#### 6.2.3.2.2.2.1 Test purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 2% for the indicated transport format.

#### 6.2.3.2.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of E-UTRA UE release 15 and forward supporting EN-DC.

#### 6.2.3.2.2.2.3 Minimum conformance requirements

The purpose of the requirements is to verify that the preferred sub-bands can be used for frequency-selective scheduling under the frequency-selective fading conditions.

The accuracy of sub-band channel CQI reporting under the frequency-selective fading conditions is determined by a double-sided percentile of the reported differential CQI offset level 0 per sub-band, and the relative increase of the throughput obtained when transmitting the transport format indicated by the corresponding reported sub-band CQI on a randomly selected sub-band among the sub-bands with the highest reported differential CQI offset level compared to the throughput when transmitting a fixed transport format according to the wideband CQI median on a randomly selected sub-band among all the sub-bands.

For the parameters specified in Table 6.2.3.2.2.2.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified by the following:

a) A sub-band differential CQI offset level of 0 shall be reported at least α% of the time but less than β% of the time for each sub-band, where α and β are specified in Table 6.2.3.2.2.2.3-2;

b) The ratio of the throughput obtained when transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level and that obtained when transmitting the transport format indicated by the reported wideband CQI median on a randomly selected sub-band among all the sub-bands shall be $\geq \gamma$, where $\gamma$ is specified in Table 6.2.3.2.2.2.3-2;

c) When transmitting the corresponding transport format on a randomly selected sub-band among the sub-bands with the highest differential CQI offset level, the average BLER for the indicated transport format shall be greater than or equal to 0.02.

The requirements only apply for sub-bands of full size and the random scheduling across the sub-bands is done by selecting a new sub-band in each available downlink transmission instance for TDD.

**Table 6.2.3.2.2.2.3-1: Sub-band CQI reporting test under frequency-selective fading conditions**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | | | |
| Subcarrier spacing | | kHz | 30 | | | |
| Duplex Mode | | | TDD | | | |
| TDD UL-DL pattern | | | FR1.30-1 | | | |
| SNR | | dB | 5 | 6 | 11 | 12 |
| Propagation channel | | | Two tap model specified in Annex B.2.4 with $a$=1, $f_D$ = 5Hz, and $\tau_d$=0.1125μs | | | |
| Antenna configuration | | | 2×4 | | | |
| Correlation configuration | | | As per Annex B.1 | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 4 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5,4 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 | | | |
| | CSI-RS periodicity and offset | slot | 10/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | | | |
| | Number of CSI-RS ports ($X$) | | 2 | | | |
| | CDM Type | | FD-CDM2 | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3,(6,-) | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 | | | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | | | |
| | CSI-IM RE pattern | | 0 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) | | | |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 | | | |
| ReportConfigType | | | Aperiodic | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | cri-RI-PMI-CQI | | | |
| timeRestrictionForChannelMeasurements | | | Not configured | | | |
| timeRestrictionForInterferenceMeasurements | | | Not configured | | | |
| cqi-FormatIndicator | | | Subband | | | |
| pmi-FormatIndicator | | | Wideband | | | |
| Sub-band Size | | RB | 16 | | | |
| csi-ReportingBand | | | 1111111 | | | |
| CSI-Report interval and offset | | slot | Not configured | | | |
| Aperiodic Report Slot Offset | | | 8 | | | |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 | | | |
| reportTriggerSize | | | 1 | | | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | | | |
| aperiodicTriggeringOffset | | | 0 | | | |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | | | |
| | Codebook Mode | | 1 | | | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | Not configured | | | |
| | CodebookSubsetRestriction | | 000001 | | | |
| | RI Restriction | | N/A | | | |

| | | |
|---|---|---|
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 9.5 |
| Maximum number of HARQ transmission | | 1 |
| Measurement channel | | As specified in Table A.4-2, TBS.2-6 |

**Table 6.2.3.2.2.2.3-2: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.2.3.2.2.2.

### 6.2.3.2.2.2.4 Test description

#### 6.2.3.2.2.2.4.1 Initial conditions

Same initial conditions as specified in clause 6.2.2.1.2.2.4.1 with the following exceptions:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 for TE diagram and section A.3.2.5 for UE diagram.

Instead of Table 6.2.2.1.2.2.3-1 → use Table 6.2.3.2.2.2.3-1.

Instead of clause 6.2.2.1.2.2.4.3 → use clause 6.2.3.2.2.2.4.3.

#### 6.2.3.2.2.2.4.2 Test procedure

Same test procedure as specified in clause 6.2.2.1.2.2.4.2 with the following exceptions:

Instead of Table 6.2.2.1.2.2.3-1 → use Table 6.2.3.2.2.2.3-1.

#### 6.2.3.2.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

##### 6.2.3.2.2.2.4.3_1 Message exceptions for SA

Same as 6.2.2.1.2.2.4.3_1 with following exceptions:

**Table 6.2.3.2.2.2.4.3_1-1: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| Slots10 | 1 | | |
| } | | | |

##### 6.2.3.2.2.2.4.3_2 Message exceptions for NSA

Same as 6.2.3.2.2.2.4.3_1.

### 6.2.3.2.2.2.5 Test requirement

The pass/fail decision is as specified in the test procedure in clause 6.2.3.2.2.2.4.2.

**Table 6.2.3.2.2.2.5-1: Minimum requirements**

| Parameters | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\beta$ [%] | 55 | 55 |
| $\gamma$ | 1.04 | 1.04 |
| Note 1: TT = 0.01 | | |

# 6.2A Reporting of Channel Quality Indicator (CQI) for CA

## 6.2A.1 General

This clause includes the requirements for the reporting of channel quality indicator (CQI) with the UE configured for CA. The purpose is to verify that the CQI is correctly reported in accordance with the CQI definition given in TS 38.214 [12] for each CC with multiple cells configured for periodic reporting.

## 6.2A.2 1RX requirements

(Void)

## 6.2A.3 2RX requirements

### 6.2A.3.1 CQI reporting definition under AWGN conditions

#### 6.2A.3.1.0 Minimum requirement for periodic CQI reporting

For each CA CQI reporting test defined in Table 6.2A.3.1.0-6, the test requirements and the test parameters are defined as below.

For each CC, the test parameters are specified in Table 6.2A.3.1.0-1. The additional parameters specified in Table 6.2A.3.1.0-2 are applicable for tests on FDD CC. The additional parameters specified in Table 6.2A.3.1.0-3 are applicable for tests on TDD CC.

For CA with 2 DL CC, for the SNR configuration specified in Table 6.2A.3.1.0-4, and using the downlink physical channels specified in Annex C.3.1 on each CC, the difference between the wideband CQI indices of PCell and SCell reported shall be such that

$$\text{wideband CQI}_{\text{PCell}} - \text{wideband CQI}_{\text{SCell}} \geq 2$$

for more than 90% of the time.

For CA with 3 or more DL CC, for the SNR configuration specified in Table 6.2A.3.1.0-5, and using the downlink physical channels specified in Annex C.3.1 on each cell, the difference between the wideband CQI indices of PCell and SCell1 reported, and the difference between the wideband CQI indices of SCell1 and SCell2, 3… reported shall be such that

$$\text{wideband CQI}_{\text{PCell}} - \text{wideband CQI}_{\text{SCell1}} \geq 2$$

$$\text{wideband CQI}_{\text{SCell1}} - \text{wideband CQI}_{\text{SCell2, 3...}} \geq 2$$

for more than 90% of the time.

**Table 6.2A.3.1.0-1: CA CQI reporting test parameters for FDD and TDD CC**

| Parameter | | Unit | Value |
|---|---|---|---|
| Propagation channel | | | AWGN |
| Antenna configuration | | | 1×2 with static channel specified in Annex B.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 5, 4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 9 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 1 |
| | CDM Type | | No CDM |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | Row 2, 6 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 13 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4, 9) |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 2 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Csi-ReportingBand | | | 1111111 |
| aperiodicTriggeringOffset | | | Not configured |
| Physical channel for CSI report | | | PUCCH |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | Derived as per section 5.1.3.2 of TS 38.214 [12] |

**Table 6.2A.3.1.0-2: Additional test parameters for FDD CC**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex Mode | | | FDD |
| Subcarrier spacing | | kHz | 15 |
| ZP CSI-RS configuration | CSI-RS periodicity and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 |
| CSI-IM configuration | CSI-IM timeConfig periodicity and offset | slot | 5/1 |
| CSI-Report periodicity and offset | | slot | 5/0 |
| CQI/RI/PMI delay | | ms | 8 |
| Sub-band Size | | RB | 8 for 5MHz and 10MHz,<br>16 for 15MHz, 20MHz and 25MHz,<br>32 for 30MHz, 35MHz, 40MHz, 45MHz and 50MHz |

**Table 6.2A.3.1.0-3: Additional test parameters for TDD CC**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex Mode | | | TDD |
| Subcarrier spacing | | kHz | 30 |
| TDD UL-DL pattern | | | FR1.30-1 |
| ZP CSI-RS configuration | CSI-RS periodicity and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 |
| CSI-IM configuration | CSI-IM timeConfig periodicity and offset | slot | 10/1 |
| CSI-Report periodicity and offset | | slot | 10/9 |
| CQI/RI/PMI delay | | ms | 9.5 |
| Sub-band Size | | RB | 8 for 10MHz, 15MHz, 20MHz and 25MHz,<br>16 for 30MHz, 40MHz and 50MHz,<br>32 for 60MHz, 80MHz, 90MHz and 100MHz |

**Table 6.2A.3.1.0-4: SNR configurations for 2 DL CA**

| Parameter | PCell | SCell |
|---|---|---|
| SNR (dB) | 10.0 | 4.0 |

**Table 6.2A.3.1.0-5: SNR configurations for 3 or more DL CA**

| Parameter | PCell | SCell1 | SCell2, 3… |
|---|---|---|---|
| SNR (dB) | 12.0 | 6.0 | 0.0 |

**Table 6.2A.3.1.0-6: List of CA CQI reporting test**

| Test number | CA duplex mode and SCS combination |
|---|---|
| 1 | FDD 15 kHz + TDD 30 kHz |
| 2 | FDD 15 kHz + FDD 15 kHz |
| 3 | TDD 30 kHz + TDD 30 kHz |
| Note 1: The applicability of requirements for different CA duplex modes, SCSs, is defined in 6.1.1.5.1.<br>Note 2: The applicability of requirements for different CA configurations and bandwidth combination sets is defined in 6.1.1.5.2. | |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.2A.3.1.1.

### 6.2A.3.1.1 2Rx CQI reporting accuracy under AWGN conditions for CA (2DL CA)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure is TBD

#### 6.2A.3.1.1.1 Test Purpose

To verify that the CQI is correctly reported in accordance with the CQI definition given in TS 38.214 [12] for each CC with multiple cells configured for periodic reporting.

#### 6.2A.3.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 2DL CA.

#### 6.2A.3.1.1.3 Test description

##### 6.2A.3.1.1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

CA capability to be tested: test any one of the supported CA capabilities with largest aggregated CA bandwidth combination, as specified in 6.1.1.5.2.

CA configuration to be tested: For the selected CA capability, test any one of the supported CA configurations with largest aggregated CA bandwidth combination, as specified in 6.1.1.5.2.

**Table 6.2A.3.1.1.3.1-1: Test point selection table**

| Test number | CA duplex mode | Configuration | PCell CC configuration |
|---|---|---|---|
| 1 | FDD 15 kHz + TDD 30 kHz | As defined in Table 6.2A.3.1.0-1 to Table 6.2A.3.1.0-3 | TDD CC if supported, otherwise FDD CC |
| 2 (Note 2) | FDD 15 kHz + FDD 15 kHz | As defined in Table 6.2A.3.1.0-1 to Table 6.2A.3.1.0-2 | Any of CCs |
| 3 | TDD 30 kHz + TDD 30 kHz | As defined in Table 6.2A.3.1.0-1 and Table 6.2A.3.1.0-3 | Any of CCs |
| NOTE 1: The test coverage can be considered fulfilled if UE passes one of the CC as PCell in Test 1.<br>NOTE 2: These scenarios are only tested for UEs which are not verified with Test 1. | | | |

1. Connect the SS and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for UE supporting only 2Rx RF bands on all CC. Annex A, Figure TBD for UE supporting 4Rx on some or all the CCs.
2. The parameter settings for the cell are set up according to Table 6.1.2-1, and Table 6.2A.3.1.0-1 to Table 6.2A.3.1.0-3 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.1.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.2A.3.1.1.3.3.

##### 6.2A.3.1.1.3.2 Test Procedure

1. Configure SCC according to Annex C.0, C.1 and C.2 for all downlink physical channels.
2. The SS shall configure SCC as per TS 38.508-1 [6] clause 5.5.1. Message contents are defined in clause 6.2A.3.1.1.3.3.
3. SS activates SCC by sending the activation MAC-CE (Refer TS 38.321 [18], clauses 5.9, 6.1.3.10). Wait for at least 1 second (Refer TS 38.133[19], clause9.3).

4. Set the parameters of bandwidth, reference channel, propagation condition and antenna configuration according to Table 6.2A.3.1.0-1 to Table 6.2A.3.1.0-3 as appropriate. Set the SNR according to Table 6.2A.3.1.0-4 as appropriate for PCC and SCC (For UE supporting 4Rx antenna ports on a given CC, the SNR is reduced by 3dB for that CC).

5. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.

6. The SS shall start gathering CQI reports, and will continue gathering CQI reports until 2000 wideband CQI reports have been gathered for each PCC and SCC. For each CSI report calculate the respective difference $CQI_{P-S}$ = wideband $CQI_{PCell}$ – wideband $CQI_{SCell}$.

7. If more than 1800 values of $CQI_{P-S}$ are $\geq 2$ pass the UE. Otherwise fail the UE.

8. Repeat steps from 1 to 7 for each test point in Table 6.2A.3.1.1.3.1-1 as appropriate.

#### 6.2A.3.1.1.1.3.3 Message contents

Message contents are according to TS 38.508 [6] clause 5.4.2 with the following exceptions:

#### 6.2A.2.1.1.1.3.3_1 Message exceptions for SA

**Table 6.2A.2.1.1.1.3.3_1-1: CSI-RS-ResourceMapping**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row2 | 0000 1000 0000 | k0= 4, row2 | |
| } | | | |
| density CHOICE { | | | |
| one | NULL | | |
| } | | | |
| } | | | |

**Table 6.2A.2.1.1.1.3.3_1-2: NZP CSI-ResourcePeriodicityAndOffset**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-2a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots5 | 1 | | For FDD CC |
| slot10 | 1 | | For TDD CC |
| } | | | |

**Table 6.2A.2.1.1.1.3.3_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-IM-Resource ::= SEQUENCE { | | | |
| periodicityAndOffset SEQUENCE { | | | |
| slots5 | 1 | | For FDD CC |
| slots10 | 1 | | For TDD CC |
| } | | | |
| } | | | |

**Table 6.2A.2.1.1.1.3.3_1-4: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-12 | | |
|---|---|---|
| **Information Element** | **Value/remark** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | |
| subbandSize | 8 | For the CC with FDD 15kHz SCS 5MHz and 10MHz CHBW; For the CC with 30kHz SCS 10MHz, 15MHz, 20MHz and 25MHz CHBW. |
| | 16 | For the CC with FDD 15kHz SCS 15MHz, 20MHz and 25MHz CHBW; For the CC with 30kHz SCS 30MHz, 40MHz and 50MHz CHBW. |
| | 32 | For the CC with FDD 15kHz SCS 30MHz, 35MHz, 40MHz, 45MHz and 50MHz CHBW; For the CC with 30kHz SCS 60MHz, 80MHz, 90MHz and 100MHz CHBW. |
| } | | |

#### 6.2A.2.1.1.1.3.3_2 Message exceptions for NSA

Same as specified in 6.2A.2.1.1.1.3.3_2.

#### 6.2A.3.1.1.1.3.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2A.3.1.1.1.3.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.6.2A.3.1.2 2Rx CQI reporting accuracy under AWGN conditions for CA (3DL CA)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure is TBD

#### 6.2A.3.1.2.1 Test Purpose

Same with 6.2A.3.1.1.1.

#### 6.2A.3.1.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 3DL CA.

#### 6.2A.3.1.2.3 Test description

#### 6.2A.3.1.2.3.1 Initial conditions

Same with 6.2A.3.1.1.3.1.

#### 6.2A.3.1.1.2.3.2 Test Procedure

1. Configure SCCs according to Annex C.0, C.1 and C.2 for all downlink physical channels.
2. The SS shall configure SCCs as per TS 38.508-1 [6] clause 5.5.1. Message contents are defined in clause 6.2A.3.1.1.2.3.3.

3. SS activates SCCs by sending the activation MAC-CE (Refer TS 38.321 [18], clauses 5.9, 6.1.3.10). Wait for at least 1 second (Refer TS 38.133[19], clause9.3).

4. Set the parameters of bandwidth, reference channel, propagation condition and antenna configuration according to Table 6.2A.3.1.0-1 to Table 6.2A.3.1.0-3 as appropriate. Set the SNR according to Table 6.2A.3.1.0-5 as appropriate for PCC and SCCs (For a UE supporting 4Rx antenna ports on a given CC, the SNR is reduced by 3dB for that CC).

5. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC on both PCC and SCCs. The SS sends downlink MAC padding bits on the DL RMC.

6. The SS shall start gathering CQI reports, and will continue gathering CQI reports until 2000 wideband CQI reports have been gathered for each PCC and SCCs. For each CSI report calculate the respective difference $CQI_{P\text{-}S1}$ = wideband $CQI_{PCell}$ – wideband $CQI_{SCell1}$ and the respective difference $CQI_{S1\text{-}S2}$ = wideband $CQI_{SCell1}$ – wideband $CQI_{SCell2}$.

7. If more than 1800 values of $CQI_{P\text{-}S1}$ are $\geq 2$ and more than 1800 values of $CQI_{S1\text{-}S2}$ are $\geq 2$, pass the UE. Otherwise fail the UE.

8. Repeat steps from 1 to 7 for each test point in Table 6.2A.3.1.0-6 as appropriate.

#### 6.2A.3.1.1.2.3.3 Message contents

Same with 6.2A.3.1.1.1.3.3.

#### 6.2A.3.1.1.2.3.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2A.3.1.1.2.3.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

### 6.2A.3.1.3 2Rx CQI reporting accuracy under AWGN conditions for CA (4DL CA)

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure is TBD

#### 6.2A.3.1.3.1 Test Purpose

Same with 6.2A.3.1.1.1.

#### 6.2A.3.1.3.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 4DL CA.

#### 6.2A.3.1.3.3 Test description

#### 6.2A.3.1.3.3.1 Initial conditions

Same with 6.2A.3.1.1.3.1.

#### 6.2A.3.1.1.3.3.2 Test Procedure

1. Configure SCCs according to Annex C.0, C.1 and C.2 for all downlink physical channels.

2. The SS shall configure SCCs as per TS 38.508-1 [6] clause 5.5.1. Message contents are defined in clause 6.2A.3.1.1.3.3.3.

3. SS activates SCCs by sending the activation MAC-CE (Refer TS 38.321 [18], clauses 5.9, 6.1.3.10). Wait for at least 1 second (Refer TS 38.133[19], clause9.3).

4. Set the parameters of bandwidth, reference channel, propagation condition and antenna configuration according to Table 6.2A.3.1.0-1 to Table 6.2A.3.1.0-3 as appropriate. Set the SNR according to Table 6.2A.3.1.0-5 as appropriate for PCC and SCCs (For a UE supporting 4Rx antenna ports on a given CC, the SNR is reduced by 3dB for that CC).

5. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC on both PCC and SCCs. The SS sends downlink MAC padding bits on the DL RMC.

6. The SS shall start gathering CQI reports, and will continue gathering CQI reports until 2000 wideband CQI reports have been gathered for each PCC and SCCs. For each CSI report calculate the respective difference $CQI_{P\text{-}S1}$ = wideband $CQI_{PCell}$ – wideband $CQI_{SCell1,}$ the respective difference $CQI_{S1\text{-}S2}$ = wideband $CQI_{SCell1}$ – wideband $CQI_{SCell2}$ and the respective difference $CQI_{S1\text{-}S3}$ = wideband $CQI_{SCell1}$ – wideband $CQI_{SCell3}$.

7. If more than 1800 values of $CQI_{P\text{-}S1}$ are $\geq 2$, more than 1800 values of $CQI_{S1\text{-}S2}$ are $\geq 2$ and more than 1800 values of $CQI_{S1\text{-}S3}$ are $\geq 2$, pass the UE. Otherwise fail the UE.

8. Repeat steps from 1 to 7 for each test point defined in Table 6.2A.3.1.0-6 as appropriate.

#### 6.2A.3.1.1.3.3.3 Message contents

Same with 6.2A.3.1.1.1.3.3.

#### 6.2A.3.1.1.3.3.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 6.2A.3.1.1.3.3.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

# 6.3 Reporting of Precoding Matrix Indicator (PMI)

## 6.3.0 General

The minimum performance requirements of PMI reporting are defined based on the precoding gain, expressed as the relative increase in throughput when the transmitter is configured according to the UE reported PMI compared to the case when the transmitter is using random precoding, respectively. When the transmitter uses random precoding, for each PDSCH allocation a precoder is randomly generated with equal probability of each applicable $i_1$ and $i_2$ combination and applied to the PDSCH. A fixed transport format (FRC) is configured for all requirements.

The requirements for transmission scheme 1 with higher layer parameter *codebookType* set to 'typeI-SinglePanel' are specified in terms of the ratio:

$$\gamma = \frac{t_{ue}}{t_{rnd}}$$

In the definition of $\gamma$, for 4TX, 8TX, 16TX, and 32TX PMI requirements, $t_{ue}$ is 90 % of the maximum throughput obtained at $SNR_{ue}$ using the precoders configured according to the UE reports, and $t_{rnd}$ is the throughput measured at $SNR_{ue}$ with random precoding.

The requirements for transmission scheme 1 with higher layer parameter *codebookType* set to 'typeII' or 'typeII-r16' are specified in terms of the ratio:

$$\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$$

In the definition of $\gamma$, for 16TX PMI requirements, $t_{ue,\,follow1,\,follow2}$ is 90 % of the maximum throughput obtained at $SNR_{follow1,\,follow2}$ using the precoders configured according to the UE reports, and $t_{rnd1,\,rnd2}$ is the throughput measured at $SNR_{follow1,\,follow2}$ with random precoding.

## 6.3.1 1RX requirements (Void)

## 6.3.2 2RX requirements

### 6.3.2.1 FDD

#### 6.3.2.1.1 2Rx FDD FR1 Single PMI with 4TX TypeI-SinglePanel codebook for both SA and NSA

##### 6.3.2.1.1.1 Test purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

##### 6.3.2.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 6.3.2.1.1.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.1.1-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.1.1-2.

**Table 6.3.2.1.1.3-1: Test parameters (single layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 4 x 2<br>(N1,N2) = (2,1) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 4, (0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) |
| | CSI-RS interval and offset | | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Patten 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 4 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |

| | | | |
|---|---|---|---|
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (2,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 11111111 |
| | RI Restriction | | 00000001 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.1 FDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (1 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-3), this reported PMI cannot be applied at the eNB downlink before slot#(n+3).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.2.1.1-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.3 |

### 6.3.2.1.1.4 Test description

#### 6.3.2.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.1.1.3_1 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause6.3.2.1.1.4.3.

#### 6.3.2.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.1.1.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.2.1.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.2.1.1.4.3.1 Message exceptions for SA

**Table 6.3.2.1.1.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | |
| } | | | |

**Table 6.3.2.1.1.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row4 | 001 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.3.2.1.1.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row5 | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.2.1.1.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.3.2.1.1.4.3.1-5: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| Slots5 | 1 | | |
| } | | | |

**Table 6.3.2.1.1.4.3.1-6: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| two-one-TypeI-SinglePanel-Restriction | 11111111 | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000001 | | |

**Table 6.3.2.1.1.4.3.1-7: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 0 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | [1111111] | | |
| } | | | |
| } | | | |
| subbandSize | 8 | | |
| } | | | |

6.3.2.1.1.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.1.1.4.3.1.

6.3.2.1.1.5 Test requirement

**Table 6.3.2.1.1.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.29 |

## 6.3.2.1.2 2Rx FDD FR1 Single PMI with 8TX TypeI-SinglePanel codebook for both SA and NSA

### 6.3.2.1.2.1 Test purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.1.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.1.2.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.1.2.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.1.2.3-2.

**Table 6.3.2.1.2.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 8 x 2<br>(N1,N2) = (4,1) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 8 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 8, (4,6) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |

| | | | |
|---|---|---|---|
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.2 |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (1 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the eNB downlink before slot#(n+4).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.2.1.2.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.5 |

### 6.3.2.1.2.4 Test description

#### 6.3.2.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.1.2.3-1 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.2.4.3.

#### 6.3.2.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.1.2.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.2.1.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.2.1.2.4.3.1 Message exceptions for SA

**Table 6.3.2.1.2.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | aperiodic | | |
| } | | | |

**Table 6.3.2.1.2.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001100 | | |
| } | | | |
| nrofPorts | p8 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.1.2.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.2.1.2.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.2.1.2.4.3.1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-one-TypeI-SinglePanel-Restriction | FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.2.1.2.4.3.1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.2.1.2.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.1.2.4.3.1.6.3.2.1.2.5Test requirement

**Table 6.3.2.1.2.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.49 |

## 6.3.2.1.3 2Rx FDD FR1 Multiple PMI with 16Tx Type I – SinglePanel Codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

### 6.3.2.1.3.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.1.3.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.1.3.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.1.3.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.1.3.3-2.

**Table 6.3.2.1.3.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLC300-5 |
| Antenna configuration | | | High XP 16 x 2<br>(N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports ($X$) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density ($\rho$) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI- | | 16 |

| | | | |
|---|---|---|---|
| | RS ports (*X*) | | |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |

| | | |
|---|---|---|
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 8 |
| Maximum number of HARQ transmission | | 4 |
| Measurement channel | | R.PDSCH.1-6.3 FDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | |

**Table 6.3.2.1.3.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.5 |

### 6.3.2.1.3.4 Test description

#### 6.3.2.1.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.1.3.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.3.4.3.

#### 6.3.2.1.3.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.1.3.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.1.3.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.2.1.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.2.1.3.4.3.1 Message exceptions for SA

**Table 6.3.2.1.3.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | P16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.1.3.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-two-TypeI-SinglePanel-Restriction | FFFF FFFF FFFF FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.2.1.3.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| | | | |
| } | | | |

#### 6.3.2.1.3.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.1.3.4.3.1.

### 6.3.2.1.3.5 Test requirement

**Table 6.3.2.1.3.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.49 |

## 6.3.2.1.4 2Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

### 6.3.2.1.4.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.1.4.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.1.4.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.1.4.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.1.4.3-2.

**Table 6.3.2.1.4.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 32 x 2<br>(N1,N2) = (4,4) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |

| | | | |
|---|---|---|---|
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 32 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 17, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, 12) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,4) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |

| | | |
|---|---|---|
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 8 |
| Maximum number of HARQ transmission | | 4 |
| Measurement channel | | R.PDSCH.1-6.3 FDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | |

**Table 6.3.2.1.4.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 5.0 |

### 6.3.2.1.4.4 Test description

#### 6.3.2.1.4.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.1.4.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.4.4.3.

#### 6.3.2.1.4.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.1.4.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,follow1,follow2}$ and $SNR_{follow1,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,follow1,follow2}}{t_{rnd1,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.2.1.4.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.2.1.4.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.2.1.4.4.3.1 Message exceptions for SA

**Table 6.3.2.1.4.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | P32 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.1.4.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-four-TypeI-SinglePanel-Restriction | FFFF FFFF FFFF FFFF<br>FFFF FFFF FFFF FFFF<br>FFFF FFFF FFFF FFFF<br>FFFF FFFF FFFF FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.2.1.4.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| } | | | |

6.3.2.1.4.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.1.4.4.3.1.

6.3.2.1.4.5 Test requirement

**Table 6.3.2.1.4.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 4.99 |

## 6.3.2.1.5 2Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

### 6.3.2.1.5.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.1.5.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.1.5.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.1.5.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.1.5.3-2.

**Table 6.3.2.1.5.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 2 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports ($X$) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density ($\rho$) | | 1 |
| | First subcarrier index in the PRB used for | | Row 5, (4,-) |

| | | | |
|---|---|---|---|
| | CSI-RS ($k_0$, $k_1$) | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII |
| | L (*numberOfBeams*) | | 2 |
| | $N_{PSK}$ (*phaseAlphabetSize*) | | 8 |
| | *subbandAmplitude* | | True |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII- | | 10 |

| | | |
|---|---|---|
| RI-Restriction) | | |
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 8 |
| Maximum number of HARQ transmission | | 4 |
| Measurement channel | | R.PDSCH.1-6.3 FDD |

Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.2.1.3-1.

Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).

Note 3: Randomization of the dual-cluster beam directions shall be used as specified in Annex B.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test.

**Table 6.3.2.1.5.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.9 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.3.2.1.5.

### 6.3.2.1.5.4 Test description

#### 6.3.2.1.5.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.1.5.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.4.4.3.

#### 6.3.2.1.5.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.1.5.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities, and the random precoder generation shall follow the codebook configuration as specified in Table 6.3.2.1.3.3-1. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.1.5.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.2.1.5.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.2.1.5.4.3.1 Message exceptions for SA

**Table 6.3.2.1.5.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | '011110'B | | |
| } | | | |
| nrofPorts | P16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.1.5.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII SEQUENCE { | | | |
| n1-n2-codebookSubsetRestriction CHOICE { | | | |
| four-two | 0x 7FF FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction | '10'B | | |
| } | | | |
| } | | | |
| phaseAlphabetSize | 8 | | |
| subbandAmplitude | TRUE | | |
| numberOfBeams | 2 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.2.1.5.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| } | | | |

6.3.2.1.5.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.1.5.4.3.1.

6.3.2.1.5.5 Test requirement

**Table 6.3.2.1.5.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.89 |

## 6.3.2.1.6 2Rx FDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA

### 6.3.2.1.6.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.1.6.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

### 6.3.2.1.6.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.1.6.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.1.6.3-2.

**Table 6.3.2.1.6.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 2<br>(N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |

| | | | |
|---|---|---|---|
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Not configured |
| Sub-band Size | | RB | 4 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII-r16 |
| | *paramCombination-r16* | | 6 (L =4, $p_v$ =1/2, β=1/2 ) |
| | R*(numberOfPMISubbandsPerCQISubband-r16)* | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig- | | (4,4) |

| | | | |
|---|---|---|---|
| | O2) | | |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction-r16) | | 0010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.3 |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.2.1.3-1.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).<br>Note 3: Randomization of the dual-cluster beam directions shall be used as specified in AnnexB.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test. | | | |

**Table 6.3.2.1.6.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.2 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.2.1.6.

### 6.3.2.1.6.4 Test description

#### 6.3.2.1.6.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.10 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.1.6.3-1 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.6.4.3.

#### 6.3.2.1.6.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.1.6.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio $\geq \gamma$ which is specified in table 6.3.2.1.6.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.2.1.6.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2.

##### 6.3.2.1.6.4.3.1 Message exceptions for SA

**Table 6.3.2.1.6.4.3.1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | Not present | | |
| } | | | |
| codebookConfig | Not present | | |
| subbandSize | Value1 | | |
| codebookConfig-r16 | CodebookConfig-r16 | | |
| } | | | |

**Table 6.3.2.1.6.4.3.1-2: *CodebookConfig-r16* (Table 6.3.2.1.6.4.3.1-1)**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig-r16 ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII-r16 SEQUENCE { | | | |
| N1-n2-codebookSubsetRestriction-r16 | | | |
| Four-two | 0x 7FF FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction-r16 | 0010 | | |
| } | | | |
| } | | | |
| numberOfPMI-SubbandsPerCQI-Subband-r16 | 1 | | |
| paramCombinatin-r16 | 6 | (L =4, $p_v$ =1/2, β=1/2 ) | |
| } | | | |
| } | | | |
| } | | | |

6.3.2.1.6.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.1.6.4.3.1.

6.3.2.1.6.5 Test requirement

**Table 6.3.2.1.6.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.19 |

## 6.3.2.2 TDD

### 6.3.2.2.1 2Rx TDD FR1 Single PMI with 4TX TypeI-SinglePanel codebook for both SA and NSA

#### 6.3.2.2.1.1 Test Purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

#### 6.3.2.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 6.3.2.2.1.3 Minimum Conformance Requirements

For the parameters specified in Table 6.3.2.2.1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.2.1.3-2.

**Table 6.3.2.2.1.3-1: Test parameters (single layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configuration | | | FR1.30-1 as specified in Annex A |
| DL BWP configuration #1 | First PRB | | 0 |
| | Number of contiguous PRB | | 106 |
| | Subcarrier spacing | kHz | 30 |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 4 x 2 (N1,N2) = (2,1) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 4, (0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggering Offset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Patten 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$, $l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (2,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubsetRestriction | | 11111111 |
| | RI Restriction | | 00000001 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 5.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.1 TDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| NOTE 1: For random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity).<br>NOTE 2: If the UE reports in an available uplink reporting instance at slot #n based on PMI estimation at a downlink slot not later than slot#[(n-4)], this reported PMI cannot be applied at the eNB downlink before slot#[(n+4)].<br>NOTE 3: Randomization of the principle beam direction shall be used as specified inAnnex B.2.3.2.3. | | | |

**Table 6.3.2.2.1.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.3 |

#### 6.3.2.2.1.4 Test Description

##### 6.3.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of TS 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.2.1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.2.1.4.3.

##### 6.3.2.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.1.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.2.2.1.5-1, then the test is pass. Otherwise, the test is fail.

##### 6.3.2.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

6.3.2.2.1.4.3_1 Message exceptions for SA

**Table 6.3.2.2.1.4.3_1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | aperiodic | | |
| } | | | |

**Table 6.3.2.2.1.4.3_1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause5.4.2, Table **5.4.2.0-15** | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row4 | 001 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.3.2.2.1.4.3_1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause5.4.2, Table5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.2.2.1.4.3_1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |

**Table 6.3.2.2.1.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| two-one-TypeI-SinglePanel-Restriction | 11111111 | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000001 | | |

**Table 6.3.2.2.1.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 0 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| subbandSize | value2 | | |
| } | | | |

6.3.2.2.1.4.3_2 Message exceptions for NSA

Same as in clause 6.2.2.1.2.1.4.3_1.

6.3.2.2.1.5 Test Requirements

**Table 6.3.2.2.1.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.29 |

## 6.3.2.2.2 2Rx TDD FR1 Single PMI with 8TX TypeI-SinglePanel codebook for both SA and NSA

### 6.3.2.2.2.1 Test purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.2.2.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.2.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.2.2.3-2.

**Table 6.3.2.2.2.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| DL BWP configuration #1 | First PRB | | 0 |
| | Number of contiguous PRB | | 106 |
| | Subcarrier spacing | kHz | 30 |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 8 x 2 (N1,N2) = (4,1) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 8 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 8, (4,6) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Patten 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |

| | | | |
|---|---|---|---|
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.2 TDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#[(n-6)], this reported PMI cannot be applied at the eNB downlink before slot#[(n+6)].<br>Note 3: Randomization of the principle beam direction shall be used as specified inAnnex B.2.3.2.3. | | | |

**Table 6.3.2.2.2.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.5 |

### 6.3.2.2.2.4 Test description

#### 6.3.2.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.2.2.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.2.2.4.3.

### 6.3.2.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.2.2.2.5-1, then the test is pass. Otherwise, the test is fail.

### 6.3.2.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

### 6.3.2.2.2.4.3_1 Message exceptions for SA

**Table 6.3.2.2.2.4.3_1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | aperiodic | | |
| } | | | |

**Table 6.3.2.2.2.4.3_1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001100 | | |
| } | | | |
| nrofPorts | p8 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.2.2.4.3_1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.2.2.2.4.3_1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.2.2.2.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-one-TypeI-SinglePanel-Restriction | FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.2.2.2.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 8 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.2.2.2.4.3_2 Message exceptions for NSA

Same as in clause 6.3.2.2.2.4.3_1.

6.3.2.2.2.5 Test requirement

**Table 6.3.2.2.2.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.49 |

## 6.3.2.2.3 2Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA

### 6.3.2.2.3.1 Test purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.2.3.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.2.3.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.2.3.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.2.3.3-2.

**Table 6.3.2.2.3.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLC300-5 |
| Antenna configuration | | | High XP 16 x 2 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |

| | | | |
|---|---|---|---|
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.3 TDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.2.2.3.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.5 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.2.2.3.

### 6.3.2.2.3.4 Test description

#### 6.3.2.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.10 for TE diagram and section A.3.2.3 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.2.3.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.2.3.4.3.

#### 6.3.2.2.3.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.3.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.2.3.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.2.2.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.2.2.3.4.3_1 Message exceptions for SA

**Table 6.3.2.2.3.4.3_1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | aperiodic | | |
| } | | | |

**Table 6.3.2.2.3.4.3_1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | p16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.2.3.4.3_1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table 5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.2.2.3.4.3_1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.2.2.3.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-one-TypeI-SinglePanel-Restriction | FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.2.2.3.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 8 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.2.2.3.4.3_2 Message exceptions for NSA

Same as in clause 6.3.2.2.3.4.3_1.

6.3.2.2.3.5 Test requirement

**Table 6.3.2.2.3.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.49 |

## 6.3.2.2.4 2Rx TDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 32 Tx is missing

### 6.3.2.2.4.1 Test purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.2.4.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC.

### 6.3.2.2.4.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.2.4.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.2.4.3-2.

**Table 6.3.2.2.4.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 32 x 2 (N1,N2) = (4,4) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 32 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 17, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, 12) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |

| | | | |
|---|---|---|---|
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,4) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.3 TDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.2.2.4.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 5.0 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.2.2.4.

### 6.3.2.2.4.4 Test description

#### 6.3.2.2.4.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.2.4.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.2.4.4.3.

### 6.3.2.2.4.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.4.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.2.4.5-1, then the test is pass. Otherwise, the test is fail.

### 6.3.2.2.4.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

#### 6.3.2.2.4.4.3_1 Message exceptions for SA

**Table 6.3.2.2.4.4.3_1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | aperiodic | | |
| } | | | |

**Table 6.3.2.2.4.4.3_1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | p32 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.2.4.4.3_1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table 5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.2.2.4.4.3_1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.2.2.4.4.3_1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-one-TypeI-SinglePanel-Restriction | FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.2.2.4.4.3_1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 8 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.2.2.4.4.3_2 Message exceptions for NSA

Same as in clause 6.3.2.2.4.4.3_1.

6.3.2.2.4.5 Test requirement

**Table 6.3.2.2.4.5-1: Test requirement (TDD)**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 4.99 |

## 6.3.2.2.5 2Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

### 6.3.2.2.5.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.2.5.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.2.2.5.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.2.5.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.2.5.3-2.

**Table 6.3.2.2.5.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 2 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |

| | | | |
|---|---|---|---|
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII |
| | L (*numberOfBeams*) | | 2 |
| | $N_{PSK}$ (*phaseAlphabetSize*) | | 8 |
| | *subbandAmplitude* | | True |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction) | | 10 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.3 TDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.2.2.3-1.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6).<br>Note 3: Randomization of the dual-cluster beam directions shall be used as specified in Annex B.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test. | | | |

**Table 6.3.2.2.5.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.9 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.3.2.2.5.

#### 6.3.2.2.5.4 Test description

##### 6.3.2.2.5.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.2.5.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.4.4.3.

##### 6.3.2.2.5.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.5.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities, and the random precoder generation shall follow the codebook configuration as specified in Table 6.3.2.2.3.3-1. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.2.2.5.5-1, then the test is pass. Otherwise, the test is fail.

##### 6.3.2.2.5.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

6.3.2.2.5.4.3.1 Message exceptions for SA

**Table 6.3.2.2.5.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | '011110'B | | |
| } | | | |
| nrofPorts | P16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.2.2.5.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII SEQUENCE { | | | |
| n1-n2-codebookSubsetRestriction CHOICE { | | | |
| four-two | 0x 7FF FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction | '10'B | | |
| } | | | |
| } | | | |
| phaseAlphabetSize | 8 | | |
| subbandAmplitude | TRUE | | |
| numberOfBeams | 2 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.2.2.5.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| } | | | |

6.3.2.2.5.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.2.5.4.3.1.

6.3.2.2.5.5 Test requirement

**Table 6.3.2.2.5.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.89 |

## 6.3.2.2.6 2Rx TDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA

### 6.3.2.2.6.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.2.2.6.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

### 6.3.2.2.6.3 Minimum conformance requirements

For the parameters specified in Table 6.3.2.2.6.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.2.2.6.3-2.

**Table 6.3.2.2.6.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 2 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |

| | | | |
|---|---|---|---|
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Not configured |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII-r16 |
| | *paramCombination-r16* | | 6 (L =4, $p_v$ =1/2, β=1/2 ) |
| | R*(numberOfPMISubbandsPerCQISubband-r16)* | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction-r16) | | 0010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.3 TDD |

Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.2.2.3-1.

Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6).

Note 3: Randomization of the dual-cluster beam directions shall be used as specified in Annex B.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test.

**Table 6.3.2.2.6.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.2 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.2.2.6.

### 6.3.2.2.6.4 Test description

#### 6.3.2.2.6.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.10 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.2.2.6.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.2.6.4.3.

#### 6.3.2.2.6.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.6.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Establish $t_{ue,follow1,follow2}$ and $SNR_{follow1,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Measure $t_{rnd1,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,follow1,follow2}}{t_{rnd1,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.2.2.6.5-1, then the test is pass. Otherwise, the test is fail.

6.3.2.2.6.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2.

6.3.2.2.6.4.3.1 Message exceptions for SA

**Table 6.3.2.2.6.4.3.1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | Not present | | |
| } | | | |
| codebookConfig | Not present | | |
| subbandSize | Value1 | | |
| codebookConfig-r16 | CodebookConfig-r16 | | |
| } | | | |

**Table 6.3.2.2.6.4.3.1-2: *CodebookConfig-r16* (Table 6.3.2.2.6.4.3.1-1)**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig-r16 ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII-r16 SEQUENCE { | | | |
| N1-n2-codebookSubsetRestriction-r16 | | | |
| Four-two | 0x 7FF<br>FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction-r16 | 0010 | | |
| } | | | |
| } | | | |
| numberOfPMI-SubbandsPerCQI-Subband-r16 | 1 | | |
| paramCombinatin-r16 | 6 | (L =4, $p_v$ =1/2, β=1/2 ) | |
| } | | | |
| } | | | |
| } | | | |

6.3.2.2.6.4.3.2 Message exceptions for NSA

Same as in clause 6.3.2.2.6.4.3.1.

6.3.2.2.6.5 Test requirement

**Table 6.3.2.2.6.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.19 |

## 6.3.3 4RX requirements

### 6.3.3.1 FDD

#### 6.3.3.1.1 4Rx FDD FR1 Single PMI with 4TX TypeI-SinglePanel codebook for both SA and NSA

##### 6.3.3.1.1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

##### 6.3.3.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 6.3.3.1.1.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.1.1-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.1.1-2.

**Table 6.3.3.1.1.3-1: Test parameters (single layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 4 x 4<br>(N1,N2) = (2,1) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 4, (0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 4 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |

| | | | |
|---|---|---|---|
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (2,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 11111111 |
| | RI Restriction | | 00000001 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.1 FDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (1 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-3), this reported PMI cannot be applied at the eNB downlink before slot#(n+3).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.3.1.1.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.3 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.3.3.1.1.

### 6.3.3.1.1.4 Test description

#### 6.3.3.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.1.1_1 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.1.1.4.3.

#### 6.3.3.1.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.2.2.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.2.1.1.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.3.1.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.1.1.4.3.1 Message exceptions for SA

**Table 6.3.3.1.1.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | |
| } | | | |

**Table 6.3.3.1.1.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row4 | 001 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.3.3.1.1.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row5 | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.3.1.1.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.3.3.1.1.4.3.1-5: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| Slots5 | 1 | | |
| } | | | |

**Table 6.3.3.1.1.4.3.1-6: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| two-one-TypeI-SinglePanel-Restriction | 11111111 | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000001 | | |

**Table 6.3.3.1.1.4.3.1-7: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 0 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | [1111111] | | |
| } | | | |
| } | | | |
| subbandSize | 8 | | |
| } | | | |

#### 6.3.3.1.1.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.1.1.4.3.1.

### 6.3.3.1.1.5 Test requirement

**Table 6.3.3.1.1.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.29 |

## 6.3.3.1.2 4Rx FDD FR1 Single PMI with 8TX TypeI-SinglePanel codebook for both SA and NSA

### 6.3.3.1.2.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.3.1.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.3.1.2.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.1.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.1.2.3-2.

**Table 6.3.3.1.2.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 8 x 4<br>(N1,N2) = (4,1) |
| Beamforming Model | | | As specified in Section Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 8 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 8, (4,6) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Patten 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, |

| | | | |
|---|---|---|---|
| | | | otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.2 FDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (1 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#[(n-4)], this reported PMI cannot be applied at the eNB downlink before slot#[(n+4)].<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3 | | | |

**Table 6.3.3.1.2.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.5 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.3.3.1.2.

### 6.3.3.1.2.4 Test description

#### 6.3.3.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.1.2.3-1and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.1.2.4.3.

#### 6.3.3.1.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.1.2.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.3.1.2.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.3.1.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.1.2.4.3.1 Message exceptions for SA

**Table 6.3.3.1.2.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | aperiodic | | |
| } | | | |

**Table 6.3.3.1.2.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001100 | | |
| } | | | |
| nrofPorts | p8 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.1.2.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.3.1.2.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.3.1.2.4.3.1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-one-TypeI-SinglePanel-Restriction | FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.3.1.2.4.3.1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.3.1.2.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.1.2.4.3.

6.3.3.1.2.5 Test requirement

**Table 6.3.3.1.2.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.49 |

## 6.3.3.1.3 4Rx FDD FR1 Multiple PMI with 16Tx Type I – SinglePanel Codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

### 6.3.3.1.3.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.3.1.3.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.3.1.3.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.1.3.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.1.3.3-2.

**Table 6.3.3.1.3.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLC300-5 |
| Antenna configuration | | | High XP 16 x 4<br>(N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |

| | | | |
|---|---|---|---|
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.3 FDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.3.1.3.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 3.0 |

### 6.3.3.1.3.4 Test description

#### 6.3.3.1.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.1.3.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.1.3.4.3.

### 6.3.3.1.3.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.1.3.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.3.1.3.5-1, then the test is pass. Otherwise, the test is fail.

### 6.3.3.1.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

#### 6.3.3.1.3.4.3.1 Message exceptions for SA

**Table 6.3.3.1.3.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | P16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.1.3.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-two-TypeI-SinglePanel-Restriction | FFFF FFFF FFFF FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.3.1.3.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| | | | |
| } | | | |

6.3.3.1.3.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.1.3.4.3.1.

6.3.3.1.3.5 Test requirement

**Table 6.3.3.1.3.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.99 |

## 6.3.3.1.4 4Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

### 6.3.3.1.4.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.3.1.4.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.3.1.4.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.1.4.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.1.4.3-2.

**Table 6.3.3.1.4.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 32 x 4<br>(N1,N2) = (4,4) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 32 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 17, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, 12) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |

| | | | |
|---|---|---|---|
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,4) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.3 FDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |

Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.

Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).

Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3.

**Table 6.3.3.1.4.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 7.0 |

#### 6.3.3.1.4.4 Test description

##### 6.3.3.1.4.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.1.4.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.1.4.4.3.

##### 6.3.3.1.4.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.1.4.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.3.1.4.5-1, then the test is pass.
   Otherwise, the test is fail.

##### 6.3.3.1.4.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

6.3.3.1.4.4.3.1 Message exceptions for SA

**Table 6.3.3.1.4.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping:: = SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | P32 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.1.4.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-four-TypeI-SinglePanel-Restriction | FFFF FFFF FFFF FFFF<br>FFFF FFFF FFFF FFFF<br>FFFF FFFF FFFF FFFF<br>FFFF FFFF FFFF FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.3.1.4.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| } | | | |

6.3.3.1.4.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.1.4.4.3.1.

6.3.3.1.4.5 Test requirement

**Table 6.3.3.1.4.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 6.99 |

6.3.3.1.5 4Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

#### 6.3.3.1.5.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

#### 6.3.3.1.5.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 6.3.3.1.5.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.1.5.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.1.5.3-2.

**Table 6.3.3.1.5.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 4 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |

| | | | |
|---|---|---|---|
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII |
| | L (*numberOfBeams*) | | 2 |
| | $N_{PSK}$ (*phaseAlphabetSize*) | | 8 |
| | *subbandAmplitude* | | True |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction) | | 10 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 8 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.1-6.3 FDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.3.1.3-1.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4).<br>Note 3: Randomization of the dual-cluster beam directions shall be used as specified in Annex B.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test. | | | |

**Table 6.3.3.1.5.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.9 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.3.3.1.5.

#### 6.3.3.1.5.4 Test description

##### 6.3.3.1.5.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.1.5.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.4.4.3.

##### 6.3.3.1.5.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.1.5.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities, and the random precoder generation shall follow the codebook configuration as specified in Table 6.3.3.1.3.3-1. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio $\geq \gamma$ which is specified in table 6.3.3.1.5.5-1, then the test is pass. Otherwise, the test is fail.

##### 6.3.3.1.5.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

6.3.3.1.5.4.3.1 Message exceptions for SA

**Table 6.3.3.1.5.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | '011110'B | | |
| } | | | |
| nrofPorts | P16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.1.5.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII SEQUENCE { | | | |
| n1-n2-codebookSubsetRestriction CHOICE { | | | |
| four-two | 0x 7FF FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction | '10'B | | |
| } | | | |
| } | | | |
| phaseAlphabetSize | 8 | | |
| subbandAmplitude | TRUE | | |
| numberOfBeams | 2 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.3.1.5.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| } | | | |

6.3.3.1.5.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.1.5.4.3.1.

6.3.3.1.5.5 Test requirement

**Table 6.3.3.1.5.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.89 |

## 6.3.3.1.6 4Rx FDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA

### 6.3.3.1.6.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.3.1.6.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

### 6.3.3.1.6.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.1.6.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.1.6.3-2.

**Table 6.3.3.1.6.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 10 |
| Subcarrier spacing | | kHz | 15 |
| Duplex Mode | | | FDD |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 2<br>(N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |

| | | | |
|---|---|---|---|
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Not configured |
| Sub-band Size | | RB | 4 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 5 |
| CSI request | | | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII-r16 |
| | *paramCombination-r16* | | 6 (L =4, $p_v$ =1/2, β=1/2 ) |
| | R*(numberOfPMISubbandsPerCQISubband-r16)* | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction-r16) | | 0010 |

| | | |
|---|---|---|
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 8 |
| Maximum number of HARQ transmission | | 4 |
| Measurement channel | | R.PDSCH.1-6.3 |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (1 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.3.1.3-1. | | |
| Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4), this reported PMI cannot be applied at the gNB downlink before slot#(n+4). | | |
| Note 3: Randomization of the dual-cluster beam directions shall be used as specified in AnnexB.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test. | | |

**Table 6.3.3.1.6.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.2 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.3.1.6.

### 6.3.3.1.6.4 Test description

#### 6.3.3.1.6.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.10 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.1.6.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.1.6.4.3.

#### 6.3.3.1.6.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.1.6.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio $\geq \gamma$ which is specified in table 6.3.3.1.6.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.3.1.6.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.1.6.4.3.1 Message exceptions for SA

**Table 6.3.3.1.6.4.3.1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| codebookConfig | Not present | | |
| subbandSize | Value1 | | |
| codebookConfig-r16 | CodebookConfig-r16 | | |
| } | | | |

**Table 6.3.3.1.6.4.3.1-2: *CodebookConfig-r16* (Table 6.3.3.1.6.4.3.1-1)**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig-r16 ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII-r16 SEQUENCE { | | | |
| N1-n2-codebookSubsetRestriction-r16 | | | |
| Four-two | 0x 7FF FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction-r16 | 0010 | | |
| } | | | |
| } | | | |
| numberOfPMI-SubbandsPerCQI-Subband-r16 | 1 | | |
| paramCombinatin-r16 | 6 | (L =4, $p_v$ =1/2, β=1/2 ) | |
| } | | | |
| } | | | |
| } | | | |

#### 6.3.3.1.6.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.1.6.4.3.1.

### 6.3.3.1.6.5 Test requirement

**Table 6.3.3.1.6.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.19 |

## 6.3.3.2 TDD

### 6.3.3.2.1 4Rx TDD FR1 Single PMI with 4TX TypeI-SinglePanel codebook for both SA and NSA

#### 6.3.3.2.1.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

#### 6.3.3.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 6.3.3.2.1.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.2.1.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.2.1.3-2.

**Table 6.3.3.2.1.3-1: Test parameters (single layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configuration | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 4 x 4 (N1,N2) = (2,1) |
| Beamforming Model | | | As specified in Section Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 4, (0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) |
| | CSI-RS interval and offset | | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Patten 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |

| | | | |
|---|---|---|---|
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (2,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 11111111 |
| | RI Restriction | | 00000001 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 5.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.1 TDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#[(n-4)], this reported PMI cannot be applied at the eNB downlink before slot#[(n+4)].<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3 | | | |

**Table 6.3.3.2.1.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.3 |

### 6.3.3.2.1.4 Test description

#### 6.3.3.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.2.1.3-1 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release On for SA or (EN-DC, DC bearer MCG and SCG, Connected without Release On) for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.2.1.4.3.

#### 6.3.3.2.1.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.2.1.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.3.1.1.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.3.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.2.1.4.3.1 Message exceptions for SA

**Table 6.3.3.2.1.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | |
| } | | | |

**Table 6.3.3.2.1.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause5.4.2, Table5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row4 | 001 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.3.3.2.1.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| Row5 | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.3.2.1.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |

**Table 6.3.3.2.1.4.3.1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| two-one-TypeI-SinglePanel-Restriction | 11111111 | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000001 | | |

**Table 6.3.3.2.1.4.3.1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 0 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | [1111111] | | |
| } | | | |
| } | | | |
| subbandSize | value2 | | |
| } | | | |

6.3.3.2.1.4.3.2 Message exception for NSA

Same as in 6.3.3.2.1.4.3.1.

6.3.3.2.1.5 Test requirement

**Table 6.3.3.2.1.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.29 |

## 6.3.3.2.2 4Rx TDD FR1 Single PMI with 8TX TypeI-SinglePanel codebook for both SA and NSA

### 6.3.3.2.2.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.3.2.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 6.3.3.2.2.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.2.2.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.2.2.3-2.

**Table 6.3.3.2.2.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | High XP 8 x 4 (N1,N2) = (4,1) |
| Beamforming Model | | | As specified in Section Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 8 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 8, (4,6) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Patten 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannnelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |

| | | | |
|---|---|---|---|
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration<br>Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,1) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,1) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.2 TDD |
| PDSCH & PDSCH DMRS Precoding configuration for random Precoding | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Note 1: For random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity).<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#[(n-6)], this reported PMI cannot be applied at the eNB downlink before slot#[(n+6)].<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3 | | | |

**Table 6.3.3.2.2.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.5 |

### 6.3.3.2.2.4 Test description

#### 6.3.3.2.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7 for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.2.1-2 and Table 6.3.3.2.2.3-1and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.2.2.4.3.

#### 6.3.3.2.2.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.2.2.3-1 as appropriate.
2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.
3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.
4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.3.2.2.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.3.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.2.2.4.3.1 Message contents for SA

**Table 6.3.3.2.2.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | |
| } | | | |

**Table 6.3.3.2.2.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause5.4.2, Table5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001100 | | |
| } | | | |
| nrofPorts | p8 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.2.2.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.3.2.2.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.3.2.2.4.3.1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-one-TypeI-SinglePanel-Restriction | FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.3.2.2.4.3.1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 8 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.3.2.2.4.3.2 Message contents for NSA

Same as in clause 6.3.3.2.2.4.3.1.

6.3.3.2.2.5 Test requirement

**Table 6.3.3.2.2.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.49 |

## 6.3.3.2.3 4Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

6.3.3.2.3.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

6.3.3.2.3.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and 4 Rx antenna ports.

6.3.3.2.3.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.2.3.3-1 and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.2.3.3-2.

**Table 6.3.3.2.3.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLC300-5 |
| Antenna configuration | | | High XP 16 x 4 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannnelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |

| | | | |
|---|---|---|---|
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubset Restriction | | 0x FFFF |
| | RI Restriction | | 00000010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.3 TDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6).<br>Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3. | | | |

**Table 6.3.3.2.3.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 3.0 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.3.2.3.

### 6.3.3.2.3.4 Test description

#### 6.3.3.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.5 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.2.1-2 and Table 6.3.3.2.3.3-1and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.2.3.4.3.

#### 6.3.3.2.3.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.2.3.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow\,1,\,follow\,2}}{t_{rnd\,1,\,rnd\,2}}$. If the ratio ≥ γ which is specified in table 6.3.3.2.3.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.3.2.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.2.3.4.3.1 Message contents for SA

**Table 6.3.3.2.3.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | |
| } | | | |

**Table 6.3.3.2.3.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause5.4.2, Table 5.4.2.0-15 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 011110 | | |
| } | | | |
| nrofPorts | p16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.2.3.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2, Table 5.4.2.0-21 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.3.3.2.3.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| | | | |

**Table 6.3.3.2.3.4.3.1-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| moreThanTwo SEQUENCE { | | | |
| n1-n2 CHOICE { | | | |
| four-two-TypeI-SinglePanel-Restriction | FFFF FFFF FFFF FFFF | | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 00000010 | | |

**Table 6.3.3.2.3.4.3.1-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 8 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| | | | |
| } | | | |

6.3.3.2.3.4.3.2 Message contents for NSA

Same as in clause 6.3.3.2.3.4.3.1.

6.3.3.2.3.5 Test requirement

**Table 6.3.3.2.3.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.99 |

6.3.3.2.4 FFS

6.3.3.2.5 4Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Connection figure for 16 Tx is missing

6.3.3.2.5.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

6.3.3.2.5.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

6.3.3.2.5.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.2.5.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.2.5.3-2.

**Table 6.3.3.2.5.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 4 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |

| | | | |
|---|---|---|---|
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Subband |
| Sub-band Size | | RB | 16 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII |
| | L (*numberOfBeams*) | | 2 |
| | $N_{PSK}$ (*phaseAlphabetSize*) | | 8 |
| | *subbandAmplitude* | | True |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction) | | 10 |

| | | |
|---|---|---|
| Physical channel for CSI report | | PUSCH |
| CQI/RI/PMI delay | ms | 6.5 |
| Maximum number of HARQ transmission | | 4 |
| Measurement channel | | R.PDSCH.2-8.3 TDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.3.2.3-1. | | |
| Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6). | | |
| Note 3: Randomization of the dual-cluster beam directions shall be used as specified in Annex B.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test. | | |

**Table 6.3.3.2.5.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.8 |

The normative reference for this requirement is TS 38.101-4 [5] clause 6.3.3.2.5.

### 6.3.3.2.5.4 Test description

#### 6.3.3.2.5.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure TBD for TE diagram and section A.3.2.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.2.5.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG*, *Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.2.1.4.4.3.

#### 6.3.3.2.5.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.2.5.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue,follow1,follow2}$ and $SNR_{follow1,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities, and the random precoder generation shall follow the codebook configuration as specified in Table 6.3.3.2.3.3-1. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd1,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,follow1,follow2}}{t_{rnd1,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.3.2.5.5-1, then the test is pass. Otherwise, the test is fail.

#### 6.3.3.2.5.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

##### 6.3.3.2.5.4.3.1 Message exceptions for SA

**Table 6.3.3.2.5.4.3.1-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | '011110'B | | |
| } | | | |
| nrofPorts | P16 | | |
| firstOFDMSymbolInTimeDomain | 5 | | |
| cdm-Type | cdm4-FD2-TD2 | | |
| } | | | |

**Table 6.3.3.2.5.4.3.1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-14 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII SEQUENCE { | | | |
| n1-n2-codebookSubsetRestriction CHOICE { | | | |
| four-two | 0x 7FF FFFF<br>FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction | '10'B | | |
| } | | | |
| } | | | |
| phaseAlphabetSize | 8 | | |
| subbandAmplitude | TRUE | | |
| numberOfBeams | 2 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 6.3.3.2.5.4.3.1-3: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 5 | | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | subbandPMI | | |
| } | | | |
| } | | | |

6.3.3.2.5.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.2.5.4.3.1.

6.3.3.2.5.5 Test requirement

**Table 6.3.3.2.5.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 1.79 |

## 6.3.3.2.6 4Rx TDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA

### 6.3.3.2.6.1 Test purpose

To test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

### 6.3.3.2.6.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and Enhanced Type II codebook with at least 16 ports per CSI-RS resource.

### 6.3.3.2.6.3 Minimum conformance requirements

For the parameters specified in Table 6.3.3.2.6.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.3.3.2.6.3-2.

**Table 6.3.3.2.6.3-1: Test parameters (dual-layer)**

| Parameter | | Unit | Test 1 |
|---|---|---|---|
| Bandwidth | | MHz | 40 |
| Subcarrier spacing | | kHz | 30 |
| Duplex Mode | | | TDD |
| TDD DL-UL configurations | | | FR1.30-1 as specified in Annex A |
| Propagation channel | | | TDLA30-5 |
| Antenna configuration | | | XP Medium 16 x 2 (N1,N2) = (4,2) |
| Beamforming Model | | | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports ($X$) | | 4 |

| | | | |
|---|---|---|---|
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) |
| | CSI-RS interval and offset | slot | Not configured |
| | ZP CSI-RS trigger | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 16 |
| | CDM Type | | CDM4 (FD2, TD2) |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$, $k_2$, $k_3$) | | Row 12, (2, 4, 6, 8) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (5, -) |
| | CSI-RS interval and offset | slot | Not configured |
| | aperiodicTriggeringOffset | | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic |
| | CSI-IM RE pattern | | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) |
| | CSI-IM timeConfig interval and offset | slot | Not configured |
| ReportConfigType | | | Aperiodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForIChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Not configured |
| Sub-band Size | | RB | 8 |
| csi-ReportingBand | | | 1111111 |
| CSI-Report interval and offset | | slot | Not configured |
| Aperiodic Report Slot Offset | | | 8 |
| CSI request | | | 1 in slots i, where mod(i, 10) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeII-r16 |
| | *paramCombination-r16* | | 6 (L =4, $p_v$ =1/2, β=1/2 ) |
| | R*(numberOfPMISubbandsPerCQISubband-r16)* | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | (4,2) |
| | (CodebookConfig-O1,CodebookConfig-O2) | | (4,4) |

| | | | |
|---|---|---|---|
| | CodebookSubsetRestriction | | 0x 7FF FFFF FFFF FFFF FFFF |
| | RI Restriction (typeII-RI-Restriction-r16) | | 0010 |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 6.5 |
| Maximum number of HARQ transmission | | | 4 |
| Measurement channel | | | R.PDSCH.2-8.3 TDD |
| Note 1: When Throughput is measured using random precoder selection, the precoder shall be updated in each slot (0.5 ms granularity) with equal probability of each applicable $i_1$, $i_2$ combination. The random precoder generation shall follow 'typeI-SinglePanel' codebook configuration as specified in table 6.3.3.2.3-1.<br>Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-6), this reported PMI cannot be applied at the gNB downlink before slot#(n+6).<br>Note 3: Randomization of the dual-cluster beam directions shall be used as specified in Annex B.2.3.2.3A. The value of relative power ratio (p) shall be fixed as 1 during the test. | | | |

**Table 6.3.3.2.6.3-2: Minimum requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.2 |

The normative reference for this requirement is TS 38.101-4 [5], clause 6.3.3.2.6.

### 6.3.3.2.6.4 Test description

#### 6.3.3.2.6.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.10 for TE diagram and section A.3.2.2 for UE diagram.

2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.3.3.2.6.3-1 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, Connected without release *On* for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without Release On)* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 6.3.3.2.6.4.3.

#### 6.3.3.2.6.4.2 Test procedure

1. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 6.3.3.2.6.3-1 as appropriate.

2. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Establish $t_{ue,\,follow1,\,follow2}$ and $SNR_{follow1,\,follow2}$ according to Annex G.3.2.

3. Set SNR to $SNR_{follow1,\,follow2}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-6 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format 0_1 with aperiodic CSI request triggered. Measure $t_{rnd1,\,rnd2}$ according to Annex G.3.3.

4. Calculate $\gamma = \frac{t_{ue,\,follow1,\,follow2}}{t_{rnd1,\,rnd2}}$. If the ratio ≥ γ which is specified in table 6.3.3.2.6.5-1, then the test is pass.
   Otherwise, the test is fail.

#### 6.3.3.2.6.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2.

##### 6.3.3.2.6.4.3.1 Message exceptions for SA

**Table 6.3.3.2.6.4.3.1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 5.4.2.5, Table 5.4.2.5-13 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| pmi-FormatIndicator | Not present | | |
| } | | | |
| codebookConfig | Not present | | |
| subbandSize | Value1 | | |
| codebookConfig-r16 | CodebookConfig-r16 | | |
| } | | | |

**Table 6.3.3.2.6.4.3.1-2: *CodebookConfig-r16* (Table 6.3.3.2.6.4.3.1-1)**

| Derivation Path: TS 38.331 [6], clause 6.3.2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig-r16 ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type2 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeII-r16 SEQUENCE { | | | |
| N1-n2-codebookSubsetRestriction-r16 | | | |
| Four-two | 0x 7FF<br>FFFF FFFF FFFF FFFF | | |
| } | | | |
| typeII-RI-Restriction-r16 | 0010 | | |
| } | | | |
| } | | | |
| numberOfPMI-SubbandsPerCQI-Subband-r16 | 1 | | |
| paramCombinatin-r16 | 6 | (L =4, $p_v$ =1/2, β=1/2 ) | |
| } | | | |
| } | | | |
| } | | | |

6.3.3.2.6.4.3.2 Message exceptions for NSA

Same as in clause 6.3.3.2.6.4.3.1.

6.3.3.2.6.5 Test requirement

**Table 6.3.3.2.6.5-1: Test requirement**

| Parameter | Test 1 |
|---|---|
| $\gamma$ | 2.19 |

# 6.4 Reporting of Rank Indicator (RI)

## 6.4.1 1RX requirements (Void)

## 6.4.2 2RX requirements

### 6.4.2.1 FDD

#### 6.4.2.1_1 2Rx FDD FR1 RI reporting for both SA and NSA

##### 6.4.2.1_1.1 Test Purpose

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission.

##### 6.4.2.1_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

6.4.2.1_1.3 Minimum Conformance Requirements

The minimum performance requirement in Table 6.4.2.1_1.3-2 is defined as:

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

For the parameters specified in Table 6.4.2.1_1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.4.2.1_1.3-2.

**Table 6.4.2.1_1.3-1: RI Test (FDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | 10 | 10 |
| Subcarrier spacing | | kHz | 15 | 15 | 15 |
| Duplex Mode | | | FDD | FDD | FDD |
| SNR | | dB | 0 | 20 | 20 |
| Propagation channel | | | TDLA30-5 | TDLA30-5 | TDLA30-5 |
| Antenna configuration | | | ULA Low 2x2 | ULA Low 2x2 | ULA High 2x2 |
| Beamforming Model | | | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 | 4 | 4 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) | Row 5, (4,-) | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) | (9,-) | (9,-) |
| | CSI-RS periodicity and offset | slot | 5/1 | 5/1 | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 2 | 2 | 2 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3 (6,-) | Row 3 (6,-) | Row 3 (6,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) | (13,-) |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | 5/1 | 5/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | Periodic | Periodic |
| | CSI-IM RE pattern | | Pattern 0 | Pattern 0 | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) | (4,9) | (4,9) |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | 5/1 | 5/1 |
| ReportConfigType | | | Periodic | Periodic | Periodic |
| CQI-table | | | Table 2 | Table 2 | Table 2 |
| reportQuantity | | | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | not configured | not configured | not configured |
| timeRestrictionForInterferenceMeasurements | | | not configured | not configured | not configured |
| cqi-FormatIndicator | | | Wideband | Wideband | Wideband |
| pmi-FormatIndicator | | | Wideband | Wideband | Wideband |
| Sub-band Size | | RB | 8 | 8 | 8 |
| csi-ReportingBand | | | 1111111 | 1111111 | 1111111 |
| CSI-Report periodicity and offset | | slot | 5/0 | 5/0 | 5/0 |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel |
| | Codebook Mode | | 1 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| | (CodebookConfig-N1,CodebookConfig-N2) | | N/A | N/A | N/A |
| | CodebookSubsetRestriction | | 010000 for fixed rank 2, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank |
| | RI Restriction | | N/A | N/A | N/A |
| Physical channel for CSI report | | | PUCCH | PUCCH | PUCCH |
| CQI/RI/PMI delay | | ms | 8 | 8 | 8 |
| Maximum number of HARQ transmission | | | 1 | 1 | 1 |
| RI Configuration | | | Fixed RI = 2 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 1 and follow RI |

**Table 6.4.2.1_1.3-2: Minimum requirement (FDD)**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | 0.9 |
| $\gamma_2$ | 1.0 | N/A | N/A |

The normative reference for this requirement is TS 38.101-4 [2] clause 6.4.2.1.

### 6.4.2.1_1.4 Test Description

#### 6.4.2.1_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state. The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex D.

Test Environment: Normal as defined in TS 38.508 [6] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 38.508 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram and section A.3.2.3 for UE diagram.

2. The parameter settings for the cell are set up according to Table 5.2-1, Table 5.2.2.1.1.0-2 and Table 5.2.2.1.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].

4. Propagation conditions for the NR cell are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, *Connected without release On* and Test Mode ON for SA or EN-DC, DC bearer *MCG* and *SCG, Connected without release On* and Test Mode ON for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 6.4.2.1_1.4.3.

#### 6.4.2.1_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference channel, the propagation condition, antenna configuration, antenna correlation, Codebook configuration, Beamforming Model, RI configuration and SNR according to Table 6.4.2.1_1.3-1 as appropriate.

2. The SS shall send PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and fixed RI as defined in Table 6.4.2.1_1.3-1. The SS sends downlink MAC padding bits on the DL RMC. Measure the $t_{fix}$ according to Annex G.3. 3.

3. Propagation conditions are set according to Annex B.2.

4. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 6.4.2.1_1.3-1.

5. The UE shall transmit RRC Connection Reconfiguration Complete message.

6. Propagation conditions are set according to Table 6.4.2.1_1.3-1.

7. The SS shall send PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC. Measure $t_{reported}$ according to Annex G.3.3.

   If the ratio ($t_{reported}$ / $t_{fix}$) satisfies the requirement in Table 6.4.2.1_1.5-1, then pass the UE for this test and go to step 8. Otherwise, declare a FAIL verdict.

8. If all tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the Table 6. 4.2. 1_1.3-2 for the other Tests as appropriate. Otherwise, declare a PASS verdict.

### 6.4.2.1_1.4.3 Message Contents

#### 6.4.2.1_1.4.3.1 Message exceptions for SA

**Table 6.4.2.1_1.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | periodic | | |
| } | | | |

**Table 6.4.2.1_1.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | row3, $k_0$=6 | |
| } | | | |
| nrofPorts | p2 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.4.2.1_1.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | row5, $k_0$=4 | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.4.2.1_1.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.4.2.1_1.4.3.1-5: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| Slots5 | 1 | | |
| } | | | |

**Table 6.4.2.1_1.4.3.1-6: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| | | | |
| twoTX-CodebookSubsetRestriction | 010000 | Fixed rank 2 | |
| | 000011 | Fixed rank 1 | |
| | 010011 | Following rank | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | Non restriction | |

**Table 6.4.2.1_1.4.3.1-7: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| reportSlotConfig CHOICE { | slots5 | | |
| slots5 | 0 | | |
| } | | | |
| pucch-CSI-ResourceList | 9 | PUCCH format Id=9 | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| } | | | |

6.4.2.1_1.4.3.2 Message exceptions for NSA

Same as in clause 6.4.2.1_1.4.3.1.

6.4.2.1_1.5 Test Requirements

**Table 6.4.2.1_1.5-1: Test Requirement (FDD)**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | 0.89 |
| $\gamma_2$ | 0.99 | N/A | N/A |

## 6.4.2.2 TDD

### 6.4.2.2_1 2Rx TDD FR1 RI reporting for both SA and NSA

#### 6.4.2.2_1.1 Test Purpose

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission.

#### 6.4.2.2_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 6.4.2.2_1.3 Minimum Conformance Requirements

The minimum performance requirement in Table 6.4.2.2_1.3-2 is defined as:

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

For the parameters specified in Table 6.4.2.2_1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.4.2.2_1.3-2.

**Table 6.4.2.2_1.3-1: RI Test (TDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| Bandwidth | | MHz | 40 | 40 | 40 |
| Subcarrier spacing | | kHz | 30 | 30 | 30 |
| Duplex Mode | | | TDD | TDD | TDD |
| TDD Slot Configuration | | | FR1.30-1 | FR1.30-1 | FR1.30-1 |
| SNR | | dB | 0 | 20 | 20 |
| Propagation channel | | | TDLA30-5 | TDLA30-5 | TDLA30-5 |
| Antenna configuration | | | ULA Low 2x2 | ULA Low 2x2 | ULA High 2x2 |
| Beamforming Model | | | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 | 4 | 4 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) | Row 5, (4,-) | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) | (9,-) | (9,-) |
| | CSI-RS periodicity and offset | slot | 10/1 | 10/1 | 10/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 2 | 2 | 2 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3 (6,-) | Row 3 (6,-) | Row 3 (6,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) | (13,-) |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | 10/1 | 10/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | Periodic | Periodic |
| | CSI-IM RE pattern | | Pattern 0 | Pattern 0 | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) | (4,9) | (4,9) |
| | CSI-IM timeConfig periodicity and offset | slot | 10/1 | 10/1 | 10/1 |
| ReportConfigType | | | Periodic | Periodic | Periodic |
| CQI-table | | | Table 2 | Table 2 | Table 2 |
| reportQuantity | | | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | not configured | not configured | not configured |
| timeRestrictionForInterferenceMeasurements | | | not configured | not configured | not configured |
| cqi-FormatIndicator | | | Wideband | Wideband | Wideband |
| pmi-FormatIndicator | | | Wideband | Wideband | Wideband |
| Sub-band Size | | RB | 16 | 16 | 16 |
| csi-ReportingBand | | | 1111111 | 1111111 | 1111111 |
| CSI-Report periodicity and offset | | slot | 10/9 | 10/9 | 10/9 |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel |
| | Codebook Mode | | 1 | 1 | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | N/A | N/A | N/A |
| | CodebookSubsetRestriction | | 010000 for fixed rank 2, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank |
| | RI Restriction | | N/A | N/A | N/A |
| Physical channel for CSI report | | | PUCCH | PUCCH | PUCCH |
| CQI/RI/PMI delay | | ms | 9.5 | 9.5 | 9.5 |
| Maximum number of HARQ transmission | | | 1 | 1 | 1 |

| | | | | |
|---|---|---|---|---|
| RI Configuration | | Fixed RI = 2 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 1 and follow RI |

**Table 6.4.2.2_1.3-2: Minimum requirement (TDD)**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | 0.9 |
| $\gamma_2$ | 1.0 | N/A | N/A |

The normative reference for this requirement is TS 38.101-4 [2] clause 6.4.2.2.

### 6.4.2.2_1.4 Test Description

#### 6.4.2.2_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state. The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex D.

Test Environment: Normal as defined in TS 38.508 [6] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 38.508 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7 for TE diagram and section A.3.2.3 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.4.2.2_1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions for the NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, *Connected without release On* and Test Mode ON for SA or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On* and Test Mode ON for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 6.4.2.2_1.4.3.

#### 6.4.2.2_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference channel, the propagation condition, antenna configuration, antenna correlation, Codebook configuration, Beamforming Model, RI configuration and SNR according to Table 6.4.2.2_1.3-1 as appropriate.
2. The SS shall send PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and fixed RI as defined in Table 6.4.2.2_1.3-1. The SS sends downlink MAC padding bits on the DL RMC. Measure the $t_{fix}$ according to Annex G.3. 3.
3. Propagation conditions are set according to Annex B.2.
4. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 6.4.2.2_1.3-1.
5. The UE shall transmit RRC Connection Reconfiguration Complete message.
6. Propagation conditions are set according to Table 6.4.2.2_1.3-1.

7. The SS shall send PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC. Measure $t_{reported}$ according to Annex G.3.3.

   If the ratio ( $t_{reported}$ / $t_{fix}$ ) satisfies the requirement in Table 6.4.2.2_1.5-1, then pass the UE for this test and go to step 8. Otherwise, declare a FAIL verdict.

8. If all tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the Table 6.4.2.2_1.3-2 for the other Tests as appropriate. Otherwise, declare a PASS verdict.

### 6.4.2.2_1.4.3 Message Contents

#### 6.4.2.2_1.4.3.1 Message Contents for SA

**Table 6.4.2.2_1.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | periodic | | |
| } | | | |

**Table 6.4.2.2_1.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | row3, $k_0$=6 | |
| } | | | |
| nrofPorts | p2 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.4.2.2_1.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | row5, $k_0$=4 | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.4.2.2_1.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.4.2.2_1.4.3.1-5: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.2-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| Slots10 | 1 | | |
| } | | | |

**Table 6.4.2.2_1.4.3.1-6: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| | | | |
| twoTX-CodebookSubsetRestriction | 010000 | Fixed rank 2 | |
| | 000011 | Fixed rank 1 | |
| | 010011 | Following rank | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | Non restriction | |

**Table 6.4.2.2_1.4.3.1-7: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| reportSlotConfig CHOICE { | slots10 | | |
| slots10 | 9 | | |
| } | | | |
| pucch-CSI-ResourceList | 9 | PUCCH format Id=9 | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| } | | | |

6.4.2.2_1.4.3.2 Message exceptions for NSA

Same as in clause 6.4.2.1.2.1.4.3_1.

6.4.2.2_1.5 Test Requirements

**Table 6.4.2.2_1.5-1: Test Requirement (TDD)**

| | **Test 1** | **Test 2** | **Test 3** |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | 0.89 |
| $\gamma_2$ | 0.99 | N/A | N/A |

# 6.4.3 4RX requirements

## 6.4.3.1 FDD

### 6.4.3.1_1 4Rx FDD FR1 RI reporting for both SA and NSA

#### 6.4.3.1_1.1 Test Purpose

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission.

#### 6.4.3.1_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

#### 6.4.3.1_1.3 Minimum Conformance Requirements

The minimum performance requirement in Table 6.4.3.1_1.3-2 is defined as

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

For the parameters specified in Table 6.4.3.1_1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.4.3.1_1.3-2.

**Table 6.4.3.1_1.3-1: RI Test (FDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 10 | 10 | 10 | 10 |
| Subcarrier spacing | | kHz | 15 | 15 | 15 | 15 |
| Duplex Mode | | | FDD | FDD | FDD | FDD |
| SNR | | dB | -2 | 16 | 16 | 22 |
| Propagation channel | | | TDLA30-5 | TDLA30-5 | TDLA30-5 | TDLA30-5 |
| Antenna configuration | | | ULA Low 2x4 | ULA Low 2x4 | ULA High 2x4 | ULA Low 4x4 |
| Beamforming Model | | | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 | 4 | 4 | 4 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) | Row 5, (4,-) | Row 5, (4,-) | Row 5, (4,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) | (9,-) | (9,-) | (9,-) |
| | CSI-RS periodicity and offset | slot | 5/1 | 5/1 | 5/1 | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 2 | 2 | 2 | 4 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3 (6,-) | Row 3 (6,-) | Row 3 (6,-) | Row 4 (0,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) | (13,-) | (13,-) |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 5/1 | 5/1 | 5/1 | 5/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | Periodic | Periodic | Periodic |
| | CSI-IM RE pattern | | Pattern 0 | Pattern 0 | Pattern 0 | Pattern 0 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) | (4,9) | (4,9) | (4,9) |
| | CSI-IM timeConfig periodicity and offset | slot | 5/1 | 5/1 | 5/1 | 5/1 |
| ReportConfigType | | | Periodic | Periodic | Periodic | Periodic |
| CQI-table | | | Table 2 | Table 2 | Table 2 | Table 2 |
| reportQuantity | | | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | not configured | not configured | not configured | not configured |
| timeRestrictionForInterferenceMeasurements | | | not configured | not configured | not configured | not configured |
| cqi-FormatIndicator | | | Wideband | Wideband | Wideband | Wideband |
| pmi-FormatIndicator | | | Wideband | Wideband | Wideband | Wideband |
| Sub-band Size | | RB | 8 | 8 | 8 | 8 |
| csi-ReportingBand | | | 1111111 | 1111111 | 1111111 | 1111111 |
| CSI-Report periodicity and offset | | slot | 5/0 | 5/0 | 5/0 | 5/0 |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel |
| | Codebook Mode | | 1 | 1 | 1 | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | N/A | N/A | N/A | (2,1) |
| | CodebookSubsetRestriction | | 010000 for fixed rank 2, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 11111111 |
| | RI Restriction | | N/A | N/A | N/A | 00000010 for fixed Rank 2 and 00001111 for follow RI |
| Physical channel for CSI report | | | PUCCH | PUCCH | PUCCH | PUCCH |
| CQI/RI/PMI delay | | ms | 8 | 8 | 8 | 8 |

| | | | | | |
|---|---|---|---|---|---|
| Maximum number of HARQ transmission | | 1 | 1 | 1 | 1 |
| RI Configuration | | Fixed RI = 2 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 2 and follow RI |

**Table 6.4.3.1_1.3-2: Minimum requirement (FDD)**

| | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | 0.9 | N/A |
| $\gamma_2$ | 0.9 | N/A | N/A | 0.9 |

The normative reference for this requirement is TS 38.101-4 [2] clause 6.4.3.1.

### 6.4.3.1_1.4 Test Description

#### 6.4.3.1_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal as defined in TS 38.508 [6] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 38.508 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.4 or A.3.1.7.5 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1, Table 6.4.3.1_1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions for the NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, *Connected without release On* and Test Mode ON for SA or (EN-DC, DC bearer *MCG* and *SCG, Connected without release On* for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 6.4.3.1_1.4.3.

#### 6.4.3.1_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference channel, the propagation condition, antenna configuration, antenna correlation, Codebook configuration, Beamforming Model, RI configuration and SNR according to Table 6.4.3.1_1.3-1 as appropriate.
2. The SS shall send PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and fixed RI as defined in Table 6.4.3.1_1.3-1. The SS sends downlink MAC padding bits on the DL RMC. Measure the $t_{fix}$ according to Annex G.3. 3.
3. Propagation conditions are set according to Annex B.2. 4
4. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 6.4.3.1_1.3-1.
5. The UE shall transmit RRC Connection Reconfiguration Complete message.
6. Propagation conditions are set according to Table 6.4.3.1_1.3-1.

7. The SS shall send PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC.

   Measure $t_{reported}$ according to Annex G.3.3.

   If the ratio ($t_{reported}$ / $t_{fix}$) satisfies the requirement in Table 6.4.3.1_1.5-1, then pass the UE for this test and go to step 8. Otherwise, declare a FAIL verdict.

8. If all tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the Table 6.4.3.1_1.3-2 for the other Tests as appropriate. Otherwise, declare a PASS verdict.

### 6.4.3.1_1.4.3 Message Contents

#### 6.4.3.1_1.4.3.1 Message exceptions for SA

**Table 6.4.3.1_1.4.3.1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | periodic | | |
| } | | | |

**Table 6.4.3.1_1.4.3.1-2: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | row3, $k_0$=6 for test 1,2,3 | |
| row 4 | 001 | row4, $k_0$=0 for test 4 | |
| } | | | |
| nrofPorts | p2 | Test 1,2,3 | |
| | p4 | Test 4 | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.4.3.1_1.4.3.1-3: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | row5, $k_0$=4 | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.4.3.1_1.4.3.1-4: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.4.3.1_1.4.3.1-5: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| Slots5 | 1 | | |
| } | | | |

**Table 6.4.3.1_1.4.3.1-6: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| | | | |
| twoTX-CodebookSubsetRestriction | 010000 | Fixed rank 2 | |
| | 000011 | Fixed rank 1 | |
| | 010011 | Following rank for test 1,2,3 | |
| | 11111111 | Test 4 | |
| } | | | |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | Non restriction for test 1,2,3 | |
| | 00000010 | For fixed Rank2 for test 4 | |
| | 00001111 | For follow RI for test 4 | |

**Table 6.4.3.1_1.4.3.1-7: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| reportSlotConfig CHOICE { | slots5 | | |
| slots5 | 0 | | |
| } | | | |
| pucch-CSI-ResourceList | 8 | PUCCH format Id=8 | |
| } | | | |
| | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| } | | | |

6.4.3.1_1.4.3.2 Message exceptions for NSA

Same as in clause 6.4.3.1_1.4.3.1.

6.4.3.1_1.5 Test Requirements

**Table 6.4.3.1_1.5-1: Minimum requirement (FDD)**

| | **Test 1** | **Test 2** | **Test 3** | **Test 4** |
|---|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | 0.89 | N/A |
| $\gamma_2$ | 0.89 | N/A | N/A | 0.89 |

## 6.4.3.2 TDD

### 6.4.3.2_1 4Rx TDD FR1 RI reporting for both SA and NSA

6.4.3.2_1.1 Test Purpose

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission.

6.4.3.2_1.2 Test applicability

This test applies to all types of NR UE release 15 and forward supporting 4 Rx antenna ports.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and 4 Rx antenna ports.

6.4.3.2_1.3 Minimum Conformance Requirements

The minimum performance requirement in Table 6.4.3.2_1.3-2 is defined as

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

For the parameters specified in Table 6.4.3.2_1.3-1, and using the downlink physical channels specified in Annex C.3.1, the minimum requirements are specified in Table 6.4.3.2_1.3-2.

**Table 6.4.3.2_1.3-1: RI Test (TDD)**

| Parameter | | | Unit | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|---|---|---|
| Bandwidth | | | MHz | 40 | 40 | 40 | 40 |
| Duplex Mode | | | | TDD | TDD | TDD | TDD |
| TDD Slot Configuration | | | | FR1.30-1 | FR1.30-1 | FR1.30-1 | FR1.30-1 |
| DL BWP configuration #1 | | First PRB | | 0 | 0 | 0 | 0 |
| | | Number of contiguous PRB | | 106 | 106 | 106 | 106 |
| | | Subcarrier spacing | kHz | 30 | 30 | 30 | 30 |
| SNR | | | dB | -2 | 16 | 16 | 22 |
| Propagation channel | | | | TDLA30-5 | TDLA30-5 | TDLA30-5 | TDLA30-5 |
| Antenna configuration | | | | ULA Low 2x4 | ULA Low 2x4 | ULA High 2x4 | ULA Low 4x4 |
| Beamforming Model | | | | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 |
| ZP CSI-RS configuration | | CSI-RS resource Type | | Periodic | Periodic | Periodic | Periodic |
| | | Number of CSI-RS ports (*X*) | | 4 | 4 | 4 | 4 |
| | | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | | Density (ρ) | | 1 | 1 | 1 | 1 |
| | | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 5, (4,-) | Row 5, (4,-) | Row 5, (4,-) | Row 5, (4,-) |
| | | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (9,-) | (9,-) | (9,-) | (9,-) |
| | | CSI-RS periodicity and offset | slot | 10/1 | 10/1 | 10/1 | 10/1 |
| NZP CSI-RS for CSI acquisition | | CSI-RS resource Type | | Periodic | Periodic | Periodic | Periodic |
| | | Number of CSI-RS ports (*X*) | | 2 | 2 | 2 | 4 |
| | | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | | Density (ρ) | | 1 | 1 | 1 | 1 |
| | | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3 (6,-) | Row 3 (6,-) | Row 3 (6,-) | Row 4 (0,-) |
| | | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) | (13,-) | (13,-) |
| | | NZP CSI-RS-timeConfig periodicity and offset | slot | 10/1 | 10/1 | 10/1 | 10/1 |
| CSI-IM configuration | | CSI-IM resource Type | | Periodic | Periodic | Periodic | Periodic |
| | | CSI-IM RE pattern | | Pattern 0 | Pattern 0 | Pattern 0 | Pattern 0 |
| | | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (4,9) | (4,9) | (4,9) | (4,9) |
| | | CSI-IM timeConfig periodicity and offset | slot | 10/1 | 10/1 | 10/1 | 10/1 |
| ReportConfigType | | | | Periodic | Periodic | Periodic | Periodic |
| CQI-table | | | | Table 2 | Table 2 | Table 2 | Table 2 |
| reportQuantity | | | | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | | not configured | not configured | not configured | not configured |
| timeRestrictionForInterferenceMeasurements | | | | not configured | not configured | not configured | not configured |
| cqi-FormatIndicator | | | | Wideband | Wideband | Wideband | Wideband |
| pmi-FormatIndicator | | | | Wideband | Wideband | Wideband | Wideband |
| Sub-band Size | | | RB | 16 | 16 | 16 | 16 |
| csi-ReportingBand | | | | 1111111 | 1111111 | 1111111 | 1111111 |
| CSI-Report periodicity and offset | | | slot | 10/9 | 10/9 | 10/9 | 10/9 |
| Codebook configuration | Codebook Type | | | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel |
| | Codebook Mode | | | 1 | 1 | 1 | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | | N/A | N/A | N/A | (2,1) |
| | CodebookSubsetRestriction | | | 010000 for fixed rank 2, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 11111111 |
| | RI Restriction | | | N/A | N/A | N/A | 00000010 for fixed Rank 2 and |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | 00001111 for follow RI |
| Physical channel for CSI report | | | PUCCH | PUCCH | PUCCH | PUCCH |
| CQI/RI/PMI delay | | ms | 9.5 | 9.5 | 9.5 | 9.5 |
| Maximum number of HARQ transmission | | | 1 | 1 | 1 | 1 |
| RI Configuration | | | Fixed RI = 2 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 2 and follow RI |

**Table 6.4.3.2_1.3-2: Minimum requirement (TDD)**

| | Test 1 | Test 2 | Test 3 | Test 4 |
|---|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | 0.9 | N/A |
| $\gamma_2$ | 0.9 | N/A | N/A | 0.9 |

The normative reference for this requirement is TS 38.101-4 [2] clause 6.4.3.2.

### 6.4.3.2_1.4 Test Description

#### 6.4.3.2_1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1[7].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7 for TE diagram and section A.3.2.5 for UE diagram.
2. The parameter settings for the cell are set up according to Table 6.1.2-1 and Table 6.4.3.2_1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR, *Connected without release On* and Test Mode ON for SA or EN-DC, DC bearer *MCG* and *SCG*, *Connected without release On* and Test Mode ON for NSA according to TS 38.508-1 [6] clause 4.5. Message contents are defined in clause 6.4.3.2_1.4.3.

#### 6.4.3.2_1.4.2 Test procedure

1. Set the parameters of bandwidth, reference channel, the propagation condition, antenna configuration, antenna correlation, Codebook configuration, Beamforming Model, RI configuration and SNR according to Table 6.4.3.2_1.3-1 as appropriate.
2. The SS shall send PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and fixed RI as defined in Table 6.4.3.2_1.3-1. The SS sends downlink MAC padding bits on the DL RMC. Measure the $t_{fix}$ according to Annex G.3.3.
3. Propagation conditions are set according to Annex B.2.

4. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 6.4.3.2_1.3-1.

5. The UE shall transmit RRC Connection Reconfiguration Complete message.

6. Propagation conditions are set according to Table 6.4.3.2_1.3-1.

7. The SS shall send PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC. Measure $t_{reported}$ according to Annex G.3.3.

   If the ratio ( $t_{reported}$ / $t_{fix}$ ) satisfies the requirement in Table 6.4.3.2_1.5-1, then pass the UE for this test and go to step 8. Otherwise, declare a FAIL verdict.

8. If all tests have not been done, then repeat the same procedure (steps 1 to 7) with test conditions according to the Table 6.4.3.2_1.3-2 for the other Tests as appropriate. Otherwise, declare a PASS verdict.

### 6.4.3.2_1.4.3 Message Contents

Message contents are according to TS 38.508-1 [6] clause 4.6 with the following exceptions:

**Table 6.4.3.2_1.4.3-1: CSI-RS-ResourceMapping for NZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 001000 | | Test1, 2, 3 |
| row4 | 001 | | Test4 |
| } | | | |
| nrofPorts | p2 | | Test1, 2, 3 |
| | p4 | | Test4 |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 6.4.3.2_1.4.3-2: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| other | 000100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 9 | | |
| } | | | |

**Table 6.4.3.2_1.4.3-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern | | | |
| pattern0 SEQUENCE { | | | |
| subcarrierLocation-p0 | s4 | | |
| symbolLocation-p0 | 9 | | |
| } | | | |
| periodicityAndOffset | CSI-ResourcePeriodicityAndOffset | | |

**Table 6.4.3.2_1.4.3-4: *CSI-ResourcePeriodicityAndOffset***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset CHOICE { | | | |
| slots10 | 1 | | |
| } | | | |

**Table 6.4.3.2_1.4.3-5: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| two SEQUENCE { | | | |
| twoTX-CodebookSubsetRestriction | 010000 | | Fixed rank 2 |
| | 000011 | | Fixed rank 1 |
| | 010011 | | Following rank |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | | |

**Table 6.4.3.2_1.4.3-6: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | | |
| periodic SEQUENCE { | | | |
| reportSlotConfig CHOICE { | slot10 | | |
| slot10 | 9 | | |
| } | | | |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands7 | 1111111 | | |
| } | | | |
| } | | | |
| } | | | |

6.4.3.2_1.5 Test Requirements

**Table 6.4.3.2_1.5-1: Test Requirement (TDD)**

| | **Test 1** | **Test 2** | **Test 3** | **Test 4** |
|---|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | 0.89 | N/A |
| $\gamma_2$ | 0.89 | N/A | N/A | 0.89 |

# 7 Demodulation performance requirements (Radiated requirements)

## 7.1 General

For conformance testing involving FR2 test cases in this specification, the UE under test shall be pre-configured with UL Tx diversity schemes disabled to account for single polarization System Simulator (SS) in the test environment. The UE under test may transmit with dual polarization.

### 7.1.1 Applicability of requirements

#### 7.1.1.1 General

The minimum performance requirements are applicable to the FR2 operating bands defined in TS 38.101-2 [3] with $F_{DL_high}$ not exceeding 40000 MHz.

The minimum performance requirements in Clause 7 are mandatary for UE supporting NR operation, except test cases listed in Clause 7.1.1.3, 7.1.1.4.

#### 7.1.1.2 Applicability of requirements for different number of RX antenna ports

UE shall support 2 RX ports for different RF operating bands. The UE requirements applicability is defined in Table 7.1.1.2-1.

**Table 7.1.1.2-1: Requirements applicability**

| Supported RX antenna ports | Test type | Test list |
|---|---|---|
| UE supports 2RX antenna ports | PDSCH | All tests in Clause 7.2.2 |
| | PDCCH | All tests in Clause 7.3.2 |
| | PBCH | All tests in Clause 7.4.2 |

#### 7.1.1.3 Applicability of requirements for optional UE features

The performance requirements in Table 7.1.1.3-1 shall apply for UEs which support optional UE features only.

**Table 7.1.1.3-1: Requirements applicability for optional UE features**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| SU-MIMO Interference Mitigation advanced receiver | FR2 TDD | PDSCH | Clause 7.2.2.2.1 (Test 3-1) | |
| Basic DL NR-NR CA operation (*supportedBandCombinationList*) | NR CA | SDR | Clause 7.5A.1 | 1)Up to 16 DL carriers<br>2)Same numerology across carrier for data/control channel at a given time |
| PDSCH repetitions over multiple slots *(pdsch-RepetitionMultiSlots)* | FR2 TDD | PDSCH | Clause 7.2.2.2.2 | |
| 256QAM for PDSCH (*pdsch-256QAM-FR2*) | FR2 TDD | PDSCH | Clause 7.2.2.2.1 (Test 1-4) | |
| 256QAM for PDSCH (*pdsch-256QAM-FR2*) | FR2 TDD | SDR | Clause 7.5A.1 | For UE capable of *pdsch-256QAM-FR2* for certain band(s), *mcs-Table* is configured to '64QAM' for SDR test. |

### 7.1.1.4 Applicability of requirements for mandatory UE features with capability signaling

The performance requirements in Table 7.1.1.4-1 shall apply for UEs which support mandatory UE features with capability signalling only.

**Table 7.1.1.4-1: Requirements applicability for mandatory features with UE capability signalling**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| Supported maximum number of PDSCH MIMO layers (*maxNumberMIMO-LayersPDSCH)* | FR2 TDD | PDSCH | Clause 7.2.2.2.1 (Tests from 2-1 to 2-6) | The requirements apply only in case the PDSCH MIMO rank in the test case does not exceed UE PDSCH MIMO layers capability |
| Support of PT-RS with one antenna port for DL reception (*onePortsPTRS*) | FR2 TDD | PDSCH | Clause 7.2 | |
| | | SDR | Clause 7.5.1<br>Clause 7.5A.1 | |
| PCell operation on FR2 (*pCell-FR2*) | FR2 TDD | SDR | Clause 7.5A.1 | |
| PDSCH mapping type B (*pdsch-MappingTypeB*) | FR2 TDD | PDSCH | Clause 7.2.2.2.3 | |

### 7.1.1.5 Applicability of CA requirements

#### 7.1.1.5.1 Definition of CA capability

The definition with respect to CA capabilities is given as in Table 7.1.1.5.1-1.

**Table 7.1.1.5.1-1: Definition of CA capability**

| CA Capability | CA Capability Description |
|---|---|
| CA_C | Intra-band contiguous CA |
| CA_N | Intra-band non-contiguous CA |
| CA_AX | Inter-band CA (X bands) |
| NOTE 1: CA_C corresponds to NR CA configurations and bandwidth combination sets defined in Section 5.5A.1 of TS 38.101-2 [3].<br>CA_N corresponds to NR CA configurations and bandwidth combination sets defined in Section 5.5A.2 of TS 38.101-2 [3].<br>CA_AX corresponds to NR CA configurations and bandwidth combination sets defined in Clause 5.5A.3 of TS 38.101-2 [3]. | |

#### 7.1.1.5.2 Applicability and test rules for different CA configurations and bandwidth combination sets

The performance requirement for CA UE demodulation tests in Clause 7.2A are defined independent of CA configurations and bandwidth combination sets specified in Section 5.5A of TS 38.101-2 [3]. For UEs supporting different CA configurations and bandwidth combination sets, the applicability and test rules are defined in Table 7.1.1.5.2-1 and Table 7.1.1.5.2-2. For simplicity, CA configuration below refers to combination of CA configuration and bandwidth combination set.

**Table 7.1.1.5.2-1: Applicability and test rules for CA UE demodulation tests**

| Tests | CA capability where the tests apply | CA configuration from the selected CA capability where the tests apply | CA Bandwidth combination to be tested in priority order | PCell CC configuration |
|---|---|---|---|---|
| Test 1 in Section 7.2A.2.1 | CA_C, CA_N, CA_AX | Table 7.1.1.5.2-2 | Largest aggregated CA bandwidth combination | Any of CCs |

**Table 7.1.1.5.2-2: Selection of CA configurations**

| CA capability | Step 1 | Step 2 | Step 3 |
|---|---|---|---|
| CA_C or CA_N or CA_AX | Select CA configuration(s), which contain all CA bandwidth combinations requiring SNR below test equipment maximum achievable SNR | Select the CA configurations with the maximum number of CCs, for which the supported maximum number of MIMO layers is not lower than 2, among all the selected CA configurations from Step 1. | Select any one of CA configurations, which contain CA bandwidth combination with the largest aggregated channel bandwidth and supported maximum data rate is not lower than the tested date rate, among all the selected CA configurations from Step 2. |
| NOTE 1: Maximum supported data rate for Step 3 is calculated based clause 4.1.2 of TS 38.306 [14] | | | |
| NOTE 2: Tested data rate for Step 3 is calculated based on the equation $DataRate = 10^{-3}\sum_{j=1}^{J} TBS_j 2^{\mu_j}$ and FRCs used in the test. | | | |

## 7.1.1_1 Applicability of test requirements due to maximum achievable SNR

Editor's note: The applicability of test requirements with a modulation above 64 QAM is FFS.

Table 7.1.1_1-1 specifies the current assumption of maximum testable $SNR_{BB}$ for indirect farfield (IFF), PC3, Max device size ≤ 30 cm under fading conditions.

**Table 7.1.1_1-1: maximum testable $SNR_{BB}$ under fading conditions for modulations up to 64 QAM**

| Operating Band / Frequency | Maximum testable $SNR_{BB}$ (dB) | | |
|---|---|---|---|
| | CHBW 50 MHz | CHBW 100 MHz | CHBW 200 MHz |
| n257 mid | 30.6 | 27.5 | 24.4 |
| n258 mid | 30.6 | 27.5 | 24.4 |
| n259 mid | 20.4 | 17.2 | 14.1 |
| n260 mid | 24.4 | 21.2 | 18.2 |
| n261 mid | 30.6 | 27.5 | 24.4 |

Based on the current assumption of maximum testable $SNR_{BB}$, the applicability of test points is defined in Table 7.1.1_1-2 for indirect farfield (IFF), PC3, Max device size size ≤ 30 cm under fading conditions.

**Table 7.1.1_1-2: Testability of test requirements due to maximum achievable SNR per band**

| Test Case | Test point | CHBW / MHz | Fading | SNR test requirement | Test Point Applicability | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | n257 | n258 | n259 | n260 | n261 |
| 7.2.2.2.1_1 | 1-1 | 100 | Yes | 1.4 | x | x | x | x | x |
| | 1-2 | 100 | Yes | 3.6 | x | x | x | x | x |
| | 1-3 | 100 | Yes | 14.2 | x | x | x | x | x |
| | 2-1 | 100 | Yes | 5.8 | x | x | x | x | x |
| | 2-2 | 100 | Yes | 16.0 | x | x | x | x | x |
| | 2-3 | 50 | Yes | 15.7 | x | x | x | x | x |
| | 2-4 | 200 | Yes | 15.8 | x | x | - | x | x |
| | 2-5 | 50 | Yes | 16 | x | x | x | x | x |
| | 2-6 | 100 | Yes | 20.3 | x | x | - | x | x |
| 7.2.2.2.1_2 | 3-1 | 100 | Yes | 20.7 | x | x | - | x | x |
| 7.2.2.2.1_3 | 1-4 | 50 | Yes | 22.0 | TBD | TBD | TBD | TBD | TBD |
| 7.2.2.2.2_1 | 1-1 | 100 | Yes | TBD | TBD | TBD | TBD | TBD | TBD |
| 7.2.2.2.3_1 | 1-1 | 100 | Yes | 3.1 | x | x | x | x | X |
| 7.3.2.2.1 | 1-1 | 100 | Yes | 7.7 | x | x | x | x | x |
| | 1-2 | 100 | Yes | 4.3 | x | x | x | x | x |
| 7.3.2.2.2 | 2-1 | 100 | Yes | 3.2 | x | x | x | x | x |
| | 2-2 | 100 | Yes | 0.2 | x | x | x | x | x |
| 7.3.2.2.3 | 1-1 | 100 | Yes | 4.7 | x | x | x | x | x |

# 7.2 PDSCH demodulation requirements

The parameters specified in Table 7.2-1 are valid for all PDSCH demodulation tests unless otherwise stated.

**Table 7.2-1: Common Test Parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| PTRS *epre-Ratio* | | | 0 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 2) | RBs | 0 |
| | Subcarrier spacing | kHz | 60 or 120 |
| DL BWP configuration #1 | Cyclic prefix | | Normal |
| | RB offset | RBs | 0 |
| | Number of contiguous PRB | PRBs | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | 1 |
| | SSB periodicity | ms | 20 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | 0 |
| | Number of PRBs in CORESET | | Table 7.2-2 for tested channel bandwidth and subcarrier spacing |
| | Number of PDCCH candidates and aggregation levels | | 1/AL8 |
| | CCE-to-REG mapping type | | Non-interleaved |
| | DCI format | | 1_1 |
| | TCI state | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | Single Panel Type I, Random per slot with equal probability of each applicable $i_1$, $i_2$ combination, and with REG bundling granularity for number of Tx larger than 1 |
| Cross carrier scheduling | | | Not configured |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | 0 for CSI-RS resource 1,2,3,4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | 6 for CSI-RS resource 1 and 3<br>10 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports ($X$) | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | 'No CDM' for CSI-RS resource 1,2,3,4 |
| | Density ($\rho$) | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | Slots | 60 kHz SCS: 80 for CSI-RS resource 1,2,3,4<br>120 kHz SCS: 160 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | Slots | 60 kHz SCS:<br>40 for CSI-RS resource 1 and 2<br>41 for CSI-RS resource 3 and 4<br><br>120 kHz SCS:<br>80 for CSI-RS resource 1 and 2<br>81 for CSI-RS resource 3 and 4 |

| | | | | |
|---|---|---|---|---|
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | | 0 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | | 12 |
| | Number of CSI-RS ports ($X$) | | | 2 |
| | CDM Type | | | FD-CDM2 |
| | Density ($\rho$) | | | 1 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80<br>120 kHz SCS: 160 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | | 4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | | 12 |
| | Number of CSI-RS ports ($X$) | | | 4 |
| | CDM Type | | | FD-CDM2 |
| | Density ($\rho$) | | | 1 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80<br>120 kHz SCS: 160 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| CSI-RS for beam refinement | First subcarrier index in the PRB used for CSI-RS | | | $k_0$=0 for CSI-RS resource 1,2 |
| | First OFDM symbol in the PRB used for CSI-RS | | | $l_0$ = 8 for CSI-RS resource 1<br>$l_0$ = 9 for CSI-RS resource 2 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2 |
| | Density (ρ) | | | 3 for CSI-RS resource 1,2 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80 for CSI-RS resource 1,2<br>120 kHz SCS: 160 for CSI-RS resource 1,2 |
| | CSI-RS offset | | Slots | 0 for CSI-RS resource 1,2 |
| | Repetition | | | ON |
| | QCL info | | | TCI state #1 |
| PDSCH DMRS configuration | Antenna ports indexes | | | {1000} for Rank 1 tests<br>{1000, 1001} for Rank 2 tests |
| | Position of the first DMRS for PDSCH mapping type A | | | 2 |
| | Number of PDSCH DMRS CDM group(s) without data | | | 1 |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type D |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |

| | | | | |
|---|---|---|---|---|
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type D |
| PTRS configuration | Frequency density ($K_{PT-RS}$) | | | 2 |
| | Time density ($L_{PT-RS}$) | | | 1 |
| | Resource Element Offset | | | 2 |
| Maximum number of code block groups for ACK/NACK feedback | | | | 1 |
| Maximum number of HARQ transmission | | | | 4 |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| PDSCH & PDSCH DMRS Precoding configuration | | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination, and with Wideband granularity |
| Symbols for all unused REs | | | | OCNG in Annex A.5 |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |
| Note 1: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission.<br>Note 2: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing. | | | | |

**Table 7.2-2: Number of PRBs in CORESET**

| SCS (kHz) | 50 MHz | 100 MHz | 200 MHz | 400 MHz |
|---|---|---|---|---|
| 60 | 66 | 132 | 264 | N.A |
| 120 | 30 | 66 | 132 | 264 |

## 7.2.1 1RX requirements (Void)

## 7.2.2 2RX requirements

### 7.2.2.1 FDD (Void)

### 7.2.2.2 TDD

#### 7.2.2.2.1 2Rx TDD FR2 PDSCH mapping Type A performance

##### 7.2.2.2.1_0 Minimum conformance requirements

For PDSCH Type-A scheduling, the requirements are specified in Table 7.2.2.2.1_0-3, 7.2.2.2.1_0-4 and 7.2.2.2.1_0-5, with the addition of the parameters in Table 7.2.2.2.1_0-2 and the downlink physical channel setup according to Annex C.5.1. The purpose is to verify the performance of PDSCH Type-A scheduling.

The test purposes are specified in Table 7.2.2.2.1_0-1.

**Table 7.2.2.2.1_0-1: Tests purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH mapping Type A normal performance under 2 receive antenna conditions and with different channel models, MCSs and number of MIMO layers | 1-1, 1-3, 1-4, 2-1, 2-2, 2-3, 2-4, 2-5, 2-6 |
| Verify the PDSCH mapping Type A HARQ soft combining performance under 2 receive antenna conditions. | 1-2 |
| Verify the PDSCH mapping Type A enhanced performance requirement Type 1 under 2 receive antenna conditions and with 2 MIMO layers. | 3-1 |

**Table 7.2.2.2.1_0-2: Test Parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| CSI-RS for tracking | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | For Test 1-1 and 1-2:<br>3 for CSI-RS resource 1 and 3<br>7 for CSI-RS resource 2 and 4 |
| | CSI-RS offset | Slots | For Test 1-2:<br>82 for CSI-RS resource 1 and 2<br>83 for CSI-RS resource 3 and 4 |
| PDCCH configuration | Number of PDCCH candidates and aggregation levels | | 1/AL4 for Test 1-4 and 2-3<br>1/AL8 for other tests |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 1 |
| | Length (L) | | Specific to each Reference channel as defined in A.3.2.2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | WB for 1-1,<br>2 for other tests |
| | Resource allocation type | | Test 2-1: Type 1 with start RB = 30, $L_{RBs}$ = 6<br>Other tests: Type 0 |
| | RBG size | | Test 2-1: N/A<br>Other tests: Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 8 for Test 1-1, 1-3, 1-4, 2-2, 2-4<br>10 for Test 2-1, 2-3, 2-5, 2-6, 3-1<br>16 for Test 1-2 |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | As defined in Annex A.1.3 |

**Table 7.2.2.2.1_0-3: Minimum performance for Rank 1 (FRC)**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | $SNR_{BB}$ (dB) |
| 1-1 | R.PDSCH.5-1.1TDD | 100/120 | QPSK, 0.30 | FR2.120-1. A | TDLC60-300 | 2x2 ULA Low | 70 | -0.4 |
| 1-2 | R.PDSCH.5-2.1 TDD | 100/120 | 16QAM, 0.48 | FR2.120-1 | TDLA30-300 | 2x2 ULA Low | 30 | 1.7 |
| 1-3 | R.PDSCH.5-3.1TDD | 100/120 | 64QAM, 0.46 | FR2.120-1 | TDLA30-300 | 2x2 XPL Med | 70 | 12.4 |
| 1-4 | R.PDSCH.5-10.1 TDD | 50/120 | 256QAM 0.67 | FR2.120-1 | TDLD30-75 | 2x2 ULA Low | 70 | 20.2 |

**Table 7.2.2.2.1_0-4: Minimum performance for Rank 2 (FRC)**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | $SNR_{BB}$ (dB) |
| 2-1 | R.PDSCH.5-4.1 TDD | 100/120 | QPSK, 0.30 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 70 | 4.1 |
| 2-2 | R.PDSCH.5-2.2 TDD | 100/120 | 16QAM, 0.48 | FR2.120-1 | TDLA30-300 | 2x2 ULA Low | 70 | 14.4 |
| 2-3 | R.PDSCH.5-5.2 TDD | 50/120 | 16QAM,0.48 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 70 | 14.0 |
| 2-4 | R.PDSCH.5-2.3 TDD | 200/120 | 16QAM, 0.48 | FR2.120-1 | TDLA30-300 | 2x2 ULA Low | 70 | 14.2 |
| 2-5 | R.PDSCH.4-1.1 TDD | 50/60 | 16QAM, 0.48 | FR2.60-1 | TDLA30-75 | 2x2 ULA Low | 70 | 14.3 |
| 2-6 | R.PDSCH.5-6.1 TDD | 100/120 | 64QAM, 0.43 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 70 | 18.6 |

**Table 7.2.2.2.1_0-5: Minimum performance for Rank 2 (FRC) for Enhanced Type X Receiver**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | $SNR_{BB}$ (dB) |
| 3-1 | R.PDSCH.5-5.1TDD | 100/120 | 16QAM, 0.48 | FR2.120-2 | TDLA30-75 | 2x2 ULA Medium | 70 | 19.0 |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.2.2.2.1.

### 7.2.2.2.1_1 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for SA and NSA

#### 7.2.2.2.1_1.1 Test Purpose

Verify the PDSCH mapping Type A normal performance with different channel models, MCSs and number of MIMO layers.

#### 7.2.2.2.1_1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

### 7.2.2.2.1_1.3 Test Description

#### 7.2.2.2.1_1.3.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 7.2-1 and Table 7.2.2.2.1.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.2.2.2.1_1.3.3.

#### 7.2.2.2.1_1.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0.If no direction found, mark the test as inconclusive.
2. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 7.2.2.2.1_1.4-1 and 7.2.2.2.1_1.4-2.. The SS sends downlink MAC padding bits on the DL RMC.
3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 7.2.2.2.1_1.4-1 and 7.2.2.2.1_1.4-2 as appropriate.
4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.1.5-1 in Annex G.
5. Repeat steps from 1 to 4 for each subtest in Tables 7.2.2.2.1_1.4-1 and 7.2.2.2.1_1.4-2 as appropriate.

#### 7.2.2.2.1_1.3.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

7.2.2.2.1_1.3.3_1 Message exceptions for SA

**Table 7.2.2.2.1_1.3.3_1-1: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl80 | 7 | Test point 2-1, 2-3, 2-6 | |
| } | | | |
| } | | | |

**Table 7.2.2.2.1_1.3.3_1-2: CSI-RS-ResourceMapping for TRS**

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| firstOFDMSymbolInTimeDomain | 3 | $l_0$ = 3 for CSI-RS resource 1 and 3 | TRS, Test 1-1, 1-2 |
| | 7 | $l_0$ = 7 for CSI-RS resource 2 and 4 | TRS, Test 1-1, 1-2 |
| nrofPorts | p1 | 1 for CSI-RS resource 1,2,3,4 | TRS |
| } | | | |

**Table 7.2.2.2.1_1.3.3_1-3: CSI-ResourcePeriodicityAndOffset for TRS**

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| Slots160 | 82 | Periodicity 20 slots and offset 10 for CSI-RS resource 1 and 2 | |
| Slots160 | 83 | Periodicity 20 slots and offset 11 for CSI-RS resource 3 and 4 | |
| } | | | |

**Table 7.2.2.2.1_1.3.3_1-4: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-162 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel1 | n0 | | |
| aggregationLevel2 | n0 | | |
| aggregationLevel4 | n1 | | Test 2-3 |
| aggregationLevel8 | n1 | AL8 | Other than test 2-3 |
| aggregationLevel16 | n0 | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.1_1.3.3_1-5: DMRS-DownlinkConfig**

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-50 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DMRS-DownlinkConfig ::= SEQUENCE { | | | |
| dmrs-Type | Type 1 | | |
| dmrs-AdditionalPosition | pos1 | | |
| maxLength | len1 | | |
| } | | | |

**Table 7.2.2.2.1_1.3.3_1-6: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| vrb-ToPRB-Interleaver | Not present | | |
| resourceAllocation | resourceAllocationType0 | | |
| pdsch-AggregationFactor | Not present | | |
| prb-BundlingType CHOICE { | | | |
| staticBundling SEQUENCE { | | | |
| bundleSize | Not present | PRB Bundling size of 2 | Other than test 1-1 |
| | Wideband | | Test 1-1 |
| } | | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.1_1.3.3_1-7: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-102 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | Set according to the test id | | 8 for Test 1-1, 1-3, 2-2, 2-4<br>10 for Test 2-1, 2-3, 2-5, 2-6, 3-1<br>16 for Test 1-2 |
| } | | | |

### 7.2.2.2.1_1.3.3_2 Message exceptions for NSA

Same as 7.2.2.2.1_1.3.3_1.

### 7.2.2.2.1_1.4 Test Requirements

Tables 7.2.2.2.1_1.4-1 and 7.2.2.2.1_1.4-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2 for each throughput test shall meet or exceed the specified value in Tables 7.2.2.2.1_1.4-1 and 7.2.2.2.1_1.4-2 for the specified SNR including test tolerances for all throughput tests.

**Table 7.2.2.2.1_1.4-1: Test Requirement for Rank 1 (FRC)**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | $SNR_{BB}$ (dB) |
| 1-1 | R.PDSCH.5-1.1TDD | 100/120 | QPSK, 0.30 | FR2.120-1 A | TDLC60-300 | 2x2 ULA Low | 70 | 1.4 |
| 1-2 | R.PDSCH.5-2.1TDD | 100/120 | 16QAM, 0.48 | FR2.120-1 | TDLA30-300 | 2x2 ULA Low | 30 | 3.6 |
| 1-3 | R.PDSCH.5-3.1TDD | 100/120 | 64QAM, 0.46 | FR2.120-1 | TDLA30-300 | 2x2 XPL Medium | 70 | 14.2 |

**Table 7.2.2.2.1_1.4-2: Test Requirement for Rank 2 (FRC)**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | $SNR_{BB}$ (dB) |
| 2-1 | R.PDSCH.5-4.1TDD | 100/120 | QPSK, 0.30 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 70 | 5.8 |
| 2-2 | R.PDSCH.5-2.2TDD | 100/120 | 16QAM, 0.48 | FR2.120-1 | TDLA30-300 | 2x2 ULA Low | 70 | 16.0 |
| 2-3 | R.PDSCH.5-5.2TDD | 50/120 | 16QAM,0.48 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 70 | 15.7 |
| 2-4 | R.PDSCH.5-2.3TDD | 200/120 | 16QAM, 0.48 | FR2.120-1 | TDLA30-300 | 2x2 ULA Low | 70 | 15.8 |
| 2-5 | R.PDSCH.4-1.1TDD | 50/60 | 16QAM, 0.48 | FR2.60-1 | TDLA30-75 | 2x2 ULA Low | 70 | 16 |
| 2-6 | R.PDSCH.5-6.1TDD | 100/120 | 64QAM, 0.43 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 70 | 20.3 |

### 7.2.2.2.1_2 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with enhanced type 1 receiver for SA and NSA

#### 7.2.2.2.1_2.1 Test Purpose

Verify the PDSCH mapping Type A normal performance with different channel models, MCSs and number of MIMO layers.

#### 7.2.2.2.1_2.2 Test Applicability

This test applies to all types of NR UE release 15 and forward supporting NR enhanced receiver type 1.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC and NR enhanced receiver type 1.

#### 7.2.2.2.1_2.3 Test Description

Same test description as in clause 7.2.2.2.1_1.3 with following exception:

- Table 7.2.2.2.1_2.4-1 instead of Tables 7.2.2.2.1_1.4-1 and 7.2.2.2.1_1.4-2

##### 7.2.2.2.1_2.3.1 Message contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

7.2.2.2.1_2.3.1_1 Message exceptions for SA

Same as 7.2.2.2.1_1.3.3_1 with following exceptions:

**Table 7.2.2.2.1_2.3.1_1-1: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| Sl80 | 7 | | Test 3-1 |
| } | | | |
| } | | | |

7.2.2.2.1_2.3.1_2 Message exceptions for NSA

Same as 7.2.2.2.1_2.3.1_1.

7.2.2.2.1_2.4 Test Requirements

Table 7.2.2.2.1_2.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2 for each throughput test shall meet or exceed the specified value in Table 7.2.2.2.1_2.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 7.2.2.2.1_2.4-1: Test Requirement for Rank 2 (FRC) for Enhanced Type 1 Receiver**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | $SNR_{BB}$ (dB) |
| 3-1 | R.PDSCH.5-5.1TDD | 100/120 | 16QAM, 0.48 | FR2.120-2 | TDLA30-75 | 2x2 ULA Med | 70 | 20.7 |

7.2.2.2.1_3 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with 256QAM for SA and NSA (Rel-16 and forward)

7.2.2.2.1_3.1 Test Purpose

Verify the PDSCH mapping Type A normal performance with different channel models, MCSs and number of MIMO layers.

7.2.2.2.1_3.2 Test Applicability

This test applies to all types of NR UE release 16 and forward supporting 256QAM for PDSCH.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and NR 256QAM for PDSCH.

7.2.2.2.1_3.3 Test Description

Same test description as in clause 7.2.2.2.1_1.3 with following exception:

- Table 7.2.2.2.1_3.4-1 instead of Table 7.2.2.2.1_1.4-1

7.2.2.2.1_3.3.1 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions:

7.2.2.2.1_3.3.1_1 Message exceptions for SA

Same as 7.2.2.2.1_1.3.3_1 with following exceptions:

**Table 7.2.2.2.1_3.3.1_1-1: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl80 | 7 | | Test 1-4 |
| } | | | |
| } | | | |

**Table 7.2.2.2.1_3.3.1_1-2: *SearchSpace***

| Derivation Path: TS 38.508-1 Table 4.6.3-162 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel1 | n0 | | |
| aggregationLevel2 | n0 | | |
| aggregationLevel4 | n1 | AL4 | Test 1-4 |
| aggregationLevel8 | n0 | | |
| aggregationLevel16 | n0 | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.1_3.3.1_1-3: PDSCH-Config**

| Derivation Path: TS 38.508-1 Table 5.4.2.0-26 with condition DEMOD_FR2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| mcs-Table | qam256 | | Test 1-4 |
| } | | | |

**Table 7.2.2.2.1_3.3.1_1-4: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 Table 5.4.2.0-25 with condition DEMOD_FR2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | 8 | | Test 1-4 |
| } | | | |

7.2.2.2.1_3.3.1_2 Message exceptions for NSA

Same as 7.2.2.2.1_3.3.1_1.

7.2.2.2.1_3.4 Test Requirements

Table 7.2.2.2.1.0-2 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2 for each throughput test shall meet or exceed the specified value in Table 7.2.2.2.1_3.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 7.2.2.2.1_3.4-1: Test Requirement for Rank 1 (FRC) for 256QAM**

| Test num. | Reference channel | Bandwidth (MHz)/Subcarrier | Modulation and code | TDD UL-DL | Propagation condition | Correlation matrix and | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of | $SNR_{BB}$ |

| | | spacing (kHz) | rate | pattern | | antenna configuration | maximum throughput (%) | (dB) |
|---|---|---|---|---|---|---|---|---|
| 1-4 | R.PDSCH.5-10.1 TDD | 50/120 | 256QAM 0.67 | FR2.120-1 | TDLD30-75 | 2x2 ULA Low | 70 | 22.0 |

### 7.2.2.2.2 2Rx TDD FR2 PDSCH repetitions over multiple slots

#### 7.2.2.2.2_0 Minimum conformance requirements

For PDSCH with slot aggregation, the requirements are specified in Table 7.2.2.2.2_0-3, additional parameters in Table 7.2.2.2.2_0-2 and the downlink physical channel setup according to Annex C.5.1.

The test purpose is specified in Table 7.2.2.2.2_0-1.

**Table 7.2.2.2.2_0-1: Test purpose**

| Purpose | Test index |
|---|---|
| Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions | 1-1 |

**Table 7.2.2.2.2_0-2: Test Parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | *k0* | | 0 |
| | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| | PDSCH aggregation factor | | 2 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 2 |
| The number of slots between final repetition of PDSCH and corresponding HARQ-ACK information | | | As defined in Annex A.1.3 (Note 1) |
| Note 1: ACK/NACK feedback is generated for PDSCH on slot i, where mod(i,4) = 1, where i is the slot index per frame; i = {0,...,79} | | | |

**Table 7.2.2.2.2_0-3: Minimum performance for Rank 1 (FRC)**

| Test num | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.5-11.1 TDD | 100 / 120 | 16QAM, 0.37 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 1% (Note 1) | -1.1 |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.2.2.2.2.

### 7.2.2.2.2_1 2Rx TDD FR2 PDSCH repetitions over multiple slots - 2x2 MIMO with baseline receiver for SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- MU factor due to finite test time needs further study

#### 7.2.2.2.2_1.1 Test Purpose

Verify the PDSCH repetitions over multiple slots performance under 2 receive antenna conditions.

#### 7.2.2.2.2_1.2 Test Applicability

This test applies to all types of NR UE release 16 and forward supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC supporting capability IE *pdsch-RepetitionMultiSlots-r16*.

#### 7.2.2.2.2_1.3 Test Description

##### 7.2.2.2.2_1.3.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 7.2-1 and Table 7.2.2.2.2.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.2.2.2.2_1.3.3.

#### 7.2.2.2.2_1.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0.If no direction found, mark the test as inconclusive.
2. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 7.2.2.2.2_1.4-1. The SS sends downlink MAC padding bits on the DL RMC. The UE may expect that the TB is repeated with same symbol allocation among each of the *pdsch-AggregationFactor* consecutive slots.
3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 7.2.2.2.2_1.4-1 as appropriate.
4. Measure the BLER for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of correctly and incorrectly received transport blocks based on ACK/NACK feedback on the UL during each subtest and decide pass or fail according to Tables G.1.5-1a in Annex G.

#### 7.2.2.2.2_1.3.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

##### 7.2.2.2.2_1.3.3_1 Message exceptions for SA

**Table 7.2.2.2.2_1.3.3_1-1: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl80 | 7 | | |
| } | | | |
| } | | | |

**Table 7.2.2.2.2_1.3.3_1-2: *PDSCH-Config***

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-100 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-Config ::= SEQUENCE { | | | |
| pdsch-AggregationFactor | 2 | | |
| } | | | |

**Table 7.2.2.2.2_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-102 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | 4 | | |
| } | | | |

##### 7.2.2.2.2_1.3.3_2 Message exceptions for NSA

Same as 7.2.2.2.2_1.3.3_1.

#### 7.2.2.2.2_1.4 Test Requirements

Table 7.2.2.2.2_1.4-1 defines the primary level settings.

The target BLER percentage for the downlink reference measurement channels specified in Annex A.3.2.2 for each BLER test shall meet or exceed the specified value in Table 7.2.2.2.2_1.4-1 for the specified SNR including test tolerances for all BLER tests.

**Table 7.2.2.2.2_1.4-1: Test Requirement for Rank 1**

| Test num | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Target BLER | SNR (dB) |
| 1-1 | R.PDSCH.5-11.1 TDD | 100 / 120 | 16QAM, 0.37 | FR2.120-2 | TDLA30-75 | 2x2 ULA Low | 1% (Note 1) | -1.1+TT |
| Note 1: BLER is defined as residual BLER; i.e. ratio of incorrectly received transport blocks / sent transport blocks, independently of the number HARQ transmission(s) for each transport block. | | | | | | | | |

## 7.2.2.2.3 2Rx TDD FR2 PDSCH Mapping Type B

### 7.2.2.2.3_0 Minimum conformance requirements

The performance requirements are specified in Table 7.2.2.2.3.0-3, with the addition of test parameters in Table 7.2.2.2. 3.0-2 and the downlink physical channel setup according to Annex C.5.1. The purpose is to verify the performance of PDSCH Type B scheduling.

The test purposes are specified in Table 7.2.2.2.3.0-1.

**Table 7.2.2.2.3.0-1: Test purpose**

| Purpose | Test index |
|---|---|
| Verify PDSCH mapping Type B performance under 2 receive antenna conditions | 1-1 |

**Table 7.2.2.2.3.0-2: Test parameters**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDCCH configuration | Number of PDCCH candidates and aggregation levels | | 1/AL8 |
| PDSCH configuration | Mapping type | | Type B |
| | k0 | | 0 |
| | Starting symbol (S) | | 1 |
| | Length (L) | | 2 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 0 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 8 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.3 |

**Table 7.2.2.2.3.0-3: Minimum performance for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.5-1.2 TDD | 100 / 120 | QPSK, 0.30 | FR2.120-1 | [TDLA30-75] | 2x2, ULA Low | 70 | 1.3 |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.2.2.2.3.

### 7.2.2.2.3_1 2Rx TDD FR2 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for SA and NSA

Editor's note: This clause is incomplete. The following aspects are either missing or not yet determined:

- Fading scenario is within brackets.
- MU and TT are TBD

#### 7.2.2.2.3_1.1 Test Purpose

Verify the PDSCH mapping Type B performance under 2 receive antenna conditions.

#### 7.2.2.2.3_1.2 Test Applicability

This test applies to all types of NR UE release 16 and forward supporting PDSCH mapping type B.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and PDSCH mapping type B.

#### 7.2.2.2.3_1.3 Test Description

##### 7.2.2.2.3_1.3.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 7.2-1 and Table 7.2.2.2.3.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.2.2.2.3_1.3.3.

### 7.2.2.2.3_1.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0.If no direction found, mark the test as inconclusive.

2. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Tables 7.2.2.2.3_1.4-1. The SS sends downlink MAC padding bits on the DL RMC.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 7.2.2.2.3_1.4-1 as appropriate.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.1.5-1 in Annex G.

### 7.2.2.2.3_1.3.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

#### 7.2.2.2.3_1.3.3_1 Message exceptions for SA

**Table 7.2.2.2.3_1.3.3_1-3: PDSCH-ServingCellConfig**

| Derivation Path: TS 38.508-1 [6] Table 4.6.3-102 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-ServingCellConfig ::= SEQUENCE { | | | |
| nrofHARQ-ProcessesForPDSCH | 8 | | |
| } | | | |

**Table 7.2.2.2.3_1.3.3_1-4: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-103 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF { | 2 entry | | |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | | |
| K0 | Not present | | |
| mappingType | typeB | | |
| startSymbolAndLength | 15 | Start symbol(S)=1, Length(L)=2 | |
| } | | | |
| } | | | |

#### 7.2.2.2.3_1.3.3_2 Message exceptions for NSA

Same as 7.2.2.2.3_1.3.3_1.

### 7.2.2.2.3_1.4 Test Requirements

Table 7.2.2.2.3_1.4-1 defines the primary level settings.

The fraction of maximum throughput percentage for the downlink reference measurement channels specified in Annex A.3.2 for each throughput test shall meet or exceed the specified value in Table 7.2.2.2.3_1.4-1 for the specified SNR including test tolerances for all throughput tests.

**Table 7.2.2.2.3_1.4-1: Test Requirement for Rank 1**

| Test num. | Reference channel | Bandwidth (MHz) / Subcarrier spacing (kHz) | Modulation format and code rate | TDD UL-DL pattern | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 1-1 | R.PDSCH.5-1.2 TDD | 100 / 120 | QPSK, 0.30 | FR2.120-1 | [TDLA30-75] | 2x2, ULA Low | 70 | 3.1 |

# 7.2A PDSCH demodulation requirements for CA

The parameters specified in Table 7.2-1 for PDSCH single carrier tests are reused for PDSCH CA test unless otherwise stated.

## 7.2A.1 1RX requirements (Void)

## 7.2A.2 2RX requirements

### 7.2A.2.0 Minimum conformance requirements

For CA with different numbers of DL component carriers, the requirements are defined in Table 7.2A.2.0-3-3 based on the single carrier requirements for different bandwidth specified in Table 7.2A.2.0-2, with the parameters in Table 7.2A.2.0-1 and the downlink physical channel setup according to Annex C.2.2. The performance requirements specified in this sub-clause do not apply for UE single carrier test.

**Table 7.2A.2.0-1: Test parameters for CA**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| Active DL BWP index | | | 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | Starting symbol (S) | | 1 |
| | Length (L) | | Specific to each Reference channel |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| Number of HARQ Processes | | | 8 |
| TDD UL-DL pattern | | | 120kHz SCS: FR2.120-1 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | As defined in Annex A.1.3 |

**Table 7.2A.2.0-2: Single carrier performance for TDD 120 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 50 | R.PDSCH.5-9.1 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.4] |
| 100 | R.PDSCH.5-9.2 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.2] |
| 200 | R.PDSCH.5-9.3 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.3] |
| 400 | R.PDSCH.5-9.4 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.3] |

**Table 7.2A.2.0-3: Minimum performance for multiple CA configurations**

| Test number | CA duplex mode | Minimum performance requirements |
|---|---|---|
| 1 | TDD 120 kHz + TDD 120 kHz | As defined in Table 7.2A.2.1_0-2 |
| Note 1: The applicability of requirements for different CA duplex modes, SCSs, CA configurations and bandwidth combination sets is defined in 7.1.1.5. | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.2A.2.1

## 7.2A.2.1 2Rx TDD FR2 CA requirements for normal PDSCH Demodulation Performance for both SA and NSA (2DLCA)

Editor's Note: Following aspects in this clause are incomplete.

- TT analysis pending

### 7.2A.2.1.1 Test Purpose

Verify the PDSCH mapping Type A normal performance with different channel models, MCSs and number of MIMO layers.

### 7.2A.2.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward that support NR 2DL CA.

### 7.2A.2.1.3 Test Description

#### 7.2A.2.1.3.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-2.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

CA capability to be tested: Any one of the supported CA capability as per the selection criteria specified in Table 7.1.1.5.2-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 7.2-1, 7.2A.2.0-1 and Table 7.2A.2.0-2 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].

4. Propagation conditions for NR cell are set according to Annex B.0.

6. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.2A.2.1.3.3.

#### 7.2A.2.1.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found, mark the test as inconclusive.

2. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to Table 7.2.2.2.1_1.3-3 and 7.2.2.2.1_1.3-4. The SS sends downlink MAC padding bits on the DL RMC.

3. Set the parameters of the bandwidth, MCS, reference channel, the propagation condition, the correlation matrix and the SNR according to Tables 7.2.2.2.1_1.4.4-1 and 7.2.2.2.1_1.4.4-2 as appropriate.

4. Measure the average throughput for a duration sufficient to achieve statistical significance according to Annex G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL during each subtest and decide pass or fail according to Tables G.1.5-1 in Annex G.

5. Repeat steps from 1 to 4 for each subtest in Table 7.2A.2.1.4-1 and Table 7.2.2.2.1_1.4.4-2 as appropriate.

#### 7.2A.2.1.3.3 Message Contents

### 7.2A.2.1.4 Test Requirements

**Table 7.2A.2.1.4-1: Single carrier performance for TDD 120 kHz SCS for CA configurations**

| Bandwidth (MHz) | Reference channel | Modulation format and code rate | Propagation condition | Correlation matrix and antenna configuration | Reference value | |
|---|---|---|---|---|---|---|
| | | | | | Fraction of maximum throughput (%) | SNR (dB) |
| 50 | R.PDSCH.5-9.1 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.4+TT] |
| 100 | R.PDSCH.5-9.2 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.2+TT] |
| 200 | R.PDSCH.5-9.3 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.3+TT] |
| 400 | R.PDSCH.5-9.4 TDD | 16QAM, 0.33 | TDLA30-75 | 2x2, ULA Low | 70 | [10.3+TT] |

## 7.3 PDCCH demodulation requirements

The receiver characteristics of the PDCCH are determined by the probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg).

The parameters specified in Table 7.3-1 are valid for all PDCCH tests unless otherwise stated.

**Table 7.3-1: Common test Parameters**

| Parameter | Unit | | Value | Parameter |
|---|---|---|---|---|
| Carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 1) | | | 0 |
| DL BWP configuration #1 | Cyclic prefix | | | Normal |
| Common serving cell parameters | Physical Cell ID | | | 0 |
| | SSB position in burst | | | 1 |
| | SSB periodicity | | ms | 20 |
| PDCCH configuration | Slots for PDCCH monitoring | | | Each slot |
| | Number of PDCCH candidates | | | 1 |
| | Frequency domain resource allocation for CORESET | | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | | TCI state #1 |
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS (k0) | | | 0 |
| | First OFDM symbol in the PRB used for CSI-RS (l0) | | | CSI-RS resource 1: 4<br>CSI-RS resource 2: 8<br>CSI-RS resource 3: 4<br>CSI-RS resource 4: 8 |
| | Number of CSI-RS ports (X) | | | 1 |
| | CDM Type | | | No CDM |
| | Density (ρ) | | | 3 |
| | CSI-RS periodicity | | Slots | 160 |
| | CSI-RS offset | | Slots | 80 for CSI-RS resource 1 and 2<br>81 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| NZP CSI-RS for beam management | First subcarrier index in the PRB used for CSI-RS (k0) | | | 0 |
| | First OFDM symbol in the PRB used for CSI-RS (l0) | | | CSI-RS resource 1: 8<br>CSI-RS resource 2: 9 |
| | Number of CSI-RS ports (X) | | | 1 |
| | CDM Type | | | No CDM |
| | Density (ρ) | | | 3 |
| | CSI-RS periodicity | | Slots | 120 kHz SCS: 160 for CSI-RS resource 1,2 |
| | CSI-RS offset | | Slots | 0 for CSI-RS resource 1,2 |
| | Repetition | | | ON |
| | QCL info | | | TCI state #1 |
| PDCCH & PDCCH DMRS Precoding configuration | | | | Single Panel Type I, Random per slot with equal probability of each applicable $i_1$, $i_2$ combination, and with REG bundling granularity for number of Tx larger than 1 |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type D |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for |

| | | | | |
|---|---|---|---|---|
| | | | | tracking' configuration |
| | | QCL Type | | Type D |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |
| Symbols for all unused REs | | | | OP.1 FDD as defined in Annex A.5.1.1 OP.1 TDD as defined in Annex A.5.2.1 |
| The number of slots between PDSCH and corresponding HARQ-ACK information | | | | Specific to each TDD UL-DL pattern and as defined in Annex A.1.3. |
| Note 1: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing. | | | | |

## 7.3.1 1RX requirements

(Void)

## 7.3.2 2RX requirements

### 7.3.2.1 FDD

(Void)

### 7.3.2.2 TDD

#### 7.3.2.2.1 2Rx TDD FR2 PDCCH 1 Tx antenna performance for both SA and NSA

##### 7.3.2.2.1.1 Test Purpose

This test verifies the demodulation performance of PDCCH for a single-antenna port with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 7.3.2.2.1.3-1.

##### 7.3.2.2.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 7.3.2.2.1.3 Minimum conformance requirements

For the parameters specified in Table 7.3.2.2.1.3-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 7.3.2.2.1.3-2. The downlink physical setup is in accordance with Annex C.2.2.

**Table 7.3.2.2.1.3-1: Test Parameters**

| Parameter | Unit | 1 Tx Antenna | 2 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | FR2.120-1 | |
| CCE to REG mapping type | | Interleaved | |
| REG bundle size | | 2 for test 1-1<br>6 for test 1-2 | 2 |
| Interleaver size | | 3 for test 1-1<br>2 for test 1-2 | 3 |
| Shift index | | 0 | |

**Table 7.3.2.2.1.3-2: Minimum performance requirements with 120 kHz SCS for 1Tx antenna**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | $SNR_{BB}$ (dB) |
| 1-1 | 100 MHz | 60 | 1 | 2 CCE | R.PDCCH.5-1.1 TDD | TDLA30-75 | 1x2 Low | 1 | 6.0 |
| 1-2 | 100 MHz | 60 | 1 | 4 CCE | R.PDCCH.5-1.2 TDD | TDLA30-300 | 1x2 Low | 1 | 2.6 |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.3.2.2.1.

### 7.3.2.2.1.4 Test Description

#### 7.3.2.2.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 7.3-1 and Table 7.3.2.2.1.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.3.2.2.1.4.3.

#### 7.3.2.2.1.4.2 Test procedure

1. Set the UE in a direction found using one of the test procedures defined in Annex H. If no direction found, mark the test as inconclusive.
2. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 7.3.2.2.1.4.4-1. The details of PDCCH are specified in Table 7.3.2.2.1.3-1 and Table 7.3.2.2.1.3-2. The SS sends downlink MAC padding bits on the DL RMC.
3. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 7.3.2.2.1.4.4-1 as appropriate.
4. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 7.3.2.2.1.4.4-1, pass the UE. Otherwise fail the UE.

5. Repeat steps from 1 to 4 for each subtest in Table 7.3.2.2.1.4.4-1 as appropriate.

#### 7.3.2.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

##### 7.3.2.2.1.4.3.1 Message exceptions for SA

**Table 7.3.2.2.1.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 11000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 60 RBs of the BWP | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n2 | | 2 for test 1-1 |
| | n6 | | 6 for test 1-2 |
| interleaverSize | n3 | | 3 for test 1-1 |
| } | n2 | | 2 for test 1-2 |
| } | | | |
| } | | | |

**Table 7.3.2.2.1.4.3.1-2: PDCCH Search Space**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel2 | n1 | AL2 | Test 1-1 |
| aggregationLevel4 | n1 | AL4 | Test 1-2 |
| } | | | |
| } | | | |

##### 7.3.2.2.1.4.3.2 Message exceptions for NSA

Same as 7.3.2.2.1.4.3.1.

#### 7.3.2.2.1.4.4 Test requirement

Table 7.3.2.2.1.4.4-1 defines the primary level settings.

For the parameters specified in Table 7.3.2.2.1.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 7.3.2.2.1.4.4-1.

**Table 7.3.2.2.1.4.4-1: Test requirements with 120 kHz SCS for 1Tx antenna**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | $SNR_{BB}$ (dB) |
| 1-1 | 100 MHz | 60 | 1 | 2 CCE | R.PDCCH.5-1.1 TDD | TDLA30-75 | 1x2 Low | 1 | 7.7 |
| 1-2 | 100 MHz | 60 | 1 | 4 CCE | R.PDCCH.5-1.2 TDD | TDLA30-300 | 1x2 Low | 1 | 4.3 |

### 7.3.2.2.2 2Rx TDD FR2 PDCCH 2 Tx antenna performance for both SA and NSA

#### 7.3.2.2.2.1 Test Purpose

This test verifies the demodulation performance of PDCCH for two-antenna port with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 7.3.2.2.2.3-1.

#### 7.3.2.2.2.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 7.3.2.2.2.3 Minimum conformance requirements

For the parameters specified in Table 7.3.2.2.2.3-1, the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 7.3.2.2.2.3-2. The downlink physical setup is in accordance with Annex C.2.2.

**Table 7.3.2.2.2.3-1: Test Parameters**

| Parameter | Unit | 1 Tx Antenna | 2 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | FR2.120-1 | |
| CCE to REG mapping type | | Interleaved | |
| REG bundle size | | 2 for test 1-1<br>6 for test 1-2 | 2 |
| Interleaver size | | 3 for test 1-1<br>2 for test 1-2 | 3 |
| Shift index | | 0 | |

**Table 7.3.2.2.2.3-2: Minimum performance requirements with 120 kHz SCS for 2Tx Antenna**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | $SNR_{BB}$ (dB) |
| 2-1 | 100 MHz | 60 | 1 | 8 CCE | R.PDCCH.5-1.3 TDD | TDLA30-75 | 2x2 Low | 1 | 1.4 |
| 2-2 | 100 MHz | 60 | 2 | 16 CCE | R.PDCCH.5-2.1 TDD | TDLA30-75 | 2x2 Low | 1 | -1.6 |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.3.2.2.2.

#### 7.3.2.2.2.4 Test Description

##### 7.3.2.2.2.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 7.3-1 and Table 7.3.2.2.2.3-1 as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.3.2.2.1.4.3.

#### 7.3.2.2.2.4.2 Test procedure

1. Set the UE in a direction found using one of the test procedures defined in Annex H If no direction found, mark the test as inconclusive.
2. SS transmits PDCCH with DCI format as specified in PDCCH Reference Channel for C_RNTI to transmit the DL RMC according to Table 7.3.2.2.2.4.4-1. The details of PDCCH are specified in Table 7.3.2.2.2.3-1 and Table 7.3.2.2.2.3-2. The SS sends downlink MAC padding bits on the DL RMC.
3. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 7.3.2.2.2.4.4-1 as appropriate.
4. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 7.3.2.2.2.4.4-1, pass the UE. Otherwise fail the UE.
5. Repeat steps from 1 to 4 for each subtest in Table 7.3.2.2.2.4.4-1 as appropriate.

#### 7.3.2.2.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

#### 7.3.2.2.2.4.3.1 Message exceptions for SA

**Table 7.3.2.2.2.4.3.1-1: PDCCH-ControlResourceSet**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 11000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 60 RBs of the BWP | |
| Duration | 2 | SearchSpace Duration of 2 symbols | Test 2-2 |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n2 | | |
| interleaverSize | n3 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.2.4.3.1-2: PDSCH-TimeDomainResourceAllocationList**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-27 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDSCH-TimeDomainResourceAllocationList ::= SEQUENCE(SIZE(1..maxNrofDL-Allocations)) OF PDSCH-TimeDomainResourceAllocation { | 2 entries | | Test 2-2 |
| PDSCH-TimeDomainResourceAllocation[1] SEQUENCE { | | entry 1 | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 53 | Start symbol(S)=2, Length(L)=12 | Test 2-2 |
| } | | | |
| PDSCH-TimeDomainResourceAllocation[2] SEQUENCE { | | entry 2 | |
| K0 | Not present | | |
| mappingType | typeA | | |
| startSymbolAndLength | 100 | Start symbol(S)=2, Length(L)=8 | Test 2-2 |
| } | | | |
| } | | | |

**Table 7.3.2.2.2.4.3.1-3: PDCCH Search Space**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel8 | n1 | AL8 | Test 2-1 |
| aggregationLevel16 | n1 | AL16 | Test 2-2 |
| } | | | |
| } | | | |

### 7.3.2.2.2.4.3.2 Message exceptions for NSA

Same as 7.3.2.2.2.4.3.1.

### 7.3.2.2.2.4.4 Test requirement

Table 7.3.2.2.2.4.4-1 defines the primary level settings.

For the parameters specified in Table 7.3.2.2.2.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 7.3.2.2.2.4.4-1.

**Table 7.3.2.2.2.4.4-1: Test requirements with 120 kHz SCS**

| Test number | Bandwidth | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | $SNR_{BB}$ (dB) |
| 2-1 | 100 MHz | 60 | 1 | 8 CCE | R.PDCCH.5-1.3 TDD | TDLA30-75 | 2x2 Low | 1 | 3.2 |
| 2-2 | 100 MHz | 60 | 2 | 16 CCE | R.PDCCH.5-2.1 TDD | TDLA30-75 | 2x2 Low | 1 | 0.2 |

### 7.3.2.2.3 2Rx TDD FR2 PDCCH 1 Tx antenna performance for power saving

#### 7.3.2.2.3.1 Test Purpose

This test verifies the demodulation performance of PDCCH under 2 receive antenna conditions and with a given SNR for which the average probability of miss-detection of the Downlink Scheduling Grant (Pm-dsg), shall be below the specified value in Table 7.3.2.2.3.3-2 after receipt wake-up indication in the *DCI format 2_6* PDCCH in DRX off state. The downlink physical setup is in accordance with Annex C.2.1.

#### 7.3.2.2.3.2 Test applicability

This test applies to all types of NR UE release 16 and forward supporting Long DRX Cycle and DRX adaptation.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and Long DRX Cycle and DRX adaptation.

#### 7.3.2.2.3.3 Minimum conformance requirements

During the test the UE shall monitor the *DCI format 2_6* PDCCH in DRX off state and decide whether to receive the following PDCCH in DRX on period.

The parameters specified in Table 7.3.2.2.3.3-1 are valid for normal PDCCH in DRX on period and PDCCH in DRX off period.

**Table 7.3.2.2.3.3-1: Test Parameters**

| Parameter | | Unit | 1 Tx Antenna |
|---|---|---|---|
| TDD UL-DL pattern | | | FR2.120-1 |
| CCE to REG mapping type | | | Interleaved |
| REG bundle size | | | 6 |
| Interleaver size | | | 2 |
| Shift index | | | 0 |
| | | | |
| DRX cycle | | ms | 10 |
| *ps-WakeUp-r16* | | | absent |
| Wake-up indication bit in DCI format 2_6 | | | 1 |
| PDCCH DCI format 2_6 configuration | PS-offset | | $(T_{minimumTimeGap}+1)/2^{\mu}/0.125$ |
| | Number of PDCCH candidates | | 1 |
| | Frequency domain resource allocation for CORESET | | Start from RB = 0 with contiguous RB allocation |
| | TCI state | | TCI state #1 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot during DRX-on period |
| | | | |
| Note: $T_{minimumTimeGap}$ is signalled as a part of *drx-Adaptation-r16* UE capability. | | | |

For the parameters specified in Table 7.3.2.2.3.3-2, the average probability of a missed downlink scheduling grant (Pm-dsg) observed on PDCCH during DRX on shall be below the specified value in Table 7.3.2.2.3.3-2. The downlink physical setup is in accordance with Annex C.5.1.

**Table 7.3.2.2.3.3-2: Minimum performance requirements with 120 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | $SNR_{BB}$ (dB) |
| 3-1 | 100 | 60 | 1 | 4 | R.PDCCH.5-1.2 TDD | TDLA30-300 | 1x2 Low | 1 | 3.0 |
| | | | | 8 | R.PDCCH.5-1.4 TDD | | | | |

The normative reference for this requirement is TS 38.101-4 [5] clause 7.3.2.2.3.

#### 7.3.2.2.3.4 Test description

##### 7.3.2.2.3.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 and Table 5.3.6-1 of 38.521-1 [7].

Configurations of DRX, DCP, PDCCH before measurement are specified in 7.3.2.2.3.4.3.1 and Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR1 operation, setup the LTE link according to Annex D:

1. Connect the SS, the faders and AWGN noise source to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the cell are set up according to Table 7.3-1 and Table 7.3.2.2.3.3-1as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2 and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without Release On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 7.3.2.2.3.4.3.

##### 7.3.2.2.3.4.2 Test procedure

1. SS transmits PDCCH with DCI format2_6 as specified in PDCCH Reference Channel for C_RNTI within DRX off state. The Wake-up indication bit in PDCCH is set to 1.
2. Set the UE in a direction found using one of the test procedures defined in Annex H. If no direction found, mark the test as inconclusive.
3. SS transmits PDCCH with DCI format1_1 as specified in PDCCH Reference Channel R.PDCCH. 5-1.2 TDD for C_RNTI to transmit the DL RMC according to Table 7.3.2.2.3.4.4-1. The details of PDCCH are specified in Table 7.3.2.2.3.3-1 and Table 7.3.2.2.3.3-2. The SS sends downlink MAC padding bits on the DL RMC.
4. Set the parameters of the propagation condition, antenna configuration, the correlation matrix and the SNR according to Table 7.3.2.2.3.4.4-1 as appropriate.
5. Measure the Pm-dsg for a duration sufficient to achieve statistical significance according to Annex G clause G.1.5. Count the number of NACKs, ACKs and statDTXs on the UL PUCCH during each subtest interval. Pm-dsg is the ratio (statDTX)/(NACK+ACK+statDTX). If Pm-dsg is less than the value specified in table 7.3.2.2.3.4.4-1, pass the UE. Otherwise fail the UE.
6. Repeat steps from 2 to 5 for each subtest in Table 7.3.2.2.3.4.4-1 as appropriate.

##### 7.3.2.2.3.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 4.6.1.

7.3.2.2.3.4.3.1 Message exceptions for SA

**Table 7.3.2.2.3.4.3.1-1: DRX-Config**

| Derivation Path: TS 38.508-1 [4],Table 4.6.3-56 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DRX-Config ::= SEQUENCE { | | | |
| drx-onDurationTimer CHOICE { | | | |
| milliSeconds | ms1 | | |
| } | | | |
| drx-InactivityTimer | ms1 | | |
| drx-HARQ-RTT-TimerDL | 0 | | |
| drx-HARQ-RTT-TimerUL | 0 | | |
| drx-RetransmissionTimerDL | sl1 | | |
| drx-RetransmissionTimerUL | sl1 | | |
| drx-LongCycleStartOffset CHOICE { | | | |
| ms10 | 0 | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.4.3.1-2: DCP-Config**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-106 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| dcp-Config-r16 CHOICE { | | | |
| setup SEQUENCE { | | | |
| ps-Offset-r16 | 40 | | |
| sizeDCI-2-6-r16 | 2 | | |
| ps-PositionDCI-2-6-r16 | 0 | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.4.3.1-3: PDCCH-Config**

| Derivation Path: TS 38.508-1 [6],Table 4.6.3-95 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCCH-Config::= SEQUENCE { | | | |
| controlResourceSetToAddModList SEQUENCE(SEQUENCE(SIZE (1..3)) OF ControlResourceSet ::= SEQUENCE { | 2 entry | | |
| ControlResourceSet[1] | ControlResourceSet1 | | |
| ControlResourceSet[2] | ControlResourceSet2 | | |
| } | | | |
| searchSpacesToAddModList SEQUENCE(SIZE (1..10)) OF SearchSpace ::= SEQUENCE { | 1 entry | | |
| SearchSpace[1] | SearchSpace | | |
| } | | | |
| searchSpacesToAddModListExt-r16 SEQUENCE(SIZE (1..10)) OF SearchSpace { | 1 entry | | |
| searchSpaceExt-r16[1] | SearchSpaceExt | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.4.3.1-4: PDCCH-ControlResourceSet1**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | 11111111 11000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 60 RBs of the BWP | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | Null | | |
| reg-BundleSize | n6 | | 2 |
| interleaverSize | n2 | | 3 |
| } | | | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.4.3.1-5: PDCCH-ControlResourceSet2**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| controlResourceSetId | 2 | | |
| frequencyDomainResources | 11111111 11000000 00000000 00000000 00000000 00000 | CORESET to use the least significant 60 RBs of the BWP | |
| Duration | 1 | SearchSpace duration of 1 symbol | |
| cce-REG-MappingType CHOICE { | | | |
| Interleaved SEQUENCE { | | | |
| reg-BundleSize | n6 | | |
| interleaverSize | n2 | | |
| } | | | |
| } | | | |
| tci-StatesPDCCH-ToAddList { | 1 | TCI State #1 | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.4.3.1-6: PDCCH Search Space**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel4 | n1 | AL4 | |
| } | | | |
| } | | | |

**Table 7.3.2.2.3.4.3.1-7: PDCCH *Search Space Ext***

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.0-7a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpaceExt-r16 ::= SEQUENCE { | | | |
| controlResourceSetId-r16 | 2 | | |
| searchSpaceType-r16 SEQUENCE { | | | |
| common SEQUENCE { | | | |
| dci-Format2-6-r16 SEQUENCE { | NULL | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

7.3.2.2.3.4.3.2 Message exceptions for NSA

Same as 7.3.2.2.3.4.3.1.

7.3.2.2.3.4.4 Test requirement

Table 7.3.2.2.3.4.4-1 defines the primary level settings.

For the parameters specified in Table 5.3-1 the average probability of a missed downlink scheduling grant (Pm-dsg) shall be below the specified value in Table 7.3.2.2.3.4.4-1.

**Table 7.3.2.2.3.4.4-1: Minimum performance with 120 kHz SCS**

| Test number | Bandwidth (MHz) | CORESET RB | CORESET duration | Aggregation level | Reference Channel | Propagation Condition | Antenna configuration and correlation Matrix | Reference value | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Pm-dsg (%) | $SNR_{BB}$ (dB) |
| 3-1 | 100 | 60 | 1 | 4 | R.PDCCH.5-1.2 TDD | TDLA30-300 | 1x2 Low | 1 | 4.7 |
| | | | | 8 | R.PDCCH.5-1.4 TDD | | | | |

# 7.4 PBCH demodulation requirements

TS 38.101-4 shall specify the PBCH performance requirements and has recommended that these requirements do not need to be tested.

# 7.5 Sustained downlink data rate provided by lower layers

# 8 CSI reporting requirements (Radiated requirements)

## 8.1 General

This clause includes radiated requirements for the reporting of channel state information (CSI).

For conformance testing involving FR2 test cases in this specification, the UE under test shall be pre-configured with UL Tx diversity schemes disabled to account for single polarization System Simulator (SS) in the test environment. The UE under test may transmit with dual polarization.

## 8.1.1 Applicability of requirements

### 8.1.1.1 General

The minimum performance requirements are applicable to the FR2 operating bands defined in TS 38.101-2 [3] with $F_{DL_high}$ not exceeding 40000 MHz.

The minimum performance requirements in Clause 8 are mandatory for UE supporting NR operation, except test cases listed in Clause 8.1.1.3, 8.1.1.4.

### 8.1.1.2 Applicability of requirements for different number of RX antenna ports

UE shall support 2 RX ports for different RF operating bands. The UE requirements applicability is defined in Table 8.1.1.2-1.

**Table 8.1.1.2-1: Requirements applicability**

| Supported RX antenna ports | Test type | Test list |
|---|---|---|
| UE supports 2RX antenna | CQI | All tests in Clause 8.2.2 |
| | PMI | All tests in Clause 8.3.2 |
| | RI | All tests in Clause 8.4.2 |

### 8.1.1.3 Applicability of requirements for optional UE features

The performance requirements in Table 8.1.1.3-1 shall apply for UEs which support optional UE features only.

**Table 8.1.1.3-1: Requirements applicability for optional UE features**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| 256QAM modulation scheme for PDSCH for FR2 (*pdsch-256QAM-FR2*) | FR2 TDD | CQI | Clause 8.2.2.2.2.1_1 (Tests 3 and 4) | The test coverage can be considered fulfilled without executing of Test 1 and 2 from Clause 8.2.2.2.2.1 if UE passes Test 3 and 4 from Clause 8.2.2.2.2.1_1 |

### 8.1.1.4 Applicability of requirements for mandatory UE features with capability signalling

The performance requirements in Table 8.1.1.4-1 shall apply for UEs which support mandatory UE features with capability signalling only.

**Table 8.1.1.4-1: Requirements applicability for mandatory features with UE capability signalling**

| UE feature/capability [14] | Test type | | Test list | Applicability notes |
|---|---|---|---|---|
| Supported maximum number of PDSCH MIMO layers (*maxNumberMIMO-LayersPDSCH)* | FR2 TDD | CQI | Clause 8.2.2.2.1.1 | The requirements apply only in case the PDSCH MIMO rank in the test case does not exceed UE PDSCH MIMO layers capability |
| | | RI | Clause 8.4.2.2 | |
| Support of 1 port PTRS (*onePortsPTRS*) | FR2 TDD | CQI | Clause 8.2 | |
| | | PMI | Clause 8.3 | |
| | | RI | Clause 8.4 | |

### 8.1.1.5 Applicability of Channel Quality Indicator (CQI) reporting requirements for CA

#### 8.1.1.5.1 Applicability and test rules for different CA configurations and bandwidth combination sets

The performance requirement for CA CQI tests in clause 8.2A are defined independent of CA configurations and bandwidth combination sets specified in clause 5.5A in TS 38.101-2 [7].

For UEs supporting multiple CA capabilities, test any one of the supported CA capabilities with largest aggregated CA bandwidth combination. The categorization of CA capability is specified in clause 7.1.1.5.1.

For UEs supporting multiple CA configurations from the selected CA capability, test any one of the supported CA configurations with largest aggregated CA bandwidth combination. For simplicity, the CA configuration refers to combination of CA configuration and bandwidth combination set.

A single uplink CC is configured for all tests.

#### 8.1.1.5.2 Test coverage for different number of component carriers

For CA CQI tests specified in clause 8.2A, among all supported CA capabilities, if corresponding CA tests with the largest number of CCs supported by the UE are tested, the test coverage can be considered fulfilled without executing the CA tests with less than the largest number of CCs supported by the UE.

## 8.1.1_1 Applicability of test requirements due to maximum achievable SNR

Editor's note: The applicability of test requirements with a modulation above 64 QAM is FFS.

The current assumption of maximum testable $SNR_{BB}$ for PC3, Max device size ≤ 30 cm under fading conditions is specified in Table 7.1.1_1-1.

The current assumption of maximum testable $SNR_{BB}$ for indirect far field (IFF), PC3, Max device size ≤ 30 cm without fading conditions is specified in Table 8.1.1_1-1.

**Table 8.1.1_1-1: Current assumption of maximum testable $SNR_{BB}$ without fading for modulations up to 64 QAM**

| Operating Band / Frequency | Maximum testable $SNR_{BB}$ (dB) | | |
|---|---|---|---|
| | CHBW 50 MHz | CHBW 100 MHz | CHBW 200 MHz |
| n257 mid | [28.7] | [25.5] | [22.5] |
| n258 mid | [28.7] | [25.5] | [22.5] |
| n259 mid | [18.4] | [15.2] | [12.1] |
| n260 mid | [22.5] | [19.3] | [16.3] |
| n261 mid | [28.7] | [25.5] | [22.5] |

Based on the current assumption of maximum testable $SNR_{BB}$, the applicability of test points is defined in Table 8.1.1_1-2 for indirect far field (IFF), PC3, Max device size ≤ 30 cm under fading conditions.

**Table 8.1.1_1-2: Testability of test requirements due to maximum achievable SNR per band**

| Test Case | Test point | CHBW / MHz | Fading | SNR test requirement | Test Point Applicability | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | n257 | n258 | n259 | n260 | n261 |
| 8.2.2.2.1.1 | 1 | 100 | No | 9 | x | x | x | x | x |
| | 2 | 100 | No | 15 | x | x | x | x | x |
| 8.2.2.2.2.1 | 1 | 100 | Yes | 7 | x | x | x | x | x |
| | 2 | 100 | Yes | 13 | x | x | x | x | x |
| 8.2.2.2.2.1_1 | 3 | 50 | Yes | 8 | TBD | TBD | TBD | TBD | TBD |
| | 4 | 50 | Yes | 21 | TBD | TBD | TBD | TBD | TBD |
| 8.4.2.2.1 | 1 | 100 | Yes | 0 | x | x | X | x | x |
| | 2 | 100 | Yes | 16 | x | x | x | x | x |
| | 3 | 100 | Yes | 16 | x | x | x | x | x |

## 8.1.2 Common test parameters

Parameters specified in Table 8.1.2-1 are applied for all test cases in this section unless otherwise stated.

**Table 8.1.2-1: Test parameters for CSI test cases**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| Duplex Mode | | | TDD |
| PTRS *epre-Ratio* | | | 0 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 3) | RBs | 0 |
| | Subcarrier spacing | kHz | 120 |
| DL BWP configuration #1 | Cyclic prefix | | Normal |
| | RB offset | RBs | 0 |
| | Number of contiguous PRB | PRBs | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing |
| Active DL BWP index | | | 1 |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | 0,1 |
| | Number of PDCCH candidates and aggregation levels | | 1/AL8 |
| | DCI format | | 1_1 |
| | TCI state | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | Multi-path fading propagation conditions:<br>Single Panel Type I, Random per slot with equal probability of each applicable i1, i2 combination, and with REG bundling granularity for number of Tx larger than 1<br><br>Static propagation conditions:<br>Single Panel Type I, Random precoder chosen from precoder index 0 and 2, selection updated per slot |
| Cross carrier scheduling | | | Not configured |
| PDSCH configuration | Mapping type | | Type A |
| | *k0* | | 0 |
| | Starting symbol (S) | | 2 |
| | Length (L) | | 12 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | 2 |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | DMRS ports indexes | | {1000} for Rank1<br>{1000,1001} for Rank2 |
| | Maximum number of OFDM symbols for DL front loaded DMRS | | 1 |
| | Number of PDSCH DMRS CDM group(s) without data | | 2 |
| PTRS configuration | Frequency density ($K_{PT-RS}$) | | 2 |
| | Time density ($L_{PT-RS}$) | | 1 |
| | Resource Element Offset | | 2 |

| | | | | |
|---|---|---|---|---|
| CSI-RS for tracking | First subcarrier index in the PRB used for CSI-RS ($k_0$) | | | 0 for CSI-RS resource 1,2,3,4 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$) | | | 4 for CSI-RS resource 1 and 3<br>8 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports ($X$) | | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | | No CDM for CSI-RS resource 1,2,3,4 |
| | Density ($\rho$) | | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | | slot | 120kHz SCS: 160 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | | slot | 120 kHz SCS:<br>80 for CSI-RS resource 1 and 2<br>81 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| CSI-RS for beam refinement | First subcarrier index in the PRB used for CSI-RS | | | $k_0$=0 for CSI-RS resource 1,2 |
| | First OFDM symbol in the PRB used for CSI-RS | | | $l_0$ = 8 for CSI-RS resource 1<br>$l_0$ = 9 for CSI-RS resource 2 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2 |
| | Density (ρ) | | | 3 for CSI-RS resource 1,2 |
| | CSI-RS periodicity | | Slots | 120 kHz SCS: 160 for CSI-RS resource 1,2 |
| | CSI-RS offset | | Slots | 0 for CSI-RS resource 1,2 |
| | Repetition | | | ON |
| | QCL info | | | TCI state #1 |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type D |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type D |
| Number of HARQ Processes | | | | 8 |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | | For FR2.120-1:<br>3 if mod (i.5) = 0,<br>6 if mod(i,5) = 2<br>For FR2.120-2:<br>11 if mod(i,8) = 0,<br>7]if mod(i,8) = 4,<br>6]if mod(i,8) = 5,<br>where i is slot index per radio fame with values 0-79. |
| Symbols for unused REs | | | | OCNG as specified in A.5 |
| Physical signals, channels mapping and precoding | | | | As specified in Annex B.4.1 |

Note 1: PDSCH is not scheduled on slots containing CSI-RS or slots which are not full DL.
Note 2: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission.
Note 3: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing.

# 8.2 Reporting of Channel Quality Indicator (CQI)

## 8.2.1 1RX requirements

TBD

## 8.2.2 2RX requirements

### 8.2.2.1 FDD

TBD

### 8.2.2.2 TDD

#### 8.2.2.2.1 CQI reporting under AWGN conditions

##### 8.2.2.2.1.1 2Rx TDD FR2 periodic wideband CQI reporting under AWGN performance for both SA and NSA

###### 8.2.2.2.1.1.1 Test Purpose

The purpose of this test is to verify the variance of the wideband CQI reports is within the limits defined and a PDSCH BLER of 10% falls between the transport format based median CQI-1 and median CQI or the transport format based median CQI and median CQI +1.

###### 8.2.2.2.1.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

###### 8.2.2.2.1.1.3 Minimum requirement for periodic CQI reporting

The purpose is to verify that the reported CQI values are in accordance with the CQI definition given in TS 38.214 [12]. To account for sensitivity of the input SNR the reporting definition is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

For the parameters specified in Table 8.2.2.2.1.1.3-1, and using the downlink physical channels specified in Annex C.5.1, the minimum requirements are specified by the following:

a) the reported CQI value shall be in the range of ±1 of the reported median more than 90% of the time;

b) if the PDSCH BLER using the transport format indicated by median CQI is less than or equal to 0.1, the BLER using the transport format indicated by the (median CQI + 1) shall be greater than 0.1. If the PDSCH BLER using the transport format indicated by the median CQI is greater than 0.1, the BLER using transport format indicated by (median CQI – 1) shall be less than or equal to 0.1.

**Table 8.2.2.2.1.1.3-1 Test parameters**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Bandwidth | | MHz | 100 | |
| Subcarrier spacing | | kHz | 120 | |
| Duplex Mode | | | TDD | |
| TDD Slot Configuration | | | FR2.120-2 Annex A.1.3 | |
| $SNR_{BB}$ | | dB | 8 \| 9 | 14 \| 15 |
| Propagation channel | | | AWGN | |
| Antenna configuration | | | 2×2 with static channel specified in Annex B.1 | |
| Beamforming Model | | | As specified in Annex B.4.1 | |
| ZP CSI-RS configuration | CSI-RS resource Type | | *Periodic* | |
| | Number of CSI-RS ports (*X*) | | 4 | |
| | CDM Type | | *FD-CDM2* | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 8 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | |
| | CSI-RS periodicity and offset | slot | 8/1 | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | *Periodic* | |
| | Number of CSI-RS ports (*X*) | | 2 | |
| | CDM Type | | *fd-CDM2* | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 6 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 8/1 | |
| CSI-IM configuration | CSI-IM resource Type | | Periodic | |
| | CSI-IM RE pattern | | 1 | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8, 13) | |
| | CSI-IM timeConfig periodicity and offset | slot | 8/1 | |
| ReportConfigType | | | *Periodic* | |
| CQI-table | | | Table 1 | |
| reportQuantity | | | *cri-RI-PMI-CQI* | |
| timeRestrictionForChannelMeasurements | | | *Not configured* | |
| timeRestrictionForInterferenceMeasurements | | | *Not configured* | |
| cqi-FormatIndicator | | | *Wideband* | |
| pmi-FormatIndicator | | | *Wideband* | |
| Sub-band Size | | RB | 8 | |
| csi-ReportingBand | | | 111111111 | |
| CSI-Report periodicity and offset | | slot | 8/3 | |
| aperiodicTriggeringOffset | | | *Not configured* | |
| Codebook configuration | Codebook Type | | *typeI-SinglePanel* | |
| | Codebook Mode | | 1 | |
| | (CodebookConfig-N1,CodebookConfig-N2) | | *Not configured* | |
| | CodebookSubsetRestriction | | 010000 | |
| | RI Restriction | | N/A | |
| Physical channel for CSI report | | | PUCCH | |
| CQI/RI/PMI delay | | ms | 8.375 | |
| Maximum number of HARQ transmission | | | 1 | |
| Measurement channel | | | As specified in Table A.4-1, TBS.1-2 | |

The normative reference for this requirement is TS 38.101-4 [5] clause 8.2.2.2.1.1.

#### 8.2.2.2.1.1.4 Test Description

##### 8.2.2.2.1.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, AWGN noise source and the UE antenna is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 8.1.2-1 and Table 8.2.2.2.1.1.3-1 and as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].

4. Propagation conditions for NR cell are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 8.2.2.2.1.1.4.3.

##### 8.2.2.2.1.1.4.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found mark the test as inconclusive.

2. Set the parameters of bandwidth, SCS, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 8.2.2.2.1.1.3-1.

3. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 2000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 1 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

4. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as wideband Median CQI value.

5. If Median CQI is not equal to 1 or 15 and 1800 or more of the wideband CQI values are in the range (Median CQI - 1) $\leq$ Median CQI $\leq$ ( Median CQI + 1) then continue with step 6, otherwise go to step 9.

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000.

For the filtered ACK and NACK responses if the ratio (NACK / ACK + NACK) ≤ 0.1 then go to step 7, otherwise go to step 8.

7. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI+1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 8 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) > 0.1

   then pass the UE for this test and go to step 10, otherwise go to step 9.

8. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the wideband median-CQI-1 value and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH, transmitted by the SS, record and filter the ACK, NACK and statDTX responses as in step 6 until 1000 filtered ACK+NACK responses are gathered.

   If the ratio (NACK /ACK + NACK) ≤ 0.1

   then pass the UE for this test and go to step 10, otherwise go to step 9.

9. If both SNR points of the test have not been tested, then repeat the same procedure (steps 1 to 8) for the other SNR point as appropriate. Otherwise fail the UE.

10. Repeat step 1 to 9 for Test2.

#### 8.2.2.2.1.1.4.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

#### 8.2.2.2.1.1.4.3_1 Message exceptions for SA

**Table 8.2.2.2.1.1.4.3_1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | Periodic | |
| periodic SEQUENCE { | | | |
| reportSlotConfig | CSI-ReportPeriodicityAndOffset | 8/1 | |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE{ | | | |
| Subbands9 | 111111111 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.2.1.1.4.3_1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 010000 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.2.1.1.4.3_1-3: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| Sl80 | 7 | | |
| } | | | |
| } | | | |

#### 8.2.2.2.1.1.4.3_2 Message exceptions for NSA

Same as 8.2.2.2.1.1.4.3_1.

#### 8.2.2.2.1.1.5 Test Requirements

The pass/fail decision is as specified in the test procedure in clause 8.2.2.2.1.1.4.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

### 8.2.2.2.2 CQI reporting under fading conditions

#### 8.2.2.2.2.1 2Rx TDD FR2 aperiodic wideband CQI reporting under fading performance for both SA and NSA

##### 8.2.2.2.2.1.1 Test Purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 1% for the indicated transport format.

##### 8.2.2.2.2.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 8.2.2.2.2.1.3 Minimum requirement for aperiodic CQI reporting

The reporting accuracy of CQI under frequency non-selective fading conditions is determined by the reporting variance, the relative increase of the throughput obtained when the transport format is indicated by the reported CQI compared to

the throughput obtained when a fixed transport format is configured according to the reported median CQI, and a minimum BLER using the transport formats indicated by the reported CQI. To account for sensitivity of the input SNR the CQI reporting under frequency non-selective fading conditions is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

For the parameters specified in Table 8.2.2.2.2.1.3-1 and using the downlink physical channels specified in Annex C.5.1, the minimum requirements are specified by the following:

a) a CQI index not in the set {median CQI -1, median CQI, median CQI +1} shall be reported at least α % of the time, where α% is specified in Table 8.2.2.2.2.1.3-2;

b) the ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be $\geq \gamma$, where γ is specified in Table 8.2.2.2.2.1.3-2;

c) when transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater or equal to 0.01.

**Table 8.2.2.2.2.1.3-1: Test parameters**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Bandwidth | | MHz | 100 | |
| Subcarrier spacing | | kHz | 120 | |
| Duplex Mode | | | TDD | |
| TDD Slot Configuration | | | FR2.120-2 Annex A.1.3 | |
| $SNR_{BB}$ | | dB | 6 7 12 13 | |
| Propagation channel | | | TDLA30-35 | |
| Antenna configuration | | | 2×2 ULA High | |
| Beamforming Model | | | As specified in Annex B.4.1 | |
| ZP CSI-RS configuration | CSI-RS resource Type | | *Periodic* | |
| | Number of CSI-RS ports (*X*) | | 4 | |
| | CDM Type | | *FD-CDM2* | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 8 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | |
| | CSI-RS interval and offset | slot | 8/1 | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | *Aperiodic* | |
| | Number of CSI-RS ports (*X*) | | 2 | |
| | CDM Type | | *fd-CDM2* | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 6 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | |
| | NZP CSI-RS-timeConfig interval and offset | slot | Not configured | |
| | aperiodicTriggeringOffset | | 0 | |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic | |
| | CSI-IM RE pattern | | 1 | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8, 13) | |
| | CSI-IM timeConfig interval and offset | slot | Not configured | |
| ReportConfigType | | | *Aperiodic* | |
| CQI-table | | | Table 1 | |
| reportQuantity | | | *cri-RI-PMI-CQI* | |
| timeRestrictionForChannelMeasurements | | | *Not configured* | |
| timeRestrictionForInterferenceMeasurements | | | *Not configured* | |
| cqi-FormatIndicator | | | *Wideband* | |
| pmi-FormatIndicator | | | *Wideband* | |
| Sub-band Size | | RB | 8 | |
| csi-ReportingBand | | | 111111111 | |
| CSI-Report periodicity and offset | | slot | Not configured | |
| Aperiodic Report Slot Offset | | | 6 | |
| CSI request | | | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | |
| reportTriggerSize | | | 1 | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | |
| Codebook configuration | Codebook Type | | *typeI-SinglePanel* | |
| | Codebook Mode | | 1 | |

| | | | |
|---|---|---|---|
| | (CodebookConfig-N1,CodebookConfig-N2) | | *Not configured* |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 1.375 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-1, TBS.1-1 |

**Table 8.2.2.2.2.1.3-2 Minimum requirements**

| | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 8.2.2.2.2.1.

### 8.2.2.2.2.1.4 Test Description

#### 8.2.2.2.2.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE antenna is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 8.1.2-1 and Table 8.2.2.2.2.1.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 8.2.2.2.2.1.4.3.

#### 8.2.2.2.2.1.4.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found mark the test as inconclusive.
2. Set the parameters of bandwidth, reference Channel, the propagation condition, antenna configuration and the SNR according to Table 8.2.2.2.2.1.5-1.

3. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to CQI value 2 and keep it regardless of the wideband CQI value sent by the UE. The SS sends downlink MAC padding bits on the DL RMC. Continue transmission of the PDSCH until 6000 wideband CQI reports have been gathered. In this process the SS collects wideband CQI reports every 1 ms and also cases where UE transmits nothing in its CQI timing are also counted as wideband CQI reports.

4. Set up a relative frequency distribution for the reported wideband CQI-values, Calculate the median value (wideband Median CQI is the wideband CQI that is at or crosses 50% distribution from the lower wideband CQI side). This CQI-value is declared as Median CQI value.

5. If Median CQI value is not equal to 1 or 15 and 120 ($\alpha$%) or more of the wideband CQI values are outside the range (Median CQI - 1) ≤ Median CQI ≤ (Median CQI + 1) then continue with step 6, otherwise go to step 8.

6. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the Median CQI value from step 4 and shall not react to the UE's wideband CQI reports. The SS sends downlink MAC padding bits on the DL RMC. Measure the average throughput according to Annex G.3.3 and G.3.4.

   Declare the throughput as $t_{median}$

7. The SS shall transmit PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the UE's reported wideband CQI value. The SS sends downlink MAC padding bits on the DL RMC. For any PDSCH transmitted by the SS, record the associated ACK, NACK and statDTX responses. The responses are then filtered as follows: for the sequence of responses for each HARQ process, discard all the statDTX responses. Continue to gather data until the number of filtered ACK+NACK responses reaches 1000. Record the BLER (NACK / ACK + NACK) and measure the average throughput according to Annex G.3.3 and G.3.4. Declare the throughput as t.

   If the recorded BLER ≥ 0.01 and t / $t_{median} \geq \gamma$ then pass the UE for this test and go to step 9.

8. If both SNR points of the test have not been tested, then repeat the same procedure (steps 2 to 7) for the other SNR point as appropriate. Otherwise fail the UE.

9. Repeat step 2 to 8, with test conditions according to the table 8.2.2.2.2.1.5-1, for Test2 as appropriate.

#### 8.2.2.2.2.1.4.3 Message Contents

Message contents are according to TS 38.508-1 [6] subclause 4.6.1 and 5.4.2 with the following exceptions:

#### 8.2.2.2.2.1.4.3_1 Message exceptions for SA

**Table 8.2.2.2.2.1.4.3_1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | Aperiodic | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 6 | | |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE{ | | | |
| Subbands9 | 111111111 | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.2.2.1.4.3_1-2: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 000001 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.2.2.2.2.1.4.3_1-3: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl80 | 7 | | |
| } | | | |
| } | | | |

8.2.2.2.2.1.4.3_2 Message exceptions for NSA

Same as in 8.2.2.2.2.1.4.3_1.

8.2.2.2.2.1.5 Test Requirements

**Table 8.2.2.2.2.1.5-1 Test parameters**

| Parameter | | Unit | Test 1 | | Test 2 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 100 | | | |
| Subcarrier spacing | | kHz | 120 | | | |
| Duplex Mode | | | TDD | | | |
| TDD Slot Configuration | | | FR2.120-2 Annex A.1.3 | | | |
| $SNR_{BB}$ | | dB | 6+ TT | 7+ TT | 12 +T T | 13 +T T |
| Propagation channel | | | TDLA30-35 | | | |
| Antenna configuration | | | 2×2 ULA High | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | *Periodic* | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | *FD-CDM2* | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 8 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | | | |
| | CSI-RS interval and offset | slot | 8/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | *Aperiodic* | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | *fd-CDM2* | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 6 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | | | |
| | NZP CSI-RS-timeConfig interval and offset | slot | Not configured | | | |
| | aperiodicTriggeringOffset | | 0 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic | | | |
| | CSI-IM RE pattern | | 1 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8, 13) | | | |
| | CSI-IM timeConfig interval and offset | slot | Not configured | | | |
| ReportConfigType | | | *Aperiodic* | | | |
| CQI-table | | | Table 1 | | | |
| reportQuantity | | | *cri-RI-PMI-CQI* | | | |
| timeRestrictionForChannelMeasurements | | | *Not configured* | | | |
| timeRestrictionForInterferenceMeasurements | | | *Not configured* | | | |
| cqi-FormatIndicator | | | *Wideband* | | | |
| pmi-FormatIndicator | | | *Wideband* | | | |
| Sub-band Size | | RB | 8 | | | |
| csi-ReportingBand | | | 111111111 | | | |
| CSI-Report periodicity and offset | | slot | Not configured | | | |
| Aperiodic Report Slot Offset | | | 6 | | | |
| CSI request | | | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | | | |
| reportTriggerSize | | | 1 | | | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains | | | |

| | | | |
|---|---|---|---|
| | | | pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | *typeI-SinglePanel* |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | *Not configured* |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 1.375 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-1, TBS.1-1 |
| Note 1: TT = 0dB | | | |

**Table 8.2.2.2.2.1.5-2 Test requirements**

| | Test 1 | Test 2 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\gamma$ | 1.05 - TT | 1.05 - TT |
| Note 1: TT = 0.01 | | |

### 8.2.2.2.2.1_1 2Rx TDD FR2 aperiodic CQI wideband reporting under fading performance for both SA and NSA – 256QAM (Rel-16 and forward)

#### 8.2.2.2.2.1_1.1 Test Purpose

To verify the variance of the wideband CQI reports is within the limits defined, that the ratio of the throughput is within the limits defined and that the average PDSCH BLER is greater than or equal to 1% for the indicated transport format.

#### 8.2.2.2.2.1_1.2 Test Applicability

This test applies to all types of NR UE release 16 and forward supporting DL 256QAM.

This test also applies to all types of EUTRA UE release 16 and forward supporting EN-DC and NR DL 256QAM.

#### 8.2.2.2.2.1_1.3 Minimum requirement for aperiodic CQI reporting

The reporting accuracy of CQI under frequency non-selective fading conditions is determined by the reporting variance, the relative increase of the throughput obtained when the transport format is indicated by the reported CQI compared to the throughput obtained when a fixed transport format is configured according to the reported median CQI, and a minimum BLER using the transport formats indicated by the reported CQI. To account for sensitivity of the input SNR the CQI reporting under frequency non-selective fading conditions is considered to be verified if the reporting accuracy is met for at least one of two SNR levels separated by an offset of 1 dB.

For the parameters specified in Table 8.2.2.2.2.1_1.3-1 and using the downlink physical channels specified in Annex C.5.1, the minimum requirements are specified by the following:

a) a CQI index not in the set {median CQI -1, median CQI, median CQI +1} shall be reported at least α % of the time, where α% is specified in Table 8.2.2.2.2.1_1.3-2;

b) the ratio of the throughput obtained when transmitting the transport format indicated by each reported wideband CQI index and that obtained when transmitting a fixed transport format configured according to the wideband CQI median shall be ≥ γ, where γ is specified in Table 8.2.2.2.2.1_1.3-2;

c) when transmitting the transport format indicated by each reported wideband CQI index, the average BLER for the indicated transport formats shall be greater or equal to 0.01.

**Table 8.2.2.2.2.1_1.3-1: Test parameters**

| Parameter | | Unit | Test 3 | Test 4 |
|---|---|---|---|---|
| Bandwidth | | MHz | 50 | |
| Subcarrier spacing | | kHz | 120 | |
| Duplex Mode | | | TDD | |
| TDD Slot Configuration | | | FR2.120-2 Annex A.1.3 | |
| $SNR_{BB}$ | | dB | 7 \| 8 | 20 \| 21 |
| Propagation channel | | | TDLA30-35 | |
| Antenna configuration | | | 2×2 ULA High | |
| Beamforming Model | | | As specified in Annex B.4.1 | |
| ZP CSI-RS configuration | CSI-RS resource Type | | *Periodic* | |
| | Number of CSI-RS ports (*X*) | | 4 | |
| | CDM Type | | *FD-CDM2* | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | 8 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | |
| | CSI-RS interval and offset | slot | 8/1 | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | *Aperiodic* | |
| | Number of CSI-RS ports (*X*) | | 2 | |
| | CDM Type | | *fd-CDM2* | |
| | Density (ρ) | | 1 | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$) | | 6 | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | |
| | NZP CSI-RS-timeConfig interval and offset | slot | Not configured | |
| | aperiodicTriggeringOffset | | 0 | |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic | |
| | CSI-IM RE pattern | | 1 | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8, 13) | |
| | CSI-IM timeConfig interval and offset | slot | Not configured | |
| ReportConfigType | | | *Aperiodic* | |
| CQI-table | | | Table 2 | |
| reportQuantity | | | *cri-RI-PMI-CQI* | |
| timeRestrictionForChannelMeasurements | | | *Not configured* | |
| timeRestrictionForInterferenceMeasurements | | | *Not configured* | |
| cqi-FormatIndicator | | | *Wideband* | |
| pmi-FormatIndicator | | | *Wideband* | |
| Sub-band Size | | RB | 8 | |
| csi-ReportingBand | | | 111111111 | |
| CSI-Report periodicity and offset | | slot | Not configured | |
| Aperiodic Report Slot Offset | | | 6 | |
| CSI request | | | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | |
| reportTriggerSize | | | 1 | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | |
| Codebook configuration | Codebook Type | | *typeI-SinglePanel* | |
| | Codebook Mode | | 1 | |

| | | | |
|---|---|---|---|
| | (CodebookConfig-N1,CodebookConfig-N2) | | *Not configured* |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 1.375 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-7 |

**Table 8.2.2.2.2.1_1.3-2 Minimum requirements**

| | Test 3 | Test 4 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\gamma$ | 1.05 | 1.05 |

The normative reference for this requirement is TS 38.101-4 [5] clause 8.2.2.2.2.1.

### 8.2.2.2.2.1_1.4 Test Description

Same test description as in clause 8.2.2.2.2.1.4 with the following exception:

- Table 8.2.2.2.2.1_1.3-1 instead of 8.2.2.2.2.1.3-1
- Table 8.2.2.2.2.1_1.4.1- instead of 8.2.2.2.2.1.4.3_1-1
- Table 8.2.2.2.2.1_1.5-1 instead of 8.2.2.2.2.1.5-1
- In test procedures, Test 3 and Test 4 are configured and tested.

**Table 8.2.2.2.2.1_1.4.1-1: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | Aperiodic | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 6 | | |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE{ | | | |
| Subbands9 | 111111111 | | |
| } | | | |
| } | | | |
| cqi-Table | table2 | | Test 3 and 4 |
| } | | | |

8.2.2.2.2.1_1.5 Test Requirements

**Table 8.2.2.2.2.1_1.5-1 Test parameters**

| Parameter | | Unit | Test 3 | | Test 4 | |
|---|---|---|---|---|---|---|
| Bandwidth | | MHz | 50 | | | |
| Subcarrier spacing | | kHz | 120 | | | |
| Duplex Mode | | | TDD | | | |
| TDD Slot Configuration | | | FR2.120-2 Annex A.1.3 | | | |
| $SNR_{BB}$ | | dB | 7+ TT | 8+ TT | 20 +T T | 21 +T T |
| Propagation channel | | | TDLA30-35 | | | |
| Antenna configuration | | | 2×2 ULA High | | | |
| Beamforming Model | | | As specified in Annex B.4.1 | | | |
| ZP CSI-RS configuration | CSI-RS resource Type | | *Periodic* | | | |
| | Number of CSI-RS ports (*X*) | | 4 | | | |
| | CDM Type | | *FD-CDM2* | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 8 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | | | |
| | CSI-RS interval and offset | slot | 8/1 | | | |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | *Aperiodic* | | | |
| | Number of CSI-RS ports (*X*) | | 2 | | | |
| | CDM Type | | *fd-CDM2* | | | |
| | Density (ρ) | | 1 | | | |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 6 | | | |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 | | | |
| | NZP CSI-RS-timeConfig interval and offset | slot | Not configured | | | |
| | aperiodicTriggeringOffset | | 0 | | | |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic | | | |
| | CSI-IM RE pattern | | 1 | | | |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8, 13) | | | |
| | CSI-IM timeConfig interval and offset | slot | Not configured | | | |
| ReportConfigType | | | *Aperiodic* | | | |
| CQI-table | | | Table 2 | | | |
| reportQuantity | | | *cri-RI-PMI-CQI* | | | |
| timeRestrictionForChannelMeasurements | | | *Not configured* | | | |
| timeRestrictionForInterferenceMeasurements | | | *Not configured* | | | |
| cqi-FormatIndicator | | | *Wideband* | | | |
| pmi-FormatIndicator | | | *Wideband* | | | |
| Sub-band Size | | RB | 8 | | | |
| csi-ReportingBand | | | 111111111 | | | |
| CSI-Report periodicity and offset | | slot | Not configured | | | |
| Aperiodic Report Slot Offset | | | 6 | | | |
| CSI request | | | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | | | |
| reportTriggerSize | | | 1 | | | |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains | | | |

| | | | |
|---|---|---|---|
| | | | pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | *typeI-SinglePanel* |
| | Codebook Mode | | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | *Not configured* |
| | CodebookSubsetRestriction | | 000001 |
| | RI Restriction | | N/A |
| Physical channel for CSI report | | | PUSCH |
| CQI/RI/PMI delay | | ms | 1.375 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | As specified in Table A.4-2, TBS.2-7 |
| Note 1: TT = 0dB | | | |

**Table 8.2.2.2.2.1_1.5-2 Test requirements**

| | Test 3 | Test 4 |
|---|---|---|
| $\alpha$ [%] | 2 | 2 |
| $\gamma$ | 1.05 - TT | 1.05 - TT |
| Note 1: TT = 0.01 | | |

# 8.2A Reporting of Channel Quality Indicator (CQI) for CA

## 8.2A.1 General

This clause includes the requirements for the reporting of channel quality indicator (CQI) with the UE configured for CA. The purpose is to verify that the CQI is correctly reported in accordance with the CQI definition given in TS 38.214 [12] for each CC with multiple cells configured for periodic reporting.

## 8.2A.2 1RX requirements

(Void)

## 8.2A.3 2RX requirements

### 8.2A.3.1 CQI reporting definition under AWGN conditions

#### 8.2A.3.1.0 Minimum requirement for periodic CQI reporting

For the CA CQI reporting test defined in Table 8.2A.3.1.0-4, the test requirements and the test parameters are defined as below.

For each CC, the test parameters are specified in Table 8.2A.3.1.0-1.

For CA with 2 DL CC, for the SNR configuration specified in Table 8.2A.3.1.0-2, and using the downlink physical channels specified in Annex C.5.1 on each CC, the difference between the wideband CQI indices of PCell and SCell reported shall be such that

$$\text{wideband CQI}_{\text{PCell}} - \text{wideband CQI}_{\text{SCell}} \geq 2$$

for more than 90% of the time.

For CA with 3 or more DL CC, for the SNR configuration specified in Table 8.2A.3.1.0-3, and using the downlink physical channels specified in Annex C.5.1 on each cell, the difference between the wideband CQI indices of PCell and SCell1 reported, and the difference between the wideband CQI indices of SCell1 and SCell2, 3… reported shall be such that

$$\text{wideband CQI}_{\text{PCell}} - \text{wideband CQI}_{\text{SCell1}} \geq 2$$

$$\text{wideband CQI}_{\text{SCell1}} - \text{wideband CQI}_{\text{SCell2, 3...}} \geq 2$$

for more than 90% of the time.

**Table 8.2A.3.1.0-1: CA CQI reporting test parameters for each CC**

| Parameter | | Unit | Value |
|---|---|---|---|
| Subcarrier spacing | | kHz | 120 |
| Duplex Mode | | | TDD |
| TDD Slot Configuration | | | FR2.120-2 Annex A.1.3 |
| Propagation channel | | | AWGN |
| Antenna configuration | | | 1×2 with static channel specified in Annex B.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 |
| | CDM Type | | FD-CDM2 |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 8 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 |
| | CSI-RS periodicity and offset | slot | 8/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Periodic |
| | Number of CSI-RS ports (*X*) | | 1 |
| | CDM Type | | No CDM |
| | Density (ρ) | | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | 6 |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | 13 |
| | NZP CSI-RS-timeConfig periodicity and offset | slot | 8/1 |
| CSI-IM configuration | CSI-IM resource Type | | Periodic |
| | CSI-IM RE pattern | | 1 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8, 13) |
| | CSI-IM timeConfig periodicity and offset | slot | 8/1 |
| ReportConfigType | | | Periodic |
| CQI-table | | | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured |
| cqi-FormatIndicator | | | Wideband |
| pmi-FormatIndicator | | | Wideband |
| Sub-band Size | | RB | 8 for 50MHz, 100MHz, 16 for 200MHz, 32 for 400MHz |
| csi-ReportingBand | | | 111111111 |
| CSI-Report periodicity and offset | | slot | 8/3 |
| aperiodicTriggeringOffset | | | Not configured |
| Physical channel for CSI report | | | PUCCH |
| CQI/RI/PMI delay | | ms | 8.375 |
| Maximum number of HARQ transmission | | | 1 |
| Measurement channel | | | Derived as per section 5.1.3.2 of TS 38.214 [12] |

**Table 8.2A.3.1.0-2: SNR configurations for 2 DL CA**

| Parameter | PCell | SCell |
|---|---|---|
| SNR (dB) | 10.0 | 4.0 |

**Table 8.2A.3.1.0-3: SNR configurations for 3 or more DL CA**

| Parameter | PCell | SCell1 | SCell2, 3… |
|---|---|---|---|
| SNR (dB) | 12.0 | 6.0 | 0.0 |

**Table 8.2A.3.1.0-4: List of CA CQI reporting test**

| Test number | CA duplex mode and SCS combination |
|---|---|
| 1 | TDD 120 kHz + TDD 120 kHz |
| Note 1: The applicability of requirements for different CA configurations and bandwidth combination sets is defined in 8.1.1.5.1. | |

The normative reference for this requirement is TS 38.101-4 [5], clause 8.2A.3.1.1.

### 8.2A.3.1.1 2Rx CQI reporting accuracy under AWGN conditions for CA (2DL CA)

#### 8.2A.3.1.1.1 Test Purpose

To verify that the CQI is correctly reported in accordance with the CQI definition given in TS 38.214 [12] for each CC with multiple cells configured for periodic reporting.

#### 8.2A.3.1.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 2DL CA.

#### 8.2A.3.1.1.3 Test description

##### 8.2A.3.1.1.3.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR2 operation, setup the LTE link according to Annex D

CA capability to be tested: test any one of the supported CA capabilities with largest aggregated CA bandwidth combination, as specified in 8.1.1.5.1.

CA configuration to be tested: For the selected CA capability, test any one of the supported CA configurations with largest aggregated CA bandwidth combination, as specified in 8.1.1.5.1.

1. Connection between SS, the AWGN noise source and the UE is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.

2. The parameter settings for the NR cell are set up according to Table 8.1.2-1, and Table 8.2A.3.1.0-1 to Table 6.2A.3.1.0-3 as appropriate.

3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].

4. Propagation conditions are set according to Annex B.1.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release*

*On, Test Mode On* for NSA according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 8.2A.3.1.1.3.3.

#### 8.2A.3.1.1.1.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found, mark the test as inconclusive.
2. Set the parameters of bandwidth, reference channel, propagation condition, antenna configuration and SNR configuration according to Table 8.2A.3.1.0-1 to Table 8.2A.3.1.0-4 as appropriate.
3. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.
4. The SS shall start gathering CQI reports, and will continue gathering CQI reports until 2000 wideband CQI reports have been gathered for each PCC and SCC. For each CSI report calculate the respective difference $CQI_{P-S}$ = wideband $CQI_{PCell}$ – wideband $CQI_{SCell}$.
5. If more than 1800 values of $CQI_{P-S}$ are $\geq 2$ pass the UE. Otherwise fail the UE.

#### 8.2A.3.1.1.1.3.3 Message contents

#### 8.2A.2.1.1.1.3.3_1 Message exceptions for SA

**Table 8.2A.2.1.1.1.3.3_1-1: CSI-RS-ResourceMapping**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-2 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row2 | 0000 0010 0000 | K0= 6, row2 | |
| } | | | |
| density CHOICE { | | | |
| one | NULL | | |
| } | | | |
| } | | | |

**Table 8.2A.2.1.1.1.3.3_1-2: NZP CSI-ResourcePeriodicityAndOffset**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-2a | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slot8 | 3 | | |
| } | | | |

**Table 8.2A.2.1.1.1.3.3_1-3: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-6 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-IM-Resource ::= SEQUENCE { | | | |
| periodicityAndOffset SEQUENCE { | | | |
| slot8 | 1 | | |
| } | | | |
| } | | | |

**Table 6.2A.2.1.1.1.3.3_1-4: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], Table 5.4.2.4-12 | | |
|---|---|---|
| **Information Element** | **Value/remark** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | |
| subbandSize | 8 | For 50MHz and 100MHz CHBW. |
| | 16 | For 200MHz CHBW. |
| | 32 | For 400MHz CHBW. |
| } | | |

#### 8.2A.2.1.1.1.3.3_2 Message exceptions for NSA

Same as specified in 8.2A.2.1.1.1.3.3_2.

#### 8.2A.3.1.1.1.3.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 8.2A.3.1.1.1.3.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

### 8.2A.3.1.2 2Rx CQI reporting accuracy under AWGN conditions for CA (3DL CA)

#### 8.2A.3.1.2.1 Test Purpose

Same with 8.2A.3.1.1.1.

#### 8.2A.3.1.2.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 3DL CA.

#### 8.2A.3.1.2.3 Test description

##### 8.2A.3.1.2.3.1 Initial conditions

Same with 8.2A.3.1.1.3.1.

##### 8.2A.3.1.1.2.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found, mark the test as inconclusive.
2. Set the parameters of bandwidth, reference channel, propagation condition, antenna configuration and SNR configuration according to Table 8.2A.3.1.0-1 to Table 8.2A.3.1.0-4 as appropriate.
3. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.
4. The SS shall start gathering CQI reports, and will continue gathering CQI reports until 2000 wideband CQI reports have been gathered for each PCC and SCCs. For each CSI report calculate the respective difference $CQI_{P\text{-}S1}$ = wideband $CQI_{PCell}$ – wideband $CQI_{SCell1}$ and the respective difference $CQI_{S1\text{-}S2}$ = wideband $CQI_{SCell1}$ – wideband $CQI_{SCell2}$.
5. If more than 1800 values of $CQI_{P\text{-}S1}$ are $\geq 2$ and more than 1800 values of $CQI_{S1\text{-}S2}$ are $\geq 2$, pass the UE. Otherwise fail the UE.

##### 8.2A.3.1.1.2.3.3 Message contents

Same with 8.2A.3.1.1.1.3.3.

#### 8.2A.3.1.1.2.3.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 8.2A.3.1.1.2.3.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

### 8.2A.3.1.3 2Rx CQI reporting accuracy under AWGN conditions for CA (4DL CA)

#### 8.2A.3.1.3.1 Test Purpose

Same with 8.2A.3.1.1.1.

#### 8.2A.3.1.3.2 Test applicability

This test applies to all types of NR UE release 15 and forward that supports 4DL CA.

#### 8.2A.3.1.3.3 Test description

#### 8.2A.3.1.3.3.1 Initial conditions

Same with 8.2A.3.1.1.3.1.

#### 8.2A.3.1.1.3.3.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found, mark the test as inconclusive.
2. Set the parameters of bandwidth, reference channel, propagation condition, antenna configuration and SNR configuration according to Table 8.2A.3.1.0-1 to Table 8.2A.3.1.0-4 as appropriate.
3. SS transmits PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC on both PCC and SCC. The SS sends downlink MAC padding bits on the DL RMC.
4. The SS shall start gathering CQI reports, and will continue gathering CQI reports until 2000 wideband CQI reports have been gathered for each PCC and SCCs. For each CSI report calculate the respective difference $CQI_{P\text{-}S1}$ = wideband $CQI_{PCell}$ – wideband $CQI_{SCell1,}$ the respective difference $CQI_{S1\text{-}S2}$ = wideband $CQI_{SCell1}$ – wideband $CQI_{SCell2}$ and the respective difference $CQI_{S1\text{-}S3}$ = wideband $CQI_{SCell1}$ – wideband $CQI_{SCell3}$.
5. If more than 1800 values of $CQI_{P\text{-}S1}$ are $\geq 2$, more than 1800 values of $CQI_{S1\text{-}S2}$ are $\geq 2$ and more than 1800 values of $CQI_{S1\text{-}S3}$ are $\geq 2$, pass the UE. Otherwise fail the UE.

#### 8.2A.3.1.1.3.3.3 Message contents

Same with 8.2A.3.1.1.1.3.3.

#### 8.2A.3.1.1.3.3.4 Test Requirements

The pass fail decision is as specified in the test procedure in clause 8.2A.3.1.1.3.3.2.

There are no parameters in the test setup or measurement process whose variation impacts the results so there are no applicable test tolerances for this test.

# 8.3 Reporting of Precoding Matrix Indicator (PMI)

## 8.3.0 General

The minimum performance requirements of PMI reporting are defined based on the precoding gain, expressed as the relative increase in throughput when the transmitter is configured according to the UE reports compared to the case when the transmitter is using random precoding, respectively. When the transmitter uses random precoding, for each

PDSCH allocation a precoder is randomly generated and applied to the PDSCH. A fixed transport format (FRC) is configured for all requirements.

The requirements for transmission scheme 1 with 2TX and higher layer parameter *codebookType* set to 'typeI-SinglePanel' are specified in terms of the ratio

$$\gamma = \frac{t_{ue}}{t_{rnd}}$$

In the definition of $\gamma$, for 2TX PMI requirements, $t_{ue}$ is 90 % of the maximum throughput obtained at $SNR_{ue}$ using the precoders configured according to the UE reports, and $t_{rnd}$ is the throughput measured at $SNR_{ue}$ with random precoding.

## 8.3.1 1RX requirements (Void)

## 8.3.2 2RX requirements

### 8.3.2.1 FDD

TBD

### 8.3.2.2 TDD

#### 8.3.2.2.1 2Rx TDD FR2 Single PMI with 2TX TypeI-SinglePanel codebook for both SA and NSA

##### 8.3.2.2.1.1 Test purpose

The purpose of this test is to test the accuracy of the Precoding Matrix Indicator (PMI) reporting such that the system throughput is maximized based on the precoders configured according to the UE reports.

##### 8.3.2.2.1.2 Test applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

##### 8.3.2.2.1.3 Minimum conformance requirements

For the parameters specified in Table 8.3.2.2.1.3-1, and using the downlink physical channels specified in Annex C.5.1, the minimum requirements are specified in Table 8.3.2.2.1.3-2.

**Table 8.3.2.2.1.3-1: Test parameters (single layer)**

| Parameter | | Unit | Test 1 | Test 2 |
|---|---|---|---|---|
| Bandwidth | | MHz | 100 | 100 |
| Subcarrier spacing | | kHz | 120 | 120 |
| TDD DL-UL configuration | | | FR2.120-2 as specified in Annex A.1.3 | FR2.120-1 as specified in Annex A.1.3 |
| Propagation channel | | | TDLA30-35 | TDLA30-35 |
| Antenna configuration | | | 2 x 2 ULA Low | 2 x 2 ULA Low |
| Beamforming Model | | | As specified in Annex B.4.1 | As specified in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 | 4 |
| | CDM Type | | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 4, (8,-) | Row 4, (8,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) |
| | CSI-RS interval and offset | slot | 8/1 | 5/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 2 | 2 |
| | CDM Type | | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3, (6,-) | Row 3, (6,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) |
| | CSI-RS interval and offset | slot | Not configured | Not configured |
| | aperiodicTriggering Offset | | 0 | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic | Aperiodic |
| | CSI-IM RE pattern | | Pattern 1 | Pattern 1 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8,13) | (8,13) |
| | CSI-IM timeConfig interval and offset | slot | Not configured | Not configured |
| ReportConfigType | | | Aperiodic | Aperiodic |
| CQI-table | | | Table 1 | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | Not configured | Not configured |
| timeRestrictionForInterferenceMeasurements | | | Not configured | Not configured |
| cqi-FormatIndicator | | | Wideband | Wideband |

| | | | | |
|---|---|---|---|---|
| pmi-FormatIndicator | | | Wideband | Wideband |
| Sub-band Size | | RB | 8 | 8 |
| csi-ReportingBand | | | 111111111 | 111111111 |
| CSI-Report interval and offset | | slot | Not configured | Not configured |
| Aperiodic Report Slot Offset | | | 6 | 8 |
| CSI request | | | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | 1 in slots i, where mod(i, 5) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM | One State with one Associated Report Configuration Associated Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | typeI-SinglePanel |
| | Codebook Mode | | 1 | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | N/A | N/A |
| | CodebookSubsetRestriction | | 001111 | 001111 |
| | RI Restriction | | N/A | N/A |
| Physical channel for CSI report | | | PUSCH | PUSCH |
| CQI/RI/PMI delay | | ms | 1.375 | 1.75 |
| Maximum number of HARQ transmission | | | 4 | 4 |
| Measurement channel | | | R.PDSCH.5-8.1 TDD | R.PDSCH.5-7.1 TDD |

Note 1: For random precoder selection, the precoder shall be updated in each slot (0.125 ms granularity).
Note 2: If the UE reports in an available uplink reporting instance at slot#n based on PMI estimation at a downlink slot not later than slot#(n-4)], this reported PMI cannot be applied at the gNB downlink before slot#(n+4)].
Note 3: Randomization of the principle beam direction shall be used as specified in Annex B.2.3.2.3.

**Table 8.3.2.2.1.3-2: Minimum requirement**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\gamma$ | 1.05 | 1.05 |

### 8.3.2.2.1.4 Test description

#### 8.3.2.2.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE antenna is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 8.1.2-1 and Table 8.3.2.2.1.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 8.3.2.2.1.4.3.

#### 8.3.2.2.1.4.2 Test procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0.If no direction found, mark the test as inconclusive.
2. Set the parameters of bandwidth, the propagation condition, antenna configuration and measurement channel according to Table 8.3.2.2.1.3-1 as appropriate.
3. The SS shall transmit PDSCH via PDCCH DCI format [1_1] for C_RNTI to transmit the DL RMC with precoding matrix according to PMI report from the UE. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CQI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Establish $t_{ue}$ and $SNR_{ue}$ according to Annex G.3.2.
4. Set SNR to $SNR_{ue}$. The SS shall transmit PDSCH with randomly selected precoding matrix from codebook (Table 5.2.2.2.1-5 in TS 38.214 [12]) every slot regardless of PMI reports from the UE. Note that each precoding matrix shall be selected in equal probabilities. The SS sends downlink MAC padding bits on the DL RMC. SS schedules the UL transmission to carry the PUSCH CSI feedback via PDCCH DCI format [0_1] with aperiodic CSI request triggered. Measure $t_{rnd}$ according to Annex G.3.3.
5. Calculate $\gamma = \frac{t_{ue}}{t_{rnd}}$. If the ratio ≥ γ which is specified in table 8.3.2.2.1.5-1, then the test is pass. Otherwise, the test is fail.
6. Repeat steps from 3 to 5 for each subtest in Table 8.3.2.2.1.3-1 as appropriate.

#### 8.3.2.2.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2 with the following exceptions:

8.3.2.2.1.4.3_1 Message exceptions for SA

**Table 8.3.2.2.1.4.3_1-1: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | CSI-RS for CSI Acquisition |
| | Periodic | | CSI-RS for Tracking or Beam Refinement |
| } | | | |

**Table 8.3.2.2.1.4.3_1-2: CSI-RS-ResourceMapping for NZP-CSI-RS for Tracking**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0001 | | |
| } | | | |
| nrofPorts | p1 | 1 for CSI-RS resource 1,2,3,4 | |
| firstOFDMSymbolInTimeDomain | 4 | $l_0$ = 4 for CSI-RS resource 1 and 3 | |
| | 8 | $l_0$ = 8 for CSI-RS resource 2 and 4 | |
| } | | | |

**Table 8.3.2.2.1.4.3_1-3: CSI-RS-ResourceMapping for NZP-CSI-RS for Acquisition**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row3 | 001000 | | |
| } | | | |
| nrofPorts | p2 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 8.3.2.2.1.4.3_1-4: CSI-RS-ResourceMapping for NZP-CSI-RS for Beam Refinement**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row1 | 0001 | | |
| } | | | |
| nrofPorts | p1 | | |
| firstOFDMSymbolInTimeDomain | 8 | $l_0$ = 8 for CSI-RS resource 1 | |
| | 9 | $l_0$ = 9 for CSI-RS resource 2 | |
| } | | | |

**Table 8.3.2.2.1.4.3_1-5: CSI-RS-ResourceMapping for ZP-CSI-RS**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-45 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-RS-ResourceMapping ::= SEQUENCE { | | | |
| frequencyDomainAllocation CHOICE { | | | |
| row4 | 100 | | |
| } | | | |
| nrofPorts | p4 | | |
| firstOFDMSymbolInTimeDomain | 13 | | |
| } | | | |

**Table 8.3.2.2.1.4.3_1-6: ZP CSI-ResourcePeriodicityAndOffset**

| Derivation Path: Table 4.6.3-43 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourcePeriodicityAndOffset ::= CHOICE { | | | |
| slots8 | 1 | | Test 1 |
| slots5 | 1 | | Test 2 |
| } | | | |

**Table 8.3.2.2.1.4.3_1-7: CSI-IM-Resource**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-34 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| csi-IM-ResourceElementPattern CHOICE { | | | |
| pattern1 SEQUENCE { | | | |
| subcarrierLocation-p1 | s8 | | |
| symbolLocation-p1 | 13 | | |
| } | | | |
| } | | | |

**Table 8.3.2.2.1.4.3_1-8: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CodebookConfig ::= SEQUENCE { | | | |
| codebookType CHOICE { | | | |
| type1 SEQUENCE { | | | |
| subType CHOICE { | | | |
| typeI-SinglePanel SEQUENCE { | | | |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-codebookSubsetRestriction | 001111 | | |
| } | | | |
| } | | | |
| TypeI-SinglePanel-ri-Restriction | 11111111 | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 8.3.2.2.1.4.3_1-9: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ReportConfig ::= SEQUENCE { | | | |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 6 | | Test 1 |
| | 8 | | Test 2 |
| } | | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands9 | 111111111 | | |
| } | | | |
| } | | | |
| subbandSize | value2 | | |
| } | | | |

8.3.2.2.1.4.3_2 Message exceptions for NSA

Same as 8.3.2.2.1.4.3_1.

8.3.2.2.1.5 Test requirement

**Table 8.3.2.2.1.5-1: Test requirement (TDD)**

| Parameter | Test 1 | Test 2 |
|---|---|---|
| $\gamma$ | 1.04 | 1.04 |

# 8.4 Reporting of Rank Indicator (RI)

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission.

## 8.4.1 1RX requirements

(Void)

## 8.4.2 2RX requirements

### 8.4.2.1 FDD

(Void)

### 8.4.2.2 TDD

#### 8.4.2.2.1 2Rx TDD FR2 RI reporting for both SA and NSA

-8.4.2.2.1.1 Test Purpose

The purpose of this test is to verify that the reported rank indicator accurately represents the channel rank. The accuracy of RI reporting is determined by the relative increase of the throughput obtained when transmitting based on the reported rank compared to the case for which a fixed rank is used for transmission.

#### 8.4.2.2.1.2 Test Applicability

This test applies to all types of NR UE release 15 and forward.

This test also applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 8.4.2.2.1.3 Minimum requirement

The minimum performance requirement in Table 8.4.2.2.1.3-2 is defined as

a) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 1 shall be $\geq \gamma_1$;

b) The ratio of the throughput obtained when transmitting based on UE reported RI and that obtained when transmitting with fixed rank 2 shall be $\geq \gamma_2$;

For the parameters specified in Table 8.4.2.2.1.3-1, and using the downlink physical channels specified in Annex C.2.2, the minimum requirements are specified in Table 8.4.2.2.1.3-2.

**Table 8.4.2.2.1.3-1: RI Test (TDD)**

| Parameter | | Unit | Test 1 | Test 2 | Test 3 |
|---|---|---|---|---|---|
| Bandwidth | | MHz | 100 | 100 | 100 |
| Subcarrier spacing | | kHz | 120 | 120 | 120 |
| Duplex Mode | | | TDD | TDD | TDD |
| TDD Slot Configuration | | | FR2.120-2 | FR2.120-2 | FR2.120-2 |
| SNR | | dB | 0 | 16 | 16 |
| Propagation channel | | | TDLA30-35 | TDLA30-35 | TDLA30-35 |
| Antenna configuration | | | ULA Low 2x2 | ULA Low 2x2 | XP High 2x2 |
| Beamforming Model | | | As defined in Annex B.4.1 | As defined in Annex B.4.1 | As defined in Annex B.4.1 |
| ZP CSI-RS configuration | CSI-RS resource Type | | Periodic | Periodic | Periodic |
| | Number of CSI-RS ports (*X*) | | 4 | 4 | 4 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 4, (8,-) | Row 4, (8,-) | Row 4, (8,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) | (13,-) |
| | CSI-RS interval and offset | slot | 8/1 | 8/1 | 8/1 |
| NZP CSI-RS for CSI acquisition | CSI-RS resource Type | | Aperiodic | Aperiodic | Aperiodic |
| | Number of CSI-RS ports (*X*) | | 2 | 2 | 2 |
| | CDM Type | | FD-CDM2 | FD-CDM2 | FD-CDM2 |
| | Density (ρ) | | 1 | 1 | 1 |
| | First subcarrier index in the PRB used for CSI-RS ($k_0$, $k_1$ ) | | Row 3 (6,-) | Row 3 (6,-) | Row 3 (6,-) |
| | First OFDM symbol in the PRB used for CSI-RS ($l_0$, $l_1$) | | (13,-) | (13,-) | (13,-) |
| | NZP CSI-RS-timeConfig interval and offset | slot | Not configured | Not configured | Not configured |
| | aperiodicTriggeringOffset | | 0 | 0 | 0 |
| CSI-IM configuration | CSI-IM resource Type | | Aperiodic | Aperiodic | Aperiodic |
| | CSI-IM RE pattern | | Pattern 1 | Pattern 1 | Pattern 1 |
| | CSI-IM Resource Mapping ($k_{CSI-IM}$,$l_{CSI-IM}$) | | (8,13) | (8,13) | (8,13) |
| | CSI-IM timeConfig interval and offset | slot | Not configured | Not configured | Not configured |
| ReportConfigType | | | Aperiodic | Aperiodic | Aperiodic |
| CQI-table | | | Table 1 | Table 1 | Table 1 |
| reportQuantity | | | cri-RI-PMI-CQI | cri-RI-PMI-CQI | cri-RI-PMI-CQI |
| timeRestrictionForChannelMeasurements | | | not configured | not configured | not configured |
| timeRestrictionForInterferenceMeasurements | | | not configured | not configured | not configured |
| cqi-FormatIndicator | | | Wideband | Wideband | Wideband |
| pmi-FormatIndicator | | | Wideband | Wideband | Wideband |
| Sub-band Size | | RB | 8 | 8 | 8 |
| csi-ReportingBand | | | 111111111 | 111111111] | 111111111 |
| CSI-Report interval and offset | | slot | Not configured | Not configured | Not configured |
| Aperiodic Report Slot Offset | | | 6 | 6 | 6 |
| CSI request | | | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 | 1 in slots i, where mod(i, 8) = 1, otherwise it is equal to 0 |
| reportTriggerSize | | | 1 | 1 | 1 |
| CSI-AperiodicTriggerStateList | | | One State with one Associated Report Configuration Associated Report | One State with one Associated Report Configuration Associated | One State with one Associated Report Configuration Associated |

| | | | | | |
|---|---|---|---|---|---|
| | | | Configuration contains pointers to NZP CSI-RS and CSI-IM | Report Configuration contains pointers to NZP CSI-RS and CSI-IM | Report Configuration contains pointers to NZP CSI-RS and CSI-IM |
| Codebook configuration | Codebook Type | | typeI-SinglePanel | typeI-SinglePanel | typeI-SinglePanel |
| | Codebook Mode | | 1 | 1 | 1 |
| | (CodebookConfig-N1,CodebookConfig-N2) | | N/A | N/A | N/A |
| | CodebookSubsetRestriction | | 010000 for fixed rank 2, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank | 000011 for fixed rank 1, 010011 for following rank |
| | RI Restriction | | N/A | N/A | N/A |
| Physical channel for CSI report | | | PUSCH | PUSCH | PUSCH |
| CQI/RI/PMI delay | | ms | 1.375 | 1.375 | 1.375 |
| Maximum number of HARQ transmission | | | 1 | 1 | 1 |
| RI Configuration | | | Fixed RI = 2 and follow RI | Fixed RI = 1 and follow RI | Fixed RI = 1 and follow RI |
| Note 1: Measurements channels are specified in Table A.4-1. TBS.1-1 is used for Rank 1 case. TBS.1-2 is used for Rank 2 case. | | | | | |

**Table 8.4.2.2.1.3-2: Minimum requirement (TDD)**

| | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.05 | 1.05 |
| $\gamma_2$ | 1.0 | N/A | N/A |

The normative reference for this requirement is TS 38.101-4 [5] clause 8.4.2.2.

### 8.4.2.2.1.4 Test Description

#### 8.4.2.2.1.4.1 Initial Conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.2-1 of TS 38.521-2 [8].

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.2.

Only sub tests shall be tested which are testable according to Table 7.1.1_1-2.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE antenna is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 8.1.2-1 and Table 8.4.2.2.1.3-1 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.

5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity NR for SA with *Connected without release On, Test Mode On* or EN-DC, DC bearer *MCG* and *SCG, Connected without release On, Test Mode On for NSA* according to TS 38.508-1 [6] clause 4.5. Message content are defined in clause 8.4.2.2.1.4.3.

#### 8.4.2.2.1.4.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found mark the test as inconclusive.
2. Set the parameters of bandwidth, reference channel, the propagation condition, antenna configuration, antenna correlation, Codebook configuration, Beamforming Model, RI configuration and SNR according to Table 8.4.2.2.1.3-1 as appropriate. Measure the $t_{fix}$ according to Annex G.3.3.
3. The SS shall send PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and fixed RI as defined in Table 8.4.2.2.1.3-1. The SS sends downlink MAC padding bits on the DL RMC.
4. Propagation conditions are set according to Annex B.2.
5. The SS shall transmit an RRC Connection Reconfiguration message to set codebookSubsetRestriction as for UE reported RI according to Table 8.4.2.2.1.3-1.
6. The UE shall transmit RRC Connection Reconfiguration Complete message.
7. Propagation conditions are set according to Table 8.4.2.2.1.3-1.
8. The SS shall send PDSCH via PDCCH DCI format 1_1 for C_RNTI to transmit the DL RMC according to the UE reported CQI (wideband CQI), PMI and RI. The SS sends downlink MAC padding bits on the DL RMC. Measure $t_{reported}$ according to Annex G.3.3.

   If the ratio ($t_{reported}$ / $t_{fix}$) satisfies the requirement in Table 8.4.2.2.1.5-1, then pass the UE for this test and go to step 9. Otherwise, declare a FAIL verdict.
9. If all tests have not been done, then repeat the same procedure (steps 1 to 8) with test conditions according to the Table 8.4.2.2.1.3-2 for the other Tests as appropriate. Otherwise, declare a PASS verdict.

#### 8.4.2.2.1.4.3 Message Contents

Message contents are according to TS 38.508-1 [6] clauses 4.6.1 and 5.4.2 with the following exceptions:

##### 8.4.2.2.1.4.3_1 Message exceptions for SA

**Table 8.4.2.2.1.4.3_1-1: SchedulingRequestResourceConfig**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-157 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SchedulingRequestResourceConfig ::= SEQUENCE { | | | |
| periodicityAndOffset CHOICE { | | | |
| sl80 | 7 | | |
| } | | | |
| } | | | |

**Table 8.4.2.2.1.4.3_1-2: CSI-ResourceConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-41 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CSI-ResourceConfig ::= SEQUENCE { | | | |
| resourceType | Aperiodic | | CSI-RS for CSI Acquisition, CSI-IM-Resource |
| | Periodic | | CSI-RS for Tracking or Beam Refinement |
| } | | | |

**Table 8.4.2.2.1.4.3_1-3: *CodebookConfig***

| Derivation Path: TS 38.508-1 [6], clause 4.6.2, Table 4.6.3-25 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| nrOfAntennaPorts CHOICE { | | | |
| Two SEQUENCE { | | | |
| twoTX-CodebookSubsetRestriction | 010000 | Fixed rank 2 | Test 1 |
| | 000011 | Fixed rank 1 | Test 2, Test 3 |
| | 010011 | Following rank | Test 1, Test 2, Test 3 |
| } | | | |
| } | | | |
| typeI-SinglePanel-ri-Restriction | 11111111 | Non restriction | |

**Table 8.4.2.2.1.4.3_1-4: CSI-ReportConfig**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3, Table 4.6.3-39 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| reportConfigType CHOICE { | | | |
| aperiodic SEQUENCE { | | | |
| reportSlotOffsetList | 6 | | |
| } | | | |
| reportFreqConfiguration SEQUENCE { | | | |
| csi-ReportingBand CHOICE { | | | |
| subbands9 | 111111111 | | |
| } | | | |
| } | | | |
| subbandSize | value2 | | |
| } | | | |

8.4.2.2.1.4.3_2 Message exceptions for NSA

Same as 8.4.2.2.1.4.3_1.

8.4.2.2.1.5 Test Requirements

**Table 8.4.2.2.1.5-1: Test Requirement (TDD)**

| | **Test 1** | **Test 2** | **Test 3** |
|---|---|---|---|
| $\gamma_1$ | N/A | 1.04 | 1.04 |
| $\gamma_2$ | 0.99 | N/A | N/A |

# 9 Demodulation performance requirements for interworking

## 9.1 General

This clause covers the UE demodulation performance requirements for EN-DC, NE-DC, inter-band NR-DC between FR1 and FR2, and inter-band NR CA between FR1 and FR2.

For conformance testing involving FR2 test cases in this specification, the UE under test shall be pre-configured with UL Tx diversity schemes disabled to account for single polarization System Simulator (SS) in the test environment. The UE under test may transmit with dual polarization.

### 9.1.1 Applicability of requirements

The following applicability rules are specified for demodulation performance requirements for interworking:

- For UEs supporting NR/5GC, EN-DC and NE-DC,
  - The performance requirements specified in Clause 5 will be verified only for NR/5GC except for the sustained downlink data rate test specified in Clause 5.5 and 5.5A.
  - The performance requirements specified in Clause 7 will be verified only for NR/5GC except for the sustained downlink data rate test specified in Clause 7.5.
  - The sustained downlink data rate tests specified in Clauses 5.5, 5.5A and 7.5 for NR/5GC and in Clause 9.4B for EN-DC and NE-DC are verified separately.
- The FR1 EN-DC test cases with the NR TDD DL-UL configurations which are not aligned with LTE's can be tested on the corresponding EN-DC band combinations where UE supports simultaneous transmission and reception.
- For UEs supporting NR FR1 CA and/or NR CA including FR1 and FR2, the requirements applicability is specified in Table 9.1.1-1.

**Table 9.1.1-1: Requirements applicability for UEs supporting NR FR2 CA and NR CA including FR1 and FR2**

| Supported scenarios | Requirements |
|---|---|
| NR FR2 CA | Clause 7.5A |
| NR CA including FR1 and FR2 | Clause 9.4A.1 |
| Both NR FR2 CA and NR CA including FR1 and FR2 | Clause 7.5A |

- For UEs supporting EN-DC including FR2 and/or EN-DC including FR1 and FR2, the requirements applicability is specified in Table 9.1.1-2.

**Table 9.1.1-2: Requirements applicability for UEs supporting EN-DC including FR2 and EN-DC including FR1 and FR2**

| Supported scenarios | SDR requirements | PDSCH requirements | PDCCH requirements |
|---|---|---|---|
| EN-DC including FR2 | Clause 9.4B.1.2 | Clause 9.2B.1.2 | Clause 9.3B.1.2 |
| EN-DC including FR1 and FR2 | Clause 9.4B.1.3 | Clause 9.2B.1.3 | Clause 9.3B.1.3 |
| Both EN-DC including FR2 and EN-DC including FR1 and FR2 | Clause 9.4B.1.2 | Clause 9.2B.1.2 | Clause 9.3B.1.2 |

- For UEs supporting NR-DC including FR1 and FR2, if the FR2 requirements in Clause 7.2 and Clause 7.3 are tested, the test coverage can be considered fulfilled without executing requirements in Clause 9.2B.2 and Clause 9.3B.2.

- For UEs supporting NR-DC between FR1 and FR2, if requirements in Clause 9.4A.1 are tested under same or higher data rate as in Clause 9.4B.2, the test coverage can be considered fulfilled without executing the requirements in Clause 9.4B.2.

- For UEs supporting NE-DC and EN-DC, the test coverage of demodulation performance requirements can be considered fulfilled, if the demodulation requirements in Clause 5 and Clause 9.4B.1 are executed for UE under test in the standalone mode.

- For UEs supporting NE-DC and not supporting EN-DC, the test coverage of demodulation performance requirements can be considered fulfilled, if the demodulation requirements in Clause 5 and Clause 9.4B.3 are executed for UE under test.

- For UEs supporting NGEN-DC, the test coverage of demodulation performance requirements can be considered fulfilled, if the demodulation requirements in Clause 5 and Clause 9.4B.1 are executed for UE under test.

#### 9.1.1.1 Applicability of requirements for optional UE features

The applicability rule defined in Clause 5.1.1.3 shall be applied for performance requirements in Clauses 9.2B.1.1 and 9.4B.1.1.

The applicability rule defined in Clause 7.1.1.3 shall be applied for performance requirements in Clauses 9.2B.1.2, 9.4A.1, 9.4B.1.2 and 9.4B.1.3.

#### 9.1.1.2 Applicability of requirements for mandatory UE features with capability signalling

The applicability rule defined in Clause 5.1.1.4 shall be applied for performance requirements in Clauses 9.2B.1.1 and 9.4B.1.1.

The applicability rule defined in Clause 7.1.1.4 shall be applied for performance requirements in Clauses 9.2B.1.2, 9.4A.1, 9.4B.1.2 and 9.4B.1.3.

### 9.1.2 E-UTRA Cell setup

This subclause provides the parameters for E-UTRA cell during the demodulation performance test for EN-DC unless otherwise stated. For EN-DC with multiple E-UTRA carriers or bands, randomly selected one carrier or band can be used as E-UTRA Pcell for the connection setup unless otherwise stated.

#### 9.1.2.1 FDD

The parameters specified in Table 9.1.2.1-1 and Table 9.1.2.1-2 are used to setup E-UTRA cell. One of test setup in Table 9.1.2.1-2 will be selected for the E-UTRA Cell depending on the maximum bandwidth of an E-UTRA carrier for all the EN-DC band combinations supported by the UE.

The measurement channels in Table 9.1.2.1-2 and OCNG pattern OP.1 FDD are specified in TS 36.521-1 [16]. The physical channel setup with downlink power allocation is according to Annex C.3.2 of TS 36.521-1 [16].

**Table 9.1.2.1-1: Common Test Parameters (FDD)**

| Parameter | Unit | Value |
|---|---|---|
| Cyclic prefix | | Normal |
| Physical Cell ID | | 0 |
| Number of PDCCH symbols | symbols | 1 |
| PHICH Ng (Note 1) | | 1 |
| PHICH duration | | Normal |
| Number of HARQ processes per component carrier | Processes | 8 |
| Maximum number of HARQ transmission | | 4 |
| Redundancy version coding sequence | | {0,0,1,2} for 64QAM |
| Propagation condition | | Static propagation condition<br>No external noise sources are applied |
| Transmission mode | | 1 |
| Transmission time difference between E-UTRA cell and NR cell(s) | µs | 0 |
| Antenna configuration | | All NR cells are in FR1: 1x2<br>Any NR cell is in FR2: 1 Tx[Note 1] |
| Codebook subset restriction | | 10 |
| Symbols for all unused REs | | OCNG in Annex A.5 |
| Note 1: As the link can be provided over the air, the UE Rx antenna configuration is not relevant for the test configuration and has no impact on the test implementation. | | |

**Table 9.1.2.1-2: Specific Test Parameters (FDD [64QAM])**

| Test setup | Bandwidth (MHz) | Downlink power allocation (dB) | | |
|---|---|---|---|---|
| | | $\rho_A$ | $\rho_B$ | σ |
| 1 | 5 | 0 | 0 | 0 |
| 2 | 10 | 0 | 0 | 0 |
| 3 | 15 | 0 | 0 | 0 |
| 4 | 20 | 0 | 0 | 0 |

## 9.1.2.2 TDD

The parameters specified in Table 9.1.2.2-1 and Table 9.1.2.2-2 are used to setup an E-UTRA cell. One of test setup in Table 9.1.2.2-2 will be selected for the E-UTRA Cell depending on the maximum bandwidth of an E-UTRA carrier for all the EN-DC band combinations supported by the UE.

The measurement channels in Table 9.1.2.2-2 and OCNG pattern OP.1 TDD are specified in TS 36.521-1 [16]. The physical channel setup with downlink power allocation is according to Annex C.3.2 of TS 36.521-1 [16].

**Table 9.1.2.2-1: Common Test Parameters (TDD)**

| Parameter | Unit | Value |
|---|---|---|
| UL DL configuration | | 2 (Note1) |
| Special subframe configuration | | 7 |
| Number of PDCCH symbols | symbols | 1 |
| PHICH Ng (Note 3) | | 1 |
| PHICH duration | | Normal |
| Cyclic prefix | | Normal |
| Cell ID | | 0 |
| Maximum number of HARQ transmission | | 4 |
| Redundancy version coding sequence | | {0,0,1,2} for 64QAM |
| Propagation condition | | Static propagation condition<br>No external noise sources are applied |
| Transmission mode | | 1 |
| Transmission time difference between E-UTRA cell and NR cell(s) | µs | 0 |
| Antenna configuration | | All NR cells are in FR1: 1x2<br>Any NR cell is in FR2: 1 Tx Note 2 |
| Codebook subset restriction | | 10 |
| Symbols for all unused REs | | OCNG in Annex A.5 |

NOTE 1: The start of transmission of LTE frame is delayed by 2 LTE subframes with respect to the start of transmission of NR frame when TDD-TDD EN-DC configuration is configured during the test.
NOTE 2: As the link can be provided over the air, the UE Rx antenna configuration is not relevant for the test configuration and has no impact on the test implementation.

**Table 9.1.2.2-2: Specific Test Parameters (FDD 64QAM)**

| Test setup | Bandwidth (MHz) | Downlink power allocation (dB) | | |
|---|---|---|---|---|
| | | $\rho_A$ | $\rho_B$ | σ |
| 1 | 10 | 0 | 0 | 0 |
| 2 | 15 | 0 | 0 | 0 |
| 3 | 20 | 0 | 0 | 0 |

# 9.2 Void

# 9.2A PDSCH Demodulation for CA

## 9.2A.1 NR CA between FR1 and FR2

FFS

# 9.2B PDSCH Demodulation for DC

## 9.2B.1 EN-DC

### 9.2B.1.1 EN-DC within FR1

The NR PDSCH demodulation performance requirements and test case details for this test case are specified in Section 5.2.

During the test, only the PDSCH performance on the NR cell(s) shall be verified

### 9.2B.1.2 EN-DC including FR2 NR carrier only

The NR PDSCH demodulation performance requirements and test case details for this test case are specified in Section 7.2.

During the test, only the PDSCH performance on the NR cell(s) on FR2 carriers shall be verified.

### 9.2B.1.3 EN-DC including FR1 and FR2 NR carriers

The demodulation performance requirements are verified according to Section 9.2B.1.1 for EN-DC with FR1 NR carrier only and Section 9.2B.1.2 for EN-DC with FR2 NR carrier only.

During the test for EN-DC with FR2 NR carriers, only demodulation performance requirements on the FR2 carriers are verified.

No demodulation requirement for FR1 NR or LTE carriers is specified for EN-DC including FR2 carrier(s).

## 9.2B.2 NR DC between FR1 and FR2

FFS

# 9.3 Void

# 9.3A PDCCH Demodulation for CA

## 9.3A.1 NR CA between FR1 and FR2

FFS

# 9.3B PDCCH Demodulation for DC

## 9.3B.1 EN-DC

### 9.3B.1.1 EN-DC within FR1

The NR PDCCH demodulation performance requirements and test case details for this test case are specified in Section 5.3.

During the test, only the PDCCH performance on the single NR cell shall be verified.

### 9.3B.1.2 EN-DC including FR2 NR carrier only

The NR PDCCH demodulation performance requirements and test case details for this test case are specified in Section 7.3.

During the test, only the PDCCH performance on the single NR cell shall be verified.

### 9.3B.1.3 EN-DC including FR1 and FR2 NR carriers

The demodulation performance requirements are verified according to Section 9.3B.1.1 for EN-DC with FR1 NR carrier only and Section 9.3B.1.2 for EN-DC with FR2 NR carrier only.

During the test for EN-DC with FR2 NR carriers, only demodulation performance requirements on the FR2 carriers are verified.

No demodulation requirement for FR1 NR or LTE carriers is specified for EN-DC including FR2 carrier(s).

## 9.3B.2 NR DC between FR1 and FR2

FFS

# 9.4 Void

# 9.4A SDR test for CA

FFS

# 9.4B SDR test for DC

## 9.4B.1 EN-DC

### 9.4B.1.1 Sustained downlink data rate performance for EN-DC within FR1

#### 9.4B.1.1.1 Test Purpose

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum data rate indicated by UE capabilities. The sustained downlink data rate shall be verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the RF conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement

#### 9.4B.1.1.2 Test Applicability

This test applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 9.4B.1.1.3 Minimum conformance requirements

During the test, the PDSCH performance on both the NR cell(s) and LTE cell(s) shall be verified.

The TB success rate shall be higher than 85% when NR PDSCH is scheduled with MCS defined for the selected EN-DC bandwidth combination and with the downlink physical channel setup according to Annex C.3.1 and when E-UTRA PDSCH is scheduled with FRC defined for the selected EN-DC bandwidth combination and with the downlink physical channel setup according to Annex C.3.2 from TS 36.101 [X].

The TB success rate is defined as $100\%*N_{DL_correct_rx} / (N_{DL_newtx} + N_{DL_retx})$, where $N_{DL_newtx}$ is the number of newly transmitted DL transport blocks, $N_{DL_retx}$ is the number of retransmitted DL transport blocks, and $N_{DL_correct_rx}$ is the number of correctly received DL transport blocks.

The common test parameters for NR cell are specified in Table 9.4B.1.1.3-1. The parameters specified in Table 9.4B.1.1.3-2 are applicable for tests on FDD NR cell and parameters specified in Table 9.4B.1.1.3-3 are applicable for tests on TDD NR cell.

Unless otherwise stated, no user data is scheduled on slot #0, 10 and 11 within 20 ms for SCS 15 kHz for NR cell.

Unless otherwise stated, no user data is scheduled on slot #0, 20 and 21 within 20 ms for SCS 30 kHz for NR cell.

**Table 9.4B.1.1.3-1: Common test parameters for FDD or TDD NR band**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| EPRE ratio of PTRS to PDSCH | | dB | N/A |
| Channel bandwidth | | MHz | Channel bandwidth from selected CA bandwidth combination |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| | First DMRS position for Type A PDSCH mapping | | 2 |
| Cross carrier scheduling | | | Not configured |
| Active DL BWP index | | | 1 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 2) | RBs | 0 |
| | Subcarrier spacing | kHz | 15 or 30 |
| DL BWP configuration #1 | RB offset | RBs | 0 |
| | Number of contiguous PRB | | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing |
| | Subcarrier spacing | kHz | 15 or 30 |
| | Cyclic prefix | | Normal |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | Symbols #0 |
| | Number of PRBs in CORESET | | Table 9.4B.1.1.3-4 |
| | Number of PDCCH candidates and aggregation levels | | 2/AL2 for 15 kHz / 5 MHz and 30 kHz / 15 MHz<br>2/AL4 for 15 kHz / 10 MHz, 30 kHz / 10 MHz and 30 kHz / 20 MHz<br>2/AL8 for other greater combinations |
| | CCE-to-REG mapping type | | Non-interleaved |
| | DCI format | | 1_1 |
| | TCI State | | TCI state #1 |
| | PDCCH & PDCCH DMRS Precoding configuration | | For 2Tx:<br>Single Panel Type I, Random precoder chosen from precoder index 0 and 2, selection updated per slot<br><br>For 4Tx:<br>Single Panel Type I, Random precoder chosen from precoders with i_1,1 in {1,2,3,5,6,7} and i_2 in {0,2}, selection updated per slot |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | WB |
| | Resource allocation type | | Type 0 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Length | | 1 |
| | Antenna ports indexes | | {1000} for 1 Layer CCs<br>{1000, 1001} for 2 Layers CCs<br>{1000 – 1003} for 4 Layers CCs |
| | Number of PDSCH DMRS CDM group(s) without data | | 1 for 1 layer and 2 layers CCs<br>2 for 4 Layers CCs |
| PTRS configuration | | | PTRS is not configured |

| | | | | |
|---|---|---|---|---|
| CSI-RS for tracking | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0$ = 3 for CSI-RS resource 1,2,3,4 |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0$ = 6 for CSI-RS resource 1 and 3<br>$l_0$ = 10 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2,3,4 |
| | Density (ρ) | | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | | Slots | 15 kHz SCS: 20 for CSI-RS resource 1,2,3,4<br>30 kHz SCS: 40 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | | Slots | 15 kHz SCS:<br>10 for CSI-RS resource 1 and 2<br>11 for CSI-RS resource 3 and 4<br><br>30 kHz SCS:<br>20 for CSI-RS resource 1 and 2<br>21 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0$ = 4 |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0$ = 12 |
| | Number of CSI-RS ports (X) | | | Same as number of transmit antenna |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0$ = 0 |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0$ = 12 |
| | Number of CSI-RS ports (X) | | | 4 |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | | 15 kHz SCS: 20<br>30 kHz SCS: 40 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| TCI state #0 | Type 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Type 2 QCL information | SSB index | | N/A |
| | | QCL Type | | N/A |
| TCI state #1 | Type 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Type 2 QCL information | CSI-RS resource | | N/A |
| | | QCL Type | | N/A |
| Maximum number of code block groups for ACK/NACK feedback | | | | 1 |
| Maximum number of HARQ transmission | | | | 4 |
| HARQ ACK/NACK bundling | | | | Multiplexed |
| Redundancy version coding sequence | | | | {0,2,3,1} |
| PDSCH & PDSCH DMRS Precoding configuration | | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable $i_1$, $i_2$ combination with PRB bundling granularity |
| Symbols for all unused REs | | | | OCNG Annex A.5 |
| Propagation condition | | | | Static propagation condition |

| | | | |
|---|---|---|---|
| | | | No external noise sources are applied |
| Antenna configuration | 1 layer CCs | | 1x2 or 1x4 |
| | 2 layers CCs | | 2x2 or 2x4 |
| | 4 layers CCs | | 4x4 |
| Physical signals, channels mapping and precoding | | | As specified in Annex B.4.1 |
| Note 1: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission<br>Note 2: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-1 [2] for tested channel bandwidth and subcarrier spacing | | | |

**Table 9.4B.1.1.3-2: Additional test parameters for NR FDD band**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | FDD |
| PDSCH configuration | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| Number of HARQ Processes | | | 4 |
| K1 value | | | 2 |

**Table 9.4B.1.1.3-3: Additional test parameters for NR TDD band**

| Parameter | | Unit | Value |
|---|---|---|---|
| Duplex mode | | | TDD |
| PDSCH configuration | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| Number of HARQ Processes | | | 8 |
| K1 value | | | Specific to each UL-DL pattern |
| TDD UL-DL pattern | | | 15 kHz SCS: FR1.15-1<br>30 kHz SCS: FR1.30-1 |
| Note 1: PDSCH is scheduled only on full DL slots | | | |

**Table 9.4B.1.1.3-4: Number of PRBs in CORESET for NR cell**

| SCS (kHz) | 5MHz | 10MHz | 15MHz | 20 MHz | 25 MHz | 30 MHz | 40 MHz | 50MHz | 60 MHz | 80 MHz | 100 MHz |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 24 | 48 | 78 | 102 | 132 | 156 | 216 | 270 | N/A | N/A | N/A |
| 30 | 6 | 24 | 36 | 48 | 60 | 78 | 102 | 132 | 162 | 216 | 270 |

**Table 9.4B.1.1.3-5: MCS indexes for indicated UE capabilities for NR cell**

| Maximum number of PDSCH MIMO layers | Maximum modulation format | Scaling factor | MCS |
|---|---|---|---|
| 1 | 8 | 1 | 26 |
| 1 | 8 | 0.8 | 21 |
| 1 | 8 | 0.75 | 20 |
| 1 | 8 | 0.4 | 11 |
| 1 | 6 | 1 | 27 |
| 1 | 6 | 0.8 | 23 |
| 1 | 6 | 0.75 | 22 |
| 1 | 6 | 0.4 | 14 |
| 1 | 4 | 1 | 16 |
| 1 | 4 | 0.8 | 16 |
| 1 | 4 | 0.75 | 16 |
| 1 | 4 | 0.4 | 10 |
| 1 | 2 | 1 | 9 |
| 1 | 2 | 0.8 | 9 |
| 1 | 2 | 0.75 | 9 |
| 1 | 2 | 0.4 | 4 |
| 2 | 8 | 1 | 26 |
| 2 | 8 | 0.8 | 21 |
| 2 | 8 | 0.75 | 20 |
| 2 | 8 | 0.4 | 11 |
| 2 | 6 | 1 | 27 |
| 2 | 6 | 0.8 | 23 |
| 2 | 6 | 0.75 | 22 |
| 2 | 6 | 0.4 | 14 |
| 2 | 4 | 1 | 16 |
| 2 | 4 | 0.8 | 16 |
| 2 | 4 | 0.75 | 16 |
| 2 | 4 | 0.4 | 10 |
| 2 | 2 | 1 | 9 |
| 2 | 2 | 0.8 | 9 |
| 2 | 2 | 0.75 | 9 |
| 2 | 2 | 0.4 | 4 |
| 4 | 8 | 1 | 26 |
| 4 | 8 | 0.8 | 23 |
| 4 | 8 | 0.75 | 22 |
| 4 | 8 | 0.4 | 12 |
| 4 | 6 | 1 | 27 |
| 4 | 6 | 0.8 | 24 |
| 4 | 6 | 0.75 | 23 |
| 4 | 6 | 0.4 | 14 |
| 4 | 4 | 1 | 16 |
| 4 | 4 | 0.8 | 16 |
| 4 | 4 | 0.75 | 16 |
| 4 | 4 | 0.4 | 11 |
| 4 | 2 | 1 | 9 |
| 4 | 2 | 0.8 | 9 |
| 4 | 2 | 0.75 | 9 |
| 4 | 2 | 0.4 | 5 |

**Table 9.4B.1.1.3-6: Additional test setup for E-UTRA CC**

| Parameter | Unit | Value |
|---|---|---|
| Inter-TTI Distance | | 1 |
| Number of OFDM symbols for PDCCH per component carrier | OFDM symbols | 1 |
| Cross carrier scheduling | | Not configured |
| Propagation condition | | Static propagation condition<br>No external noise sources are applied |
| $\hat{E}_s$ at antenna port | dBm/15kHz | -85 |
| Antenna configuration | 2 layer CC | 2x2 or 2x4 |
| | 4 layer CC | 4x4 |
| Codebook subset restriction | 2 layer CC | 10 |
| | 4 layer CC | 1000 |
| Downlink power allocation | 2 layer CC | $\rho_A$ = -3dB, $\rho_B$ = -3dB, σ = 0dB |
| | 4 layer CC | $\rho_A$ = -6dB, $\rho_B$ = -6dB, σ = 3dB |

**Table 9.4B.1.1.3-7: E-UTRA FRC for SDR test (FDD)**

| MIMO layer | Bandwidth | Reference channel | | |
|---|---|---|---|---|
| | | 64QAM | 256QAM | 1024QAM |
| 2 layer | 5 | R.PDSCH.4-1.1 FDD | R.PDSCH.4-3.1 FDD | R.PDSCH.4-5.1 FDD |
| | 10 | R.PDSCH.4-1.2 FDD | R.PDSCH.4-3.2 FDD | R.PDSCH.4-5.2 FDD |
| | 15 | R.PDSCH.4-1.3 FDD | R.PDSCH.4-3.3 FDD | R.PDSCH.4-5.3 FDD |
| | 20 | R.PDSCH.4-1.4 FDD | R.PDSCH.4-3.4 FDD | R.PDSCH.4-5.4 FDD |
| 4 layer | 5 | R.PDSCH.4-2.1 FDD | R.PDSCH.4-4.1 FDD | R.PDSCH.4-6.1 FDD |
| | 10 | R.PDSCH.4-2.2 FDD | R.PDSCH.4-4.2 FDD | R.PDSCH.4-6.2 FDD |
| | 15 | R.PDSCH.4-2.3 FDD | R.PDSCH.4-4.3 FDD | R.PDSCH.4-6.3 FDD |
| | 20 | R.PDSCH.4-2.4 FDD | R.PDSCH.4-4.4 FDD | R.PDSCH.4-6.4 FDD |

**Table 9.4B.1.1.3-8: E-UTRA FRC for SDR test (TDD)**

| MIMO layer | Bandwidth | Reference channel | | |
|---|---|---|---|---|
| | | 64QAM | 256QAM | 1024QAM |
| 2 layer | 10 | R.PDSCH.6-1.1 TDD | R.PDSCH.6-3.1 TDD | R.PDSCH.6-5.1 TDD |
| | 15 | R.PDSCH.6-1.2 TDD | R.PDSCH.6-3.2 TDD | R.PDSCH.6-5.2 TDD |
| | 20 | R.PDSCH.6-1.3 TDD | R.PDSCH.6-3.3 TDD | R.PDSCH.6-5.3 TDD |
| 4 layer | 10 | R.PDSCH.6-2.1 TDD | R.PDSCH.6-4.1 TDD | R.PDSCH.6-6.1 TDD |
| | 15 | R.PDSCH.6-2.2 TDD | R.PDSCH.6-4.2 TDD | R.PDSCH.6-6.2 TDD |
| | 20 | R.PDSCH.6-2.3 TDD | R.PDSCH.6-4.3 TDD | R.PDSCH.6-6.3 TDD |

#### 9.4B.1.1.3.1 Procedure for test parameter selection

The test parameters are determined by the following procedure:

- Select one EN-DC bandwidth combination among all supported EN-DC configurations and set of per component carrier (CC) UE capabilities among all supported UE capabilities that provides the largest data rate [TS 38.306 [14, Section 4.1.2]].

  - Set of per NR CC UE capabilities include channel bandwidth, subcarrier spacing, number of PDSCH MIMO layers, modulation format and scaling factor TS 38.306 [14] Section 4.1.2]].

  - Set of per E-UTRA CC UE capabilities includes channel bandwidth, number of PDSCH MIMO layers and modulation format [TS 38.306 [14] Section 4.1.2]].

  - When there are multiple sets of EN-DC bandwidth combinations and UE capabilities with same largest data rate, select one among sets with the smallest aggregated channel bandwidth.

- For each NR FR1 CC in EN-DC bandwidth combination, use Table 9.4B.1.1.3-5 to determine MCS based on test parameters and indicated UE capabilities.

- For each E-UTRA CC in EN-DC bandwidth combination, use Table 9.4B.1.1.3-7 and Table 9.4B.1.1.3-8 to determine FRC based on test parameters and indicated UE capabilities.

Pasting relevant portion of max data rate equation from TS 38.306 [14] section 4.1

For NR, the approximate data rate for a given number of aggregated carriers in a band or band combination is computed as follows.

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left( v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

wherein

J is the number of aggregated component carriers in a band or band combination

$R_{max}$ = 948/1024

For the j-th CC,

$v_{Layers}^{(j)}$ is the maximum number of supported layers given by higher layer parameter *maxNumberMIMO-LayersPDSCH* for downlink and maximum of higher layer parameters *maxNumberMIMO-LayersCB-PUSCH* and *maxNumberMIMO-LayersNonCB-PUSCH* for uplink.

$Q_m^{(j)}$ is the maximum supported modulation order given by higher layer parameter *supportedModulationOrderDL* for downlink and higher layer parameter *supportedModulationOrderUL* for uplink.

$f^{(j)}$ is the scaling factor given by higher layer parameter *scalingFactor* and can take the values 1, 0.8, 0.75, and 0.4.

$\mu$ is the numerology (as defined in TS 38.211 [6])

$T_s^{\mu}$ is the average OFDM symbol duration in a subframe for numerology $\mu$, i.e. $T_s^{\mu} = \frac{10^{-3}}{14 \cdot 2^{\mu}}$. Note that normal cyclic prefix is assumed.

$N_{PRB}^{BW(j),\mu}$ is the maximum RB allocation in bandwidth $BW^{(j)}$ with numerology $\mu$, as defined in 5.3 TS 38.101-1 [2] and 5.3 TS 38.101-2 [3], where $BW^{(j)}$ is the UE supported maximum bandwidth in the given band or band combination.

$OH^{(j)}$ is the overhead and takes the following values

0.14, for frequency range FR1 for DL

0.18, for frequency range FR2 for DL

0.08, for frequency range FR1 for UL

0.10, for frequency range FR2 for UL

NOTE: Only one of the UL or SUL carriers (the one with the higher data rate) is counted for a cell operating SUL.

For EUTRA in case of MR-DC, the approximate data rate for a given number of aggregated carriers in a band or band combination is computed as follows.

Data rate (in Mbps) = $10^{-3} \cdot \sum_{j=1}^{J} TBS_j$

wherein

J is the number of aggregated EUTRA component carriers in MR-DC band combination

$TBS_j$ is the total maximum number of DL-SCH transport block bits received within a 1ms TTI for j-th CC, as derived from TS36.213 [22] based on the UE supported maximum MIMO layers for the j-th carrier, and based on the modulation order and number of PRBs based on the bandwidth of the j-th carrier.

The approximate maximum data rate can be computed as the maximum of the approximate data rates computed using the above formula for each of the supported band or band combinations.

For MR-DC, the approximate maximum data rate is computed as the sum of the approximate maximum data rates from NR and EUTRA

The normative reference for this requirement is TS 38.101-4 [5], clause 9.4B.1.1.

### 9.4B.1.1.4 Test description

#### 9.4B.1.1.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR and E-UTRA operating bands specified in Table 5.3.5-1 of TS 38.521-1.

Configurations of NR PDSCH and NR PDCCH before measurement are specified in Annex C.

E-UTRA configurations before measurement are specified in at Table 9.4B.1.1.3-6.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

1. Connect the SS to the UE antenna connectors as shown in TS 38.508-1 [6] Annex A, in Figure A.3.1.7.1 for TE diagram (without fader and AWGN) and clause A.3.2.2 for UE diagram.
2. Downlink signals for the NR cell are initially set up according to Annexes C.0, C.1, C.2, C.3.1, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-1 [7].
3. Downlink signals for E-UTRA cell are initially set up according to TS 36.521-1 [16] Annex C.0 and uplink signals according to TS 36.521-1 [16] Annex H
4. Propagation conditions are set according to TS 36.521-1 [16] and TS 38.521-1 [7] Annex B.0 for E-UTRA CG and NR CG respectively.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters Connectivity EN-DC, DC bearer *MCG(s)* and *SCG*, Connected without release *On, Test Loop Function On with UE Test Loop Mode A with UL_PDCP_SDU_SIZE = 0* for MCG DRB and SCG DRB according to TS 38.508-1 [6] clause 4.5.4. Message content are defined in clause 5.5.1.4.3.
6. SS sends a RRCConnectionReconfiguration message to change PDCP version of MCG DRB to NR PDCP.
7. SS shall transmit UECapabilityEnquiry message containing *UE-CapabilityRAT-Request* with *rat-Type* set to *eutra-nr* and *eutra*.
8. The UE shall transmit UECapabilityInformation message.
9. Using the UE capabilities advertised in the *UE-CapabilityRAT-Container* of the type *UE-MRDC-Capability and UE-EUTRA-Capability*, and the procedure outlined in 9.4B.1.1.3.1 determine one EN-DC bandwidth combination that would provide the largest aggregated data rate.
10. Setup up the NR CG and E-UTRA CG using these parameters for the test.

11. Configure the NR CG TBsize, NR CG DL RMC, NR CG UL RMC from Annex A.3.2_1 and Annex A.2.2 for UL as appropriate. Configure the E-UTRA CG TBsize, DL RMC and UL RMC from Table 9.4B.1.1.3-7, Table 9.4B.1.1.3-8 as appropriate.

#### 9.4B.1.1.4.2 Test procedure

1. SS configures T-reordering timer to be infinity for both E-UTRA MCG DRB and NR SCG DRB.
2. SS sends a PDCP reestablishment via RRCConnectionReconfigurationmessage requesting for PDCP Status Report for both E-UTRA MCG DRB and NR SCG DRB.
3. SS sets the counters $N_{DL_newtx}$ $N_{DL_retx}$ per NR CG and E-UTRA CG to 0.
4. For each new DL HARQ transmission the SS generates sufficient NR PDCP SDUs (max PDCP SDU size and minimum number of consecutive PDCP SDUs) to fill up the TB in accordance with Annex A.3.2_1 for both E-UTRA MCG DRB and NR SCG DRB. The SS ciphers the PDCP SDUs, concatenates the resultant PDCP PDUs to form an RLC PDU and then a MAC PDU. The SS transmits the MAC PDU per NR CG and E-UTRA CG. The SS increments then $N_{DL_newtx}$ by one per CG.
5. If PHY requests a DL HARQ retransmission, the SS performs a HARQ retransmission and increments $N_{DL_retx}$ by one for that CG accordingly.
6. Steps 5 to 6 are repeated at every TTI for at least 300 frames and the SS waits for 300ms to let any HARQ retransmissions and RLC retransmissions to finish.
7. SS sends a PDCP reestablishment via RRCConnectionReconfigurationmessage requesting for PDCP Status Report for both E-UTRA MCG and NR SCG DRB.
8. The SS calculates the TB success rate per NR CG and E-UTRA CG as A = 100% $N_{DL_correct_rx}$ */ ($N_{DL_newtx}$ + $N_{DL_retx}$).
9. SS computes the PDCP SDU loss by looking into the FMC and Bitmap field in the PDCP Status Report. PDCP SDU loss B = COUNT reported in the Bitmap field of PDCP Status Report.
10. The UE passes the test if A $\geq$ 85% TB success rates for both NR CG and E-UTRA CG and B = 0.

NOTE 1: In case of RLC PDU retransmission, the number of new required PDCP SDUs is as many as to fill the rest of TB.

#### 9.4B.1.1.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions

**Table 9.4B.1.1.4.3-0: CLOSE UE TEST LOOP (MCG and SCG DRB in the preamble)**

| Derivation Path: 38.509 clause 6.3.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 1 1 1 1 | | |
| Skip indicator | 0 0 0 0 | | |
| Message type | 1 0 0 0 0 0 0 0 | | |
| UE test loop mode | 0 0 0 0 0 0 0 0 | UE test loop mode A | |
| UE test loop mode A LB setup | | | |
| Length of UE test loop mode A LB setup list in bytes | 0 0 0 0 1 1 0 0 | Length of two LB setup DRB (6 bytes) | |
| LB setup DRB[1] | 0 0 0 0 0 0 0 0,<br>0 0 0 0 0 0 0 0,<br>0 0 0 Q4 Q3 Q2 Q1 Q0 | UL PDCP SDU size = 0<br>Q4..Q0 = MCG Data Radio Bearer identity number -1 for the radio bearer. See 38.509 clause 6.3.1 | |
| LB setup DRB[2] | 0 0 0 0 0 0 0 0,<br>0 0 0 0 0 0 0 0,<br>0 0 0 Q4 Q3 Q2 Q1 Q0 | UL PDCP SDU size = 0<br>Q4..Q0 = SCG Data Radio Bearer identity number -1 for the radio bearer. See 38.509 clause 6.3.1 | |
| UE test loop mode B LB setup | Not present | | |

**Table 9.4B.1.1.4.3-1 to -7: Void**

**Table 9.4B.1.1.4.3-8: RadioBearerConfig (Initial Conditions, Step 5)**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3-132 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioBearerConfig ::= SEQUENCE { | | | |
| drb-ToAddModList SEQUENCE (SIZE (1..maxDRB)) OF DRB-ToAddMod { | 1 entry | | |
| DRB-ToAddMod[1] SEQUENCE { | | entry 1 | |
| cnAssociation CHOICE { | | | |
| eps-BearerIdentity | Dedicated EPS bearer ID | | |
| } | | | |
| drb-Identity | DRB-Identity of the SCG DRB | | |
| reestablishPDCP | Not Present | | |
| pdcp-Config | PDCP-Config | Table 9.4B.1.1.4.3-8A | |
| } | | | |
| } | | | |

**Table 9.4B.1.1.4.3-8A: *PDCP-Config***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-99 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCP-Config ::= SEQUENCE { | | | |
| drb SEQUENCE { | | | |
| discardTimer | infinity | | |
| pdcp-SN-Size-UL | len18bits | | |
| pdcp-SN-Size-DL | len18bits | | |
| headerCompression CHOICE { | | | |
| notUsed | Null | | |
| } | | | |
| integrityProtection | Not present | | |
| statusReportRequired | true | | |
| outOfOrderDelivery | Not present | | |
| } | | | |
| t-Reordering | Not present | | |
| } | | | |

**Table 9.4B.1.1.4.3-9: *RRCConnectionReconfiguration* (Initial conditions, step6)**

| Derivation Path: TS 36.508 [7], Table 4.6.1-8 with condition HO | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RRCConnectionReconfiguration ::= SEQUENCE { | | | |
| criticalExtensions CHOICE { | | | |
| c1 CHOICE { | | | |
| rrcConnectionReconfiguration-r8 SEQUENCE { | | | |
| mobilityControlInfo | As per Table 4.6.5-1 of TS 36.508 [19] | | |
| radioResourceConfigDedicated | RadioResourceConfigDedicated-MCG-DRB-NR-PDCP | As per Table 9.4B.1.1.4.3-10 | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nonCriticalExtension SEQUENCE { | | | |
| nr-Config-r15 | Not present | | |
| nr-RadioBearerConfig1-r15 | OCTET STRING containing RadioBearerConfig according to TS 38.508-1 [6], Table 4.6.3-132 with conditions MCG_NR_PDCP | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |
| } | | | |

**Table 9.4B.1.1.4.3-10: *RadioResourceConfigDedicated-MCG-DRB-NR-PDCP***

| Derivation Path: TS 36.508 [7], Table 4.6.3-19 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioResourceConfigDedicated-MCG-DRB-NR-PDCP ::= SEQUENCE { | | | |
| drb-ToAddModList SEQUENCE (SIZE (1..maxDRB)) OF DRB-ToAddMod { | 1 entry | | |
| DRB-ToAddMod[1] | DRB-ToAddMod-MCG-DRB-NR-PDCP | entry 1<br>As per Table 9.4B.1.1.4.3-11 | |
| } | | | |
| drb-ToReleaseList SEQUENCE (SIZE (1..maxDRB)) OF DRB-Identity { | 1 entry | | |
| DRB-Identity[1] | Same as the DRB identity associated with the default EPS bearer | entry 1 | |
| } | | | |
| physicalConfigDedicated | PhysicalConfigDedicated-DEFAULT with condition RBC-HO | | |
| } | | | |

**Table 9.4B.1.1.4.3-11: DRB-ToAddMod-MCG-DRB-NR-PDCP**

| Derivation Path: TS 36.508 [19], Table 4.8.2.1.7-1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| DRB-ToAddMod ::= SEQUENCE { | | | |
| eps-BearerIdentity | Same as the default EPS bearer Identity | | |
| drb-Identity | Same as the DRB identity associated with the default EPS bearer | | |
| pdcp-Config | Not present | | |
| reestablishPDCP | Not present | | |
| } | | | |

**Table 9.4B.1.1.4.3-12: RadioBearerConfig (Test procedure, step 2 and step 7)**

| Derivation Path: TS 38.508-1 [6], clause 4.6.3-132 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioBearerConfig ::= SEQUENCE { | | | |
| drb-ToAddModList SEQUENCE (SIZE (1..maxDRB)) OF DRB-ToAddMod { | 2 entries | | |
| DRB-ToAddMod[1] SEQUENCE { | | entry 1 | |
| cnAssociation CHOICE { | | | |
| eps-BearerIdentity | Default EPS bearer ID | | |
| } | | | |
| drb-Identity | DRB-Identity of the MCG DRB | | |
| reestablishPDCP | true | | |
| pdcp-Config | PDCP-Config | | |
| } | | | |
| DRB-ToAddMod[2] SEQUENCE { | | entry 2 | |
| cnAssociation CHOICE { | | | |
| eps-BearerIdentity | Dedicated EPS bearer ID | | |
| } | | | |
| drb-Identity | DRB-Identity of the SCG DRB | | |
| reestablishPDCP | true | | |
| pdcp-Config | PDCP-Config | | |
| } | | | |

#### 9.4B.1.1.5 Test requirement

The PDCP SDU success rate of greater than 85% shall be sustained during at least 300 frames.

### 9.4B.1.2 Sustained downlink data rate performance for EN-DC including FR2 NR carrier

Editor's Note: MU analysis is complete for up to 100 MHz ChBW.

#### 9.4B.1.2.1 Test Purpose

The purpose of the test is to verify that the Layer 1 and Layer 2 correctly process in a sustained manner the received packets corresponding to the maximum data rate indicated by UE capabilities. The sustained downlink data rate shall be verified in terms of the success rate of delivered PDCP SDU(s) by Layer 2. The test case below specifies the conditions and the required success rate of delivered TB by Layer 1 to meet the sustained data rate requirement.

#### 9.4B.1.2.2 Test Applicability

This test applies to all types of EUTRA UE release 15 and forward supporting EN-DC.

#### 9.4B.1.2.3 Minimum conformance requirements

The test setup for E-UTRA Pcell is specified in Clause 9.1.2 and Table 9.4B.1.1.1-1. During the test, only the PDSCH performance on the NR cell(s) on FR2 carriers is verified.

The TB success rate shall be higher than 85% when NR PDSCH is scheduled with MCS defined for the selected EN-DC bandwidth combination and with the downlink physical channel setup according to Annex C.2.2.

The TB success rate of delivered PDCP SDU(s) by Layer2 is defined as TB success rate = 100%*NDL_correct_rx/ (NDL_newtx + NDL_retx), where NDL_newtx is the number of newly transmitted DL transport blocks, NDL_retx is the number of retransmitted DL transport blocks, and DL_correct_rx is the number of correctly received DL transport blocks. All the above numbers of transmitted, retransmitted or correctly received DL transport blocks are calculated as the sum of the numbers of DL transport blocks per CG used for DC.

The test parameters are specified in Tables 9.4B.1.2.3-1, 9.4B.1.2.3-2.

Unless otherwise stated, no user data is scheduled on slot #0, 40 and 41 within 20 ms for SCS 60 kHz.

Unless otherwise stated, no user data is scheduled on slot #0, 80 and 81 within 20 ms for SCS 120 kHz.

**Table 9.4B.1.2.3-1: Test parameters for FR2 TDD**

| Parameter | | Unit | Value |
|---|---|---|---|
| PDSCH transmission scheme | | | Transmission scheme 1 |
| PTRS epre-Ratio | | | 0 |
| Channel bandwidth | | MHz | Channel bandwidth from selected CA bandwidth combination |
| Common serving cell parameters | Physical Cell ID | | 0 |
| | SSB position in burst | | First SSB in Slot #0 |
| | SSB periodicity | ms | 20 |
| | First DMRS position for Type A PDSCH mapping | | 2 |
| Cross carrier scheduling | | | Not configured |
| Active DL BWP index | | | 1 |
| Actual carrier configuration | Offset between Point A and the lowest usable subcarrier on this carrier (Note 3) | RBs | 0 |
| | Subcarrier spacing | kHz | 60 or 120 |
| DL BWP configuration #1 | RB Offset | | 0 |
| | Number of contiguous PRB | | Maximum transmission bandwidth configuration as specified in clause 5.3.2 of TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing |
| | Subcarrier spacing | kHz | 60 or 120 |
| | Cyclic prefix | | Normal |
| PDCCH configuration | Slots for PDCCH monitoring | | Each slot |
| | Symbols with PDCCH | | Symbols #0 |
| | Number of PRBs in CORESET | | Table 7.5A.1-2 |
| | Number of PDCCH candidates and aggregation levels | | 1/8 |
| | CCE-to-REG mapping type | | Non-interleaved |
| | DCI format | | 1-1 |
| | TCI State | | TCI state #1 |
| | PDCCH &PDCCH DMRS Precoding configuration | | Single Panel Type I, Random per slot with equal probability of precoder index 0 and 2, and with REG bundling granularity for number of Tx larger than 1 |
| PDSCH configuration | Mapping type | | Type A |
| | k0 | | 0 |
| | PDSCH aggregation factor | | 1 |
| | PRB bundling type | | Static |
| | PRB bundling size | | WB |
| | Resource allocation type | | Type 0 |
| | RBG size | | Config2 |
| | VRB-to-PRB mapping type | | Non-interleaved |
| | VRB-to-PRB mapping interleaver bundle size | | N/A |
| | Starting symbol (S) | | 1 |
| | Length (L) | | 13 |
| PDSCH DMRS configuration | DMRS Type | | Type 1 |
| | Number of additional DMRS | | 1 |
| | Length | | 1 |
| | Antenna ports indexes | | {1000} for 1 Layer CCs<br>{1000, 1001} for 2 Layers CCs |
| | Number of PDSCH DMRS CDM group(s) without data | | 1 |
| PTRS configuration | Frequency density ($K_{PT-RS}$) | | 2 |
| | Time density ($L_{PT-RS}$) | | 1 |
| CSI-RS for tracking | Subcarrier indexes in the PRB used for CSI-RS | | $k_0$ = 3 for CSI-RS resource 1,2,3,4 |
| | OFDM symbols in the PRB used for CSI-RS | | $l_0$ = 6 for CSI-RS resource 1 and 3<br>$l_0$ = 10 for CSI-RS resource 2 and 4 |
| | Number of CSI-RS ports (X) | | 1 for CSI-RS resource 1,2,3,4 |
| | CDM Type | | 'No CDM' for CSI-RS resource |

| | | | | |
|---|---|---|---|---|
| | | | | 1,2,3,4 |
| | Density (ρ) | | | 3 for CSI-RS resource 1,2,3,4 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80 for CSI-RS resource 1,2,3,4<br>120 kHz SCS: 160 for CSI-RS resource 1,2,3,4 |
| | CSI-RS offset | | Slots | 60 kHz SCS:<br>40 for CSI-RS resource 1 and 2<br>41 for CSI-RS resource 3 and 4<br><br>120 kHz SCS:<br>80 for CSI-RS resource 1 and 2<br>81 for CSI-RS resource 3 and 4 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #0 |
| NZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0 = 4$ |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0 = 13$ |
| | Number of CSI-RS ports (X) | | | Same as number of transmit antenna |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80<br>120 kHz SCS: 160 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| | QCL info | | | TCI state #1 |
| ZP CSI-RS for CSI acquisition | Subcarrier indexes in the PRB used for CSI-RS | | | $k_0 = 0$ |
| | OFDM symbols in the PRB used for CSI-RS | | | $l_0 = 12$ |
| | Number of CSI-RS ports (X) | | | 4 |
| | CDM Type | | | 'FD-CDM2' |
| | Density (ρ) | | | 1 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80<br>120 kHz SCS: 160 |
| | CSI-RS offset | | | 0 |
| | Frequency Occupation | | | Start PRB 0<br>Number of PRB = BWP size |
| CSI-RS for beam refinement | First subcarrier index in the PRB used for CSI-RS | | | $k_0$=0 for CSI-RS resource 1,2 |
| | First OFDM symbol in the PRB used for CSI-RS | | | $l_0 = 8$ for CSI-RS resource 1<br>$l_0 = 9$ for CSI-RS resource 2 |
| | Number of CSI-RS ports (X) | | | 1 for CSI-RS resource 1,2 |
| | CDM Type | | | 'No CDM' for CSI-RS resource 1,2 |
| | Density (ρ) | | | 3 for CSI-RS resource 1,2 |
| | CSI-RS periodicity | | Slots | 60 kHz SCS: 80 for CSI-RS resource 1,2<br>120 kHz SCS: 160 for CSI-RS resource 1,2 |
| | CSI-RS offset | | Slots | 0 for CSI-RS resource 1,2 |
| | Repetition | | | ON |
| | QCL info | | | TCI state #1 |
| TCI state #0 | Tyoe 1 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type C |
| | Tyoe 2 QCL information | SSB index | | SSB #0 |
| | | QCL Type | | Type D |
| TCI state #1 | Tyoe 1 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type A |
| | Tyoe 2 QCL information | CSI-RS resource | | CSI-RS resource 1 from 'CSI-RS for tracking' configuration |
| | | QCL Type | | Type D |
| Maximum number of code block groups for ACK/NACK | | | | 1 |

| | | | |
|---|---|---|---|
| feedback | | | |
| Number of HARQ Processes | | | 10 for FR2.60-1 and 8 for FR2.120-1 |
| K1 value | | | Specific to each UL-DL pattern |
| Maximum number of HARQ transmission | | | 4 |
| HARQ ACK/NACK bundling | | | Multiplexed |
| Redundancy version coding sequence | | | {0,2,3,1} |
| TDD UL-DL pattern | | | 60 kHz SCS: FR2.60-1<br>120 kHz SCS: FR2.120-1 |
| PDSCH & PDSCH DMRS Precoding configuration | | | Single Panel Type I, Random precoder selection updated per slot, with equal probability of each applicable i1, i2 combination, and with Wideband granularity for Rank 2 |
| Symbols for all unused REs | | | OCNG Annex A.5 |
| Propagation condition | | | Static propagation condition<br>No external noise sources are applied |
| Antenna configuration | 1 layer CCs | | 1x2 or 1x4 |
| | 2 layers CCs | | 2x2 or 2x4 |
| Physical signals, channels mapping and precoding | | | As specified in Annex B.4.1 |
| Note 1: PDSCH is scheduled only on full DL slots not containing SSB or TRS.<br>Note 2: UE assumes that the TCI state for the PDSCH is identical to the TCI state applied for the PDCCH transmission.<br>Note 3: Point A coincides with minimum guard band as specified in Table 5.3.3-1 from TS 38.101-2 [3] for tested channel bandwidth and subcarrier spacing. | | | |

**Table 9.4B.1.2.3-2: Number of PRBs in CORESET**

| SCS (kHz) | 50 MHz | 100 MHz | 200 MHz | 400 MHz |
|---|---|---|---|---|
| 60 | 66 | 132 | 264 | N.A |
| 120 | 30 | 66 | 132 | 264 |

**Table 9.4B.1.2.3-3: MCS indexes for indicated UE capabilities**

| Maximum number of PDSCH MIMO layers | Maximum modulation format | Scaling factor | MCS |
|---|---|---|---|
| 1 | 6 | 1 | 27 |
| 1 | 6 | 0.8 | 23 |
| 1 | 6 | 0.75 | 22 |
| 1 | 6 | 0.4 | 14 |
| 1 | 4 | 1 | 16 |
| 1 | 4 | 0.8 | 16 |
| 1 | 4 | 0.75 | 16 |
| 1 | 4 | 0.4 | 10 |
| 1 | 2 | 1 | 9 |
| 1 | 2 | 0.8 | 9 |
| 1 | 2 | 0.75 | 9 |
| 1 | 2 | 0.4 | 4 |
| 2 | 6 | 1 | 27 |
| 2 | 6 | 0.8 | 23 |
| 2 | 6 | 0.75 | 22 |
| 2 | 6 | 0.4 | 14 |
| 2 | 4 | 1 | 16 |
| 2 | 4 | 0.8 | 16 |
| 2 | 4 | 0.75 | 16 |
| 2 | 4 | 0.4 | 10 |
| 2 | 2 | 1 | 9 |
| 2 | 2 | 0.8 | 9 |
| 2 | 2 | 0.75 | 9 |
| 2 | 2 | 0.4 | 4 |

**Table 9.4B.1.2.3-4: SNR required to achieve 85% of peak throughput under AWGN conditions**

| MCS Index (Note 1) | $SNR_{BB}$(dB) for maximum number of PDSCH MIMO Layers = 1 | $SNR_{BB}$(dB) for maximum number of PDSCH MIMO Layers = 2 |
|---|---|---|
| 13 | 6.2 | 9.0 |
| 14 | 7.2 | 9.9 |
| 15 | 8.2 | 10.9 |
| 16 | 8.7 | 11.6 |
| 17 | 10.1 | 13.2 |
| 18 | 10.7 | 13.7 |
| 19 | 11.7 | 14.7 |
| 20 | 12.7 | 15.6 |
| 21 | 13.6 | 16.5 |
| 22 | 14.8 | 17.6 |
| 23 | 15.6 | 18.6 |
| 24 | 16.9 | 19.7 |
| 25 | 18.3 | 21.2 |
| 26 | 19.3 | 22.3 |
| 27 | 20.5 | 23.3 |
| Note 1: MCS Index is based on MCS Table defined in clause 5.1.3 of TS 38.214 [12] when 256QAM is not enabled. | | |

The normative reference for this requirement is TS 38.101-4 [5], clause 9.4B.1.2.

9.4B.1.2.3.1 Procedure for test parameter selection

The test parameters are determined by the following procedure:

- Step 1: Calculate the NR FR2 data rate for EN-DC bandwidth combinations, using a procedure from Clause 7.5A, for all supported EN-DC configurations and set of per NR component carrier (CC) UE capabilities among all supported UE capabilities:
  - Set of per NR CC UE capabilities includes a channel bandwidth, subcarrier spacing, number of PDSCH MIMO layers, modulation format and scaling factor as defined in clause 4.1.2 of TS 38.306 [14].
- Step 2: Calculate the E-UTRA data rate for EN-DC bandwidth combinations, using a procedure from clause 4.1.2 of TS 38.306 [14], for all supported EN-DC configurations and set of per E-UTRA component carrier (CC) UE capabilities among all supported UE capabilities:
  - Set of per E-UTRA CC UE capabilities includes a channel bandwidth, number of PDSCH MIMO layers and modulation format as defined in clause 4.1.2 of TS 38.306 [14].
- Step 3: Select the EN-DC bandwidth combination among all supported EN-DC configurations that achieves maximum total data rate in steps 1 and 2 among all UE capabilities:
  - When there are multiple sets of EN-DC bandwidth combinations and UE capabilities with the same largest data rate, select a single set with the smallest aggregated channel bandwidth.
- Step 4: For each NR FR2 CC in the selected EN-DC bandwidth combination, use MCS determined in step 1 for that EN-DC bandwidth combination based on test parameters and indicated UE capabilities.

Pasting relevant portion of max data rate equation from TS 38.306 [14] section 4.1

For NR, the approximate data rate for a given number of aggregated carriers in a band or band combination is computed as follows.

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^{J} \left( v_{Layers}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\max} \cdot \frac{N_{PRB}^{BW(j),\mu} \cdot 12}{T_s^{\mu}} \cdot \left(1 - OH^{(j)}\right) \right)$$

wherein

J is the number of aggregated component carriers in a band or band combination

$R_{max}$ = 948/1024

For the j-th CC,

$v_{Layers}^{(j)}$ is the maximum number of supported layers given by higher layer parameter *maxNumberMIMO-LayersPDSCH* for downlink and maximum of higher layer parameters *maxNumberMIMO-LayersCB-PUSCH* and *maxNumberMIMO-LayersNonCB-PUSCH* for uplink.

$Q_m^{(j)}$ is the maximum supported modulation order given by higher layer parameter *supportedModulationOrderDL* for downlink and higher layer parameter *supportedModulationOrderUL* for uplink.

$f^{(j)}$ is the scaling factor given by higher layer parameter *scalingFactor* and can take the values 1, 0.8, 0.75, and 0.4.

$\mu$ is the numerology (as defined in TS 38.211 [6])

$T_s^\mu$ is the average OFDM symbol duration in a subframe for numerology $\mu$, i.e. $T_s^\mu = \frac{10^{-3}}{14 \cdot 2^\mu}$. Note that normal cyclic prefix is assumed.

$N_{PRB}^{BW(j),\mu}$ is the maximum RB allocation in bandwidth $BW^{(j)}$ with numerology $\mu$, as defined in 5.3 TS 38.101-1 [2] and 5.3 TS 38.101-2 [3], where $BW^{(j)}$ is the UE supported maximum bandwidth in the given band or band combination.

$OH^{(j)}$ is the overhead and takes the following values

0.14, for frequency range FR1 for DL

0.18, for frequency range FR2 for DL

0.08, for frequency range FR1 for UL

0.10, for frequency range FR2 for UL

NOTE: Only one of the UL or SUL carriers (the one with the higher data rate) is counted for a cell operating SUL.

For EUTRA in case of MR-DC, the approximate data rate for a given number of aggregated carriers in a band or band combination is computed as follows.

Data rate (in Mbps) = $10^{-3} \cdot \sum_{j=1}^{J} TBS_j$

wherein

J is the number of aggregated EUTRA component carriers in MR-DC band combination

$TBS_j$ is the total maximum number of DL-SCH transport block bits received within a 1ms TTI for j-th CC, as derived from TS36.213 [22] based on the UE supported maximum MIMO layers for the j-th carrier, and based on the modulation order and number of PRBs based on the bandwidth of the j-th carrier.

The approximate maximum data rate can be computed as the maximum of the approximate data rates computed using the above formula for each of the supported band or band combinations.

For MR-DC, the approximate maximum data rate is computed as the sum of the approximate maximum data rates from NR and EUTRA

#### 9.4B.1.2.4 Test description

##### 9.4B.1.2.4.1 Initial conditions

Initial conditions are a set of test configurations the UE needs to be tested in and the steps for the SS to take with the UE to reach the correct measurement state.

The initial test configurations consist of environmental conditions, test frequencies, test channel bandwidths and sub-carrier spacing based on NR operating bands specified in Table 5.3.5-1 of 38.521-1.

Configurations of PDSCH and PDCCH before measurement are specified in Annex C.

Test Environment: Normal, as defined in TS 38.508-1 [6] clause 4.1.

Frequencies to be tested: Mid Range, as defined in TS 38.508-1 [6] clause 4.3.1.1.

For EN-DC within FR2 operation, setup the LTE radiated link according to Annex D:

1. Connection between SS, the faders, AWGN noise source and the UE is shown in TS 38.508-1 [6] Annex A, Figure A.3.3.2 for TE diagram and Figure A.3.4.2 for UE diagram.
2. The parameter settings for the NR cell are set up according to Table 7.2-1 and Table 7.2.2.2.1.0-2 and as appropriate.
3. Downlink signals for NR cell are initially set up according to Annexes C.0, C.1, C.2, and uplink signals according to Annexes G.0, G.1, G.2, G.3.1 of TS 38.521-2 [8].
4. Propagation conditions for NR cell are set according to Annex B.0.
5. Ensure the UE is in state RRC_CONNECTED with generic procedure parameters *Test Mode On*, (EN-DC, DC bearer *MCG* and *SCG), Connected without release On, Test Loop Function On with UE Test Loop Mode A with UL_PDCP_SDU_SIZE = 0* according to TS 38.508-1 [6] clause 4.5.4. Message content are defined in clause 9.4B.1.2.4.3.
6. SS shall transmit UECapabilityEnquiry message containing *UE-CapabilityRAT-Request* with *rat-Type* set to *eutra-nr* and *eutra*.
7. The UE shall transmit UECapabilityInformation message.
8. Using the UE capabilities advertised in the *UE-CapabilityRAT-Container* of the type *UE-MRDC-Capability and UE-EUTRA-Capability*, and the procedure outlined in 9.4B.1.2.3.1 determine one EN-DC bandwidth combination that would provide the largest aggregated data rate.
9. Setup up the NR CG for these parameters for the test.

##### 9.4B.1.2.4.2 Test Procedure

1. Set the UE in a direction that satisfies the 3 normative criteria specified in Annex H.0. If no direction found, mark the test as inconclusive.
2. Based on the maximum SNR capability of the FR2 chamber, determine the max MCS index from table 9.4B.1.2.3-4 to be configured for this test.
3. Configure the NR CG TBsize, NR CG DL RMC, NR CG UL RMC from Annex A.3.2_1 and Annex A.2.2 for UL as appropriate based on the MCS index chosen in step 2.
4. SS configures T-reordering timer to be infinity for NR SCG DRB.
5. SS sends a PDCP reestablishment via RRC Reconfiguration message requesting for PDCP Status Report for NR SCG DRB.
6. SS sets the counters $N_{DL_newtx}$ $N_{DL_retx}$ per NR CG to 0.
7. For each new DL HARQ transmission the SS generates sufficient NR PDCP SDUs (max PDCP SDU size and minimum number of consecutive PDCP SDUs) to fill up the TB in accordance with Annex A.3.2_1 for NR SCG

DRB. The SS ciphers the PDCP SDUs, concatenates the resultant PDCP PDUs to form an RLC PDU and then a MAC PDU. The SS transmits the MAC PDU per NR CG. The SS increments then $N_{DL_newtx}$ by one per CG.

8. If PHY requests a DL HARQ retransmission, the SS performs a HARQ retransmission and increments $N_{DL_retx}$ by one for that CG accordingly.

9. Steps 7 and 8 are repeated at every TTI for at least 300 frames and the SS waits for 300ms to let any HARQ retransmissions and RLC retransmissions to finish.

10. SS sends a PDCP reestablishment via RRC Reconfiguration message requesting for PDCP Status Report for NR SCG DRB.

11. The SS calculates the TB success rate per NR CG as A = 100% $N_{DL_correct_rx}$ */ ($N_{DL_newtx}$ + $N_{DL_retx}$).

12. SS computes the PDCP SDU loss by looking into the FMC and Bitmap field in the PDCP Status Report. PDCP SDU loss B = COUNT reported in the Bitmap field of PDCP Status Report.

13. The UE passes the test if A ≥ 85% TB success rates for NR CG and B = 0.

NOTE 1: In case of RLC PDU retransmission, the number of new required PDCP SDUs is as many as to fill the rest of TB.

#### 9.4B.1.2.4.3 Message contents

Message contents are according to TS 38.508-1 [6] clause 5.4.2 with the following exceptions

**Table 9.4B.1.2.4.3-0: CLOSE UE TEST LOOP (in the preamble)**

| Derivation Path: 38.509 clause 6.3.1 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| Protocol discriminator | 1 1 1 1 | | |
| Skip indicator | 0 0 0 0 | | |
| Message type | 1 0 0 0 0 0 0 0 | | |
| UE test loop mode | 0 0 0 0 0 0 0 0 | UE test loop mode A | |
| UE test loop mode A LB setup | | | |
| Length of UE test loop mode A LB setup list in bytes | 0 0 0 0 0 0 1 1 | Length of one LB setup DRB (3 bytes) | |
| LB setup DRB | 0 0 0 0 0 0 0 0,<br>0 0 0 0 0 0 0 0,<br>0 0 0 Q4 Q3 Q2 Q1 Q0 | UL PDCP SDU size = 0<br>Q4..Q0 = Data Radio Bearer identity number for the default radio bearer. See 38.509 clause 6.3.1 | |
| UE test loop mode B LB setup | Not present | | |

**Table 9.4B.1.1.4.3-1: PDCCH-ControlResourceSet-spCellConfigDedicated**

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-28 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| ControlResourceSet ::= SEQUENCE { | | | |
| frequencyDomainResources | CORESET value according to Table 9.4B.1.2.3-2 as applicable | | |
| } | | | |
| } | | | |

**Table 9.4B.1.1.4.3-2: PDCCH *Search Space***

| Derivation Path: TS 38.508-1 [6], Table 4.6.3-162 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| SearchSpace ::= SEQUENCE { | | | |
| monitoringSymbolsWithinSlot | 10000000000000 | Symbols 0 | |
| nrofCandidates SEQUENCE { | | | |
| aggregationLevel1 | n0 | | |
| aggregationLevel2 | n0 | | |
| aggregationLevel4 | n0 | | |
| aggregationLevel8 | n1 | AL8 | |
| aggregationLevel16 | n0 | | |
| } | | | |
| } | | | |

**Table 9.4B.1.1.4.3-3: RadioBearerConfig**

| Derivation Path: TS 38.508 [6], clause 4.6.3-132 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| RadioBearerConfig ::= SEQUENCE { | | | |
| drb-ToAddModList SEQUENCE (SIZE (1..maxDRB)) OF SEQUENCE { | 1 entry | | EN-DC_DRB |
| cnAssociation CHOICE { | | | |
| eps-BearerIdentity | 6 | | |
| } | | | |
| drb-Identity | DRB-Identity using condition DRB2 | | |
| reestablishPDCP | true | | EN-DC_DRB AND Re-establish_PDCP |
| pdcp-Config | PDCP-Config | | |
| } | | | |

**Table 9.4B.1.1.4.3-4: *PDCP-Config***

| Derivation Path: TS 38.508 [6], Table 4.6.3-99 | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PDCP-Config ::= SEQUENCE { | | | |
| drb SEQUENCE { | | | |
| discardTimer | infinity | | |
| pdcp-SN-Size-UL | len18bits | | |
| pdcp-SN-Size-DL | len18bits | | |
| headerCompression CHOICE { | | | |
| notUsed | Null | | |
| } | | | |
| integrityProtection | Not present | | |
| statusReportRequired | true | | |
| outOfOrderDelivery | Not present | | |
| } | | | |
| t-Reordering | Not present | | |
| } | | | |

### 9.4B.1.2.5 Test requirement

The PDCP SDU success rate of greater than 85% shall be sustained during at least 300 frames.

## 9.4B.2

## 9.4B.3 NE-DC

### 9.4B.3.1 Sustained downlink data rate performance for NE-DC within FR1

The sustained downlink data rate performance for NR CC and E-UTRA CC along with test case details for this test case are specified in clause 9.4B.1.1.

# 10 CSI reporting requirements for interworking

## 10.1 General

This clause specifies CSI performance requirements for EN-DC, NE-DC, inter-band NR-DC between FR1 and FR2, and inter-band NR CA between FR1 and FR2.

The definition of frequency ranges (FR1 and FR2) are specified in table 5.1-1 of TS 38.101-3 [4].

For conformance testing involving FR2 test cases in this specification, the UE under test shall be pre-configured with UL Tx diversity schemes disabled to account for single polarization System Simulator (SS) in the test environment. The UE under test may transmit with dual polarization.

### 10.1.1 Applicability of requirements

The following applicability rules are specified for demodulation performance requirements for interworking:

- For UEs supporting NR/5GC, EN-DC and NE-DC,
  - The performance requirements specified in Clause 6 will be verified only for NR/5GC mode.
  - The performance requirements specified in Clause 8 will be verifiedonly for NR/5GC mode.
- The FR1 EN-DC test cases with the NR TDD DL-UL configurations which are not aligned with LTE's can be tested on the corresponding EN-DC band combinations where UE supports simultaneous transmission and reception.
- For UEs supporting NR-DC including FR1 and FR2, if the FR2 requirements in Clause 8.2, Clause 8.3 and Clause 8.4 are tested, the test coverage can be considered fulfilled without executing requirements in Clause 10.2B.2, Clause 10.3B.2 and Clause 10.4B.2.
- For UEs supporting NE-DC, the test coverage of CSI reporting requirements can be considered fulfilled, if the CSI reporting requirements in Clause 6 are executed for UE under test in the standalone mode.
- For UEs supporting NGEN-DC, the test coverage of CSI reporting requirements can be considered fulfilled, if the CSI reporting requirements in Clause 6 are executed for UE under test.
- For UEs supporting EN-DC including FR2 and/or EN-DC including FR1 and FR2, the requirements applicability is specified in Table 10.1.1-1.

**Table 10.1.1-1: Requirements applicability for UEs supporting EN-DC including FR2 and/or EN-DC including FR1 and FR2**

| Supported scenarios | CQI requirements | PMI requirements | RI requirements |
|---|---|---|---|
| EN-DC including FR2 | Clause 10.2B.1.2 | Clause 10.3B.1.2 | Clause 10.4B.1.2 |
| EN-DC including FR1 and FR2 | Clause 10.2B.1.3 | Clause 10.3B.1.3 | Clause 10.4B.1.3 |
| Both EN-DC including FR2 and EN-DC including FR1 and FR2 | Clause 10.2B.1.2 | Clause 10.3B.1.2 | Clause 10.4B.1.2 |

#### 10.1.1.1 Applicability of requirements for optional UE features

#### 10.1.1.2 Applicability of requirements for mandatory UE features with capability signalling

The applicability rule defined in Clause 6.1.1.4 shall be applied for performance requirements in Clauses 10.2B.1.1, 10.3B.1.1 and 10.4B.1.1.

The applicability rule defined in Clause 8.1.1.4 shall be applied for performance requirements in Clauses 10.2B.1.2, 10.3B.1.2 and 10.4B.1.2.

## 10.2 Void

## 10.2A Reporting of Channel Quality Indicator (CQI) for CA

FFS

## 10.2B Reporting of Channel Quality Indicator (CQI) for DC

### 10.2B.1 EN-DC

#### 10.2B.1.1 EN-DC within FR1

The NR CQI requirements and test case details for this test case are specified in Section 6.2.

During the test, only the CQI requirements on the NR cell shall be verified.

#### 10.2B.1.2 EN-DC including FR2 NR carrier

The NR CQI requirements and test case details for this test case are specified in Section 8.2.

During the test, only the CQI performance on the NR cell(s) on FR2 carriers shall be verified.

#### 10.2B.1.3 EN-DC including FR1 and FR2 NR carriers

The CSI performance requirements are verified according to section 10.2B.1.1 for EN-DC with FR1 NR carrier only and section 10.2B.1.2 for EN-DC with FR2 NR carrier only.

During the test for EN-DC with FR2 NR carriers, only the CSI performance requirements on the FR2 carriers are verified.

No CSI requirement for FR1 NR or LTE carriers is specified for EN-DC including FR2 carrier(s).

### 10.2B.2 NR DC between FR1 and FR2

FFS

## 10.3A Reporting of Precoding Matrix Indicator (PMI) for CA

FFS

# 10.3B Reporting of Precoding Matrix Indicator (PMI) for DC

## 10.3B.1 EN-DC

### 10.3B.1.1 EN-DC within FR1

The NR PMI requirements and test case details for this test case are specified in Section 6.3.

During the test, only the PMI requirements on the NR cell shall be verified.

### 10.3B.1.2 EN-DC including FR2 NR carrier

The NR PMI requirements and test case details for this test case are specified in Section 8.3.

During the test, only the PMI performance on the NR cell(s) on FR2 carriers shall be verified.

### 10.3B.1.3 EN-DC including FR1 and FR2 NR carriers

The PMI performance requirements are verified according to section 10.3B.1.1 for EN-DC with FR1 NR carrier only and section 10.3B.1.2 for EN-DC with FR2 NR carrier only.

During the test for EN-DC with FR2 NR carriers, only the PMI performance requirements on the FR2 carriers are verified.

No PMI requirement for FR1 NR or LTE carriers is specified for EN-DC including FR2 carrier(s).

## 10.3B.2 NR DC between FR1 and FR2

FFS

# 10.4A Reporting of Rank Indicator (RI) for CA

FFS

# 10.4B Reporting of Rank Indicator (RI) for DC

## 10.4B.1 EN-DC

### 10.4B.1.1 EN-DC within FR1

The NR RI requirements and test case details for this test case are specified in Section 6.4.

During the test, only the RI requirements on the NR cell shall be verified.

### 10.4B.1.2 EN-DC including FR2 NR carrier

The NR RI requirements and test case details for this test case are specified in Section 8.4.

During the test, only the RI performance on the NR cell(s) on FR2 carriers shall be verified.

### 10.4B.1.3 EN-DC including FR1 and FR2 NR carriers

The RI performance requirements are verified according to section 10.4B.1.1 for EN-DC with FR1 NR carrier only and section 10.4B.1.2 for EN-DC with FR2 NR carrier only.

During the test for EN-DC with FR2 NR carriers, only the RI performance requirements on the FR2 carriers are verified.

No RI requirement for FR1 NR or LTE carriers is specified for EN-DC including FR2 carrier(s).

### 10.4B.2 NR DC between FR1 and FR2

FFS

# Annex A (normative): Measurement channels

## A.1 General

### A.1.1 Throughput definition

The throughput values defined in the measurement channels specified in Annex A, are calculated and are valid per codeword. For multi-codeword transmissions, the throughput referenced in the minimum requirements is the sum of throughputs of all codewords.

### A.1.2 TDD UL-DL configurations for FR1

TDD UL-DL configurations for performance requirements are provided in Tables A.1.2-1, A.1.2-2, and A.1.2-3.

**Table A.1.2-1: TDD UL-DL configuration for SCS 15 kHz**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| | | | FR1.15-1 |
| TDD Slot Configuration pattern (Note 1) | | | DDDSU |
| Special Slot Configuration (Note 2) | | | 10D+2G+2U |
| *referenceSubcarrierSpacing* | | kHz | 15 |
| pattern1 | *dl-UL-TransmissionPeriodicity* | ms | 5 |
| | *nrofDownlinkSlots* | | 3 |
| | *nrofDownlinkSymbols* | | 10 |
| | *nrofUplinkSlot* | | 1 |
| | *nrofUplinkSymbols* | | 2 |
| The number of slots between PDSCH and corresponding HARQ-ACK information (Note 3) | | | 4 if mod(i,5) = 0<br>3 if mod(i,5) = 1<br>2 if mod(i,5) = 2<br>6 if mod(i,5) = 3 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,9}. | | | |

**Table A.1.2-2: TDD UL-DL configuration for SCS 30 kHz**

| Parameter | | Unit | UL-DL pattern | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | **FR1.30-1** | **FR1.30-2** | **FR1.30-3** | **FR1.30-4** | **FR1.30-5** | **FR1.30-6** |
| TDD Slot Configuration pattern (Note 1) | | | 7DS2U | DDDSU | DDDSUDDSUU | DDDSUUDDDD | DSUU | $DS_1S_2U$ |
| Special Slot Configuration (Note 2) | | | 6D+4G+4U | 10D+2G+2U | 10D+2G+2U | 6D+4G+4U | 12D+2G | S1: 10D+2G+2U S2: 12D+2G+0U |
| *referenceSubcarrierSpacing* | | kHz | 30 | 30 | 30 | 30 | 30 | 30 |
| pattern1 | | | | | | | | |
| | *dl-UL-TransmissionPeriodicity* | ms | 5 | 2.5 | 2.5 | 3 | 2 | 1 |
| | *nrofDownlinkSlots* | | 7 | 3 | 3 | 3 | 1 | 1 |
| | *nrofDownlinkSymbols* | | 6 | 10 | 10 | 6 | 12 | 10 |
| | *nrofUplinkSlot* | | 2 | 1 | 1 | 2 | 2 | 0 |
| | *nrofUplinkSymbols* | | 4 | 2 | 2 | 4 | 0 | 2 |
| pattern2 | | | | | | | | |
| | *dl-UL-TransmissionPeriodicity* | ms | N/A | N/A | 2.5 | 2 | N/A | 1 |
| | *nrofDownlinkSlots* | | N/A | N/A | 2 | 4 | N/A | 0 |
| | *nrofDownlinkSymbols* | | N/A | N/A | 10 | 0 | N/A | 12 |
| | *nrofUplinkSlot* | | N/A | N/A | 2 | 0 | N/A | 1 |
| | *nrofUplinkSymbols* | | N/A | N/A | 2 | 0 | N/A | 0 |
| The number of slots between PDSCH and corresponding HARQ-ACK information (Note 3) | | | 8 if mod(i,10) = 0<br>7 if mod(i,10) = 1<br>6 if mod(i,10) = 2<br>5 if mod(i,10) = 3<br>5 if mod(i,10) = 4<br>4 if mod(i,10) = 5<br>3 if mod(i,10) = 6<br>2 if | 4 if mod(i,5) = 0<br>3 if mod(i,5) = 1<br>2 if mod(i,5) = 2<br>6 if mod(i,5) = 3 | 4 if mod(i,10) = 0<br>3 if mod(i,10) = 1<br>2 if mod(i,10) = 2<br>5 if mod(i,10) = 3<br>3 if mod(i,10) = 5<br>3 if mod(i,10) = 6<br>2 if mod(i,10) = 7 | 5 if mod(i,10) = 0<br>4 if mod(i,10) = 1<br>3 if mod(i,10) = 2<br>2 if mod(i,10) = 3<br>8 if mod(i,10) = 6<br>7 if mod(i,10) = 7<br>6 if mod(i,10) = 8<br>5 if mod(i,10) = 9 | 3 if mod(i,4) = 0<br>2 if mod(i,4) = 1 | 3 if mod(i,4) = 0<br>2 if mod(i,4) = 1<br>3 if mod(i,4) = 2 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | mod(i,10) = 7 | | | | | |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,19} | | | | | | | |

**Table A.1.2-2a: TDD UL-DL configuration for SCS 30 kHz for DCI-based dynamic UL/DL detection**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| | | | FR1.30-1A |
| TDD Slot Configuration pattern (Note 1) | | | 7DS2U |
| Special Slot Configuration (Note 2) | | | 6D+4G+4U |
| *referenceSubcarrierSpacing* | | kHz | N/A |
| pattern1 (Note 4) | | | |
| | *dl-UL-TransmissionPeriodicity* | ms | N/A |
| | *nrofDownlinkSlots* | | N/A |
| | *nrofDownlinkSymbols* | | N/A |
| | *nrofUplinkSlot* | | N/A |
| | *nrofUplinkSymbols* | | N/A |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| PDCCH DCI Configuration | DCI Format | | 1-1 for slot indices with mod(i,10) = 0,1,2,3,4,5,6,7 |
| | Scheduled Grant | | Symbol 2-13 for slot indices with mod(i,10) = 0,1,2,3,4,5,6 and Symbol 2-5 for slot indices with mod(i,10) = 7 |
| The number of slots between PDSCH and corresponding HARQ-ACK information (Note 3) (PDSCH-to-HARQ-timing-indicator) | | | 8 if mod(i,10) = 0<br>7 if mod(i,10) = 1<br>6 if mod(i,10) = 2<br>5 if mod(i,10) = 3<br>5 if mod(i,10) = 4<br>4 if mod(i,10) = 5<br>3 if mod(i,10) = 6<br>2 if mod(i,10) = 7 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G and U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,19}<br>Note 4: Do not configure *tdd-UL-DL-ConfigurationCommon* using RRC configuration. | | | |

## A.1.3 TDD UL-DL configurations for FR2

TDD UL-DL configurations for performance requirements are provided in Tables A.1.3-1, A.1.3-2.

**Table A.1.3-1: TDD UL-DL configuration for SCS 60 kHz**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| | | | FR2.60-1 |
| TDD Slot Configuration pattern (Note 1) | | | DDSU |
| Special Slot Configuration (Note 2) | | | 11D+3G+0U |
| *referenceSubcarrierSpacing* | | kHz | 60 |
| pattern1 | *dl-UL-TransmissionPeriodicity* | ms | 1 |
| | *nrofDownlinkSlots* | | 2 |
| | *nrofDownlinkSymbols* | | 11 |
| | *nrofUplinkSlot* | | 1 |
| | *nrofUplinkSymbols* | | 0 |

| | | |
|---|---|---|
| The number of slots between PDSCH and corresponding HARQ-ACK information (Note 3) | | 3 if mod(i,4) = 0<br>2 if mod(i,4) = 1<br>5 if mod(i,4) = 2 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,39} | | |

**Table A.1.3-2: TDD UL-DL configuration for SCS 120 kHz**

| Parameter | | Unit | UL-DL pattern | |
|---|---|---|---|---|
| | | | FR2.120-1 | FR2.120-2 |
| TDD Slot Configuration pattern (Note 1) | | | DDDSU | DDSU |
| Special Slot Configuration (Note 2) | | | 10D+2G+2U | 11D+3G+0U |
| *referenceSubcarrierSpacing* | | kHz | 120 | 120 |
| pattern1 | *dl-UL-TransmissionPeriodicity* | ms | 0.625 | 0.5 |
| | *nrofDownlinkSlots* | | 3 | 2 |
| | *nrofDownlinkSymbols* | | 10 | 11 |
| | *nrofUplinkSlot* | | 1 | 1 |
| | *nrofUplinkSymbols* | | 2 | 0 |
| The number of slots between PDSCH and corresponding HARQ-ACK information(Note 3) | | | 4 if mod(i,5) = 0<br>3 if mod(i,5) = 1<br>2 if mod(i,5) = 2<br>6 if mod(i,5) = 3 | 3 if mod(i,4) = 0<br>2 if mod(i,4) = 1<br>5 if mod(i,4) = 2 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,79} | | | | |

**Table A.1.3-2a: TDD UL-DL configuration for SCS 120 kHz for DCI-based dynamic UL/DL detection**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| | | | FR2.120-1A |
| TDD Slot Configuration pattern (Note 1) | | | DDDSU |
| Special Slot Configuration (Note 2) | | | 10D+2G+2U |
| *referenceSubcarrierSpacing* | | kHz | N/A |
| pattern1 (Note 4) | *dl-UL-TransmissionPeriodicity* | ms | N/A |
| | *nrofDownlinkSlots* | | N/A |
| | *nrofDownlinkSymbols* | | N/A |
| | *nrofUplinkSlot* | | N/A |
| | *nrofUplinkSymbols* | | N/A |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| PDCCH DCI Configuration | DCI Format | | 1-1 for slot indices with mod(i,5) = 0,1,2,3 |
| | Scheduled Grant | | Symbol 1-13 for slot indices with mod(i,5) = 0,1,2 and Symbol 1-9 for slot indices with mod(i,5) = 3 |
| The number of slots between PDSCH and corresponding HARQ-ACK information(Note 3) | | | 4 if mod(i,5) = 0<br>3 if mod(i,5) = 1<br>2 if mod(i,5) = 2<br>6 if mod(i,5) = 3 |

Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.
Note 2: D, G and U denote DL, guard and UL symbols, respectively. The field is for information.
Note 3: i is the slot index per frame; i = {0,…,79}
Note 4: Do not configure *tdd-UL-DL-ConfigurationCommon* using RRC configuration.

# A.2 UL Reference measurement channels

## A.2.1 General

The measurement channels in the following subclauses are defined to test the performance requirements where PUSCH is required. The measurement channels represent example configurations of physical channels for different data rates.

# A.2.2 Reference measurement channels for FDD

## A.2.2.1 RMC for Sustained downlink data rate

### A.2.2.1.1 CP-OFDM 64QAM

**Table A.2.2.1.1-1: Reference Channels for CP-OFDM 64QAM for 15kHz SCS**

| Parameter | Channel bandwidth | Subcarrier Spacing | Allocated resource blocks | CP-OFDM Symbols per slot (Note 1) | Modulation | MCS Index (Note 2) | Target Coding Rate | Payload size | Transport block CRC | LDPC Base Graph | Number of code blocks per slot (Note 3) | Total number of bits per slot | Total modulated symbols per slot |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit | MHz | KHz | | | | | | Bits | Bits | | | Bits | |
| | 5 | 15 | 25 | 11 | 64QAM | 19 | 1/2 | 9992 | 24 | 1 | 2 | 19800 | 3300 |
| | 10 | 15 | 52 | 11 | 64QAM | 19 | 1/2 | 21000 | 24 | 1 | 3 | 41184 | 6864 |
| | 15 | 15 | 79 | 11 | 64QAM | 19 | 1/2 | 31752 | 24 | 1 | 4 | 62568 | 10428 |
| | 20 | 15 | 106 | 11 | 64QAM | 19 | 1/2 | 42016 | 24 | 1 | 5 | 83952 | 13992 |
| | 25 | 15 | 133 | 11 | 64QAM | 19 | 1/2 | 53288 | 24 | 1 | 7 | 105336 | 17556 |
| | 30 | 15 | 160 | 11 | 64QAM | 19 | 1/2 | 63528 | 24 | 1 | 8 | 126720 | 21120 |
| | 40 | 15 | 216 | 11 | 64QAM | 19 | 1/2 | 86040 | 24 | 1 | 11 | 171072 | 28512 |
| | 50 | 15 | 270 | 11 | 64QAM | 19 | 1/2 | 108552 | 24 | 1 | 13 | 213840 | 35640 |

Note 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data.
Note 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [12].
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.2.2.1.1-2: Reference Channels for CP-OFDM 64QAM for 30kHz SCS**

| Parameter | Channel bandwidth | Subcarrier Spacing | Allocated resource blocks | CP-OFDM Symbols per slot (Note 1) | Modulation | MCS Index (Note 2) | Target Coding Rate | Payload size | Transport block CRC | LDPC Base Graph | Number of code blocks per slot (Note 3) | Total number of bits per slot | Total modulated symbols per slot |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit | MHz | KHz | | | | | | Bits | Bits | | | Bits | |
| | 5 | 30 | 11 | 11 | 64QAM | 19 | 1/2 | 4352 | 24 | 1 | 1 | 8712 | 1452 |
| | 10 | 30 | 24 | 11 | 64QAM | 19 | 1/2 | 9480 | 24 | 1 | 2 | 19008 | 3168 |
| | 15 | 30 | 38 | 11 | 64QAM | 19 | 1/2 | 15112 | 24 | 1 | 2 | 30096 | 5016 |
| | 20 | 30 | 51 | 11 | 64QAM | 19 | 1/2 | 20496 | 24 | 1 | 3 | 40392 | 6732 |
| | 25 | 30 | 65 | 11 | 64QAM | 19 | 1/2 | 26120 | 24 | 1 | 4 | 51480 | 8580 |
| | 30 | 30 | 78 | 11 | 64QAM | 19 | 1/2 | 31240 | 24 | 1 | 4 | 61776 | 10296 |
| | 40 | 30 | 106 | 11 | 64QAM | 19 | 1/2 | 42016 | 24 | 1 | 5 | 83952 | 13992 |
| | 50 | 30 | 133 | 11 | 64QAM | 19 | 1/2 | 53288 | 24 | 1 | 7 | 105336 | 17556 |
| | 60 | 30 | 162 | 11 | 64QAM | 19 | 1/2 | 64552 | 24 | 1 | 8 | 128304 | 21384 |
| | 80 | 30 | 217 | 11 | 64QAM | 19 | 1/2 | 86040 | 24 | 1 | 11 | 171864 | 28644 |
| | 90 | 30 | 245 | 11 | 64QAM | 19 | 1/2 | 98376 | 24 | 1 | 12 | 194040 | 32340 |
| | 100 | 30 | 273 | 11 | 64QAM | 19 | 1/2 | 108552 | 24 | 1 | 13 | 216216 | 36036 |

Note 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data.
Note 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [12].
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

# A.2.3 Reference measurement channels for TDD

## A.2.3.1 RMC for Sustained downlink data rate

### A.2.3.1.1 CP-OFDM 16QAM

**Table A.2.3.1.1-1: Reference Channels for CP-OFDM 16QAM for 15kHz SCS**

| Parameter | Channel bandwidth | Subcarrier Spacing | Allocated resource blocks | CP-OFDM Symbols per slot (Note 1) | Modulation | MCS Index (Note 2) | Target Coding Rate | Payload size for slots 4 and 9 | Transport block CRC | LDPC Base Graph | Number of code blocks per slot for slots 4 and 9 (Note 3) | Total number of bits per slot for slots 4 and 9 | Total modulated symbols per slot for slots 4 and 9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit | MHz | KHz | | | | | | Bits | Bits | | | Bits | |
| | 5-50 | 15 | 1 | 11 | 16QAM | 10 | 1/3 | 176 | 16 | 2 | 1 | 528 | 132 |
| | 5 | 15 | 13 | 11 | 16QAM | 10 | 1/3 | 2280 | 16 | 2 | 1 | 6864 | 1716 |
| | 5 | 15 | 25 | 11 | 16QAM | 10 | 1/3 | 4352 | 24 | 1 | 1 | 13200 | 3300 |
| | 10 | 15 | 26 | 11 | 16QAM | 10 | 1/3 | 4480 | 24 | 1 | 1 | 13728 | 3432 |
| | 10 | 15 | 52 | 11 | 16QAM | 10 | 1/3 | 9224 | 24 | 1 | 2 | 27456 | 6864 |
| | 15 | 15 | 40 | 11 | 16QAM | 10 | 1/3 | 7040 | 24 | 1 | 1 | 21120 | 5280 |
| | 15 | 15 | 79 | 11 | 16QAM | 10 | 1/3 | 13832 | 24 | 1 | 2 | 41712 | 10428 |
| | 20 | 15 | 53 | 11 | 16QAM | 10 | 1/3 | 9224 | 24 | 1 | 2 | 27984 | 6996 |
| | 20 | 15 | 106 | 11 | 16QAM | 10 | 1/3 | 18432 | 24 | 1 | 3 | 55968 | 13992 |
| | 25 | 15 | 67 | 11 | 16QAM | 10 | 1/3 | 11784 | 24 | 1 | 2 | 35376 | 8844 |
| | 25 | 15 | 133 | 11 | 16QAM | 10 | 1/3 | 23040 | 24 | 1 | 3 | 70224 | 17556 |
| | 30 | 15 | 80 | 11 | 16QAM | 10 | 1/3 | 14088 | 24 | 1 | 2 | 42240 | 10560 |
| | 30 | 15 | 160 | 11 | 16QAM | 10 | 1/3 | 28168 | 24 | 1 | 4 | 84480 | 21120 |
| | 40 | 15 | 108 | 11 | 16QAM | 10 | 1/3 | 18960 | 24 | 1 | 3 | 57024 | 14256 |
| | 40 | 15 | 216 | 11 | 16QAM | 10 | 1/3 | 37896 | 24 | 1 | 5 | 114048 | 28512 |
| | 50 | 15 | 135 | 11 | 16QAM | 10 | 1/3 | 23568 | 24 | 1 | 3 | 71280 | 17820 |
| | 50 | 15 | 270 | 11 | 16QAM | 10 | 1/3 | 47112 | 24 | 1 | 6 | 142560 | 35640 |

Note 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data.
Note 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [12].
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.2.3.1.1-2: Reference Channels for CP-OFDM 16QAM for 30kHz SCS**

| Parameter | Channel bandwidth | Subcarrier Spacing | Allocated resource blocks | CP-OFDM Symbols per slot (Note 1) | Modulation | MCS Index (Note 2) | Target Coding Rate | Payload size for slots 8, 9, 18 and 19 | Transport block CRC | LDPC Base Graph | Number of code blocks per slot for slots 8, 9, 18 and 19 (Note 3) | Total number of bits per slot for slots 8, 9, 18 and 19 | Total modulated symbols per slot for slots 8, 9, 18 and 19 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit | MHz | KHz | | | | | | Bits | Bits | | | Bits | |
| | 5-50 | 30 | 1 | 11 | 16QAM | 10 | 1/3 | 176 | 16 | 2 | 1 | 528 | 132 |
| | 5 | 30 | 6 | 11 | 16QAM | 10 | 1/3 | 1064 | 16 | 2 | 1 | 3168 | 792 |
| | 5 | 30 | 11 | 11 | 16QAM | 10 | 1/3 | 1928 | 16 | 2 | 1 | 5808 | 1452 |
| | 10 | 30 | 12 | 11 | 16QAM | 10 | 1/3 | 2088 | 16 | 2 | 1 | 6336 | 1584 |
| | 10 | 30 | 24 | 11 | 16QAM | 10 | 1/3 | 4224 | 24 | 1 | 1 | 12672 | 3168 |
| | 15 | 30 | 19 | 11 | 16QAM | 10 | 1/3 | 3368 | 16 | 2 | 1 | 10032 | 2508 |
| | 15 | 30 | 38 | 11 | 16QAM | 10 | 1/3 | 6656 | 24 | 1 | 1 | 20064 | 5016 |
| | 20 | 30 | 26 | 11 | 16QAM | 10 | 1/3 | 4480 | 24 | 1 | 1 | 13728 | 3432 |
| | 20 | 30 | 51 | 11 | 16QAM | 10 | 1/3 | 8968 | 24 | 1 | 2 | 26928 | 6732 |
| | 25 | 30 | 33 | 11 | 16QAM | 10 | 1/3 | 5760 | 24 | 1 | 1 | 17424 | 4356 |
| | 25 | 30 | 65 | 11 | 16QAM | 10 | 1/3 | 11272 | 24 | 1 | 2 | 34320 | 8580 |
| | 30 | 30 | 39 | 11 | 16QAM | 10 | 1/3 | 6784 | 24 | 1 | 1 | 20592 | 5148 |
| | 30 | 30 | 78 | 11 | 16QAM | 10 | 1/3 | 13576 | 24 | 1 | 2 | 41184 | 10296 |
| | 40 | 30 | 53 | 11 | 16QAM | 10 | 1/3 | 9224 | 24 | 1 | 2 | 27984 | 6996 |
| | 40 | 30 | 106 | 11 | 16QAM | 10 | 1/3 | 18432 | 24 | 1 | 3 | 55968 | 13992 |
| | 50 | 30 | 67 | 11 | 16QAM | 10 | 1/3 | 11784 | 24 | 1 | 2 | 35376 | 8844 |
| | 50 | 30 | 133 | 11 | 16QAM | 10 | 1/3 | 23040 | 24 | 1 | 3 | 70224 | 17556 |
| | 60 | 30 | 81 | 11 | 16QAM | 10 | 1/3 | 14088 | 24 | 1 | 2 | 42768 | 10692 |
| | 60 | 30 | 162 | 11 | 16QAM | 10 | 1/3 | 28168 | 24 | 1 | 4 | 85536 | 21384 |
| | 80 | 30 | 109 | 11 | 16QAM | 10 | 1/3 | 18960 | 24 | 1 | 3 | 57552 | 14388 |
| | 80 | 30 | 217 | 11 | 16QAM | 10 | 1/3 | 37896 | 24 | 1 | 5 | 114576 | 28644 |
| | 90 | 30 | 123 | 11 | 16QAM | 10 | 1/3 | 21504 | 24 | 1 | 3 | 64944 | 16236 |
| | 90 | 30 | 245 | 11 | 16QAM | 10 | 1/3 | 43032 | 24 | 1 | 6 | 129360 | 32340 |
| | 100 | 30 | 137 | 11 | 16QAM | 10 | 1/3 | 24072 | 24 | 1 | 3 | 72336 | 18084 |
| | 100 | 30 | 273 | 11 | 16QAM | 10 | 1/3 | 48168 | 24 | 1 | 6 | 144144 | 36036 |

Note 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data.
Note 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [12].
Note 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

# A.3 DL reference measurement channels

## A.3.1 General

The transport block size (TBS) determination procedure is described in clause 5.1.3.2 of TS 38.214 [12].

Unless otherwise stated, no user data is scheduled on slot #0 within 20 ms in order to avoid SSB and PDSCH transmissions in one slot and simplify test configuration.

## A.3.2 Reference measurement channels for PDSCH performance requirements

For PDSCH reference channels if more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).

### A.3.2.1 FDD

#### A.3.2.1.1 Reference measurement channels for SCS 15 kHz FR1

**Table A.3.2.1.1-1: PDSCH Reference Channel for FDD (QPSK)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-1.1 FDD | R.PDSCH.1-1.2 FDD | R.PDSCH.1-1.3 FDD | R.PDSCH.1-1.4 FDD | |
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | |
| Subcarrier spacing | kHz | 15 | 15 | 15 | 15 | |
| Number of allocated resource blocks | PRBs | 52 | 6 | 52 | 52 | |
| Number of consecutive PDSCH symbols | | 12 | 12 | 7 | 12 | |
| Allocated slots per 2 frames | Slots | 19 | 19 | 19 | 19 | |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAMLowSE | |
| MCS index | | 4 | 4 | 4 | 14 | |
| Modulation | | QPSK | QPSK | QPSK | QPSK | |
| Target Coding Rate | | 0.30 | 0.30 | 0.30 | 0.59 | |
| Number of MIMO layers | | 1 | 1 | 1 | 1 | |
| Number of DMRS REs | | 18 | 12 | 12 | 12 | |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | Bits | 3904 | 480 | 2280 | 8064 | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | Bits | 24 | 16 | 16 | 24 | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | CBs | 1 | 1 | 1 | 1 | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 10, 11 | Bits | 12480 | 1512 | 6864 | 13104 | |
| For Slots i = 1,…, 9, 12, …, 19 | Bits | 13104 | 1584 | 7488 | 13728 | |
| Max. Throughput averaged over 2 frames | Mbps | 3.709 | 0.456 | 2.166 | 7.661 | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.1.1-2: PDSCH Reference Channel for FDD (16QAM)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-2.1 FDD | R.PDSCH.1-2.2 FDD | R.PDSCH.1-2.3 FDD | R.PDSCH.1-2.4 FDD | R.PDSCH.1-2.5 FDD | R.PDSCH.1-2.6 FDD |
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | 10 | 10 |
| Subcarrier spacing | kHz | 15 | 15 | 15 | 15 | 15 | 15 |
| Number of allocated resource blocks | PRBs | 52 | 52 | 52 | 52 | 52 | 52 |
| Number of consecutive PDSCH symbols | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated slots per 2 frames | Slots | 19 | 19 | 19 | 19 | 19 | 19 |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAMLowSE | 64QAM |
| MCS index | | 13 | 13 | 13 | 13 | 19 | 16 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding Rate | | 0.48 | 0.48 | 0.48 | 0.48 | 0.54 | 0.64 |
| Number of MIMO layers | | 1 | 2 | 3 | 4 | 2 | 1 |
| Number of DMRS REs | | 12 | 12 | 24 | 24 | 12 | 12 |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | 0 | 0 |
| Information Bit Payload per Slot | | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slots i = 1,…, 19 | Bits | 13064 | 26120 | 35856 | 48168 | 29704 | 17424 |
| Transport block CRC per Slot | | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slots i = 1,…, 19 | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Slot | | | | | | | |
| For Slot i = 0 | CBs | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slots i = 1,…, 19 | CBs | 2 | 4 | 5 | 6 | 4 | 3 |
| Binary Channel Bits Per Slot | | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slots i = 10, 11 | Bits | 26208 | 52416 | 71136 | 94848 | 49920 | 26208 |
| For Slots i = 1,…, 9, 12, …, 19 | Bits | 27456 | 54912 | 74880 | 99840 | 54912 | 27456 |
| Max. Throughput averaged over 2 frames | Mbps | 12.411 | 24.814 | 34.063 | 45.760 | 28.219 | 16.553 |
| NOTE 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>NOTE 2: Slot i is slot index per 2 frames. | | | | | | | |

**Table A.3.2.1.1-3: PDSCH Reference Channel for FDD (64QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-3.1 FDD | R.PDSCH.1-3.2 FDD | R.PDSCH.1-3.3 FDD | R.PDSCH.1-3.4 FDD | |
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | |
| Subcarrier spacing | kHz | 15 | 15 | 15 | 15 | |
| Number of allocated resource blocks | PRBs | 52 | 52 | 26 (Note 3) | 26 (Note 4) | |
| Number of consecutive PDSCH symbols | | 12 | 12 | 12 | 12 | |
| Allocated slots per 2 frames | Slots | 19 | 19 | 19 | 19 | |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAM | |
| MCS index | | 19 | 19 | 19 | 19 | |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM | |
| Target Coding Rate | | 0.51 | 0.51 | 0.51 | 0.51 | |
| Number of MIMO layers | | 2 | 2 | 2 | 2 | |
| Number of DMRS REs | | 12 | 24 | 24 | 24 | |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | Bits | 42016 | 37896 | 18960 | 18960 | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | Bits | 24 | 24 | 24 | 24 | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | CBs | 5 | 5 | 3 | 3 | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 10, 11 | Bits | 78624 | 67392 | 33696 | 33696 | |
| For Slots i = 1,…, 9, 12, …, 19 | Bits | 82368 | 74880 | 37440 | 37440 | |
| Max. Throughput averaged over 2 frames | Mbps | 39.915 | 36.001 | 18.012 | 18.012 | |

Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.
Note 2: Slot i is slot index per 2 frames.
Note 3: PDSCH is scheduled in PRB numbers from 0 to 25.

Note 4: PDSCH is scheduled in PRB numbers from 26 to 51.

**Table A.3.2.1.1-4: PDSCH Reference Channel for FDD (256QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-4.1 FDD | | | | |
| Channel bandwidth | MHz | 10 | | | | |
| Subcarrier spacing | kHz | 15 | | | | |
| Number of allocated resource blocks | PRBs | 52 | | | | |
| Number of consecutive PDSCH symbols | | 12 | | | | |
| Allocated slots per 2 frames | Slots | 19 | | | | |
| MCS table | | 256QAM | | | | |
| MCS index | | 24 | | | | |
| Modulation | | 256QAM | | | | |
| Target Coding Rate | | 0.82 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 19 | Bits | 45096 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 19 | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | | | | |
| For Slots i = 1,…, 19 | CBs | 6 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 10, 11 | Bits | 52416 | | | | |
| For Slots i = 1,…, 9, 12, …, 19 | Bits | 54912 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 42.841 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.1.1-5: PDSCH Reference Channel for FDD and CSI-RS overlapped with PDSCH**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-5.1 FDD | | | | |
| Channel bandwidth | MHz | 10 | | | | |
| Subcarrier spacing | kHz | 15 | | | | |
| Number of allocated resource blocks | PRBs | 52 | | | | |
| Number of consecutive PDSCH symbols | | 12 | | | | |
| Allocated slots per 2 frames | Slots | 19 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 13 | | | | |
| Modulation | | 16QAM | | | | |
| Target Coding Rate | | 0.48 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 19 | Bits | 26120 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 19 | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | | | | |
| For Slots i = 1,…, 19 | CBs | 4 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 5, 15 | Bits | 50752 | | | | |
| For Slots i = 10 | Bits | 48256 | | | | |
| For Slots i = 11 | Bits | 52416 | | | | |
| For Slots i = 1,…,4,6,…, 9,12,…14,16,…,19 | Bits | 54912 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 24.814 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.1.1-6: PDSCH Reference Channel for FDD PMI reporting requirements**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1 -6.1 FDD | R.PDSCH.1 -6.2 FDD | R.PDSCH.1 -6.3 FDD | | |
| Channel bandwidth | MHz | 10 | 10 | 10 | | |
| Subcarrier spacing | kHz | 15 | 15 | 15 | | |
| Number of allocated resource blocks | PRBs | 52 | 52 | 52 | | |
| Number of consecutive PDSCH symbols | | 12 | 12 | 12 | | |
| Allocated slots per 2 frames | Slots | 15 | 15 | 15 | | |
| MCS table | | 64QAM | 64QAM | 64QAM | | |
| MCS index | | 13 | 13 | 20 | | |
| Modulation | | 16QAM | 16QAM | 64QAM | | |
| Target Coding Rate | | 0.48 | 0.48 | 0.55 | | |
| Number of MIMO layer | | 1 | 2 | 2 | | |
| Number of DMRS REs (Note 3) | | 24 | 24 | 24 | | |
| Overhead for TBS determination | | 0 | 0 | 0 | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | | |
| For CSI Slots i, if mod (i,5) =1, i={0,…,19} | | N/A | N/A | N/A | | |
| For Non CSI-RS Slot i, if mod (i,5) ={0,2,3,4}, i={1,..19} | Bits | 12040 | 24072 | 40976 | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | | |
| For CSI Slots i, if mod (i,5) =1, i={0,…,19} | | N/A | N/A | N/A | | |
| For Non CSI-RS Slot i, if mod (i,5) ={0,2,3,4}, i={1,..19} | Bits | 24 | 24 | 24 | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | N/A | N/A | | |
| For CSI Slots i, if mod (i,5) =1, i={0,…,19} | | N/A | N/A | N/A | | |
| For Non CSI-RS Slot i, if mod (i,5) ={0,2,3,4}, i={1,..,19} | CBs | 2 | 3 | 5 | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | | |
| For CSI Slots i, if mod (i,5) =1, i={0,…,19} | | N/A | N/A | N/A | | |
| For Slots i = 10 | Bits | 23712 | 47424 | 71136 | | |
| For Non CSI-RS Slot i, if mod (i,5) ={0,2,3,4}, i={1,..9,11,…,19} | Bits | 24960 | 49920 | 74880 | | |
| Max. Throughput averaged over 2 frames | Mbps | 9.030 | 18.054 | 30.732 | | |

Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.
Note 2: Slot i is slot index per 2 frames.
Note 3: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data.

**Table A.3.2.1.1-7: PDSCH Reference Channel for FDD LTE-NR coexistence scenario**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-7.1 FDD | R.PDSCH.1-7.2 FDD | | | |
| Channel bandwidth | MHz | 10 | 10 | | | |
| Subcarrier spacing | kHz | 15 | 15 | | | |
| Number of allocated resource blocks | PRBs | 52 | 52 | | | |
| Number of consecutive PDSCH symbols | | 9 | 11 | | | |
| Allocated slots per 2 frames | Slots | 16 | 16 | | | |
| MCS table | | 64QAM | 64QAM | | | |
| MCS index | | 4 | 4 | | | |
| Modulation | | QPSK | QPSK | | | |
| Target Coding Rate | | 0.30 | 0.30 | | | |
| Number of MIMO layers | | 1 | 1 | | | |
| Number of DMRS REs | | 12 | 12 | | | |
| Overhead for TBS determination | | 18 | 18 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots i = 0,5,10,15 | Bits | N/A | N/A | | | |
| For Slots i, if mod(i, 5) = {1,2,3,4} for i from {0,…,19} | Bits | 2472 | 3240 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots i = 0,5,10,15 | Bits | N/A | N/A | | | |
| For Slots i, if mod(i, 5) = {1,2,3,4} for i from {0,…,19} | Bits | 16 | 16 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots i = 0,5,10,15 | CBs | N/A | N/A | | | |
| For Slots i, if mod(i, 5) = {1,2,3,4} for i from {0,…,19} | CBs | 1 | 1 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots i = 0,5,10,15 | Bits | N/A | N/A | | | |
| For Slots i = 11 | Bits | 7760 | 10256 | | | |
| For Slots i, if mod(i, 5) = {1,2,3,4} for i from {1,…, 9, 12, …, 19} | Bits | 8384 | 10880 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 1.978 | 2.592 | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames<br>Note 3: No user data is scheduled on slots with LTE PBCH/PSS/SSS | | | | | | |

**Table A.3.2.1.1-8: PDSCH Reference Channel for FDD HST scenario**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-8.1 FDD | R.PDSCH.1-8.2 FDD | R.PDSCH.1-8.3 FDD | R.PDSCH.1-8.4 FDD | |
| Channel bandwidth | MHz | 10 | 10 | 10 | 10 | |
| Subcarrier spacing | kHz | 15 | 15 | 15 | 15 | |
| Number of allocated resource blocks | PRBs | 52 | 52 | 52 | 52 | |
| Number of consecutive PDSCH symbols | | 12 | 12 | 12 | 12 | |
| Allocated slots per 2 frames | Slots | 19 | 19 | 19 | 19 | |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAM | |
| MCS index | | 13 | 17 | 13 | 17 | |
| Modulation | | 16QAM | 64QAM | 16QAM | 64QAM | |
| Target Coding Rate | | 0.48 | 0.43 | 0.48 | 0.43 | |
| Number of MIMO layers | | 1 | 1 | 2 | 2 | |
| Number of DMRS REs | | 18 | 18 | 18 | 18 | |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | Bits | 12552 | 16896 | 25104 | 28680 | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | Bits | 24 | 24 | 24 | 24 | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | N/A | N/A | N/A | |
| For Slots i = 1,…, 19 | CBs | 2 | 3 | 3 | 4 | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 1,2,11,12 | Bits | 24960 | 37440 | 51168 | 76752 | |
| For Slots i = 3,…, 10, 13, …, 19 | Bits | 26208 | 39312 | 52416 | 78624 | |
| Max. Throughput averaged over 2 frames | Mbps | 11.924 | 16.0512 | 23.8488 | 27.246 | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

**Table A.3.2.1.1-9: PDSCH Reference Channel for FDD CC and CA scenario**

FFS

**Table A.3.2.1.1-10: PDSCH Reference Channel for FDD CC and CA scenario**

FFS

**Table A.3.2.1.1-11: PDSCH Reference Channel for FDD**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-11.1 FDD | R.PDSCH.1-11.2 FDD | | | |
| Channel bandwidth | MHz | 10 | 10 | | | |
| Subcarrier spacing | kHz | 15 | 15 | | | |
| Number of allocated resource blocks | PRBs | 52 | 52 | | | |
| Number of consecutive PDSCH symbols | | 12 | 12 | | | |
| Allocated slots per 2 frames | Slots | 18 | 18 | | | |
| MCS table | | 64QAMLowSE | 64QAMLowSE | | | |
| MCS index | | 19 | 19 | | | |
| Modulation | | 16QAM | 16QAM | | | |
| Target Coding Rate | | 0.54 | 0.54 | | | |
| Number of MIMO layers | | 1 | 1 | | | |
| Number of DMRS REs | | 12 | 12 | | | |
| Overhead for TBS determination | | 0 | 0 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0,1 | Bits | N/A | N/A | | | |
| For Slots i = 2,…, 19 | Bits | 14856 | 14856 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0,1 | Bits | N/A | N/A | | | |
| For Slots i = 2,…, 19 | Bits | 24 | 24 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0,1 | CBs | N/A | N/A | | | |
| For Slots i = 2,…, 19 | CBs | 2 | 2 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0,1 | Bits | N/A | N/A | | | |
| For Slots i = 10, 11 | Bits | 26208 | 24960 | | | |
| For Slots i =2,…, 9, 12, …, 19 | Bits | 27456 | 27456 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 6.685 (NOTE 3) | 6.685 (NOTE 4) | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames<br>Note 3: Throughput is calculated under assumption of aggregation factor 2.<br>Note 4: Throughput is calculated under assumption of repetition number 2. | | | | | | |

**Table A.3.2.1.1-12: PDSCH Reference Channel for FDD**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-12.1 FDD | | | | |
| Channel bandwidth | MHz | 10 | | | | |
| Subcarrier spacing | kHz | 15 | | | | |
| Number of allocated resource blocks | PRBs | 52 | | | | |
| Number of consecutive PDSCH symbols | | 2 | | | | |
| Allocated slots per 2 frames | Slots | 19 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.3 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | 6 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 19 | Bits | 576 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 19 | Bits | 16 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | | | | |
| For Slots i = 1,…, 19 | CBs | 1 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 10, 11 | Bits | 1872 | | | | |
| For Slots i =1,…, 9, 12, …, 19 | Bits | 1872 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 0.547 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

### A.3.2.1.2 Reference measurement channels for SCS 30 kHz FR1

**Table A.3.2.1.2-1: PDSCH Reference Channel for FDD (64QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-1.1 FDD | | | | |
| Channel bandwidth | MHz | 20 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Number of allocated resource blocks | PRBs | 51 | | | | |
| Number of consecutive PDSCH symbols | | 12 | | | | |
| Allocated slots per 2 frames | Slots | 39 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 19 | | | | |
| Modulation | | 64QAM | | | | |
| Target Coding Rate | | 0.51 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 39 | Bits | 40976 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 1,…, 39 | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i = 0 | CBs | N/A | | | | |
| For Slots i = 1,…, 39 | CBs | 5 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i = 0 | Bits | N/A | | | | |
| For Slots i = 20, 21 | Bits | 77112 | | | | |
| For Slots i = 1,…, 19, 22, …, 39 | Bits | 80784 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 79.903 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

### A.3.2.1.3 Reference measurement channels for SCS 60 kHz FR1

### A.3.2.1.4 Reference measurement channels for E-UTRA

**Table A.3.2.1.4-1: PDSCH Reference Channel for sustained data-rate test (64QAM, 2 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-1.1 FDD | R.PDSCH.4-1.2 FDD | R.PDSCH.4-1.3 FDD | R.PDSCH.4-1.4 FDD |
| Channel bandwidth | MHz | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 |
| Allocated subframes per Radio Frame | | 9 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9, | | 0.85 | 0.85 | 0.85 | 0.88 |
| For Sub-Frame 5 | | N/A | 0.89 | 0.91 | 0.87 |
| For Sub-Frame 0 | | 0.83 | 0.90 | 0.88 | 0.90 |
| Information Bit Payload (Note 3) | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 18336 | 36696 | 55056 | 75376 |
| For Sub-Frame 5 | Bits | N/A | 35160 | 52752 | 71112 |
| For Sub-Frame 0 | Bits | 15840 | 36696 | 55056 | 75376 |
| Number of Code Blocks (Notes 3 and 4) | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | CBs | 3 | 6 | 9 | 13 |
| For Sub-Frame 5 | CBs | N/A | 6 | 9 | 12 |
| For Sub-Frame 0 | CBs | 3 | 6 | 9 | 13 |
| Binary Channel Bits (Note 3) | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 21600 | 43200 | 64800 | 86400 |
| For Sub-Frame 5 | Bits | N/A | 39744 | 60480 | 82080 |
| For Sub-Frame 0 | Bits | 19152 | 40752 | 62352 | 83952 |
| Number of layers | | 2 | 2 | 2 | 2 |
| Max. Throughput averaged over 1 frame (Note 3) | Mbps | 16.253 | 36.542 | 54.826 | 74.950 |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: Given per component carrier per codeword.
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 6: Resource blocks $n_{PRB}$ = 0..24 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 8: Resource blocks nPRB = 4..74 are allocated for the user data in sub-frame 5, and resource blocks nPRB = 0..74 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9.

**Table A.3.2.1.4-2: PDSCH Reference Channel for sustained data-rate test (64QAM, 4 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-2.1 FDD | R.PDSCH.4-2.2 FDD | R.PDSCH.4-2.3 FDD | R.PDSCH.4-2.4 FDD |
| Channel bandwidth | MHz | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 |
| Allocated subframes per Radio Frame | | 9 | 10 | 10 | 10 |
| Modulation | | 64QAM | 64QAM | 64QAM | 64QAM |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9, | | 0.78 | 0.78 | 0.77 | 0.79 |
| For Sub-Frame 5 | | N/A | 0.80 | 0.79 | 0.81 |
| For Sub-Frame 0 | | 0.85 | 0.83 | 0.8 | 0.81 |
| Information Bit Payload (Note 3) | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 31704 | 63776 | 93800 | 128496 |
| For Sub-Frame 5 | Bits | N/A | 59256 | 90816 | 124464 |
| For Sub-Frame 0 | Bits | 30576 | 63776 | 93800 | 128496 |
| Number of Code Blocks (Notes 3 and 4) | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | CBs | 6 | 11 | 16 | 21 |
| For Sub-Frame 5 | CBs | N/A | 10 | 15 | 21 |
| For Sub-Frame 0 | CBs | 5 | 11 | 16 | 21 |
| Binary Channel Bits (Note 3) | | | | | |
| For Sub-Frames 1,2,3,4,6,7,8,9 | Bits | 40800 | 81600 | 122400 | 163200 |
| For Sub-Frame 5 | Bits | N/A | 74976 | 114144 | 154944 |
| For Sub-Frame 0 | Bits | 36192 | 76992 | 117792 | 158592 |
| Number of layers | | 4 | 4 | 4 | 4 |
| Max. Throughput averaged over 1 frame (Note 3) | Mbps | 28.421 | 63.324 | 93.502 | 128.093 |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: Given per component carrier per codeword.
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 6: Resource blocks $n_{PRB}$ = 0..24 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 8: Resource blocks nPRB = 4..74 are allocated for the user data in sub-frame 5, and resource blocks nPRB = 0..74 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9.

**Table A.3.2.1.4-3: PDSCH Reference Channel for sustained data-rate test (256QAM, 2 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-3.1 FDD | R.PDSCH.4-3.2 FDD | R.PDSCH.4-3.3 FDD | R.PDSCH.4-3.4 FDD |
| Channel bandwidth | MHz | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 |
| Modulation | | 256QAM | 256QAM | 256QAM | 256QAM |
| Coding Rate | | | | | |
| For Sub-Frames 3,4,8,9 | | 0.85 | 0.85 | 0.88 | 0.85 |
| For Sub-Frames 1,2,6,7 | | 0.77 | 0.74 | 0.74 | 0.74 |
| For Sub-Frame 5 | | 0.79 | 0.77 | 0.77 | 0.75 |
| For Sub-Frame 0 | | 0.84 | 0.78 | 0.77 | 0.76 |
| Information Bit Payload (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 24496 | 48936 | 75376 | 97896 |
| For Sub-Frames 1,2,6,7 | Bits | 21384 | 42368 | 63776 | 84760 |
| For Sub-Frame 5 | Bits | 19848 | 40576 | 61664 | 81176 |
| For Sub-Frame 0 | Bits | 21384 | 42368 | 63776 | 84760 |
| Number of Code Blocks (Notes 3 and 4) | | | | | |
| For Sub-Frames 3,4,8,9 | CBs | 4 | 8 | 13 | 16 |
| For Sub-Frames 1,2,6,7 | CBs | 4 | 7 | 11 | 14 |
| For Sub-Frame 5 | CBs | 4 | 7 | 11 | 14 |
| For Sub-Frame 0 | CBs | 4 | 7 | 11 | 14 |
| Binary Channel Bits (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 28800 | 57600 | 86400 | 115200 |
| For Sub-Frames 1,2,6,7 | Bits | 28800 | 57600 | 86400 | 115200 |
| For Sub-Frame 5 | Bits | 25344 | 52992 | 80640 | 109440 |
| For Sub-Frame 0 | Bits | 25536 | 54336 | 83136 | 111936 |
| Number of layers | | 2 | 2 | 2 | 2 |
| Max. Throughput averaged over 1 frame (Note 3) | Mbps | 22.475 | 44.816 | 68.205 | 89.656 |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: Given per component carrier per codeword.
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 6: Resource blocks $n_{PRB}$ = 2..24 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..24 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 8: Resource blocks nPRB = 4..74 are allocated for the user data in sub-frame 5, and resource blocks nPRB = 0..74 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9.

**Table A.3.2.1.4-4: PDSCH Reference Channel for sustained data-rate test (256QAM, 4 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-4.1 FDD | R.PDSCH.4-4.2 FDD | R.PDSCH.4-4.3 FDD | R.PDSCH.4-4.4 FDD |
| Channel bandwidth | MHz | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 |
| Modulation | | 256QAM | 256QAM | 256QAM | 256QAM |
| Coding Rate | | | | | |
| For Sub-Frames 3,4,8,9 | | 0.85 | 0.78 | 0.79 | 0.78 |
| For Sub-Frames 1,2,6,7 | | 0.77 | 0.78 | 0.79 | 0.78 |
| For Sub-Frame 5 | | 0.79 | 0.82 | 0.82 | 0.786 |
| For Sub-Frame 0 | | 0.84 | 0.83 | 0.82 | 0.80 |
| Information Bit Payload (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 42368 | 84760 | 128496 | 169544 |
| For Sub-Frames 1,2,6,7 | Bits | 42368 | 84760 | 128496 | 169544 |
| For Sub-Frame 5 | Bits | 39232 | 81176 | 124464 | 161760 |
| For Sub-Frame 0 | Bits | 39232 | 84760 | 128496 | 169544 |
| Number of Code Blocks (Notes 3 and 4) | | | | | |
| For Sub-Frames 3,4,8,9 | CBs | 7 | 14 | 21 | 28 |
| For Sub-Frames 1,2,6,7 | CBs | 7 | 14 | 21 | 28 |
| For Sub-Frame 5 | CBs | 7 | 14 | 21 | 27 |
| For Sub-Frame 0 | CBs | 7 | 14 | 21 | 28 |
| Binary Channel Bits (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 54400 | 108800 | 163200 | 217600 |
| For Sub-Frames 1,2,6,7 | Bits | 54400 | 108800 | 163200 | 217600 |
| For Sub-Frame 5 | Bits | 47744 | 99968 | 152192 | 206592 |
| For Sub-Frame 0 | Bits | 48256 | 102656 | 157056 | 211456 |
| Number of layers | | 4 | 4 | 4 | 4 |
| Max. Throughput averaged over 1 frame (Note 3) | Mbps | 41.741 | 84.4016 | 128.093 | 168.766 |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: Given per component carrier per codeword.
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 6: Resource blocks $n_{PRB}$ = 2..24 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..24 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 8: Resource blocks nPRB = 4..74 are allocated for the user data in sub-frame 5, and resource blocks nPRB = 0..74 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9.

**Table A.3.2.1.4-5: PDSCH Reference Channel for sustained data-rate test (1024QAM, 2 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-5.1 FDD | R.PDSCH.4-5.2 FDD | R.PDSCH.4-5.3 FDD | R.PDSCH.4-5.4 FDD |
| Channel bandwidth | MHz | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 |
| Modulation | | 1024QAM | 1024QAM | 1024QAM | 1024QAM |
| Coding Rate | | | | | |
| For Sub-Frames 3,4,8,9 | | 0.76 | 0.73 | 0.75 | 0.76 |
| For Sub-Frames 1,2,6,7 | | 0.76 | 0.73 | 0.75 | 0.76 |
| For Sub-Frame 5 | | 0.80 | 0.77 | 0.78 | 0.77 |
| For Sub-Frame 0 | | 0.86 | 0.78 | 0.78 | 0.79 |
| Information Bit Payload (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 27376 | 52752 | 81176 | 110136 |
| For Sub-Frames 1,2,6,7 | Bits | 27376 | 52752 | 81176 | 110136 |
| For Sub-Frame 5 | Bits | 25456 | 51024 | 78704 | 105528 |
| For Sub-Frame 0 | Bits | 27376 | 52752 | 81176 | 110136 |
| Number of Code Blocks (Notes 3 and 4) | | | | | |
| For Sub-Frames 3,4,8,9 | CBs | 5 | 9 | 14 | 18 |
| For Sub-Frames 1,2,6,7 | CBs | 5 | 9 | 14 | 18 |
| For Sub-Frame 5 | CBs | 5 | 9 | 13 | 18 |
| For Sub-Frame 0 | CBs | 5 | 9 | 14 | 18 |
| Binary Channel Bits (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 36000 | 72000 | 108000 | 144000 |
| For Sub-Frames 1,2,6,7 | Bits | 36000 | 72000 | 108000 | 144000 |
| For Sub-Frame 5 | Bits | 31680 | 66240 | 100800 | 136800 |
| For Sub-Frame 0 | Bits | 31920 | 67920 | 103920 | 139920 |
| Number of layers | | 2 | 2 | 2 | 2 |
| Max. Throughput averaged over 1 frame (Note 3) | Mbps | 27.18 | 52.58 | 80.93 | 109.68 |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: Given per component carrier per codeword.
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 6: Resource blocks $n_{PRB}$ = 2..24 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..24 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 8: Resource blocks nPRB = 4..74 are allocated for the user data in sub-frame 5, and resource blocks nPRB = 0..74 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9.

**Table A.3.2.1.4-6: PDSCH Reference Channel for sustained data-rate test (1024QAM, 4 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-6.1 FDD | R.PDSCH.4-6.2 FDD | R.PDSCH.4-6.3 FDD | R.PDSCH.4-6.4 FDD |
| Channel bandwidth | MHz | 5 | 10 | 15 | 20 |
| Allocated resource blocks | | Note 6 | Note 7 | Note 8 | Note 9 |
| Allocated subframes per Radio Frame | | 10 | 10 | 10 | 10 |
| Modulation | | 1024QAM | 1024QAM | 1024QAM | 1024QAM |
| Coding Rate | | | | | |
| For Sub-Frames 3,4,8,9 | | 0.78 | 0.81 | 0.79 | 0.81 |
| For Sub-Frames 1,2,6,7 | | 0.78 | 0.81 | 0.79 | 0.81 |
| For Sub-Frame 5 | | 0.82 | 0.81 | 0.83 | 0.82 |
| For Sub-Frame 0 | | 0.87 | 0.86 | 0.82 | 0.83 |
| Information Bit Payload (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 52752 | 110136 | 161760 | 220296 |
| For Sub-Frames 1,2,6,7 | Bits | 52752 | 110136 | 161760 | 220296 |
| For Sub-Frame 5 | Bits | 48936 | 101840 | 157432 | 211936 |
| For Sub-Frame 0 | Bits | 52752 | 110136 | 161760 | 220296 |
| Number of Code Blocks (Notes 3 and 4) | | | | | |
| For Sub-Frames 3,4,8,9 | CBs | 9 | 18 | 27 | 36 |
| For Sub-Frames 1,2,6,7 | CBs | 9 | 18 | 27 | 36 |
| For Sub-Frame 5 | CBs | 8 | 17 | 26 | 35 |
| For Sub-Frame 0 | CBs | 9 | 18 | 27 | 36 |
| Binary Channel Bits (Note 3) | | | | | |
| For Sub-Frames 3,4,8,9 | Bits | 68000 | 136000 | 204000 | 272000 |
| For Sub-Frames 1,2,6,7 | Bits | 68000 | 136000 | 204000 | 272000 |
| For Sub-Frame 5 | Bits | 59680 | 124960 | 190240 | 258240 |
| For Sub-Frame 0 | Bits | 60320 | 128320 | 196320 | 264320 |
| Number of layers | | 4 | 4 | 4 | 4 |
| Max. Throughput averaged over 1 frame (Note 3) | Mbps | 52.37 | 109.31 | 161.33 | 219.46 |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: Given per component carrier per codeword.
Note 4: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 5: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 6: Resource blocks $n_{PRB}$ = 2..24 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..24 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 8: Resource blocks nPRB = 4..74 are allocated for the user data in sub-frame 5, and resource blocks nPRB = 0..74 in sub-frames 0,1,2,3,4,6,7,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,1,2,3,4,6,7,8,9.

## A.3.2.2 TDD

### A.3.2.2.1 Reference measurement channels for SCS 15 kHz FR1

**Table A.3.2.2.1-1: PDSCH Reference Channel for TDD UL-DL pattern FR1.15-1 and LTE-NR coexistence scenario**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-1.1 TDD | R.PDSCH.1-1.2 TDD | | | |
| Channel bandwidth | MHz | 10 | 10 | | | |
| Subcarrier spacing | kHz | 15 | 15 | | | |
| Allocated resource blocks | PRBs | 52 | 52 | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,19} | | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,19} | | 9 | 11 | | | |
| Allocated slots per 2 frames | | 7 | 7 | | | |
| MCS table | | 64QAM | 64QAM | | | |
| MCS index | | 4 | 4 | | | |
| Modulation | | QPSK | QPSK | | | |
| Target Coding Rate | | 0.30 | 0.30 | | | |
| Number of MIMO layers | | 1 | 1 | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,19} | | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,19} | | 12 | 12 | | | |
| Overhead for TBS determination | | 18 | 18 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,19} | Bits | 2472 | 3240 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,19} | Bits | 16 | 16 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | CBs | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,19} | CBs | 1 | 1 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | Bits | N/A | N/A | | | |
| For Slots i = 10, 11 | Bits | 7760 | 10256 | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,9,12,…,19} | Bits | 8384 | 10880 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 0.865 | 1.134 | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames<br>Note 3: No user data is scheduled on slots with LTE PBCH/PSS/SSS | | | | | | |

Editor's note: this old replacd table shall be deleted

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.1-1.1 TDD | R.PDSCH.1-1.2 TDD | | | |
| Channel bandwidth | MHz | 10 | 10 | | | |
| Subcarrier spacing | kHz | 15 | 15 | | | |
| Allocated resource blocks | PRBs | 52 | 52 | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,19} | | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,19} | | 9 | 11 | | | |
| Allocated slots per 2 frames | | 7 | 7 | | | |
| MCS table | | 64QAM | 64QAM | | | |
| MCS index | | 4 | 4 | | | |
| Modulation | | QPSK | QPSK | | | |
| Target Coding Rate | | 0.30 | 0.30 | | | |
| Number of MIMO layers | | 1 | 1 | | | |
| Number of DMRS REs | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,19} | | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,19} | | 12 | 12 | | | |
| Overhead for TBS determination | | 18 | 18 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,19} | Bits | 2472 | 3240 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,19} | Bits | 16 | 16 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | CBs | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,19} | CBs | 1 | 1 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = {2,3,4} for i from {0,…,19} | Bits | N/A | N/A | | | |
| For Slots i = 10, 11 | Bits | 7760 | 10256 | | | |
| For Slot i, if mod(i, 5) = {0,1} for i from {1,…,9,12,…,19} | Bits | 8384 | 10880 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 0.865 | 1.134 | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames.<br>Note 3: No user data is scheduled on slots with LTE PBCH/PSS/SSS. | | | | | | |

### A.3.2.2.2 Reference measurement channels for SCS 30 kHz FR1

**Table A.3.2.2.2-1: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1 and FR1.30-1A (QPSK)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-1.1 TDD | R.PDSCH.2-1.2 TDD | R.PDSCH.2-1.3 TDD | R.PDSCH.2-1.4 TDD | |
| Channel bandwidth | MHz | 40 | 40 | 40 | 40 | |
| Subcarrier spacing | kHz | 30 | 30 | 30 | 30 | |
| Allocated resource blocks | PRBs | 106 | 6 | 106 | 106 | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | 4 | N/A | N/A | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | 7 | 12 | |
| Allocated slots per 2 frames | | 31 | 31 | 27 | 27 | |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAMLowSE | |
| MCS index | | 4 | 4 | 4 | 14 | |
| Modulation | | QPSK | QPSK | QPSK | QPSK | |
| Target Coding Rate | | 0.30 | 0.30 | 0.30 | 0.59 | |
| Number of MIMO layers | | 1 | 1 | 1 | 1 | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | 6 | N/A | N/A | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 18 | 12 | 12 | 12 | |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 2664 | 144 | N/A | N/A | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 8064 | 480 | 4608 | 16392 | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 16 | 16 | N/A | N/A | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 24 | 16 | 24 | 24 | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 1 | 1 | N/A | N/A | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | CBs | 1 | 1 | 1 | 2 | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 20, 21 | Bits | 25440 | 1512 | 13992 | 26712 | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 8904 | 504 | N/A | N/A | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 26712 | 1584 | 15264 | 27984 | |
| Max. Throughput averaged over 2 frames | Mbps | 11.419 | 0.677 | 6.221 | 22.129 | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.2-2: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1 (16QAM)**

| Parameter | | Unit | Value | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-2.1 TDD | R.PDSCH.2-2.2 TDD | R.PDSCH.2-2.3 TDD | R.PDSCH.2-2.4 TDD | R.PDSCH.2-2.5 TDD | R.PDSCH.2-2.6 TDD |
| Channel bandwidth | MHz | 40 | 40 | 40 | 40 | 40 | 40 |
| Subcarrier spacing | kHz | 30 | 30 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | PRBs | 106 | 106 | 106 | 106 | 106 | 106 |
| Number of consecutive PDSCH symbols | | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | 4 | 4 | 4 | 4 | 4 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | 12 | 12 | 12 | 12 |
| Allocated slots per 2 frames | | 31 | 31 | 31 | 31 | 31 | 31 |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAMLowSE | 64QAM |
| MCS index | | 13 | 13 | 13 | 13 | 19 | 16 |
| Modulation | | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM | 16QAM |
| Target Coding Rate | | 0.48 | 0.48 | 0.48 | 0.48 | 0.54 | 0.64 |
| Number of MIMO layers | | 1 | 2 | 3 | 4 | 2 | 1 |
| Number of DMRS REs | | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | 6 | 12 | 12 | 6 | 6 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | 24 | 24 | 12 | 12 |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | 0 | 0 |
| Information Bit Payload per Slot | | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 8456 | 16896 | 22032 | 29192 | 19464 | 11528 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 26632 | 53288 | 73776 | 98376 | 60456 | 35856 |
| Transport block CRC per Slot | | | | | | | |
| For Slots 0 and | Bits | N/A | N/A | N/A | N/A | N/A | N/A |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | | | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6}for i from {1,…,39} | Bits | 24 | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Slot | | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 2 | 3 | 3 | 4 | 3 | 2 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | CBs | 4 | 7 | 9 | 12 | 8 | 5 |
| Binary Channel Bits Per Slot | | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | N/A | N/A |
| For Slots i = 20, 21 | Bits | 53424 | 106848 | 144008 | 193344 | 101760 | 53424 |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 17808 | 35616 | 45792 | 61056 | 35616 | 17808 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 55968 | 111936 | 152640 | 203520 | 111936 | 55968 |
| Max. Throughput averaged over 2 frames | Mbps | 37.644 | 75.318 | 104.004 | 138.646 | 85.508 | 50.711 |
| NOTE 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>NOTE 2: Slot i is slot index per 2 frames. | | | | | | | |

**Table A.3.2.2.2-3: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1 (64QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-3.1 TDD | R.PDSCH.2-3.2 TDD | R.PDSCH.2-3.3 TDD | R.PDSCH.2-3.4 TDD | |
| Channel bandwidth | MHz | 40 | 40 | 40 | 40 | |
| Subcarrier spacing | kHz | 30 | 30 | 30 | 30 | |
| Allocated resource blocks | PRBs | 106 | 106 | 53 (Note 3) | 53 (Note 4) | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | 4 | 4 | 4 | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | 12 | 12 | |
| Allocated slots per 2 frames | | 31 | 31 | 31 | 31 | |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAM | |
| MCS index | | 19 | 19 | R.PDSCH.2-3.3 TDD | R.PDSCH.2-3.4 TDD | |
| Modulation | | 64QAM | 64QAM | 40 | 40 | |
| Target Coding Rate | | 0.51 | 0.51 | 30 | 30 | |
| Number of MIMO layers | | 2 | 2 | 53 (Note 3) | 53 (Note 4) | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | 12 | 4 | 4 | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 24 | 12 | 12 | |
| Overhead for TBS determination | | 0 | 0 | 31 | 31 | |
| Information Bit Payload per Slot | | | | 64QAM | 64QAM | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 27144 | 23040 | 11528 | 11528 | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 83976 | 77896 | 38936 | 38936 | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 24 | 24 | 24 | 24 | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 24 | 24 | 24 | 24 | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | N/A | N/A | N/A | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 4 | 3 | 2 | 2 | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | CBs | 10 | 10 | 5 | 5 | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | |
| For Slots i = 20, 21 | Bits | 160272 | 137376 | 68688 | 68688 | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 53424 | 45792 | 22896 | 22896 | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 167904 | 152640 | 76320 | 76320 | |
| Max. Throughput averaged over 2 frames | Mbps | 118.796 | 109.768 | 54.869 | 54.869 | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames.<br>Note 3: PDSCH is scheduled in PRB numbers from 0 to 52.<br>Note 4: PDSCH is scheduled in PRB numbers from 53 to 105. | | | | | | |

**Table A.3.2.2.2-4: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1 (256QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-4.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | | | | |
| Allocated slots per 2 frames | | 31 | | | | |
| MCS table | | 256QAM | | | | |
| MCS index | | 24 | | | | |
| Modulation | | 256QAM | | | | |
| Target Coding Rate | | 0.82 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 29192 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 92200 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 4 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | CBs | 11 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slots i = 20, 21 | Bits | 106848 | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 35616 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 111936 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 130.308 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.2-5: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-2**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-5.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | | 8 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | | 12 | | | | |
| Allocated slots per 2 frames | | 31 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.30 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | | 12 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | Bits | 5376 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | Bits | 8456 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | CBs | 1 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | CBs | 2 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i = 20, 21 | Bits | 26712 | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | Bits | 17808 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,19,22,…,39} | Bits | 27984 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 11.875 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.2-6: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-3**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-6.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot 0 and Slot i, if mod(i, 10) = {4,8,9} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = {3,7} for i from {0,…,39} | | 8 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,5,6} for i from {1,…,39} | | 12 | | | | |
| Allocated slots per 2 frames | | 27 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.30 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot 0 and Slot i, if mod(i, 10) = {4,8,9} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = {3,7} for i from {0,…,39} | | 12 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,5,6} for i from {1,…,39} | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 10) = {4,8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = {3,7} for i from {0,…,39} | Bits | 5376 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,5,6} for i from {1,…,39} | Bits | 8456 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 10) = {4,8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = {3,7} for i from {0,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,5,6} for i from {1,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 10) = {4,8,9} for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 10) = {3,7} for i from {0,…,39} | CBs | 1 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,5,6} for i from {1,…,39} | CBs | 2 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 10) = {4,8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i = 20, 21 | Bits | 26712 | | | | |
| For Slot i, if mod(i, 10) = {3,7} for i from {0,…,39} | Bits | 17808 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,5,6} for i from {1,…,19,22,…,39} | Bits | 27984 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 10.184 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.2-7: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1 and CSI-RS overlapped with PDSCH**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-7.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | | | | |
| Allocated slots per 2 frames | | 31 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 13 | | | | |
| Modulation | | 16QAM | | | | |
| Target Coding Rate | | 0.48 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 16896 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 53288 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 3 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | CBs | 7 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = {0,5} for i from {1,…,19,22,…,39} | Bits | 103456 | | | | |
| For Slots i = 20 | Bits | 98368 | | | | |
| For Slots i = 21 | Bits | 106848 | | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 35616 | | | | |
| For Slot i, if mod(i, 10) = {1,2,3,4,6} for i from {1,…,19,22,…,39} | Bits | 111936 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 75.318 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.2-8: PDSCH Reference Channel for TDD PMI reporting requirements with UL-DL pattern FR1.30-1 (16QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-8.1 TDD | R.PDSCH.2-8.2 TDD | R.PDSCH.2-8.3 TDD | | |
| Channel bandwidth | MHz | 40 | 40 | 40 | | |
| Subcarrier spacing | kHz | 30 | 30 | 30 | | |
| Allocated resource blocks | PRBs | 106 | 106 | 106 | | |
| Number of consecutive PDSCH symbols | | 12 | 12 | 12 | | |
| Allocated slots per 2 frames | | 23 | 23 | 23 | | |
| MCS table | | 64QAM | 64QAM | 64QAM | | |
| MCS index | | 13 | 13 | 20 | | |
| Modulation | | 16QAM | 16QAM | 64QAM | | |
| Target Coding Rate | | 0.48 | 0.48 | 0.55 | | |
| Number of MIMO layers | | 1 | 2 | 2 | | |
| Number of DMRS REs (Note 3) | | 24 | 24 | 24 | | |
| Overhead for TBS determination | | 0 | 0 | 0 | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | | |
| For CSI-RS Slot i, if mod(i,10) =1 for i from {0,…,39} | Bits | N/A | N/A | N/A | | |
| For Slot i = 20 | Bits | 24576 | 49176 | 83976 | | |
| For Slot i, if mod(i, 10) = {0,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 24576 | 49176 | 83976 | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | | |
| For CSI-RS Slot i, if mod(i,10) =1 for i from {0,…,39} | Bits | N/A | N/A | N/A | | |
| For Slot i = 20 | Bits | 24 | 24 | 24 | | |
| For Slot i, if mod(i, 10) = {0,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 24 | 24 | 24 | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,…,39} | CBs | N/A | N/A | N/A | | |
| For CSI-RS Slot i, if mod(i,10) =1 for i from {0,…,39} | CBs | N/A | N/A | N/A | | |
| For Slot i = 20 | CBs | 3 | 6 | 10 | | |
| For Slot i, if mod(i, 10) = {0,2,3,4,5,6} for i from {1,…,19,22,…,39} | CBs | 3 | 6 | 10 | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | | |
| For CSI-RS Slot i, if mod(i,10) =1 for i from {0,…,39} | Bits | N/A | N/A | N/A | | |
| For Slot i = 20 | Bits | 48336 | 96672 | 145008 | | |
| For Slot i, if mod(i, 10) = {0,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 50880 | 101760 | 152640 | | |
| Max. Throughput averaged over 2 frames | Mbps | 28.2624 | 56.5524 | 96.5724 | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames.<br>Note 3: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data. | | | | | | |

**Table A.3.2.2.2-9: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-4 (64QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-9.1 TDD | | | | |
| Channel bandwidth | MHz | 20 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 51 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {4,5} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = 3 for i from {0,…,39} | | 4 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,6,7,8,9} for i from {1,…,39} | | 12 | | | | |
| Allocated slots per 2 frames | | 31 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 19 | | | | |
| Modulation | | 64QAM | | | | |
| Target Coding Rate | | 0.51 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {4,5} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 10) = 3 for i from {0,…,39} | | 6 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,6,7,8,9} for i from {1,…,39} | | 12 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {4,5} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = 3 for i from {0,…,39} | Bits | 13064 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,6,7,8,9} for i from {1,…,39} | Bits | 40976 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {4,5} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 10) = 3 for i from {0,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,6,7,8,9} for i from {1,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {4,5} for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 10) = 3 for i from {0,…,39} | CBs | 2 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,6,7,8,9} for i from {1,…,39} | CBs | 5 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {4,5} for i from {0,…,39} | Bits | N/A | | | | |
| For Slots i = 20, 21 | Bits | 77112 | | | | |
| For Slot i, if mod(i, 10) = 3 for i from {0,…,39} | Bits | 25704 | | | | |
| For Slot i, if mod(i, 10) = {0,1,2,6,7,8,9} for i from {1,…,19,22,…,39} | Bits | 80784 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 57.930 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

**Table A.3.2.2.2-10: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1 and HST scenario**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-10.1 TDD | R.PDSCH.2-10.2 TDD | R.PDSCH.2-10.3 TDD | R.PDSCH.2-10.4 TDD | R.PDSCH.2-10.5 TDD |
| Channel bandwidth | MHz | 40 | 40 | 40 | 40 | 40 |
| Subcarrier spacing | kHz | 30 | 30 | 30 | 30 | 30 |
| Allocated resource blocks | PRBs | 106 | 106 | 106 | 106 | 106 |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | N/A | 4 | N/A | 4 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | 12 | 12 | 12 |
| Allocated slots per 2 frames | | 31 | 27 | 31 | 27 | 31 |
| MCS table | | 64QAM | 64QAM | 64QAM | 64QAM | 64QAM |
| MCS index | | 13 | 13 | 17 | 13 | 17 |
| Modulation | | 16QAM | 16QAM | 64QAM | 16QAM | 64QAM |
| Target Coding Rate | | 0.48 | 0.48 | 0.43 | 0.48 | 0.43 |
| Number of MIMO layers | | 1 | 1 | 1 | 2 | 2 |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | N/A | 6 | N/A | 6 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 18 | 18 | 18 | 18 | 18 |
| Overhead for TBS determination | | 0 | 0 | 0 | 0 | 0 |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 8456 | N/A | 11528 | N/A | 19464 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 25608 | 25608 | 33816 | 51216 | 58384 |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 24 | N/A | 24 | N/A | 24 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | Bits | 24 | 24 | 24 | 24 | 24 |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | N/A | N/A | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 2 | N/A | 2 | N/A | 3 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | CBs | 4 | 4 | 5 | 7 | 7 |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A | N/A | N/A | N/A |
| For Slots i = 1,2,21,22 | Bits | 52176 | 50880 | 76320 | 104304 | 156456 |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 17808 | N/A | 26712 | N/A | 53424 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {3,…,20,23,…,39} | Bits | 53424 | 53424 | 80136 | 106848 | 160272 |
| Max. Throughput averaged over 2 frames | Mbps | 36.262 | 34.5708 | 47.9572 | 69.1416 | 82.7112 |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

**Table A.3.2.2.2-11: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-5**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-11.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | | 12 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | | 10 | | | | |
| Allocated slots per 2 frames | | 31 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.30 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | | 18 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | | 18 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | Bits | 8064 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | Bits | 6528 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | CBs | 1 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | CBs | 1 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i = 20 | Bits | 25440 | | | | |
| For Slot i = 21 | Bits | 20352 | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,19,22,…,39} | Bits | 26712 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,19,22,…,39} | Bits | 21624 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 6.893 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

**Table A.3.2.2.2-12: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-6**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-12.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | | 12 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | | 8 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {0,…,39} | | 10 | | | | |
| Allocated slots per 2 frames | | 31 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.30 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,39} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | | 18 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | | 18 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {0,…,39} | | 18 | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | Bits | 8064 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | Bits | 4992 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {0,…,39} | Bits | 6528 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | Bits | 24 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {0,…,39} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,39} | CBs | N/A | | | | |
| For Slot i, if mod(i, 4) = 0 for i from {1,…,39} | CBs | 1 | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {0,…,39} | CBs | 1 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {0,…,39} | CBs | 1 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,39} | Bits | N/A | | | | |
| For Slot i = 20 | Bits | 25440 | | | | |
| For Slot i = 21 | Bits | 15264 | | | | |
| For Slot i, if mod(i, 4) = 0 for i from | Bits | 26712 | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| {1,…,19,22,…,39} | | | | | | |
| For Slot i, if mod(i, 4) = 1 for i from {1,…,19,22,…,39} | Bits | 16536 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {0,…,39} | Bits | 21624 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 9.389 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

**Table A.3.2.2.2-13: PDSCH Reference Channel for TDD CC with UL-DL pattern FR1.30-1 and CA scenario**

FFS

**Table A.3.2.2.2-14: PDSCH Reference Channel for TDD CC with UL-DL pattern FR1.30-1 and CA scenario**

FFS

**Table A.3.2.2.2-15: PDSCH Reference Channel for TDD CC with UL-DL pattern FR1.30-1 and CA scenario**

FFS

**Table A.3.2.2.2-16: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-1**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-16.1 TDD | R.PDSCH.2-16.2 TDD | | | |
| Channel bandwidth | MHz | 40 | 40 | | | |
| Subcarrier spacing | kHz | 30 | 30 | | | |
| Allocated resource blocks | PRBs | 106 | 106 | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot i, if mod(i, 10) = {0, 7} for i from {0,…,39} | | N/A | N/A | | | |
| For Slot i, if mod(i, 10) = {1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | | | |
| Allocated slots per 2 frames | | 24 | 24 | | | |
| MCS table | | 64QAMLowSE | 64QAMLowSE | | | |
| MCS index | | 19 | 19 | | | |
| Modulation | | 16QAM | 16QAM | | | |
| Target Coding Rate | | 0.54 | 0.54 | | | |
| Number of MIMO layers | | 1 | 1 | | | |
| Number of DMRS REs | | | | | | |
| For Slot i, if mod(i, 10) = {0, 7} for i from {0,…,39} | | N/A | N/A | | | |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {1,…,39} | | 12 | 12 | | | |
| Overhead for TBS determination | | 0 | 0 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i, if mod(i, 10) = {0,7,8,9} for i from {0,…,39} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 10) = {1,2,3,4,5,6} for i from {1,…,39} | Bits | 30216 | 30216 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i, if mod(i, 10) = {0,7,8,9} for i from {0,…,39} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 10) = {1,2,3,4,5,6} for i from {1,…,39} | Bits | 24 | 24 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i, if mod(i, 10) = {0,7,8,9} for i from {0,…,39} | CBs | N/A | N/A | | | |
| For Slot i, if mod(i, 10) = {1,2,3,4,5,6} for i from {1,…,39} | CBs | 2 | 4 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i, if mod(i, 10) = {0,7,8,9} for i from {0,…,39} | Bits | N/A | N/A | | | |
| For Slot i = 21 | Bits | 53424 | 50880 | | | |
| For Slot i, if mod(i, 10) = {1,2,3,4,5,6} for i from {1,…,19,22,…,39} | Bits | 55968 | 55968 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 18.130 (NOTE 3) | 18.130 (NOTE 3) | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames<br>Note 3: Throughput is calculated under assumption of aggregation factor 2.<br>Note 4: Throughput is calculated under assumption of repetition number 2. | | | | | | |

**Table A.3.2.2.2-17: PDSCH Reference Channel for TDD UL-DL pattern FR1.30-2**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.2-17.1 TDD | | | | |
| Channel bandwidth | MHz | 40 | | | | |
| Subcarrier spacing | kHz | 30 | | | | |
| Allocated resource blocks | PRBs | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | | 2 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | | N/A | | | | |
| Allocated slots per 2 frames | | 8 | | | | |
| MCS table | | | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.3 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | | 6 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | | N/A | | | | |
| Overhead for TBS determination | | 0 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | Bits | 1160 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | Bits | N/A | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | Bits | 16 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | Bits | N/A | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | CBs | 1 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | CBs | N/A | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…,39} | Bits | 3816 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,39} | Bits | N/A | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 0.464 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

### A.3.2.2.3 Reference measurement channels for SCS 60 kHz FR1

### A.3.2.2.4 Reference measurement channels for SCS 60 kHz FR2

**Table A.3.2.2.4-1: PDSCH Reference Channel for TDD UL-DL pattern FR2.60-1 (16QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.4-1.1 TDD | | | | |
| Channel bandwidth | MHz | 50 | | | | |
| Subcarrier spacing | kHz | 60 | | | | |
| Allocated resource blocks | PRBs | 66 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,79} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 79} | | 10 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79} | | 13 | | | | |
| Allocated slots per 2 frames | | 59 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 13 | | | | |
| Modulation | | 16QAM | | | | |
| Target Coding Rate | | 0.48 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,79} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 79} | | 12 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79} | | 12 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,79} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 79} | Bits | 25608 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79} | Bits | 34816 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,79} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 79} | Bits | 24 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,79} | CBs | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 79} | CBs | 4 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79} | CBs | 5 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,79} | Bits | N/A | | | | |
| For Slot i = 40, 41 | Bits | 69960 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {4,…, 79} | Bits | 54912 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,39,42,…,79} | Bits | 73128 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 93.499 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

### A.3.2.2.5 Reference measurement channels for SCS 120 kHz FR2

**Table A.3.2.2.5-1: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-1 and FR2.120-1A (QPSK)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-1.1 TDD | R.PDSCH.5-1.2 TDD | | | |
| Channel bandwidth | MHz | 100 | 100 | | | |
| Subcarrier spacing | kHz | 120 | 120 | | | |
| Allocated resource blocks | PRBs | 66 | 66 | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 9 | 2 | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 13 | 2 | | | |
| Allocated slots per 2 frames | | 127 | 127 | | | |
| MCS table | | 64QAM | 64QAM | | | |
| MCS index | | 4 | 4 | | | |
| Modulation | | QPSK | QPSK | | | |
| Target Coding Rate | | 0.30 | 0.30 | | | |
| Number of MIMO layers | | 1 | 1 | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 12 | 6 | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 12 | 6 | | | |
| Overhead for TBS determination | | 6 | 0 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 3624 | 736 | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 5504 | 736 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 16 | 16 | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 24 | 16 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | CBs | N/A | N/A | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | CBs | 1 | 1 | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | CBs | 1 | 1 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | N/A | | | |
| For Slots i = 80, 81 | Bits | 17490 | 2310 | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 12210 | 2310 | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,79,82,…,159} | Bits | 18282 | 2310 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 31.942 | 4.673 | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.5-2: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-1 (16QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-2.1 TDD | R.PDSCH.5-2.2 TDD | R.PDSCH.5-2.3 TDD | | |
| Channel bandwidth | MHz | 100 | 100 | 200 | | |
| Subcarrier spacing | kHz | 120 | 120 | 120 | | |
| Allocated resource blocks | PRBs | 66 | 66 | 132 | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | | N/A | N/A | N/A | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 9 | 9 | 9 | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 13 | 13 | 13 | | |
| Allocated slots per 2 frames | | 127 | 127 | 127 | | |
| MCS table | | 64QAM | 64QAM | 64QAM | | |
| MCS index | | 13 | 13 | 13 | | |
| Modulation | | 16QAM | 16QAM | 16QAM | | |
| Target Coding Rate | | 0.48 | 0.48 | 0.48 | | |
| Number of MIMO layers | | 1 | 2 | 2 | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | | N/A | N/A | N/A | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 12 | 12 | 12 | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 12 | 12 | 12 | | |
| Overhead for TBS determination | | 6 | 6 | 6 | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | N/A | N/A | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 11272 | 22536 | 45096 | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 17424 | 34816 | 69672 | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | N/A | N/A | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 24 | 24 | 24 | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 24 | 24 | 24 | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | CBs | N/A | N/A | N/A | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | CBs | 2 | 3 | 6 | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | CBs | 3 | 5 | 9 | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | N/A | N/A | | |
| For Slots i = 80, 81 | Bits | 36564 | 69960 | 139920 | | |
| For Slots i = 82, 83 | Bits | 34980 | 73128 | 146256 | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 24420 | 48840 | 97680 | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,79,84,…,159} | Bits | 36564 | 73128 | 146256 | | |
| Max. Throughput averaged over 2 frames | Mbps | 100.799 | 201.434 | 403.096 | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.5-3: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-1 (64QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-3.1 TDD | | | | |
| Channel bandwidth | MHz | 100 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 66 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 9 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 13 | | | | |
| Allocated slots per 2 frames | | 127 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 18 | | | | |
| Modulation | | 64QAM | | | | |
| Target Coding Rate | | 0.46 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 12 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 12 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 16136 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 25104 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 24 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | CBs | 2 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | CBs | 3 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | | | | |
| For Slots i = 80, 81 | Bits | 52470 | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 36630 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,79,82,…,159} | Bits | 54846 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 145.062 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.5-4: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-2 (QPSK)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-4.1 TDD | | | | |
| Channel bandwidth | MHz | 100 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 6 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | | 10 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | | 13 | | | | |
| Allocated slots per 2 frames | | 119 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 4 | | | | |
| Modulation | | QPSK | | | | |
| Target Coding Rate | | 0.30 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | | 12 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | | 12 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | Bits | 736 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | Bits | 1032 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | Bits | 16 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | Bits | 16 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | CBs | 1 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | CBs | 1 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80, 81 | Bits | 3180 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {4,…, 159} | Bits | 2496 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79,82,…,159} | Bits | 3324 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 5.548 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.5-5: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-2 (16QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-5.1 TDD | R.PDSCH.5-5.2 TDD | | | |
| Channel bandwidth | MHz | 100 | 50 | | | |
| Subcarrier spacing | kHz | 120 | 120 | | | |
| Allocated resource blocks | PRBs | 66 | 32 | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | | N/A | N/A | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | | 10 | 10 | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | | 13 | 13 | | | |
| Allocated slots per 2 frames | | 119 | 119 | | | |
| MCS table | | 64QAM | 64QAM | | | |
| MCS index | | 13 | 13 | | | |
| Modulation | | 16QAM | 16QAM | | | |
| Target Coding Rate | | 0.48 | 0.48 | | | |
| Number of MIMO layers | | 2 | 2 | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | | N/A | N/A | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | | 12 | 12 | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | | 12 | 12 | | | |
| Overhead for TBS determination | | 6 | 6 | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | Bits | 25608 | 12552 | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | Bits | 34816 | 16896 | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | N/A | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | Bits | 24 | 24 | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | Bits | 24 | 24 | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | CBs | N/A | N/A | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | CBs | 4 | 2 | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | CBs | 5 | 3 | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | N/A | | | |
| For Slot i = 80, 81 | Bits | 69960 | 33920 | | | |
| For Slot i, if mod(i, 4) = 2 for i from {4,…, 159} | Bits | 54912 | 26624 | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79,82,…,159} | Bits | 73128 | 35456 | | | |
| Max. Throughput averaged over 2 frames | Mbps | 188.739 | 91.843 | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.5-6: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-2 (64QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-6.1 TDD | | | | |
| Channel bandwidth | MHz | 100 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 66 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | | 10 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | | 13 | | | | |
| Allocated slots per 2 frames | | 119 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 17 | | | | |
| Modulation | | 64QAM | | | | |
| Target Coding Rate | | 0.43 | | | | |
| Number of MIMO layers | | 2 | | | | |
| Number of DMRS REs | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | | 12 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | | 12 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | Bits | 34816 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | Bits | 47112 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | Bits | 24 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {1,…, 159} | CBs | 5 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,159} | CBs | 6 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = 3 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80, 81 | Bits | 114940 | | | | |
| For Slot i, if mod(i, 4) = 2 for i from {4,…, 159} | Bits | 82368 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {1,…,79,82,…,159} | Bits | 109692 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 255.724 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | | | | |

**Table A.3.2.2.5-7: PDSCH Reference Channel for TDD PMI reporting requirements with UL-DL pattern FR2.120-1 (16QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-7.1 TDD | | | | |
| Channel bandwidth | MHz | 100 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 66 | | | | |
| Number of consecutive PDSCH symbols | | 12 | | | | |
| Allocated slots per 2 frames | | 63 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 13 | | | | |
| Modulation | | 16QAM | | | | |
| Target Coding Rate | | 0.48 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs (Note 3) | | 24 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = {3,4} for i from {0,…,159} | Bits | N/A | | | | |
| For CSI-RS Slot i, if mod(i,5) =1 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80 | Bits | 14344 | | | | |
| For Slot i, if mod(i, 5) = {0,2} for i from {1,…, 79,82,…,159} | Bits | 14344 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = {3,4} for i from {0,…,159} | Bits | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 5) =1 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80 | Bits | 24 | | | | |
| For Slot i, if mod(i, 5) = {0,2} for i from {1,…,79,82,…,159} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = {3,4} for i from {0,…,159} | CBs | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 5) =1 for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i = 80 | CBs | 2 | | | | |
| For Slot i, if mod(i, 5) = {0,2} for i from {1,…,79,82,…,159} | CBs | 2 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = {3,4} for i from {0,…,159} | Bits | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 5) =1 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80 | Bits | 28776 | | | | |
| For Slot i, if mod(i, 5) = {0,2} for i from {1,…,79,82,…,159} | Bits | 30360 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 45.1836 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames.<br>Note 3: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data. | | | | | | |

**Table A.3.2.2.5-8: PDSCH Reference Channel for TDD PMI reporting requirements with UL-DL pattern FR2.120-2 (16QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-8.1 TDD | | | | |
| Channel bandwidth | MHz | 100 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 66 | | | | |
| Number of consecutive PDSCH symbols | | 12 | | | | |
| Allocated slots per 2 frames | | 59 | | | | |
| MCS table | | 64QAM | | | | |
| MCS index | | 13 | | | | |
| Modulation | | 16QAM | | | | |
| Target Coding Rate | | 0.48 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs (Note 3) | | 24 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | Bits | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 8) =1 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80 | Bits | 14344 | | | | |
| For Slot i, if mod(i, 8) = {0,4,5} for i from {1,…,79,82,…,159} | Bits | 14344 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | Bits | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 8) =1 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80 | Bits | 24 | | | | |
| For Slot i, if mod(i, 8) = {0,4,5,} for i from {1,…,79,82,…,159} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | CBs | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 8) =1 for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i = 80 | CBs | 2 | | | | |
| For Slot i, if mod(i, 8) = {0,4,5} for i from {1,…,79,82,…,159} | CBs | 2 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | Bits | N/A | | | | |
| For CSI-RS Slot i, if mod(i, 8) =1 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80 | Bits | 28776 | | | | |
| For Slot i, if mod(i, 8) = {0,4,5}for i from {1,…,79,82,…,159} | Bits | 30360 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 42.3148 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames.<br>Note 3: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data. | | | | | | |

**Table A.3.2.2.5-9: PDSCH Reference Channel for TDD CC with UL-DL pattern FR2.120-1 and CA scenario**

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**Table A.3.2.2.5-10: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-1 (256QAM)**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-10.1 TDD | | | | |
| Channel bandwidth | MHz | 50 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 32 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 9 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 13 | | | | |
| Allocated slots per 2 frames | | 127 | | | | |
| MCS table | | 256QAM | | | | |
| MCS index | | 20 | | | | |
| Modulation | | 256QAM | | | | |
| Target Coding Rate | | 0.67 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | | 12 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | | 12 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 15368 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 23568 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 24 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | CBs | 2 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,159} | CBs | 3 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0 and Slot i, if mod(i, 5) = 4 for i from {0,…,159} | Bits | N/A | | | | |
| For Slots i = 80, 81 | Bits | 33920 | | | | |
| For Slot i, if mod(i, 5) = 3 for i from {0,…, 159} | Bits | 23680 | | | | |
| For Slot i, if mod(i, 5) = {0,1,2} for i from {1,…,79,82,…,159} | Bits | 35456 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 136.537 | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames | | | | | | |

**Table A.3.2.2.5-11: PDSCH Reference Channel for TDD UL-DL pattern FR2.120-2**

| Parameter | Unit | Value | | | | |
|---|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.5-11.1 TDD | | | | |
| Channel bandwidth | MHz | 100 | | | | |
| Subcarrier spacing | kHz | 120 | | | | |
| Allocated resource blocks | PRBs | 66 | | | | |
| Number of consecutive PDSCH symbols | | | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {2,…,159} | | 13 | | | | |
| Allocated slots per 2 frames | | 78 | | | | |
| MCS table | | 64QAMLowSE | | | | |
| MCS index | | 16 | | | | |
| Modulation | | 16QAM | | | | |
| Target Coding Rate | | 0.37 | | | | |
| Number of MIMO layers | | 1 | | | | |
| Number of DMRS REs | | | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {2,…,159} | | 12 | | | | |
| Overhead for TBS determination | | 6 | | | | |
| Information Bit Payload per Slot | | | | | | |
| For Slots 0, 1 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {2,…,159} | Bits | 13320 | | | | |
| Transport block CRC per Slot | | | | | | |
| For Slots 0, 1 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {2,…,159} | Bits | 24 | | | | |
| Number of Code Blocks per Slot | | | | | | |
| For Slots 0, 1 and Slot i, if mod(i, 4) = {2,3} for i from {0,…,159} | CBs | N/A | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {2,…,159} | CBs | 2 | | | | |
| Binary Channel Bits Per Slot | | | | | | |
| For Slots 0,1 and Slot i, if mod(i, 4) = {2, 3} for i from {0,…,159} | Bits | N/A | | | | |
| For Slot i = 80, 81 | Bits | 34980 | | | | |
| For Slot i, if mod(i, 4) = {0,1} for i from {2,…,159} | Bits | 36564 | | | | |
| Max. Throughput averaged over 2 frames | Mbps | 25.974 (Note 3) | | | | |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms<br>Note 2: Slot i is slot index per 2 frames<br>Note 3: Throughput is calculated under assumption of aggregation factor 2. | | | | | | |

### A.3.2.2.6 Reference measurement channels for E-UTRA

**Table A.3.2.2.6-1: PDSCH Reference Channel for sustained data-rate test (64QAM, 2 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.6-1.1 TDD | R.PDSCH.6-1.2 TDD | R.PDSCH.6-1.3 TDD | |
| Channel bandwidth | MHz | 10 | 15 | 20 | |
| Allocated resource blocks | | Note 7 | Note 8 | Note 9 | |
| Uplink-Downlink Configuration (Note 3) | | 2 | 2 | 2 | |
| Number of HARQ Processes per component carrier | | 10 | 10 | 10 | |
| Allocated subframes per Radio Frame (D+S) | | 6 | 6 | 6 | |
| Modulation | | 64QAM | 64QAM | 64QAM | |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,6,7 | | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | | 0.85 | 0.85 | 0.88 | |
| For Sub-Frame 5 | | 0.88 | 0.87 | 0.87 | |
| For Sub-Frame 0 | | 0.90 | 0.88 | 0.90 | |
| Information Bit Payload (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | Bits | 36696 | 55056 | 75376 | |
| For Sub-Frame 5 | Bits | 35160 | 52752 | 71112 | |
| For Sub-Frame 0 | Bits | 36696 | 55056 | 75376 | |
| Number of Code Blocks (Notes 4 and 5) | | | | | |
| For Sub-Frames 1,2,6,7 | CBs | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | CBs | 6 | 9 | 13 | |
| For Sub-Frame 5 | CBs | 6 | 9 | 12 | |
| For Sub-Frame 0 | CBs | 6 | 9 | 13 | |
| Binary Channel Bits (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | Bits | 43200 | 64800 | 86400 | |
| For Sub-Frame 5 | Bits | 40176 | 60912 | 82512 | |
| For Sub-Frame 0 | Bits | 41184 | 62784 | 84384 | |
| Number of layers | | 2 | 2 | 2 | |
| Max. Throughput averaged over 1 frame (Note 4) | Mbps | 21.864 | 32.803 | 44.799 | |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: As per Table 4.2-2 in TS 36.211 [15].
Note 4: Given per component carrier per codeword.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,8,9.
Note 8: Resource blocks $n_{PRB}$ = 4..74 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..74 in sub-frames 0,3,4,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,8,9.

**Table A.3.2.2.6-2: PDSCH Reference Channel for sustained data-rate test (64QAM, 4 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.6-2.1 TDD | R.PDSCH.6-2.2 TDD | R.PDSCH.6-2.3 TDD | |
| Channel bandwidth | MHz | 10 | 15 | 20 | |
| Allocated resource blocks | | Note 7 | Note 8 | Note 9 | |
| Uplink-Downlink Configuration (Note 3) | | 2 | 2 | 2 | |
| Number of HARQ Processes per component carrier | | 10 | 10 | 10 | |
| Allocated subframes per Radio Frame (D+S) | | 6 | 6 | 6 | |
| Modulation | | 64QAM | 64QAM | 64QAM | |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,6,7 | | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | | 0.78 | 0.77 | 0.79 | |
| For Sub-Frame 5 | | 0.79 | 0.79 | 0.80 | |
| For Sub-Frame 0 | | 0.82 | 0.79 | 0.81 | |
| Information Bit Payload (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | Bits | 63776 | 93800 | 128496 | |
| For Sub-Frame 5 | Bits | 59256 | 90816 | 124464 | |
| For Sub-Frame 0 | Bits | 63776 | 93800 | 128496 | |
| Number of Code Blocks (Notes 4 and 5) | | | | | |
| For Sub-Frames 1,2,6,7 | CBs | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | CBs | 11 | 16 | 21 | |
| For Sub-Frame 5 | CBs | 10 | 15 | 21 | |
| For Sub-Frame 0 | CBs | 11 | 16 | 21 | |
| Binary Channel Bits (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4,8,9 | Bits | 81600 | 122400 | 163200 | |
| For Sub-Frame 5 | Bits | 75840 | 115008 | 155808 | |
| For Sub-Frame 0 | Bits | 77856 | 118656 | 159456 | |
| Number of layers | | 4 | 4 | 4 | |
| Max. Throughput averaged over 1 frame (Note 4) | Mbps | 37.813 | 55.981 | 76.694 | |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: As per Table 4.2-2 in TS 36.211 [15].
Note 4: Given per component carrier per codeword.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,8,9.
Note 8: Resource blocks $n_{PRB}$ = 4..74 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..74 in sub-frames 0,3,4,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,8,9.

**Table A.3.2.2.6-3: PDSCH Reference Channel for sustained data-rate test (256QAM, 2 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.6-3.1 TDD | R.PDSCH.6-3.2 TDD | R.PDSCH.6-3.3 TDD | |
| Channel bandwidth | MHz | 10 | 15 | 20 | |
| Allocated resource blocks | | Note 7 | Note 8 | Note 9 | |
| Uplink-Downlink Configuration (Note 3) | | 2 | 2 | 2 | |
| Number of HARQ Processes per component carrier | | 10 | 10 | 10 | |
| Allocated subframes per Radio Frame (D+S) | | 6 | 6 | 6 | |
| Modulation | | 256QAM | 256QAM | 256QAM | |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,6,7 | | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | | 0.74 | 0.79 | 0.74 | |
| For Sub-Frames 8,9 | | 0.85 | 0.88 | 0.85 | |
| For Sub-Frame 5 | | 0.76 | 0.76 | 0.74 | |
| For Sub-Frame 0 | | 0.78 | 0.77 | 0.76 | |
| Information Bit Payload (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 42368 | 63776 | 84760 | |
| For Sub-Frames 8,9 | Bits | 48936 | 75376 | 97896 | |
| For Sub-Frame 5 | Bits | 40576 | 61664 | 81176 | |
| For Sub-Frame 0 | Bits | 42368 | 63776 | 84760 | |
| Number of Code Blocks (Notes 4 and 5) | | | | | |
| For Sub-Frames 1,2,6,7 | CBs | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | CBs | 7 | 11 | 14 | |
| For Sub-Frames 8,9 | CBs | 8 | 13 | 16 | |
| For Sub-Frame 5 | CBs | 7 | 11 | 14 | |
| For Sub-Frame 0 | CBs | 7 | 11 | 14 | |
| Binary Channel Bits (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 57600 | 86400 | 115200 | |
| For Sub-Frames 8,9 | Bits | 57600 | 86400 | 115200 | |
| For Sub-Frame 5 | Bits | 53568 | 81216 | 110016 | |
| For Sub-Frame 0 | Bits | 54912 | 83712 | 112512 | |
| Number of layers | | 2 | 2 | 2 | |
| Max. Throughput averaged over 1 frame (Note 4) | Mbps | 26.555 | 40.374 | 53.125 | |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: As per Table 4.2-2 in TS 36.211 [15].
Note 4: Given per component carrier per codeword.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,8,9.
Note 8: Resource blocks $n_{PRB}$ = 4..74 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..74 in sub-frames 0,3,4,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,8,9.

**Table A.3.2.2.6-4: PDSCH Reference Channel for sustained data-rate test (256QAM, 4 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.6-4.1 TDD | R.PDSCH.6-4.2 TDD | R.PDSCH.6-4.3 TDD | |
| Channel bandwidth | MHz | 10 | 15 | 20 | |
| Allocated resource blocks | | Note 7 | Note 8 | Note 9 | |
| Uplink-Downlink Configuration (Note 3) | | 2 | 2 | 2 | |
| Number of HARQ Processes per component carrier | | 10 | 10 | 10 | |
| Allocated subframes per Radio Frame (D+S) | | 6 | 6 | 6 | |
| Modulation | | 256QAM | 256QAM | 256QAM | |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,6,7 | | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | | 0.78 | 0.79 | 0.78 | |
| For Sub-Frames 8,9 | | 0.78 | 0.79 | 0.78 | |
| For Sub-Frame 5 | | 0.81 | 0.82 | 0.78 | |
| For Sub-Frame 0 | | 0.82 | 0.82 | 0.80 | |
| Information Bit Payload (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 84760 | 128496 | 169544 | |
| For Sub-Frames 8,9 | Bits | 84760 | 128496 | 169544 | |
| For Sub-Frame 5 | Bits | 81176 | 124464 | 161760 | |
| For Sub-Frame 0 | Bits | 84760 | 128496 | 169544 | |
| Number of Code Blocks (Notes 4 and 5) | | | | | |
| For Sub-Frames 1,2,6,7 | CBs | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | CBs | 14 | 21 | 28 | |
| For Sub-Frames 8,9 | CBs | 14 | 21 | 28 | |
| For Sub-Frame 5 | CBs | 14 | 21 | 27 | |
| For Sub-Frame 0 | CBs | 14 | 21 | 28 | |
| Binary Channel Bits (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 108800 | 163200 | 217600 | |
| For Sub-Frames 8,9 | Bits | 108800 | 163200 | 217600 | |
| For Sub-Frame 5 | Bits | 101120 | 153344 | 207744 | |
| For Sub-Frame 0 | Bits | 103808 | 158208 | 212608 | |
| Number of layers | | 4 | 4 | 4 | |
| Max. Throughput averaged over 1 frame (Note 4) | Mbps | 50.498 | 76.694 | 100.948 | |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: As per Table 4.2-2 in TS 36.211 [15].
Note 4: Given per component carrier per codeword.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,8,9.
Note 8: Resource blocks $n_{PRB}$ = 4..74 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..74 in sub-frames 0,3,4,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,8,9.

**Table A.3.2.2.6-5: PDSCH Reference Channel for sustained data-rate test (1024QAM, 2 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.6-5.1 TDD | R.PDSCH.6-5.2 TDD | R.PDSCH.6-5.3 TDD | |
| Channel bandwidth | MHz | 10 | 15 | 20 | |
| Allocated resource blocks | | Note 7 | Note 8 | Note 9 | |
| Uplink-Downlink Configuration (Note 3) | | 2 | 2 | 2 | |
| Number of HARQ Processes per component carrier | | 10 | 10 | 10 | |
| Allocated subframes per Radio Frame (D+S) | | 6 | 6 | 6 | |
| Modulation | | 1024QAM | 1024QAM | 1024QAM | |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,6,7 | | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | | 0.76 | 0.75 | 0.76 | |
| For Sub-Frames 8,9 | | 0.76 | 0.75 | 0.76 | |
| For Sub-Frame 5 | | 0.76 | 0.78 | 0.77 | |
| For Sub-Frame 0 | | 0.80 | 0.78 | 0.78 | |
| Information Bit Payload (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 55056 | 81176 | 110136 | |
| For Sub-Frames 8,9 | Bits | 55056 | 81176 | 110136 | |
| For Sub-Frame 5 | Bits | 51024 | 78704 | 105528 | |
| For Sub-Frame 0 | Bits | 55056 | 81176 | 110136 | |
| Number of Code Blocks (Notes 4 and 5) | | | | | |
| For Sub-Frames 1,2,6,7 | CBs | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | CBs | 9 | 14 | 18 | |
| For Sub-Frames 8,9 | CBs | 9 | 14 | 18 | |
| For Sub-Frame 5 | CBs | 9 | 13 | 18 | |
| For Sub-Frame 0 | CBs | 9 | 14 | 18 | |
| Binary Channel Bits (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 72000 | 108000 | 144000 | |
| For Sub-Frames 8,9 | Bits | 72000 | 108000 | 144000 | |
| For Sub-Frame 5 | Bits | 66960 | 101520 | 137520 | |
| For Sub-Frame 0 | Bits | 68640 | 104640 | 140640 | |
| Number of layers | | 2 | 2 | 2 | |
| Max. Throughput averaged over 1 frame (Note 4) | Mbps | 32.630 | 48.458 | 65.621 | |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: As per Table 4.2-2 in TS 36.211 [15].
Note 4: Given per component carrier per codeword.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,8,9.
Note 8: Resource blocks $n_{PRB}$ = 4..74 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..74 in sub-frames 0,3,4,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,8,9.

**Table A.3.2.2.6-6: PDSCH Reference Channel for sustained data-rate test (1024QAM, 4 MIMO layers)**

| Parameter | Unit | Value | | | |
|---|---|---|---|---|---|
| Reference channel | | R.PDSCH.6-6.1 TDD | R.PDSCH.6-6.2 TDD | R.PDSCH.6-6.3 TDD | |
| Channel bandwidth | MHz | 10 | 15 | 20 | |
| Allocated resource blocks | | Note 7 | Note 8 | Note 9 | |
| Uplink-Downlink Configuration (Note 3) | | 2 | 2 | 2 | |
| Number of HARQ Processes per component carrier | | 10 | 10 | 10 | |
| Allocated subframes per Radio Frame (D+S) | | 6 | 6 | 6 | |
| Modulation | | 1024QAM | 1024QAM | 1024QAM | |
| Coding Rate | | | | | |
| For Sub-Frames 1,2,6,7 | | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | | 0.81 | 0.79 | 0.81 | |
| For Sub-Frames 8,9 | | 0.81 | 0.79 | 0.81 | |
| For Sub-Frame 5 | | 0.81 | 0.82 | 0.82 | |
| For Sub-Frame 0 | | 0.85 | 0.82 | 0.83 | |
| Information Bit Payload (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 110136 | 161760 | 220296 | |
| For Sub-Frames 8,9 | Bits | 110136 | 161760 | 220296 | |
| For Sub-Frame 5 | Bits | 101840 | 157432 | 211936 | |
| For Sub-Frame 0 | Bits | 110136 | 161760 | 220296 | |
| Number of Code Blocks (Notes 4 and 5) | | | | | |
| For Sub-Frames 1,2,6,7 | CBs | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | CBs | 18 | 27 | 36 | |
| For Sub-Frames 8,9 | CBs | 18 | 27 | 36 | |
| For Sub-Frame 5 | CBs | 17 | 26 | 35 | |
| For Sub-Frame 0 | CBs | 18 | 27 | 36 | |
| Binary Channel Bits (Note 4) | | | | | |
| For Sub-Frames 1,2,6,7 | Bits | N/A | N/A | N/A | |
| For Sub-Frames 3,4 | Bits | 136000 | 204000 | 272000 | |
| For Sub-Frames 8,9 | Bits | 136000 | 204000 | 272000 | |
| For Sub-Frame 5 | Bits | 126400 | 191680 | 259680 | |
| For Sub-Frame 0 | Bits | 129760 | 197760 | 265760 | |
| Number of layers | | 2 | 2 | 2 | |
| Max. Throughput averaged over 1 frame (Note 4) | Mbps | 65.252 | 96.623 | 131.342 | |

Note 1: 1 symbol allocated to PDCCH for all tests.
Note 2: Reference signal, synchronization signals and PBCH allocated as per TS 36.211 [17].
Note 3: As per Table 4.2-2 in TS 36.211 [15].
Note 4: Given per component carrier per codeword.
Note 5: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
Note 6: Resource blocks $n_{PRB}$ = 0..2 are allocated for SIB transmissions in sub-frame 5 for all bandwidths.
Note 7: Resource blocks $n_{PRB}$ = 3..49 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..49 in sub-frames 0,3,4,8,9.
Note 8: Resource blocks $n_{PRB}$ = 4..74 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..74 in sub-frames 0,3,4,8,9.
Note 9: Resource blocks $n_{PRB}$ = 4..99 are allocated for the user data in sub-frame 5, and resource blocks $n_{PRB}$ = 0..99 in sub-frames 0,3,4,8,9.

# A.3.2_1 Reference measurement channels for Sustained downlink data rate performance requirements

## A.3.2_1.1 FDD

### A.3.2_1.1.1 Reference measurement channels for SCS 15 kHz FR1

**Table A.3.2_1.1.1-1: Sustained Downlink Data Rate Reference Channel for FDD 15kHz SCS FR1 (64QAM)**

| Channel bandwidth | Subcarrier spacing | Allocated resource blocks | Number of consecutive PDSCH symbols for allocated full DL slots (Note 1) | MCS Index (Note 2) | Modulation | Target Coding Rate | Number of MIMO layers | LDPC Base Graph | Information Bit Payload per Slot for allocated full DL slots (Note 1) | Transport block CRC per Slot for allocated full DL slots (Note 1) | Number of Code Blocks per Slot for allocated full DL slots (Note 1, 6) | Binary Channel Bits per Slot for allocated full DL slots (Note 1) | Th a |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MHz | kHz | PRBs | Symbols | | | | | | Bits | Bits | CBs | Bits | |
| 10 | 15 | 52 | 13 | 18 | 64QAM | 0.46 | 1 | 1 | 20496 | 24 | 3 | 44928 | |
| 20 | 15 | 106 | 13 | 18 | 64QAM | 0.46 | 1 | 1 | 42016 | 24 | 5 | 91584 | |
| 10 | 15 | 52 | 13 | 22 | 64QAM | 0.65 | 1 | 1 | 29192 | 24 | 4 | 44928 | |
| 20 | 15 | 106 | 13 | 22 | 64QAM | 0.65 | 1 | 1 | 59432 | 24 | 8 | 91584 | |
| 10 | 15 | 52 | 13 | 23 | 64QAM | 0.7 | 1 | 1 | 31752 | 24 | 4 | 44928 | |
| 20 | 15 | 106 | 13 | 23 | 64QAM | 0.7 | 1 | 1 | 64552 | 24 | 8 | 91584 | |
| 10 | 15 | 52 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 39936 | 24 | 5 | 44928 | |
| 20 | 15 | 106 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 81976 | 24 | 10 | 91584 | |
| 10 | 15 | 52 | 13 | 18 | 64QAM | 0.46 | 2 | 1 | 40976 | 24 | 5 | 89856 | |
| 20 | 15 | 106 | 13 | 18 | 64QAM | 0.46 | 2 | 1 | 83976 | 24 | 10 | 183168 | |
| 10 | 15 | 52 | 13 | 22 | 64QAM | 0.65 | 2 | 1 | 58384 | 24 | 7 | 89856 | |
| 20 | 15 | 106 | 13 | 22 | 64QAM | 0.65 | 2 | 1 | 118896 | 24 | 15 | 183168 | |
| 10 | 15 | 52 | 13 | 23 | 64QAM | 0.7 | 2 | 1 | 63528 | 24 | 8 | 89856 | |
| 20 | 15 | 106 | 13 | 23 | 64QAM | 0.7 | 2 | 1 | 129128 | 24 | 16 | 183168 | |
| 10 | 15 | 52 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 79896 | 24 | 10 | 89856 | |
| 20 | 15 | 106 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 163976 | 24 | 20 | 183168 | |
| 10 | 15 | 52 | 13 | 19 | 64QAM | 0.5 | 4 | 1 | 83976 | 24 | 10 | 164736 | |
| 20 | 15 | 106 | 13 | 19 | 64QAM | 0.5 | 4 | 1 | 167976 | 24 | 20 | 335808 | |
| 10 | 15 | 52 | 13 | 23 | 64QAM | 0.7 | 4 | 1 | 114776 | 24 | 14 | 164736 | |
| 20 | 15 | 106 | 13 | 23 | 64QAM | 0.7 | 4 | 1 | 237776 | 24 | 29 | 335808 | |
| 10 | 15 | 52 | 13 | 24 | 64QAM | 0.75 | 4 | 1 | 125016 | 24 | 15 | 164736 | |
| 20 | 15 | 106 | 13 | 24 | 64QAM | 0.75 | 4 | 1 | 254176 | 24 | 31 | 335808 | |
| 10 | 15 | 52 | 13 | 27 | 64QAM | 0.89 | 4 | 1 | 147576 | 24 | 18 | 164736 | |
| 20 | 15 | 106 | 13 | 27 | 64QAM | 0.89 | 4 | 1 | 295176 | 24 | 36 | 335808 | |

ocated full DL slots are with slot index i, if i is not in {0,10,11} for i = 0,1,...,19. So total number of allocated slots per 2 frames is 17.
CS Index is based on MCS Table defined in TS38.214 when 256QAM is not enabled. MCS 18 and 19 are equivalent to MCS 11 and 12 in 256Q
ly.
mber of DMRS REs per RB = 12,12,24,24 for number of MIMO layers = 1,2,3,4, respectively
S/PBCH block is transmitted in slot #0 with periodicity 20 ms.
erhead parameter for TBS determination is 0.

nore than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2_1.1.1-2: Sustained Downlink Data Rate Reference Channel for FDD 15kHz SCS FR1 (256QAM)**

| ı | Subcarrier spacing | Allocated resource blocks | Number of consecutive PDSCH symbols for allocated full DL slots (Note 1) | MCS Index (Note 2) | Modulation | Target Coding Rate | Number of MIMO layers | LDPC Base Graph | Information Bit Payload per Slot for allocated full DL slots (Note 1) | Transport block CRC per Slot for allocated full DL slots (Note 1) | Number of Code Blocks per Slot for allocated full DL slots (Note 1, 6) | a |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | kHz | PRBs | Symbols | | | | | | Bits | Bits | CBs | |
| | 15 | 52 | 13 | 20 | 256QAM | 0.67 | 1 | 1 | 39936 | 24 | 5 | |
| | 15 | 106 | 13 | 20 | 256QAM | 0.67 | 1 | 1 | 81976 | 24 | 10 | |
| | 15 | 52 | 13 | 21 | 256QAM | 0.69 | 1 | 1 | 42016 | 24 | 5 | |
| | 15 | 106 | 13 | 21 | 256QAM | 0.69 | 1 | 1 | 83976 | 24 | 10 | |
| | 15 | 52 | 13 | 26 | 256QAM | 0.9 | 1 | 1 | 53288 | 24 | 7 | |
| | 15 | 106 | 13 | 26 | 256QAM | 0.9 | 1 | 1 | 108552 | 24 | 13 | |
| | 15 | 52 | 13 | 20 | 256QAM | 0.67 | 2 | 1 | 79896 | 24 | 10 | |
| | 15 | 106 | 13 | 20 | 256QAM | 0.67 | 2 | 1 | 163976 | 24 | 20 | |
| | 15 | 52 | 13 | 21 | 256QAM | 0.69 | 2 | 1 | 83976 | 24 | 10 | |
| | 15 | 106 | 13 | 21 | 256QAM | 0.69 | 2 | 1 | 167976 | 24 | 20 | |
| | 15 | 133 | 13 | 21 | 256QAM | 0.69 | 2 | 1 | 213176 | 24 | 26 | |
| | 15 | 52 | 13 | 26 | 256QAM | 0.9 | 2 | 1 | 106576 | 24 | 13 | |
| | 15 | 106 | 13 | 26 | 256QAM | 0.9 | 2 | 1 | 217128 | 24 | 26 | |
| | 15 | 52 | 13 | 22 | 256QAM | 0.74 | 4 | 1 | 159880 | 24 | 19 | |
| | 15 | 106 | 13 | 22 | 256QAM | 0.74 | 4 | 1 | 327888 | 24 | 39 | |
| | 15 | 52 | 13 | 23 | 256QAM | 0.78 | 4 | 1 | 172176 | 24 | 21 | |
| | 15 | 106 | 13 | 23 | 256QAM | 0.78 | 4 | 1 | 352440 | 24 | 42 | |
| | 15 | 133 | 13 | 23 | 256QAM | 0.78 | 4 | 1 | 434280 | 24 | 52 | |
| | 15 | 52 | 13 | 26 | 256QAM | 0.9 | 4 | 1 | 196776 | 24 | 24 | |
| | 15 | 106 | 13 | 26 | 256QAM | 0.9 | 4 | 1 | 401640 | 24 | 48 | |

slots are with slot index i, if i is not in {0,10,11} for i = 0,1,...,19. So total number of allocated slots per 2 frames is 17.
ısed on MCS Table defined in TS38.214 when 256QAM is enabled.
S REs per RB = 12,12,24,24 for number of MIMO layers = 1,2,3,4, respectively
is transmitted in slot #0 with periodicity 20 ms.
ıeter for TBS determination is 0.
Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

## A.3.2_1.2 TDD

### A.3.2_1.2.1 Reference measurement channels for SCS 30 kHz FR1

**Table A.3.2_1.2.1-1: Sustained Downlink Data Rate Reference Channel for TDD 30kHz SCS FR1 (64QAM)**

| ı | Channel | Subcarrier | Allocated | Number of | MCS | Modulation | Target | Number | LDPC | Information | Transport | Number | Binary | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| bandwidth | spacing | resource blocks | consecutive PDSCH symbols for allocated full DL slots (Note 1) | Index (Note 2) | | Coding Rate | of MIMO layers | Base Graph | Bit Payload per Slot for allocated full DL slots (Note 1) | block CRC per Slot for allocated full DL slots (Note 1) | of Code Blocks per Slot for allocated full DL slots (Note 1, 6) | Channel Bits per Slot for allocated full DL slots (Note 1) | Th a |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MHz | kHz | PRBs | Symbols | | | | | | Bits | Bits | CBs | Bits | |
| 20 | 30 | 51 | 13 | 18 | 64QAM | 0.46 | 1 | 1 | 19968 | 24 | 3 | 44064 | |
| 100 | 30 | 273 | 13 | 18 | 64QAM | 0.46 | 1 | 1 | 106576 | 24 | 13 | 235872 | |
| 20 | 30 | 51 | 13 | 22 | 64QAM | 0.65 | 1 | 1 | 28680 | 24 | 4 | 44064 | |
| 100 | 30 | 273 | 13 | 22 | 64QAM | 0.65 | 1 | 1 | 151608 | 24 | 18 | 235872 | |
| 20 | 30 | 51 | 13 | 23 | 64QAM | 0.7 | 1 | 1 | 30728 | 24 | 4 | 44064 | |
| 100 | 30 | 273 | 13 | 23 | 64QAM | 0.7 | 1 | 1 | 163976 | 24 | 20 | 235872 | |
| 20 | 30 | 51 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 38936 | 24 | 5 | 44064 | |
| 100 | 30 | 273 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 208976 | 24 | 25 | 235872 | |
| 20 | 30 | 51 | 13 | 18 | 64QAM | 0.46 | 2 | 1 | 39936 | 24 | 5 | 88128 | |
| 100 | 30 | 273 | 13 | 18 | 64QAM | 0.46 | 2 | 1 | 213176 | 24 | 26 | 471744 | |
| 20 | 30 | 51 | 13 | 22 | 64QAM | 0.65 | 2 | 1 | 57376 | 24 | 7 | 88128 | |
| 100 | 30 | 273 | 13 | 22 | 64QAM | 0.65 | 2 | 1 | 303240 | 24 | 36 | 471744 | |
| 20 | 30 | 51 | 13 | 23 | 64QAM | 0.7 | 2 | 1 | 61480 | 24 | 8 | 88128 | |
| 100 | 30 | 273 | 13 | 23 | 64QAM | 0.7 | 2 | 1 | 327888 | 24 | 39 | 471744 | |
| 20 | 30 | 51 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 77896 | 24 | 10 | 88128 | |
| 100 | 30 | 273 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 417976 | 24 | 50 | 471744 | |
| 20 | 30 | 51 | 13 | 19 | 64QAM | 0.5 | 4 | 1 | 81976 | 24 | 10 | 161568 | |
| 100 | 30 | 273 | 13 | 19 | 64QAM | 0.5 | 4 | 1 | 434280 | 24 | 52 | 864864 | |
| 20 | 30 | 51 | 13 | 23 | 64QAM | 0.7 | 4 | 1 | 112648 | 24 | 14 | 161568 | |
| 100 | 30 | 273 | 13 | 23 | 64QAM | 0.7 | 4 | 1 | 606504 | 24 | 72 | 864864 | |
| 20 | 30 | 51 | 13 | 24 | 64QAM | 0.75 | 4 | 1 | 120936 | 24 | 15 | 161568 | |
| 100 | 30 | 273 | 13 | 24 | 64QAM | 0.75 | 4 | 1 | 655800 | 24 | 78 | 864864 | |
| 20 | 30 | 51 | 13 | 27 | 64QAM | 0.89 | 4 | 1 | 143400 | 24 | 18 | 161568 | |
| 100 | 30 | 273 | 13 | 27 | 64QAM | 0.89 | 4 | 1 | 770568 | 24 | 92 | 864864 | |

ocated full DL slots are with slot index i, if mod(i,10) = 0,1,2,3,4,5,6 and i is not in {0,20,21} for i = 0,1,...,39. So total number of allocated slots p
25.
CS Index is based on MCS Table defined in TS38.214 when 256QAM is not enabled. MCS 18 and 19 are equivalent to MCS 11 and 12 in 256Q
ly.
mber of DMRS REs per RB = 12,12,24,24 for number of MIMO layers = 1,2,3,4, respectively
S/PBCH block is transmitted in slot #0 with periodicity 20 ms.
erhead parameter for TBS determination is 0.
nore than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2_1.2.1-2: Sustained Downlink Data Rate Reference Channel for TDD 30kHz SCS FR1(256QAM)**

| r | Channel bandwidth | Subcarrier spacing | Allocated resource blocks | Number of consecutive PDSCH symbols for allocated full DL slots (Note 1) | MCS Index (Note 2) | Modulation | Target Coding Rate | Number of MIMO layers | LDPC Base Graph | Information Bit Payload per Slot for allocated full DL slots (Note 1) | Transport block CRC per Slot for allocated full DL slots | Number of Code Blocks per Slot for allocated full DL | Binary Channel Bits per Slot for allocated full DL slots | Th a |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | | (Note 1) | slots (Note 1, 6) | (Note 1) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MHz | kHz | PRBs | Symbols | | | | | | Bits | Bits | CBs | Bits | |
| 20 | 30 | 51 | 13 | 20 | 256QAM | 0.67 | 1 | 1 | 38936 | 24 | 5 | 58752 | |
| 100 | 30 | 273 | 13 | 20 | 256QAM | 0.67 | 1 | 1 | 208976 | 24 | 25 | 314496 | |
| 20 | 30 | 51 | 13 | 21 | 256QAM | 0.69 | 1 | 1 | 40976 | 24 | 5 | 58752 | |
| 100 | 30 | 273 | 13 | 21 | 256QAM | 0.69 | 1 | 1 | 217128 | 24 | 26 | 314496 | |
| 20 | 30 | 51 | 13 | 26 | 256QAM | 0.9 | 1 | 1 | 52224 | 24 | 7 | 58752 | |
| 100 | 30 | 273 | 13 | 26 | 256QAM | 0.9 | 1 | 1 | 278776 | 24 | 34 | 314496 | |
| 20 | 30 | 51 | 13 | 20 | 256QAM | 0.67 | 2 | 1 | 77896 | 24 | 10 | 117504 | |
| 100 | 30 | 273 | 13 | 20 | 256QAM | 0.67 | 2 | 1 | 417976 | 24 | 50 | 628992 | |
| 20 | 30 | 51 | 13 | 21 | 256QAM | 0.69 | 2 | 1 | 81976 | 24 | 10 | 117504 | |
| 100 | 30 | 273 | 13 | 21 | 256QAM | 0.69 | 2 | 1 | 434280 | 24 | 52 | 628992 | |
| 20 | 30 | 51 | 13 | 26 | 256QAM | 0.9 | 2 | 1 | 104496 | 24 | 13 | 117504 | |
| 100 | 30 | 273 | 13 | 26 | 256QAM | 0.9 | 2 | 1 | 557416 | 24 | 67 | 628992 | |
| 20 | 30 | 51 | 13 | 22 | 256QAM | 0.74 | 4 | 1 | 159880 | 24 | 19 | 215424 | |
| 100 | 30 | 273 | 13 | 22 | 256QAM | 0.74 | 4 | 1 | 852696 | 24 | 102 | 1153152 | |
| 20 | 30 | 51 | 13 | 23 | 256QAM | 0.78 | 4 | 1 | 167976 | 24 | 20 | 215424 | |
| 100 | 30 | 273 | 13 | 23 | 256QAM | 0.78 | 4 | 1 | 901344 | 24 | 107 | 1153152 | |
| 20 | 30 | 51 | 13 | 26 | 256QAM | 0.9 | 4 | 1 | 192624 | 24 | 23 | 215424 | |
| 100 | 30 | 273 | 13 | 26 | 256QAM | 0.9 | 4 | 1 | 1032192 | 24 | 123 | 1153152 | |

ocated full DL slots are with slot index i, if mod(i,10) = 0,1,2,3,4,5,6 and i is not in {0,20,21} for i = 0,1,...,39. So total number of allocated slots p
25.
CS Index is based on MCS Table defined in TS38.214 when 256QAM is enabled.
mber of DMRS REs per RB = 12,12,24,24 for number of MIMO layers = 1,2,3,4, respectively
S/PBCH block is transmitted in slot #0 with periodicity 20 ms.
erhead parameter for TBS determination is 0.
nore than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2_1.2.1-3: Sustained Downlink Data Rate Reference Channel for TDD 120kHz SCS FR2 (QPSK)**

| | Subcarrier spacing | Allocated resource blocks | Number of consecutive PDSCH symbols for allocated full DL slots (Note 1) | MCS Index (Note 2) | Modulation | Target Coding Rate | Number of MIMO layers | LDPC Base Graph | Information Bit Payload per Slot for allocated full DL slots (Note 1) | Transport block CRC per Slot for allocated full DL slots (Note 1) | Number of Code Blocks per Slot for allocated full DL slots (Note 1, 6) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | kHz | PRBs | Symbols | | | | | | Bits | Bits | CBs | |
| | 120 | 32 | 13 | 9 | QPSK | 0.66 | 1 | 1 | 5888 | 24 | 1 | |
| | 120 | 66 | 13 | 9 | QPSK | 0.66 | 1 | 1 | 12040 | 24 | 2 | |
| | 120 | 132 | 13 | 9 | QPSK | 0.66 | 1 | 1 | 24072 | 24 | 3 | |
| | 120 | 32 | 13 | 9 | QPSK | 0.66 | 2 | 1 | 11784 | 24 | 2 | |
| | 120 | 66 | 13 | 9 | QPSK | 0.66 | 2 | 1 | 24072 | 24 | 3 | |
| | 120 | 132 | 13 | 9 | QPSK | 0.66 | 2 | 1 | 48168 | 24 | 6 | |

slots are with slot index i, if mod(i,5) = 0,1,2 and i is not in {0,80,81} for i = 0,1,...,159. So total number of allocated slots per 2 frames is 93.
sed on MCS Table defined in TS38.214 when 256QAM is not enabled.
S REs per RB is 12.
is transmitted in slot #0 with periodicity 20 ms.
ieter for TBS determination is 6.
Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2_1.2.1-4: Sustained Downlink Data Rate Reference Channel for TDD 120kHz SCS FR2 (16QAM)**

| | Subcarrier spacing | Allocated resource blocks | Number of consecutive PDSCH symbols for allocated full DL slots (Note 1) | MCS Index (Note 2) | Modulation | Target Coding Rate | Number of MIMO layers | LDPC Base Graph | Information Bit Payload per Slot for allocated full DL slots (Note 1) | Transport block CRC per Slot for allocated full DL slots (Note 1) | Number of Code Blocks per Slot for allocated full DL slots (Note 1, 6) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **kHz** | **PRBs** | **Symbols** | | | | | | **Bits** | **Bits** | **CBs** | |
| | 120 | 32 | 13 | 16 | 16QAM | 0.64 | 1 | 1 | 11272 | 24 | 2 | |
| | 120 | 66 | 13 | 16 | 16QAM | 0.64 | 1 | 1 | 23568 | 24 | 3 | |
| | 120 | 132 | 13 | 16 | 16QAM | 0.64 | 1 | 1 | 47112 | 24 | 6 | |
| | 120 | 32 | 13 | 16 | 16QAM | 0.64 | 2 | 1 | 22536 | 24 | 3 | |
| | 120 | 66 | 13 | 16 | 16QAM | 0.64 | 2 | 1 | 47112 | 24 | 6 | |
| | 120 | 132 | 13 | 16 | 16QAM | 0.64 | 2 | 1 | 94248 | 24 | 12 | |

slots are with slot index i, if mod(i,5) = 0,1,2 and i is not in {0,80,81} for i = 0,1,...,159. So total number of allocated slots per 2 frames is 93.
sed on MCS Table defined in TS38.214 when 256QAM is not enabled.
S REs per RB is 12.
is transmitted in slot #0 with periodicity 20 ms.
ieter for TBS determination is 6.
Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

**Table A.3.2_1.2.1-5: Sustained Downlink Data Rate Reference Channel for TDD 120kHz SCS FR2 (64QAM)**

| | Subcarrier spacing | Allocated resource blocks | Number of consecutive PDSCH symbols for allocated full DL slots (Note 1) | MCS Index (Note 2) | Modulation | Target Coding Rate | Number of MIMO layers | LDPC Base Graph | Information Bit Payload per Slot for allocated full DL slots (Note 1) | Transport block CRC per Slot for allocated full DL slots (Note 1) | Number of Code Blocks per Slot for allocated full DL slots (Note 1, 6) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | KHz | PRBs | Symbols | | | | | | Bits | Bits | CBs | |
| | 120 | 32 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 23568 | 24 | 3 | |
| | 120 | 66 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 48168 | 24 | 6 | |
| | 120 | 132 | 13 | 27 | 64QAM | 0.89 | 1 | 1 | 96264 | 24 | 12 | |
| | 120 | 32 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 47112 | 24 | 6 | |
| | 120 | 66 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 96264 | 24 | 12 | |
| | 120 | 132 | 13 | 27 | 64QAM | 0.89 | 2 | 1 | 192624 | 24 | 23 | |

slots are with slot index i, if mod(i,5) = 0,1,2 and i is not in {0,80,81} for i = 0,1,...,159. So total number of allocated slots per 2 frames is 93.
sed on MCS Table defined in TS38.214 when 256QAM is not enabled.
S REs per RB is 12.
is transmitted in slot #0 with periodicity 20 ms.
neter for TBS determination is 6.
Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

## A.3.3 Reference measurement channels for PDCCH performance requirements

### A.3.3.1 FDD

#### A.3.3.1.1 Reference measurement channels for SCS 15 kHz FR1

**Table A.3.3.1.1-1: PDCCH Reference Channels (Time domain allocation 1 symbol)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.1-1.1 FDD | R.PDCCH.1-1.2 FDD | R.PDCCH.1-1.3 FDD | | | |
| Subcarrier spacing | kHz | 15 | 15 | 15 | | | |
| CORESET frequency domain allocation | | 48 | 48 | 48 | | | |
| CORESET time domain allocation | | 1 | 1 | 1 | | | |
| Aggregation level | | 4 | 4 | 8 | | | |
| DCI Format | | 1_0 | 1_1 | 1_1 | | | |
| Payload (without CRC) | Bits | 39 | 52 | 52 | | | |

**Table A.3.3.1.1-2: PDCCH Reference Channel (Time domain allocation 2 symbols)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.1-2.1 FDD | R.PDCCH.1-2.2 FDD | R.PDCCH.1-2.3 FDD | R.PDCCH.1-2.4 FDD | R.PDCCH.1-2.5 FDD | R.PDCCH.1-2.6 FDD |
| Subcarrier spacing | kHz | 15 | 15 | 15 | 15 | 15 | 15 |
| CORESET frequency domain allocation | | 24 | 24 | 24 | 48 | 48 | 48 |
| CORESET time domain allocation | | 2 | 2 | 2 | 2 | 2 | 2 |
| Aggregation level | | 2 | 4 | 2 | 4 | 8 | 16 |
| DCI Format | | 1_0 | 1_0 | 1_1 | 1_1 | 1_1 | 1_0 |
| Payload (without CRC) | Bits | 39 | 39 | 52 | 52 | 52 | 39 |

**Table A.3.3.1.1-3: Additional PDSCH Reference Channel FDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| DCI Format | | 1_0 | 1_1 |
| Channel bandwidth | MHz | 10 | 10 |
| Subcarrier spacing | kHz | 15 | 15 |
| Number of allocated resource blocks | PRBs | 52 | 52 |
| Number of consecutive PDSCH symbols | | 12 | 12 |
| Allocated slots per 2 frames | Slots | 19 | 19 |
| MCS table | | 64QAM | 64QAM |
| MCS index | | 4 | 4 |
| Modulation | | QPSK | QPSK |
| Target Coding Rate | | 0.30 | 0.30 |
| Number of MIMO layers | | 1 | 1 |
| Number of DMRS REs | | 12 | 12 |
| Overhead for TBS determination | | 0 | 0 |
| Information Bit Payload per Slot | | | |
| For Slot i = 0 | Bits | N/A | N/A |
| For Slots i = 1,…, 19 | Bits | 3368 | 4096 |
| Transport block CRC per Slot | | | |
| For Slot i = 0 | Bits | N/A | N/A |
| For Slots i = 1,…, 19 | Bits | 16 | 24 |
| Number of Code Blocks per Slot | | | |
| For Slot i = 0 | CBs | N/A | N/A |
| For Slots i = 1,…, 19 | CBs | 1 | 1 |
| Binary Channel Bits Per Slot | | | |
| For Slot i = 0 | Bits | N/A | N/A |
| For Slots i = 10, 11 | Bits | 9984 | 13104 |
| For Slots i = 1,…, 9, 12, …, 19 | Bits | 11232 | 13728 |
| Max. Throughput averaged over 2 frames | Mbps | 3.1996 | 3.8912 |
| Note 1: SS/PBCH block is transmitted in slot #0 with periodicity 20 ms.<br>Note 2: Slot i is slot index per 2 frames. | | | |

#### A.3.3.1.2 Reference measurement channels for SCS 30 kHz FR1

**Table A.3.3.1.2-1: PDCCH Reference Channels (Time domain allocation 1 symbol)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.2-1.1 FDD | R.PDCCH.2-1.2 FDD | R.PDCCH.2-1.3 FDD | | | |
| Subcarrier spacing | kHz | 30 | 30 | 30 | | | |
| CORESET frequency domain allocation | | 102 | 102 | 90 | | | |
| CORESET time domain allocation | | 1 | 1 | 1 | | | |
| Aggregation level | | 2 | 4 | 8 | | | |
| DCI Format | | 1_0 | 1_1 | 1_1 | | | |
| Payload (without CRC) | Bits | 41 | 53 | 53 | | | |

**Table A.3.3.1.2-2: PDCCH Reference Channel (Time domain allocation 2 symbols)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.2-2.1 FDD | | | | | |
| Subcarrier spacing | kHz | 30 | | | | | |
| CORESET frequency domain allocation | | 48 | | | | | |
| CORESET time domain allocation | | 2 | | | | | |
| Aggregation level | | 16 | | | | | |
| DCI Format | | 1_0 | | | | | |
| Payload (without CRC) | Bits | 41 | | | | | |

### A.3.3.2 TDD

#### A.3.3.2.1 Reference measurement channels for SCS 15 kHz FR1

**Table A.3.3.2.1-1: PDCCH Reference Channels (Time domain allocation 1 symbol)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.1-1.1 TDD | R.PDCCH.1-1.2 TDD | R.PDCCH.1-1.3 TDD | | | |
| Subcarrier spacing | kHz | 15 | 15 | 15 | | | |
| CORESET frequency domain allocation | | 48 | 48 | 48 | | | |
| CORESET time domain allocation | | 1 | 1 | 1 | | | |
| Aggregation level | | 4 | 4 | 8 | | | |
| DCI Format | | 1_0 | 1_1 | 1_1 | | | |
| Payload (without CRC) | Bits | 39 | 52 | 52 | | | |

**Table A.3.3.2.1-2: PDCCH Reference Channel (Time domain allocation 2 symbols)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.1-2.1 TDD | R.PDCCH.1-2.2 TDD | R.PDCCH.1-2.3 TDD | R.PDCCH.1-2.4 TDD | R.PDCCH.1-2.5 TDD | R.PDCCH.1-2.6 TDD |
| Subcarrier spacing | kHz | 15 | 15 | 15 | 15 | 15 | 15 |
| CORESET frequency domain allocation | | 24 | 24 | 24 | 48 | 48 | 48 |
| CORESET time domain allocation | | 2 | 2 | 2 | 2 | 2 | 2 |
| Aggregation level | | 2 | 4 | 2 | 4 | 8 | 16 |
| DCI Format | | 1_0 | 1_0 | 1_1 | 1_1 | 1_1 | 1_0 |
| Payload (without CRC) | Bits | 39 | 39 | 52 | 52 | 52 | 39 |

### A.3.3.2.2 Reference measurement channels for SCS 30 kHz FR1

**Table A.3.3.2.2-1: PDCCH Reference Channels (Time domain allocation 1 symbol)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.2-1.1 TDD | R.PDCCH.2-1.2 TDD | R.PDCCH.2-1.3 TDD | | | |
| Subcarrier spacing | kHz | 30 | 30 | 30 | | | |
| CORESET frequency domain allocation | | 102 | 102 | 90 | | | |
| CORESET time domain allocation | | 1 | 1 | 1 | | | |
| Aggregation level | | 2 | 4 | 8 | | | |
| DCI Format | | 1_0 | 1_1 | 1_1 | | | |
| Payload (without CRC) | Bits | 41 | 53 | 53 | | | |

**Table A.3.3.2.2-2: PDCCH Reference Channel (Time domain allocation 2 symbols)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.2-2.1 TDD | | | | | |
| Subcarrier spacing | kHz | 30 | | | | | |
| CORESET frequency domain allocation | | 48 | | | | | |
| CORESET time domain allocation | | 2 | | | | | |
| Aggregation level | | 16 | | | | | |
| DCI Format | | 1_0 | | | | | |
| Payload (without CRC) | Bits | 41 | | | | | |

**Table A.3.3.2.2-3: Additional PDSCH Reference Channel TDD**

| Parameter | Unit | Value | |
|---|---|---|---|
| DCI Format | | 1-0 | 1-1 |
| TDD UL/DL pattern | | FR1.30-1 | FR1.30-1 |
| Channel bandwidth | MHz | 40 | 40 |
| Subcarrier spacing | kHz | 30 | 30 |
| Allocated resource blocks | PRBs | 106 | 106 |
| Number of consecutive PDSCH symbols | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 4 | 4 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6) for i from {1,…,39} | | 12 | 12 |
| Allocated slots per 2 frames | | 31 | 31 |
| MCS table | | 64QAM | 64QAM |
| MCS index | | 4 | 4 |
| Modulation | | QPSK | QPSK |
| Target Coding Rate | | 0.30 | 0.3 |
| Number of MIMO layers | | 1 | 1 |
| Number of DMRS rEs | | | |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | | 6 | 6 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6) for i from {1,…,39} | | 12 | 12 |
| Overhead for TBS determination | | 0 | 0 |
| Information Bit Payload per Slot | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 2280 | 2664 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6) for i from {1,…,39} | Bits | 6912 | 8456 |
| Transport block CRC per Slot | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 16 | 16 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6) for i from {1,…,39} | Bits | 24 | 24 |
| Number of Code Blocks per Slot | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | CBs | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | CBs | 1 | 1 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6) for i from {1,…,39} | CBs | 1 | 2 |
| Binary Channel Bits Per Slot | | | |
| For Slots 0 and Slot i, if mod(i, 10) = {8,9} for i from {0,…,39} | Bits | N/A | N/A |
| For Slot i, if mod(i, 10) = 7 for i from {0,…,39} | Bits | 7488 | 8904 |
| For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6) for i from {1,…,39} | Bits | 22896 | 27984 |
| Max. Throughput averaged over 2 frames | Mbps | 9.78 | 11.94 |

#### A.3.3.2.3 Reference measurement channels for SCS 60 kHz FR1

#### A.3.3.2.4 Reference measurement channels for SCS 60 kHz FR2

#### A.3.3.2.5 Reference measurement channels for SCS 120 kHz FR2

**Table A.3.3.2.5-1: PDCCH Reference Channels (Time domain allocation 1 symbol)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.5-1.1 TDD | R.PDCCH.5-1.2 TDD | R.PDCCH.5-1.3 TDD | | | |
| Subcarrier spacing | kHz | 120 | 120 | 120 | | | |
| CORESET frequency domain allocation | | 60 | 60 | 60 | | | |
| CORESET time domain allocation | | 1 | 1 | 1 | | | |
| Aggregation level | | 2 | 4 | 8 | | | |
| DCI Format | | 1_0 | 1_1 | 1_1 | | | |
| Payload (without CRC) | Bits | 40 | 56 | 56 | | | |

**Table A.3.3.2.5-2: PDCCH Reference Channel (Time domain allocation 2 symbols)**

| Parameter | Unit | Value | | | | | |
|---|---|---|---|---|---|---|---|
| Reference channel | | R.PDCCH.5-2.1 TDD | | | | | |
| Subcarrier spacing | kHz | 120 | | | | | |
| CORESET frequency domain allocation | | 60 | | | | | |
| CORESET time domain allocation | | 2 | | | | | |
| Aggregation level | | 16 | | | | | |
| DCI Format | | 1_0 | | | | | |
| Payload (without CRC) | Bits | 40 | | | | | |

## A.3.4 Reference measurement channels for PBCH demodulation requirements

### A.3.4.1 Reference measurement channels for FR1

**Table A.3.4.1-1: PBCH Reference Channel**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channel | | R.PBCH.1 | R.PBCH.2 |
| SS/PBCH block subcarrier spacing | kHz | 15 | 30 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 56/864 | 56/864 |
| Payload (without CRC and timing related PBCH payload bits) | bits | 24 | 24 |

### A.3.4.2 Reference measurement channels for FR2

**Table A.3.4.2-1: PBCH Reference Channel**

| Parameter | Unit | Value | |
|---|---|---|---|
| Reference channels | | R.PBCH.5 | R.PBCH.6 |
| SS/PBCH block subcarrier spacing | kHz | 120 | 240 |
| Modulation | | QPSK | QPSK |
| Target coding rate | | 56/864 | 56/864 |
| Payload (without CRC and timing related PBCH payload bits) | bits | 24 | 24 |

## A.4 CSI reference measurement channels

This section defines the DL signal applicable to the reporting of channel status information (Clause X).

Tables in this section specifies the mapping of CQI index to Information Bit payload, which complies with the CQI definition specified in clause 5.2.2.1 of TS 38.214 [12] and with MCS definition specified in clause 5.1.3 of TS 38.214 [12]

**Table A.4-1: Mapping of CQI Index to Information Bit payload (CQI table 1)**

| TBS Scheme | | | | TBS.1-1 | TBS.1-2 | | | | |
|---|---|---|---|---|---|---|---|---|---|
| MCS table | | | | 64QAM | | | | | |
| Number of allocated PDSCH resource blocks | | | | 66 | 66 | | | | |
| Number of consecutive PDSCH symbols | | | | 12 | 12 | | | | |
| Number of PDSCH MIMO layers | | | | 1 | 2 | | | | |
| Number of DMRS REs (Note 1) | | | | 24 | 24 | | | | |
| Overhead for TBS determination | | | | 6 | 6 | | | | |
| Available RE-s | | | | 7920 | 7920 | | | | |
| CQI index | Spectral efficiency | MCS index | Modulation | Information Bit Payload per Slot | | | | | |
| 0 | OOR | OOR | OOR | N/A | N/A | | | | |
| 1 | 0.2344 | 0 | QPSK | 1800 | 3624 | | | | |
| 2 | 0.2344 | 0 | | 1800 | 3624 | | | | |
| 3 | 0.3770 | 2 | | 2856 | 5640 | | | | |
| 4 | 0.6016 | 4 | | 4480 | 8968 | | | | |
| 5 | 0.8770 | 6 | | 6528 | 13064 | | | | |
| 6 | 1.1758 | 8 | | 8712 | 17928 | | | | |
| 7 | 1.4766 | 11 | 16QAM | 11016 | 22032 | | | | |
| 8 | 1.9141 | 13 | | 14343 | 28680 | | | | |
| 9 | 2.4063 | 15 | | 17928 | 35856 | | | | |
| 10 | 2.7305 | 18 | 64QAM | 20496 | 40976 | | | | |
| 11 | 3.3223 | 20 | | 25104 | 50184 | | | | |
| 12 | 3.9023 | 22 | | 29192 | 58384 | | | | |
| 13 | 4.5234 | 24 | | 33816 | 67584 | | | | |
| 14 | 5.1152 | 26 | | 38936 | 77896 | | | | |
| 15 | 5.5547 | 28 | | 42016 | 83976 | | | | |
| Note 1: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data<br>Note 2: PDSCH is not scheduled on slots containing CSI-RS or slots which are not full DL<br>Note 3: PDSCH is not scheduled on slots containing PBCH, i.e. slot#0 per 20ms periodicity<br>Note 4: Spectral efficiency is based on MCS Table defined in Table 5.1.3.1-1 of TS 38.214 [12] | | | | | | | | | |

**Table A.4-2: Mapping of CQI Index to Information Bit payload (CQI table 2)**

| TBS Scheme | | | | TBS.2-1 | TBS.2-2 | TBS.2-3 | TBS.2-4 | TBS.2-5 | TBS.2-6 | TBS.2- |
|---|---|---|---|---|---|---|---|---|---|---|
| MCS table | | | | 256QAM | | | | | | |
| Number of allocated PDSCH resource blocks | | | | 52 | 52 | 106 | 106 | 8 | 16 | 32 |
| Number of consecutive PDSCH symbols | | | | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| Number of PDSCH MIMO layers | | | | 1 | 2 | 1 | 2 | 1 | 1 | 1 |
| Number of DMRS REs (Note 1) | | | | 24 | 24 | 24 | 24 | 24 | 24 | 24 |
| Overhead for TBS determination | | | | 0 | 0 | 0 | 0 | 0 | 0 | 6 |
| Available RE-s for PDSCH | | | | 6240 | 6240 | 12720 | 12720 | 960 | 1920 | 3680 |
| CQI index | Spectral efficiency | MCS index | Modulation | Information Bit Payload per Slot | | | | | | |
| 0 | OOR | OOR | OOR | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 1 | 0.2344 | 0 | QPSK | 1480 | 2976 | 2976 | 5896 | 224 | 456 | 848 |
| 2 | 0.3770 | 1 | | 2408 | 4744 | 4744 | 9480 | 368 | 736 | 1416 |
| 3 | 0.8770 | 3 | | 5504 | 11016 | 11016 | 22536 | 848 | 1736 | 3240 |
| 4 | 1.4766 | 5 | 16QAM | 9224 | 18432 | 18960 | 37896 | 1416 | 2856 | 5376 |
| 5 | 1.9141 | 7 | | 12040 | 24072 | 24576 | 49176 | 1864 | 3752 | 6912 |
| 6 | 2.4063 | 9 | | 15112 | 30216 | 30728 | 61480 | 2408 | 4608 | 8712 |
| 7 | 2.7305 | 11 | 64QAM | 16896 | 33816 | 34816 | 69672 | 2600 | 5248 | 9992 |
| 8 | 3.3223 | 13 | | 20496 | 40976 | 42016 | 83976 | 3240 | 6400 | 12040 |
| 9 | 3.9023 | 15 | | 24576 | 49176 | 49176 | 98376 | 3752 | 7424 | 14344 |
| 10 | 4.5234 | 17 | | 28168 | 56368 | 57376 | 114776 | 4352 | 8712 | 16392 |
| 11 | 5.1152 | 19 | | 31752 | 63528 | 65576 | 131176 | 4864 | 9736 | 18432 |
| 12 | 5.5547 | 21 | 256QAM | 34816 | 69672 | 69672 | 139376 | 5248 | 10760 | 20496 |
| 13 | 6.2266 | 23 | | 38936 | 77896 | 79896 | 159880 | 6016 | 12040 | 22536 |
| 14 | 6.9141 | 25 | | 43032 | 86040 | 88064 | 176208 | 6656 | 13320 | 25104 |
| 15 | 7.4063 | 27 | | 46104 | 92200 | 94248 | 188576 | 7040 | 14088 | 27144 |

Note 1: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data
Note 2: PDSCH is not scheduled on slots containing CSI-RS or slots which are not full DL
Note 3: PDSCH is not scheduled on slots containing PBCH, i.e. slot#0 per 20ms periodicity
Note 4: Spectral efficiency is based on MCS Table defined in Table 5.1.3.1-2 of TS 38.214 [12]

**Table A.4-3: Mapping of CQI Index to Information Bit payload (CQI table 2, Rank 3 and Rank 4)**

| TBS Scheme | | | | TBS.3-1 | TBS.3-2 | TBS.3-3 | TBS.3-4 | | |
|---|---|---|---|---|---|---|---|---|---|
| MCS table | | | | 256QAM | | | | | |
| Number of allocated PDSCH resource blocks | | | | 52 | 52 | 106 | 106 | | |
| Number of consecutive PDSCH symbols | | | | 12 | 12 | 12 | 12 | | |
| Number of PDSCH MIMO layers | | | | 3 | 4 | 3 | 4 | | |
| Number of DMRS REs (Note 1) | | | | 24 | 24 | 24 | 24 | | |
| Overhead for TBS determination | | | | 0 | 0 | 0 | 0 | | |
| Available RE-s for PDSCH | | | | 6240 | 6240 | 12720 | 12720 | | |
| CQI index | Spectral efficiency | MCS index | Modulation | Information Bit Payload per Slot | | | | | |
| 0 | OOR | OOR | OOR | N/A | N/A | N/A | N/A | | |
| 1 | 0.2344 | 0 | QPSK | 4360 | 5896 | 8976 | 11784 | | |
| 2 | 0.3770 | 1 | | 7048 | 9480 | 14344 | 18976 | | |
| 3 | 0.8770 | 3 | | 16392 | 22032 | 33816 | 45096 | | |
| 4 | 1.4766 | 5 | 16QAM | 27656 | 36896 | 56368 | 75792 | | |
| 5 | 1.9141 | 7 | | 35856 | 48168 | 73776 | 98376 | | |
| 6 | 2.4063 | 9 | | 45096 | 60456 | 92200 | 122976 | | |
| 7 | 2.7305 | 11 | 64QAM | 51216 | 67584 | 104496 | 139376 | | |
| 8 | 3.3223 | 13 | | 62504 | 81976 | 127080 | 167976 | | |
| 9 | 3.9023 | 15 | | 73776 | 98376 | 147576 | 196776 | | |
| 10 | 4.5234 | 17 | | 83976 | 112648 | 172176 | 229576 | | |
| 11 | 5.1152 | 19 | | 96264 | 127080 | 196776 | 262376 | | |
| 12 | 5.5547 | 21 | 256QAM | 104496 | 139376 | 213176 | 278776 | | |
| 13 | 6.2266 | 23 | | 116792 | 155776 | 237776 | 319784 | | |
| 14 | 6.9141 | 25 | | 129128 | 172176 | 262376 | 352440 | | |
| 15 | 7.4063 | 27 | | 139376 | 184424 | 278776 | 376896 | | |
| Note 1: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data<br>Note 2: PDSCH is not scheduled on slots containing CSI-RS or slots which are not full DL<br>Note 3: PDSCH is not scheduled on slots containing PBCH, i.e. slot#0 per 20ms periodicity<br>Note 4: Spectral efficiency is based on MCS Table defined in Table 5.1.3.1-2 of TS 38.214 [12] | | | | | | | | | |

**Table A.4-4: Mapping of CQI Index to Information Bit payload (CQI table 3)**

| TBS Scheme | | | | TBS.4-1 | TBS.4-2 | | | | |
|---|---|---|---|---|---|---|---|---|---|
| MCS table | | | | 64QAMLowSE | | | | | |
| Number of allocated PDSCH resource blocks | | | | 52 | 106 | | | | |
| Number of consecutive PDSCH symbols | | | | 12 | 12 | | | | |
| Number of PDSCH MIMO layers | | | | 1 | 1 | | | | |
| Number of DMRS REs (Note 1) | | | | 24 | 24 | | | | |
| Overhead for TBS determination | | | | 0 | 0 | | | | |
| Available RE-s for PDSCH | | | | 6240 | 12720 | | | | |
| CQI index | Spectral efficiency | MCS index | Modulation | Information Bit Payload per Slot | | | | | |
| 0 | OOR | OOR | OOR | N/A | N/A | | | | |
| 1 | 0.0586 | 0 | QPSK | 368 | 768 | | | | |
| 2 | 0.0977 | 2 | | 608 | 1256 | | | | |
| 3 | 0.1523 | 4 | | 984 | 2024 | | | | |
| 4 | 0.2344 | 6 | | 1480 | 2976 | | | | |
| 5 | 0.3770 | 8 | | 2408 | 4744 | | | | |
| 6 | 0.6016 | 10 | | 3752 | 7680 | | | | |
| 7 | 0.8770 | 12 | | 5504 | 11016 | | | | |
| 8 | 1.1758 | 14 | | 7296 | 14856 | | | | |
| 9 | 1.4766 | 16 | 16QAM | 9224 | 18960 | | | | |
| 10 | 1.9141 | 18 | | 12040 | 24576 | | | | |
| 11 | 2.4063 | 20 | | 15112 | 30728 | | | | |
| 12 | 2.7305 | 22 | 64QAM | 16896 | 34816 | | | | |
| 13 | 3.3223 | 24 | | 20496 | 42016 | | | | |
| 14 | 3.9023 | 26 | | 24576 | 49176 | | | | |
| 15 | 4.5234 | 28 | | 28168 | 57376 | | | | |
| Note 1: Number of DMRS REs includes the overhead of the DM-RS CDM groups without data.<br>Note 2: PDSCH is not scheduled on slots containing CSI-RS or slots which are not full DL.<br>Note 3: PDSCH is not scheduled on slots containing PBCH, i.e. slot#0 per 20ms periodicity. | | | | | | | | | |

# A.5 OFDMA Channel Noise Generator (OCNG)

## A.5.1 OCNG Patterns for FDD

### A.5.1.1 OCNG FDD pattern 1: Generic OCNG FDD Pattern for all unused REs

**Table A.5.1.1-1: OP.1 FDD: Generic OCNG FDD Pattern for all unused REs**

| OCNG Appliance / OCNG Parameters | Control Region (CORESET) | Data Region |
|---|---|---|
| Resources allocated | All unused REs (Note 1) | All unused REs (Note 2) |
| Structure | PDCCH | PDSCH |
| Content | Uncorrelated pseudo random QPSK modulated data | Uncorrelated pseudo random QPSK modulated data |
| Transmission scheme for multiple antennas ports transmission | Single Tx port transmission | Spatial multiplexing using any precoding matrix with dimensions same as the precoding matrix for PDSCH |
| Subcarrier Spacing | Same as for RMC PDCCH in the active BWP | Same as for RMC PDSCH in the active BWP |
| Power Level | Same as for RMC PDCCH | Same as for RMC PDSCH |
| Note 1: All unused REs in the active CORESETS appointed by the search spaces in use.<br>Note 2: Unused available REs refer to REs in PRBs not allocated for any physical channels, CORESETs, synchronization signals or reference signals, and excluding REs in all the available PDSCH DMRS CDM groups, in channel bandwidth. | | |

## A.5.2 OCNG Patterns for TDD

### A.5.2.1 OCNG TDD pattern 1: Generic OCNG TDD Pattern for all unused REs

**Table A.5.2.1-1: OP.1 TDD: Generic OCNG TDD Pattern for all unused REs**

| OCNG Appliance / OCNG Parameters | Control Region (CORESET) | Data Region |
|---|---|---|
| Resources allocated | All unused REs (Note 1) | All unused REs (Note 2) |
| Structure | PDCCH | PDSCH |
| Content | Uncorrelated pseudo random QPSK modulated data | Uncorrelated pseudo random QPSK modulated data |
| Transmission scheme for multiple antennas ports transmission | Single Tx port transmission | Spatial multiplexing using any precoding matrix with dimensions same as the precoding matrix for PDSCH |
| Subcarrier Spacing | Same as for RMC PDCCH in the active BWP | Same as for RMC PDSCH in the active BWP |
| Power Level | Same as for RMC PDCCH | Same as for RMC PDSCH |
| Note 1: All unused REs in the active CORESETS appointed by the search spaces in use.<br>Note 2: Unused available REs refer to REs in PRBs not allocated for any physical channels, CORESETs, synchronization signals or reference signals, and excluding REs in all the available PDSCH DMRS CDM groups, in channel bandwidth. | | |

# Annex B (normative): Propagation conditions

## B.0 No interference

The downlink connection between the System Simulator and the UE is without Additive White Gaussian Noise, and has no fading or multipath effects.

## B.1 Static propagation condition

### B.1.1 UE Receiver with 2Rx

For 1 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For 2 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{pmatrix} 1 & j \\ 1 & -j \end{pmatrix}.$$

For 4 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & j & j \\ 1 & 1 & -j & -j \end{bmatrix}$$

For 8 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 & j & j & j & j \\ 1 & 1 & 1 & 1 & -j & -j & -j & -j \end{bmatrix}$$

### B.1.2 UE Receiver with 4Rx

For 1 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

For 2 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{bmatrix} 1 & j \\ 1 & -j \\ 1 & j \\ 1 & -j \end{bmatrix}.$$

For 4 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & j & j \\ 1 & 1 & -j & -j \\ 1 & -1 & j & -j \\ 1 & -1 & -j & j \end{bmatrix}.$$

For 8 port transmission the channel matrix is defined in the frequency domain by:

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 & j & j & j & j \\ 1 & 1 & 1 & 1 & -j & -j & -j & -j \\ 1 & 1 & -1 & -1 & j & j & -j & -j \\ 1 & 1 & -1 & -1 & -j & -j & j & j \end{bmatrix}$$

# B.2 Multi-path fading propagation conditions

The multipath propagation conditions consist of several parts:

- A delay profile in the form of a "tapped delay-lin", characterized by a number of taps at fixed positions on a sampling grid. The profile can be further characterized by the r.m.s. delay spread and the maximum delay spanned by the taps.
- A combination of channel model parameters that include the Delay profile and the Doppler spectrum that is characterized by a classical spectrum shape and a maximum Doppler frequency.
- Different models are used for FR1 and FR2.

Initial channel matrix for LOS component of TDL-D channel model is equal to channel matrix of Static propagation conditions in Clause B.1.

## B.2.1 Delay profiles

The delay profiles are simplified from the TR 38.901 [15] TDL models. The simplification steps are shown below for information. These steps are only used when new delay profiles are created. Otherwise, the delay profiles specified in B.2.1.1 and B.2.1.2 can be used as such.

Step 1: Use the original TDL model from TR38.901 [15].

Step 2: Re-order the taps in ascending delays

Step 3: Perform delay scaling according to the procedure described in subclause 7.7.3 in TR 38.901 [15].

Step 4: Apply the quantization to the delay resolution 5 ns. This is done simply by rounding the tap delays to the nearest multiple of the delay resolution.

Step 5: If multiple taps are rounded to the same delay bin, merge them by calculating their linear power sum.

Step 6: If there are more than 12 taps in the quantized model, merge the taps as follows:

- Find the weakest tap from all taps (both merged and unmerged taps are considered)
  - If there are two or more taps having the same value and are the weakest, select the tap with the smallest delay as the weakest tap.
- When the weakest tap is the first delay tap, merge taps as follows:
  - Update the power of the first delay tap as the linear power sum of the weakest tap and the second delay tap.
  - Remove the second delay tap.

- When the weakest tap is the last delay tap, merge taps as follows:
  - Update the power of the last delay tap as the linear power sum of the second-to-last tap and the last tap.
  - Remove the second-to-last tap.
- Otherwise
  - For each side of the weakest tap, identify the neighbour tap that has the smaller delay difference to the weakest tap.
    - When the delay difference between the weakest tap and the identified neighbour tap on one side equals the delay difference between the weakest tap and the identified neighbour tap on the other side.
      - Select the neighbour tap that is weaker in power for merging.
    - Otherwise, select the neighbour tap that has smaller delay difference for merging.
  - To merge, the power of the merged tap is the linear sum of the power of the weakest tap and the selected tap.
  - When the selected tap is the first tap, the location of the merged tap is the location of the first tap. The weakest tap is removed.
  - When the selected tap is the last tap, the location of the merged tap is the location of the last tap. The weakest tap is removed.
  - Otherwise, the location of the merged tap is based on the average delay of the weakest tap and selected tap. If the average delay is on the sampling grid, the location of the merged tap is the average delay. Merge two parallel taps with different delays (average delay, sum power) starting from the weakest ones. Otherwise, the location of the merged tap is rounded towards the direction of the selected tap (e.g. 10 ns & 20 ns → 15 ns, 10 ns & 25 ns → 20 ns, if 25 ns had higher or equal power; 15 ns, if 10 ns had higher power). The weakest tap and the selected tap are removed.
- Repeat step 6 until the final number of taps is 12.

Step 7: Round the amplitudes of taps to one decimal (e.g. -8.78 dB → -8.8 dB)

Step 8: If the delay spread has slightly changed due to the tap merge, adjust the final delay spread by increasing or decreasing the power of the last tap so that the delay spread is corrected.

Step 9: Re-normalize tap powers such that the strongest tap is at 0dB.

Note 1: Some values of the delay profile created by the simplification steps may differ from the values in tables B.2.1.1-2, B.2.1.1-3, B.2.1.1-4, B.2.1.2-2, and B.2.1.1-3 for the corresponding model.

Note 2: For Step 5 and Step 6, the power values are expressed in the linear domain using 6 digits of precision. The operations are in the linear domain.

Note 3: Delay profile for TDLD30 is generated under assumption that Steps 1-8 are applied for taps with Rayleigh distribution.

## B.2.1.1 Delay profiles for FR1

The delay profiles for FR1 are selected to be representative of low, medium and high delay spread environment. The resulting model parameters are specified in B.2.1.1-1 and the tapped delay line models are specified in Tables B.2.1.1-2 ~ Table B.2.1.1-4.

**Table B.2.1.1-1: Delay profiles for NR channel models**

| Model | Number of channel taps | Delay spread (r.m.s.) | Maximum excess tap delay (span) | Delay resolution |
|---|---|---|---|---|
| TDLA30 | 12 | 30 ns | 290 ns | 5 ns |
| TDLB100 | 12 | 100 ns | 480 ns | 5 ns |
| TDLC300 | 12 | 300 ns | 2595 ns | 5 ns |

**Table B.2.1.1-2: TDLA30 (DS = 30 ns)**

| Tap # | Delay [ns] | Power [dB] | Fading distribution |
|---|---|---|---|
| 1 | 0 | -15.5 | Rayleigh |
| 2 | 10 | 0 | Rayleigh |
| 3 | 15 | -5.1 | Rayleigh |
| 4 | 20 | -5.1 | Rayleigh |
| 5 | 25 | -9.6 | Rayleigh |
| 6 | 50 | -8.2 | Rayleigh |
| 7 | 65 | -13.1 | Rayleigh |
| 8 | 75 | -11.5 | Rayleigh |
| 9 | 105 | -11.0 | Rayleigh |
| 10 | 135 | -16.2 | Rayleigh |
| 11 | 150 | -16.6 | Rayleigh |
| 12 | 290 | -26.2 | Rayleigh |

**Table B.2.1.1-3: TDLB100 (DS = 100ns)**

| Tap # | Delay [ns] | Power [dB] | Fading distribution |
|---|---|---|---|
| 1 | 0 | 0 | Rayleigh |
| 2 | 10 | -2.2 | Rayleigh |
| 3 | 20 | -0.6 | Rayleigh |
| 4 | 30 | -0.6 | Rayleigh |
| 5 | 35 | -0.3 | Rayleigh |
| 6 | 45 | -1.2 | Rayleigh |
| 7 | 55 | -5.9 | Rayleigh |
| 8 | 120 | -2.2 | Rayleigh |
| 9 | 170 | -0.8 | Rayleigh |
| 10 | 245 | -6.3 | Rayleigh |
| 11 | 330 | -7.5 | Rayleigh |
| 12 | 480 | -7.1 | Rayleigh |

**Table B.2.1.1-4: TDLC300 (DS = 300 ns)**

| Tap # | Delay [ns] | Power [dB] | Fading distribution |
|---|---|---|---|
| 1 | 0 | -6.9 | Rayleigh |
| 2 | 65 | 0 | Rayleigh |
| 3 | 70 | -7.7 | Rayleigh |
| 4 | 190 | -2.5 | Rayleigh |
| 5 | 195 | -2.4 | Rayleigh |
| 6 | 200 | -9.9 | Rayleigh |
| 7 | 240 | -8.0 | Rayleigh |
| 8 | 325 | -6.6 | Rayleigh |
| 9 | 520 | -7.1 | Rayleigh |
| 10 | 1045 | -13.0 | Rayleigh |
| 11 | 1510 | -14.2 | Rayleigh |
| 12 | 2595 | -16.0 | Rayleigh |

## B.2.1.2 Delay profiles for FR2

The delay profiles for FR2 are specified in B.2.1.2-1 and the tapped delay line models are specified in Tables B.2.1.2-2 and B.2.1.2-3.

**Table B.2.1.2-1: Delay profiles for NR channel models**

| Model | Number of channel taps | Delay spread (r.m.s.) | Maximum excess tap delay (span) | Delay resolution |
|---|---|---|---|---|
| TDLA30 | 12 | 30 ns | 290 ns | 5 ns |
| TDLC60 | 12 | 60 ns | 520 ns | 5 ns |
| TDLD30 | 10 | 30 ns | 375 ns | 5 ns |

**Table B.2.1.2-2: TDLA30 (DS = 30 ns)**

| Tap # | Delay [ns] | Power [dB] | Fading distribution |
|---|---|---|---|
| 1 | 0 | -15.5 | Rayleigh |
| 2 | 10 | 0 | Rayleigh |
| 3 | 15 | -5.1 | Rayleigh |
| 4 | 20 | -5.1 | Rayleigh |
| 5 | 25 | -9.6 | Rayleigh |
| 6 | 50 | -8.2 | Rayleigh |
| 7 | 65 | -13.1 | Rayleigh |
| 8 | 75 | -11.5 | Rayleigh |
| 9 | 105 | -11.0 | Rayleigh |
| 10 | 135 | -16.2 | Rayleigh |
| 11 | 150 | -16.6 | Rayleigh |
| 12 | 290 | -26.2 | Rayleigh |

**Table B.2.1.2-3: TDLC60 (DS = 60 ns)**

| Tap # | Delay [ns] | Power [dB] | Fading distribution |
|---|---|---|---|
| 1 | 0 | -7.8 | Rayleigh |
| 2 | 15 | -0.3 | Rayleigh |
| 3 | 40 | 0 | Rayleigh |
| 4 | 50 | -8.9 | Rayleigh |
| 5 | 55 | -14.5 | Rayleigh |
| 6 | 75 | -8.5 | Rayleigh |
| 7 | 80 | -10.2 | Rayleigh |
| 8 | 130 | -12.1 | Rayleigh |
| 9 | 210 | -13.9 | Rayleigh |
| 10 | 300 | -15.2 | Rayleigh |
| 11 | 360 | -16.9 | Rayleigh |
| 12 | 520 | -19.4 | Rayleigh |

**Table B.2.1.2-4: TDLD30 (DS = 30 ns)**

| Tap # | Delay [ns] | Power [dB] | Fading distribution |
|---|---|---|---|
| 1 | 0 | -0.2 | LOS path |
| | 0 | -12.4 | Rayleigh |
| 2 | 20 | -21 | Rayleigh |
| 3 | 40 | -16.7 | Rayleigh |
| 4 | 55 | -18.3 | Rayleigh |
| 5 | 80 | -21.9 | Rayleigh |
| 6 | 120 | -27.8 | Rayleigh |
| 7 | 240 | -23.6 | Rayleigh |
| 8 | 285 | -24.8 | Rayleigh |
| 9 | 290 | -30.0 | Rayleigh |
| 10 | 375 | -27.6 | Rayleigh |
| Note 1: Tap #1 follows a Ricean distribution. | | | |

## B.2.2 Combinations of channel model parameters

The propagation conditions used for the performance measurements in multi-path fading environment are indicated as a combination of a channel model name and a maximum Doppler frequency, i.e. TDLA<DS>-<Doppler>, TDLB<DS>-<Doppler> or TDLC<DS>-<Doppler> where '<DS>' indicates the desired delay spread and '<Doppler>' indicates the maximum Doppler frequency (Hz).

Table B.2.2-1 and Table B.2.2-2 show the propagation conditions that are used for the performance measurements in multi-path fading environment for low, medium and high Doppler frequencies for FR1 and FR2, respectively.

**Table B.2.2-1: Channel model parameters for FR1**

| Combination name | Model | Maximum Doppler frequency |
|---|---|---|
| TDLA30-5 | TDLA30 | 5 Hz |
| TDLA30-10 | TDLA30 | 10 Hz |
| TDLB100-400 | TDLB100 | 400 Hz |
| TDLC300-100 | TDLC300 | 100 Hz |
| TDLC300-600 | TDLC300 | 600 Hz |
| TDLC300-1200 | TDLC300 | 1200 Hz |

**Table B.2.2-2: Channel model parameters for FR2**

| Combination name | Model | Maximum Doppler frequency |
|---|---|---|
| TDLA30-35 | TDLA30 | 35 Hz |
| TDLA30-75 | TDLA30 | 75 Hz |
| TDLA30-300 | TDLA30 | 300 Hz |
| TDLC60-300 | TDLC60 | 300 Hz |
| TDLD30-75 | TDLD30 | 75 Hz |

## B.2.3 MIMO Channel Correlation Matrices

The MIMO channel correlation matrices defined in B.2.3 apply for the antenna configuration using uniform linear arrays at both gNB and UE and for the antenna configuration using cross polarized antennas.

### B.2.3.1 MIMO Correlation Matrices using Uniform Linear Array (ULA)

The MIMO channel correlation matrices defined in B.2.3.1 apply for the antenna configuration using uniform linear array (ULA) at both gNB and UE.

#### B.2.3.1.1 Definition of MIMO Correlation Matrices

Table B.2.3.1.1-1 defines the correlation matrix for the gNB.

**Table B.2.3.1.1-1: gNB correlation matrix**

| | One antenna | Two antennas | Four antennas |
|---|---|---|---|
| gNB Correlation | $R_{gNB} = 1$ | $R_{gNB} = \begin{pmatrix} 1 & \alpha \\ \alpha^* & 1 \end{pmatrix}$ | $R_{gNB} = \begin{pmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{pmatrix}$ |

Table B.2.3.1.1-2 defines the correlation matrix for the UE:

**Table B.2.3.1.1-2 UE correlation matrix**

| | One antenna | Two antennas | Four antennas |
|---|---|---|---|
| UE Correlation | $R_{UE}=1$ | $R_{UE}=\begin{pmatrix}1 & \beta\\ \beta^{*} & \mathbf{1}\end{pmatrix}$ | $R_{UE}=\begin{pmatrix}1 & \beta^{1/9} & \beta^{4/9} & \beta\\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9}\\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9}\\ \beta^{*} & \beta^{4/9*} & \beta^{1/9*} & 1\end{pmatrix}$ |

Table B.2.3.1.1-3 defines the channel spatial correlation matrix $R_{spat}$ . The parameters, $\alpha$ and $\beta$ in Table B.2.3.1-3 defines the spatial correlation between the antennas at the gNB and UE.

**Table B.2.3.1.1-3: $R_{spat}$ correlation matrices**

| | |
|---|---|
| **1x2 case** | $R_{spat} = R_{UE} = \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix}$ |
| **1x4 case** | $R_{spat} = R_{UE} = \begin{pmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^* & \beta^{4/9*} & \beta^{1/9*} & 1 \end{pmatrix}$ |
| **2x1 case** | $R_{spat} = R_{gNB} = \begin{bmatrix} 1 & \alpha \\ \alpha^* & 1 \end{bmatrix}$ |
| **2x2 case** | $R_{spat} = R_{gNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha \\ \alpha^* & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix} = \begin{bmatrix} 1 & \beta & \alpha & \alpha\beta \\ \beta^* & 1 & \alpha\beta^* & \alpha \\ \alpha^* & \alpha^*\beta & 1 & \beta \\ \alpha^*\beta^* & \alpha^* & \beta^* & 1 \end{bmatrix}$ |
| **2x4 case** | $R_{spat} = R_{gNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha \\ \alpha^* & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^* & \beta^{4/9*} & \beta^{1/9*} & 1 \end{bmatrix}$ |
| **4x1 case** | $R_{spat} = R_{gNB} = \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{bmatrix}$ |
| **4x2 case** | $R_{spat} = R_{gNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & \beta \\ \beta^* & 1 \end{bmatrix}$ |
| **4x4 case** | $R_{spat} = R_{gNB} \otimes R_{UE} = \begin{bmatrix} 1 & \alpha^{1/9} & \alpha^{4/9} & \alpha \\ \alpha^{1/9*} & 1 & \alpha^{1/9} & \alpha^{4/9} \\ \alpha^{4/9*} & \alpha^{1/9*} & 1 & \alpha^{1/9} \\ \alpha^* & \alpha^{4/9*} & \alpha^{1/9*} & 1 \end{bmatrix} \otimes \begin{pmatrix} 1 & \beta^{1/9} & \beta^{4/9} & \beta \\ \beta^{1/9*} & 1 & \beta^{1/9} & \beta^{4/9} \\ \beta^{4/9*} & \beta^{1/9*} & 1 & \beta^{1/9} \\ \beta^* & \beta^{4/9*} & \beta^{1/9*} & 1 \end{pmatrix}$ |

For cases with more antennas at either gNB or UE or both, the channel spatial correlation matrix can still be expressed as the Kronecker product of $R_{gNB}$ and $R_{UE}$ according to $R_{spat} = R_{gNB} \otimes R_{UE}$.

### B.2.3.1.2 MIMO Correlation Matrices at High, Medium and Low Level

The $\alpha$ and $\beta$ for different correlation types are given in Table B.2.3.1.2-1.

**Table B.2.3.1.2-1: The $\alpha$ and $\beta$ parameters for ULA MIMO correlation matrices**

| Correlation Model | $\alpha$ | $\beta$ |
|---|---|---|
| **Low correlation** | 0 | 0 |
| **Medium Correlation** | 0.3 | 0.9 |
| **Medium Correlation A** | 0.3 | 0.3874 |
| **High Correlation** | 0.9 | 0.9 |

The correlation matrices for high, medium, medium A and low correlation are defined in Tables B.2.3.1.2-2, B.2.3.1.2-3, B.2.3.1.2-4 and B.2.3.1.2-5 as below.

The values in Table B.2.3.1.2-2 have been adjusted for the 4x2 and 4x4 high correlation cases to insure the correlation matrix is positive semi-definite after round-off to 4 digit precision. This is done using the equation:

$$R_{high} = [R_{spat} + aI_n]/(1+a)$$

Where the value "*a*" is a scaling factor such that the smallest value is used to obtain a positive semi-definite result. For the 4x2 high correlation case, *a*=0.00010. For the 4x4 high correlation case, *a*=0.00012.

The same method is used to adjust the 2x4 and 4x4 medium correlation matrix in Table B.2.3.1.2-3 to insure the correlation matrix is positive semi-definite after round-off to 4 digit precision with *a* = 0.00010 and *a* = 0.00012.

**Table B.2.3.1.2-2: MIMO correlation matrices for high correlation**

| | |
|---|---|
| **1x2 case** | $R_{high} = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$ |
| **2x1 case** | $R_{high} = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$ |
| **2x2 case** | $R_{high} = \begin{pmatrix} 1 & 0.9 & 0.9 & 0.81 \\ 0.9 & 1 & 0.81 & 0.9 \\ 0.9 & 0.81 & 1 & 0.9 \\ 0.81 & 0.9 & 0.9 & 1 \end{pmatrix}$ |
| **4x2 case** | $R_{high} = \begin{bmatrix} 1.0000 & 0.8999 & 0.9883 & 0.8894 & 0.9542 & 0.8587 & 0.8999 & 0.8099 \\ 0.8999 & 1.0000 & 0.8894 & 0.9883 & 0.8587 & 0.9542 & 0.8099 & 0.8999 \\ 0.9883 & 0.8894 & 1.0000 & 0.8999 & 0.9883 & 0.8894 & 0.9542 & 0.8587 \\ 0.8894 & 0.9883 & 0.8999 & 1.0000 & 0.8894 & 0.9883 & 0.8587 & 0.9542 \\ 0.9542 & 0.8587 & 0.9883 & 0.8894 & 1.0000 & 0.8999 & 0.9883 & 0.8894 \\ 0.8587 & 0.9542 & 0.8894 & 0.9883 & 0.8999 & 1.0000 & 0.8894 & 0.9883 \\ 0.8999 & 0.8099 & 0.9542 & 0.8587 & 0.9883 & 0.8894 & 1.0000 & 0.8999 \\ 0.8099 & 0.8999 & 0.8587 & 0.9542 & 0.8894 & 0.9883 & 0.8999 & 1.0000 \end{bmatrix}$ |
| **4x4 case** | $R_{high} = \begin{bmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.8999 & 0.8894 & 0.8587 & 0.8099 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.8894 & 0.8999 & 0.8894 & 0.8587 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.8587 & 0.8894 & 0.8999 & 0.8894 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8099 & 0.8587 & 0.8894 & 0.8999 \\ 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 0.9541 & 0.9430 & 0.9105 & 0.8587 \\ 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9430 & 0.9541 & 0.9430 & 0.9105 \\ 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9105 & 0.9430 & 0.9541 & 0.9430 \\ 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8587 & 0.9105 & 0.9430 & 0.9541 \\ 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.9882 & 0.9767 & 0.9430 & 0.8894 \\ 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.9767 & 0.9882 & 0.9767 & 0.9430 \\ 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.9430 & 0.9767 & 0.9882 & 0.9767 \\ 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.8894 & 0.9430 & 0.9767 & 0.9882 \\ 0.8999 & 0.8894 & 0.8587 & 0.8099 & 0.9541 & 0.9430 & 0.9105 & 0.8587 & 0.9882 & 0.9767 & 0.9430 & 0.8894 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.8894 & 0.8999 & 0.8894 & 0.8587 & 0.9430 & 0.9541 & 0.9430 & 0.9105 & 0.9767 & 0.9882 & 0.9767 & 0.9430 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.8587 & 0.8894 & 0.8999 & 0.8894 & 0.9105 & 0.9430 & 0.9541 & 0.9430 & 0.9430 & 0.9767 & 0.9882 & 0.9767 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.8099 & 0.8587 & 0.8894 & 0.8999 & 0.8587 & 0.9105 & 0.9430 & 0.9541 & 0.8894 & 0.9430 & 0.9767 & 0.9882 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{bmatrix}$ |

**Table B.2.3.1.2-3: MIMO correlation matrices for medium correlation**

| | |
|---|---|
| **1x2 case** | N/A |
| **2x1 case** | N/A |
| **2x2 case** | $R_{medium} = \begin{pmatrix} 1 & 0.9 & 0.3 & 0.27 \\ 0.9 & 1 & 0.27 & 0.3 \\ 0.3 & 0.27 & 1 & 0.9 \\ 0.27 & 0.3 & 0.9 & 1 \end{pmatrix}$ |
| **2x4 case** | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.3000 & 0.2965 & 0.2862 & 0.2700 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.2965 & 0.3000 & 0.2965 & 0.2862 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.2862 & 0.2965 & 0.3000 & 0.2965 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.2700 & 0.2862 & 0.2965 & 0.3000 \\ 0.3000 & 0.2965 & 0.2862 & 0.2700 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.2965 & 0.3000 & 0.2965 & 0.2862 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.2862 & 0.2965 & 0.3000 & 0.2965 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.2700 & 0.2862 & 0.2965 & 0.3000 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{pmatrix}$ |
| **4x2 case** | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9000 & 0.8748 & 0.7873 & 0.5856 & 0.5271 & 0.3000 & 0.2700 \\ 0.9000 & 1.0000 & 0.7873 & 0.8748 & 0.5271 & 0.5856 & 0.2700 & 0.3000 \\ 0.8748 & 0.7873 & 1.0000 & 0.9000 & 0.8748 & 0.7873 & 0.5856 & 0.5271 \\ 0.7873 & 0.8748 & 0.9000 & 1.0000 & 0.7873 & 0.8748 & 0.5271 & 0.5856 \\ 0.5856 & 0.5271 & 0.8748 & 0.7873 & 1.0000 & 0.9000 & 0.8748 & 0.7873 \\ 0.5271 & 0.5856 & 0.7873 & 0.8748 & 0.9000 & 1.0000 & 0.7873 & 0.8748 \\ 0.3000 & 0.2700 & 0.5856 & 0.5271 & 0.8748 & 0.7873 & 1.0000 & 0.9000 \\ 0.2700 & 0.3000 & 0.5271 & 0.5856 & 0.7873 & 0.8748 & 0.9000 & 1.0000 \end{pmatrix}$ |
| **4x4 case** | $R_{medium} = \begin{pmatrix} 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.3000 & 0.2965 & 0.2862 & 0.2700 \\ 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.2965 & 0.3000 & 0.2965 & 0.2862 \\ 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.2862 & 0.2965 & 0.3000 & 0.2965 \\ 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.2700 & 0.2862 & 0.2965 & 0.3000 \\ 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 0.5855 & 0.5787 & 0.5588 & 0.5270 \\ 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.5787 & 0.5855 & 0.5787 & 0.5588 \\ 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.5588 & 0.5787 & 0.5855 & 0.5787 \\ 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.5270 & 0.5588 & 0.5787 & 0.5855 \\ 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 & 0.8747 & 0.8645 & 0.8347 & 0.7872 \\ 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 & 0.8645 & 0.8747 & 0.8645 & 0.8347 \\ 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 & 0.8347 & 0.8645 & 0.8747 & 0.8645 \\ 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 & 0.7872 & 0.8347 & 0.8645 & 0.8747 \\ 0.3000 & 0.2965 & 0.2862 & 0.2700 & 0.5855 & 0.5787 & 0.5588 & 0.5270 & 0.8747 & 0.8645 & 0.8347 & 0.7872 & 1.0000 & 0.9882 & 0.9541 & 0.8999 \\ 0.2965 & 0.3000 & 0.2965 & 0.2862 & 0.5787 & 0.5855 & 0.5787 & 0.5588 & 0.8645 & 0.8747 & 0.8645 & 0.8347 & 0.9882 & 1.0000 & 0.9882 & 0.9541 \\ 0.2862 & 0.2965 & 0.3000 & 0.2965 & 0.5588 & 0.5787 & 0.5855 & 0.5787 & 0.8347 & 0.8645 & 0.8747 & 0.8645 & 0.9541 & 0.9882 & 1.0000 & 0.9882 \\ 0.2700 & 0.2862 & 0.2965 & 0.3000 & 0.5270 & 0.5588 & 0.5787 & 0.5855 & 0.7872 & 0.8347 & 0.8645 & 0.8747 & 0.8999 & 0.9541 & 0.9882 & 1.0000 \end{pmatrix}$ |

**Table B.2.3.1.2-4: MIMO correlation matrices for medium correlation A**

| | |
|---|---|
| **2x4 case** | $R_{medium\ A} = \begin{pmatrix} 1.0000 & 0.9000 & 0.6561 & 0.3874 & 0.3000 & 0.2700 & 0.1968 & 0.1162 \\ 0.9000 & 1.0000 & 0.9000 & 0.6561 & 0.2700 & 0.3000 & 0.2700 & 0.1968 \\ 0.6561 & 0.9000 & 1.0000 & 0.9000 & 0.1968 & 0.2700 & 0.3000 & 0.2700 \\ 0.3874 & 0.6561 & 0.9000 & 1.0000 & 0.1162 & 0.1968 & 0.2700 & 0.3000 \\ 0.3000 & 0.2700 & 0.1968 & 0.1162 & 1.0000 & 0.9000 & 0.6561 & 0.3874 \\ 0.2700 & 0.3000 & 0.2700 & 0.1968 & 0.9000 & 1.0000 & 0.9000 & 0.6561 \\ 0.1968 & 0.2700 & 0.3000 & 0.2700 & 0.6561 & 0.9000 & 1.0000 & 0.9000 \\ 0.1162 & 0.1968 & 0.2700 & 0.3000 & 0.3874 & 0.6561 & 0.9000 & 1.0000 \end{pmatrix}$ |
| **4x4 case** | $R_{medium\ A} = \begin{pmatrix} 1.0000 & 0.9000 & 0.6561 & 0.3874 & 0.8748 & 0.7873 & 0.5739 & 0.3389 & 0.5856 & 0.5270 & 0.3842 & 0.2269 & 0.3000 & 0.2700 & 0.1968 & 0.1162 \\ 0.9000 & 1.0000 & 0.9000 & 0.6561 & 0.7873 & 0.8748 & 0.7873 & 0.5739 & 0.5270 & 0.5856 & 0.5270 & 0.3842 & 0.2700 & 0.3000 & 0.2700 & 0.1968 \\ 0.6561 & 0.9000 & 1.0000 & 0.9000 & 0.5739 & 0.7873 & 0.8748 & 0.7873 & 0.3842 & 0.5270 & 0.5856 & 0.5270 & 0.1968 & 0.2700 & 0.3000 & 0.2700 \\ 0.3874 & 0.6561 & 0.9000 & 1.0000 & 0.3389 & 0.5739 & 0.7873 & 0.8748 & 0.2269 & 0.3842 & 0.5270 & 0.5856 & 0.1162 & 0.1968 & 0.2700 & 0.3000 \\ 0.8748 & 0.7873 & 0.5739 & 0.3389 & 1.0000 & 0.9000 & 0.6561 & 0.3874 & 0.8748 & 0.7873 & 0.5739 & 0.3389 & 0.5856 & 0.5270 & 0.3842 & 0.2269 \\ 0.7873 & 0.8748 & 0.7873 & 0.5739 & 0.9000 & 1.0000 & 0.9000 & 0.6561 & 0.7873 & 0.8748 & 0.7873 & 0.5739 & 0.5270 & 0.5856 & 0.5270 & 0.3842 \\ 0.5739 & 0.7873 & 0.8748 & 0.7873 & 0.6561 & 0.9000 & 1.0000 & 0.9000 & 0.5739 & 0.7873 & 0.8748 & 0.7873 & 0.3842 & 0.5270 & 0.5856 & 0.5270 \\ 0.3389 & 0.5739 & 0.7873 & 0.8748 & 0.3874 & 0.6561 & 0.9000 & 1.0000 & 0.3389 & 0.5739 & 0.7873 & 0.8748 & 0.2269 & 0.3842 & 0.5270 & 0.5856 \\ 0.5856 & 0.5270 & 0.3842 & 0.2269 & 0.8748 & 0.7873 & 0.5739 & 0.3389 & 1.0000 & 0.9000 & 0.6561 & 0.3874 & 0.8748 & 0.7873 & 0.5739 & 0.3389 \\ 0.5270 & 0.5856 & 0.5270 & 0.3842 & 0.7873 & 0.8748 & 0.7873 & 0.5739 & 0.9000 & 1.0000 & 0.9000 & 0.6561 & 0.7873 & 0.8748 & 0.7873 & 0.5739 \\ 0.3842 & 0.5270 & 0.5856 & 0.5270 & 0.5739 & 0.7873 & 0.8748 & 0.7873 & 0.6561 & 0.9000 & 1.0000 & 0.9000 & 0.5739 & 0.7873 & 0.8748 & 0.7873 \\ 0.2269 & 0.3842 & 0.5270 & 0.5856 & 0.3389 & 0.5739 & 0.7873 & 0.8748 & 0.3874 & 0.6561 & 0.9000 & 1.0000 & 0.3389 & 0.5739 & 0.7873 & 0.8748 \\ 0.3000 & 0.2700 & 0.1968 & 0.1162 & 0.5856 & 0.5270 & 0.3842 & 0.2269 & 0.8748 & 0.7873 & 0.5739 & 0.3389 & 1.0000 & 0.9000 & 0.6561 & 0.3874 \\ 0.2700 & 0.3000 & 0.2700 & 0.1968 & 0.5270 & 0.5856 & 0.5270 & 0.3842 & 0.7873 & 0.8748 & 0.7873 & 0.5739 & 0.9000 & 1.0000 & 0.9000 & 0.6561 \\ 0.1968 & 0.2700 & 0.3000 & 0.2700 & 0.3842 & 0.5270 & 0.5856 & 0.5270 & 0.5739 & 0.7873 & 0.8748 & 0.7873 & 0.6561 & 0.9000 & 1.0000 & 0.9000 \\ 0.1162 & 0.1968 & 0.2700 & 0.3000 & 0.2269 & 0.3842 & 0.5270 & 0.5856 & 0.3389 & 0.5739 & 0.7873 & 0.8748 & 0.3874 & 0.6561 & 0.9000 & 1.0000 \end{pmatrix}$ |

**Table B.2.3.1.2-5: MIMO correlation matrices for low correlation**

| | |
|---|---|
| **1x2 case** | $R_{low} = \mathbf{I}_2$ |
| **1x4 case** | $R_{low} = \mathbf{I}_4$ |
| **2x1 case** | $R_{low} = \mathbf{I}_2$ |
| **2x2 case** | $R_{low} = \mathbf{I}_4$ |
| **2x4 case** | $R_{low} = \mathbf{I}_8$ |
| **4x1 case** | $R_{low} = \mathbf{I}_4$ |
| **4x2 case** | $R_{low} = \mathbf{I}_8$ |
| **4x4 case** | $R_{low} = \mathbf{I}_{16}$ |

In Table B.2.3.1.2-5, $\mathbf{I}_d$ is the $d \times d$ identity matrix.

## B.2.3.2 MIMO Correlation Matrices using Cross Polarized Antennas (X-pol)

The MIMO channel correlation matrices defined in B.2.3.2 apply for the antenna configuration using cross polarized (XP/X-pol) antennas at both gNB and UE. The cross-polarized antenna elements with +/-45 degrees polarization slant angles are deployed at gNB and cross-polarized antenna elements with +90/0 degrees polarization slant angles are deployed at UE.

For the 2D cross-polarized antenna array at eNodeB, the *N* antennas are indexed by $(N_1, N_2, P)$, and total number of antennas is $N = P \cdot N_1 \cdot N_2$, where

- $N_1$ is the number of antenna elements in first dimension with same polarization,
- $N_2$ is the number of antenna elements in second dimension with same polarization, and
- $P$ is the number of polarization groups.

For the 2D cross-polarized antennas at gNB, the *N* antennas are labelled such that antennas shall be in increasing order of the second dimension firstly, then the first dimension, and finally the polarization group. For a specific antenna element at *p*-th polarization, $n_1$-th row, and $n_2$-th column within the 2D antenna array, the following index number is used for antenna labelling:

$$Index(p,n_1,n_2) = p \cdot N_1 \cdot N_2 + n_1 \cdot N_2 + n_2 + 1; \qquad p = 0,1;\ n_1 = 0,\cdots,N_1 - 1;\ n_2 = 0,\cdots,N_2 - 1.$$

where *N* is the number of transmit antennas, *p* is the polarization group index, $n_1$ is the row index, and $n_2$ is the column index of the antenna element.

For the linear (single dimension, 1D) cross-polarized antenna, the *N* antennas are labelled following the above equations with $N_2$=1.

### B.2.3.2.1 Definition of MIMO Correlation Matrices using cross polarized antennas

For the channel spatial correlation matrix, the following is used:

$$R_{spat} = P\left(R_{gNB} \otimes \Gamma \otimes R_{UE}\right)P^T$$

where

- $R_{UE}$ is the spatial correlation matrix at the UE with same polarization,
- $R_{gNB}$ is the spatial correlation matrix at the gNB with same polarization,
- $\Gamma$ is a polarization correlation matrix, and
- $(\bullet)^T$ denotes transpose.

The matrix $\Gamma$ is defined as:

$$\Gamma = \begin{bmatrix} 1 & 0 & -\gamma & 0 \\ 0 & 1 & 0 & \gamma \\ -\gamma & 0 & 1 & 0 \\ 0 & \gamma & 0 & 1 \end{bmatrix}$$

A permutation matrix P elements are defined as:

$$P(a,b) = \begin{cases} 1 & for \quad a = (j-1)Nr + i \quad and \quad b = 2(j-1)Nr + i, \qquad i = 1,\cdots,Nr,\ j = 1,\cdots Nt/2 \\ 1 & for \quad a = (j-1)Nr + i \quad and \quad b = 2(j - Nt/2)Nr - Nr + i, \quad i = 1,\cdots,Nr,\ j = Nt/2+1,\cdots,Nt. \\ 0 & otherwise \end{cases}$$

where *Nt* and *Nr* is the number of transmitter and receiver respectively. This is used to map the spatial correlation coefficients in accordance with the antenna element labelling system described in B.2.3.2.

For the 2D cross-polarized antenna array at gNB, the spatial correlation matrix at the gNB is further expressed as following for 2D cross-polarized antenna array at gNB:

$$R_{gNB} = R_{gNB_Dim,1} \otimes R_{gNB_Dim,2}$$

where

- $R_{gNB_Dim,1}$ is the correlation matrix of antenna elements in first dimension with same polarization, and
- $R_{gNB_Dim,2}$ is the correlation matrix of antenna elements in second dimension with same polarization.

For the 2D cross polarized antenna array at gNB side, the spatial correlation matrices in one direction of antenna array are as follows:

- For 1 antenna element with the same polarization in one direction,

$$R_{gNB_Dim,i} = 1.$$

- For 2 antenna elements with the same polarization in one direction,

$$R_{gNB_Dim,i} = \begin{pmatrix} 1 & \alpha_i \\ \alpha_i^* & 1 \end{pmatrix}.$$

- For 3 antenna elements with the same polarization in one direction,

$$R_{gNB_Dim,i} = \begin{pmatrix} 1 & \alpha_i^{1/4} & \alpha_i \\ \alpha_i^{1/4*} & 1 & \alpha_i^{1/4} \\ \alpha_i^* & \alpha_i^{1/4*} & 1 \end{pmatrix}.$$

- For 4 antenna elements with the same polarization in one direction,

$$R_{gNB_Dim,i} = \begin{pmatrix} 1 & \alpha_i^{1/9} & \alpha_i^{4/9} & \alpha_i \\ \alpha_i^{1/9*} & 1 & \alpha_i^{1/9} & \alpha_i^{4/9} \\ \alpha_i^{4/9*} & \alpha_i^{1/9*} & 1 & \alpha_i^{1/9} \\ \alpha_i^* & \alpha_i^{4/9*} & \alpha_i^{1/9*} & 1 \end{pmatrix}.$$

where the index $i$ = 1,2 stands for first dimension and second dimension respectively.

For the 1D cross-polarized antenna array at gNB, the matrix of $R_{gNB}$ is determined by follow the equations for 2D cross-polarized antenna array and letting $R_{gNB_Dim,2} = 1$, i.e.

$$R_{gNB} = R_{gNB_Dim,1}$$

The spatial correlation matrices at UE side are as follows:

- For 1 antenna element with the same polarization,

$$R_{UE} = 1.$$

- For 2 antenna elements with the same polarization,

$$R_{UE} = \begin{pmatrix} 1 & \beta \\ \beta^* & 1 \end{pmatrix}.$$

### B.2.3.2.2 MIMO Correlation Matrices using cross polarized antennas

The values for parameters $\alpha_1$, $\alpha_2$. $\beta$ and $\gamma$ for the cross polarized antenna models are given in Table B.2.3.2.2-1.

**Table B.2.3.2.2-1: The $\alpha$ and $\beta$ parameters for cross-polarized MIMO correlation matrices**

| Correlation Model | $\alpha_1$ | $\alpha_2$ | $\beta$ | $\gamma$ |
|---|---|---|---|---|
| **Medium Correlation** | 0.3 | 0.3 | 0.6 | 0.2 |
| **High Correlation** | 0.9 | 0.9 | 0.9 | 0.3 |
| NOTE 1: Value of $\alpha_1$ applies when more than one pair of cross-polarized antenna elements in first dimension at gNB side. | | | | |
| NOTE 2: Value of $\alpha_2$ applies when more than one pair of cross-polarized antenna elements in second dimension at gNB side. | | | | |
| NOTE 3: Value of $\beta$ applies when more than one pair of cross-polarized antenna elements at UE side. | | | | |

For the 1D cross polarized antenna array at gNB side, the correlation matrices for high spatial correlation and medium correlation are defined in Table B.2.3.2.2-2 and Table B.2.3.2.2-3 as below.

The values in Table B.2.3.2.2-2 have been adjusted to ensure the correlation matrix is positive semi-definite after round-off to 4 digit precision. This is done using the equation:

$$R_{high} = [R_{spat} + aI_n]/(1+a) \text{ or } R_{medium} = [R_{spat} + aI_n]/(1+a)$$

Where the value "$a$" is a scaling factor such that the smallest value is used to obtain a positive semi-definite result. For the 8(4,1,2)x2 high spatial correlation case, $a$=0.00010.

**Table B.2.3.2.2-2: MIMO correlation matrices for high spatial correlation**

**4(2,1,2)x 2 case**

$$R_{high} = \begin{bmatrix} 1.0000 & 0.0000 & 0.9000 & 0.0000 & -0.3000 & 0.0000 & -0.2700 & 0.0000 \\ 0.0000 & 1.0000 & 0.0000 & 0.9000 & 0.0000 & 0.3000 & 0.0000 & 0.2700 \\ 0.9000 & 0.0000 & 1.0000 & 0.0000 & -0.2700 & 0.0000 & -0.3000 & 0.0000 \\ 0.0000 & 0.9000 & 0.0000 & 1.0000 & 0.0000 & 0.2700 & 0.0000 & 0.3000 \\ -0.3000 & 0.0000 & -0.2700 & 0.0000 & 1.0000 & 0.0000 & 0.9000 & 0.0000 \\ 0.0000 & 0.3000 & 0.0000 & 0.2700 & 0.0000 & 1.0000 & 0.0000 & 0.9000 \\ -0.2700 & 0.0000 & -0.3000 & 0.0000 & 0.9000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.2700 & 0.0000 & 0.3000 & 0.0000 & 0.9000 & 0.0000 & 1.0000 \end{bmatrix}$$

**2(1,1,2)x 4 case**

$$R_{high} = \begin{bmatrix} 1.0000 & 0.9000 & 0.0000 & 0.0000 & -0.3000 & -0.2700 & 0.0000 & 0.0000 \\ 0.9000 & 1.0000 & 0.0000 & 0.0000 & -0.2700 & -0.3000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 & 0.3000 & 0.2700 \\ 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 & 0.2700 & 0.3000 \\ -0.3000 & -0.2700 & 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 \\ -0.2700 & -0.3000 & 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.3000 & 0.2700 & 0.0000 & 0.0000 & 1.0000 & 0.9000 \\ 0.0000 & 0.0000 & 0.2700 & 0.3000 & 0.0000 & 0.0000 & 0.9000 & 1.0000 \end{bmatrix}$$

**4(2,1,2)x 4 case**

$$R_{high} = \begin{bmatrix} 1.0000 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 0.8100 & 0.0000 & 0.0000 & -0.3000 & -0.2700 & 0.0000 & 0.0000 & -0.2700 & -0.2430 & 0.0000 & 0.0000 \\ 0.9000 & 1.0000 & 0.0000 & 0.0000 & 0.8100 & 0.9000 & 0.0000 & 0.0000 & -0.2700 & -0.3000 & 0.0000 & 0.0000 & -0.2430 & -0.2700 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 0.8100 & 0.0000 & 0.0000 & 0.3000 & 0.2700 & 0.0000 & 0.0000 & 0.2700 & 0.2430 \\ 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 & 0.8100 & 0.9000 & 0.0000 & 0.0000 & 0.2700 & 0.3000 & 0.0000 & 0.0000 & 0.2430 & 0.2700 \\ 0.9000 & 0.8100 & 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 & -0.2700 & -0.2430 & 0.0000 & 0.0000 & -0.3000 & -0.2700 & 0.0000 & 0.0000 \\ 0.8100 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 & -0.2430 & -0.2700 & 0.0000 & 0.0000 & -0.2700 & -0.3000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.9000 & 0.8100 & 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 & 0.2700 & 0.2430 & 0.0000 & 0.0000 & 0.3000 & 0.2700 \\ 0.0000 & 0.0000 & 0.8100 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 & 0.2430 & 0.2700 & 0.0000 & 0.0000 & 0.2700 & 0.3000 \\ -0.3000 & -0.2700 & 0.0000 & 0.0000 & -0.2700 & -0.2430 & 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 0.8100 & 0.0000 & 0.0000 \\ -0.2700 & -0.3000 & 0.0000 & 0.0000 & -0.2430 & -0.2700 & 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 & 0.8100 & 0.9000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.3000 & 0.2700 & 0.0000 & 0.0000 & 0.2700 & 0.2430 & 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 0.8100 \\ 0.0000 & 0.0000 & 0.2700 & 0.3000 & 0.0000 & 0.0000 & 0.2430 & 0.2700 & 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 & 0.8100 & 0.9000 \\ -0.2700 & -0.2430 & 0.0000 & 0.0000 & -0.3000 & -0.2700 & 0.0000 & 0.0000 & 0.9000 & 0.8100 & 0.0000 & 0.0000 & 1.0000 & 0.9000 & 0.0000 & 0.0000 \\ -0.2430 & -0.2700 & 0.0000 & 0.0000 & -0.2700 & -0.3000 & 0.0000 & 0.0000 & 0.8100 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 1.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.2700 & 0.2430 & 0.0000 & 0.0000 & 0.3000 & 0.2700 & 0.0000 & 0.0000 & 0.9000 & 0.8100 & 0.0000 & 0.0000 & 1.0000 & 0.9000 \\ 0.0000 & 0.0000 & 0.2430 & 0.2700 & 0.0000 & 0.0000 & 0.2700 & 0.3000 & 0.0000 & 0.0000 & 0.8100 & 0.9000 & 0.0000 & 0.0000 & 0.9000 & 1.0000 \end{bmatrix}$$

**8(4,1,2)x 2 case**

$$R_{high} = \begin{bmatrix} 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 & 0.8999 & 0.0000 & -0.3000 & 0.0000 & -0.2965 & 0.0000 & -0.2862 & 0.0000 & -0.2700 & 0.0000 \\ 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 & 0.8999 & 0.0000 & 0.3000 & 0.0000 & 0.2965 & 0.0000 & 0.2862 & 0.0000 & 0.2700 \\ 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 & -0.2965 & 0.0000 & -0.3000 & 0.0000 & -0.2965 & 0.0000 & -0.2862 & 0.0000 \\ 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 & 0.2965 & 0.0000 & 0.3000 & 0.0000 & 0.2965 & 0.0000 & 0.2862 \\ 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & -0.2862 & 0.0000 & -0.2965 & 0.0000 & -0.3000 & 0.0000 & -0.2965 & 0.0000 \\ 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.2862 & 0.0000 & 0.2965 & 0.0000 & 0.3000 & 0.0000 & 0.2965 \\ 0.8999 & 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & -0.2700 & 0.0000 & -0.2862 & 0.0000 & -0.2965 & 0.0000 & -0.3000 & 0.0000 \\ 0.0000 & 0.8999 & 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.2700 & 0.0000 & 0.2862 & 0.0000 & 0.2965 & 0.0000 & 0.3000 \\ -0.3000 & 0.0000 & -0.2965 & 0.0000 & -0.2862 & 0.0000 & -0.2700 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 & 0.8999 & 0.0000 \\ 0.0000 & 0.3000 & 0.0000 & 0.2965 & 0.0000 & 0.2862 & 0.0000 & 0.2700 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 & 0.8999 \\ -0.2965 & 0.0000 & -0.3000 & 0.0000 & -0.2965 & 0.0000 & -0.2862 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 & 0.0000 \\ 0.0000 & 0.2965 & 0.0000 & 0.3000 & 0.0000 & 0.2965 & 0.0000 & 0.2862 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 & 0.9542 \\ -0.2862 & 0.0000 & -0.2965 & 0.0000 & -0.3000 & 0.0000 & -0.2965 & 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 & 0.0000 \\ 0.0000 & 0.2862 & 0.0000 & 0.2965 & 0.0000 & 0.3000 & 0.0000 & 0.2965 & 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 & 0.9883 \\ -0.2700 & 0.0000 & -0.2862 & 0.0000 & -0.2965 & 0.0000 & -0.3000 & 0.0000 & 0.8999 & 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.2700 & 0.0000 & 0.2862 & 0.0000 & 0.2965 & 0.0000 & 0.3000 & 0.0000 & 0.8999 & 0.0000 & 0.9542 & 0.0000 & 0.9883 & 0.0000 & 1.0000 \end{bmatrix}$$

**Table B.2.3.2.2-3: MIMO correlation matrices for medium spatial correlation**

| | |
|---|---|
| **2(1,1,2)x2 case** | $R_{medium} = \begin{bmatrix} 1.0000 & 0.0000 & -0.2000 & 0.0000 \\ 0.0000 & 1.0000 & 0.0000 & 0.2000 \\ -0.2000 & 0.0000 & 1.0000 & 0.0000 \\ 0.0000 & 0.2000 & 0.0000 & 1.0000 \end{bmatrix}$ |

### B.2.3.2.3 Beam steering approach

For the 2D cross-polarized antenna array at gNB, given the channel spatial correlation matrix in B.2.3.2.1 and B.2.3.2.2, the corresponding random channel matrix *H* can be calculated. The signal model for the *k*-th slot is denoted as:

$$y = HD_{\theta_{k,1},\theta_{k,2}}Wx + n$$

And the steering matrix is further expressed as following:

$$D_{\theta_{k,1},\theta_{k,2}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \left( D_{\theta_{k,1}}(N_1) \otimes D_{\theta_{k,2}}(N_2) \right)$$

Where:

- *H* is the *Nr*×*Nt* channel matrix per subcarrier.
- $D_{\theta_{k,1},\theta_{k,2}}$ is the steering matrix,
- $D_{\theta_{k,1}}(N_1)$ is the steering matrix in first dimension with same polarization,
- $D_{\theta_{k,2}}(N_2)$ is the steering matrix in second dimension with same polarization,
- $N_1$ is the number of antenna elements in first dimension with same polarization,
- $N_2$ is the number of antenna elements in second dimension with same polarization,

For antenna array with only one direction, number of antenna element in second direction $N_2$ equals 1.

For 1 antenna element with the same polarization in one direction,

$$D_{\theta_{k,i}}(1) = 1.$$

For 2 antenna elements with the same polarization in one direction,

$$D_{\theta_{k,i}}(2) = \begin{bmatrix} 1 & 0 \\ 0 & e^{j3\theta_{k,i}} \end{bmatrix}.$$

For 3 antenna elements with the same polarization in one direction,

$$D_{\theta_{k,i}}(3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{j1.5\theta_{k,i}} & 0 \\ 0 & 0 & e^{j3\theta_{k,i}} \end{bmatrix}.$$

For 4 antenna elements with the same polarization in one direction,

$$D_{\theta_{k,i}}(4) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{j\theta_{k,i}} & 0 & 0 \\ 0 & 0 & e^{j2\theta_{k,i}} & 0 \\ 0 & 0 & 0 & e^{j3\theta_{k,i}} \end{bmatrix}.$$

where the index $i = 1,2$ stands for first dimension and second dimension respectively.

$\theta_{k,i}$ controls the phase variation in first dimension and second dimension respectively, and the phase for k-th subframe is denoted by $\theta_{k,i} = \theta_{0,i} + \Delta\theta \cdot k$, where $\theta_{0,i}$ is the random start value with the uniform distribution, i.e. $\theta_{0,i} \in [0, 2\pi]$, $\Delta\theta$ is the step of phase variation, which is defined in Table B.2.3.2.3-1, and k is the linear increment of $2^{-\mu}$ for every slot throughout the simulation, the index $i = 1,2$ stands for first dimension and second dimension respectively.

- $W$ is the precoding matrix for Nt transmission antennas,
- $y$ is the received signal, $x$ is the transmitted signal, and $n$ is AWGN.
- $\mu$ corresponds to subcarrier spacing configuration, $\Delta f = 2^{\mu} \cdot 15[\text{kHz}]$

For the 1D cross-polarized antenna array at gNB, the corresponding random channel matrix $H$ can be calculated by letting $N_2$=1, i.e.

$$D_{\theta_{k,1}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes D_{\theta_{k,1}}(N_1)$$

**Table B.2.3.2.3-1: The step of phase variation**

| Variation Step | Value (rad/ms) |
|---|---|
| $\Delta\theta$ | $1.2566 \times 10^{-3}$ |

## B.2.3.2.3A Beam steering approach with dual cluster beams

For the 2D cross-polarized antenna array at gNB, given the channel spatial correlation matrix in B.2.3.2.1 and B.2.3.2.2, the corresponding random channel matrix $H$ can be calculated. The signal model for the $k$-th slot is denoted as

$$y = \left[ \sqrt{\frac{1}{1+p^2}} H_m D^{(m)}_{\theta_{k,1},\theta_{k,2}} + \sqrt{\frac{p^2}{1+p^2}} H_s D^{(s)}_{\theta_{k,1},\theta_{k,2}} \right] W x + n$$

And the steering matrix is further expressed as following:

$$D_{\theta_{k,1},\theta_{k,2}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \left( D_{\theta_{k,1}}(N_1) \otimes D_{\theta_{k,2}}(N_2) \right)$$

where

- $H_m$, $H_s$ are independent channels for the first beam and second beam with the Nr xNt channel matrix per subcarrier.
- $D^{(m)}_{\theta_{k,1},\theta_{k,2}}$, $D^{(s)}_{\theta_{k,1},\theta_{k,2}}$ are the steering matrix for first beam and second beam
- $D_{\theta_{k,1}}(N_1)$ is the steering matrix in first dimension with same polarization,
- $D_{\theta_{k,2}}(N_2)$ is the steering matrix in second dimension with same polarization,
- $N_1$ is the number of antenna elements in first dimension with same polarization,
- $N_2$ is the number of antenna elements in second dimension with same polarization,
- For antenna array with only one direction, number of antenna element in second direction $N_2$ equals 1,
- $p$ is the relative power ratio of the second beam to the first beam, the value of $p$ is specific to a test case,

For 1 antenna element of the same polarization in one direction, $D_{\theta_{k,i}}(1)=1$.

For 2 antenna elements of the same polarization in one direction, $D_{\theta_{k,i}}(2)=\begin{bmatrix}1 & 0\\ 0 & e^{j3\theta_{k,i}}\end{bmatrix}$.

For 3 antenna elements of the same polarization in one direction, $D_{\theta_{k,i}}(3)=\begin{bmatrix}1 & 0 & 0\\ 0 & e^{j1.5\theta_{k,i}} & 0\\ 0 & 0 & e^{j3\theta_{k,i}}\end{bmatrix}$.

For 4 antenna elements of the same polarization in one direction, $D_{\theta_{k,i}}(4)=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & e^{j\theta_{k,i}} & 0 & 0\\ 0 & 0 & e^{j2\theta_{k,i}} & 0\\ 0 & 0 & 0 & e^{j3\theta_{k,i}}\end{bmatrix}$.

where the index $i=1,2$ stands for first dimension and second dimension respectively.

- $\theta_{k,i}$ controls the phase variation in first dimension and second dimension respectively, and the phase for k-th subframe is denoted by $\theta_{k,i}=\theta_{0,i}+\Delta\theta\cdot k$, where $\theta_{0,i}$ is the random start value with the uniform distribution, i.e., $\theta_{0,i}\in[0,2\pi]$, $\Delta\theta$ is the step of phase variation, which is defined in Table B.2.3.2.3A-1, and k is the linear increment of $2^{-\mu}$ for every slot throughout the simulation, the index $i=1,2$ stands for first dimension and second dimension respectively.
- $W$ is the precoding matrix for Nt transmission antennas,
- $y$ is the received signal, $x$ is the transmitted signal, and $n$ is AWGN.
- $\mu$ corresponds to subcarrier spacing configuration, $\Delta f=2^{\mu}\cdot 15[\text{kHz}]$

For the 1D cross-polarized antenna array at gNB, the corresponding random channel matrix $H$ can be calculated by letting $N_2$=1, i.e.,

$$D_{\theta_{k,1}}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\otimes D_{\theta_{k,1}}(N_1)$$

**Table B.2.3.2.3A-1: The step of phase variation**

| Variation Step | Value (rad/subframe) |
|---|---|
| $\Delta\theta^{(m)}$ | $1.2566\times10^{-3}$ |
| $\Delta\theta^{(s)}$ | $2.5132\times10^{-3}$ |

## B.2.4 Two-tap propagation conditions for CQI tests

For Channel Quality Indication (CQI) tests, the following additional multi-path profile is used:

$$h(t,\tau)=\delta(\tau)+a\exp(-i2\pi f_D t)\delta(\tau-\tau_d)$$

in continuous time $(t,\tau)$ representation, with $\tau_d$ the delay, a constant value of $a$ and $f_D$ the Doppler frequency. The same $h(t,\tau)$ is used to describe the fading channel between every pair of Tx and Rx.

# B.3 High Speed Train Scenario

## B.3.1 Single Tap Channel Profile

The high speed train condition for the test of the baseband performance is a non-fading propagation channel with one tap. Doppler shift is given by

$$f_s(t) = f_d \cos\theta(t) \quad \text{(B.3.1.1)}$$

where $f_s(t)$ is the Doppler shift and $f_d$ is the maximum Doppler frequency. The cosine of angle $\theta(t)$ is given by

$$\cos\theta(t) = \frac{D_s/2 - vt}{\sqrt{D_{\min}{}^2 + (D_s/2 - vt)^2}},\ 0 \le t \le D_s/v \quad \text{(B.3.1.2)}$$

$$\cos\theta(t) = \frac{-1.5D_s + vt}{\sqrt{D_{\min}{}^2 + (-1.5D_s + vt)^2}},\ D_s/v < t \le 2D_s/v \quad \text{(B.3.1.3)}$$

$$\cos\theta(t) = \cos\theta(t \mod (2D_s/v)),\ t > 2D_s/v \quad \text{(B.3.1.4)}$$

where $D_s/2$ is the initial distance of the train from gNB, and $D_{\min}$ is gNB Railway track distance, both in meters; $v$ is the velocity of the train in m/s, $t$ is time in seconds.

Doppler shift and cosine angle are given by equation B.3.1.1 and B.3.1.2-B.3.1.4 respectively, where the required input parameters listed in table B.3.1-1 and the resulting Doppler shift shown in Figures B.3.1-1, B.3.1-2, B.3.1-3, B.3.1-4 are applied for all frequency bands.

**Table B.3.1-1: High speed train scenario**

| Parameter | Value | | | |
|---|---|---|---|---|
| | **HST-750** | **HST-972** | **HST-1000** | **HST-1667** |
| $D_s$ | 300 m | 300 m | 300 m | 300 m |
| $D_{\min}$ | 2 m | 2 m | 2 m | 2 m |
| $v$ | 300 km/h | 500 km/h | 300 km/h | 500 km/h |
| $f_d$ | 750 Hz for 15 kHz SCS test | 972 Hz for 15 kHz SCS test | 1000 Hz for 30 kHz SCS test | 1667 Hz for 30 kHz SCS test |

Note 1: Parameters for HST conditions in table B.3.1-1 including $f_d$ and Doppler shift trajectories presented on figures B.3.1-1 for 750 Hz and B.3.1-3 for 972 Hz for 15 kHz SCS and figures B.3.1-2 for 1000 Hz and B.3.1-4 for 1667 Hz for 30 kHz SCS are applied for performance verification in all frequency bands.

Note 2: The propagation conditions used for the performance requirements under high speed train condition are indicated as a combination of "HST" and Doppler shift $f_d$, i.e. HST-<Doppler shift>, where '<Doppler shift>'indicates the maximum Doppler shift (Hz).

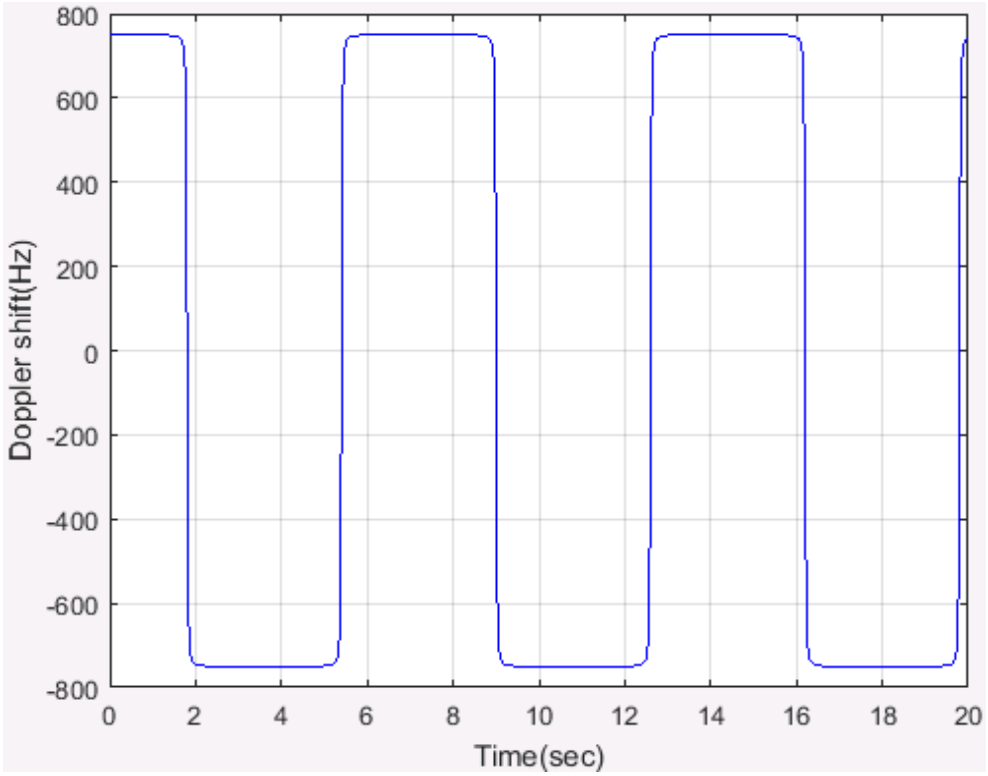



**Figure B.3.1-1: Doppler shift trajectory ( $f_d$ = 750 Hz)**

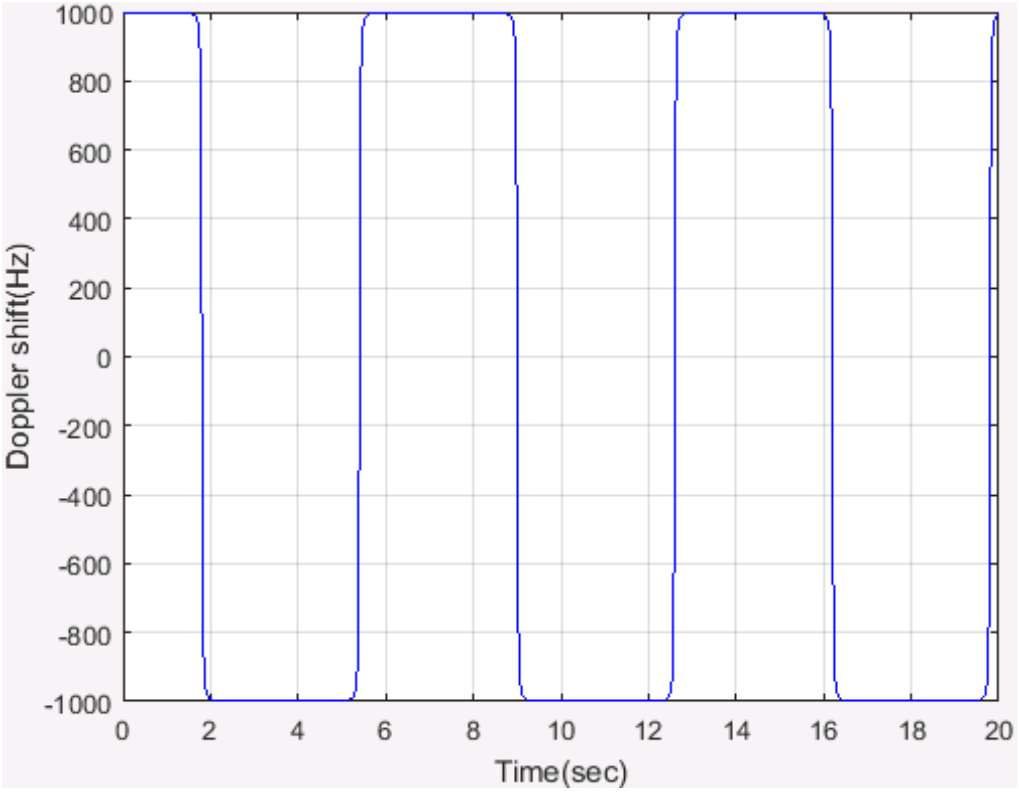



**Figure B.3.1-2: Doppler shift trajectory ( $f_d$ = 1000 Hz)**

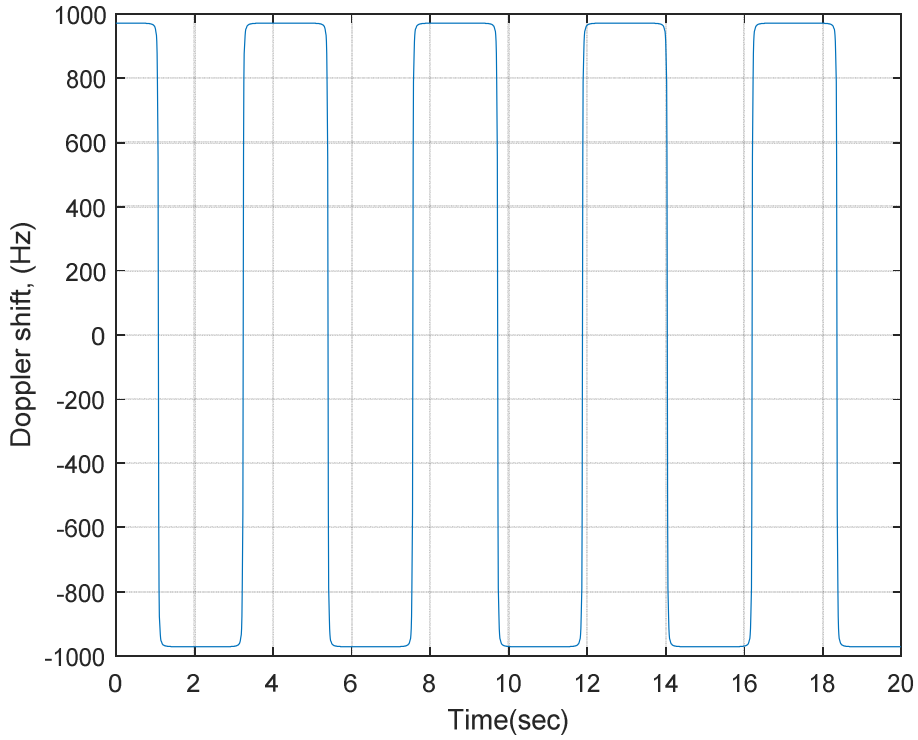



**Figure B.3.1-3: Doppler shift trajectory ( $f_d$ = 972 Hz)**

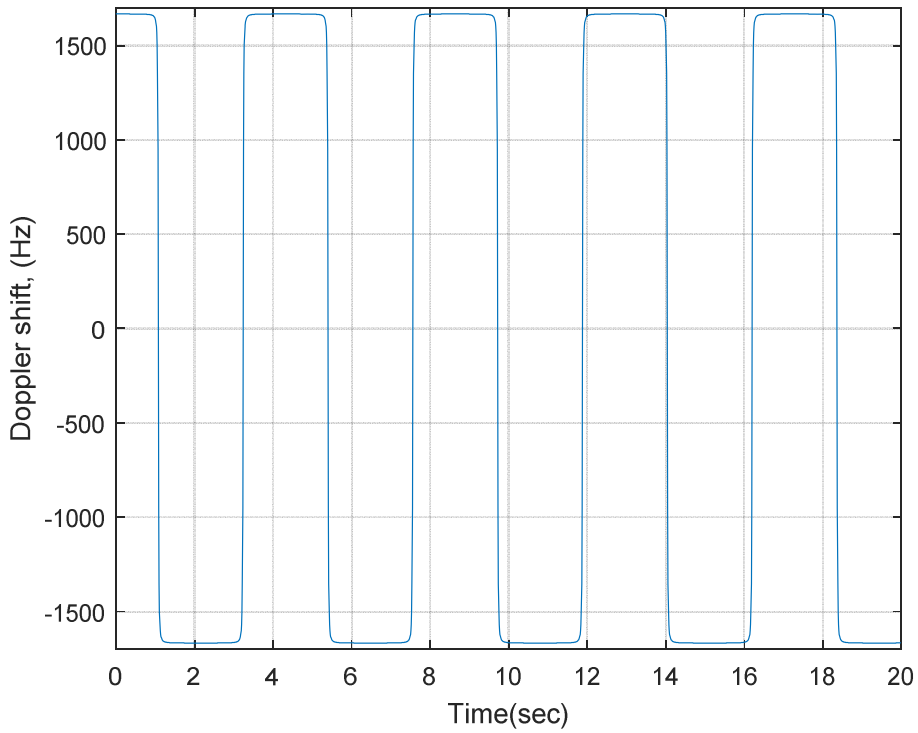



**Figure B.3.1-4: Doppler shift trajectory ( $f_d$ = 1667 Hz)**

For 1x2 antenna configuration, the same $h(t,\tau)$ is used to describe the channel between every pair of Tx and Rx.

For 1x4 antenna configuration, the same $h(t,\tau)$ is used to describe the channel between every pair of Tx and Rx.

## B.3.2 HST-SFN Channel Profile

There is an infinite number of RRHs distributed equidistantly along the track with the same Cell ID as depicted in figure B.3.2-1.

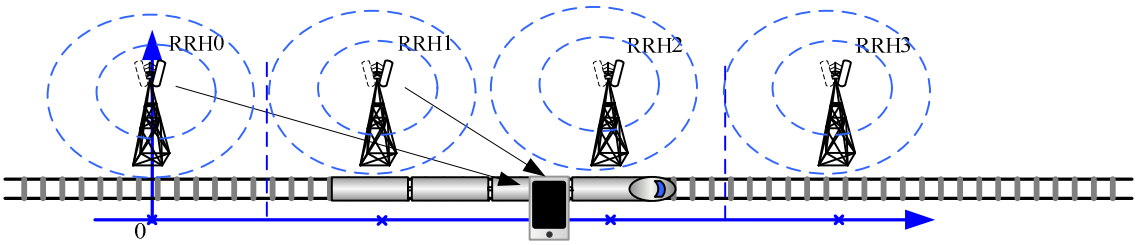



**Figure B.3.2-1: Deployment of HST-SFN**

The location of RRH *k* is given as:

$$x_k = k*D_s + j*D_{min} \quad \text{(B.3.2.1)}$$

where: $k \in [-\infty, \infty]$, $j = sqrt(-1)$ and $D_{min}$ is the distance between the RRHs and railway track, while $D_s$ is the distance of two RRHs, both in meters.

The train location is denoted as:

$$y = a + j * 0 \quad \text{(B.3.2.2)}$$

where: $a \in [0, \infty]$ and *a* means distance in meters, which means the train is right on the track.

The HST-SFN scenario for the test of the baseband performance is a non-fading propagation channel with four taps, namely the four nearest RRHs. Thus, RRH *k* is visible for the train only in the range:

$$k*D_s - 2*D_s \le a < k*D_s + 2*D_s \quad \text{(B.3.2.3)}$$

Power level $P_k$ (dB) for the signal from $k^{th}$ RRH, normalized to the total power received from all visible RRHs, is given by:

$$P_k = -20\lg(|y - x_k|) - 10\lg\left(\sum_{i \in \{i | i*D_S - 2*D_S \le a < i*D_S + 2*D_S\}} \frac{1}{|y - x_i|^2}\right) \text{ for } k*D_s - 2*D_s \le a < k*D_s + 2*D_s \quad \text{(B.3.2.4)}$$

Doppler shift $F_{D,k}$ (Hz) from $k^{th}$ RRH is given by:

$$F_{D,k} = f_C \times real\left[-v \times \frac{y - x_k}{|y - x_k| \times C}\right] \text{ for } k*D_s - 2*D_s \le a < k*D_s + 2*D_s \quad \text{(B.3.2.5)}$$

The relative delay $T_k$ (s) for the signal from $k^{th}$ RRH can be derived as:

$$T_k = \frac{|y - x_k|}{C} \text{ for } k*D_s - 2*D_s \le a < k*D_s + 2*D_s \quad \text{(B.3.2.6)}$$

In the above $v$ (m/s) is the moving speed of the train, $f_C$ (Hz) is the centre frequency, and $C$ (m/s) is the velocity of light.

Power level, Doppler shift and relative delay are given by equations B.3.2.4 ~ B.3.2.6 respectively, where the required input parameters listed in table B.3.2-1 and the resulting Doppler shift shown in Figures B.3.2-3 and B.3.2-4 are applied for all frequency bands.

**Table B.3.2-1: HST-SFN scenario**

| Parameter | Value |
|---|---|
| $D_s$ | 700 m |
| $D_{min}$ | 150 m |
| $v$ | 500 km/h |
| $f_d$ | 870 Hz for 15 kHz SCS test;<br>1667 Hz for 30 kHz SCS test |

NOTE 1: The trajectories of relative power, Doppler shifts and absolute delays presented in Figures B.3.2-2, B.3.2-3, B.3.2-4 and B.3.2-5 are derived from the equations B.3.2.4 ~ B.3.2.6 respectively.

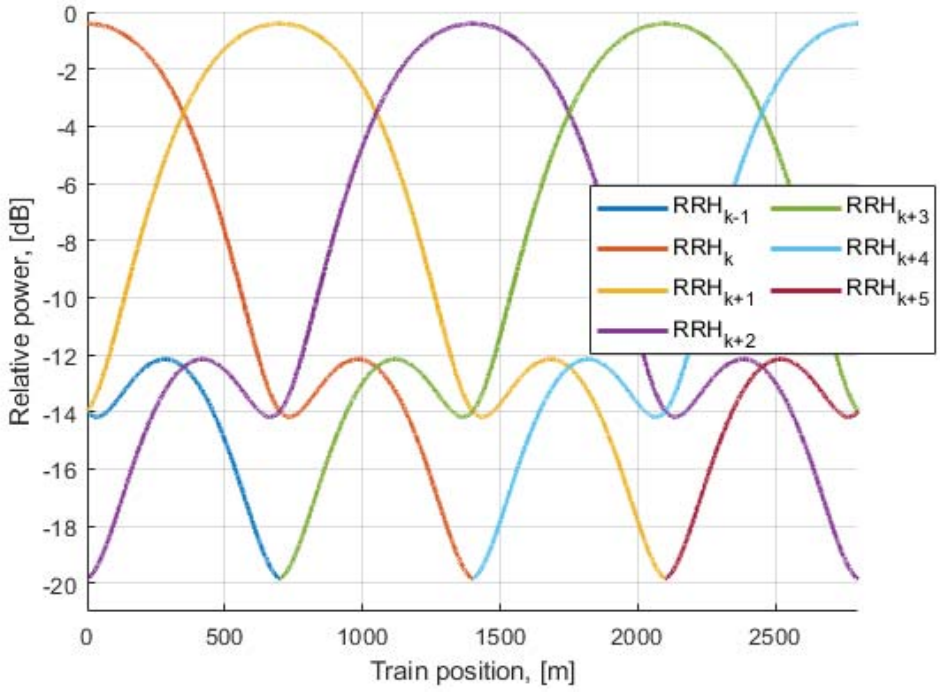



**Figure B.3.2-2 Relative power level trajectories**

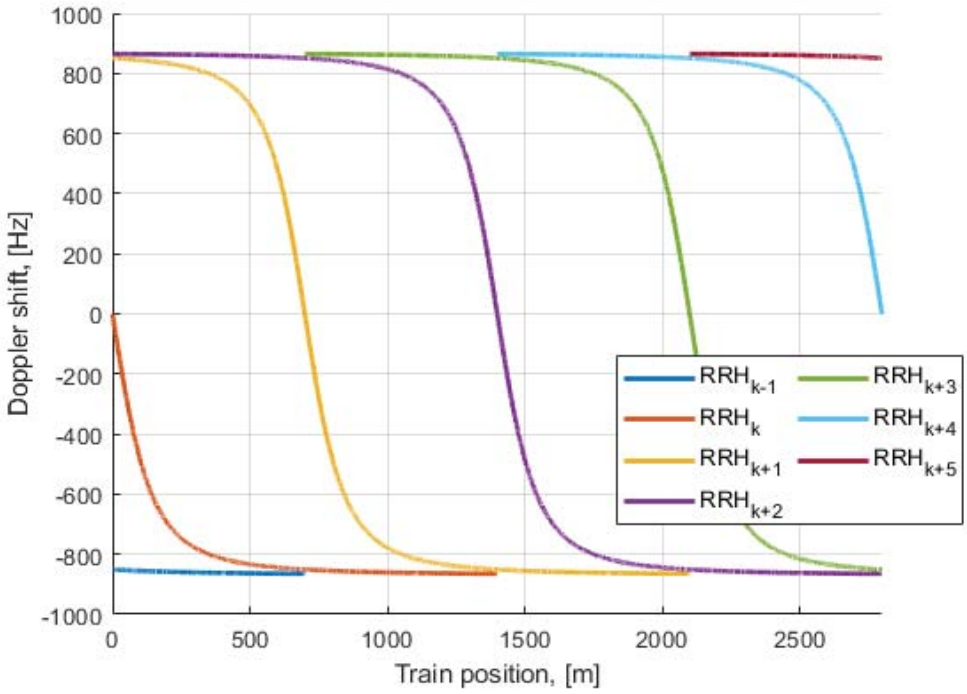



**Figure B.3.2-3 Doppler shift trajectories ( $f_d$ = 870 Hz)**

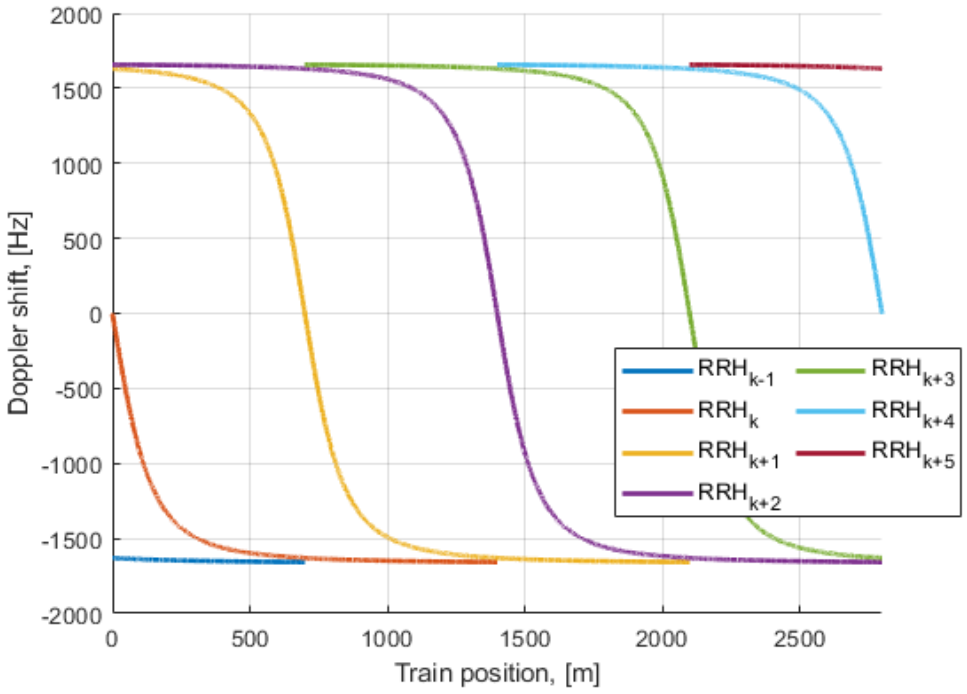



**Figure B.3.2-4 Doppler shift trajectories ( $f_d$ = 1667 Hz)**

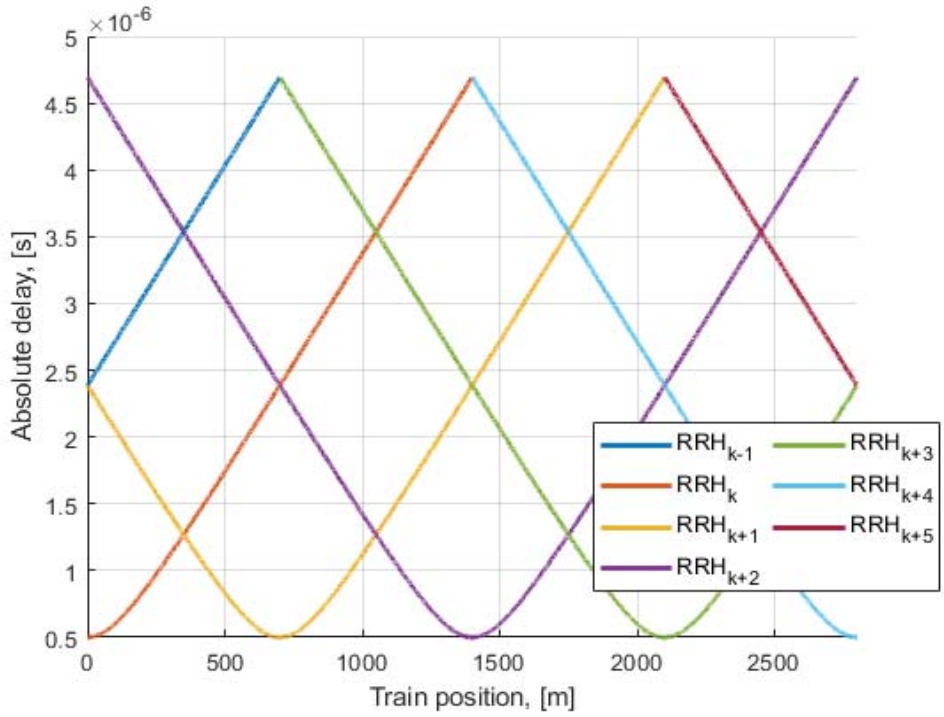



**Figure B.3.2-5 Absolute delay trajectories**

Static channel matrix will be used as defined in Annex B.1.

## B.3.3 HST-DPS Channel Profile

There is an infinite number of RRHs distributed equidistantly along the railway track with the same Cell ID as illustrated in Figure B.3.3-1.

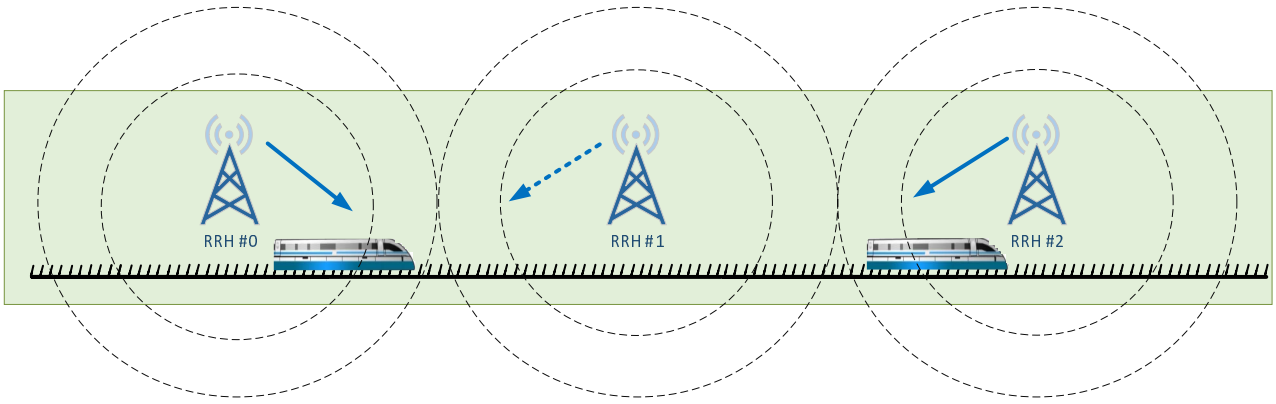



**Figure B.3.3-1: Deployment of HST-DPS**

The location of RRH *k* is given as:

$$x_k = k * D_s + j * D_{\min} \quad \text{(B.3.3.1)}$$

where: $k \in [-\infty, \infty]$, $j = sqrt(-1)$ and $D_{\min}$ is the distance between the RRHs and railway track, while $D_s$ is the distance of two RRHs, both in meters.

The train location is denoted as:

$$y = a + j*0 \quad \text{(B.3.3.2)}$$

where: $a \in [0, \infty]$ and $a$ means distance in meters, which means the train is right on the track.

The HST DPS multi-RRH scenario for the test of the baseband performance is a single tap propagation channel at each time with switching of transmission point in the middle point between two RRHs. Thus, RRH $k$ is visible for the train only in the range:

$$k * D_s - \frac{D_s}{2} \le a < k * D_s + \frac{D_s}{2} \quad \text{(B.3.3.3)}$$

Power level $P_k$ (dB) for the signal from $k^{th}$ RRH equals to 0. Doppler shift $F_{D,k}$ (Hz) from $k^{th}$ RRH is given by:

$$F_{D,k} = f_C \times real\left[-v \times \frac{y - x_k}{|y - x_k| \times C}\right] \text{ for } k * D_s - \frac{D_s}{2} \le a < k * D_s + \frac{D_s}{2} \quad \text{(B.3.3.4)}$$

In the above v (m/s) is the moving speed of the train, $f_C$ (Hz) is the centre frequency, and C (m/s) is the velocity of light.

Doppler shift is given by equation B.3.3.4, where the required input parameters listed in table B.3.3-1 and the resulting Doppler shift shown in Figures B.3.3-2 and B.3.3-3 are applied for all frequency bands.

**Table B.3.3-1: HST-DPS scenario**

| Parameter | Value |
|---|---|
| $D_s$ | 700 m |
| $D_{min}$ | 150 m |
| $v$ | 500 km/h |
| $f_d$ | 870 Hz for 15 kHz SCS test;<br>1667 Hz for 30 kHz SCS test |

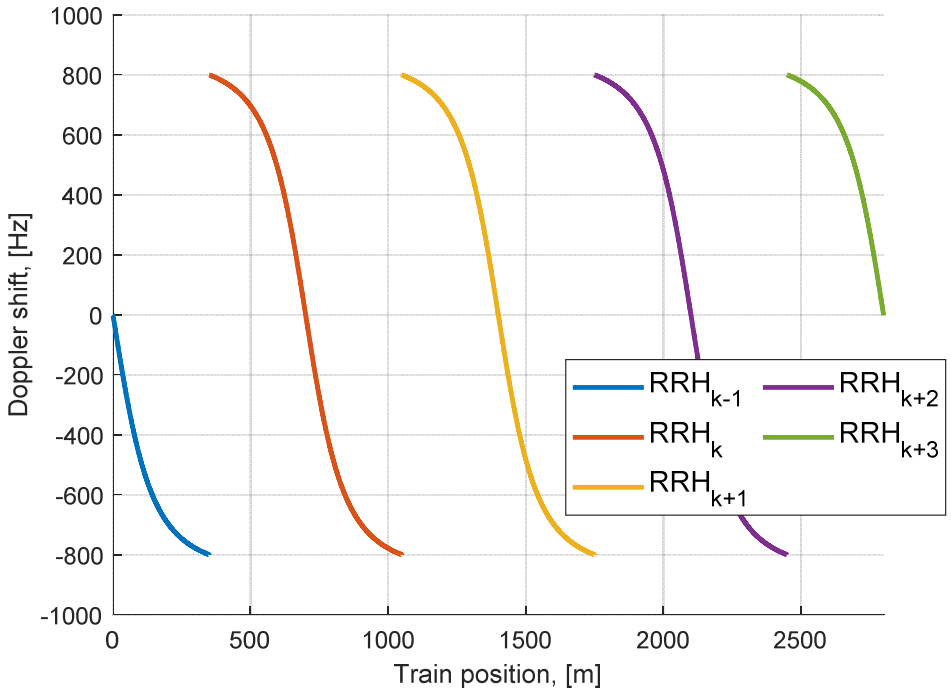



**Figure B.3.3-2 Doppler shift trajectory ( $f_d$ = 870 Hz)**

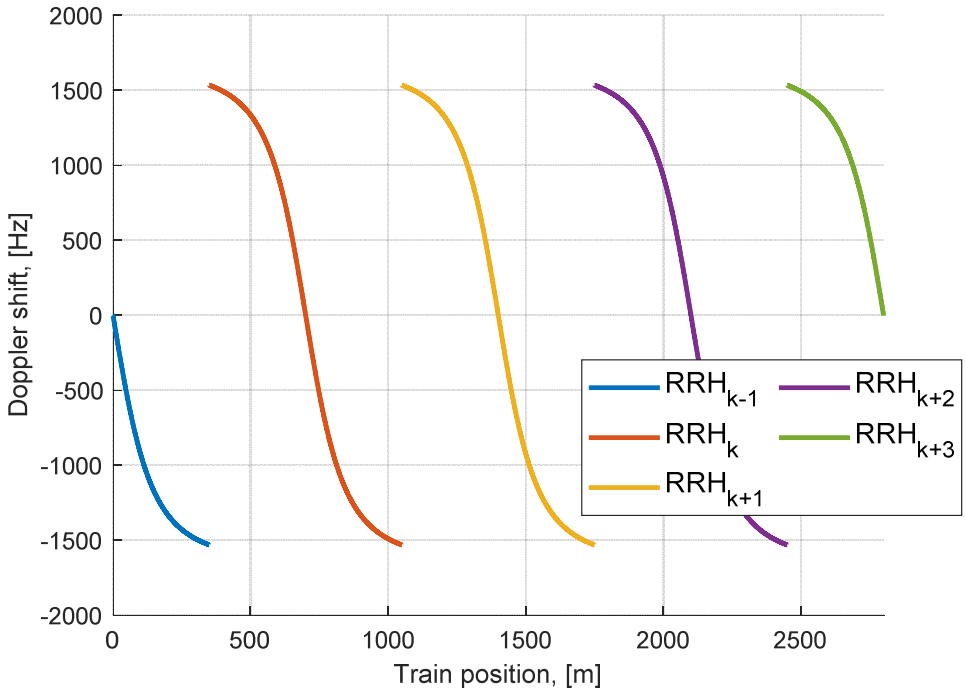



**Figure B.3.3-3 Doppler shift trajectory ( $f_d$ = 1667 Hz)**

Static channel matrix will be used as defined in Annex B.1.

# B.4 Physical signals, channels mapping and precoding

## B.4.1 General

Unless otherwise stated, the transmission on antenna port(s) $p = p_0, p_0 + 1, \ldots, p_0 + N_p - 1$ is defined by using a precoder matrix $W(i)$ of size $N_{ANT} \times N_p$, where $N_{ANT}$ is the number of physical transmit antenna elements configured per test , $N_p$ is the number of ports for a reference signal or physical channel configured per test, and $p_0$ is the first port for that reference signal or physical channel as defined in clauses 7.3 and 7.4 in TS 38.211 [9]. This precoder takes as an input a block of signals for antenna port(s) $p = p_0, p_0 + 1, \ldots, p_0 + N_p - 1$, $y^{(p)}(i) = \left[y^{(p_0)}(i)\ y^{(p_0+1)}(i)\ \ldots\ y^{(p_0+N_p-1)}(i)\right]^T$, $i = 0,1,\ldots, M^{ap}_{symb} - 1$, with $M^{ap}_{symb}$ being the number of modulation symbols per antenna port including the reference signal symbols, and generates a block of signals $y^{(q)}_{bf}(i) = \left[y^{(0)}_{bf}(i)\ y^{(1)}_{bf}(i)\ \ldots\ y^{(N_{ANT}-1)}_{bf}(i)\right]^T$ the elements of which are to be mapped onto the frequency-time index pair $(k,l)$ as per the test configuration but transmitted on different physical antenna elements:

$$y^{(q)}_{bf}(i) = W(i)y^{(p)}(i)$$

For Clause 6 and 8, the transmission of PDCCH and PDCCH DMRS on antenna port $p = p_0$ is defined by using a precoder matrix $W(i)$ of size 2x1. This precoder takes as an input a block of signals for antenna port(s) $p = p_0$, $y^{(p)}(i) = y^{(p_0)}(i)$ and generates a block of signals $y^{(q)}_{bf}(i) = \left[y^{(q)}_{bf}(i)\ y^{\left(\frac{N_{ANT}}{2}\right)}_{bf}(i)\right]^T$ the elements of which are to be mapped onto the frequency-time index pair $(k,l)$ as per the test configuration but transmitted on different physical antenna elements:

$$y_{BF}^{(p)}(i) = W(i)y^{(p)}(i)$$

The precoder matrix $W(i)$ is specific to the test case configuration $W(i)$ is defined in Clause 5.2.2.2 of TS 38.214 [12].

The transmission on PT-RS antenna port is associated (using same precoder) with the lowest indexed DM-RS antenna port among the DM-RS antenna ports assigned for the PDSCH.

The physical antenna elements are identified by indices $j = 0,1,...,N_{ANT} - 1$, where $N_{ANT}$ is the number of physical antenna elements configured per test.

Modulation symbols $y^{(p)}(i)$ with $p \in \{4000\}$ (i.e. PSS, SSS, PBCH and DM-RS for PBCH) are directly mapped to first physical antenna element.

Modulation symbols $a_{k,l}$ for CSI-RS resources which configured for tracking with one port are directly mapped to first physical antenna element.

Modulation symbols $a_{k,l}$ for CSI-RS resources which configured for beam refinement with one port are directly mapped to first physical antenna element.

Modulation symbols $a_{k,l}^{(p)}$ for NZP CSI-RS which configured for CSI acquisition with $p \in \{p_0, p_0 + 1,..., p_0 + N_{CSI} - 1\}$ are mapped to the physical antenna index $j = p - p_0$ where $N_{CSI}$ is the number of NZP CSI-RS ports configured per test.

# Annex C (normative): Downlink physical channels

## C.0 Downlink signal levels

Downlink power settings to be configured for connection setup has been defined in this clause covering both FR1 and FR2.

### C.0.1 FR1 Downlink Signal Levels (Conducted)

The downlink power settings in Table C.0.1-1 is used for FR1 conducted unless otherwise specified in a test case.

If the UE has more than one Rx antenna, the downlink signal is applied to each one. All UE Rx antennas shall be connected.

**Table C.0.1-1: Default Downlink power levels for NR FR1**

| SCS (kHz) | | Unit | Channel bandwidth | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 5 MHz | 10 MHz | 15 MHz | 20 MHz | 25 MHz | 30 MHz | 40 MHz | 50 MHz | 60 MHz | 80 MHz | 90 MHz | 100 MHz |
| 15 | Number of RBs | | 25 | 50 | 75 | 100 | 128 | 160 | 215 | 270 | N/A | N/A | N/A | N/A |
| | Channel BW power | dBm | -60 | -57 | -55 | -54 | -53 | -52 | -51 | -50 | N/A | N/A | N/A | N/A |
| 30 | Number of RBs | | 10 | 24 | 36 | 50 | 64 | 75 | 100 | 128 | 162 | 216 | 243 | 270 |
| | Channel BW power | dBm | -61 | -57 | -55 | -54 | -53 | -52 | -51 | -50 | -49 | -48 | -47 | -47 |
| 60 | Number of RBs | | N/A | 10 | 18 | 24 | 30 | 36 | 50 | 64 | 75 | 100 | 120 | 135 |
| | Channel BW power | dBm | N/A | -58 | -56 | -54 | -53 | -52 | -51 | -50 | -49 | -48 | -47 | -47 |
| | SSS EPRE | dBm/ 15 kHz | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 | -85 |
| | NOTE 1: The channel bandwidth powers are informative, based on -85dBm/15kHz SS/PBCH SSS EPRE, then scaled according to the number of RBs and rounded to the nearest integer dBm value. Full RE allocation with no boost or deboost is assumed.<br>NOTE 2: The power level is specified at each UE Rx antenna.<br>NOTE 3: DL level is applied for any of the Subcarrier Spacing configuration ( ) with the same power spectrum density of -85 dBm/15 kHz. | | | | | | | | | | | | | |

The default signal level uncertainty is [+/-3] dB at each test port, for any level specified. If the uncertainty value is critical for the test purpose, a tighter uncertainty is specified for the related test case in [Annex F]

### C.0.2 FR2 Downlink Signal Levels (Radiated)

The downlink power settings in Table C.0.2-1 is used unless otherwise specified in a test case.

**Table C.0.2-1: Default Downlink power levels for NR FR2**

| SCS (kHz) | | Unit | Channel Bandwidth | | | |
|---|---|---|---|---|---|---|
| | | | 50 MHz | 100 MHz | 200 MHz | 400 MHz |
| 60 | Number of RBs | | 66 | 132 | 264 | N/A |
| | Channel BW power | dBm | -70 | -67 | -64 | N/A |
| 120 | Number of RBs | | 32 | 66 | 132 | 264 |
| | Channel BW power | dBm | -70 | -67 | -64 | -61 |
| | SS/PBCH SSS EPRE | dBm/60 kHz | [-99] | [-99] | [-99] | [-99] |

NOTE 1: The channel bandwidth powers are informative, based on [-99] dBm/60 kHz SS/PBCH SSS EPRE, then scaled according to the number of RBs and rounded to the nearest integer dBm value. Full RE allocation with no boost or deboost is assumed.
NOTE 2: The power level is specified at the centre of quiet zone.
NOTE 3: DL level is applied for any of the Subcarrier Spacing configuration ( $\mu$ ) with the same power spectrum density of [-99]dBm/60kHz.

The default downlink signal level uncertainty is +/- TBD dB, for any level specified. If the uncertainty value is critical for the test purpose, a tighter uncertainty is specified for the related test case in Annex F.

# C.1 Setup

The following clause describes the downlink Physical Channels that are transmitted during connection setup.

## C.1.1 FR1 Setup

Table C.1.1-1 describes the downlink Physical Channels that are required for FR1 connection set up.

**Table C.1.1-1: Downlink Physical Channels required for FR1 connection setup**

| Physical Channel |
|---|
| PBCH |
| SSS |
| PSS |
| PDCCH |
| PDSCH |
| PBCH DMRS |
| PDCCH DMRS |
| PDSCH DMRS |
| CSI-RS |

The following common PDSCH and PDCCH configuration parameters shall be used to bring up the connection setup for FR1 NR cell.

**Table C.1.1-2: Common reference channel parameters for FR1**

| Parameter | Unit | Value |
|---|---|---|
| CORESET frequency domain allocation | | Full BW,number of RB's to be in multiple of 6 |
| CORESET time domain allocation | | 2 OFDM symbols at the begin of each slot |
| PDSCH mapping type | | Type A |
| PDSCH start symbol index (S) | | 2 |
| Number of consecutive PDSCH symbols (L) | | 12 |
| PDSCH PRB bundling | PRBs | 2 |
| Dynamic PRB bundling | | false |
| Overhead value for TBS determination | | 0 |
| First DMRS position for Type A PDSCH mapping | | 2 |
| DMRS type | | Type 1 |
| Number of additional DMRS | | 1 |
| FDM between DMRS and PDSCH | | Enable |
| TRS configuration | | 2 slots, periodicity 20 ms, offset 10 |
| PTRS configuration | | PTRS is not configured |
| Num of HARQ processes | | 8 (TDD) |

**Table C.1.1-3: Additional reference channels parameters for FDD**

| Parameter | Unit | Value |
|---|---|---|
| Number of HARQ Processes | | 4 |
| K1 value | | 2 for all slots |

**Table C.1.1-4: TDD UL-DL pattern for SCS 15 KHz**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| | | | FR1.15-1 |
| TDD Slot Configuration pattern (Note 1) | | | DDDSU |
| Special Slot Configuration (Note 2) | | | 10D+2G+2U |
| UL-DL configuration (*tdd-UL-DL-ConfigurationCommon*) | *referenceSubcarrierSpacing* | kHz | 15 |
| | *dl-UL-TransmissionPeriodicity* | ms | 5 |
| | *nrofDownlinkSlots* | | 3 |
| | *nrofDownlinkSymbols* | | 10 |
| | *nrofUplinkSlot* | | 1 |
| | *nrofUplinkSymbols* | | 2 |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | [4] if mod(I,5) = 0<br>[3] if mod(i,5) = 1<br>[2] if mod(i,5) = 2<br>[6] if mod(i,5) = 3 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,9} | | | |

**Table C.1.1-5: TDD UL-DL pattern for SCS 30 KHz**

| Parameter | | Unit | UL-DL Pattern |
|---|---|---|---|
| TDD Slot Configuration pattern (Note 1) | | | 7DS2U |
| Special Slot Configuration (Note 2) | | | 6D+4G+4U |
| UL-DL configuration (*tdd-UL-DL-ConfigurationCommon*) | *referenceSubcarrierSpacing* | 30 | kHz |
| | *dl-UL-TransmissionPeriodicity* | 5 | |
| | *nrofDownlinkSlots* | 7 | |
| | *nrofDownlinkSymbols* | 6 | |
| | *nrofUplinkSlot* | 2 | |
| | *nrofUplinkSymbols* | 4 | |
| UL-DL configuration2 (*tdd-UL-DL-ConfigurationCommon2*) | *referenceSubcarrierSpacing* | N/A | |
| | *dl-UL-TransmissionPeriodicity* | N/A | |
| | *nrofDownlinkSlots* | N/A | |
| | *nrofDownlinkSymbols* | N/A | |
| | *nrofUplinkSlot* | N/A | |
| | *nrofUplinkSymbols* | N/A | |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | 8 if mod(i,10) = 0<br>7 if mod(i,10) = 1<br>6 if mod(i,10) = 2<br>5 if mod(i,10) = 3<br>5 if mod(i,10) = 4<br>4 if mod(i,10) = 5<br>3 if mod(i,10) = 6<br>2 if mod(i,10) = 7 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,19} | | | |

## C.1.2 FR2 Setup

Table C.1.2-1 describes the downlink Physical Channels that are required for FR2 connection set up.

**Table C.1.2-1: Downlink Physical Channels required for FR2 connection set-up**

| Physical Channel |
|---|
| PBCH |
| SSS |
| PSS |
| PDCCH |
| PDSCH |
| PBCH DMRS |
| PDCCH DMRS |
| PDSCH DMRS |
| CSI-RS |
| PTRS |

The following common PDSCH and PDCCH configuration parameters shall be used to bring up the connection setup for FR2 NR cell.

**Table C.1.2-2: Common reference channel parameters for FR2**

| Parameter | Unit | Value |
|---|---|---|
| CORESET frequency domain allocation | | Full BW, number of RB's to be in multiple of 6 |
| CORESET time domain allocation | | 1 OFDM symbols at the begin of each slot |
| PDSCH mapping type | | Type A |
| PDSCH start symbol index (S) | | 1 |
| Number of consecutive PDSCH symbols (L) | | 13 |
| PDSCH PRB bundling | PRBs | 2 |
| Dynamic PRB bundling | | false |
| MCS table for TBS determination | | 64QAM |
| Overhead value for TBS determination | | 0 |
| First DMRS position for Type A PDSCH mapping | | 2 |
| DMRS type | | Type 1 |
| Number of additional DMRS | | 1 |
| FDM between DMRS and PDSCH | | Enable |
| TRS configuration | | 2 slots, periodicity 20 ms, offset 10 |
| PTRS configuration | | Single port, every other RB, every symbol (K=2, L=1) |
| Num of HARQ processes | | 8 |

**Table C.1.2-3: Additional test parameters for TDD for SCS 60 KHz**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| TDD Slot Configuration pattern (Note 1) | | | DDSU |
| Special Slot Configuration (Note 2) | | | 11D+3G+0U |
| UL-DL configuration (*tdd-UL-DL-ConfigurationCommon*) | *referenceSubcarrierSpacing* | kHz | 60 |
| | *dl-UL-TransmissionPeriodicity* | ms | 1 |
| | *nrofDownlinkSlots* | | 2 |
| | *nrofDownlinkSymbols* | | 11 |
| | *nrofUplinkSlot* | | 1 |
| | *nrofUplinkSymbols* | | 0 |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | K1 = 3 if mod(i,4) = 0<br>K1 = 2 if mod(i,4) = 1<br>K1 = 5 if mod(i,4) = 2 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,39} | | | |

**Table C.1.2-4: Additional test parameters for TDD for SCS 120 KHz**

| Parameter | | Unit | UL-DL pattern |
|---|---|---|---|
| TDD Slot Configuration pattern (Note 1) | | | DDDSU |
| Special Slot Configuration (Note 2) | | | 10D+2G+2U |
| UL-DL configuration (*tdd-UL-DL-ConfigurationCommon*) | *referenceSubcarrierSpacing* | kHz | 120 |
| | *dl-UL-TransmissionPeriodicity* | ms | 0.625 |
| | *nrofDownlinkSlots* | | 3 |
| | *nrofDownlinkSymbols* | | 10 |
| | *nrofUplinkSlot* | | 1 |
| | *nrofUplinkSymbols* | | 2 |
| K1 value (PDSCH-to-HARQ-timing-indicator) | | | K1 = [4] if mod(i,5) = 0<br>K1 = [3] if mod(i,5) = 1<br>K1 = [2] if mod(i,5) = 2<br>K1 = [6] if mod(i,5) = 3 |
| Note 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information.<br>Note 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information.<br>Note 3: i is the slot index per frame; i = {0,…,79} | | | |

# C.2 Connection

## C.2.1 FR1 Measurement of Performance Characteristics

Unless otherwise stated, Table C.2.1-1 is applicable for measurements in which uniform RS-to-EPRE boosting for all downlink physical channels is used.

**Table C.2.1-1: Downlink Physical Channels transmitted during a connection (FDD and TDD) for FR1**

| Parameter | Unit | Value (NOTE 2) |
|---|---|---|
| SSS transmit power | W | Test specific |
| EPRE ratio of PSS to SSS | dB | 0 |
| EPRE ratio of PBCH to SSS | dB | 0 |
| EPRE ratio of PBCH to PBCH DMRS | dB | 0 |
| EPRE ratio of PDCCH to SSS | dB | 0 |
| EPRE ratio of PDCCH to PDCCH DMRS | dB | 0 |
| EPRE ratio of PDSCH to SSS | dB | 0 |
| EPRE ratio of PDSCH to PDSCH DMRS | dB | Test specific (Note 1) |
| EPRE ratio of NZP CSI-RS to SSS | dB | -10*log10(L) (Note 3) |
| EPRE ratio of PDSCH OCNG to SSS | dB | 0 |
| EPRE ratio of PDCCH OCNG to SSS | dB | 0 |
| EPRE ratio of LTE CRS to NR SSS | dB | 0 (Note 4) |
| NOTE 1: Value is derived from Table 4.1-1 in TS 38.214 [X] based on “Number of DM-RS CDM groups without data” and “DMRS Type” parameters specified for each test.<br>NOTE 2: The value is the energy of per RE for a single antenna port before pre-coding.<br>NOTE 3: $L \in \{1,2,4,8\}$ is the CDM group size of NZP CSI-RS specified for each test.<br>NOTE 4: It is only applicable to LTE-NR coexistence tests. | | |

## C.2.2 FR2 Measurement of Performance Characteristics

Unless otherwise stated, Table C.2.2-1 is applicable for measurements on the Performance Characteristics.

**Table C.2.2-1: Downlink Physical Channels transmitted during a connection (TDD) for FR2**

| Parameter | Unit | Value (Note 2) |
|---|---|---|
| SSS transmit power | W | Test specific |
| EPRE ratio of PSS to SSS | dB | 0 |
| EPRE ratio of PBCH to SSS | dB | 0 |
| EPRE ratio of PBCH to PBCH DMRS | dB | 0 |
| EPRE ratio of PDCCH to SSS | dB | 0 |
| EPRE ratio of PDCCH to PDCCH DMRS | dB | 0 |
| EPRE ratio of PDSCH to SSS | dB | 0 |
| EPRE ratio of PDSCH to PDSCH DMRS | dB | Test specific (Note 1) |
| EPRE ratio of NZP CSI-RS to SSS | dB | -10*log10(L) (Note 3) |
| EPRE ratio of PTRS to PDSCH | dB | Test specific (Note 4) |
| EPRE ratio of PDSCH OCNG to SSS | dB | 0 |
| EPRE ratio of PDCCH OCNG to SSS | dB | 0 |
| Note 1: Value is derived from Table 4.1-1 in TS 38.214 [12] based on "Number of DM-RS CDM groups without data" and "DMRS Type" parameters specified for each test<br>Note 2: The value is the energy of per RE for a single antenna port before pre-coding.<br>Note 3: $L \in \{1,2,4,8\}$ is the CDM group size of NZP CSI-RS specified for each test.<br>Note 4: Value is derived from Table 4.1-2 in TS 38.214 [12] based on “The number of PDSCH layers” and “*epre-Ratio*” parameters specified for each test. | | |

# Annex D (normative): E-UTRA link setup config for NSA testing

## D.0 General

Below clauses define the E-UTRA link setup config for NSA Demodulation and CSI tests cases unless otherwise specified within the main test case.

## D.1 E-UTRA test parameters

Below are the common test parameters to be configured for E-UTRA link.

**Table D.1-1: Common Test Parameters (FDD)**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| Inter-TTI Distance | | 1 | |
| Number of HARQ processes | Processes | 8 | For FDD, 8 HARQ processes in the DL, as specified in TS 36.213 [10] clause 7. All 8 HARQ processes are used. |
| Scheduling of retransmissions | | | 1. Retransmissions use the same Transport Block Size (TBS) as the initial transmission.<br>2. HARQ processes are scheduled consecutively, independent of the fact, whether retransmissions (for negatively acknowledged HARQ processes) or new transmissions (for positively acknowledged HARQ processes) occur. |
| Maximum number of HARQ transmission | | 4 | It is always 4 for FDD, as specified in TS 36.213 [10] clause 8 |
| Redundancy version coding sequence | | {0,1,2,3} for QPSK | |
| Number of OFDM symbols for PDCCH | OFDM symbols | 3 for 5 MHz bandwidths,<br>2 for 10 MHz, 20MHz | The PCFICH carries information about the number of OFDM symbols used for transmission of PDCCHs in a subframe, as specified in TS 36.211 [8] clause 6.7 |
| Cyclic Prefix | | Normal | CP consist of the following physical resource blocks (RBs) parameters: 12 consecutive subcarriers at a 15 kHz spacing and 7 OFDM symbols, as specified in TS 36.211 [8] clause 6.2.3 |
| Cell ID | | 0 (Note 1) | The Cell ID is uniquely defined by a number in the range of 0 to 503, representing the physical-layer cell identity, as specified in TS 36.211 [8] clause 6.11. |
| DCI format for PDSCH | Format 1A | | |
| DCI format for PUSCH | Format 0 | | |

**Table D.1-2: Common Test Parameters (TDD)**

| Parameter | Unit | Value | Comments |
|---|---|---|---|
| Uplink downlink configuration (Note 1) | | 2 | |
| Special subframe configuration (Note 2) | | 5 | |
| Inter-TTI Distance | | 1 | |
| Number of HARQ processes | Processes | 7 | For TDD, 7 HARQ processes in the DL, as specified in TS 36.213 [10] clause 7.<br>All 7 HARQ processes are used. |
| Scheduling of retransmissions | | | 1. Retransmissions use the same Transport Block Size (TBS) as the initial transmission.<br>2. HARQ processes are scheduled consecutively, independent of the fact, whether retransmissions (for negatively acknowledged HARQ processes) or new transmissions (for positively acknowledged HARQ processes) occur.<br>3. In case when the initial transmission and the retransmissions are scheduled in subframes with a different $N_{PRB}$ (in terms of TS 36.213 [10] subclause 7.1.7) $29 \le I_{MCS} \le 31$ according to TS 36.213 [10] subclause 7.1.7.2 and the appropriate modulation is used. |
| Maximum number of HARQ transmission | | 4 | It is always 4 for TDD, as specified in TS 36.213 [10] clause 8 |
| Redundancy version coding sequence | | {0,1,2,3} for QPSK | |
| Number of OFDM symbols for PDCCH | OFDM symbols | 3 for 5 MHz bandwidths, 2 for 10 MHz | The PCFICH carries information about the number of OFDM symbols used for transmission of PDCCHs in a subframe, as specified in TS 36.211 [8] clause 6.7 |
| Cyclic Prefix | | Normal | CP consist of the following physical resource blocks (RBs) parameters: 12 consecutive subcarriers at a 15 kHz spacing and 7 OFDM symbols, as specified in TS 36.211 [8] clause 6.2.3 |
| Cell ID | | 0 (Note 3) | The Cell ID is uniquely defined by a number in the range of 0 to 503, representing the physical-layer cell identity, as specified in TS 36.211 [8] clause 6.11. |
| DCI format for PDSCH | Format 1A | | |
| DCI format for PUSCH | Format 0 | | |

NOTE 1: as specified in Table 4.2-2 in TS 36.211 [8].
NOTE 2: as specified in Table 4.2-1 in TS 36.211 [8].
NOTE 3: For CA tests, Cell ID = 0 applies only to P-Cell. For (n)*th* S-Cell, Cell ID = n is used.

# D.2 E-UTRA configuration

This clause defines the E-UTRA link settings for the test cases defined in clauses 5 and 6. The LTE link is supposed to be a functional link. The configuration defined in this clause ensures establishment of LTE link. Unless otherwise stated, ensure the UE is in state 3A-RF on the E-UTRA cell as defined in TS 36.508 [19].

**Table D.2-1: E-UTRA configuration for EN-DC tests**

| Parameter | Value | Comments |
|---|---|---|
| Test Frequency during and after connection setup | Mid | As defined in TS 36.508 [19] for inter band test cases and as defined in TS 38.508-1 [6] clause 4.3.1 for intra band test cases, with NR SCS as per the test case for the LTE band under test |
| Bandwidth during and after connection setup | 5 MHz (Note 1) | Supported by all LTE bands |
| PDSCH transmission mode and antenna config | TM1 1x2 | |
| OCNG pattern | OP.1 for FDD<br>OP.1 for TDD | These physical resource blocks are assigned to an arbitrary number of virtual UE′s with one PDSCH per virtual UE; the data transmitted over the OCNG PDSCHs shall be uncorrelated pseudo random data, which is QPSK modulated. |
| DL RMC | According to table A.3.2-1 in TS 36.521-1 [16] for FDD<br>According to table A.3.1.1-1 in TS 38.521-3 [21] for TDD | Note 1 |
| DL RB allocation | 25 | Full RB allocation assuming 5 MHz ChBW. 100 RB for 20 MHz ChBW as applicable |
| UL Signal levels during connection setup | PUSCH Power | Attained by enabling open loop power control and setting up UL signal levels according to Annexes H.0, H.2 and H.3 of TS 36.521-1 [16] |
| TA adjustments | *TimeAlignmentTimerDedicated* IE to be set to infinity | *TimeAlignmentTimerDedicated* IE to be set to infinity to ensure UE doesn't look for TA adjustments (See Table D.2-4) |
| CQI reports and SRS after connection setup | Disabled (See Table D.2-2 and D.2-3) | Disable periodic and aperiodic CQI reports to ensure none of these transmissions occur on the LTE uplink. |
| NOTE 1: If none of the UE supported EN-DC band combos support 5MHz E-UTRA carrier, configure 20 MHz channel BW. | | |

**Table D.2-2: *CQI-ReportConfig-DEFAULT:* Additional E-UTRA Anchor Configuration**

| Derivation Path: TS 36.508 [7] clause 4.6.3, Table 4.6.3-2 CQI-ReportConfig-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| CQI-ReportConfig-DEFAULT ::= SEQUENCE { | | | |
| cqi-ReportModeAperiodic | NOT PRESENT | | |
| cqi-ReportPeriodic | NOT PRESENT | | |
| } | | | |

**Table D.2-3: *PhysicalConfigDedicated-DEFAULT:* Additional E-UTRA Anchor Configuration**

| Derivation Path: TS 36.508 [7] clause 4.8.2, Table 4.8.2.1.6-1 PhysicalConfigDedicated-DEFAULT | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| PhysicalConfigDedicated-DEFAULT ::= SEQUENCE { | | | |
| soundingRS-UL-ConfigDedicated | Not present | | RBC |
| } | | | |

**Table D.2-4: *MAC-MainConfig-RBC*: Additional E-UTRA Anchor Configuration**

| Derivation Path: TS 36.508 [7] clause 4.8.2.1.5, Table 4.8.2.1.5-1 MAC-MainConfig-RBC | | | |
|---|---|---|---|
| **Information Element** | **Value/remark** | **Comment** | **Condition** |
| timeAlignmentTimerDedicated | Infinity | | |

# D.3 E-UTRA link common physical channel setup

Table D.3-1 describes the downlink Physical Channels that are required for E-UTRA connection set up.

**Table D.3-1: Downlink Physical Channels required for E-UTRA connection set-up**

| Physical Channel | EPRE Ratio | Note |
|---|---|---|
| **PBCH** | PBCH_RA = 0 dB | |
| | PBCH_RB = 0 dB | |
| **PSS** | PSS_RA = 0 dB | |
| **SSS** | SSS_RA = 0 dB | |
| **PCFICH** | PCFICH_RB = 0 dB | |
| **PDCCH** | PDCCH_RA = 0 dB | |
| | PDCCH_RB = 0 dB | |
| **PDSCH** | PDSCH_RA = 0 dB | |
| | PDSCH_RB = 0 dB | |
| **PHICH** | PHICH_RA = 0 dB | |
| | PHICH_RB = 0 dB | |
| NOTE 1: $P_B = 0$. | | |
| NOTE 2: PHICH group power, i.e. the total power of all active PHICH sequences within a PHICH group. | | |

# D.4 E-UTRA power level

## D.4.1 E-UTRA power level (conducted)

**Table D.4.1-1: DL power level for E-UTRA (conducted)**

| Parameter | Value | Comments |
|---|---|---|
| DL signal level | RS EPRE -85.0 dBm/15 kHz | The power level is specified at each UE Rx antenna |

## D.4.2 E-UTRA power level (radiated)

**Table D.4.2-1: Downlink power levels for E-UTRA (radiated)**

| Parameter | Value | Comments |
|---|---|---|
| DL signal level | RS EPRE -100 dBm/15 kHz | The power level is specified at each UE Rx antenna |

# Annex E (normative): Environmental conditions

FFS

# Annex F (normative): Measurement uncertainties and test tolerances

The requirements of this clause apply to all tests in the present document.

## F.1 Measurement uncertainties and test tolerances for FR1

### F.1.1 Acceptable uncertainty of test system (normative)

The maximum acceptable uncertainty of the Test System is specified below for each test, where appropriate. The Test System shall enable the stimulus signals in the test case to be adjusted to within the specified range, and the equipment under test to be measured with an uncertainty not exceeding the specified values. All ranges and uncertainties are absolute values, and are valid for a confidence level of 95 %, unless otherwise stated.

A confidence level of 95 % is the measurement uncertainty tolerance interval for a specific measurement that contains 95 % of the performance of a population of test equipment.

For RF tests it should be noted that the uncertainties in clause F.1 apply to the Test System operating into a nominal 50 ohm load and do not include system effects due to mismatch between the DUT and the Test System.

The downlink signal uncertainties apply at each receiver antenna connector.

#### F.1.1.1 Measurement of test environments

The measurement accuracy of the UE test environments defined in TS 38.508-1 [5] subclause 4.1, Test environments shall be

- Pressure ±5 kPa.
- Temperature ±2 degrees.
- Relative Humidity ±5 %.
- DC Voltage ±1,0 %.
- AC Voltage ±1,5 %.
- Vibration 10 %.
- Vibration frequency 0,1 Hz.

The above values shall apply unless the test environment is otherwise controlled and the specification for the control of the test environment specifies the uncertainty for the parameter.

#### F.1.1.2 Measurement of Demod Performance requirements

This clause defines the maximum test system uncertainty for Demod Performance requirements. The maximum test system uncertainty allowed for the measurement uncertainty contributors are defined in Table F.1.1.2-1.

**Table F.1.1.2-1: Maximum measurement uncertainty values for the test system for FR1 (up to 6 GHz) and Channel BW ≤ 40 MHz**

| MU contributor | Unit | Value | Comment |
|---|---|---|---|
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to | dB | ±2.0 | Same as in LTE |

| MU contributor | Unit | Value | Comment |
|---|---|---|---|
| average over $BW_{config}$ | | | |
| Signal to noise ratio uncertainty | dB | ±0.3 | Same as in LTE |
| Signal to noise ratio variation | dB | ±0.5 | Same as in LTE |
| Fading profile power uncertainty for 1Tx | dB | ±0.5 | Same as in LTE |
| Fading profile power uncertainty for 2Tx | dB | ±0.7 | Same as in LTE |
| Fading profile power uncertainty for 4Tx | dB | ±0.7 | Same as in LTE |

The maximum test system uncertainty for test cases defined in section 5 is defined in Table F.1.1.2-2.

**Table F.1.1.2-2: Maximum test system uncertainty for FR1 demodulation performance test cases**

| Subclause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 5.2.2.1.1_1 2Rx FDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for both SA and NSA | ± 0.9 dB for > 10Hz doppler<br>± 1 dB for 10Hz doppler<br>± 0.6 dB for test 1-6<br>± 0.9 dB for test 1-7 | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$) + SNR uncertainty due to finite test time$^2$<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>SNR uncertainty due to finite test time ±0.3 dB for 10Hz Doppler, otherwise ±0.0 dB<br>For test point 1-6, Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$) = 0.6 dB<br><br>For test point 1-7, Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$) = 0.9 dB |
| 5.2.2.1.1_2 2Rx FDD FR1 PDSCH Mapping Type A performance - 2x2 MIMO with enhanced receiver type X for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.2_1 2Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.3_1 2Rx FDD FR1 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.4_1 2Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |

| | | |
|---|---|---|
| 5.2.2.1.5_1 2Rx FDD FR1 PDSCH 0.001% BLER performance - 1x2 MIMO with baseline receiver for both SA and NSA | [± 0.6 dB] | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±2.0 dB |
| 5.2.2.1.6_1 2Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA | FFS | The method of calculationg the overall system uncertainty is the same as 5.2.2.1.1_1 except:<br>SNR uncertainty due to finite test time for 1% residual BLER FFS dB for 10Hz Doppler, otherwise FFS dB<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$) + SNR uncertainty due to finite test time$^2$ = FFS dB |
| 5.2.2.1.7_1 2Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.8_1 2Rx FDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.9_1 2Rx FDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA | ± 0.6 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty, ±0.3 dB<br>2. Effect of AWGN flatness and signal flatness, ±2.0 dB<br>3. SNR uncertainty due to finite test time, ±0.0 dB for >10Hz Doppler.<br><br>Items 1, 2, 3 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$) = 0.6 dB |

| | | |
|---|---|---|
| 5.2.2.1.10_1 2Rx FDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA | ± 0.6 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty, ±0.3 dB<br>2. Effect of AWGN flatness and signal flatness, ±2.0 dB<br>3. SNR uncertainty due to finite test time, ±0.0 dB for >10Hz Doppler.<br><br>Items 1, 2, 3 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br><br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$) = 0.6 dB |
| 5.2.2.1.11_1 2Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.12_1 2Rx FDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.13_1 2Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.1.14_1 2Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.1_1 2Rx TDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for both SA and NSA | ± 0.9 dB for test 1-10<br>± 0.6 dB for test 1-11<br>For other TPs, same as 5.2.2.1.1_1 | For test point 1-10, Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$) = 0.9 dB<br><br>For test point 1-11, Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$) = 0.6 dB<br><br>For other TPs, same as 5.2.2.1.1_1 |
| 5.2.2.2.1_2 2Rx TDD FR1 PDSCH Mapping Type A performance - 2x2 MIMO with enhanced receiver type X for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.2_1 2Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.3_1 2Rx TDD FR1 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.4_1 2Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |

| | | |
|---|---|---|
| 5.2.2.2.5_1 2Rx TDD FR1 PDSCH 0.001% BLER performance - 1x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.5_1 | Same as 5.2.2.1.5_1 |
| 5.2.2.2.6_1 2Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.6_1 | Same as 5.2.2.1.6_1 |
| 5.2.2.2.7_1 2Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.8_1 2Rx TDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.9_1 2Rx TDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.9_1 | Same as 5.2.2.1.9_1 |
| 5.2.2.2.10_1 2Rx TDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.10_1 | Same as 5.2.2.1.10_1 |
| 5.2.2.2.11_1 2Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.12_1 2Rx TDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.13_1 2Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.2.2.14_1 2Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2.3.1.1_1 4Rx FDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with baseline receiver for both SA and NSA | ± 0.9 dB for > 10Hz doppler<br>± 1.0 dB for 10Hz doppler | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time $^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>SNR uncertainty due to finite test time ±0.3 dB for 10Hz Doppler, otherwise ±0.0 dB |
| 5.2.3.1.1_2 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |

| | | |
|---|---|---|
| 5.2.3.1.1_4 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with enhanced receiver type 1 for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.2_1 4Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 4x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.3_1 4Rx FDD FR1 PDSCH mapping Type B performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.4_1 4Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.5_1 4Rx FDD FR1 PDSCH 0.001% BLER performance - 1x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.5_1 | Same as 5.2.2.1.5_1 |
| 5.2.3.1.6_1 4Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.6_1 | Same as 5.2.2.1.6_1 |
| 5.2.3.1.7_1 4Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.8_1 4Rx FDD FR1 PDSCH pre-emption performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.9_1 4Rx FDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.9_1 | Same as 5.2.2.1.9_1 |
| 5.2.3.1.10_1 4Rx FDD FR1 HST-DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.10_1 | Same as 5.2.2.1.10_1 |
| 5.2.3.1.11_1 4Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x4 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.12_1 4Rx FDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x4 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.13_1 4Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x4 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.1.14_1 4Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x4 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.1_1 4Rx TDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.1_2 4Rx TDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.1_4 4Rx TDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with enhanced receiver type 1 for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |

| | | |
|---|---|---|
| 5.2.3.2.2_1 4Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.3_1 4Rx TDD FR1 PDSCH mapping Type B performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.4_1 4Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.5_1 4Rx TDD FR1 PDSCH 0.001% BLER performance - 1x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.5_1 | Same as 5.2.2.1.5_1 |
| 5.2.3.2.6_1 4Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.2.1.6_1 | Same as 5.2.2.1.6_1 |
| 5.2.3.2.7_1 4Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.9_1 4Rx TDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA | Same as 5.2.3.1.9_1 | Same as 5.2.3.1.9_1 |
| 5.2.3.2.10_1 4Rx TDD FR1 HST DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA | TBD | TBD |
| 5.2.3.2.11_1 4Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.12_1 4Rx TDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.13_1 4Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2.3.2.14_1 4Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2A.2.1.1 2Rx Normal PDSCH Demodulation Performance for CA (2DL CA) | Same as 5.2.2.1.1_1 | Same as 5.2.2.1.1_1 |
| 5.2A.3.1.1 4Rx Normal PDSCH Demodulation Performance for CA (2DL CA) | Same as 5.2.3.1.1_1 | Same as 5.2.3.1.1_1 |
| 5.2A.4.1.1 2Rx-4Rx Normal PDSCH Demodulation Performance for CA (2DL CA) | Same as 5.2.2.1.1_1 for 2Rx CC and 5.2.3.1.1_1 for 4Rx CC | Same as 5.2.2.1.1_1 for 2Rx CC and 5.2.3.1.1_1 for 4Rx CC |

| | | |
|---|---|---|
| 5.3.2.1.1 2Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | ± 0.9 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$+ SNR uncertainty due to finite test time$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for 1Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>SNR uncertainty due to finite test time ±0.4 dB |
| 5.3.2.1.2 2Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | ± 1.0 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared:<br>AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$+ SNR uncertainty due to finite test time$^2$)<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2 Tx<br>AWGN flatness and signal flatness ±2.0 dB<br>SNR uncertainty due to finite test time ±0.4 dB |
| 5.3.2.1.3 2Rx FDD FR1 PDCCH 1 Tx antenna performance for power saving | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |
| 5.3.2.2.1 2Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |
| 5.3.2.2.2 2Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | Same as 5.3.2.1.2 | Same as 5.3.2.1.2 |
| 5.3.2.2.3 2Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |
| 5.3.3.1.1 4Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |
| 5.3.3.1.2 4Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | Same as 5.3.2.1.2 | Same as 5.3.2.1.2 |
| 5.3.3.1.3 4Rx FDD FR1 PDCCH 1 Tx antenna performance for power saving | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |

| | | |
|---|---|---|
| 5.3.3.2.1 4Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |
| 5.3.3.2.2 4Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | Same as 5.3.2.1.2 | Same as 5.3.2.1.2 |
| 5.3.3.2.3 4Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving | Same as 5.3.2.1.1 | Same as 5.3.2.1.1 |
| 5.5.1 FR1 Sustained downlink data rate performance for single carrier | ±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz<br>±1.5 dB, 4.2GHz < f ≤ 6GHz<br><br>Downlink EVM ≤ 3% | 3% EVM is equivalent to a Test system downlink SNR of 30.5dB. The noise from the Test system is then sufficiently below that required for the UE to demodulate the signal with the required % success rate. Under these conditions the UE throughput is limited by the Reference measurement channel and the UE capability, and not by the Test system EVM. |
| 9.4B.1.1 Sustained downlink data rate performance for EN-DC within FR1 | E-UTRA CC:<br>±0.7 dB, f ≤ 3.0GHz<br>±1.0 dB, 3.0GHz < f ≤ 4.2GHz<br><br>NR CC:<br>Same as 5.5.1 | Same as 5.5.1 |

## F.1.1.3 Measurement of Channel State Information reporting

This clause defines the maximum test system uncertainty for channel state information reporting requirements. The maximum test system uncertainty allowed for the measurement uncertainty contributors are defined in Table F.1.1.3-1.

**Table F.1.1.3-1: Maximum measurement uncertainty values for the test system for FR1 (up to 6 GHz) and Channel BW ≤ 40 MHz**

| MU contributor | Unit | Value | Comment |
|---|---|---|---|
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{config}$ | dB | Same as in table F.1.1.2-1 | |
| Signal to noise ratio uncertainty | dB | Same as in table F.1.1.2-1 | |
| Signal to noise ratio variation | dB | Same as in table F.1.1.2-1 | |
| Fading profile power uncertainty for 1Tx | dB | Same as in table F.1.1.2-1 | |
| Fading profile power uncertainty for 2Tx | dB | Same as in table F.1.1.2-1 | |

The maximum test system uncertainty for test cases defined in section 6 is defined in Table F.1.1.3-2.

**Table F.1.1.3-2: Maximum test system uncertainty for FR1 channel state information reporting test cases**

| Subclause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 6.2.2.1.1.1 2Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | +/- 0.3 dB | Overall system uncertainty for AWGN conditions comprises:<br>Signal-to-noise ratio uncertainty ±0.3 dB<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect* |
| 6.2.2.1.1.22Rx FDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.2.2.1.2.12Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | +/- 0.8 dB | Overall system uncertainty for fading conditions comprises two quantities:<br>1. Signal-to-noise ratio uncertainty ±0.3 dB<br>2. Fading profile power uncertainty for 2Tx ±0.7 dB<br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared:<br>Test System uncertainty = SQRT (Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$)<br>*AWGN flatness and signal flatness ±2.0 dB not expected to have any significant effect* |
| 6.2.2.1.2.22Rx FDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.2.2.1.1 2Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.2.2.2.1.22Rx TDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.2.2.2.2.12Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.2.2.2.22Rx TDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.3.1.1.1 4Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.2.3.1.1.24Rx FDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.2.3.1.2.14Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.3.1.2.24Rx FDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.3.2.1.1 4Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.2.3.2.2.14Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.3.2.2.24Rx TDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |

| | | |
|---|---|---|
| 6.2A.3.1.1 2Rx CQI reporting accuracy under AWGN conditions for CA (2DL CA) | Same as 6.2.2.1.1.1 for each CC | Same as 6.2.2.1.1.1 |
| 6.2A.3.1.2 2Rx CQI reporting accuracy under AWGN conditions for CA (3DL CA) | Same as 6.2.2.1.1.1 for each CC | Same as 6.2.2.1.1.1 |
| 6.2A.3.1.3 2Rx CQI reporting accuracy under AWGN conditions for CA (4DL CA) | Same as 6.2.2.1.1.1 for each CC | Same as 6.2.2.1.1.1 |
| 6.3.2.1.1 2Rx FDD FR1 Single PMI with 4Tx Type I- SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.1.2 2Rx FDD FR1 Single PMI with 8Tx Type I – SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.1.3 2Rx FDD FR1 Multiple PMI with 16Tx Type I – SinglePanel Codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.1.4 2Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.1.5 2Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.1.6 2Rx FDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.2.3.2.1.24Rx TDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | Same as 6.2.2.1.1.1 | Same as 6.2.2.1.1.1 |
| 6.3.2.2.1 2Rx TDD FR1 Single PMI with 4Tx TypeI – SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.2.2 2Rx TDD FR1 Single PMI with 8Tx TypeI – SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.2.3 2Rx TDD FR1 Single PMI with 16Tx Type1 – SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.2.4 2Rx TDD FR1 Single PMI with 32Tx Type1 – SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.2.2.6 2Rx TDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.1.1 Single PMI with 4TX TypeI-SinglePanel Codebook– SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.1.2 Single PMI with 8TX TypeI-SinglePanel Codebook– SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.1.3 4Rx FDD FR1 Multiple PMI with 16Tx Type I – SinglePanel Codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.1.4 4Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.1.5 4Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.1.6 4Rx FDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.2.1 4Rx TDD FR1 Single PMI with 4Tx Type1 - SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |

| | | |
|---|---|---|
| 6.3.2.2.5 2Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.2.2 4Rx TDD FR1 Single PMI with 8Tx Type1 - SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.2.3 4Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.2.5 4Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.3.3.2.6 4Rx TDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.4.2.1_1 2Rx FDD FR1 RI reporting for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.4.2.2_1 2Rx TDD FR1 RI reporting for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.4.3.1_1 4Rx FDD FR1 RI reporting for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |
| 6.4.3.2_1 4Rx TDD FR1 RI reporting for both SA and NSA | Same as 6.2.2.1.2.1 | Same as 6.2.2.1.2.1 |

## F.1.2 Interpretation of measurement results (normative)

The measurement results returned by the Test System are compared – without any modification – against the Test Requirements as defined by the shared risk principle.

The Shared Risk principle is defined in ETR 273-1-2 clause 6.5.

The actual measurement uncertainty of the Test System for the measurement of each parameter shall be included in the test report.

The recorded value for the Test System uncertainty shall be, for each measurement, equal to or lower than the appropriate figure in clause F.1 of the present document.

If the Test System for a test is known to have a measurement uncertainty greater than that specified in clause F.1, it is still permitted to use this apparatus provided that an adjustment is made value as follows:

Any additional uncertainty in the Test System over and above that specified in clause F.1 shall be used to tighten the Test Requirement, making the test harder to pass. For some tests, for example receiver tests, this may require modification of stimulus signals. This procedure will ensure that a Test System not compliant with clause F.1does not increase the chance of passing a device under test where that device would otherwise have failed the test if a Test System compliant with clause F.1 had been used.

## F.1.3 Test Tolerance and Derivation of Test Requirements (informative)

The Test Requirements in the present document have been calculated by relaxing the Minimum Requirements of the core specification using the Test Tolerances defined in this clause. When the Test Tolerance is zero, the Test Requirement will be the same as the Minimum Requirement. When the Test Tolerance is non-zero, the Test Requirements will differ from the Minimum Requirements, and the formula used for the relaxation is given in this clause.

The Test Tolerances are derived from Test System uncertainties, regulatory requirements and criticality to system performance. As a result, the Test Tolerances may sometimes be set to zero.

The test tolerances should not be modified for any reason e.g. to take account of commonly known test system errors (such as mismatch, cable loss, etc.).

The downlink Test Tolerances apply at each receiver antenna connector.

### F.1.3.1 Measurement of test environments

The UE test environments are set to the values defined in TS 36.508 subclause 4.1, without any relaxation. The applied Test Tolerance is therefore zero.

### F.1.3.2 Measurement of Demod Performance requirements

The derivation of the test requirements for the test cases in section 5 is defined in Table F.1.3.2-1.

**Table F.1.3.2-1: Derivation of Test Requirements (FR1 demodulation performance tests)**

| Test | Minimum Requirement in TS 38.101-4 | Test Tolerance (TT) | Test Requirement in TS 38.521-4 |
|---|---|---|---|
| 5.2.2.1.1_1 2Rx FDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler<br>0.6 dB for test 1-6<br>0.9 dB for test 1-7 | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.1_2 2Rx FDD FR1 PDSCH Mapping Type A performance - 2x2 MIMO with enhanced receiver type X for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.2_1 2Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.3_1 2Rx FDD FR1 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.4_1 2Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.6_1 2Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | FFS | Formula: SNR + TT |
| 5.2.2.1.5_1 2Rx FDD FR1 PDSCH 0.001% BLER performance - 1x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | [0.6 dB] | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.7 2Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.8_1 2Rx FDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.9_1 2Rx FDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.10_1 2Rx FDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.11_1 2Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.12_1 2Rx FDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.13_1 2Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.1.14_1 2Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |

| | | | |
|---|---|---|---|
| 5.2.2.2.1_1 2Rx TDD FR1 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler<br>0.9 dB for test 1-10<br>0.6 dB for test 1-11 | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.1_2 2Rx TDD FR1 PDSCH Mapping Type A performance - 2x2 MIMO with enhanced receiver type X for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.2_1 2Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.3_1 2Rx TDD FR1 PDSCH mapping Type B performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.4_1 2Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.5_1 2Rx FDD FR1 PDSCH 0.001% BLER performance - 1x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | [0.6 dB] | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.6_1 2Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | FFS | Formula: SNR + TT |
| 5.2.2.2.7_1 2Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.8_1 2Rx TDD FR1 PDSCH pre-emption performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.9_1 2Rx TDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.10_1 2Rx TDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.11_1 2Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.12_1 2Rx TDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.13_1 2Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.2.2.14_1 2Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.1_1 4Rx FDD FR1 PDSCH mapping Type A performance - 2x4 MIMO baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |

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| 5.2.3.1.1_2 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.1_4 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with enhanced receiver type 1 for both SA and NSA | SNRs as specified | 0.9 dB for > 10Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.2_1 4Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 4x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.3_1 4Rx FDD FR1 PDSCH mapping Type B performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.4_1 4Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB for > 10 Hz doppler<br>1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.5_1 4Rx FDD FR1 PDSCH 0.001% BLER performance - 1x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | [0.6 dB] | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.6_1 4Rx FDD FR1 PDSCH repetitions over multiple slots performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | FFS | Formula: SNR + TT |
| 5.2.3.1.7_1 4Rx FDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.8_1 4Rx FDD FR1 PDSCH pre-emption performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.9_1 4Rx FDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.10_1 4Rx FDD FR1 HST-DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.6 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.11_1 4Rx FDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x4 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.12_1 4Rx FDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x4 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.13_1 4Rx FDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x4 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.1.14_1 4Rx FDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x4 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.2_1 4Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.3_1 4Rx TDD FR1 PDSCH mapping Type B performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.4_1 4Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.5_1 4Rx TDD FR1 PDSCH 0.001% BLER performance - 1x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | [0.6 dB] | Formula: SNR + TT<br>T-put limit unchanged |

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| 5.2.3.2.6_1 4Rx TDD FR1 PDSCH repetitions over multiple slots performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | FFS | Formula: SNR + TT |
| 5.2.3.2.7_1 4Rx TDD FR1 PDSCH Mapping Type B and UE processing capability 2 performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.8_1 4Rx TDD FR1 PDSCH pre-emption performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.9_1 4Rx TDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | TBD | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.10_1, 4Rx TDD FR1 HST DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA | SNRs as specified | TBD | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.11_1 4Rx TDD FR1 PDSCH Single-DCI based SDM scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.12_1 4Rx TDD FR1 PDSCH Multiple-DCI based transmission scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.13_1 4Rx TDD FR1 PDSCH Single-DCI based FDM scheme A performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2.3.2.14_1 4Rx TDD FR1 PDSCH Single-DCI based Inter-slot TDM scheme performance - 2x2 MIMO for both SA and NSA | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2A.2.1.1 2Rx Normal PDSCH Demodulation Performance for CA (2DL CA) | SNRs as specified | 1.0 dB for 10Hz doppler | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2A.3.1.1 4Rx Normal PDSCH Demodulation Performance for CA (2DL CA) | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.2A.4.1.1 2Rx-4Rx Normal PDSCH Demodulation Performance for CA (2DL CA) | SNRs as specified | 2Rx CC:<br>1.0 dB for 10Hz doppler<br><br>4Rx CC:<br>0.9dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.2.1.1 2Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.2.1.2 2Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.2.1.3 2Rx FDD FR1 PDCCH 1 Tx antenna performance for power saving | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.2.2.1 2Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.2.2.2 2Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.2.2.3 2Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.3.1.1 4Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.3.1.2 4Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.3.1.3 4Rx FDD FR1 PDCCH 1 Tx antenna performance power saving | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.3.2.1 4Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | SNRs as specified | 0.9 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 5.3.3.2.2 4Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | SNRs as specified | 1.0 dB | Formula: SNR + TT<br>T-put limit unchanged |

### F.1.3.3 Measurement of Channel State Information reporting

The derivation of the test requirements for the test cases in section 6 is defined in Table F.1.3.3-1.

**Table F.1.3.3-1: Derivation of Test Requirements (FR1 channel state information reporting tests)**

| Test | Minimum Requirement in TS 38.101-4 | Test Tolerance (TT) | Test Requirement in TS 38.521-4 |
|---|---|---|---|
| 6.2.2.1.1.12Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.2.1.1.22Rx FDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.2.1.2.12Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 20%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ unchanged<br>$\gamma$ 1.04<br>BLER limit unchanged |
| 6.2.2.1.2.22Rx FDD FR1 periodic subband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 2%<br>$\beta$ 55%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\beta$ 0%<br>$\gamma$ 0.01<br>BLER 0 | SNR unchanged<br>$\alpha$ limit unchanged<br>$\beta$ limit unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 6.2.2.2.1.12Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.2.2.1.22Rx TDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.2.2.2.12Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 20%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 6.2.2.2.2.22Rx TDD FR1 periodic subband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 2%<br>$\beta$ 55%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\beta$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ limit unchanged<br>$\beta$ limit unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 6.2.3.1.1.14Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.3.1.1.24Rx FDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.3.1.2.14Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 5%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ unchanged<br>$\gamma$ 1.04<br>BLER limit unchanged |
| 6.2.3.1.2.24Rx FDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 2%<br>$\beta$ 55%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\beta$ 0%<br>$\gamma$ 0.01<br>BLER 0 | SNR unchanged<br>$\alpha$ limit unchanged<br>$\beta$ limit unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 6.2.3.2.1.14Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.3.2.1.24Rx TDD FR1 periodic CQI reporting with Table 3 under AWGN conditions for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 6.2.3.2.2.14Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 5%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ unchanged<br>$\gamma$1.04<br>BLER limit unchanged |

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| 6.2.3.2.2.24Rx TDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | SNRs as specified<br>$\alpha$ 2%<br>$\beta$ 55%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\beta$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ limit unchanged<br>$\beta$ limit unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 6.2A.3.1.1 2Rx CQI reporting accuracy under AWGN conditions for CA (2DL CA) | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | Test requirement unchanged |
| 6.2A.3.1.2 2Rx CQI reporting accuracy under AWGN conditions for CA (3DL CA) | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | Test requirement unchanged |
| 6.2A.3.1.3 2Rx CQI reporting accuracy under AWGN conditions for CA (4DL CA) | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | Test requirement unchanged |
| 6.3.2.1.1 2Rx FDD FR1 Single PMI with 4Tx Type I- SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.30 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.29 |
| 6.3.2.1.2 2Rx FDD FR1 Single PMI with 8Tx Type I – SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.50 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.49 |
| 6.3.2.1.3 2Rx FDD FR1 Multiple PMI with 16Tx Type I – SinglePanel Codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 2.50 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.49 |
| 6.3.2.1.4 2Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 5.0 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$4.99 |
| 6.3.2.1.5 2Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.9 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.89 |
| 6.3.2.1.6 2Rx FDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 2.2 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.19 |
| 6.3.2.2.1 2Rx TDD FR1 Single PMI with 4Tx TypeI - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.30 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.29 |
| 6.3.2.2.2 2Rx TDD FR1 Single PMI with 8Tx TypeI - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.50 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.49 |
| 6.3.2.2.3 2Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ *2*.50 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.49 |
| 6.3.2.2.4 2Rx TDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ *5*.0 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$4.99 |
| 6.3.2.2.5 2Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.9 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.89 |
| 6.3.2.2.6 2Rx TDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 2.2 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.19 |
| 6.3.3.1.1 Single PMI with 4TX TypeI-SinglePanel Codebook– SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.30 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.29 |

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| 6.3.3.1.2 Single PMI with 8TX TypeI-SinglePanel Codebook– SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.50 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.49 |
| 6.3.3.1.3 4Rx FDD FR1 Multiple PMI with 16Tx Type I – SinglePanel Codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 3.00 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.99 |
| 6.3.3.1.4 4Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 7.0 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$6.99 |
| 6.3.3.1.5 4Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ *1.9* | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.89 |
| 6.3.3.1.6 4Rx FDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 2.2 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.19 |
| 6.3.3.2.1 4Rx TDD FR1 Single PMI with 4Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.30 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.29 |
| 6.3.3.2.2 4Rx TDD FR1 Single PMI with 8Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.50 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.49 |
| 6.3.3.2.3 4Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 3.0 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.99 |
| 6.3.3.2.5 4Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.8 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$1.79 |
| 6.3.3.2.6 4Rx TDD FR1 Multiple PMI with 16Tx Enhanced TypeII codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 2.2 | SNR 0 dB<br>$\gamma$ *0.01* | SNR unchanged<br>$\gamma$2.19 |
| 6.4.2.1_1 2Rx FDD FR1 RI reporting for both SA and NSA | SNRs as specified<br>$\gamma_2$ 1.00 for Test 1<br>$\gamma_1$ 1.05 for Test 2<br>$\gamma_1$ 0.90 for Test 3 | SNR 0 dB<br>$\gamma_2$ *0.01 for Test 1*<br>$\gamma_1$ *0.01 for Test 2*<br>$\gamma_1$ *0.01 for Test 3* | SNR unchanged<br>$\gamma_2$ 0.99 for Test 1<br>$\gamma_1$ 1.04 for Test 2<br>$\gamma_1$ 0.89 for Test 3 |
| 6.4.2.2_1 2Rx TDD FR1 RI reporting for both SA and NSA | SNRs as specified<br>$\gamma_2$ 1.00 for Test 1<br>$\gamma_1$ 1.05 for Test 2<br>$\gamma_1$ 0.90 for Test 3 | SNR 0 dB<br>$\gamma_2$ *0.01 for Test 1*<br>$\gamma_1$ *0.01 for Test 2*<br>$\gamma_1$ *0.01 for Test 3* | SNR unchanged<br>$\gamma_2$ 0.99 for Test 1<br>$\gamma_1$ 1.04 for Test 2<br>$\gamma_1$ 0.89 for Test 3 |
| 6.4.3.1_1 4Rx FDD FR1 RI reporting for both SA and NSA | SNRs as specified<br>$\gamma_2$ 0.90 for Test 1<br>$\gamma_1$ 1.05 for Test 2<br>$\gamma_1$ 0.90 for Test 3<br>$\gamma_2$ 0.90 for Test 4 | SNR 0 dB<br>$\gamma_2$ *0.01 for Test 1*<br>$\gamma_1$ *0.01 for Test 2*<br>$\gamma_1$ *0.01 for Test 3*<br>$\gamma_2$ *0.01 for Test 4* | SNR unchanged<br>$\gamma_2$ 0.89 for Test 1<br>$\gamma_1$ 1.04 for Test 2<br>$\gamma_1$ 0.89 for Test 3<br>$\gamma_2$ 0.89 for Test 4 |
| 6.4.3.2_1 4Rx TDD FR1 RI reporting for both SA and NSA | SNRs as specified<br>$\gamma_2$ 0.90 for Test 1<br>$\gamma_1$ 1.05 for Test 2<br>$\gamma_1$ 0.90 for Test 3<br>$\gamma_2$ 0.90 for Test 4 | SNR 0 dB<br>$\gamma_2$ *0.01 for Test 1*<br>$\gamma_1$ *0.01 for Test 2*<br>$\gamma_1$ *0.01 for Test 3*<br>$\gamma_2$ *0.01 for Test 4* | SNR unchanged<br>$\gamma_2$ 0.89 for Test 1 as per Table G.3.4<br>$\gamma_1$ 1.04 for Test 2 as per Table G.3.4<br>$\gamma_1$ 0.89 for Test 3 as per Table G.3.4<br>$\gamma_2$ 0.89 for Test 4 as per Table G.3.4 |

# F.2 Measurement uncertainties and test tolerances for FR2

## F.2.1 Acceptable uncertainty of test system (normative)

The maximum acceptable uncertainty of the Test System is specified below for each test, where appropriate. The Test System shall enable the stimulus signals in the test case to be adjusted to within the specified range, and the equipment under test to be measured with an uncertainty not exceeding the specified values. Care should be taken to ensure that each conformance test implementation including the OTA chamber aspects meets the specified measurement uncertainty for each test case by requiring the test laboratory to maintain a detailed measurement uncertainty test report showing compliance to all the measurement uncertainty requirements. The detailed measurement uncertainty report would contain the justification for each measurement uncertainty component and its value and distribution. The derivation of these values is based on the minimum conformance requirements plus relaxation, i.e., test tolerance is not to be considered. All ranges and uncertainties are absolute values, and are valid for a confidence level of 95 %, unless otherwise stated.

A confidence level of 95 % is the measurement uncertainty tolerance interval for a specific measurement that contains 95 % of the performance of a population of test equipment.

The downlink signal uncertainties apply at the defined quiet zone with the UE properly positioned in the quiet zone. The uplink signal uncertainties apply at the measurement equipment with the UE positioned properly in the quiet zone.

### F.2.1.1 Measurement of test environments

TBD

### F.2.1.2 Measurement of Demod Performance requirements

This clause defines the maximum test system uncertainty for Demod Performance requirements. The maximum test system uncertainty allowed for the measurement uncertainty contributors are defined in Table F.2.1.2-1.

**Table F.2.1.2-1: Maximum measurement uncertainty values for the test system for FR2 (up to 40 GHz) and Channel BW ≤ 400 MHz**

| MU contributor | Unit | Value | Comment |
|---|---|---|---|
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{config}$ | dB | ±3.6 | |
| gNB emulator Signal to noise ratio uncertainty | dB | ±0.3 | |
| Impact on non-ideal isolation between branches for the wireless cable mode | dB | 0.60 for Rank1<br>0.45 for Rank2 | Systematic uncertainty |
| Fading profile power uncertainty | dB | ±0.5 for 1Tx<br>±0.7 for 2Tx | |
| SNR uncertainty due to finite test time | dB | ±0.3 for PDSCH and doppler < 100Hz<br>0.0 for PDSCH and doppler ≥<br>±0.4 for PDCCH | |

The maximum test system uncertainty for test cases defined in section 7 is defined in Table F.2.1.2-2.

**Table F.2.1.2-2: Maximum test system uncertainty for FR2 demodulation performance test cases**

| Subclause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 7.2.2.2.1_1 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for SA and NSA | 2Tx, Rank 1:<br>± 1.82 dB for Doppler < 100 Hz<br>± 1.78 dB for Doppler ≥100 Hz<br><br>2Tx, Rank 2:<br>± 1.67 dB for Doppler < 100Hz<br>± 1.63 dB for Doppler ≥100 Hz | Overall system uncertainty for fading conditions comprises four quantities:<br>1. gNB emulator Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br>5. Impact on non-ideal isolation between branches for the wireless cable mode<br>gNB emulator SNR<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (gNB emulator Signal-to-noise ratio uncertainty $^2$ + Fading profile power uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ + SNR uncertainty due to finite test time$^2$ ) + Impact on non-ideal isolation between branches for the wireless cable mode<br><br>gNB emulator Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB<br>AWGN flatness and signal flatness ±3.6 dB<br>SNR uncertainty due to finite test time ±0.3 dB for doppler < 100Hz, otherwise 0 dB<br>Impact on non-ideal isolation between branches for the wireless cable mode 0.60 dB for Rank1, 0.45 dB for Rank2 |
| 7.2.2.2.1_2 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with enhanced type 1 receiver for SA and NSA | 2Tx, Rank 2:<br>± 1.67 dB for Doppler < 100Hz<br>± 1.63 dB for Doppler ≥ 100Hz | Same as 7.2.2.2.1_1 |
| 7.2.2.2.1_3 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with 256QAM for SA and NSA (Rel-16 and forward) | 2Tx, Rank 1:<br>± 1.82 dB for Doppler < 100 Hz | Same as 7.2.2.2.1_1 |

| | | |
|---|---|---|
| 7.3.2.2.1 2Rx TDD FR2 PDCCH 1 Tx antenna performance for both SA and NSA | 1Tx, rank1:<br>± 1.74 dB | Overall system uncertainty for fading conditions comprises four quantities:<br>1. gNB emulator Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br>5. Impact on non-ideal isolation between branches for the wireless cable mode<br>gNB emulator SNR<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (gNB emulator Signal-to-noise ratio uncertainty 2 + Fading profile power uncertainty 2 + (0.25 x AWGN flatness and signal flatness) 2 + SNR uncertainty due to finite test time2 ) + Impact on non-ideal isolation between branches for the wireless cable mode<br><br>gNB emulator Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.5 dB for 1Tx, ±0.7 dB for 2Tx<br>AWGN flatness and signal flatness ±3.6 dB<br>SNR uncertainty due to finite test time ±0.4 dB<br>Impact on non-ideal isolation between branches for the wireless cable mode 0.6 for Rank1 and 0.45 for rank2 |
| 7.3.2.2.2 2Rx TDD FR2 PDCCH 2 Tx antenna performance for both SA and NSA | 2Tx, rank1:<br>± 1.84 dB | Same as 7.3.2.2.1 |
| 7.3.2.2.3 2Rx TDD FR2 PDCCH 1 Tx antenna performance for power saving | 1Tx, rank1:<br>± 1.74 dB | Same as 7.3.2.2.1 |
| 9.4B.1.2 Sustained downlink data rate performance for EN-DC including FR2 NR carrier | Downlink absolute power uncertainty (including beam peak search , averaged over $BW_{Config}$ ± 5.19 dB | Downlink absolute power uncertainty (including beam peak search error) is one of the factors used to determine the max testable SNR for a given Test System as listed in sheet “Mode2 100MHz” in 38.521-4 Spreadsheet - Demod SNR range calculator V3.xlsx of TR 38.903 |

## F.2.1.3 Measurement of Channel State Information reporting

This clause defines the maximum test system uncertainty for channel state information reporting requirements. The maximum test system uncertainty allowed for the measurement uncertainty contributors are defined in Table F.2.1.3-1.

**Table F.2.1.3-1: Maximum measurement uncertainty values for the test system for FR2 (up to 40 GHz) and Channel BW ≤ 400 MHz**

| MU contributor | Unit | Value | Comment |
|---|---|---|---|
| AWGN flatness and signal flatness, max deviation for any Resource Block, relative to average over $BW_{config}$ | dB | Same as in table F.2.1.2-1 | |
| Signal to noise ratio uncertainty | dB | Same as in table F.2.1.2-1 | |
| Impact on non-ideal isolation between branches for the wireless cable mode | dB | Same as in table F.2.1.2-1 | |

| MU contributor | Unit | Value | Comment |
|---|---|---|---|
| Fading profile power uncertainty | dB | Same as in table F.2.1.2-1 | |

The maximum test system uncertainty for test cases defined in section 8 is defined in Table F.2.1.3-2.

**Table F.2.1.3-2: Maximum test system uncertainty for FR2 channel state information reporting test cases**

| Subclause | Maximum Test System Uncertainty | Derivation of Test System Uncertainty |
|---|---|---|
| 8.2.2.2.1.12 Rx TDD FR2 periodic wideband CQI reporting under AWGN performance for both SA and NSA | ± 1.40 dB | Overall system uncertainty under AWGN conditions comprises three quantities:<br>1. gNB emulator Signal-to-noise ratio uncertainty<br>2. Effect of AWGN flatness and signal flatness<br>3. Impact on non-ideal isolation between branches for the wireless cable mode<br>gNB emulator SNR<br><br>Items 1 and 2 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x [0.25] effect on the required SNR, so use sensitivity factor of x [0.25] for the uncertainty contribution.<br>Test System uncertainty = SQRT (gNB emulator Signal-to-noise ratio uncertainty $^2$ + (0.25 x AWGN flatness and signal flatness) $^2$ ) + Impact on non-ideal isolation between branches for the wireless cable mode<br><br>gNB emulator Signal-to-noise ratio uncertainty ±0.3 dB<br>AWGN flatness and signal flatness ±3.6 dB<br>Impact on non-ideal isolation between branches for the wireless cable mode 0.45 dB for Rank2 and 0.6 for Rank1 |

| | | |
|---|---|---|
| 8.2.2.2.2.12 Rx TDD FR2 aperiodic wideband CQI reporting under fading performance for both SA and NSA | ± 1.82 dB for Doppler < 100Hz | Overall system uncertainty for fading conditions comprises five quantities:<br>1. gNB emulator Signal-to-noise ratio uncertainty<br>2. Fading profile power uncertainty<br>3. Effect of AWGN flatness and signal flatness<br>4. SNR uncertainty due to finite test time<br>5. Impact on non-ideal isolation between branches for the wireless cable mode<br>gNB emulator SNR<br><br>Items 1, 2, 3 and 4 are assumed to be uncorrelated so can be root sum squared: AWGN flatness and signal flatness has x 0.25 effect on the required SNR, so use sensitivity factor of x 0.25 for the uncertainty contribution.<br>Test System uncertainty = SQRT (gNB emulator Signal-to-noise ratio uncertainty 2 + Fading profile power uncertainty 2 + (0.25 x AWGN flatness and signal flatness) 2 + SNR uncertainty due to finite test time2 ) + Impact on non-ideal isolation between branches for the wireless cable mode<br><br>gNB emulator Signal-to-noise ratio uncertainty ±0.3 dB<br>Fading profile power uncertainty ±0.7 dB for 2Tx<br>AWGN flatness and signal flatness ±3.6 dB<br>SNR uncertainty due to finite test time ±0.3 dB<br>Impact on non-ideal isolation between branches for the wireless cable mode 0.6 for Rank1 and 0.45 for Rank2 |
| 8.2.2.2.2.1_1 2Rx TDD FR2 aperiodic wideband CQI reporting under fading performance for both SA and NSA – 256QAM (Rel-16 and forward) | Same as 8.2.2.2.2.1 | Same as 8.2.2.2.2.1 |
| 8.2A.3.1.1 2Rx CQI reporting accuracy under AWGN conditions for CA (2DL CA) | Same as 8.2.2.2.1.1 on each CC. | Same as 8.2.2.2.1.1 |
| 8.2A.3.1.2 2Rx CQI reporting accuracy under AWGN conditions for CA (3DL CA) | Same as 8.2.2.2.1.1 on each CC. | Same as 8.2.2.2.1.1 |
| 8.2A.3.1.3 2Rx CQI reporting accuracy under AWGN conditions for CA (4DL CA) | Same as 8.2.2.2.1.1 on each CC. | Same as 8.2.2.2.1.1 |
| 8.3.2.2.1 2Rx TDD FR2 Single PMI with 2TX TypeI-SinglePanel codebook for both SA and NSA | Same as 8.2.2.2.2.1 | Same as 8.2.2.2.2.1 |
| 8.4.2.2.1 2Rx TDD FR2 RI reporting for both SA and NSA | Same as 8.2.2.2.2.1 | Same as 8.2.2.2.2.1 |

## F.2.2 Interpretation of measurement results (normative)

The actual measurement uncertainty of the Test System for the measurement of each parameter shall be included in the test report.

The recorded value for the Test System uncertainty shall be, for each measurement, equal to or lower than the appropriate figure in clause F.1 of the present document.

If the Test System using one of the permitted test methods defined in TR38.903 [20] for a test is known to have a measurement uncertainty greater than that specified in clause F.1, it is still permitted to use this apparatus provided that an adjustment is made value as follows:

Any additional uncertainty in the Test System over and above that specified in clause F.1 shall be used to tighten the Test Requirement, making the test harder to pass. For some tests, for example receiver tests, this may require modification of stimulus signals. This procedure will ensure that a Test System not compliant with clause F.1does not increase the chance of passing a device under test where that device would otherwise have failed the test if a Test System compliant with clause F.1 had been used.

## F.2.3 Test Tolerance and Derivation of Test Requirements (informative)

TBD

### F.2.3.1 Measurement of test environments

TBD

### F.2.3.2 Measurement of Demod Performance requirements

The derivation of the test requirements for the test cases in section 7 is defined in Table F.2.3.2-1.

**Table F.2.3.2-1: Derivation of Test Requirements (FR2 demodulation performance tests)**

| Test | Minimum Requirement in TS 38.101-4 | Test Tolerance (TT) | Test Requirement in TS 38.521-4 |
|---|---|---|---|
| 7.2.2.2.1_1 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with baseline receiver for SA and NSA | SNRs as specified | 2Tx, Rank 1: 1.8 dB<br><br>2Tx, Rank 2: 1.7 dB for doppler < 100Hz<br>1.6 dB otherwise | Formula: SNR + TT<br>T-put limit unchanged |
| 7.2.2.2.1_2 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with enhanced type 1 receiver for SA and NSA | SNRs as specified | 2Tx, Rank 2: 1.7 dB for doppler < 100Hz<br>1.6 dB otherwise | Formula: SNR + TT<br>T-put limit unchanged |
| 7.2.2.2.1_3 2Rx TDD FR2 PDSCH mapping Type A performance - 2x2 MIMO with 256QAM for SA and NSA (Rel-16 and forward) | SNRs as specified | 2Tx, Rank 1: 1.8 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 7.3.2.2.1 2Rx TDD FR2 PDCCH 1 Tx antenna performance for both SA and NSA | SNRs as specified | 1Tx, rank1: 1.7 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 7.3.2.2.2 2Rx TDD FR2 PDCCH 2 Tx antenna performance for both SA and NSA | SNRs as specified | 2Tx, rank1: 1.8 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 7.3.2.2.3 2Rx TDD FR2 PDCCH 1 Tx antenna performance for power saving | SNRs as specified | 1Tx, rank1: 1.7 dB | Formula: SNR + TT<br>T-put limit unchanged |
| 9.4B.1.2 Sustained downlink data rate performance for EN-DC including FR2 NR carrier | Indirect far field (IFF) with 30cm QZ, PC3, 100MHz CHBW Downlink power<br>n257, n261: -79.5 dBm/120 kHz<br>n258: -79.2 dBm/120 kHz | No test tolerance applied | T-put limit unchanged |

## F.2.3.3 Measurement of Channel State Information reporting

The derivation of the test requirements for the test cases in section 8 is defined in Table F.2.3.3-1.

**Table F.2.3.3-1: Derivation of Test Requirements (FR2 channel state information reporting tests)**

| Test | Minimum Requirement in TS 38.101-4 | Test Tolerance (TT) | Test Requirement in TS 38.521-4 |
|---|---|---|---|
| 8.2.2.2.1.12 Rx TDD FR2 periodic wideband CQI reporting under AWGN performance for both SA and NSA | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | SNR unchanged |
| 8.2.2.2.2.12 Rx TDD FR2 aperiodic wideband CQI reporting under fading performance for both SA and NSA | SNRs as specified<br>$\alpha$ 2%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 8.2.2.2.2.1_1 2Rx TDD FR2 aperiodic wideband CQI reporting under fading performance for both SA and NSA | SNRs as specified<br>$\alpha$ 2%<br>$\gamma$ 1.05<br>BLER 0.02 | SNR 0 dB<br>$\alpha$ 0%<br>$\gamma$ *0.01*<br>BLER 0 | SNR unchanged<br>$\alpha$ unchanged<br>$\gamma$1.04<br>BLER limit unchanged |
| 8.2A.3.1.1 2Rx CQI reporting accuracy under AWGN conditions for CA (2DL CA) | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | Test requirement unchanged |
| 8.2A.3.1.2 2Rx CQI reporting accuracy under AWGN conditions for CA (3DL CA) | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | Test requirement unchanged |
| 8.2A.3.1.3 2Rx CQI reporting accuracy under AWGN conditions for CA (4DL CA) | SNRs as specified<br>Limits as in the Test Procedure | No test tolerances applied | Test requirement unchanged |
| 8.3.2.2.1 2Rx TDD FR2 Single PMI with 2TX TypeI-SinglePanel codebook for both SA and NSA | SNRs as specified<br>$\gamma$ 1.05 for Test 1<br>$\gamma$ 1.05 for Test 2 | SNR 0 dB<br>$\gamma$ *0.01 for Test 1*<br>$\gamma$ *0.01 for Test 2* | SNR unchanged<br>$\gamma$1.04 for Test 1<br>$\gamma$1.04 for Test 2 |
| 8.4.2.2.1 2Rx TDD FR2 RI reporting for both SA and NSA | SNRs as specified<br>$\gamma_2$ 1.00 for Test 1<br>$\gamma_1$ 1.05 for Test 2<br>$\gamma_1$ 1.05 for Test 3 | SNR 0 dB<br>$\gamma_2$ *0.01 for Test 1*<br>$\gamma_1$ *0.01 for Test 2*<br>$\gamma_1$ *0.01 for Test 3* | SNR unchanged<br>$\gamma_2$ 0.99 for Test 1<br>$\gamma_1$ 1.04 for Test 2<br>$\gamma_1$ 1.04 for Test 3 |

# Annex G (normative): Statistical Testing

## G.1 Statistical testing of Performance Requirements with throughput

### G.1.1 General

The test of receiver performance characteristics is twofold.

1. A signal or a combination of signals is offered to the RX port(s) of the receiver.
2. The ability of the receiver to demodulate /decode this signal is verified by measuring the throughput.

In (2) is the statistical aspect of the test and is treated here.

The minimum requirement for most receiver performance tests is either 70 % or 30 % of the maximum throughput.

All receiver performance tests are performed in fading conditions. In addition to the statistical considerations, this requires the definition of a minimum test time.

### G.1.2 Mapping throughput to error ratio

a) The measured information bit throughput R is defined as the sum (in kilobits) of the information bit payloads successfully received during the test interval, divided by the duration of the test interval (in seconds).

b) In measurement practice the UE indicates successfully received information bit payload by signalling an ACK to the SS.
If payload is received, but damaged and cannot be decoded, the UE signals a NACK.

c) Only the ACK and NACK signals, not the data bits received, are accessible to the SS.
The number of bits is known in the SS from knowledge of what payload was sent.

d) For the reference measurement channel, applied for testing, the number of bits is different in different slots, however in a radio frame it is fixed during one test.

e) The time in the measurement interval is composed of successfully received slots (ACK), unsuccessfully received slots (NACK) and no reception at all (DTX-slots).

f) DTX-slots may occur regularly according the applicable reference measurement channel (regDTX).
In real live networks this is the time when other UEs are served. In TDD these are the UL and special slots.
regDTX vary from test to test but are fixed within the test.

g) Additional DTX-slots occur statistically when the UE is not responding ACK or NACK where it should. (statDTX)
This may happen when the UE was not expecting data or decided that the data were not intended for it.

The pass / fail decision is done by observing the:

- number of NACKs
- number of ACKs and
- number of statDTXs (regDTX is implicitly known to the SS)

The ratio `(NACK + statDTX)/(NACK+ statDTX + ACK)`is the Error Ratio (ER). Taking into account the time consumed by the ACK, NACK, and DTX-TTIs (regular and statistical), ER can be mapped unambiguously to throughput for any single reference measurement channel test.

## G.1.3 Design of the test

The test is defined by the following design principles (see clause G.2, Theory):

1. The standard concept is applied. (not the early decision concept)
2. A second limit is introduced: The second limit is different, whether 30 % or 70 % throughput is tested.
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

   To decide the test fail:

   Customer Risk is applied based on the specified DUT quality

The test is defined by the following parameters:

1a) Limit Error Ratio = 0.3 (in case 70 % Throughput is tested) or

1b) Limit Throughput = 0.3 (in case 30 % Throughput is tested) or

1c) Limit Error Ratio = 0.01 (in case 1% BLER is tested)

2a) Bad DUT factor M=1.378 (selectivity)

2b) Bad DUT factor m=0.692 (selectivity)

2c) Bad DUT facor M=1.5 (selectivity)

justification see: TS 34.121 Clause F.6.3.3

3) Confidence level CL = 95 % (for specified DUT and Bad DUT-quality)

## G.1.4 Pass Fail limit

Testing Throughput = 30 %, then the test limit is

Number of successes (ACK) / number of samples ≥ 59 / 233

Testing Throughput = 70 % then the test limit is

Number of fails (NACK and statDTX) / number of samples ≤ 66 / 184

Testing BLER = 1% then the test limit is

Number of fails (NACK and statDTX) / number of samples ≤ 163 / 13135

There are 3 distinct cases:

a) The duration for the number of samples (233, 184 or 13135) is greater than the minimum test time:

Then the number of samples (233, 184 or 13135) is predefined and the decision is done according to the number of events (59 successes, 66 fails or 163 fails)

b) Since subframe 0 and 5 contain less bits than the remaining subframes, it is allowed to predefine a number of samples contained in an integer number of frames. In this case test-limit-ratio applies.

c) The minimum test time is greater than the duration for the number of samples:

The minimum test time is predefined and the decision is done comparing the measured ratio at that instant against the test-limit-ratio.

NOTE: The test time for most of the tests is governed by the Minimum Test Time.

## G.1.5 Minimum Test time

Editor's Note: Simulation method to derive minimum test time for FR2 needs to be evaluated.

If a pass fail decision in clause G.1.4 can be achieved earlier than the minimum test time, then the test shall not be decided, but continued until the minimum test time is elapsed.

The tables below contain the minimum number of slots for FDD and TDD.

By simulations the minimum number of active subframes (carrying DL payload) was derived (MNAS), then adding inactive subframes to the active ones. (for TDD additional subframes contain no DL payload), then rounding up to full thousand.

Simulation method to derive minimum test time:

With a level, corresponding a throughput at the test limit (here 30 % or 70 % of the max. throughput) the preliminary throughput versus time converges towards the final throughput. The allowance of ± 0.2 dB around the above mentioned level is predefined by RAN5 to find the minimum test time. The allowance of ±0.2 dB maps through the function "final throughput versus level" into a throughput corridor. The minimum test time is achieved when the preliminary throughput escapes the corridor the last time. The two functions "final throughput versus level" and "preliminary throughput versus time" are simulation results, which are done individual for each demodulation scenario.

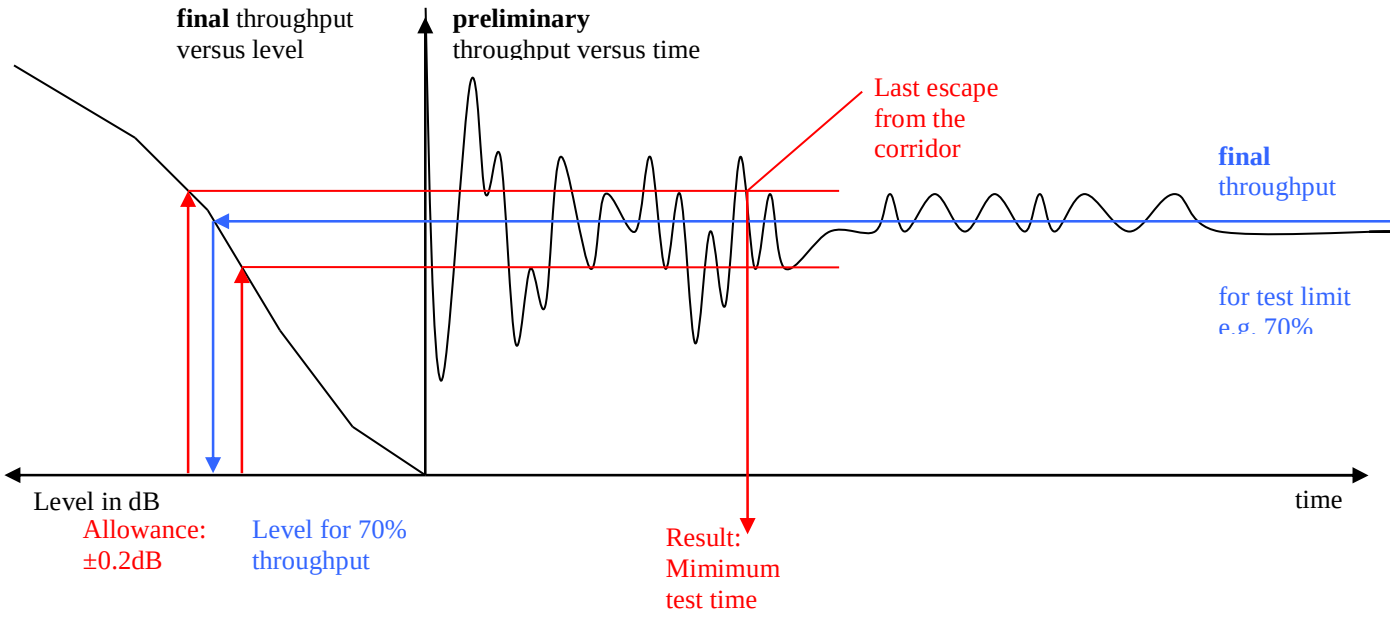



**Figure G.1.5-1: Simulation method to derive minimum test time**

**Table G.1.5-1: Minimum Test time for PDSCH demodulation**

| )D UL-DL pattern | Reference Channel | Propagation condition | Demodulation scenario (doppler speed) | Minimum number of active subframes (MNAS) | MNAS to MNS Scaling factor (Note 3) | Minimum Number of Subframes (MNS) after rounding u to nearest thousand $MNS=1000*\lceil \frac{MNS}{1000} \rceil$ |
|---|---|---|---|---|---|---|
| NA | R.PDSCH.1-8.1 FDD | HST-750 | 750 Hz | 6000 (Note 1) | 1.0526 | 6400 |
| NA | R.PDSCH.1-8.2 FDD | HST-972 | 972 Hz | 6000 (Note 1) | 1.0526 | 6400 |
| NA | R.PDSCH.1-8.1 FDD | TDLC300-600 | 600 Hz | 8000 (Note 1) | 1.0526 | 9000 |
| NA | R.PDSCH.1-1.1 FDD | TDLB100-400 | 400 Hz | 10000 (Note 1) | 1.0526 | 11000 |
| NA | R.PDSCH.1-1.2 FDD, R.PDSCH.1-2.1 FDD, R.PDSCH.1-5.1 FDD | TDLC300-100 | 100 Hz | 20000 (Note 1) | 1.0526 | 22000 |
| :R1.15-1 | R.PDSCH.1-1.1 TDD, R.PDSCH.1-1.2 TDD | TBD | 10 Hz | 75000 (Note 2) | 2.8571 | 215000 |
| NA | R.PDSCH.1-1.3 FDD, R.PDSCH.1-2.2 FDD, R.PDSCH.1-2.3 FDD, R.PDSCH.1-2.4 FDD, R.PDSCH.1-2.5 FDD, R.PDSCH.1-3.1 FDD, R.PDSCH.1-3.2 FDD, R.PDSCH.1-3.3 FDD, R.PDSCH.1-3.4 FDD, R.PDSCH.1-4.1 FDD, R.PDSCH.1-12.1 FDD, R.PDSCH.2-1.1 FDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.0526 | 79000 |
| NA | R.PDSCH.1-7.1 FDD, R.PDSCH.1-7.2 FDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.25 | 94000 |
| NA | R.PDSCH.1-8.3 FDD | HST-SFN | 870 Hz | 30000 (Note 1) | 1.0526 | 32000 |
| NA | R.PDSCH.1-8.4 FDD, R.PDSCH.1-8.4 FDD | HST-DPS | 870 Hz | 30000 (Note 1) | 1.0526 | 32000 |
| :R1.15-1 | R.PDSCH.1-1.1 TDD, R.PDSCH.1-1.2 TDD | TDLA30-10 | 10Hz | 75000 (Note 1) | 2.8571 | 215000 |
| R1.30-1A | R.PDSCH.2-1.1 TDD | TDLB100-400 | 400 Hz | 10000 (Note 1) | 1.2903 | 13000 |
| :R1.30-1 | R.PDSCH.2-1.2 TDD, R.PDSCH.2-2.1 TDD, R.PDSCH.2-7.1 TDD | TDLC300-100 | 100 Hz | 20000 (Note 1) | 1.2903 | 26000 |
| :R1.30-1 | R.PDSCH.2-2.2 TDD, R.PDSCH.2-2.3 TDD, R.PDSCH.2-2.4 TDD, R.PDSCH.2-2.5 TDD, R.PDSCH.2-3.1 TDD, R.PDSCH.2-3.2 TDD, R.PDSCH.2-3.3 TDD, R.PDSCH.2-3.4 TDD, R.PDSCH.2-4.1 TDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.2903 | 97000 |
| :R1.30-1 | R.PDSCH.2-1.3 TDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.4815 | 112000 |
| :R1.30-2 | R.PDSCH.2-5.1 TDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.2903 | 97000 |
| :R1.30-2 | R.PDSCH.2-17.1 TDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 5 | 375000 |
| :R1.30-3 | R.PDSCH.2-6.1 TDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.4815 | 112000 |
| :R1.30-4 | R.PDSCH.2-9.1 TDD | TDLA30-10 | 10 Hz | 75000 (Note 1) | 1.2903 | 97000 |
| :R1.30-5 | R.PDSCH.2-11.1 TDD | TDLB100-400 | 400Hz | 10000 (Note 1) | 1.2903 | 13000 |
| :R1.30-6 | R.PDSCH.2-12.1 TDD | TDLB100-400 | 400Hz | 10000 (Note 1) | 1.2903 | 13000 |
| :R1.30-1 | R.PDSCH.2-10.1 TDD | HST-1000 | 1000 Hz | 15000 (Note 1) | 1.4815 | 23000 |
| :R1.30-1 | R.PDSCH.2-10.1 TDD | HST-1667 | 1667 Hz | 15000 (Note 1) | 1.4815 | 23000 |
| :R1.30-1 | R.PDSCH.2-10.4 TDD | HST-SFN | 1667 Hz | 30000 (Note 1) | 1.4815 | 45000 |
| :R1.30-1 | R.PDSCH.2-10.5 TDD | HST-DPS | 1667 Hz | 30000 (Note 1) | 1.4815 | 45000 |
| :R2.60-1 | R.PDSCH.4-1.1 TDD | TDLA30-75 | 75 Hz | 20000 (Note 2) | 1.33 | 27000 |
| R2.120-1A | R.PDSCH.5-1.1 TDD | TDLC60-300 | 300 Hz | 10000 (Note 2) | 1.25 | 13000 |
| R2.120-1 | R.PDSCH.5-2.1 TDD, R.PDSCH.5-2.2 TDD, R.PDSCH.5-2.3 TDD, R.PDSCH.5-3.1 TDD | TDLA30-300 | 300 Hz | 10000 (Note 2) | 1.25 | 13000 |
| R2.120-1 | R.PDSCH.5-1.2 TDD | TDLA30-75 | 75 Hz | 20000 (Note 2) | 1.25 | 25000 |

| | | | | | | |
|---|---|---|---|---|---|---|
| R2.120-2 | R.PDSCH.5-4.1 TDD, R.PDSCH.5-5.1 TDD, R.PDSCH.5-5.2 TDD, R.PDSCH.5-6.1 TDD | TDLA30-75 | 75 Hz | 20000 (Note 2) | 1.33 | 27000 |

te 1: MNAS determined by simulations.
te 2: For cases where MNS is not determined by simulations, use same MNAS as the similar case simulated (same doppler speed)
te 3: MNS/MNAS ratio decided by scheduling pattern and is ratio of all slots to DL slots.

**Table G.1.5-1a: Minimum Test time for PDSCH demodulation with 1% BLER**

| TDD UL-DL pattern | Reference Channel | Propagation condition | Demodulation scenario (doppler speed) | Minimum number of active subframes (MNAS) | MNAS to MNS Scaling factor (Note 3) | Minimum Number of Subframes (MNS) after rounding up to nearest thousand MNS= $1000*\lceil \frac{MNS}{1000} \rceil$ |
|---|---|---|---|---|---|---|
| NA | R.PDSCH.1-11.1 FDD R.PDSCH.1-11.2 FDD | TDLA30-10 | 10 Hz | FFS (Note 1) | 1.1111 | FFS |
| FR1.30-1 | R.PDSCH.2-16.1 TDD R.PDSCH.2-16.2 TDD | TDLA30-10 | 10 Hz | FFS (Note 1) | 1.6667 | FFS |

Note 1: MNAS determined by simulations.
Note 2: For cases where MNS is not determined by simulations, use same MNAS as the similar case simulated (same doppler speed).
Note 3: MNS/MNAS ratio decided by scheduling pattern (how much time is required to collect required number of active DL SFs).

**Table G.1.5-2: Minimum Test time for PDCCH demodulation**

| Reference Channel | Demodulation scenario (doppler speed) | Minimum number of active subframes (MNAS) | MNAS to MNS Scaling factor (Note 3) | Minimum Number of Subframes (MNS) after rounding up to nearest thousand MNS= $1000*\lceil \frac{MNS}{1000} \rceil$ |
|---|---|---|---|---|
| R.PDCCH 1-1.1 FDD, R.PDCCH.1-1.3 FDD, R.PDCCH.1-2.1 FDD, R.PDCCH.1-2.2 FDD, R.PDCCH.1-2.3 FDD, R.PDCCH.1-2.4 FDD, R.PDCCH.1-2.5 FDD, R.PDCCH.1-2.6 FDD | 10, 100, 400 Hz | 100000 (Note 1) | 1.0526 | 106000 |
| R.PDCCH.2-1.1 TDD, R.PDCCH.2-1.2 TDD, R.PDCCH.2-2.1 TDD, R.PDCCH.2-1.3 TDD | 10, 100, 400 Hz | 100000 (Note 1) | 1.2903 | 130000 |
| R.PDCCH.5-1.1 TDD, R.PDCCH.5-1.2 TDD, R.PDCCH.5-1.3 TDD, R.PDCCH.5-2.1 TDD | 75, 300 Hz | 100000 (Note 2) | 1.25 | 130000 |

Note 1: MNAS determined by simulations.
Note 2: For cases where MNS is not determined by simulations, use same MNAS as the similar case simulated (same doppler speed)
Note 3: MNS/MNAS ratio decided by scheduling pattern and is ratio of all slots to DL slots.

# G.2 Theory to derive the numbers for statistical testing (informative)

Editor's note: This clause of the Annex G is for information only and it described the background theory and information for statistical testing.

## G.2.1 Error Ratio (ER)

The Error Ratio (ER) is defined as the ratio of number of errors (ne) to all results, number of samples (ns).

(1-ER is the success ratio).

## G.2.2 Test Design

A statistical test is characterized by:

Test-time, Selectivity and Confidence level.

## G.2.3 Confidence level

The outcome of a statistical test is a decision. This decision may be correct or in-correct. The Confidence Level CL describes the probability that the decision is a correct one. The complement is the wrong decision probability (risk) D = 1-CL.

## G.2.4 Introduction: Supplier Risk versus Customer Risk

There are two targets of decision:

(a) A measurement on the pass-limit shows, that the DUT has the specified quality or is better with probability CL (CL e.g.95 %). This shall lead to a "pass decision".

The pass-limit is on the good side of the specified DUT-quality. A more stringent CL (CL e.g.99 %) shifts the pass-limit farer into the good direction. Given the quality of the DUTs is distributed, a greater CL passes less and better DUTs.

A measurement on the bad side of the pass-limit is simply "not pass" (undecided or artificial fail).

(aa) Complementary:

A measurement on the fail-limit shows, that the DUT is worse than the specified quality with probability CL.

The fail-limit is on the bad side of the specified DUT-quality. A more stringent CL shifts the fail-limit farer into the bad direction. Given the quality of the DUTs is distributed, a greater CL fails less and worse DUTs.

A measurement on the good side of the fail-limit is simply "not fail".

(b) A DUT, known to have the specified quality, shall be measured and decided pass with probability CL. This leads to the test limit.

For CL e.g. 95 %, the test limit is on the bad side of the specified DUT-quality. CL e.g.99 % shifts the pass-limit farer into the bad direction. Given the DUT-quality is distributed, a greater CL passes more and worse DUTs.

(bb) A DUT, known to be an ($\varepsilon \rightarrow 0$) beyond the specified quality, shall be measured and decided fail with probability CL.

For CL e.g.95 %, the test limit is on the good side of the specified DUT-quality.

NOTE 1: The different sense for CL in (a), (aa) versus (b), (bb).

NOTE 2: For constant CL in all 4 bullets (a) is equivalent to (bb) and (aa) is equivalent to (b).

## G.2.5 Supplier Risk versus Customer Risk

The table below summarizes the different targets of decision.

**Table G.2.5-1: Equivalent statements**

| | **Equivalent statements, using different cause-to-effect-directions, and assuming CL = constant >1/2** | |
|---|---|---|
| cause-to-effect-directions | Known measurement result → estimation of the DUT's quality | Known DUT's quality → estimation of the measurement's outcome |
| Supplier Risk | A measurement on the pass-limit shows, that the DUT has the specified quality or is better (a) | A DUT, known to have an (ε→0) beyond the specified DUT-quality, shall be measured and decided fail (bb) |
| Customer Risk | A measurement on the fail-limit shall shows, that the DUT is worse than the specified quality (aa) | A DUT, known to have the specified quality, shall be measured and decided pass (b) |

The shaded area shown the direct interpretation of Supplier Risk and Customer Risk.

The same statements can be based on other DUT-quality-definitions.

## G.2.6 Introduction: Standard test versus early decision concept

In standard statistical tests, a certain number of results (ns) is predefined in advance to the test. After ns results the number of bad results (ne) is counted and the error ratio (ER) is calculated by ne/ns.

Applying statistical theory, a decision limit can be designed, against which the calculated ER is compared to derive the decision. Such a limit is one decision point and is characterized by:

- D: the wrong decision probability (a predefined parameter)
- ns: the number of results (a fixed predefined parameter)
- ne: the number of bad results (the limit based on just ns)

In the formula for the limit, D and ns can be understood as variable parameter and variable. However the standard test execution requires fixed ns and D. The property of such a test is: It discriminates between two states only, depending on the test design:

- pass (with CL) / undecided (undecided in the sense: finally undecided)
- fail (with CL) / undecided (undecided in the sense: finally undecided)
- pass(with CL) / fail (with CL) (however against two limits).

In contrast to the standard statistical tests, the early decision concept predefines a set of (ne,ns) co-ordinates, representing the limit-curve for decision. After each result a preliminary ER is calculated and compared against the limit-curve. After each result one may make the decision or not (undecided for later decision). The parameters and variables in the limit-curve for the early decision concept have a similar but not equal meaning:

- D: the wrong decision probability (a predefined parameter)
- ns: the number of results (a variable parameter)
- ne: the number of bad results (the limit. It varies together with ns)

To avoid a "final undecided" in the standard test, a second limit shall be introduced and the single decision co-ordinate (ne,ns) needs a high ne, leading to a fixed (high) test time. In the early decision concept, having the same selectivity and

the same confidence level an "undecided" need not to be avoided, as it can be decided later. A perfect DUT will hit the decision coordinate (ne,ns) with ne=0. This test time is short.

## G.2.7 Standard test versus early decision concept

For Supplier Risk:

The wrong decision probability D in the standard test is the probability, to decide a DUT in-correct in the single decision point. In the early decision concept there is a probability of in-correct decisions d at each point of the limit-curve. The sum of all those wrong decision probabilities accumulate to D. Hence d<D.

For Customer Risk:

The correct decision probability CL in the standard test is the probability, to decide a DUT correct in the single decision point. In the early decision concept there is a probability of correct decisions cl at each point of the limit-curve. The sum of all those correct decision probabilities accumulate to CL. Hence cl<CL or d>D.

## G.2.8 Selectivity

There is no statistical test which can discriminate between a limit DUT and a DUT which is an ($\varepsilon \rightarrow 0$) apart from the limit in finite time and high confidence level CL. Either the test discriminates against one limit with the results pass (with CL)/undecided or fail (with CL)/undecided, or the test ends in a result pass (with CL)/fail (with CL) but this requires a second limit.

For CL>1/2, a (measurement-result = specified-DUT-quality), generates undecided in test "supplier risk against pass limit" (a, from above) and also in the test "customer risk against the fail limit " (aa)

For CL>1/2, a DUT, known to be on the limit, will be decided pass for the test "customer risk against pass limit" (b) and also "supplier risk against fail limit" (bb).

This overlap or undecided area is not a fault or a contradiction, however it can be avoided by introducing a Bad or a Good DUT quality according to:

- Bad DUT quality: specified DUT-quality * M (M>1)
- Good DUT quality: specified DUT-quality * m (m<1)

Using e.g. M>1 and CL=95 % the test for different DUT qualities yield different pass probabilities:

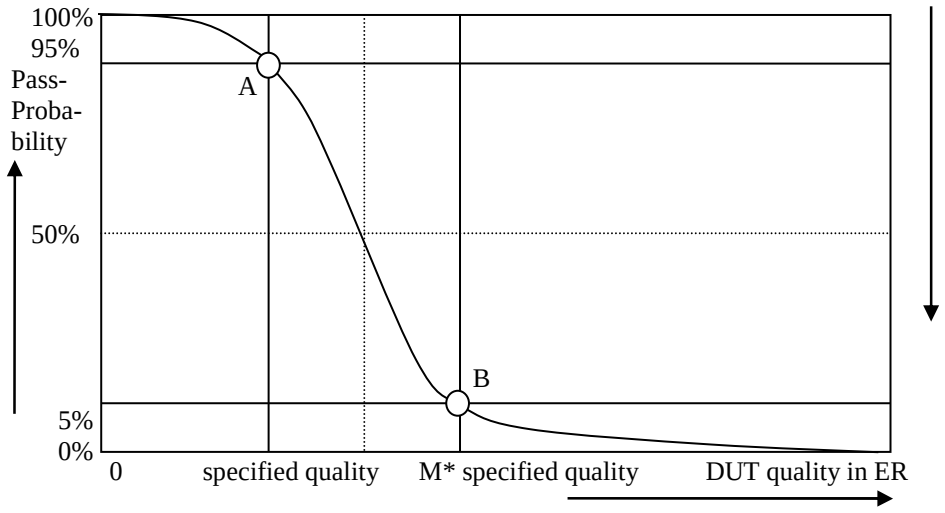



**Figure G.2.8-1: Pass probability versus DUT quality**

## G.2.9 Design of the test

The receiver characteristic test are defined by the following design principles:

1. The early decision concept is applied.
2. A second limit is introduced: Bad DUT factor M>1
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

   To decide the test fail

   Customer Risk is applied based on the specified DUT quality

The receiver characteristic test are defined by the following parameters:

1. Limit ER = 0.05
2. Bad DUT factor M=1.5 (selectivity)
3. Confidence level CL = 95 % (for specified DUT and Bad DUT-quality)

This has the following consequences:

1. A measurement on the fail limit is connected with 2 equivalent statements:

| | |
|---|---|
| A measurement on the fail-limit shows, that the DUT is worse than the specified DUT-quality | A DUT, known have the specified quality, shall be measured and decided pass |

2. A measurement on the pass limit is connected with the complementary statements:

| | |
|---|---|
| A measurement on the pass limit shows, that the DUT is better than the Bad DUT-quality. | A DUT, known to have the Bad DUT quality, shall be measured and decided fail |

   The left column is used to decide the measurement.

   The right column is used to verify the design of the test by simulation.

   The simulation is based on the two fulcrums A and B only in Figure G.2.8-1

3. Test time

   The minimum and maximum test time is fixed.

   The average test time is a function of the DUT's quality.

   The individual test time is not predictable.

4. The number of decision co-ordinates (ne,ns) in the early decision concept is responsible for the selectivity of the test and the maximum test time. Having fixed the number of decision co-ordinates there is still freedom to select the individual decision co-ordinates in many combinations, all leading to the same confidence level.

## G.2.10 Simulation to derive the pass fail limits

There is freedom to design the decision co-ordinates (ne,ns).

The binomial distribution and its inverse is used to design the pass and fail limits. Note that this method is not unique and that other methods exist.

$$fail(ne, d_f) := \frac{ne}{(ne + qnbinom(d_f, ne, ER))}$$

$$pass(ne, cl_p, M) := \frac{ne}{(ne + qnbinom(cl_p, ne, ER \cdot M))}$$

Where

- fail(..) is the error ratio for the fail limit
- pass(..) is the error ratio for the pass limit
- ER is the specified error ratio 0.05
- ne is the number of bad results. This is the variable in both equations
- M is the Bad DUT factor M=1.5
- $d_f$ is the wrong decision probability of a single (ne,ns) co-ordinate for the fail limit.
  It is found by simulation to be $d_f = 0.004$
- $cl_p$ is the confidence level of a single (ne,ns) co-ordinate for the pass limit.
  It is found by simulation to be $cl_p = 0.9975$
- qnbinom(..): The inverse cumulative function of the negative binomial distribution

The simulation works as follows:

- A large population of limit DUTs with true ER = 0.05 is decided against the pass and fail limits.
- $cl_p$ and $d_f$ are tuned such that CL (95 %) of the population passes and D (5 %) of the population fails.
- A population of Bad DUTs with true ER = M*0.05 is decided against the same pass and fail limits.
- $cl_p$ and $d_f$ are tuned such that CL (95 %) of the population fails and D (5 %) of the population passes.
- This procedure and the relationship to the measurement is justified in clause G.2.9. The number of DUTs decrease during the simulation, as the decided DUTs leave the population. That number decreases with an approximately exponential characteristics. After 169 bad results all DUTs of the population are decided.

NOTE: The exponential decrease of the population is an optimal design goal for the decision co-ordinates (ne,ns), which can be achieved with other formulas or methods as well.

# G.3 Measuring throughput ratio

## G.3.1 General

Annex G.3 is applicable for clauses 6.2, 6,3 and 6.4. Common to those clauses is, that a throughput ratio γ of the form $\gamma = \frac{\text{Throughput}_{\text{Numerator}}}{\text{Throughput}_{\text{Denominator}}}$ is measured. These clauses are tested exclusively with “slow” multipath fading profiles. Hence the test time is governed by test time due to fading, and number of samples due to statistical significance is not applicable.

The test requirement in clause 6.3 is a ratio of 2 throughput tests γ. In either numerator or denominator (depending on test case) a target throughput is desired, which is established by an approach resulting in the throughput and the reference SNR that is defined in G.3.2. This SNR is then reused when measuring the throughput of the other factor of the formula. The formulas for calculation of γ are defined directly under sections 6.3.

The test requirements in clauses 6.2 and 6.4 are a ratio of 2 throughput tests γ, where numerator and denominator are ordinary throughput tests. The formulas for calculation of γ are defined in sections 6.2 and 6.4 respectively

## G.3.2 Establishing SNR

Adjust SNR such that the measured throughput is within 2% of target value (TBD% depending on test case). The approach, leading to target throughput and reference SNR is not specified.

The resulting SNR is the reference SNR to use when measuring throughput in the other factor (numerator or denominator) of γ.

To achieve statistical significance the final throughput measurement must be done with MNS samples, given table G.3.4-1

## G.3.3 Measuring T-put

To achieve statistical significance the final throughput measurement must be done with MNS samples, given in table G.3.4 -1. Number of samples due to statistical significance is not applicable.

For measuring $t_{ue,follow1,follow2}$ and $t_{ue,rnd1,rnd2}$, the SS collects ACK, NACK and statDTX from the UE and records the time, elapsed from the beginning of the test. The payload size, received by the UE and acknowledged towards the SS, is constant. Throughput can be calculated in the SS by multiplying the payload size with the number of ACKs and dividing the accumulated payload in kilobits by the time in seconds, elapsed from the beginning of the test, being associated to the following ratio: ACK/ (ACK+NACK + DTX).

## G.3.4 Number of samples for throughput ratios

**Table G.3.4-1: Test time for testing throughput ratios**

| Demodulation scenario (doppler speed) | Minimum number of active subframes (MNAS) | Scheduling pattern | MNAS to MNS Scaling factor (Note 2) | Minimum Number of Subframes (MNS) after rounding up to nearest thousand $MNS=1000*\lceil \frac{MNS}{1000} \rceil$ |
|---|---|---|---|---|
| 5Hz | 100000 | FDD | 1.0526 | 106000 |
| 5Hz | 100000 | TDD FR1.30-1 | 1.2903 | 130000 |
| 35Hz | 100000 | TDD FR2.120-1 | 1.2598 | 126000 |
| 35Hz | 100000 | TDD FR2.120-2 | 1.3445 | 135000 |
| Note 1: MNAS determined by theoretical estimations inherited from LTE based on R5-106393. All slots in active subframe is assumed to be DL slots. | | | | |
| Note 2: MNS/MNAS ratio decided by scheduling pattern and is ratio of all slots to DL slots | | | | |
| Note 3: MNS apply for both denominator and numerator measurement | | | | |

# G.4 Statistical testing of Performance Requirements with BLER limit

## G.4.1 General

The test of receiver performance characteristics is twofold.

1. A signal or a combination of signals is offered to the RX port(s) of the receiver.
2. The ability of the receiver to demodulate /decode this signal is verified by measuring the throughput.

In (2) is the statistical aspect of the test and is treated here.

The minimum requirement for several receiver performance test cases is specified in regards of BLER: 0.001%.

## G.4.2 Design of the test

The test is defined by the following design principles (see clause G.2, Theory):

1. The early decision concept is applied.
2. A second limit is introduced: Bad DUT factor M>1.
3. To decide the test pass:

   Supplier risk is applied based on the Bad DUT quality

   To decide the test fail:

   Customer Risk is applied based on the specified DUT quality

## G.4.3 Numerical definition of the pass fail limits for 0.001% BLER

The numerical pass/fail limit is derived by the following parameters:

1a) Limit Error Ratio = 0.001%

2a) Bad DUT factor M=1.5 (selectivity)

justification see: TS 34.121 Clause F.6.3.3

3) Confidence level CL = 99.999%

**Table G.4.3-1: Pass fail limits**

| ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ | ne | $ns_p$ | $ns_f$ |
|---|---|---|---|---|---|---|---|---|
| 0 | 1074532 | 1067 | 215 | 20006574 | 14871394 | 430 | 36441701 | 33298651 |
| 1 | 1074532 | 1067 | 216 | 20085020 | 14954177 | 431 | 36516711 | 33386452 |
| 2 | 1274645 | 1067 | 217 | 20163439 | 15036999 | 432 | 36591711 | 33474268 |
| 3 | 1444583 | 1067 | 218 | 20241831 | 15119861 | 433 | 36666702 | 33562097 |
| 4 | 1599072 | 4727 | 219 | 20320196 | 15202761 | 434 | 36741683 | 33649940 |
| 5 | 1743641 | 12160 | 220 | 20398535 | 15285701 | 435 | 36816654 | 33737797 |
| 6 | 1881111 | 23683 | 221 | 20476847 | 15368679 | 436 | 36891616 | 33825668 |
| 7 | 2013164 | 39190 | 222 | 20555133 | 15451695 | 437 | 36966568 | 33913553 |
| 8 | 2140902 | 58403 | 223 | 20633393 | 15534749 | 438 | 37041511 | 34001452 |
| 9 | 2265092 | 81000 | 224 | 20711628 | 15617841 | 439 | 37116445 | 34089364 |
| 10 | 2386297 | 106667 | 225 | 20789836 | 15700971 | 440 | 37191369 | 34177291 |
| 11 | 2504945 | 135116 | 226 | 20868019 | 15784137 | 441 | 37266283 | 34265231 |
| 12 | 2621369 | 166089 | 227 | 20946177 | 15867341 | 442 | 37341189 | 34353184 |
| 13 | 2735834 | 199360 | 228 | 21024309 | 15950581 | 443 | 37416085 | 34441151 |
| 14 | 2848557 | 234730 | 229 | 21102417 | 16033858 | 444 | 37490972 | 34529132 |
| 15 | 2959718 | 272025 | 230 | 21180499 | 16117172 | 445 | 37565849 | 34617126 |
| 16 | 3069467 | 311091 | 231 | 21258557 | 16200521 | 446 | 37640718 | 34705134 |
| 17 | 3177931 | 351792 | 232 | 21336590 | 16283906 | 447 | 37715577 | 34793155 |
| 18 | 3285220 | 394009 | 233 | 21414599 | 16367326 | 448 | 37790427 | 34881189 |
| 19 | 3391428 | 437636 | 234 | 21492584 | 16450782 | 449 | 37865268 | 34969237 |
| 20 | 3496637 | 482577 | 235 | 21570545 | 16534273 | 450 | 37940100 | 35057298 |
| 21 | 3600921 | 528746 | 236 | 21648482 | 16617799 | 451 | 38014923 | 35145372 |
| 22 | 3704343 | 576068 | 237 | 21726395 | 16701360 | 452 | 38089737 | 35233459 |
| 23 | 3806960 | 624473 | 238 | 21804284 | 16784955 | 453 | 38164542 | 35321560 |
| 24 | 3908823 | 673898 | 239 | 21882150 | 16868585 | 454 | 38239338 | 35409673 |
| 25 | 4009977 | 724286 | 240 | 21959993 | 16952248 | 455 | 38314125 | 35497800 |
| 26 | 4110465 | 775585 | 241 | 22037812 | 17035945 | 456 | 38388903 | 35585939 |
| 27 | 4210324 | 827748 | 242 | 22115608 | 17119676 | 457 | 38463672 | 35674092 |
| 28 | 4309587 | 880730 | 243 | 22193382 | 17203440 | 458 | 38538432 | 35762258 |
| 29 | 4408285 | 934492 | 244 | 22271133 | 17287238 | 459 | 38613184 | 35850436 |
| 30 | 4506448 | 988997 | 245 | 22348861 | 17371068 | 460 | 38687927 | 35938627 |
| 31 | 4604101 | 1044211 | 246 | 22426567 | 17454931 | 461 | 38762661 | 36026831 |
| 32 | 4701268 | 1100101 | 247 | 22504250 | 17538827 | 462 | 38837386 | 36115048 |
| 33 | 4797972 | 1156638 | 248 | 22581911 | 17622755 | 463 | 38912102 | 36203278 |
| 34 | 4894232 | 1213795 | 249 | 22659550 | 17706716 | 464 | 38986810 | 36291520 |
| 35 | 4990069 | 1271547 | 250 | 22737168 | 17790708 | 465 | 39061510 | 36379774 |
| 36 | 5085500 | 1329869 | 251 | 22814763 | 17874733 | 466 | 39136200 | 36468042 |
| 37 | 5180542 | 1388740 | 252 | 22892337 | 17958789 | 467 | 39210882 | 36556322 |
| 38 | 5275209 | 1448137 | 253 | 22969889 | 18042876 | 468 | 39285556 | 36644614 |
| 39 | 5369517 | 1508043 | 254 | 23047420 | 18126994 | 469 | 39360221 | 36732919 |
| 40 | 5463478 | 1568438 | 255 | 23124929 | 18211144 | 470 | 39434877 | 36821237 |
| 41 | 5557107 | 1629304 | 256 | 23202418 | 18295325 | 471 | 39509525 | 36909566 |
| 42 | 5650414 | 1690627 | 257 | 23279885 | 18379536 | 472 | 39584165 | 36997908 |
| 43 | 5743410 | 1752389 | 258 | 23357331 | 18463778 | 473 | 39658796 | 37086263 |
| 44 | 5836108 | 1814577 | 259 | 23434757 | 18548050 | 474 | 39733419 | 37174629 |
| 45 | 5928516 | 1877177 | 260 | 23512162 | 18632353 | 475 | 39808033 | 37263008 |
| 46 | 6020643 | 1940175 | 261 | 23589546 | 18716685 | 476 | 39882639 | 37351399 |
| 47 | 6112500 | 2003560 | 262 | 23666910 | 18801047 | 477 | 39957237 | 37439803 |
| 48 | 6204094 | 2067319 | 263 | 23744254 | 18885439 | 478 | 40031826 | 37528218 |
| 49 | 6295434 | 2131442 | 264 | 23821577 | 18969861 | 479 | 40106407 | 37616645 |
| 50 | 6386526 | 2195916 | 265 | 23898880 | 19054311 | 480 | 40180980 | 37705085 |

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|---|---|---|---|---|---|---|---|---|
| 51 | 6477380 | 2260734 | 266 | 23976164 | 19138791 | 481 | 40255545 | 37793536 |
| 52 | 6568000 | 2325884 | 267 | 24053427 | 19223300 | 482 | 40330102 | 37882000 |
| 53 | 6658395 | 2391358 | 268 | 24130671 | 19307838 | 483 | 40404650 | 37970475 |
| 54 | 6748569 | 2457146 | 269 | 24207895 | 19392404 | 484 | 40479190 | 38058963 |
| 55 | 6838530 | 2523241 | 270 | 24285099 | 19476999 | 485 | 40553722 | 38147462 |
| 56 | 6928283 | 2589634 | 271 | 24362284 | 19561623 | 486 | 40628246 | 38235973 |
| 57 | 7017834 | 2656318 | 272 | 24439450 | 19646274 | 487 | 40702762 | 38324496 |
| 58 | 7107187 | 2723285 | 273 | 24516597 | 19730954 | 488 | 40777270 | 38413030 |
| 59 | 7196348 | 2790528 | 274 | 24593724 | 19815662 | 489 | 40851770 | 38501576 |
| 60 | 7285321 | 2858041 | 275 | 24670832 | 19900397 | 490 | 40926262 | 38590134 |
| 61 | 7374112 | 2925816 | 276 | 24747922 | 19985160 | 491 | 41000746 | 38678704 |
| 62 | 7462724 | 2993848 | 277 | 24824993 | 20069950 | 492 | 41075222 | 38767285 |
| 63 | 7551162 | 3062130 | 278 | 24902045 | 20154768 | 493 | 41149690 | 38855878 |
| 64 | 7639430 | 3130657 | 279 | 24979078 | 20239613 | 494 | 41224150 | 38944482 |
| 65 | 7727532 | 3199424 | 280 | 25056093 | 20324485 | 495 | 41298602 | 39033098 |
| 66 | 7815471 | 3268424 | 281 | 25133089 | 20409383 | 496 | 41373047 | 39121725 |
| 67 | 7903252 | 3337653 | 282 | 25210068 | 20494309 | 497 | 41447483 | 39210364 |
| 68 | 7990878 | 3407105 | 283 | 25287028 | 20579261 | 498 | 41521912 | 39299014 |
| 69 | 8078352 | 3476777 | 284 | 25363970 | 20664239 | 499 | 41596333 | 39387675 |
| 70 | 8165677 | 3546663 | 285 | 25440893 | 20749244 | 500 | 41670746 | 39476348 |
| 71 | 8252857 | 3616759 | 286 | 25517799 | 20834275 | 501 | 41745152 | 39565032 |
| 72 | 8339894 | 3687060 | 287 | 25594687 | 20919332 | 502 | 41819550 | 39653727 |
| 73 | 8426792 | 3757563 | 288 | 25671558 | 21004415 | 503 | 41893940 | 39742434 |
| 74 | 8513553 | 3828263 | 289 | 25748411 | 21089524 | 504 | 41968323 | 39831151 |
| 75 | 8600181 | 3899156 | 290 | 25825246 | 21174658 | 505 | 42042698 | 39919880 |
| 76 | 8686677 | 3970239 | 291 | 25902063 | 21259818 | 506 | 42117065 | 40008620 |
| 77 | 8773044 | 4041508 | 292 | 25978864 | 21345003 | 507 | 42191424 | 40097371 |
| 78 | 8859286 | 4112960 | 293 | 26055647 | 21430213 | 508 | 42265777 | 40186133 |
| 79 | 8945403 | 4184590 | 294 | 26132413 | 21515449 | 509 | 42340121 | 40274907 |
| 80 | 9031399 | 4256396 | 295 | 26209162 | 21600709 | 510 | 42414458 | 40363691 |
| 81 | 9117276 | 4328375 | 296 | 26285893 | 21685995 | 511 | 42488788 | 40452486 |
| 82 | 9203035 | 4400523 | 297 | 26362608 | 21771305 | 512 | 42563110 | 40541292 |
| 83 | 9288680 | 4472838 | 298 | 26439306 | 21856639 | 513 | 42637425 | 40630109 |
| 84 | 9374212 | 4545316 | 299 | 26515987 | 21941999 | 514 | 42711732 | 40718937 |
| 85 | 9459633 | 4617954 | 300 | 26592652 | 22027382 | 515 | 42786032 | 40807776 |
| 86 | 9544944 | 4690751 | 301 | 26669300 | 22112790 | 516 | 42860324 | 40896625 |
| 87 | 9630149 | 4763702 | 302 | 26745931 | 22198222 | 517 | 42934609 | 40985485 |
| 88 | 9715249 | 4836806 | 303 | 26822546 | 22283678 | 518 | 43008887 | 41074356 |
| 89 | 9800245 | 4910060 | 304 | 26899145 | 22369157 | 519 | 43083157 | 41163238 |
| 90 | 9885139 | 4983461 | 305 | 26975727 | 22454661 | 520 | 43157420 | 41252131 |
| 91 | 9969933 | 5057007 | 306 | 27052293 | 22540188 | 521 | 43231676 | 41341034 |
| 92 | 10054629 | 5130696 | 307 | 27128843 | 22625739 | 522 | 43305924 | 41429947 |
| 93 | 10139228 | 5204526 | 308 | 27205377 | 22711313 | 523 | 43380165 | 41518872 |
| 94 | 10223731 | 5278493 | 309 | 27281895 | 22796910 | 524 | 43454399 | 41607806 |
| 95 | 10308141 | 5352597 | 310 | 27358398 | 22882531 | 525 | 43528626 | 41696752 |
| 96 | 10392459 | 5426835 | 311 | 27434884 | 22968175 | 526 | 43602846 | 41785708 |
| 97 | 10476685 | 5501204 | 312 | 27511355 | 23053842 | 527 | 43677058 | 41874674 |
| 98 | 10560822 | 5575703 | 313 | 27587810 | 23139531 | 528 | 43751263 | 41963651 |
| 99 | 10644871 | 5650331 | 314 | 27664249 | 23225243 | 529 | 43825462 | 42052638 |
| 100 | 10728833 | 5725084 | 315 | 27740673 | 23310978 | 530 | 43899653 | 42141635 |
| 101 | 10812709 | 5799961 | 316 | 27817081 | 23396736 | 531 | 43973837 | 42230643 |
| 102 | 10896501 | 5874961 | 317 | 27893475 | 23482516 | 532 | 44048014 | 42319662 |
| 103 | 10980210 | 5950082 | 318 | 27969852 | 23568318 | 533 | 44122183 | 42408690 |

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|---|---|---|---|---|---|---|---|---|
| 104 | 11063837 | 6025321 | 319 | 28046215 | 23654143 | 534 | 44196346 | 42497729 |
| 105 | 11147384 | 6100677 | 320 | 28122563 | 23739989 | 535 | 44270502 | 42586778 |
| 106 | 11230851 | 6176149 | 321 | 28198895 | 23825858 | 536 | 44344651 | 42675837 |
| 107 | 11314239 | 6251735 | 322 | 28275212 | 23911748 | 537 | 44418793 | 42764907 |
| 108 | 11397550 | 6327434 | 323 | 28351515 | 23997661 | 538 | 44492928 | 42853986 |
| 109 | 11480785 | 6403243 | 324 | 28427803 | 24083595 | 539 | 44567056 | 42943076 |
| 110 | 11563945 | 6479161 | 325 | 28504075 | 24169550 | 540 | 44641177 | 43032176 |
| 111 | 11647030 | 6555187 | 326 | 28580333 | 24255527 | 541 | 44715291 | 43121286 |
| 112 | 11730042 | 6631320 | 327 | 28656577 | 24341526 | 542 | 44789399 | 43210406 |
| 113 | 11812982 | 6707558 | 328 | 28732806 | 24427546 | 543 | 44863499 | 43299535 |
| 114 | 11895850 | 6783899 | 329 | 28809020 | 24513587 | 544 | 44937593 | 43388675 |
| 115 | 11978648 | 6860343 | 330 | 28885220 | 24599649 | 545 | 45011680 | 43477825 |
| 116 | 12061377 | 6936887 | 331 | 28961405 | 24685732 | 546 | 45085760 | 43566985 |
| 117 | 12144037 | 7013532 | 332 | 29037577 | 24771836 | 547 | 45159833 | 43656155 |
| 118 | 12226629 | 7090274 | 333 | 29113734 | 24857961 | 548 | 45233900 | 43745334 |
| 119 | 12309155 | 7167114 | 334 | 29189876 | 24944107 | 549 | 45307960 | 43834523 |
| 120 | 12391614 | 7244050 | 335 | 29266005 | 25030273 | 550 | 45382013 | 43923723 |
| 121 | 12474008 | 7321081 | 336 | 29342119 | 25116460 | 551 | 45456059 | 44012932 |
| 122 | 12556338 | 7398206 | 337 | 29418220 | 25202668 | 552 | 45530099 | 44102150 |
| 123 | 12638604 | 7475422 | 338 | 29494306 | 25288896 | 553 | 45604132 | 44191379 |
| 124 | 12720808 | 7552731 | 339 | 29570379 | 25375144 | 554 | 45678159 | 44280617 |
| 125 | 12802949 | 7630129 | 340 | 29646438 | 25461412 | 555 | 45752178 | 44369865 |
| 126 | 12885029 | 7707617 | 341 | 29722483 | 25547700 | 556 | 45826192 | 44459123 |
| 127 | 12967048 | 7785194 | 342 | 29798514 | 25634009 | 557 | 45900198 | 44548390 |
| 128 | 13049007 | 7862857 | 343 | 29874532 | 25720337 | 558 | 45974198 | 44637667 |
| 129 | 13130907 | 7940606 | 344 | 29950536 | 25806685 | 559 | 46048192 | 44726953 |
| 130 | 13212749 | 8018441 | 345 | 30026527 | 25893053 | 560 | 46122179 | 44816249 |
| 131 | 13294533 | 8096360 | 346 | 30102504 | 25979441 | 561 | 46196159 | 44905555 |
| 132 | 13376259 | 8174362 | 347 | 30178468 | 26065848 | 562 | 46270133 | 44994870 |
| 133 | 13457929 | 8252446 | 348 | 30254418 | 26152274 | 563 | 46344100 | 45084194 |
| 134 | 13539543 | 8330612 | 349 | 30330355 | 26238721 | 564 | 46418061 | 45173528 |
| 135 | 13621102 | 8408859 | 350 | 30406279 | 26325186 | 565 | 46492016 | 45262871 |
| 136 | 13702605 | 8487185 | 351 | 30482190 | 26411671 | 566 | 46565964 | 45352224 |
| 137 | 13784055 | 8565589 | 352 | 30558087 | 26498174 | 567 | 46639906 | 45441586 |
| 138 | 13865452 | 8644072 | 353 | 30633972 | 26584697 | 568 | 46713841 | 45530958 |
| 139 | 13946795 | 8722632 | 354 | 30709843 | 26671239 | 569 | 46787770 | 45620339 |
| 140 | 14028086 | 8801268 | 355 | 30785702 | 26757800 | 570 | 46861692 | 45709729 |
| 141 | 14109325 | 8879979 | 356 | 30861547 | 26844380 | 571 | 46935608 | 45799128 |
| 142 | 14190513 | 8958765 | 357 | 30937380 | 26930979 | 572 | 47009518 | 45888537 |
| 143 | 14271650 | 9037625 | 358 | 31013200 | 27017596 | 573 | 47083422 | 45977955 |
| 144 | 14352737 | 9116558 | 359 | 31089007 | 27104232 | 574 | 47157319 | 46067382 |
| 145 | 14433775 | 9195563 | 360 | 31164802 | 27190886 | 575 | 47231210 | 46156818 |
| 146 | 14514763 | 9274640 | 361 | 31240584 | 27277559 | 576 | 47305094 | 46246264 |
| 147 | 14595702 | 9353788 | 362 | 31316353 | 27364250 | 577 | 47378973 | 46335719 |
| 148 | 14676593 | 9433006 | 363 | 31392110 | 27450959 | 578 | 47452845 | 46425182 |
| 149 | 14757437 | 9512294 | 364 | 31467854 | 27537687 | 579 | 47526711 | 46514655 |
| 150 | 14838233 | 9591650 | 365 | 31543586 | 27624433 | 580 | 47600570 | 46604137 |
| 151 | 14918983 | 9671074 | 366 | 31619306 | 27711197 | 581 | 47674424 | 46693628 |
| 152 | 14999686 | 9750566 | 367 | 31695013 | 27797979 | 582 | 47748271 | 46783128 |
| 153 | 15080344 | 9830124 | 368 | 31770708 | 27884779 | 583 | 47822113 | 46872637 |
| 154 | 15160956 | 9909749 | 369 | 31846390 | 27971597 | 584 | 47895948 | 46962155 |
| 155 | 15241523 | 9989439 | 370 | 31922061 | 28058432 | 585 | 47969777 | 47051682 |
| 156 | 15322045 | 10069194 | 371 | 31997719 | 28145286 | 586 | 48043599 | 47141218 |

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|---|---|---|---|---|---|---|---|---|
| 157 | 15402524 | 10149014 | 372 | 32073365 | 28232157 | 587 | 48117416 | 47230762 |
| 158 | 15482959 | 10228896 | 373 | 32149000 | 28319045 | 588 | 48191227 | 47320316 |
| 159 | 15563350 | 10308842 | 374 | 32224622 | 28405951 | 589 | 48265031 | 47409879 |
| 160 | 15643699 | 10388851 | 375 | 32300232 | 28492875 | 590 | 48338830 | 47499450 |
| 161 | 15724005 | 10468921 | 376 | 32375831 | 28579815 | 591 | 48412622 | 47589030 |
| 162 | 15804270 | 10549052 | 377 | 32451417 | 28666774 | 592 | 48486409 | 47678619 |
| 163 | 15884492 | 10629245 | 378 | 32526992 | 28753749 | 593 | 48560190 | 47768217 |
| 164 | 15964673 | 10709497 | 379 | 32602555 | 28840741 | 594 | 48633964 | 47857823 |
| 165 | 16044814 | 10789809 | 380 | 32678107 | 28927751 | 595 | 48707733 | 47947438 |
| 166 | 16124913 | 10870180 | 381 | 32753646 | 29014778 | 596 | 48781495 | 48037062 |
| 167 | 16204973 | 10950610 | 382 | 32829175 | 29101821 | 597 | 48855252 | 48126695 |
| 168 | 16284993 | 11031098 | 383 | 32904691 | 29188882 | 598 | 48929003 | 48216336 |
| 169 | 16364973 | 11111643 | 384 | 32980196 | 29275959 | 599 | 49002747 | 48305986 |
| 170 | 16444914 | 11192245 | 385 | 33055690 | 29363053 | 600 | 49076486 | 48395644 |
| 171 | 16524817 | 11272904 | 386 | 33131172 | 29450164 | 601 | 49150219 | 48485312 |
| 172 | 16604680 | 11353619 | 387 | 33206643 | 29537291 | 602 | 49223946 | 48574987 |
| 173 | 16684506 | 11434390 | 388 | 33282102 | 29624435 | 603 | 49297668 | 48664671 |
| 174 | 16764294 | 11515215 | 389 | 33357550 | 29711596 | 604 | 49371383 | 48754364 |
| 175 | 16844045 | 11596095 | 390 | 33432987 | 29798773 | 605 | 49445093 | 48844065 |
| 176 | 16923758 | 11677030 | 391 | 33508413 | 29885966 | 606 | 49518797 | 48933775 |
| 177 | 17003435 | 11758018 | 392 | 33583827 | 29973176 | 607 | 49592495 | 49023493 |
| 178 | 17083075 | 11839059 | 393 | 33659230 | 30060402 | 608 | 49666187 | 49113220 |
| 179 | 17162679 | 11920153 | 394 | 33734623 | 30147644 | 609 | 49739874 | 49202955 |
| 180 | 17242247 | 12001299 | 395 | 33810004 | 30234902 | 610 | 49813554 | 49292699 |
| 181 | 17321779 | 12082497 | 396 | 33885374 | 30322176 | 611 | 49887229 | 49382451 |
| 182 | 17401276 | 12163747 | 397 | 33960734 | 30409467 | 612 | 49960899 | 49472211 |
| 183 | 17480738 | 12245048 | 398 | 34036082 | 30496773 | 613 | 50034562 | 49561980 |
| 184 | 17560165 | 12326400 | 399 | 34111419 | 30584095 | 614 | 50108220 | 49651757 |
| 185 | 17639558 | 12407801 | 400 | 34186746 | 30671433 | 615 | 50181872 | 49741542 |
| 186 | 17718917 | 12489253 | 401 | 34262062 | 30758787 | 616 | 50255519 | 49831335 |
| 187 | 17798241 | 12570754 | 402 | 34337367 | 30846156 | 617 | 50329160 | 49921137 |
| 188 | 17877532 | 12652304 | 403 | 34412662 | 30933541 | 618 | 50402795 | 50010947 |
| 189 | 17956790 | 12733903 | 404 | 34487945 | 31020942 | 619 | 50476425 | 50100765 |
| 190 | 18036015 | 12815550 | 405 | 34563218 | 31108358 | 620 | 50550049 | 50190592 |
| 191 | 18115206 | 12897245 | 406 | 34638481 | 31195790 | 621 | 50623667 | 50280427 |
| 192 | 18194366 | 12978988 | 407 | 34713733 | 31283237 | 622 | 50697280 | 50370270 |
| 193 | 18273492 | 13060777 | 408 | 34788974 | 31370699 | 623 | 50770887 | 50460120 |
| 194 | 18352587 | 13142614 | 409 | 34864205 | 31458177 | 624 | 50844489 | 50549980 |
| 195 | 18431650 | 13224497 | 410 | 34939426 | 31545670 | 625 | 50918085 | 50639847 |
| 196 | 18510681 | 13306426 | 411 | 35014636 | 31633178 | 626 | 50991676 | 50729722 |
| 197 | 18589681 | 13388401 | 412 | 35089836 | 31720702 | 627 | 51065261 | 50819605 |
| 198 | 18668650 | 13470421 | 413 | 35165025 | 31808240 | 628 | 51138840 | 50909497 |
| 199 | 18747588 | 13552486 | 414 | 35240204 | 31895794 | 629 | 51212414 | 50999396 |
| 200 | 18826495 | 13634596 | 415 | 35315373 | 31983362 | 630 | 51285983 | 51089304 |
| 201 | 18905372 | 13716750 | 416 | 35390532 | 32070946 | 631 | 51359546 | 51179219 |
| 202 | 18984219 | 13798949 | 417 | 35465680 | 32158544 | 632 | 51433104 | 51269143 |
| 203 | 19063035 | 13881191 | 418 | 35540819 | 32246157 | 633 | 51506656 | 51359074 |
| 204 | 19141822 | 13963476 | 419 | 35615947 | 32333785 | 634 | 51580203 | 51449013 |
| 205 | 19220579 | 14045805 | 420 | 35691065 | 32421428 | 635 | 51653744 | 51538961 |
| 206 | 19299307 | 14128176 | 421 | 35766173 | 32509085 | 636 | 51727280 | 51628916 |
| 207 | 19378006 | 14210590 | 422 | 35841272 | 32596757 | 637 | 51800811 | 51718879 |
| 208 | 19456676 | 14293046 | 423 | 35916360 | 32684443 | 638 | 51874336 | 51808850 |
| 209 | 19535318 | 14375544 | 424 | 35991438 | 32772144 | 639 | 51947856 | 51898828 |

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|---|---|---|---|---|---|---|---|---|
| 210 | 19613930 | 14458083 | 425 | 36066507 | 32859859 | 640 | 52021370 | 51988815 |
| 211 | 19692515 | 14540664 | 426 | 36141565 | 32947589 | 641 | 52094880 | 52078809 |
| 212 | 19771071 | 14623286 | 427 | 36216614 | 33035333 | 642 | 52168384 | 52168811 |
| 213 | 19849600 | 14705948 | 428 | 36291653 | 33123092 | | | |
| 214 | 19928101 | 14788651 | 429 | 36366682 | 33210864 | | | |

NOTE 1: The first column is the number of errors (ne = number of NACK).
NOTE 2: The second column is the number of samples for the pass limit ($ns_p$, ns=Number of Samples= number of NACK + ACK).
NOTE 3: The third column is the number of samples for the fail limit ($ns_f$).
NOTE 4: An ideal DUT passes after 1074532 samples. The maximum test time is 52171625 samples. A DUT passes, if the maximum number of samples is reached and it did not fail before.

## G.4.4 Simulation to derive the pass-fail limits for 0.001% BLER

The binomial distribution and its inverse are used to design the pass and fail limits. Note that this method is not unique and that other methods exist.

$$\mathrm{fail}\left(\mathrm{ne},\mathrm{d_f}\right):=\frac{\mathrm{ne}}{\mathrm{ns_f}}=\frac{\mathrm{ne}}{\left(\mathrm{ne}+\mathrm{qnbinom}\left(\mathrm{d_f},\mathrm{ne},\mathrm{ER}\right)\right)}$$

$$\mathrm{pass}\left(\mathrm{ne},\mathrm{cl_p},\mathrm{M}\right):=\frac{\mathrm{ne}}{\mathrm{ns_p}}=\frac{\mathrm{ne}}{\left(\mathrm{ne}+\mathrm{qnbinom}\left(\mathrm{cl_p},\mathrm{ne},\mathrm{ER}\cdot\mathrm{M}\right)\right)}$$

Where

- fail(..) is the error ratio for the fail limit.
- pass(..) is the error ratio for the pass limit.
- ER is the specified error ratio 1e-5.
- ne is the number of bad results. This is the variable in both equations.
- M is the Bad DUT factor M=1.5.
- $d_f$ is the wrong decision probability of a single (ne, ns) co-ordinate for the fail limit.
  It is found by simulation to be $d_f$ = 2e-7.
- $cl_p$ is the confidence level of a single (ne, ns) co-ordinate for the pass limit.
  It is found by simulation to be $cl_p$ = 0.9999999.
- qnbinom(..): The inverse cumulative function of the negative binomial distribution.

The simulation works as follows:

- A large population of limit DUTs with true ER = 1e-5 is decided against the pass and fail limits.
- $cl_p$ and $d_f$ are tuned such that CL (99.999 %) of the population passes and D (0.001 %) of the population fails.
- A population of Bad DUTs with true ER = M*1e-5 is decided against the same pass and fail limits.
- $cl_p$ and $d_f$ are tuned such that CL (99.999 %) of the population fails and D (0.001 %) of the population passes.
- The number of DUTs decrease during the simulation, as the decided DUTs leave the population. That number decreases with an approximately exponential characteristics. After 642 bad results all DUTs of the population are decided.

NOTE: The exponential decrease of the population is an optimal design goal for the decision co-ordinates (ne, ns), which can be achieved with other formulas or methods as well.

# Annex H:
# Approach for finding UE direction for FR2 Demod and CSI Testing

## H.0 Normative criteria for determining UE direction for Demod and CSI

Following 3 criteria shall be satisfied for a given UE direction. Procedure for finding the UE direction is captured in Annex H.1

1. UE shall pass the REFSENS test as per TC 7.3.2 of TS 38.521-2 [8].
2. Minimum isolation requirement of 12 dB between the 2 TE polarization branches shall be met.
3. UE reported rank shall be higher or same as intended rank for a given test.

## H.1 Procedure for finding UE direction

This section provides example approaches for finding the UE direction for Demod and CSI tests. Other approaches satisfying the normative criteria listed in H.0 are not precluded.

Default approach is as defined in H.1.2.

### H.1.1 Using Rx beam peak direction search

1. For Rx beam peak direction search, please refer to procedure defined in Annex K.1.2/K.3.2 of TS 38.521-2 [8].
2. Run wireless cable mode isolation procedure as defined in H.2.
3. Ensure UE reported rank is higher or same as intended rank for a given test.

### H.1.2 RSRPB based scan with fallback option to Rx beam peak direction search

1. Enable periodic RSRPB reporting from the UE.
2. Set of grid points for the UE scan can be user defined set or entire sphere.
3. For each grid point, record RSRPB first by connecting SS to the DUT through the measurement antenna with $Pol_{Link}= \theta$ polarization to form the Rx beam towards the measurement antenna and similarly for $Pol_{Link}= \phi$ polarization.
4. Wait for BEAM_SELECT_WAIT_TIME before recording the RSRPB reports.
5. Once the grid points scan is completed, sort the grid points based on the linear sum of 4 RSRPB values (2 each for $\theta$ and $\phi$ polarization).
6. For the top [10] grid points, run the REFSENS throughput test as per the test condition defined in 38.521-2 clause 7.3.2
7. Grid points that pass the REFSENS throughput test are the potential UE direction to be used for running the tests.
8. If no grid points found in step 7, fall back to using H.1.1.
9. For running rank1 tests,

a. Pick any of the grid points obtained in step 7.

b. Run the wireless cable isolation procedure defined in H.2.

c. Exit the procedure.

10. For running rank2 tests,

a. Pick a grid point obtained in step 7.

b. Run the wireless cable mode isolation procedure defined in H.2.

c. If the grid point satisfies the minimum isolation, proceed to RI check.

Enable RI reporting from UE. If the UE reported rank = 2, exit the procedure.

If UE reported rank is not equal to 2, move to the next grid from step 7 and run step 10.

d. If no grid point meets the criteria in step 7 and step 10c, fallback to using H.1.1.

### H.1.3 Isolation based scan with fallback option to Rx beam peak direction search

1. Enable periodic RSRPB reporting from the UE.

2. Set of grid points for the UE scan can be user defined set or entire sphere.

3. For each grid point, record RSRPB first by connecting SS to the DUT through the measurement antenna with $Pol_{Link}= \theta$ polarization to form the Rx beam towards the measurement antenna and similarly for $Pol_{Link}= \phi$ polarization.

4. Wait for BEAM_SELECT_WAIT_TIME before recording the RSRPB reports.

5. Once the grid points scan is completed, sort the grid points based on the highest $ISO_{\theta,B1}$ and $ISO_{\phi,B2}$ or $ISO_{\theta,B2}$ and $ISO_{\phi,B1}$ ($ISO_{\theta,B1}$, $ISO_{\phi,B2}$, $ISO_{\theta,B2}$ and $ISO_{\phi,B1}$ are explained in H.2).

6. For the top [10] grid points, run the REFSENS throughput test as per the test condition defined in 38.521-2 clause 7.3.2

7. Grid points that pass the REFSENS throughput test are the potential UE direction to be used for running the tests.

8. If no grid points found in step 7, fall back to using H.1.1.

9. For running rank1 tests,

a. Pick any of the grid points obtained in step 7.

b. If the grid point satisfies the minimum isolation, exit the procedure.

10. For running rank2 tests,

a. Pick a grid point obtained in step 7.

b. If the grid point satisfies the minimum isolation, proceed to RI check.

Enable RI reporting from UE. If the UE reported rank = 2, exit the procedure.

If UE reported rank is not equal to 2, move to the next grid from step 7 and run step 10.

d. If no grid point meets the criteria in step 7 and step 10b, fallback to using H.1.1.

# H.2 Wireless cable mode isolation procedure

The following procedure shall be used to verify the wireless cable mode has been established and that the minimum isolation has been achieved

1. Select any of the three Alignment Options (1, 2, or 3) to mount the DUT inside the QZ.
2. If the re-positioning concept is applied to demodulation test cases, position the DUT in DUT Orientation 1 if the RX beam peak is within $0^o \leq \theta \leq 90^o$. Otherwise, position the DUT in DUT Orientation 2 (Option 1 or 2). If the re-positioning concept is not applied to demodulation test cases, position the DUT in DUT Orientation 1
3. Connect the SS (System Simulator) using static propagation conditions with the DUT through the measurement antenna with $Pol_{Link}=\theta$ polarization to form the RX beam towards the desired test direction. Allow at least BEAM_SELECT_WAIT_TIME for the UE RX beam selection to complete.
4. Adjust the DL power of the SS to obtain $P_{DL}$ defined in Table C.0.2-1 at the centre of QZ
5. Perform the isolation of the branches to achieve the wireless cable mode. The inverse channel matrix approach in [4] is one suitable approach. Alternate approaches are not precluded.
6. To verify the wireless cable mode and thus the min. isolation between branches
   a) Query SS-RSRPB($Pol_{Meas}=Pol_{Link}=\theta$) from the DUT for the θ-polarization and convert the two measurements in dBm, i.e., $SS\text{-}RSRPB_{B1}$ and $SS\text{-}RSRPB_{B2}$
   b) Calculate the isolation from θ-polarization into Branch 1, i.e., $ISO_{\theta,B1} = SS\text{-}RSRPB_{B1} - SS\text{-}RSRPB_{B2}$ and the isolation into Branch 2, i.e., $ISO_{\theta,B2} = SS\text{-}RSRPB_{B2} - SS\text{-}RSRPB_{B1}$
   c) Connect the SS (System Simulator) using static propagation conditions with the DUT through the measurement antenna with $Pol_{Link}=\phi$ polarization to form the RX beam towards desired test direction. Allow at least BEAM_SELECT_WAIT_TIME for the UE RX beam selection to complete.
   d) Adjust the DL power of the SS to obtain $P_{DL}$ defined in Table C.0.2-1 at the centre of QZ
   e) Query SS-RSRPB($Pol_{Meas}=Pol_{Link}=\phi$) from the DUT for ϕ-polarization and convert the two measurements in dBm, i.e., $SS\text{-}RSRPB_{B1}$ and $SS\text{-}RSRPB_{B2}$
   f) Calculate the isolation from ϕ-polarization into Branch 2, i.e., $ISO_{\phi,B2} = SS\text{-}RSRPB_{B2} - SS\text{-}RSRPB_{B1}$ and the isolation into Branch 1, i.e., $ISO_{\phi,B1} = SS\text{-}RSRPB_{B1} - SS\text{-}RSRPB_{B2}$

If either of the isolations pairs, $ISO_{\theta,B1}$ and $ISO_{\phi,B2}$ or $ISO_{\theta,B2}$ and $ISO_{\phi,B1}$ exceed 12dB, the wireless cable mode has been achieved.

# Annex I (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Date** | **Meeting** | **Tdoc** | **CR** | **Rev** | **Cat** | **Subject/Comment** | **New version** |
| 2018-01 | | R5-180064 | | | | Skeleton for NR Demod spec | 0.0.1 |
| 2018-04-13 | | R5-182036 | | | | Added the test procedure for FR2 Demod testing in Annex | 0.1.0 |
| 2018-10-12 | | R5-185903 | | | | Added the demod spec test case section titles to be in line with RAN4 approved skeleton for 38.101-4 | 0.1.1 |
| 2018-11-20 | RAN5 #81 | R5-188006 | | | | new TC for PDSCH FR1 demod | 0.2.0 |
| 2018-11-20 | RAN5 #81 | R5-188008 | | | | new TC for PDSCH FR2 demod | 0.2.0 |
| 2018-11-20 | RAN5 #81 | R5-187573 | | | | section 3 of 38.521-4 spec | 0.2.0 |
| 2018-11-20 | RAN5 #81 | R5-187845 | | | | section 4 of 38.521-4 spec | 0.2.0 |
| 2018-11-20 | RAN5 #81 | R5-188009 | | | | pCR for new TC addition for FR1 FDD PDSCH Demod | 0.2.0 |
| 2018-11-20 | RAN5 #81 | R5-188010 | | | | pCR for new TC addition for FR1 FDD PDCCH Demod | 0.2.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190054 | | | | update to 2Rx TDD FR1 PDSCH mapping Type A performance test case | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190926 | | | | pCR for new TC addition for FR1 4Rx FDD PDSCH Demodulation performance (2x4) | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190927 | | | | pCR for new TC addition for FR1 4Rx FDD PDSCH Demodulation performance (4x4) | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190928 | | | | pCR for new TC addition for FR1 4Rx FDD PDSCH Demodulation performance with enhanced receiver type X (4x4) | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190291 | | | | Updated to Annex A Measurement Channels for Performance tests | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190292 | | | | Updated to Annex B Propagation conditions for Performance tests | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190458 | | | | update to 2Rx TDD FR2 PDSCH mapping Type A performance test case | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190461 | | | | 2Rx TDD FR2 PDCCH performance test case | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190929 | | | | LTE link setup details for demod test cases | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190930 | | | | Annex for statistical tput calculation for demod test cases | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190931 | | | | pCR for TC addition of FR1 TDD 4Rx PDSCH | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190932 | | | | pCR for modification of FDD 2Rx FR1 PDSCH Demod | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190933 | | | | Annex for DL and UL Signal Setup | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190934 | | | | pCR for modification of FDD FR1 PDCCH Demod | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190935 | | | | PDSCH and PDCCH Config before measurement | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190986 | | | | 38.521-4 Common Section updates to clarify leverage across architecture options | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190552 | | | | Addition of 2Rx TDD FR1 Single PMI tests for both SA and NSA | 0.3.0 |
| 2019-01-25 | RAN5 5G-NR AH#4 | R5-190553 | | | | Addition of 2Rx TDD FR1 RI reporting for both SA and NSA | 0.3.0 |
| 2019-03-01 | RAN5 #82 | R5-191183 | | | | Adding relevant references to 38.521-4 | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192461 | | | | Adding of test case 6.2.2.1.2.1.2, Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192672 | | | | Introduction of New test case 5.3.2.2.1 2Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192463 | | | | Introduction of New test case 5.3.2.2.2 2Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192462 | | | | Introduction of New test case 5.3.3.1.1 4Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192464 | | | | Introduction of New test case 5.3.3.1.2 4Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192465 | | | | Introduction of New test case 5.3.3.2.1 4Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192465 | | | | Introduction of New test case 5.3.3.2.2 4Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | 0.4.0 |

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|---|---|---|---|---|---|---|---|
| 2019-03-01 | RAN5 #82 | R5-192474 | | | | Introduction of TS 38.521-4 test case 6.3.2.1.1 | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192475 | | | | Introduction of TS 38.521-4 test case 6.3.2.1.2 | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192467 | | | | Introduction of test case 5.2.2.1.2_1, 2Rx FDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - 2x2 MIMO with baseline receiver for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192840 | | | | Demod spec section 4 update | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192673 | | | | Update to TDD FR1 2Rx PDSCH Type A test case | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192103 | | | | addition of 2Rx TDD FR1 periodic CQI reporting test case | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192468 | | | | pCR for addition of 2Rx TDD FR1 TypeA and CSI-RS overlapped TC | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192866 | | | | pCR for modification of PDSCH and PDCCH Config before measurement | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192470 | | | | pCR for modification of FDD FR1 PDCCH Demod | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192471 | | | | pCR for modification of FDD 2Rx FR1 PDSCH Demod | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192472 | | | | Update to 2Rx TDD FR1 RI reporting for both SA and NSA | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192460 | | | | Minimum test time update for FR1 Demod test case | 0.4.0 |
| 2019-03-01 | RAN5 #82 | R5-192473 | | | | Addition of Annex F for Demod spec | 0.4.0 |
| 2019-03 | RAN#83 | RP-190222 | - | - | - | Presented to the RAN#83 plenary for 1-step approval | 1.0.0 |
| 2019-03 | RAN#83 | - | - | - | - | raised to v15.0.0 with editorial changes only | 15.0.0 |
| 2019-06 | RAN5#83 | R5-193544 | 0030 | - | F | Updates to test case 6.2.2.1.2.1, 2Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-193943 | 0035 | - | F | Adding test case 6.2.2.2.2.2, 2Rx TDD FR1 periodic subband CQI reporting under fading conditions for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194159 | 0048 | - | F | Alignment of Annex C with core specification | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194466 | 0056 | - | F | Introduction of FR1 CQI test case 6.2.2.2.2.1 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194622 | 0057 | - | F | Corrections TDD UL-DL configurations | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194680 | 0066 | - | F | Demod section 5 general update | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194689 | 0073 | - | F | Addition of text for FR1 PBCH demodulation test case | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194690 | 0074 | - | F | Update to 2Rx TDD FR2 PDSCH Type A test case | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194691 | 0075 | - | F | Update to FR2 PDCCH config param | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194692 | 0076 | - | F | Addition of text for FR2 PBCH demodulation test case | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194693 | 0077 | - | F | Update to section 8 CSI reporting | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194979 | 0063 | 1 | F | Further updates to 2Rx TDD FR1 PDSCH mapping Type A test case | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194980 | 0032 | 1 | F | Introduction of TC 6.4.3.2_1 4Rx TDD FR1 RI reporting for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194981 | 0034 | 1 | F | Adding test case 6.2.2.1.2.2, 2Rx FDD FR1 periodic subband CQI reporting under fading conditions for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194982 | 0053 | 1 | F | Update to 4Rx FDD FR1 PDSCH mapping Type A performance 4x4 MIMO with baseline Rx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194983 | 0054 | 1 | F | Update to 4Rx FDD FR1 PDSCH mapping Type A performance 4x4 MIMO with enhanced Rx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194984 | 0037 | 1 | F | Editorial changes to TS 38.521-4 test case 6.3.2.1.2 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194985 | 0038 | 1 | F | Introduction to TS 38.521-4 test case 6.3.3.1.1 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194986 | 0039 | 1 | F | Introduction to TS 38.521-4 test case 6.3.3.1.2 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194987 | 0040 | 1 | F | Introduction to TS 38.521-4 test case 6.3.3.2.1 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194988 | 0041 | 1 | F | Introduction to TS 38.521-4 test case 6.3.3.2.2 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194989 | 0059 | 1 | F | Modification of 2Rx FDD FR1 PDSCH mapping Type A performance - enhanced Rx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194990 | 0060 | 1 | F | Modification of 2Rx TDD FR1 PDSCH mapping Type A and CSI-RS overlapped with PDSCH performance - baseline Rx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194991 | 0061 | 1 | F | Modification of 2Rx FDD FR1 PDCCH 1 Tx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194992 | 0062 | 1 | F | Modification of 2Rx FDD FR1 PDCCH 2 Tx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194993 | 0042 | 1 | F | Update to test case 5.3.2.2.1 2Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194994 | 0043 | 1 | F | Update to test case 5.3.2.2.2 2Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194995 | 0044 | 1 | F | Update to test case 5.3.3.1.1 4Rx FDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194996 | 0045 | 1 | F | Update to test case 5.3.3.1.2 4Rx FDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194997 | 0046 | 1 | F | Update to test case 5.3.3.2.1 4Rx TDD FR1 PDCCH 1 Tx antenna performance for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194998 | 0047 | 1 | F | Update to test case 5.3.3.2.2 4Rx TDD FR1 PDCCH 2 Tx antenna performance for both SA and NSA | 15.1.0 |
| 2019-06 | RAN5#83 | R5-194999 | 0055 | 1 | F | Update to FR1 demod test case 5.2.2.1.2_1 | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195000 | 0078 | 1 | F | Update to RI Reporting Accuracy test | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195001 | 0049 | 1 | F | Updated to Annexes for performance tests | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195002 | 0068 | 1 | F | Demod section 2-4 update | 15.1.0 |

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|---|---|---|---|---|---|---|---|
| 2019-06 | RAN5#83 | R5-195003 | 0058 | 1 | F | Modification of 2Rx FDD FR1 PDSCH mapping Type A performance - baseline Rx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195088 | 0029 | 1 | F | Editorial Aligning CSI common test parameters with core specification | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195089 | 0031 | 1 | F | Updating of E-UTRA test frequency for DEMOD test cases | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195098 | 0079 | - | F | Performance implementation of FR2 UL demod OTA tests using single pol Rx TE | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195170 | 0052 | 1 | F | Update to 4Rx FDD FR1 PDSCH mapping Type A performance 2x4 MIMO with baseline Rx | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195171 | 0033 | 1 | F | Introducing MU and TT clauses in annex F for Channel State Information reporting test cases | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195172 | 0069 | 1 | F | Annex update for PDSCH PDCCH minimum test time | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195413 | 0067 | 1 | F | Update to section 9 and 10 of Demod spec | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195438 | 0050 | 2 | F | Introducing 5.2.2.1.4_1 2Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195439 | 0051 | 2 | F | Introducing 5.2.3.1.4_1 4Rx FDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195440 | 0064 | 1 | F | Addition of new test case for 2Rx FDD FR1 periodic CQI reporting under AWGN | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195441 | 0065 | 1 | F | Update to 2Rx TDD FR1 periodic CQI reporting under AWGN | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195442 | 0070 | 1 | F | Addition of SDR test case for single carrier in SA mode | 15.1.0 |
| 2019-06 | RAN5#83 | R5-195443 | 0072 | 1 | F | Addition of FR1 SDR test case for CA in NSA mode | 15.1.0 |
| 2019-06 | RAN#84 | - | - | - | - | Administrative release upgrade to match the release of 3GPP TS 38.508-1 and TS 38.521-1 which were upgraded at RAN#84 to Rel-16 due to Rel-16 relevant CR(s) | 16.0.0 |
| 2019-09 | RAN#85 | R5-195558 | 0080 | - | F | Correction to 5.2.2.1.4_1 2Rx FR1 PDSCH LTE-NR coexistence performance | 16.1.0 |
| 2019-09 | RAN#85 | R5-196245 | 0090 | - | F | Correction to 2Rx TDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-196247 | 0092 | - | F | Correction to 5.3.2.2.1 and 5.3.3.2.1 TDD FR1 PDCCH 1Tx performance | 16.1.0 |
| 2019-09 | RAN#85 | R5-196495 | 0097 | - | F | Updated to Annex A for performance tests | 16.1.0 |
| 2019-09 | RAN#85 | R5-196496 | 0098 | - | F | Updated to Annex B for performance tests | 16.1.0 |
| 2019-09 | RAN#85 | R5-196498 | 0100 | - | F | Updated to General clauses for Demod and CSI requirements | 16.1.0 |
| 2019-09 | RAN#85 | R5-196857 | 0119 | - | F | Corrections to PDSCH demod TCs | 16.1.0 |
| 2019-09 | RAN#85 | R5-197370 | 0086 | 1 | F | Updates to 6.2.2.1.2.1, 2Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197371 | 0087 | 1 | F | Updates to 6.2.2.2.2.1, 2Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197372 | 0125 | 1 | F | Modification of 4Rx FDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with baseline receiver for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197373 | 0084 | 1 | F | Clean up test cases 5.3.3.1.1, 5.3.3.1.2, 5.3.3.2.1 and 5.3.3.2.2 for 4Rx PDCCH | 16.1.0 |
| 2019-09 | RAN#85 | R5-197374 | 0099 | 1 | F | Updated to General clauses for performance tests | 16.1.0 |
| 2019-09 | RAN#85 | R5-197375 | 0123 | 1 | F | Modification of FDD FR1 2Rx TypeA baseline and TypeX Rxvr | 16.1.0 |
| 2019-09 | RAN#85 | R5-197376 | 0083 | 1 | F | Clean up test cases 5.3.2.2.1 and 5.3.2.2.2 for 2Rx PDCCH | 16.1.0 |
| 2019-09 | RAN#85 | R5-197377 | 0093 | 1 | F | Correction to FR1 FDD PDSCH mapping Type A performance test cases | 16.1.0 |
| 2019-09 | RAN#85 | R5-197378 | 0095 | 1 | F | Correction to MU and TT for FR1 demodulation test cases | 16.1.0 |
| 2019-09 | RAN#85 | R5-197379 | 0096 | 1 | F | Update to 4Rx FDD FR1 PDSCH mapping Type A performance | 16.1.0 |
| 2019-09 | RAN#85 | R5-197380 | 0117 | 1 | F | Update of Annex F to add new CSI test cases | 16.1.0 |
| 2019-09 | RAN#85 | R5-197512 | 0101 | 1 | F | Update to SA SDR test case | 16.1.0 |
| 2019-09 | RAN#85 | R5-197513 | 0102 | 1 | F | Update to NSA SDR test case | 16.1.0 |
| 2019-09 | RAN#85 | R5-197566 | 0127 | 1 | F | Modification on 2Rx TDD FR1 Single PMI with 4Tx Type1 - SinglePanel codebook for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197567 | 0128 | 1 | F | Introduce 2Rx TDD FR1 Single PMI with 8Tx Type1 - SinglePanel codebook for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197572 | 0126 | 1 | F | Modification of 4Rx FDD FR1 PDSCH mapping Type A performance - 4x4 MIMO with baseline receiver for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197573 | 0091 | 1 | F | Correction to 2Rx TDD FR1 PDSCH mapping Type A performance | 16.1.0 |
| 2019-09 | RAN#85 | R5-197574 | 0105 | 1 | F | Update to TDD FR1 2Rx TypeA Baseline and Type X receiver Demod test cases | 16.1.0 |
| 2019-09 | RAN#85 | R5-197575 | 0107 | 1 | F | Editorial and updates to TS 38.521-4 test case 6.3.2.1.1 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197576 | 0108 | 1 | F | Updates to TS 38.521-4 test case 6.3.2.1.2 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197577 | 0109 | 1 | F | Updates to TS 38.521-4 test case 6.3.3.1.1 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197578 | 0110 | 1 | F | Update to TS 38.521-4 test case 6.3.3.1.2 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197579 | 0111 | 1 | F | Editorial and update to TS 38.521-4 test case 6.3.3.2.1 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197580 | 0112 | 1 | F | Editorial and update to TS 38.521-4 test case 6.3.3.2.2 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197581 | 0120 | 1 | F | Correction of PRACH-ConfigurationIndex for TC 5.2.2.2.1_1 | 16.1.0 |
| 2019-09 | RAN#85 | R5-197582 | 0122 | 1 | F | Update to RI Reporting Accuracy test | 16.1.0 |

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|---|---|---|---|---|---|---|---|
| 2019-09 | RAN#85 | R5-197615 | 0088 | 1 | F | Updates to 6.2.2.1.2.2, 2Rx FDD FR1 periodic subband CQI reporting under fading conditions for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197616 | 0089 | 1 | F | Updates to 6.2.2.2.2.2, 2Rx TDD FR1 periodic subband CQI reporting under fading conditions for both SA and NSA | 16.1.0 |
| 2019-09 | RAN#85 | R5-197648 | 0115 | 2 | F | Update to Annex G to restructure minimum test time tables for Demodulation test cases | 16.1.0 |
| 2019-09 | RAN#85 | R5-197649 | 0116 | 2 | F | Update to Annex G to add minimum test time for CSI test cases | 16.1.0 |
| 2019-12 | RAN#86 | R5-198248 | 0141 | - | F | Updates to Annex F | 16.2.0 |
| 2019-12 | RAN#86 | R5-198281 | 0142 | - | F | Update to FR1 4Rx FDD PDSCH Type A Demodulation performance | 16.2.0 |
| 2019-12 | RAN#86 | R5-198395 | 0151 | - | F | Corrections to E-UTRA configurations for EN-DC test cases | 16.2.0 |
| 2019-12 | RAN#86 | R5-198407 | 0152 | - | F | Correction to 2Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | 16.2.0 |
| 2019-12 | RAN#86 | R5-198408 | 0153 | - | F | Correction to 2Rx and 4Rx TDD FR1 Single PMI with 4Tx Type1 - SinglePanel codebook for both SA and NSA | 16.2.0 |
| 2019-12 | RAN#86 | R5-198409 | 0154 | - | F | Correction to Sections 5.2 and 5.3 | 16.2.0 |
| 2019-12 | RAN#86 | R5-198560 | 0157 | - | F | Updated to Annex A and B for performance tests | 16.2.0 |
| 2019-12 | RAN#86 | R5-198679 | 0161 | - | F | Correction of SchedulingRequestResourceConfig periodicityAndOffset for TC 7.2.2.2.1_1 | 16.2.0 |
| 2019-12 | RAN#86 | R5-198680 | 0162 | - | F | Include PDSCH RMC for PDCCH demod FR1 test cases | 16.2.0 |
| 2019-12 | RAN#86 | R5-199079 | 0137 | 2 | F | Adding new test case 6.2.3.1.2.1, 4Rx FDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | 16.2.0 |
| 2019-12 | RAN#86 | R5-199382 | 0129 | 1 | F | Addition of 5.2.2.1.3_1 2Rx FDD PDSCH mapping Type B | 16.2.0 |
| 2019-12 | RAN#86 | R5-199383 | 0130 | 1 | F | Addition of NR test case 5.2.3.1.2_1-FDD type A CSI-RS overlap 4x4 MIMO | 16.2.0 |
| 2019-12 | RAN#86 | R5-199384 | 0134 | 1 | F | Addition of NR test case 6.2.3.1.1.1-FDD periodical CQI | 16.2.0 |
| 2019-12 | RAN#86 | R5-199385 | 0136 | 1 | F | Addition of NR test case 6.4.2.1_1-FDD RI reporting | 16.2.0 |
| 2019-12 | RAN#86 | R5-199387 | 0149 | 1 | F | Update to starting MCS index for CQI reporting test cases | 16.2.0 |
| 2019-12 | RAN#86 | R5-199388 | 0145 | 1 | F | Update to Annex G for minimum test time for FR2 Demod test cases | 16.2.0 |
| 2019-12 | RAN#86 | R5-199414 | 0131 | 1 | F | Addition of NR test case 5.2.3.1.3_1-FDD type B 2x4 MIMO | 16.2.0 |
| 2019-12 | RAN#86 | R5-199415 | 0132 | 1 | F | Addition of NR test case 5.2.3.2.2_1-TDD type A CSI-RS overlap 2x4 MIMO | 16.2.0 |
| 2019-12 | RAN#86 | R5-199416 | 0133 | 1 | F | Addition of NR test case 5.2.3.2.3_1-TDD type B 2x4 MIMO | 16.2.0 |
| 2019-12 | RAN#86 | R5-199417 | 0135 | 1 | F | Addition of NR test case 6.2.3.2.1.1-TDD periodical CQI | 16.2.0 |
| 2019-12 | RAN#86 | R5-199418 | 0138 | 1 | F | Adding new test case 6.2.3.1.2.2, 4Rx FDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | 16.2.0 |
| 2019-12 | RAN#86 | R5-199419 | 0139 | 1 | F | Adding new test case 6.2.3.2.2.1, 4Rx TDD FR1 periodic wideband CQI reporting under fading conditions for both SA and NSA | 16.2.0 |
| 2019-12 | RAN#86 | R5-199420 | 0140 | 1 | F | Adding new test case 6.2.3.2.2.2, 4Rx TDD FR1 aperiodic subband CQI reporting under fading conditions for both SA and NSA | 16.2.0 |
| 2019-12 | RAN#86 | R5-199421 | 0155 | 1 | F | Correction to chapter 5 and 6 to be aligned with core spec | 16.2.0 |
| 2019-12 | RAN#86 | R5-199422 | 0156 | 1 | F | Editorial correction to CSI reporting tests | 16.2.0 |
| 2019-12 | RAN#86 | R5-199425 | 0146 | 1 | F | Update to FR2 2Rx PDSCH Type A enhanced type X receiver test case | 16.2.0 |
| 2019-12 | RAN#86 | R5-199516 | 0160 | 1 | F | Update PrachConfigIndex in 5.2.3.2.1_1 test case | 16.2.0 |
| 2019-12 | RAN#86 | R5-199525 | 0148 | 1 | F | Clarification on PDCP SDU size for SDR SA Demod test case | 16.2.0 |
| 2019-12 | RAN#86 | R5-199526 | 0147 | 1 | F | Clarification on PDCP SDU size for SDR NSA Demod test case | 16.2.0 |
| 2019-12 | RAN#86 | R5-199527 | 0143 | 1 | F | Update to FR2 2Rx PDSCH Type A baseline receiver test case | 16.2.0 |
| 2019-12 | RAN#86 | R5-199531 | 0144 | 1 | F | Annex update for UE positioning procedure for Demod test cases | 16.2.0 |
| 2019-12 | RAN#86 | R5-199532 | 0150 | 1 | F | Update to FR2 PDCCH Demod test case | 16.2.0 |
| 2019-12 | RAN#86 | R5-199570 | 0158 | 1 | F | Introduction of FR2 CQI test cases | 16.2.0 |
| 2020-03 | RAN#87 | R5-200271 | 0165 | - | F | Update to Demod TC 5.2.3.2.1_1 | 16.3.0 |
| 2020-03 | RAN#87 | R5-200322 | 0166 | - | F | CR to 38.521-4 to introduce isolation procedure | 16.3.0 |
| 2020-03 | RAN#87 | R5-200450 | 0168 | - | F | Addition of message exceptions for Type2 QCL information | 16.3.0 |
| 2020-03 | RAN#87 | R5-201245 | 0170 | 1 | F | Core alignment to 4Rx PDCCH Demod Test Cases | 16.3.0 |
| 2020-03 | RAN#87 | R5-200453 | 0171 | - | F | Correction to FR1 2Rx PDSCH demodulation test cases | 16.3.0 |
| 2020-03 | RAN#87 | R5-200454 | 0172 | - | F | Correction to FR1 4Rx PDSCH demodulation test cases | 16.3.0 |
| 2020-03 | RAN#87 | R5-200455 | 0173 | - | F | Correction to measurement uncertainty and test tolerance for CQI test cases | 16.3.0 |
| 2020-03 | RAN#87 | R5-200456 | 0174 | - | F | Correction to PDCCH demod TCs | 16.3.0 |
| 2020-03 | RAN#87 | R5-200660 | 0175 | - | F | Correcting CQI value in test procedure | 16.3.0 |
| 2020-03 | RAN#87 | R5-200672 | 0178 | - | F | Updated to Annex A and B for performance tests | 16.3.0 |
| 2020-03 | RAN#87 | R5-200682 | 0179 | - | F | Correction to Applicability rules for Performance tests | 16.3.0 |
| 2020-03 | RAN#87 | R5-200710 | 0180 | - | F | Update of TC 5.2.2.1.3_1 2Rx FDD PDSCH mapping Type B | 16.3.0 |
| 2020-03 | RAN#87 | R5-200711 | 0181 | - | F | Update of TC 5.2.3.1.2_1 4Rx FDD PDSCH mapping Type A and CSI-RS overlapped | 16.3.0 |
| 2020-03 | RAN#87 | R5-200712 | 0182 | - | F | Update of TC 5.2.3.1.3_1 4Rx FDD PDSCH mapping Type B | 16.3.0 |

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| 2020-03 | RAN#87 | R5-200713 | 0183 | - | F | Update of TC 5.2.3.2.2_1 4Rx TDD PDSCH mapping Type A and CSI-RS overlapped | 16.3.0 |
| 2020-03 | RAN#87 | R5-200714 | 0184 | - | F | Update of TC 5.2.3.2.3_1 4Rx TDD PDSCH mapping Type B | 16.3.0 |
| 2020-03 | RAN#87 | R5-200718 | 0188 | - | F | Update of Test Tolerance in Annex F | 16.3.0 |
| 2020-03 | RAN#87 | R5-200729 | 0189 | - | F | Core spec alignment for FR1 4Rx FDD PDSCH Type A Demodulation performance | 16.3.0 |
| 2020-03 | RAN#87 | R5-200914 | 0176 | 1 | F | Correction to test case 8.2.2.2.1.1 2 Rx, TDD FR2 periodic CQI reporting under AWGN performance for both SA and NSA | 16.3.0 |
| 2020-03 | RAN#87 | R5-200915 | 0164 | 1 | F | Update of Clause 4 in TS 38.521-4 | 16.3.0 |
| 2020-03 | RAN#87 | R5-200985 | 0169 | 1 | F | Core alignment for FR2 demod test case | 16.3.0 |
| 2020-03 | RAN#87 | R5-201068 | 0187 | 1 | F | Update of TC 6.4.2.1_1 2Rx FDD RI reporting | 16.3.0 |
| 2020-03 | RAN#87 | R5-201090 | 0177 | 1 | F | Replacing derivation paths to 38.331 | 16.3.0 |
| 2020-03 | RAN#87 | R5-201180 | 0167 | 1 | F | Addition of FR2 Demod sustained data rate test case | 16.3.0 |
| 2020-06 | RAN#88 | R5-201816 | 0190 | - | F | Correction to TC 5.2.3.1.1_4 4Rx FDD FR1 PDSCH mapping Type A performance | 16.4.0 |
| 2020-06 | RAN#88 | R5-201945 | 0191 | - | F | Updated to Annex A and B for performance tests | 16.4.0 |
| 2020-06 | RAN#88 | R5-202242 | 0195 | - | F | Clarification of propagation condition for Demod test cases during call setup | 16.4.0 |
| 2020-06 | RAN#88 | R5-202297 | 0198 | - | F | Correction to 4Rx FDD FR1 periodic CQI reporting under AWGN conditions for both SA and NSA | 16.4.0 |
| 2020-06 | RAN#88 | R5-202980 | 0201 | 1 | F | Correction to CSI reporting test cases missing MIMO correlation matrixes | 16.4.0 |
| 2020-06 | RAN#88 | R5-202304 | 0205 | - | F | Correction to FR2 PDCCH demodulation tests | 16.4.0 |
| 2020-06 | RAN#88 | R5-202307 | 0208 | - | F | Editorial correction on the table numbers for Minimum Test Time | 16.4.0 |
| 2020-06 | RAN#88 | R5-202308 | 0209 | - | F | Editorial correction to 4x4 MIMO PDSCH demodulation tests | 16.4.0 |
| 2020-06 | RAN#88 | R5-202736 | 0197 | 1 | F | Message exception correction for Demod test cases | 16.4.0 |
| 2020-06 | RAN#88 | R5-202737 | 0202 | 1 | F | Correction to FR1 aperiodic subband CQI reporting under fading conditions | 16.4.0 |
| 2020-06 | RAN#88 | R5-202738 | 0203 | 1 | F | Correction to FR1 Single PMI with 8Tx TypeI - SinglePanel codebook for both SA and NSA | 16.4.0 |
| 2020-06 | RAN#88 | R5-202739 | 0207 | 1 | F | Correction to message exception and test description in RI tests | 16.4.0 |
| 2020-06 | RAN#88 | R5-202740 | 0196 | 1 | F | Update to FR2 PDSCH Demod test case | 16.4.0 |
| 2020-06 | RAN#88 | R5-202741 | 0211 | 1 | F | Introduction of 8.4.2.2.1 2Rx TDD FR2 RI reporting for both SA and NSA | 16.4.0 |
| 2020-06 | RAN#88 | R5-202742 | 0210 | 1 | F | Editorial correction to Annex C.2 | 16.4.0 |
| 2020-06 | RAN#88 | R5-202743 | 0213 | 1 | F | Update Wireless isolation procedure | 16.4.0 |
| 2020-06 | RAN#88 | R5-202766 | 0212 | 1 | F | Updates of FR2 MU and TT in TS 38.521-4 | 16.4.0 |
| 2020-06 | RAN#88 | R5-202832 | 0214 | 1 | F | Addition of message exceptions for PDSCH test cases | 16.4.0 |
| 2020-06 | RAN#88 | R5-202908 | 0193 | 1 | F | Clarification of disabling Tx diversity for FR2 UE for FR2 Demod testing | 16.4.0 |
| 2020-06 | RAN#88 | R5-202979 | 0199 | 2 | F | Correction to 4Rx TDD FR1 RI reporting | 16.4.0 |
| 2020-06 | RAN#88 | R5-202981 | 0204 | 1 | F | Correction to FR2 CQI reporting tests | 16.4.0 |
| 2020-06 | RAN#88 | R5-202989 | 0192 | 1 | F | Updates to 8.2.2.2.2.1, 2Rx TDD FR2 aperiodic CQI reporting under fading performance for both SA and NSA | 16.4.0 |
| 2020-09 | RAN#89 | R5-203298 | 0215 | - | F | Activate Test Mode in NSA Demod Test Cases | 16.5.0 |
| 2020-09 | RAN#89 | R5-203670 | 0217 | - | F | message contents correction for TC 5.2.3.1.2_1 | 16.5.0 |
| 2020-09 | RAN#89 | R5-203717 | 0219 | - | F | Correction to TC 5.2.3.1.1_1 4Rx FDD FR1 PDSCH mapping Type A performance | 16.5.0 |
| 2020-09 | RAN#89 | R5-203756 | 0220 | - | F | Removing unnecessary IE rbg-Size from message exceptions | 16.5.0 |
| 2020-09 | RAN#89 | R5-203902 | 0221 | - | F | Correction to Annex G minimum test time table | 16.5.0 |
| 2020-09 | RAN#89 | R5-204062 | 0226 | - | F | Correction to PDSCH reference channel | 16.5.0 |
| 2020-09 | RAN#89 | R5-204063 | 0227 | - | F | Correction to 2Rx FDD FR1 periodic wideband CQI reporting under fading conditions | 16.5.0 |
| 2020-09 | RAN#89 | R5-204064 | 0228 | - | F | Correction to LTE-NR coexistence performance | 16.5.0 |
| 2020-09 | RAN#89 | R5-204100 | 0232 | - | F | Update to common test parameters and channel mappings | 16.5.0 |
| 2020-09 | RAN#89 | R5-204101 | 0233 | - | F | Update E-UTRA cell configuration for NSA | 16.5.0 |
| 2020-09 | RAN#89 | R5-204261 | 0235 | - | F | Editorial correction of message exceptions | 16.5.0 |
| 2020-09 | RAN#89 | R5-204774 | 0223 | 1 | F | Test applicability update for all PDSCH mapping type B test cases | 16.5.0 |
| 2020-09 | RAN#89 | R5-204870 | 0222 | 1 | F | Addition of FR1 2Rx TDD PDSCH mapping type B test case | 16.5.0 |
| 2020-09 | RAN#89 | R5-204871 | 0224 | 1 | F | Addition of 4Rx FDD FR1 RI reporting test case | 16.5.0 |
| 2020-09 | RAN#89 | R5-204933 | 0229 | 1 | F | CR to update MU and TT in 38.521-4 | 16.5.0 |
| 2020-09 | RAN#89 | R5-204934 | 0225 | 1 | F | Correction to frequencyDomainAllocation | 16.5.0 |
| 2020-09 | RAN#89 | R5-204935 | 0230 | 1 | F | Correction to MU and TT for FR1 PMI and RI tests | 16.5.0 |
| 2020-09 | RAN#89 | R5-204936 | 0218 | 1 | F | Update to FR2 PDSCH test case | 16.5.0 |
| 2020-09 | RAN#89 | R5-204937 | 0216 | 1 | F | Annex F Update of MU and TT for FR2 PDSCH and PDCCH Demodulation scenario | 16.5.0 |
| 2020-09 | RAN#89 | R5-204938 | 0236 | 1 | F | Update of AWGN flatness in TS 38.521-4 | 16.5.0 |
| 2020-12 | RAN#90 | R5-205920 | 0243 | - | F | Introduction of new test case for FR2 CA PDSCH Demodulation | 16.6.0 |
| 2020-12 | RAN#90 | R5-205925 | 0247 | - | F | Update to FDD LTE-NR coexistence test case | 16.6.0 |
| 2020-12 | RAN#90 | R5-206090 | 0248 | - | F | Correction to 5.2.2.1.4_1 LTE NR coexistence performance | 16.6.0 |
| 2020-12 | RAN#90 | R5-206091 | 0249 | - | F | Correction to 9.4B.1.1 Sustained downlink data rate performance for EN-DC within FR1 | 16.6.0 |

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| 2020-12 | RAN#90 | R5-206092 | 0250 | - | F | Core alignment to FR1 and FR2 CSI test cases | 16.6.0 |
| 2020-12 | RAN#90 | R5-206093 | 0251 | - | F | Clean up on FR2 CQI and RI test cases | 16.6.0 |
| 2020-12 | RAN#90 | R5-206094 | 0252 | - | F | Clean up on FR1 RI test cases | 16.6.0 |
| 2020-12 | RAN#90 | R5-206097 | 0255 | - | F | Correction to incorrect parameter settings for subband CQI tests | 16.6.0 |
| 2020-12 | RAN#90 | R5-206098 | 0256 | - | F | Correction to Message contents for Sustained downlink data rate tests | 16.6.0 |
| 2020-12 | RAN#90 | R5-206163 | 0259 | - | F | Correction in message content of 5.2.2.2.1_1, 5.2.3.2.1_1 test cases | 16.6.0 |
| 2020-12 | RAN#90 | R5-206165 | 0260 | - | F | Update on TB success rate definition in Sustain data rate test cases | 16.6.0 |
| 2020-12 | RAN#90 | R5-206208 | 0262 | - | F | Editorial update of uplink signals | 16.6.0 |
| 2020-12 | RAN#90 | R5-206666 | 0237 | 1 | F | Update of LTE-NR coexistence performance test case 5.2.2.1.4 | 16.6.0 |
| 2020-12 | RAN#90 | R5-206667 | 0238 | 1 | F | Update of LTE-NR coexistence performance test case 5.2.3.1.4 | 16.6.0 |
| 2020-12 | RAN#90 | R5-206668 | 0253 | 1 | F | Correction to number of CQI and HARQ in CQI TCs under fading | 16.6.0 |
| 2020-12 | RAN#90 | R5-206669 | 0254 | 1 | F | Correction to FR1 periodic wideband CQI reporting under fading conditions | 16.6.0 |
| 2020-12 | RAN#90 | R5-206670 | 0258 | 1 | F | Correction of CSI-IM periodicity and offset in 4RX FDD wideband CQI under fading condition | 16.6.0 |
| 2020-12 | RAN#90 | R5-206671 | 0240 | 1 | F | Update to OCNG definition in DEMOD spec | 16.6.0 |
| 2020-12 | RAN#90 | R5-206775 | 0239 | 1 | F | Addition of test case 5.2.2.2.4_1 2Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x2 MIMO with baseline receiver for both SA and NSA | 16.6.0 |
| 2020-12 | RAN#90 | R5-206776 | 0241 | 1 | F | Applicability rules for section 5 CA Demodulation requirements | 16.6.0 |
| 2020-12 | RAN#90 | R5-206777 | 0242 | 1 | F | Applicability rules for section 7 CA Demodulation requirements | 16.6.0 |
| 2020-12 | RAN#90 | R5-206829 | 0263 | 1 | F | Update of Annex F | 16.6.0 |
| 2020-12 | RAN#90 | R5-206830 | 0244 | 1 | F | Update to FR2 PDSCH Demodulation test case | 16.6.0 |
| 2020-12 | RAN#90 | R5-206831 | 0245 | 1 | F | Update to FR2 PDCCH Demodulation test case | 16.6.0 |
| 2020-12 | RAN#90 | R5-206832 | 0246 | 1 | F | Update to FR2 CQI reporting under AWGN test case | 16.6.0 |
| 2020-12 | RAN#90 | R5-206833 | 0261 | 1 | F | CR on MU and testability limit for FR2 demod test case | 16.6.0 |
| 2021-03 | RAN#91 | R5-210520 | 0275 | - | F | Correction to SR config for TDD PDSCH Type A performance test cases | 16.7.0 |
| 2021-03 | RAN#91 | R5-210521 | 0276 | - | F | Correction to test applicability for LTE-NR coexistence performance test cases | 16.7.0 |
| 2021-03 | RAN#91 | R5-210522 | 0277 | - | F | Correction to wideband CQI reporting under fading test cases | 16.7.0 |
| 2021-03 | RAN#91 | R5-210523 | 0278 | - | F | Addition of 8.3.2.2.1 2Rx TDD FR2 Single PMI with 2TX TypeI-SinglePanel Codebook | 16.7.0 |
| 2021-03 | RAN#91 | R5-210770 | 0282 | - | F | Update message content in test case 7.3.2.2.2 | 16.7.0 |
| 2021-03 | RAN#91 | R5-210773 | 0283 | - | F | Correction in 6.4.2.1_1 test requirements | 16.7.0 |
| 2021-03 | RAN#91 | R5-210868 | 0284 | - | F | Correction to Table F.1.1.2-2 for FR1 test cases | 16.7.0 |
| 2021-03 | RAN#91 | R5-210869 | 0285 | - | F | Correction to Test Purpose of PDCCH test cases | 16.7.0 |
| 2021-03 | RAN#91 | R5-210993 | 0288 | - | F | Editorial, cleanup of some references in 38.521-4 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211050 | 0289 | - | F | Updating applicability in test case 5.2.2.2.4_1 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211081 | 0293 | - | F | Update to downlink physical channel EPRE level for LTE-NR coex scenario | 16.7.0 |
| 2021-03 | RAN#91 | R5-211086 | 0296 | - | F | Adding new CSI test cases to annex F | 16.7.0 |
| 2021-03 | RAN#91 | R5-211658 | 0297 | 1 | F | Addition of new test case 6.3.2.1.3 2Rx FDD FR1 Multiple PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | 16.7.0 |
| 2021-03 | RAN#91 | R5-211659 | 0298 | 1 | F | Addition of new test case 6.3.3.1.3 4Rx FDD FR1 Multiple PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | 16.7.0 |
| 2021-03 | RAN#91 | R5-211716 | 0280 | 1 | F | Correction to DCI bit size for PDSCH Type B performance and LTE coexistence tests | 16.7.0 |
| 2021-03 | RAN#91 | R5-211717 | 0281 | 1 | F | Correction to LB setup DRB in CLOSE UE TEST LOOP message | 16.7.0 |
| 2021-03 | RAN#91 | R5-211718 | 0286 | 1 | F | Correction to NR test case 6.2.2.1.2.1 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211719 | 0273 | 1 | F | Correction to E-UTRA link setup for NSA testing | 16.7.0 |
| 2021-03 | RAN#91 | R5-211813 | 0290 | 1 | F | Adding new test case 6.3.2.2.3, 2Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | 16.7.0 |
| 2021-03 | RAN#91 | R5-211814 | 0292 | 1 | F | Adding new test case 6.3.3.2.3, 4Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | 16.7.0 |
| 2021-03 | RAN#91 | R5-211816 | 0274 | 1 | F | Update of minimum conformance requirements for 4Rx FDD FR1 PDSCH in TC 5.2.3.1.1_1 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211817 | 0265 | 1 | F | Addition of Applicability of different requirements for R16 NR HST in 5.1.1.7 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211818 | 0268 | 1 | F | Update of Applicability of requirements for mandatory UE features with capability signalling for R16 NR HST in 5.1.1.4 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211819 | 0269 | 1 | F | Update of Applicability of requirements for optional UE features for R16 NR HST in 5.1.1.3 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211820 | 0264 | 1 | F | Addition of Abbreviations and References for R16 NR HST in 3.3 and References | 16.7.0 |
| 2021-03 | RAN#91 | R5-211821 | 0266 | 1 | F | Addition of HST-DPS Channel Profile in B.3.3 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211822 | 0267 | 1 | F | Addition of HST-SFN Channel Profile in B.3.2 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211823 | 0270 | 1 | F | Update of Combinations of channel model parameters for R16 NR HST in B.2.2 | 16.7.0 |

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| 2021-03 | RAN#91 | R5-211824 | 0271 | 1 | F | Update of Reference measurement channels for PDSCH performance requirements for R16 NR HST in A.3.2 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211825 | 0272 | 1 | F | Update of Single Tap Channel Profile for R16 NR HST in B.3.1 | 16.7.0 |
| 2021-03 | RAN#91 | R5-211916 | 0291 | 1 | F | Adding new test case 6.3.2.2.4, 2Rx TDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | 16.7.0 |
| 2021-03 | RAN#91 | R5-211929 | 0299 | 1 | F | Update of FR2 demod test cases | 16.7.0 |
| 2021-06 | RAN#92 | R5-212063 | 0301 | - | F | Addition of test applicability rules for UE supporting FR2 DL 256QAM | 16.8.0 |
| 2021-06 | RAN#92 | R5-212064 | 0302 | - | F | Updating on annexes for FR2 DL 256QAM test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-212067 | 0303 | - | F | Addition of new test case 6.3.2.1.4 2Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | 16.8.0 |
| 2021-06 | RAN#92 | R5-212068 | 0304 | - | F | Addition of new test case 6.3.3.1.4 4Rx FDD FR1 Single PMI with 32Tx Type1 - SinglePanel codebook for both SA and NSA | 16.8.0 |
| 2021-06 | RAN#92 | R5-212254 | 0308 | - | F | Update MU and TT for 8.4.2.2.1 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212632 | 0311 | - | F | Correction of E-UTRA link settings | 16.8.0 |
| 2021-06 | RAN#92 | R5-212635 | 0312 | - | F | Correction of DL RMC for TC 5.2.3.1.4_1 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212743 | 0314 | - | F | Update to Demod test cases title | 16.8.0 |
| 2021-06 | RAN#92 | R5-212933 | 0315 | - | F | Addition of eMIMO demod test case 5.2.2.1.11 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212934 | 0316 | - | F | Addition of eMIMO demod test case 5.2.2.2.11 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212935 | 0317 | - | F | Addition of eMIMO demod test case 5.2.3.1.11 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212936 | 0318 | - | F | Addition of eMIMO demod test case 5.2.3.2.11 | 16.8.0 |
| 2021-06 | RAN#92 | R5-212937 | 0319 | - | F | Adding FRC for eMIMO demod test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-212977 | 0327 | - | F | Updating G.1.2 for performance testing | 16.8.0 |
| 2021-06 | RAN#92 | R5-213306 | 0328 | - | F | Introduction of additional PDSCH RMC for FDD | 16.8.0 |
| 2021-06 | RAN#92 | R5-213308 | 0329 | - | F | Update of message exceptions in FR2 demod test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-213341 | 0330 | - | F | Message content update in 5.2.2.2.1_1 and 5.2.3.2.1_1 test 1-9 | 16.8.0 |
| 2021-06 | RAN#92 | R5-213342 | 0331 | - | F | Message content update in SA LTE-NR coexistence test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-213358 | 0334 | - | F | Addition of FR1 PDSCH Demodulation CA with power imbalance test case | 16.8.0 |
| 2021-06 | RAN#92 | R5-213919 | 0313 | 1 | F | Correction of derivation paths to 38.508-1 | 16.8.0 |
| 2021-06 | RAN#92 | R5-213920 | 0332 | 1 | F | TT update to FR2 CQI reporting under fading test case | 16.8.0 |
| 2021-06 | RAN#92 | R5-214012 | 0326 | 1 | F | Adding 256QAM into CQI reporting test case | 16.8.0 |
| 2021-06 | RAN#92 | R5-214016 | 0325 | 1 | F | Adding FRC for URLLC demod test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-214058 | 0300 | 1 | F | Update of FR2 demod test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-214059 | 0310 | 1 | F | Correction to TC 9.4B.1.1-SDR performance | 16.8.0 |
| 2021-06 | RAN#92 | R5-214088 | 0307 | 1 | F | Update to minimum test time | 16.8.0 |
| 2021-06 | RAN#92 | R5-214098 | 0333 | 1 | F | Addition of FR1 normal PDSCH demodulation CA test case for 2CC | 16.8.0 |
| 2021-06 | RAN#92 | R5-214099 | 0320 | 1 | F | Addition of URLLC demod test case 5.2.2.1.5 | 16.8.0 |
| 2021-06 | RAN#92 | R5-214100 | 0321 | 1 | F | Addition of URLLC demod test case 5.2.2.2.5 | 16.8.0 |
| 2021-06 | RAN#92 | R5-214101 | 0322 | 1 | F | Addition of URLLC demod test case 5.2.3.1.5 | 16.8.0 |
| 2021-06 | RAN#92 | R5-214102 | 0323 | 1 | F | Addition of URLLC demod test case 5.2.3.2.5 | 16.8.0 |
| 2021-06 | RAN#92 | R5-214103 | 0324 | 1 | F | Adding MU and TT for URLLC demod test cases | 16.8.0 |
| 2021-06 | RAN#92 | R5-214112 | 0306 | 1 | F | Core alignment of common test parameters for PDCCH demodulation tests | 16.8.0 |
| 2021-09 | RAN#93 | R5-214533 | 0338 | - | F | Updates on FRC for FR2 DL 256QAM | 16.9.0 |
| 2021-09 | RAN#93 | R5-215065 | 0353 | - | F | Core spec alignment of RMC | 16.9.0 |
| 2021-09 | RAN#93 | R5-215084 | 0357 | - | F | Addition of eMIMO demod test case 5.2.2.2.12 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215085 | 0358 | - | F | Addition of eMIMO demod test case 5.2.2.2.13 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215090 | 0363 | - | F | Addition of eMIMO demod test case 5.2.3.2.12 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215091 | 0364 | - | F | Addition of eMIMO demod test case 5.2.3.2.13 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215092 | 0365 | - | F | Addition of eMIMO demod test case 5.2.3.2.14 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215093 | 0366 | - | F | Adding FRC for eMIMO demod test cases | 16.9.0 |
| 2021-09 | RAN#93 | R5-215094 | 0367 | - | F | Adding MU and TT for eMIMO demod test cases | 16.9.0 |
| 2021-09 | RAN#93 | R5-215103 | 0372 | - | F | Addition of URLLC demod test case 5.2.3.2.7 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215342 | 0380 | - | F | Correction to reporting granularity for single PMI TCs | 16.9.0 |
| 2021-09 | RAN#93 | R5-215343 | 0381 | - | F | Correction to test time for measuring CQI in Sub-band CQI TCs | 16.9.0 |
| 2021-09 | RAN#93 | R5-215345 | 0383 | - | F | Correction to DCI bitlength for test 1-5 and 1-6 in TC 5.2.2.2.1_1 and 5.2.3.2.1_1 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215470 | 0387 | - | F | Correction of message exceptions in PDCCH test cases | 16.9.0 |
| 2021-09 | RAN#93 | R5-215609 | 0390 | - | F | MTSU and TT mapping related to Max Device Size in TS 38.521-4 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215610 | 0391 | - | F | Update 9.4B.1.1 message content | 16.9.0 |
| 2021-09 | RAN#93 | R5-215663 | 0394 | - | F | Updates to FR1 2DLCA PDSCH demodulation with power imbalance test case | 16.9.0 |
| 2021-09 | RAN#93 | R5-215665 | 0396 | - | F | Editorial correction to the section 6.2.2.2.2 title | 16.9.0 |
| 2021-09 | RAN#93 | R5-215666 | 0397 | - | F | Update to test coverage across 5G NR architecture options for Demod scenarios | 16.9.0 |
| 2021-09 | RAN#93 | R5-215901 | 0346 | 1 | F | Update FR2 RI test configuration update for TS 38.521-4 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215902 | 0382 | 1 | F | Editorial error correction in Section 7 and 8 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215934 | 0360 | 1 | F | Addition of eMIMO demod test case 5.2.3.1.12 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215937 | 0343 | 1 | F | Update of Annex F for test cases of demodulation for power | 16.9.0 |

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| 2021-09 | RAN#93 | R5-215942 | 0339 | 1 | F | Updates to PDSCH Demodulation Performance for 2DL CA | 16.9.0 |
| 2021-09 | RAN#93 | R5-215944 | 0345 | 1 | F | Update Applicability of requirement for HST-DPS and multi-TRxP test cases | 16.9.0 |
| 2021-09 | RAN#93 | R5-215946 | 0350 | 1 | F | Addition of NR HST Demod TC 5.2.2.1.9 - HST SFN | 16.9.0 |
| 2021-09 | RAN#93 | R5-215947 | 0351 | 1 | F | Addition of NR HST Demod TC 5.2.2.1.10 - HST DPS | 16.9.0 |
| 2021-09 | RAN#93 | R5-215950 | 0368 | 1 | F | Completing CQI reporting test case with 256QAM | 16.9.0 |
| 2021-09 | RAN#93 | R5-215952 | 0369 | 1 | F | Addition of URLLC demod test case 5.2.2.1.7 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215953 | 0370 | 1 | F | Addition of URLLC demod test case 5.2.2.2.7 | 16.9.0 |
| 2021-09 | RAN#93 | R5-215954 | 0374 | 1 | F | Addition of 5.2.2.1.6 2Rx FDD FR1 PDSCH repetitions over multiple slots performance | 16.9.0 |
| 2021-09 | RAN#93 | R5-215955 | 0375 | 1 | F | Addition of 5.2.2.1.8 2Rx FDD FR1 PDSCH pre-emption performance | 16.9.0 |
| 2021-09 | RAN#93 | R5-215956 | 0376 | 1 | F | Addition of 5.2.2.2.6 2Rx TDD FR1 PDSCH repetitions over multiple slots performance | 16.9.0 |
| 2021-09 | RAN#93 | R5-215957 | 0377 | 1 | F | Addition of 5.2.2.2.8 2Rx TDD FR1 PDSCH pre-emption performance | 16.9.0 |
| 2021-09 | RAN#93 | R5-215958 | 0378 | 1 | F | Addition of 5.2.3.1.6 4Rx FDD FR1 PDSCH repetitions over multiple slots performance | 16.9.0 |
| 2021-09 | RAN#93 | R5-215959 | 0379 | 1 | F | Addition of 5.2.3.2.6 4Rx TDD FR1 PDSCH repetitions over multiple slots performance | 16.9.0 |
| 2021-09 | RAN#93 | R5-216021 | 0385 | 1 | F | Clean-up of parameter settings and message contents in 8.4.2.2.1 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216040 | 0384 | 1 | F | Correction to dedicated CORESET ID setting in PDCCH-Config for Standalone | 16.9.0 |
| 2021-09 | RAN#93 | R5-216041 | 0388 | 1 | F | Update of message exceptions | 16.9.0 |
| 2021-09 | RAN#93 | R5-216071 | 0359 | 1 | F | Addition of eMIMO demod test case 5.2.2.2.14 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216072 | 0340 | 1 | F | Addition of 2Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving test case | 16.9.0 |
| 2021-09 | RAN#93 | R5-216073 | 0341 | 1 | F | Addition of 4Rx TDD FR1 PDCCH 1 Tx antenna performance for power saving test case | 16.9.0 |
| 2021-09 | RAN#93 | R5-216074 | 0342 | 1 | F | Addition of 2Rx TDD FR2 PDCCH 1 Tx antenna performance for power saving test case | 16.9.0 |
| 2021-09 | RAN#93 | R5-216075 | 0395 | 1 | F | Updates to FR2 2DLCA PDSCH demodulation test case | 16.9.0 |
| 2021-09 | RAN#93 | R5-216076 | 0337 | 1 | F | Addition of FR2 DL 256QAM demodulation test case | 16.9.0 |
| 2021-09 | RAN#93 | R5-216078 | 0371 | 1 | F | Addition of URLLC demod test case 5.2.3.1.7 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216096 | 0336 | 1 | F | Update of FR2 demod test cases | 16.9.0 |
| 2021-09 | RAN#93 | R5-216112 | 0347 | 1 | F | Addition of NR PS Demod TC 5.3.2.1.3-FR1 FDD 2Rx | 16.9.0 |
| 2021-09 | RAN#93 | R5-216113 | 0348 | 1 | F | Addition of NR PS Demod TC 5.3.3.1.3-FR1 FDD 4Rx | 16.9.0 |
| 2021-09 | RAN#93 | R5-216118 | 0392 | 1 | F | Update to FR2 NSA SDR TC 9.4B.1.2 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216126 | 0354 | 1 | F | Addition of eMIMO demod test case 5.2.2.1.12 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216127 | 0355 | 1 | F | Addition of eMIMO demod test case 5.2.2.1.13 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216128 | 0356 | 1 | F | Addition of eMIMO demod test case 5.2.2.1.14 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216129 | 0361 | 1 | F | Addition of eMIMO demod test case 5.2.3.1.13 | 16.9.0 |
| 2021-09 | RAN#93 | R5-216130 | 0362 | 1 | F | Addition of eMIMO demod test case 5.2.3.1.14 | 16.9.0 |
| 2021-12 | RAN#94 | R5-216786 | 0403 | - | F | Addition of applicability of different requirements with Multi-TRxP | 16.10.0 |
| 2021-12 | RAN#94 | R5-216787 | 0404 | - | F | Update Applicability of requirements for optional UE features | 16.10.0 |
| 2021-12 | RAN#94 | R5-216908 | 0407 | - | F | Addition of NR HST Demod TC 5.2.2.1.1_1 - 2Rx FDD type A | 16.10.0 |
| 2021-12 | RAN#94 | R5-216909 | 0408 | - | F | Addition of NR HST Demod TC 5.2.2.2.1_1 - 2Rx TDD type A | 16.10.0 |
| 2021-12 | RAN#94 | R5-216910 | 0409 | - | F | Correction to NR HST Demod TC 5.2.2.1.9_1 - HST-SFN | 16.10.0 |
| 2021-12 | RAN#94 | R5-217059 | 0412 | - | F | Correction to NR TC 5.3.2.1.1-2Rx FDD FR1 PDCCH 1 Tx antenna performance | 16.10.0 |
| 2021-12 | RAN#94 | R5-217060 | 0413 | - | F | Correction to NR TC 5.3.3.2.1-PDCCH 1 Tx antenna performance | 16.10.0 |
| 2021-12 | RAN#94 | R5-217363 | 0430 | - | F | Addition of RMC in Annex A for eMIMO enhanced typeII CSI reporting | 16.10.0 |
| 2021-12 | RAN#94 | R5-217364 | 0431 | - | F | Addition of B.2.3.2.3A Beam steering approach with dual cluster beams | 16.10.0 |
| 2021-12 | RAN#94 | R5-217370 | 0434 | - | F | Updating 5.2.x.y.7 PDSCH with UE processing capability 2 | 16.10.0 |
| 2021-12 | RAN#94 | R5-217375 | 0439 | - | F | Update to Annex F for URLLC test cases | 16.10.0 |
| 2021-12 | RAN#94 | R5-217377 | 0441 | - | F | Update to applicability of optional features for URLLC test cases | 16.10.0 |
| 2021-12 | RAN#94 | R5-217378 | 0442 | - | F | Update to URLLC RMC for demodulation testing in Annex A | 16.10.0 |
| 2021-12 | RAN#94 | R5-217437 | 0443 | - | F | Correction to frequencyDomainResources in PDCCHConfigCommon message exception | 16.10.0 |
| 2021-12 | RAN#94 | R5-217522 | 0446 | - | F | Addition of DL and UL RMC for FR2 SDR test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-217525 | 0449 | - | F | Updates to FR1 normal PDSCH CA test cases | 16.10.0 |
| 2021-12 | RAN#94 | R5-217526 | 0450 | - | F | Updates to FR2 normal PDSCH CA test cases | 16.10.0 |
| 2021-12 | RAN#94 | R5-218248 | 0424 | 1 | F | Updating minimum test time in Annex G | 16.10.0 |
| 2021-12 | RAN#94 | R5-218308 | 0425 | 1 | F | Addition of 6.3.2.1.6 2Rx FDD FR1 Multiple PMI with 16Tx Enhanced Type II codebook | 16.10.0 |
| 2021-12 | RAN#94 | R5-218309 | 0426 | 1 | F | Addition of 6.3.2.2.6 2Rx TDD FR1 Multiple PMI with 16Tx Enhanced Type II codebook | 16.10.0 |
| 2021-12 | RAN#94 | R5-218310 | 0427 | 1 | F | Addition of 6.3.3.1.6 4Rx FDD FR1 Multiple PMI with 16Tx | 16.10.0 |

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| | | | | | | Enhanced Type II codebook | |
| 2021-12 | RAN#94 | R5-218311 | 0428 | 1 | F | Addition of 6.3.3.2.6 4Rx TDD FR1 Multiple PMI with 16Tx Enhanced Type II codebook | 16.10.0 |
| 2021-12 | RAN#94 | R5-218312 | 0429 | 1 | F | Addition of applicability of optional features in 6.1.1.3 | 16.10.0 |
| 2021-12 | RAN#94 | R5-218328 | 0405 | 1 | F | Correction to PS Demod TC 5.3.2.1.3 - 2Rx | 16.10.0 |
| 2021-12 | RAN#94 | R5-218329 | 0406 | 1 | F | Correction to PS Demod TC 5.3.3.1.3 - 4Rx | 16.10.0 |
| 2021-12 | RAN#94 | R5-218339 | 0420 | 1 | F | Addition of new test case 6.3.2.1.5 2Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218340 | 0421 | 1 | F | Addition of new test case 6.3.2.2.5 2Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218341 | 0422 | 1 | F | Addition of new test case 6.3.3.1.5 4Rx FDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218342 | 0423 | 1 | F | Addition of new test case 6.3.3.2.5 4Rx TDD FR1 Multiple PMI with 16Tx TypeII codebook for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218343 | 0444 | 1 | F | Correction to TC 5.2.2.2.4_1 and editorial corrections | 16.10.0 |
| 2021-12 | RAN#94 | R5-218344 | 0447 | 1 | F | Addition of NE-DC SDR test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-218345 | 0399 | 1 | F | Addition of new test case 5.2.3.1.9_1 for NR HST | 16.10.0 |
| 2021-12 | RAN#94 | R5-218346 | 0400 | 1 | F | Addition of new test case 5.2.3.1.10_1 for NR HST | 16.10.0 |
| 2021-12 | RAN#94 | R5-218347 | 0414 | 1 | F | Addition of test case 5.2.2.2.9_1, 2Rx TDD FR1 HST-SFN performance - 2x2 MIMO with baseline receiver for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218348 | 0415 | 1 | F | Addition of test case 5.2.2.2.10_1, 2Rx TDD FR1 HST-DPS performance - 2x2 MIMO with baseline receiver for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218349 | 0416 | 1 | F | Addition of test case 5.2.3.2.9_1, 4Rx TDD FR1 HST-SFN performance - 2x4 MIMO with baseline receiver for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218350 | 0418 | 1 | F | Addition of test cases 5.2.2.2.9_1, 5.2.2.2.10_1, 5.2.3.2.9_1 to annex F | 16.10.0 |
| 2021-12 | RAN#94 | R5-218357 | 0419 | 1 | F | Update of URLLC demodulation Test Cases | 16.10.0 |
| 2021-12 | RAN#94 | R5-218358 | 0433 | 1 | F | Updating 5.2.x.y.5 PDSCH with 1e-5 BLER | 16.10.0 |
| 2021-12 | RAN#94 | R5-218359 | 0435 | 1 | F | Addition of 6.2.2.1.1.2 URLLC 2RX FDD CQI reporting test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-218360 | 0440 | 1 | F | Addition of statistical testing limit for URLLC test cases in Annex G | 16.10.0 |
| 2021-12 | RAN#94 | R5-218436 | 0453 | 1 | F | Update to LTE-NR coex test case message exception | 16.10.0 |
| 2021-12 | RAN#94 | R5-218462 | 0432 | 1 | F | Addition of MU and TT in Annex F for enhanced typeII CSI reporting | 16.10.0 |
| 2021-12 | RAN#94 | R5-218465 | 0401 | 1 | F | Addition of PDCCH Search Space Ext configuration for power saving test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-218466 | 0417 | 1 | F | Update of test case 5.2.3.2.1_1, 4Rx TDD FR1 PDSCH mapping Type A performance - 2x4 MIMO with baseline receiver for both SA and NSA | 16.10.0 |
| 2021-12 | RAN#94 | R5-218467 | 0436 | 1 | F | Addition of 6.2.2.2.1.2 URLLC 2RX TDD CQI reporting test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-218468 | 0437 | 1 | F | Addition of 6.2.3.1.1.2 URLLC 4RX FDD CQI reporting test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-218469 | 0438 | 1 | F | Addition of 6.2.3.2.1.2 URLLC 4RX TDD CQI reporting test case | 16.10.0 |
| 2021-12 | RAN#94 | R5-218486 | 0445 | 1 | F | Clarification on cl 4.6 test coverage across 5G NR architecture options for Demod | 16.10.0 |
| 2022-03 | RAN#95 | R5-220276 | 0454 | - | F | Clarifications on 5G NR connectivity options for Demod | 16.11.0 |
| 2022-03 | RAN#95 | R5-220629 | 0460 | - | F | Correction to demod TC 5.2.2.1.4_1 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220630 | 0461 | - | F | Correction to demod TC 5.2.3.2.1_1 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220634 | 0463 | - | F | Updates to HST test case 5.2.3.1.9_1 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220635 | 0464 | - | F | Updates to HST test case 5.2.3.1.10_1 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220638 | 0467 | - | F | Addition of fading profile power uncertainty for 4Tx, FR1 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220651 | 0468 | - | F | Editorial correction for test case title in Annex F | 16.11.0 |
| 2022-03 | RAN#95 | R5-220664 | 0469 | - | F | Editorial change for the position of clause 5.2.3.1.9 and 5.2.3.1.10 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220678 | 0471 | - | F | Correcting applicability part of HST test cases in 38.521-4 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220684 | 0473 | - | F | Addition of new RMCs to Annex | 16.11.0 |
| 2022-03 | RAN#95 | R5-220686 | 0475 | - | F | Correcting test applicability for EN-DC, rel-16 to rel-15 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220751 | 0476 | - | F | Correction to PS Demod TC 5.3.2.1.3 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220764 | 0477 | - | F | Updating test case 6.3.2.2.3, 2Rx TDD FR1 Single PMI with 16Tx Type1 - SinglePanel codebook for both SA and NSA | 16.11.0 |
| 2022-03 | RAN#95 | R5-220796 | 0478 | - | F | Update to eMIMO demod test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-220820 | 0487 | - | F | Adding testability description of 7.2.2.2.2 and 7.2.2.2.3 | 16.11.0 |
| 2022-03 | RAN#95 | R5-220936 | 0493 | - | F | Editorial correction to 5.3.3.1.3 and 5.3.3.2.3 | 16.11.0 |
| 2022-03 | RAN#95 | R5-221153 | 0494 | - | F | Update to FR1 CA normal PDSCH test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221154 | 0495 | - | F | Update to FR1 CA power imbalance test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221155 | 0496 | - | F | Update to FR2 CA normal PDSCH test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221707 | 0499 | 1 | F | Editorial update to PBCH demod requirements section | 16.11.0 |
| 2022-03 | RAN#95 | R5-221708 | 0500 | 1 | F | Update to testability of test requirements due to achievable SNR improvements | 16.11.0 |
| 2022-03 | RAN#95 | R5-221709 | 0501 | 1 | F | FR1 NSA SDR message contents update | 16.11.0 |
| 2022-03 | RAN#95 | R5-221710 | 0491 | 1 | F | Correction to Annex H.1.2 | 16.11.0 |

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| 2022-03 | RAN#95 | R5-221842 | 0455 | 1 | F | Correction on Type I PMI test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221843 | 0456 | 1 | F | Addition of FR1 CA CQI test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221844 | 0457 | 1 | F | Addition of applicability for FR1 CA CQI test requirements | 16.11.0 |
| 2022-03 | RAN#95 | R5-221845 | 0474 | 1 | F | Addition of test case 5.2.3.2.4_1, 4Rx TDD FR1 PDSCH Mapping Type A and LTE-NR coexistence performance - 4x4 MIMO with baseline receiver for both SA and NSA | 16.11.0 |
| 2022-03 | RAN#95 | R5-221846 | 0497 | 1 | F | Introduction of FR1 CA SDR test case | 16.11.0 |
| 2022-03 | RAN#95 | R5-221847 | 0458 | 1 | F | Addition of FR2 CA CQI test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221848 | 0459 | 1 | F | Addition of applicability for FR2 CA CQI test requirements | 16.11.0 |
| 2022-03 | RAN#95 | R5-221854 | 0470 | 1 | F | Addition of test case 5.2.3.2.10_1, 4Rx TDD FR1 HST DPS performance - 2x4 MIMO with baseline receiver for both SA and NSA | 16.11.0 |
| 2022-03 | RAN#95 | R5-221855 | 0502 | 1 | F | Update to HST Demod test cases | 16.11.0 |
| 2022-03 | RAN#95 | R5-221856 | 0465 | 1 | F | Addition of HST test case 5.2.3.1.9_1 to annex F | 16.11.0 |
| 2022-03 | RAN#95 | R5-221857 | 0466 | 1 | F | Addition of HST test case 5.2.3.1.10_1 to annex F | 16.11.0 |
| 2022-03 | RAN#95 | R5-221860 | 0480 | 1 | F | Update to 5.2.x.y.5 PDSCH with 1e-5 BLER | 16.11.0 |
| 2022-03 | RAN#95 | R5-221861 | 0481 | 1 | F | Update to 5.2.x.y.6 PDSCH with repetitions over multiple slots | 16.11.0 |
| 2022-03 | RAN#95 | R5-221862 | 0483 | 1 | F | Update to 5.2.2.y.8 PDSCH pre-emption | 16.11.0 |
| 2022-03 | RAN#95 | R5-221863 | 0484 | 1 | F | Addition of 5.2.3.1.8 PDSCH pre-emption 4Rx FDD | 16.11.0 |
| 2022-03 | RAN#95 | R5-221864 | 0485 | 1 | F | Addition of 5.2.3.2.8 PDSCH pre-emption 4Rx TDD | 16.11.0 |
| 2022-03 | RAN#95 | R5-221865 | 0488 | 1 | F | Addition of 7.2.2.2.2 FR2 PDSCH repetition | 16.11.0 |
| 2022-03 | RAN#95 | R5-221866 | 0489 | 1 | F | Addition of 7.2.2.2.3 FR2 PDSCH mapping Type B | 16.11.0 |
| 2022-03 | RAN#95 | R5-221867 | 0479 | 1 | F | Addition of minimum test time for 1% residual BLER | 16.11.0 |
| 2022-03 | RAN#95 | R5-221868 | 0486 | 1 | F | Update to Annex F for URLLC test cases | 16.11.0 |

# History

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